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Preface

It is with great enthusiasm and renewed commitment that we present the proceedings of the 2026 7th International Conference on Robotics Systems and Automation Engineering (RSAE 2026), held in Shanghai, China, from June 26 to 28, 2026.

Co-sponsored by the IEEE and Shanghai University of Engineering Science (SUES), and hosted by School of Mechanical and Automotive Engineering, SUES, China, RSAE 2026 builds upon the strong foundation of its predecessor while embracing evolving challenges and opportunities at the intersection of robotics, automation, and intelligent systems. The continued support from these esteemed institutions underscores their shared vision for advancing responsible innovation, interdisciplinary synergy, and globally inclusive scientific exchange.

The mission of RSAE 2026 remains centered on strengthening international collaboration and cultivating a vibrant, inclusive forum for researchers, engineers, educators, and industry practitioners. This year's program expanded its scope to address emerging frontiers—including path planning and motion control for mobile robots; machine vision-driven environmental perception and intelligent control for robots; control and intelligent decision-making for unmanned autonomous systems based on signal sensing.

All papers published in the proceedings underwent a rigorous double-blind peer-review process, ensuring scientific rigor, methodological soundness, and practical relevance across academic and industrial domains. We extend our deepest appreciation to the reviewers whose meticulous feedback and constructive insights significantly elevated the quality and impact of this volume.

We also express profound gratitude to the authors for their outstanding contributions; to session chairs and all the committee members for their expert guidance; to the organizing and local arrangements teams for their tireless coordination. Together, they transformed RSAE 2026 into a catalyst for dialogue, discovery, and enduring partnerships.

As we look ahead, we warmly invite you to join us again for RSAE 2027.

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Boundary-Aware Finite-Time LOS Guidance for Path Following of Underactuated USV

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***Abstract*—This paper proposes a hierarchical path-following control scheme for underactuated unmanned surface vehicles (USVs). A boundary-aware finite-time line-of-sight (BAF-LOS) guidance law is formulated to achieve fast convergence while enforcing cross-track error constraints. A composite control strategy, which integrates a finite-time disturbance observer (FTDO) with an FTDO-based adaptive controller, is designed to ensure robust tracking in the presence of time-varying disturbances. Lyapunov-based analysis establishes the constraint satisfaction and finite-time stability of the closed-loop system. The simulation results on the CyberShip II model demonstrate the effectiveness of the proposed method.**

***Index Terms*—path following, line-of-sight (LOS) guidance, unmanned surface vehicles, error constraints**

I. Introduction

The underactuated unmanned surface vehicle (USV) is increasingly utilized in marine missions due to its cost-efficiency and agility [1]. Path following, a fundamental motion control problem for USV [2], aims to steer the vessel along a predefined geometric path with specific velocity and attitude requirements, independent of temporal constraints. The achievement of path following depends critically on the design of both the guidance law and the dynamic controller.

Regarding guidance laws, the line-of-sight (LOS) method [3] is widely adopted due to its physical interpretation. Various methods have been proposed to advance LOS guidance, such as adaptive LOS [4], integral LOS [5] and predictor-based LOS [6]. To accelerate the convergence rate, a finite-time LOS guidance law incorporating fractional power terms has been developed [7], and Huang [8] proposed a finite-time convergence LOS to achieve rapid convergence of position errors by a novel cross-track error function. Meanwhile, to ensure navigation safety, Zheng [9] proposed an error-constrained LOS to strictly bound cross-track errors within a permissible range. However, the above method cannot achieve a balance between transient performance and navigation safety in path following within restricted environments.

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Regarding the controller design, common methods include backstepping control, PID control, and sliding mode control. However, traditional PID fails to handle marine nonlinearities [10], and backstepping control suffers from the explosion of complexity. Although the sliding mode control offers robustness, it introduces harmful chattering [11].

Existing schemes struggle to balance convergence speed with state constraints, and the integration of constrained guidance with robust anti-disturbance control remains a challenge. The main innovations of this paper are summarized as follows:

1) A boundary-aware finite-time LOS (BAF-LOS) guidance law is proposed, which incorporates a barrier gain and a fractional-power term. It guarantees that the cross-track error strictly remains within the prescribed safety boundary while converging to zero in finite time.
2) A finite-time disturbance observer (FTDO) is designed with additional linear error terms. This design ensures a rapid transient response even under large initial estimation errors and achieves exact disturbance estimation within a finite time.
3) A hierarchical control framework integrating the FTDO with an adaptive finite-time controller is designed to ensure the USV robustly follows the desired heading generated by the guidance law.

The remainder of this paper is organized as follows. Section II presents the USV model and path-following error dynamics. Section III describes the proposed guidance law and control design. Section IV presents simulation results, and Section V concludes the paper.

II. Problem Formulation

This section establishes the mathematical model of the USV and introduces the system control objectives.

A. USV Dynamic Model

Three coordinate frames are defined to describe the USV's motion, as shown in Fig. 1: the Earth-fixed North-East inertial frame $\{i\}$, the body-fixed frame $\{b\}$, and the Serret-Frenet

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frame $\{f\}$ aligned with the path tangent. Ignoring the model

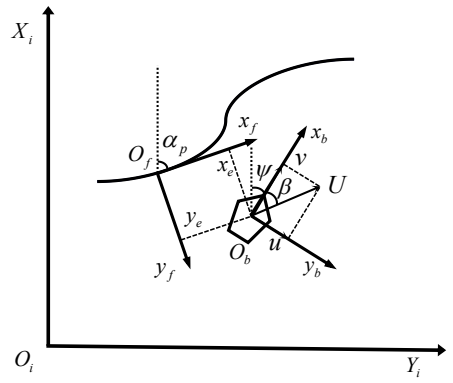

Fig. 1. Illustration of USV path following.

uncertainty and the offset of USV's center of gravity, the 3-DOF kinematics and kinetics of the USV are modeled as [12]:

$$\begin{cases} \dot{x} = u\cos\psi - v\sin\psi \\ \dot{y} = u\sin\psi + v\cos\psi \\ \dot{\psi} = r \\ \dot{u} = \dfrac{1}{m_{11}}\left(\tau_u + m_{22}vr - d_{11}u + \tau_{du}\right) \\ \dot{v} = \dfrac{1}{m_{22}}\left(-m_{11}ur - d_{22}v + \tau_{dv}\right) \\ \dot{r} = \dfrac{1}{m_{33}}\left(\tau_r + (m_{11}-m_{22})uv - d_{33}r + \tau_{dr}\right) \end{cases} \tag{1}$$

where $\eta = [x, y, \psi]^T$ denotes the position and heading in the inertial frame $\{i\}$, and $\nu = [u, v, r]^T$ represents the surge, sway, and yaw velocities in the body-fixed frame $\{b\}$. The model parameters are defined as $m_{11} = m - X_{\dot{u}}$, $m_{22} = m - Y_{\dot{v}}$, $m_{33} = I_z - N_{\dot{r}}$, $d_{11} = -X_u - X_{|u|u}|u|$, $d_{22} = -Y_v - Y_{|v|v}|v|$, and $d_{33} = -N_r - N_{|r|r}|r|$. Here, m_{ii} denotes the total inertial terms including added mass, and d_{ii} represents the hydrodynamic damping terms. τ_u and τ_r are the control input force and moment, respectively, while $\tau_{du}, \tau_{dv}, \tau_{dr}$ represent the time-varying environmental disturbances.

B. Control Objectives

The desired path to be followed is defined as $p(\omega) = [x_p(\omega), y_p(\omega)]^T$ and $\alpha_p = \mathrm{atan2}(y_p'(\omega), x_p'(\omega))$ is the tangential angle of the path. To facilitate the control design, the position error in the inertial frame $\{i\}$ is transformed into the Serret-Frenet frame $\{f\}$. Specifically, the virtual target on the path is selected as the orthogonal projection of the USV's position onto the path. The error transformation is defined as:

$$\begin{bmatrix} x_e \\ y_e \end{bmatrix} = \begin{bmatrix} \cos\alpha_p & \sin\alpha_p \\ -\sin\alpha_p & \cos\alpha_p \end{bmatrix} \begin{bmatrix} x - x_p \\ y - y_p \end{bmatrix}, \tag{2}$$

where x_e represents the along-track error, y_e denotes the cross-track error.

By means of orthogonal projection, the along-track error is negligible $x_e = 0$. Consequently, the derivative of the cross-track error y_e is derived as [10]:

$$\begin{aligned} \dot{y}_e &= \dot{y}\cos\alpha_p - \dot{x}\sin\alpha_p \\ &= U\sin(\psi+\beta)\cos\alpha_p - U\cos(\psi+\beta)\sin\alpha_p \\ &= U\sin(\psi - \alpha_p + \beta), \end{aligned} \tag{3}$$

where $U = \sqrt{u^2 + v^2}$ represents the speed of the vessel, and $\beta = \mathrm{atan2}(v, u)$ is the sideslip angle, as shown in Fig. 1.

Assumption 1: [13] The system states, including position (x, y), heading ψ, and velocities $[u, v, r]^T$, are measurable or can be estimated by onboard sensors, and the sideslip angle β is bounded.

Assumption 2: [14] The time-varying environmental disturbances are continuous and differentiable, and their time derivatives are bounded, satisfying $|\dot{\tau}_{dj}| = m_{jj}|\dot{\delta}_j| \le m_{jj}L_j$, where L_j is a positive constant for $j \in \{u, v, r\}$, the term m_{jj} corresponds to the inertia term m_{11} for $j = u$ and m_{33} for $j = r$.

Based on the aforementioned assumptions and the requirements for USV path following control, a BAF-LOS-based USV control system is designed in the next section to achieve the following tracking objectives:

1) Boundary Constraint: The cross-track error satisfies $|y_e(t)| < k_c$ for $\forall t > t_0$, where k_c is a positive constant representing the safety boundary.
2) Fast Convergence: The cross-track error y_e converges to a small neighborhood of the reference path within a finite time T, such that $|y_e(t)| < \varepsilon_1$ for all $t \ge T$, where ε_1 is a positive constant.

C. Preliminaries

To facilitate the design of the proposed control scheme, the following lemmas are introduced.

Lemma 1: [15] Consider a system as follows:

$$\dot{x}(t) = f(x(t)), \quad x(0) = 0, \quad f(0) = 0, \quad x \in U_0 \subset \mathbb{R}^n \tag{4}$$

where $x = [x_1, \ldots, x_n]^T$ and $f(\cdot)$ are continuous on an open neighborhood U_0 of the origin. If there exists a positive definite function $V(x)$ satisfying

$$\dot{V}(x) + \lambda V^{\rho}(x) \le 0 \tag{5}$$

where $\lambda > 0$ and $0 < \rho < 1$. Then, the origin of the system is stable in finite time, and $T \le \frac{V(0)^{1-\rho}}{\lambda(1-\rho)}$.

Lemma 2: [16] For any real symmetric matrix $A \in \mathbb{R}^{n\times n}$ and any non-zero vector $x \in \mathbb{R}^n$, the Rayleigh quotient satisfies the following inequality:

$$\lambda_{\min}(A)\|x\|^2 \le x^T A x \le \lambda_{\max}(A)\|x\|^2 \tag{6}$$

where $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote the minimum and maximum eigenvalues of the matrix A, respectively.

III. Control Design

This section details the design of the control system for the USV.

A. Design of Boundary-Aware Finite-Time LOS

To ensure that the USV remains within the prescribed safety bound k_c while the cross-track error y_e converges to zero smoothly and rapidly, the BAF-LOS guidance law is designed as:

$$\psi_d = \alpha_p - \arctan\left(\frac{f_{BAF}}{\Delta}\right) - \beta \tag{7}$$

where $f_{BAF} = \mathcal{K}(y_e)k_y \mathrm{sig}(y_e)^\gamma$, and $\mathcal{K}(y_e) = \left(\frac{k_c^2}{k_c^2 - y_e^2}\right)^p$ represents the barrier gain, k_c is the safety limit. Δ is the look-ahead distance. The design parameters are chosen such that $p \in (0,1)$, $k_y > 0$, $0 < \gamma < 1$. Additionally, the function $\mathrm{sig}(\cdot)$ is defined as $\mathrm{sig}(x)^\gamma = |x|^\gamma \mathrm{sgn}(x)$.

Remark 1: In practice, safe initialization requires verifying $|y_e(0)| < k_c$ before engaging the controller.

Theorem 1: Consider the USV system driven by the proposed BAF-LOS guidance law. Provided that the initial cross-track error satisfies $|y_e(0)| < k_c$, the cross-track error y_e is strictly constrained within the safety boundary and will converge to zero within a finite time T_g.

Proof: Assign the logarithmic barrier Lyapunov function:

$$V_g = \frac{1}{2}\ln\left(\frac{k_c^2}{k_c^2 - y_e^2}\right). \tag{8}$$

Substituting (7) into $\dot{y}_e$, the derivative of the cross-track error is given by:

$$\begin{aligned} \dot{y}_e &= U\sin(\psi - \alpha_p + \beta) \\ &= U\sin(-\arctan(\frac{f_{BAF}}{\Delta})) \\ &= -U\frac{f_{BAF}}{\sqrt{\Delta^2 + f_{BAF}^2}}. \end{aligned} \tag{9}$$

Taking the time derivative of V_g and substituting (9) yields:

$$\begin{aligned} \dot{V}_g &= \frac{y_e \dot{y}_e}{k_c^2 - y_e^2} \\ &= -\left(\frac{U}{(k_c^2 - y_e^2)\sqrt{\Delta^2 + f_{BAF}^2}}\right) y_e f_{BAF} \\ &= -\left(\frac{U}{\sqrt{\Delta^2 + f_{BAF}^2}}\right)\frac{k_y}{k_c^2}\left(\frac{k_c^2}{k_c^2 - y_e^2}\right)^{1+p} |y_e|^{\gamma+1} \\ &\le 0. \end{aligned} \tag{10}$$

Since $\dot{V}_g \le 0$ holds, it implies that $V_g(t) \le V_g(0) < \infty$. Therefore, $|y_e| < k_c$ holds strictly for any time.

Consequently, the composite term f_{BAF} is bounded. Let its maximum upper bound be F_{max}. Then, there exists another bound M such that:

$$\frac{U}{\sqrt{\Delta^2 + f_{BAF}^2}} \ge \frac{U}{\sqrt{\Delta^2 + F_{max}^2}} = M > 0. \tag{11}$$

Assign the other Lyapunov function $V_e = \frac{1}{2}y_e^2$, taking its time derivative yields:

$$\begin{aligned} \dot{V}_e &= y_e \dot{y}_e \\ &= -y_e \frac{U f_{BAF}}{\sqrt{\Delta^2 + f_{BAF}^2}} \\ &\le -M y_e f_{BAF} \\ &= -M\left(\frac{k_c^2}{k_c^2 - y_e^2}\right)^p k_y |y_e|^{\gamma+1} \\ &\le -M k_y |y_e|^{\gamma+1} \\ &= -\lambda_e V_e^{\rho_e}, \end{aligned} \tag{12}$$

where $\lambda_e = M k_y 2^{\frac{\gamma+1}{2}} > 0$ and $\rho_e = \frac{\gamma+1}{2} \in (\frac{1}{2}, 1)$. According to Lemma 1, the error y_e will converge to zero within a finite time bounded by $T_g \le \frac{V_e(0)^{1-\rho_e}}{\lambda_e(1-\rho_e)}$.

A second-order command filter (SOCF) is applied to ψ_d to generate smooth reference signals while mitigating high-frequency noise [17]:

$$\begin{cases} \dot{\psi}_c = r_c \\ \dot{r}_c = -2\zeta\omega_n r_c - \omega_n^2(\psi_c - \psi_d), \end{cases} \tag{13}$$

where ω_n and ζ are the natural frequency and damping ratio. These parameters are tuned to ensure sufficient bandwidth to track the guidance command.

B. Finite-Time Disturbance Observer Design

In this subsection, an FTDO is designed to exactly identify external disturbances encountered by the USV in finite time. We rewrite the dynamics model of the USV in the following decoupled form:

$$\dot{\nu}_j = \frac{1}{m_{jj}}(\tau_j + f_j) + \delta_j, \quad j \in \{u, r\} \tag{14}$$

where $\nu_u = u, \nu_r = r$. The term $\delta_j = \frac{\tau_{dj}}{m_{jj}}$ denotes the unknown lumped disturbance to be estimated. f_j represents the Coriolis and damping terms, explicitly defined as:

$$\begin{cases} f_u = m_{22} v r - d_{11} u \\ f_r = (m_{11} - m_{22}) u v - d_{33} r. \end{cases} \tag{15}$$

Then, the FTDO is proposed as follows:

$$\begin{cases} \dot{\hat{\nu}}_j = \hat{\delta}_j + \frac{1}{m_{jj}}(\tau_j + f_j) + l_{j,1}\mathrm{sig}(\nu_j - \hat{\nu}_j)^{\frac{1}{2}} + l_{j,2}(\nu_j - \hat{\nu}_j) \\ \dot{\hat{\delta}}_j = l_{j,3}\mathrm{sgn}(\nu_j - \hat{\nu}_j) + l_{j,4}(\nu_j - \hat{\nu}_j) \end{cases} \tag{16}$$

where $\hat{\nu}_j$ and $\hat{\delta}_j$ denote the estimates of the velocity state ν_j and the lumped disturbance δ_j, respectively. The parameters $l_{j,1}, l_{j,2}, l_{j,3}, l_{j,4}$ are positive observer gains, for brevity, the subscript j is omitted in the subsequent proof.

Theorem 2: According to Assumption 2. If the observer gains l_1, l_2, l_3, l_4 satisfy the condition

$$\begin{cases} l_1 l_3 + \frac{l_1^3}{2} - l_1 L > 0 \\ |Q_1| > 0 \\ l_2 l_3 + 2l_1^2 l_2 - l_2 L > 0 \\ |Q_2| > 0 \end{cases} \tag{17}$$

where $|\cdot|$ denotes the determinant of matrix $\cdot$, Q_1 and Q_2 are defined as:

$$\begin{cases} Q_1 = \frac{1}{2}\begin{bmatrix} 2l_1l_3+l_1^3-2l_1L & 0 & -l_1^2-2L \\ 0 & 2l_1l_4+5l_1l_2^2 & -3l_1l_2 \\ -l_1^2-2L & -3l_1l_2 & l_1 \end{bmatrix} \\ Q_2 = \begin{bmatrix} l_2l_3+2l_1^2l_2-l_2L & 0 & 0 \\ 0 & l_2l_4+l_2^3 & -l_2^2 \\ 0 & -l_2^2 & l_2 \end{bmatrix}, \end{cases} \tag{18}$$

then, the proposed FTDO can accurately estimate the lumped disturbances in finite time T_δ.

Proof: Define the FTDO estimation errors as $e_1 = \nu_j - \hat{\nu}_j$ and $e_2 = \delta_j - \hat{\delta}_j$. Subtracting (14) from (16), we have

$$\begin{cases} \dot{e}_1 = -l_1 \mathrm{sig}(e_1)^{\frac{1}{2}} - l_2 e_1 + e_2 \\ \dot{e}_2 = \dot{\delta} - l_3 \mathrm{sgn}(e_1) - l_4 e_1 \end{cases} \tag{19}$$

Define $\Theta = [|e_1|^{\frac{1}{2}}\mathrm{sgn}(e_1), e_1, e_2]^T$. Consider the Lyapunov function V_o:

$$\begin{aligned} V_o &= 2l_3|e_1| + l_4 e_1^2 + \frac{1}{2}e_2^2 + \frac{1}{2}\left(l_1 \mathrm{sig}(e_1)^{\frac{1}{2}} + l_2 e_1 - e_2\right)^2 \\ &= \Theta^T P \Theta, \end{aligned} \tag{20}$$

where $P = \frac{1}{2}\begin{bmatrix} 4l_3+l_1^2 & l_1l_2 & -l_1 \\ l_1l_2 & 2l_4+l_2^2 & -l_2 \\ -l_1 & -l_2 & 2 \end{bmatrix}$. V_o is continuous everywhere and differentiable everywhere except $e_1 = 0$, taking the time derivative of V_o:

$$\begin{aligned} \dot{V}_o = {} & 2l_3 \mathrm{sgn}(e_1)\dot{e}_1 + 2l_4 e_1 \dot{e}_1 + e_2 \dot{e}_2 \\ & + \left(l_1 \mathrm{sig}(e_1)^{\frac{1}{2}} + l_2 e_1 - e_2\right) \\ & \times \left(\frac{1}{2} l_1 |e_1|^{-\frac{1}{2}} \dot{e}_1 + l_2 \dot{e}_1 - \dot{e}_2\right), \end{aligned} \tag{21}$$

according to Assumption 2, $|\dot{\delta}| \le L$ and $e_1 \le |e_1|, e_2 \le |e_2|$, substituting (19) to (21) yields:

$$\dot{V}_o \le -|e_1|^{-\frac{1}{2}} \Phi^T Q_1 \Phi - \Phi^T Q_2 \Phi, \tag{22}$$

where $\Phi = [|e_1|^{\frac{1}{2}}, e_1, e_2]^T$. Note that $\Phi^T\Phi = \Theta^T\Theta$, as (17) shows, Q_1 and Q_2 are positive definite matrices. According to Lemma 2

$$\begin{aligned} \dot{V}_o &\le -|e_1|^{-\frac{1}{2}} \Phi^T Q_1 \Phi \\ &\le -|e_1|^{-\frac{1}{2}} \lambda_{\min}(Q_1) \Theta^T \Theta \\ &\le -|e_1|^{-\frac{1}{2}} \lambda_{\min}(Q_1) \frac{V_o}{\lambda_{\max}(P)}, \end{aligned} \tag{23}$$

we have:

$$\begin{aligned} V_o^{\frac{1}{2}} &\ge \sqrt{\lambda_{\min}(P)} |\Theta^T\Theta|^{\frac{1}{2}} \\ &\ge \sqrt{\lambda_{\min}(P)} |e_1|^{\frac{1}{2}}, \end{aligned} \tag{24}$$

from Lemma 2, then

$$|e_1|^{\frac{1}{2}} \le \frac{V_o^{\frac{1}{2}}}{\sqrt{\lambda_{\min}(P)}}. \tag{25}$$

Combine (23) and (25), we have:

$$\dot{V}_o \le -\lambda_o V_o^{\frac{1}{2}}, \tag{26}$$

where $\lambda_o = \frac{\lambda_{\min}(Q_1)\sqrt{\lambda_{\min}(P)}}{\lambda_{\max}(P)}$. According to Lemma 1, the errors converge to zero within a finite time $T_\delta \le \frac{2V_o^{\frac{1}{2}}(0)}{\lambda_o}$, indicating $\hat{\delta}_j = \delta_j$ for $t \ge T_\delta$.

Remark 2: The estimation errors e_1 cannot be stabilized in $e_1 = 0$, because by the first equation in (19), if $e_1 = 0$ then $\dot{e}_1 = e_2 \ne 0$. Consequently, the equilibrium point $e_1 = e_2 = 0$ can be reached in finite time.

C. Controller Design

To ensure robust control of the USV, an FTDO-based adaptive controller is designed.

Defining the error equations:

$$\begin{cases} e_u = u - u_d \\ e_r = (r - r_c) + \alpha_k(\psi - \psi_c), \end{cases} \tag{27}$$

where $\alpha_k > 0$ is a designed gain, e_u is the surge tracking error, and e_r is the heading tracking error.

The control law is as follows:

$$\begin{cases} \tau_u = -f_u - m_{11}\hat{\delta}_u + m_{11}\left(\dot{u}_d - K_{u,1}e_u - K_{u,2}\mathrm{sig}(e_u)^{\gamma_u}\right) \\ \tau_r = -f_r - m_{33}\hat{\delta}_r + m_{33}\Big(\dot{r}_c - \alpha_k(r - r_c) \\ \qquad - K_{r,1}e_r - K_{r,2}\mathrm{sig}(e_r)^{\gamma_r}\Big), \end{cases} \tag{28}$$

where $K_{u,1} = k_{u,1} + \alpha_u|e_u|, K_{u,2} = k_{u,2} + \beta_u|e_u|, K_{r,1} = k_{r,1} + \alpha_r|e_r|, K_{r,2} = k_{r,2} + \beta_r|e_r|$, $k_{u,1}, k_{u,2}, k_{r,1}, k_{r,2}, \alpha_u, \beta_u, \alpha_r, \beta_r$ are positive control gains, and $0 < \gamma_u, \gamma_r < 1$ is the designed fractional parameter.

Theorem 3: Consider the USV model given in (1). Under the control law τ_u, τ_r in (28) and satisfying Assumption 1, the tracking error e_u and e_r will converge to zero in finite time.

Proof: Define the Lyapunov functions $V_u = \frac{1}{2}e_u^2$ and $V_r = \frac{1}{2}e_r^2$. Substituting the control laws (28) into the functions yields:

$$\begin{aligned} \dot{V}_u &= e_u\Bigg[\frac{1}{m_{11}}\Big(f_u + \tau_{du} - f_u - m_{11}\hat{\delta}_u \\ &\quad + m_{11}(\dot{u}_d - K_{u,1}e_u - K_{u,2}\mathrm{sig}(e_u)^{\gamma_u})\Big) - \dot{u}_d\Bigg] \\ &= e_u\left(-K_{u,1}e_u - K_{u,2}\mathrm{sig}(e_u)^{\gamma_u}\right) \\ &= -k_{u,1}e_u^2 - k_{u,2}|e_u|^{\gamma_u+1} - \alpha_u|e_u|e_u^2 - \beta_u|e_u|^{\gamma_u+2} \\ &\le -k_{u,2}|e_u|^{\gamma_u+1} = -\lambda_u V_u^{\rho_u}, \end{aligned} \tag{29}$$

where $\lambda_u = k_{u,2} \cdot 2^{\frac{\gamma_u+1}{2}}$ and $\rho_u = \frac{\gamma_u+1}{2}$. Given the design parameter $0 < \gamma_u < 1$, the exponent satisfies $\frac{1}{2} < \rho < 1$. According to Lemma 1, the surge tracking error e_u converges to zero within a finite time $T_u \le \frac{V_u^{1-\rho_u}(0)}{\lambda_u(1-\rho_u)}$,

$$\begin{aligned} \dot{V}_r &= -k_{r,1}e_r^2 - k_{r,2}|e_r|^{\gamma_r+1} - \alpha_r|e_r|e_r^2 - \beta_r|e_r|^{\gamma_r+2} \\ &\le -k_{r,2}|e_r|^{\gamma_r+1} = -\lambda_r V_r^{\rho_r}, \end{aligned} \tag{30}$$

where $\lambda_r = k_{r,2} \cdot 2^{\frac{\gamma_r+1}{2}}$ and $\rho_r = \frac{\gamma_r+1}{2} \in (\frac{1}{2}, 1)$. Thus, the generalized error e_r converges to zero within a finite time $T_r \le \frac{V_r^{1-\rho_r}(0)}{\lambda_r(1-\rho_r)}$.

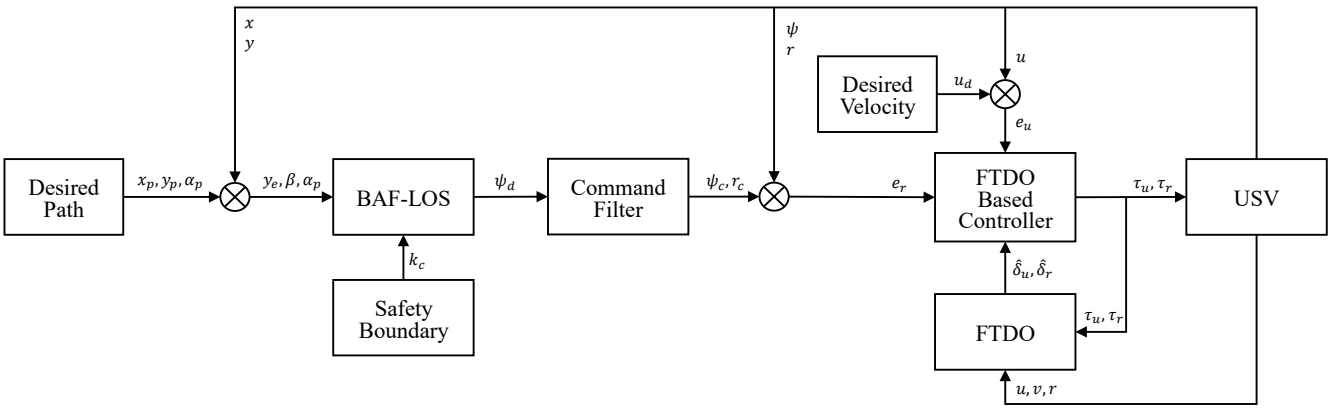

Fig. 2. Control scheme of USV

IV. Simulation Results

This section presents the simulation results of the proposed scheme, LOS, and FTLOS implemented on the CyberShip II. The control system parameters and CyberShip II model parameters [18] are listed in Table I and the complete control scheme is illustrated in Fig. 2. To verify the robustness of the proposed scheme against environmental disturbances, time-varying disturbances representing wind, waves, and currents are introduced into the dynamic model as $\tau_{du} = 2.05\sin(0.2t) + 8.0[N], \tau_{dv} = 0.8\sin(0.3t + \pi/4) + 4.0[N], \tau_{dr} = 0.6\cos(0.3t) + 1.5[N \cdot m]$.

TABLE I
CyberShip II and Control System Parameters

Param.	Value	Param.	Value	Param.	Value
m_{11}	25.8 kg	$X_{\|u\|u}$	1.33 kg/m	γ	0.5
m_{22}	33.8 kg	Y_v	0.89 kg/s	ω_n	2.0 rad/s
m_{33}	2.76 kg·m^2	$Y_{\|v\|v}$	36.5 kg/m	ζ	0.9
N_r	1.90 kg·m^2/s	Δ	10 m	u_d	2.0 m/s
$N_{\|r\|r}$	0.75 kg·m^2	k_c	10 m	α_k	0.7
X_u	0.72 kg/s	k_y	1.4	γ_u, γ_r	0.6, 0.6
p	0.5	α_u, β_u	0.5, 0.3	α_r, β_r	1.0, 0.8
$k_{u,1}, k_{u,2}$	1.0, 0.5	$k_{r,1}, k_{r,2}$	2.0, 1.5	$l_{u,1}, l_{u,2}$	1.5, 1.0
$l_{u,3}, l_{u,4}$	2.0, 1.5	$l_{r,1}, l_{r,2}$	2.5, 1.5	$l_{r,3}, l_{r,4}$	3.0, 2.0

To quantitatively evaluate the path-following performance, three metrics are introduced to compare the proposed BAF-LOS with FTLOS and traditional LOS:

1) Convergence time T_s, defined as the time required for the cross-track error to satisfy and stay within $|y_e| \leq 0.15$ m.
2) Integral Absolute Error (IAE), which is calculated as $\int_0^{t_{end}} |y_e(\tau)| d\tau$ to reflect the overall tracking accuracy.
3) Maximum transient cross-track error $\max |y_e|$, to verify the safety constraint.

The quantitative results are summarized in Table II. The proposed BAF-LOS scheme achieves the smallest IAE and the fastest convergence time among the three methods. Most importantly, only BAF-LOS strictly restricts the maximum transient error below the prescribed safety boundary ($k_c = 10$ m), while FTLOS and LOS violate this constraint.

TABLE II
Quantitative Performance of Guidance Laws

Metric	BAF-LOS	FTLOS	LOS
Convergence Time T_s (s)	12.61	17.96	-
IAE of y_e (m·s)	50.36	62.96	227.39
Max Transient $\|y_e\|$ (m)	9.92	10.20	10.45

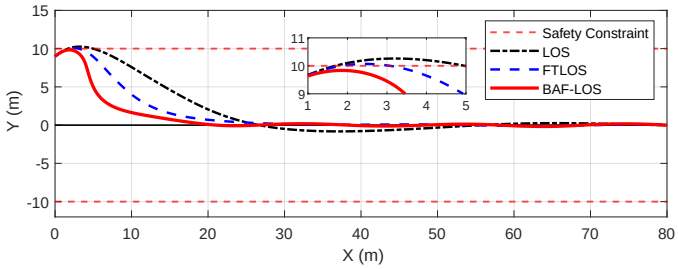

Fig. 3. Path following trajectories.

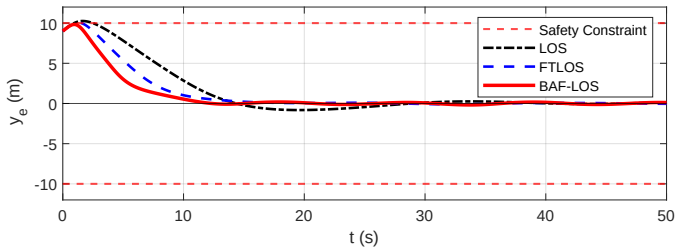

Fig. 4. Cross-track error y_e.

A comparative simulation is designed to verify the safety and agility of the BAF-LOS scheme. With the USV initialized at $(0, 9.0)$ m ($u = 2.5$ m/s, $\psi = \pi/5$ rad, $v = 1.0$ m/s), the system operates under a strict constraint of $k_c = 10$

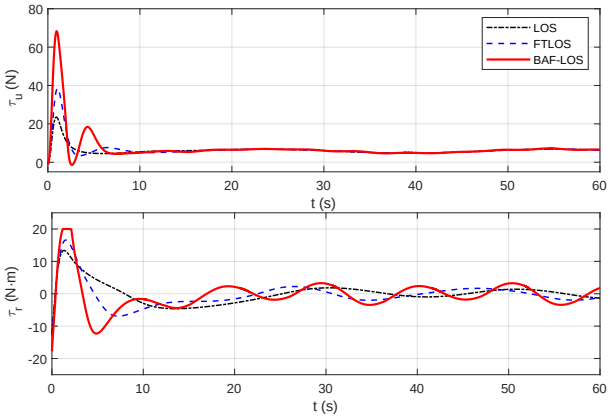

Fig. 5. Control inputs.

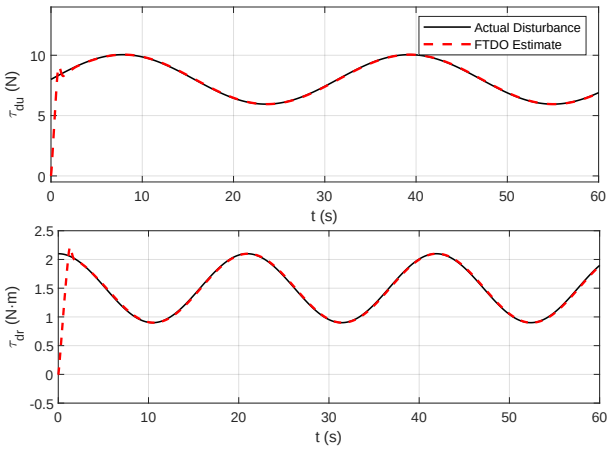

Fig. 6. Estimation of disturbances.

m. The surge speed is regulated at a constant $u_d = 2$ m/s. Based on the given USV model, the proposed BAF-LOS guidance law is compared with the existing FTLOS and LOS guidance laws. The comparison of the different guidance law is shown in Fig. 3–4. It can be observed that the proposed BAF-LOS guidance law not only exhibits a distinct advantage in convergence speed, but also strictly satisfies the predefined boundary constraints. As shown in Fig. 4, the BAF-LOS control strategy tracks the desired trajectory much faster within the same simulation duration. Fig. 5 and Fig. 6 show the control inputs and the disturbance estimation performance of FTDO, respectively. In summary, the proposed scheme provides a highly effective solution for the boundary-constrained path tracking problem of USV.

V. Conclusion

This paper studies the path-following control problem of underactuated USV operating in constrained environments under external disturbances. A BAF-LOS guidance law is proposed to achieve fast convergence while enforcing boundary constraints. A hierarchical control framework combining an SOCF, an FTDO, and an FTDO-based adaptive controller is developed to ensure smooth and robust tracking. Lyapunov-based analysis establishes forward invariance of the constrained set and finite-time stability of the closed-loop system. The simulation results on the CyberShip II model demonstrate the effectiveness and improved convergence performance of the proposed method.

References

[1] Z. Peng, J. Wang, D. Wang, and Q.-L. Han, "An overview of recent advances in coordinated control of multiple autonomous surface vehicles," *IEEE Trans. Ind. Inform.*, vol. 17, no. 2, pp. 732–745, 2021.

[2] X. Yang, H. She, M. Lou, H. Ye, J. Guan, J. Li, Z. Xiang, H. Shen, and B. Zhang, "A joint ship detection and waterway segmentation method for environment-aware of USVs in canal waterways," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 2132–2144, 2024.

[3] T. I. Fossen, M. Breivik, and R. Skjetne, "Line-of-sight path following of underactuated marine craft," in *Proceedings of IFAC*, vol. 36, no. 21, 2003, pp. 211–216.

[4] T. I. Fossen, "An adaptive line-of-sight (ALOS) guidance law for path following of aircraft and marine craft," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 6, pp. 2887–2894, 2023.

[5] W. Caharija, K. Y. Pettersen, M. Bibuli, P. Calado, E. Zereik, J. Braga, J. T. Gravdahl, A. J. Sorensen, M. Milovanovic, and G. Bruzzone, "Integral line-of-sight guidance and control of underactuated marine vehicles: Theory, simulations, and experiments," *IEEE Trans. Control Syst. Technol.*, vol. 24, no. 5, pp. 1623–1642, 2016.

[6] L. Liu, D. Wang, Z. Peng, and H. Wang, "Predictor-based LOS guidance law for path following of underactuated marine surface vehicles with sideslip compensation," *Ocean Eng.*, vol. 124, pp. 340–348, 2016.

[7] M. Li, C. Guo, H. Yu, and Y. Yuan, "Line-of-sight-based global finite-time stable path following control of unmanned surface vehicles with actuator saturation," *ISA Trans.*, vol. 125, pp. 306–317, 2022.

[8] Y. Huang, G. Lai, F. Lin, X. Shi, and D. Li, "Path following with finite-time convergence for unmanned surface vehicle," *IEEE J. Ocean. Eng.*, 2025.

[9] Z. Zheng, L. Sun, and L. Xie, "Error-constrained LOS path following of a surface vessel with actuator saturation and faults," *IEEE Trans. Syst. Man Cybern. Syst.*, vol. 48, no. 10, pp. 1794–1805, 2017.

[10] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. Chichester: John Wiley & Sons, 2011.

[11] K. D. Young, V. I. Utkin, and U. Ozguner, "A control engineer's guide to sliding mode control," *IEEE Trans. Control Syst. Technol.*, vol. 7, no. 3, pp. 328–342, 1999.

[12] N. Wang, Z. Sun, J. Yin, S.-F. Su, and S. Sharma, "Finite-time observer based guidance and control of underactuated surface vehicles with unknown sideslip angles and disturbances," *IEEE Access*, vol. 6, pp. 14 059–14 070, 2018.

[13] G. Zhang, X. Zhang, and Y. Zheng, "Adaptive neural path-following control for underactuated ships in fields of marine practice," *Ocean Eng.*, vol. 104, pp. 558–567, 2015.

[14] N. Wang, Z. Sun, Y. Jiao, and G. Han, "Surge-heading guidance-based finite-time path following of underactuated marine vehicles," *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8523–8532, 2019.

[15] S. P. Bhat and D. S. Bernstein, "Finite-time stability of continuous autonomous systems," *SIAM J. Control Optim.*, vol. 38, no. 3, pp. 751–766, 2000.

[16] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge University Press, 2012.

[17] J. A. Farrell, M. Polycarpou, M. Sharma, and W. Dong, "Command filtered backstepping," *IEEE Trans. Autom. Control*, vol. 54, no. 6, pp. 1391–1395, 2009.

[18] R. Skjetne, O. N. Smogeli, and T. I. Fossen, "A nonlinear ship manoeuvring model: Identification and adaptive control with experiments for a model ship," *Model. Identif. Control*, vol. 25, no. 1, pp. 3–27, 2004.

A High-Sensitivity Torque Sensor based on Strain Gauge for Collaborative Robot Joints

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***Abstract*—A collaborative robot needs information on the external force applied to the main body of the robot for collision safety and precise work. This function can be performed using a joint torque sensor mounted between the reducer and the link, but a torque sensor with high sensitivity is required for accurate torque measurement and various errors to minimize the performance of the robot. Through the circuit configuration and mechanical design analysis of the sensor, a torque sensor with high sensitivity and excellent interference error cancellation effect was developed in this study. The developed torque sensor can offset the interference error by constructing a Wheatstone bridge with four strain gauges. In this paper, the performance of the proposed torque sensor was verified through various experiments.**

Keywords—torque sensor, strain gauge, sensitivity

I. INTRODUCTION

Recently, research on collaborative robots capable of human-robot collaboration has been actively conducted. These collaborative robots are equipped with various safety functions, and a function to detect and respond to physical collisions is required, especially when physical collisions occur.[1] In addition, it is necessary to be able to accurately measure the external force applied to the end of the robot in order to perform precise work. To this end, many collaborative robots measure the torque applied to each joint by mounting a joint torque sensor on each joint. This enables force control and collision detection in all parts of the robot arm. In addition, the external force applied to the end may be estimated using the torque measured at each joint and the dynamic information of the robot。And, Commercial torque sensors generally provide stiffness that is 4-6 times that of the harmonic drive used with the torque sensor. [2,3] Therefore, when it is combined with the robot joint, there is a decrease in sensitivity of about 15-20%.

In general, the torque sensor is mounted between the output terminal and the output link of the reducer as shown in Fig. 1. Therefore, when the sensitivity of the torque sensor is low, the sensitivity of the robot joint is also lowered, which leads to a decrease in the overall sensitivity of the robot. When the sensitivity of the robot decreases, the position accuracy of the end is lowered, so the rotational and moment sensitivity of the torque sensor is a very important factor when designing the torque sensor. [4,5]

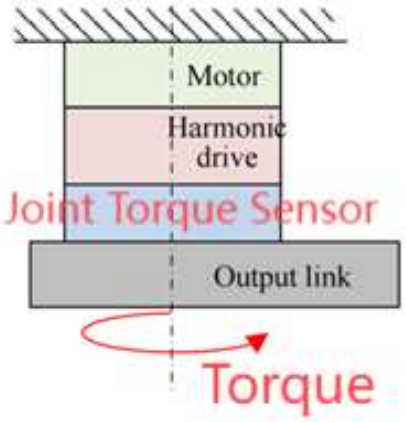


Fig. 1. Schematic diagram of joint structure

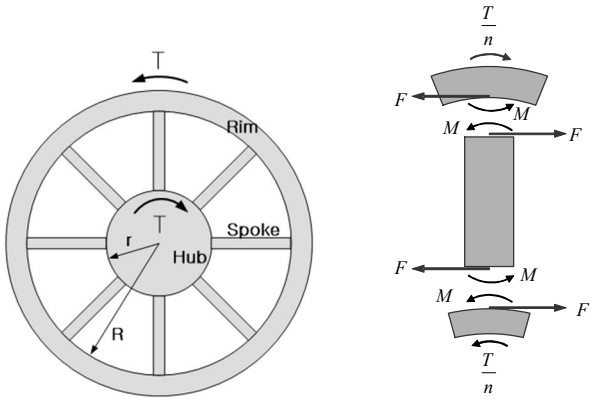
Therefore, in this study, through the circuit configuration of the strain gauge and the mechanical shape design of the sensor, mutual interference errors were eliminated, a torque sensor for collaborative robot joints with higher rigidity than commercial torque sensors were proposed, and performance evaluation was performed through experiments.

II. DESIGN AND ANALYSIS OF TORQUE SENSOR

A. Principle of torque detection

The torque sensor consists of an outer wheel and an inner wheel, and four spokes connecting them as shown in Fig. 2(a). The inner wheel is fixed to the reducer output terminal of Fig. 1, and the outer wheel is fixed to the output link. The torque load applied to the link is transmitted through the outer wheel, and accordingly, the outer wheel rotates relatively with respect to the inner wheel. In this process, the spoke connecting the outer wheel and the inner wheel is deformed as shown in Fig. 2(b). Deformation of the spoke is measured through a strain gauge attached to the side of the spoke, through which the torque is measured. [6,7]

The rotational force is transmitted through the active wheel and the driven wheel, causing subtle deformation on the spokes that transmit this force. The strain was detected using a strain gauge composed of Wheatstone bridge circuits. Fig.2 is the shear force acting on the end of the spokes.



(a) (b)

Fig. 2. (a) Schematic view of the torque meter; (b) Forces and moments acting upon spokes, hub and rim

According to the literature[8], as shown in Fig.3, the deformation of spokes is:

$$F = \frac{T}{n} \times \frac{2}{(R+r)}$$

$$\frac{T}{n} + M - Fr = 0$$

$$M = \frac{T}{n} \times \frac{(R-r)}{(R+r)}$$

$$M_x = M - Fx$$

$$\sigma_x = \frac{M_x y}{I}$$

$$where\ I = \int_{-t/2}^{t/2} y^2 b\, dy$$

$$\text{if } y = \frac{t}{2},\ I_{\max} = \frac{bt^3}{12},\ \sigma = \sigma_{\max}$$

$$\sigma_{\max} = \frac{M_0 y_{\max}}{I_{\max}} = \frac{M_0 \times t/2}{bt^3/12} = \frac{6M}{bt^3}$$

$$\varepsilon_x = \frac{I}{E}\sigma_x = \frac{t}{2EI}(M - Fx)$$

Therefore, the amount of deformation increases as the distance x from the end becomes shorter in proportion to the moment. However, it is not appropriate as the position where the strain gauge is attached because it is difficult to attach to the end and the end effect actually occurs. Therefore, the FEM analysis can be performed, it was concluded that the x=l/4 point is appropriate.

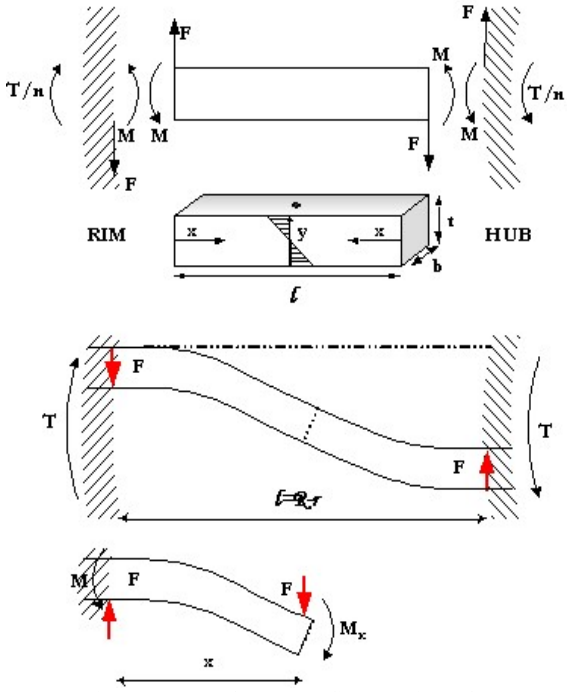
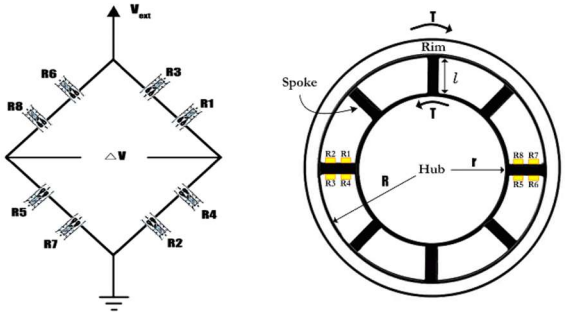


Fig. 3. Forces and moments diagram of a spoke

In order to measure the tensile strain on the side of the spoke generated by torque, attach the strain gauge to the position where the deformation occurs the most through FEM analysis. A Wheatstone bridge circuit as shown in Fig.4(a) is constructed using a total of 4 strain gauges attached to both sides of the four spokes as shown in Fig.4(a). Through this circuit, the output to the input voltage is measured. When a clockwise torque load is applied to the torque sensor as shown in Fig.4(b), tensile deformation occurs in R1,3,6,8and compressive deformation occurs in R2,4,5,7 due to the load applied to the spoke. At this time, the resistance change ΔR shown in the strain gauge is as follows



(a) (b)

Fig. 4. (a) Wheatstone bridge circuit; (b) strain gauge deformation due to torque load

$$\Delta V = \frac{\Delta R}{R_0} V_{ext}$$

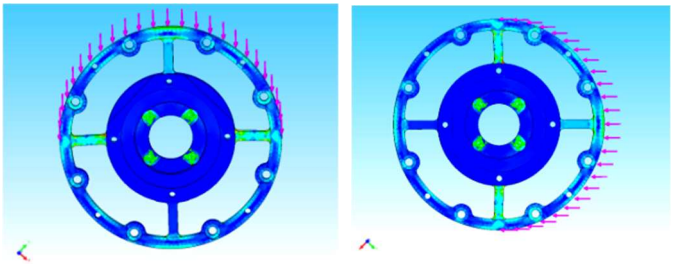
$$\Delta V = GF \times \frac{V_{ext}}{bt^2 E} \times \frac{3(R-r)}{n(R+r)} \times T$$

B. *Principle of Non-torque Load Offset*

The torque sensor applied to the robot joint is applied with torque and moment loads according to the robot's motion and posture. At this time, in order to accurately measure only the torque acting on the joint, it is important that the torque sensor is not affected. [8]

Fig.5(a) is the condition where tension acts perpendicular to the strain gauge installation surface. In this case, tension bending plays the greatest role on the surface where the strain gauge is installed, and the effects of left-right bending are symmetric; therefore, the strain gauge bridge structures cancel each other out.

On the other hand, as shown in Fig.5(b), when the strain gauge attachment surface is subjected to tension in the same direction, the left spokes are directly affected by the tension, while the right spokes transmit pressure via hub and rim. During this process, due to the deformation of the wheel rim, the pressure transmitted to the right spokes decreases, resulting in a smaller strain compared to the left spokes. From this, it can be inferred that the strain gauge bridge structures cancel each other out due to the difference in strain between the left and right spokes.



(a) (b)

Fig. 5. (a) Effects of tension perpendicular to the plane of strain gauge attachment; (b) Effects of tension in the same direction as the strain gauge attachment surface

III. Experiments and Results

In this study, the material of the torque sensor was processed using an low-carbon steel (AIS 1018), and the strain gauge used Vishay's linear type strain gauge (CEA-06-032UW-1205S-350-EL). Fig.6(a) is the shape of the work piece of the torque sensor, and the Wheatstone bridge circuit was constructed with a strain gauge as shown in Fig. 6(b).

(a) (b)

Fig. 6. (a) Torque sensor metal work; (b) The strain gauge attachment surface

The manufactured torque sensor is a device that can determine the magnitude of the torque applied by the voltage change value induced by the deformation of the spoke, so the relationship between torque and deformation must be known in advance through calibration. Calibration was experimented by configuring the test rig as shown in Fig.7.

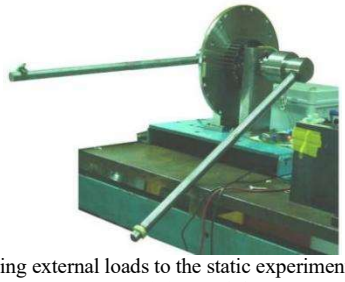


Fig. 7. Applying external loads to the static experimental setup of the robot joint module

When the sprocket is fixed to the rotating part of the test rig and the spoke inside is fixed to the fixing part of the test rig with a bolt, and a weight is attached to the 0.5m bar connected to the rotating part, a force is applied in the direction of gravity by the weight of the weight, causing relative deformation to occur in the spoke. This deformation breaks the voltage balance of the strain gauge and creates a voltage difference that is input to the amplifier built into the torque converter circuit, resulting in a change in the ADC value.

Fig.8 is a graph of repeated loading and unloading experiments while sequentially increasing the torque from 0 to 50 kg (0~245Nm). It was confirmed that it had a good linearity as a torque sensor, and the hysteresis during loading and unloading, which was concerned, was negligible.

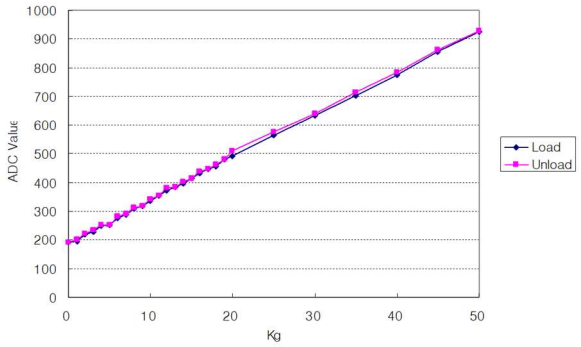


Fig. 8. Output voltage for torque

TABLE I. Summarization of Torque Sensor Sensitivity

Method	$^{*}\varepsilon_x$	Sensitivity [V/Nm]
FEM Analysis	3.16e-6	0.0083
Actual Calibration	3.82e-6	0.01
Analytical calculation	4.51e-6	0.0118

*ε_x is the strain gage actually attachaed.

Table I summarizes comparability results of sensitivity of a torque sensor.

IV. Conclusion

This study proposes a high-sensitivity torque sensor based on strain gauges. To maintain sufficient strain on the sensor detection unit, FEM analysis was conducted, and a Wheatstone bridge circuit was configured to offset loads other than those from the torque sensor. Actual torque sensors were fabricated, and their performance was confirmed through experiments.

When applied to robot joints, this can maximize sensitivity while minimizing the rigidity of the robotic arm. Additionally, due to low mutual interference errors, the correct torque at each joint can be measured in multiple poses, making it suitable for forming joints in collaborative robots.

References

[1] S. Gushi, Y. Shimabukuro and H. Higa, "A Self-Feeding Assistive Robotic Arm for People with Physical Disabilities of the Extremities," 2020 5th International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS), Okinawa, Japan, pp. 61-64,

[2] Liangjie. Tao, "A Joint Torque Sensor Based on Strain Gauge for Collaborative Robot," 2025 9th International Conference on Robotics, Control and Automation (ICRCA), Shanghai, Chana, pp. 74-77

[3] Tsetserukou, D., Tadakuma, R., Kajimoto, H. and Kawakami, N., Tachi, S., 2007, "Development of a whole-sensitive teleoperated robot arm using torque sensing technique," IEEE Int. Conf. on Intelligent Robots and Systems, pp. 476-481.

[4] Jung, J., Ryu, D., Jeong, K.S. and Chang, K. Development of Clutch Disk Torque Sensor for An Automobile. SAE Technical Paper, 2001, 2001-01-0869.

[5] Baniasad, S. M. and Emes, M.R. Design and Development of Method of Valve-Train Friction Measurement. SAE Technical Paper, 1998, No.980572.

[6] Gere, J. M. and Timoshenko, S. Mechanics of Materials. 4th edition, PWS Publishing Company.

[7] Benham, P.P. and Crawford, R.J. Mechanics of Engineering Materials. Longman, 1987.

[8] Jin, D. Z., Abe, F. and Mochizuki, H. Development of a Rotary Transformer and Its Application to SRC Connectors. Furukawa Electric, 1998

Space Robot Dual-Arm Cascaded Reconfiguration Motion Planning Method

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***Abstract*—Space robots serve as core equipment for realizing intelligent on-orbit servicing and space exploration. To address the limitations of traditional fixed-base serial manipulators — such as restricted workspace, single operational scale, and insufficient adaptability — this paper proposes a scheme in which dual 7-DOF arms are cascaded and reconfigured into a hyper-redundant combined arm, thereby significantly expanding the workspace and enhancing task adaptability without requiring overall base maneuvering. First, for a 7-DOF manipulator with joint offsets, an inverse kinematics (IK) solving method combining analytical and numerical techniques is proposed, which improves solving efficiency while supporting position-level obstacle avoidance and configuration optimization. Second, a bidirectional nearest neighbor search is employed to determine the dual-arm cooperative workspace. Particle swarm optimization (PSO) is then utilized to optimize the reconfiguration pose and configuration, ensuring favorable kinematic performance during the reconfiguration process. Smooth reconfiguration motion is achieved through segmented trajectory planning. Finally, gravity compensation strategies are implemented on a physical dual-arm experimental platform to simulate the space environment. Experiments validate the effectiveness of the proposed kinematic algorithms and reconfiguration optimization methods, providing theoretical and experimental references for the development of reconfigurable space robotic systems.**

***Index Terms*—Space robot, redundant manipulator, inverse kinematics, configuration optimization, trajectory planning**

I. Introduction

Space robotic systems have evolved from single-arm to dual-arm and multi-arm configurations. Among them, dual-arm space robots have become a research focus due to their structural symmetry and strong motion coordination capability [1], [2]. Nevertheless, existing dual-arm systems still suffer from inherent structural and systemic limitations: most adopt rigid fixed configurations, resulting in restricted degrees of freedom and functional redundancy. Moreover, their operational range is typically confined to the vicinity of the base, making it difficult to perform long-range or large-span tasks [3]. Consequently, the development of reconfigurable robots is widely regarded as a key approach to enhancing on-orbit servicing performance and overcoming current technical bottlenecks in space robotics [4].

The commonly used 7-DOF redundant manipulator configuration in space robotics improves obstacle avoidance capability but simultaneously introduces significantly more complex kinematic and dynamic problems. The mapping between end-effector pose and joint angles exhibits a highly nonlinear many-to-many relationship, making inverse kinematics (IK) solutions have to handle multiple solutions [5]. Existing approaches — including analytical methods, numerical methods [5], and intelligent optimization algorithms [6] — each have advantages but also notable limitations [5], [7]. Particularly for redundant manipulators with multiple joint offsets, analytical inverse solutions at the position level are highly configuration-dependent, while conventional numerical iterative methods are sensitive to initial guesses, exhibit poor real-time performance, and suffer from instability. Therefore, this paper aims to develop a highly general, real-time capable, and accurate inverse kinematics algorithm that can effectively exploit redundancy for manipulators with joint offsets.

Reconfiguration technology improves robotic performance by dynamically adjusting configuration or optimizing control strategies. Among various approaches, structural self-reconfiguration demonstrates significant advantages and broad application prospects [8]. Relevant research has attracted extensive attention worldwide, with notable achievements including multiple types of reconfigurable robots developed by

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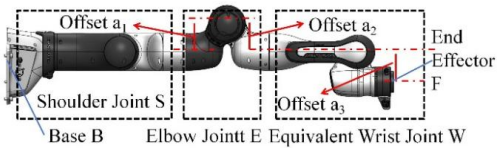
Tokyo Institute of Technology [9] and ESA [10], as well as the successful implementation of hyper-redundant combined arm reconfiguration using large and small arms on the Chinese Space Station [11]. A fundamental prerequisite for configuration reconfiguration is that the end-effectors of both arms can reach the same spatial position with good kinematic performance. The redundant degrees of freedom provide optimization space for reconfiguration. This study focuses on optimizing the kinematic performance during autonomous reconfiguration to maximize task success rate. After determining the optimal reconfiguration position and configuration, complete motion planning throughout the reconfiguration process is realized using motion/trajectory planning algorithms.

The rest of this paper is organized as follows: Section II proposes a hybrid analytical-numerical inverse kinematics algorithm for 7-DOF manipulators with multiple joint offsets. Section III presents the dual-arm reconfiguration configuration optimization and motion planning algorithms. Section IV describes the experimental setup, hardware platform, and result analysis. Finally, Section V concludes the work and discusses future directions.

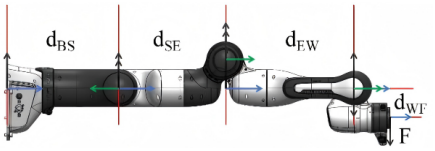
II. A Hybrid Analytical-Numerical Inverse Kinematics Algorithm for 7-DOF Manipulators with Multiple Joint Offsets

A. Kinematic Model of 7-DOF Manipulator with Multiple Offsets

The specific configuration of the 7-DOF arm used in this study is shown in Fig. 1(a), featuring offsets at both the elbow joint E and wrist joint W. The coordinate systems are established using the Modified Denavit-Hartenberg (MDH) convention, as illustrated in Fig. 1(b). The main structural parameters are listed in Table I.



(a) Joint modular division of a single robotic arm



(b) MDH coordinate frames for a single robotic arm

Fig. 1: Joint segmentation and coordinate frames of a single robotic arm

The homogeneous transformation matrix between any two adjacent coordinate systems $\{i\}$ and $\{i-1\}$ is derived through four consecutive operations: a rotation of θ_i about the $\hat{z}_{i-1}$-axis, a translation of d_i along the same $\hat{z}_{i-1}$-axis, a translation of a_i along the current $\hat{x}_i$-axis, and finally a rotation of α_i about the current $\hat{x}_i$-axis. This transformation is expressed as:

TABLE I: MDH Parameters of the Single Robotic Arm

i	d_i (m)	θ_i (rad)	a_i (m)	α_i (°)
1	0.333	θ_1	0	0
2	0	θ_2	0	-90
3	0.107	θ_3	0	90
4	0	θ_4	0.0825	90
5	0.384	θ_5	-0.0825	-90
6	0	θ_6	0	90
7	0.316	θ_7	0.088	90

$$ {}^{i-1}_{i}T = \text{Rot}(z,\theta_i)\text{Trans}(z,d_i)\text{Trans}(x,a_i)\text{Rot}(x,\alpha_i) \quad (1) $$

The forward kinematics equation of the single arm is obtained by successive multiplication of the transformation matrices from base to end-effector:

$$ {}^{B}_{F}T = {}^{B}_{1}T\,{}^{1}_{2}T\,{}^{2}_{3}T\,{}^{3}_{4}T\,{}^{4}_{5}T\,{}^{5}_{6}T\,{}^{6}_{7}T\,{}^{7}_{F}T \quad (2) $$

B. Position-Level Inverse Kinematics Combining Arm Angle Parameterization and Numerical Optimization

Inverse kinematics can be categorized into velocity-level and position-level approaches depending on the control signals. Velocity-level IK struggles to achieve precise poses under regional constraints. In particular, in complex truss-intersected space environments, pure velocity control exhibits significant disadvantages in terms of end-effector accuracy and dynamic obstacle avoidance. Therefore, designing a precise and efficient position-based kinematic algorithm is of great importance for space robots.

This paper proposes a hybrid analytical-numerical algorithm that combines arm angle parameterization with numerical optimization. By providing a high-quality analytical initial guess, the method significantly improves both the efficiency and success rate of numerical iteration. The overall solving procedure consists of two stages: initial value prediction and precision refinement.

The initial value prediction process obtains a predicted inverse solution $\hat{\theta}$ analytically. Denote the actual manipulator model as Arm A, and introduce an auxiliary model Arm B, which is identical to Arm A except for the offsets at the last three joints. By setting the wrist joint offset parameter of Arm A to $a_3 = 0$ m, the configuration of Arm B is derived. Inspired by the arm angle definition, Arm B is divided into two spherical joints at the shoulder S and wrist W, and an elbow joint E with rotational offset. The arm plane is defined by points S, E, and W; the reference plane is formed by the SW line and the first joint axis; and the arm angle φ is the angle between the reference plane and the current arm plane, with sign determined by the right-hand rule around the SW line.

The inverse solution initial value prediction based on the arm angle method involves: solving the interval of valid parameter φ_d, performing input parameter validity checks, and sequentially derive the angle of each joint in the order of $\theta_1 - \theta_2 - \theta_3 - \theta_4 - \theta_5 - \theta_6 - \theta_7$ to obtain the predicted values

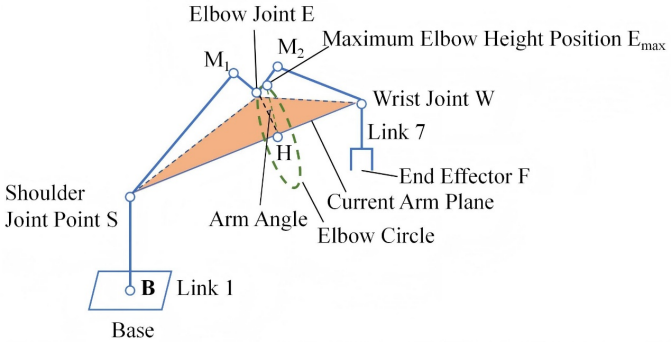


Fig. 2: Equivalent schematic of Arm B with elbow offset

$\hat{\theta}$. The predicted inverse solutions $\hat{\theta} = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7]$ can be calculated using the following equations:

$$\tan\theta_1 = \frac{-A_s(2,2)\sin\varphi_d - B_s(2,2)\cos\varphi_d - C_s(2,2)}{-A_s(1,2)\sin\varphi_d - B_s(1,2)\cos\varphi_d - C_s(1,2)} \quad (3)$$

$$\cos\theta_2 = -A_s(3,2)\sin\varphi_d - B_s(3,2)\cos\varphi_d - C_s(3,2) \quad (4)$$

$$\tan\theta_3 = \frac{A_s(3,3)\sin\varphi_d + B_s(3,3)\cos\varphi_d + C_s(3,3)}{-A_s(3,1)\sin\varphi_d - B_s(3,1)\cos\varphi_d - C_s(3,1)} \quad (5)$$

$$\theta_4 = \pi - \arccos\left(\frac{d_{SE}^2 + d_{EW}^2 - |SW|^2}{2d_{SE}\cdot d_{EW}}\right) - \angle SEM_1 - \angle WEM_2 \quad (6)$$

$$\tan\theta_5 = \frac{A_w(2,3)\sin\varphi_d + B_w(2,3)\cos\varphi_d + C_w(2,3)}{A_w(1,3)\sin\varphi_d + B_w(1,3)\cos\varphi_d + C_w(1,3)} \quad (7)$$

$$\cos\theta_6 = A_w(3,3)\sin\varphi_d + B_w(3,3)\cos\varphi_d + C_w(3,3) \quad (8)$$

$$\tan\theta_7 = \frac{A_w(3,2)\sin\varphi_d + B_w(3,2)\cos\varphi_d + C_w(3,2)}{-A_w(3,1)\sin\varphi_d - B_w(3,1)\cos\varphi_d - C_w(3,1)} \quad (9)$$

From (3)–(9), the joint angles are obtained via the inverse cosine function, which has multiple solutions, leading to two sets of solutions at certain joints. Thus, the entire process yields 8 sets of solutions, which are subsequently filtered using joint limits, singularity avoidance, and minimum joint displacement principles to select the most suitable one.

The precision refinement process uses the predicted values $\hat{\theta}$ from the previous step as initial guesses for numerical optimization to obtain precise inverse solutions θ meeting accuracy requirements. This paper employs the Levenberg-Marquardt (LM) algorithm for numerical optimization. Assuming the desired end-effector pose is T_d and the current pose is T_c, the LM algorithm minimizes the following problem to approximate the desired pose:

$$E(\theta) = \frac{1}{2}e^T W e \to \min \quad (10)$$

where $e(\theta)$ is the deviation between current and desired poses, $e(\theta) = \|T_d - T_c\|$; W is the weighting matrix for the 6 DOFs, $W = \mathrm{diag}\{\omega_{E,i}\}$ $(\omega_{E,i} \geq 0,\ i = 1 \sim n)$.

The LM algorithm takes the predicted inverse solutions $\hat{\theta}$ as input and iteratively updates the joint angles θ via the strategy shown in the equation until $E(\theta)$ is below the set threshold:

$$\theta_{k+1} = \theta_k + H_k^{-1} g_k \quad (11)$$

$$H_k = J_k^T W_E J_k + W_N \quad (12)$$

$$g_k = J_k^T W_E e_k \quad (13)$$

where θ_k and θ_{k+1} are the joint vectors at iterations k and $k+1$; J_k is the Jacobian at iteration k; H_k is the Coefficient Matrix at iteration k; g_k is the gradient at iteration k; and W_N is the damping factor.

III. Dual-Arm Reconfiguration Configuration Optimization and Motion Planning Algorithms

Fig. 3 illustrates the dual-arm space robot model adopted in this study. Two cascadable 7-DOF manipulators are symmetrically mounted on both sides of the base, enabling switching between independent dual-arm mode and hyper-redundant combined arm mode through reconfiguration. The manipulator design incorporates three radial offsets, resulting in intersecting adjacent joint axes, which complicates kinematic modeling and inverse solution.

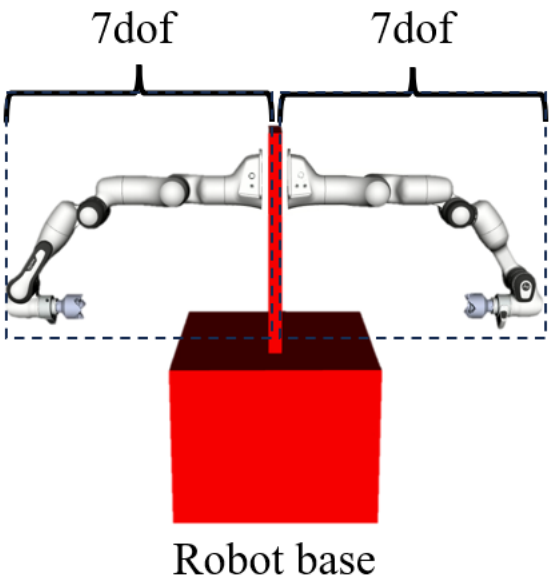


Fig. 3: The dual-arm space robot in this paper

A. Dual-Arm Cascaded Reconfiguration Task Description

The dual-arm cascaded reconfiguration task consists of two phases: first, the arms move with adapters to the vicinity of the docking position; second, force interaction and mechanical locking occur to form the hyper-redundant combined arm. This study focuses on the first phase, emphasizing optimization methods for selecting reconfiguration positions and configurations, as well as smooth motion planning strategies.

B. Dual-Arm Cooperative Workspace Determination Based on Bidirectional Nearest Neighbor Search

This paper proposes an efficient dual-arm cooperative workspace solving algorithm based on bidirectional nearest neighbor search. Traditional methods have a computational complexity of $O(N^2)$, making them unsuitable for million-level point clouds. By constructing KD-Tree indices for the left and right arm point clouds separately and performing bidirectional nearest neighbor searches, points with distances less than 0.01 m are filtered to form a candidate set, thereby defining a well-shaped "disk-like" cooperative workspace. This algorithm reduces the complexity to $O(N \log N)$, providing a precise geometric foundation for subsequent reconfiguration optimization.

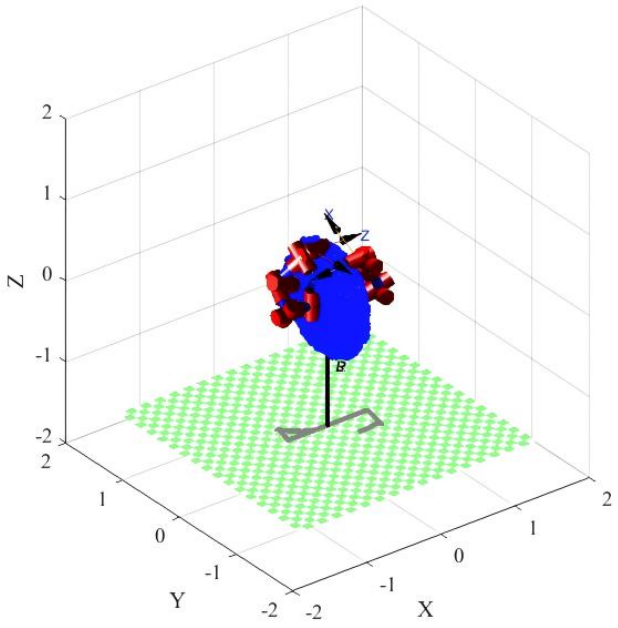


Fig. 4: Dual-arm cooperative workspace

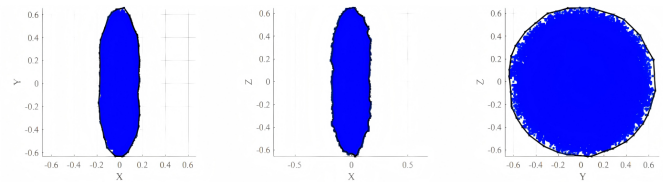


Fig. 5: Projections of the cooperative workspace on three coordinate planes

C. Reconfiguration Configuration Optimization Based on Particle Swarm Optimization

This section employs particle swarm optimization (PSO) to jointly optimize the end-effector pose and arm angle during reconfiguration, aiming to enhance kinematic performance. The optimization core maximizes the manipulability measure, which evaluates motion flexibility under a given configuration, calculated as:

$$\omega(\theta) = \sqrt{\det\left(J(\theta)J(\theta)^T\right)} \tag{14}$$

where θ is the joint angle, and $J(\theta)$ is the Jacobian matrix. Higher manipulability indicates better end-effector pose adjustment capability and motion accuracy, facilitating successful reconfiguration tasks.

PSO simulates swarm intelligence to search in high-dimensional solution spaces. Each particle represents a candidate solution $X = [x, y, z; \alpha, \beta, \gamma; \varphi_L, \varphi_R]$, comprising position, orientation, and arm angle, guided by personal best ($pBest$) and global best ($gBest$). The update equations are:

$$\begin{aligned} v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (pBest_i - x_i(t)) \\ + c_2 \cdot r_2 \cdot (gBest - x_i(t)) \end{aligned} \tag{15}$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{16}$$

where w is the inertia weight, c_1 and c_2 are learning factors, and r_1, r_2 are random numbers. The search space χ_{search} fully covers the dual-arm cooperative workspace region and satisfies constraints for co-location and orientation alignment of the end-effectors.

The objective function is defined as the weighted sum of the manipulabilities of both arms:

$$\eta(X) = -\frac{\lambda_L \omega_L + \lambda_R \omega_R}{\lambda_L + \lambda_R} \tag{17}$$

where ω_L and ω_R are the manipulabilities of the left and right arms, and λ_L, λ_R are weighting coefficients. Minimizing the negative of this function maximizes overall system flexibility. The overall optimization process is shown in Fig. 6.

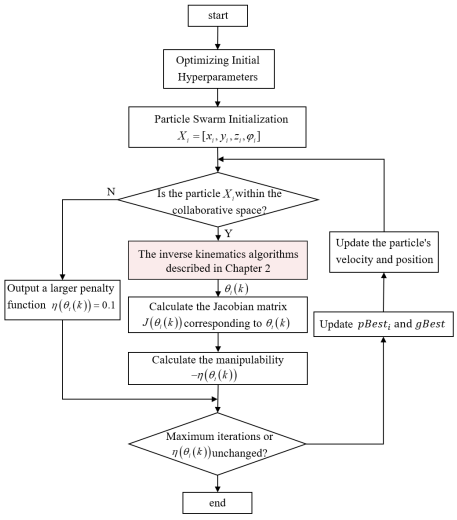


Fig. 6: Flowchart of reconfiguration position and configuration optimization using PSO

With parameters set to 40 particles, learning factors of 1.49, and weights of 0.5 each, the algorithm converges to the optimal solution after approximately 30 iterations:

$$X_{\text{best}} = [0,\ -0.38,\ 0.145;\ 0,\ 180,\ 180;\ 83.6,\ -83.6] \tag{18}$$

Namely, for dual-arm cascaded reconfiguration, the optimal reconfiguration position coordinates are $[0, -0.38, 0.045]$ m, with Euler angles $[0, 180, 180]^\circ$, and corresponding arm angles $\varphi_L = 83.6^\circ$, $\varphi_R = -83.6^\circ$. The optimized configuration aligns with human-like dual-arm collaboration characteristics, verifying the reasonableness of the results.

D. Dual-Arm Reconfiguration Motion Planning

To ensure smooth transition from the initial configuration to the optimal reconfiguration configuration, this paper adopts quintic polynomial trajectory planning $\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$. This method satisfies position, velocity, and acceleration constraints at start and end points, meeting the continuity requirements for physical adapter docking.

The planning uses a reverse segmented strategy: First, based on the cascaded docking target pose T_{end} and optimal arm angle φ_{best}, translate 0.1 m along the negative Z-axis of the end-effector frame to determine the intermediate pose $T_{\text{temp}} = T_{\text{end}} \cdot \text{Trans}(z, -0.1)$, and plan the second-stage straight-line docking trajectory along the Z-axis. Then, using T_{end} and φ_{best} as inverse kinematics inputs, solve for the joint angles q_{temp} of the intermediate configuration. Finally, plan the first-stage trajectory from the initial configuration q_{start} to q_{temp} using quintic polynomials in joint space. Both stages constrain start and end velocities to zero for smooth and continuous motion. The initial configuration is set as $q_{\text{start}} = [0, 0, 0, -1, 0, 1, 0]$ rad.

After integrating the two-stage planning, simulation verification is performed. The motion process is shown in Fig. 7, and the left arm's joint angles and velocities are depicted in Fig. 8 (right arm information can be obtained by symmetry). Results indicate continuous joint states without abrupt changes, satisfying dynamic performance requirements.

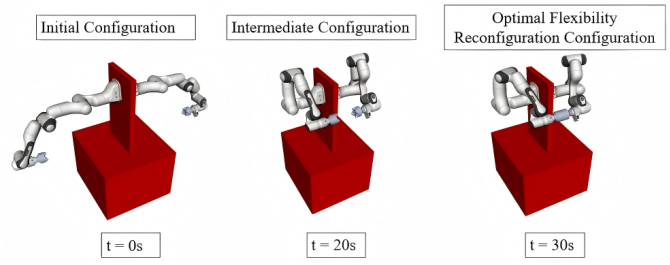
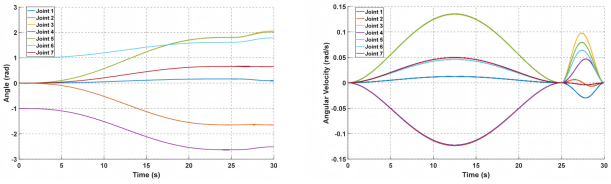


Fig. 7: The entire motion process

IV. Experiments and Analysis

A. Experimental Equipment and Environment

This experiment is based on the Franka Research 3 (FR3) robotic platform, as shown in Fig. 9. The dual-arm experimental setup consists of two Franka arms, controllers, reconfiguration docking adapters, aluminum profile trusses, and a user control computer, with the environment illustrated in Fig. 10. The reconfiguration adapter is the end-effector tool for physical dual-arm connection, with its structure shown in Fig. 11(a) and actual connection method in Fig. 11(b).



(a) joint angle variation curve **(b) joint angular velocity variation curve**

Fig. 8: The joint states of the left arm during the configuration transition from initial to reconfigured

Fig. 9: Franka manipulator configuration

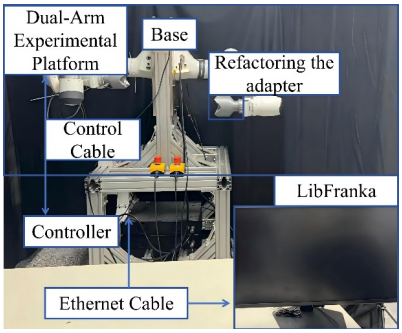
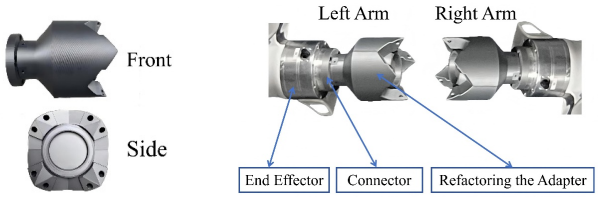


Fig. 10: Schematic of the experimental environment in this paper

During research and testing, gravity compensation is considered, including compensation for the robot's self-weight and end-effector payload, effectively replicating the dynamic characteristics of space robots in microgravity or zero-gravity environments.



(a) Docking adapter **(b) Installation method of the docking adapter**

Fig. 11: Docking adapter and its installation method

B. Experiments on the Hybrid Analytical-Numerical Inverse Kinematics Algorithm

To verify the feasibility of the algorithm in Section II, tests on solving efficiency and success rate are conducted and compared with traditional pure numerical methods. Since the algorithm has two inputs — desired end-effector pose and desired arm angle — the efficiency test scheme is designed as follows:

Randomly select 1000 desired poses T_{range} within the robot's workspace. For each pose, compute the feasible arm angle interval and uniformly interpolate 100 points within it. Traverse these arm angles, solve inverse kinematics sequentially, and use the average time as the solving time for that pose.

The traditional numerical algorithm uses the ikine function from the Robotic Toolbox for Python, based on an improved Newton iteration method. The hardware environment is a computer with an Intel i9-13900 processor and 32GB RAM (5600MHz), and the software environment is Python 3.10 with PyCharm 2023.1.2, under which solving accuracy is evaluated.

Testing shows that the proposed algorithm achieves a 96% success rate with an average time of 6.9 ms, improving efficiency by about 12% compared to the pure numerical method (8.2 ms), which enhances real-time robot control. Results indicate that the analytical predictions based on the arm angle method are close to the true inverse solutions, providing good initial values for subsequent numerical optimization and improving efficiency and stability.

C. Experiments on Dual-Arm Reconfiguration Configuration Optimization and Motion Planning

This section uses the segmented planning method from Section III to complete the full motion planning and demonstration from initial to optimal reconfiguration configuration. An intermediate configuration with aligned end-effector orientations is inserted between the initial and optimal configurations to ensure continuity and smooth transition.

This intermediate configuration is obtained by translating both arms -0.1 m along the Z-axis of the end-effector frame at the optimal reconfiguration pose. Taking the left arm as an example, the joint angles for the intermediate configuration with $\varphi = 83.6°$ are solved using inverse kinematics:

$$q_{temp} = [0.1653 \quad -1.648456 \quad 1.796928 \quad -2.632522 \quad -4.477024 \quad 1.612007] \text{ rad} \tag{19}$$

With $q_{start} = [0, 0, 0, -1, 0, 1, 0]\,\text{rad}$, the first stage uses quintic polynomial trajectory planning in joint space to achieve smooth transition from initial joint vector q_{start} to intermediate joint vector q_{temp}, with the motion process shown in Fig. 12(a), (b), and (c).

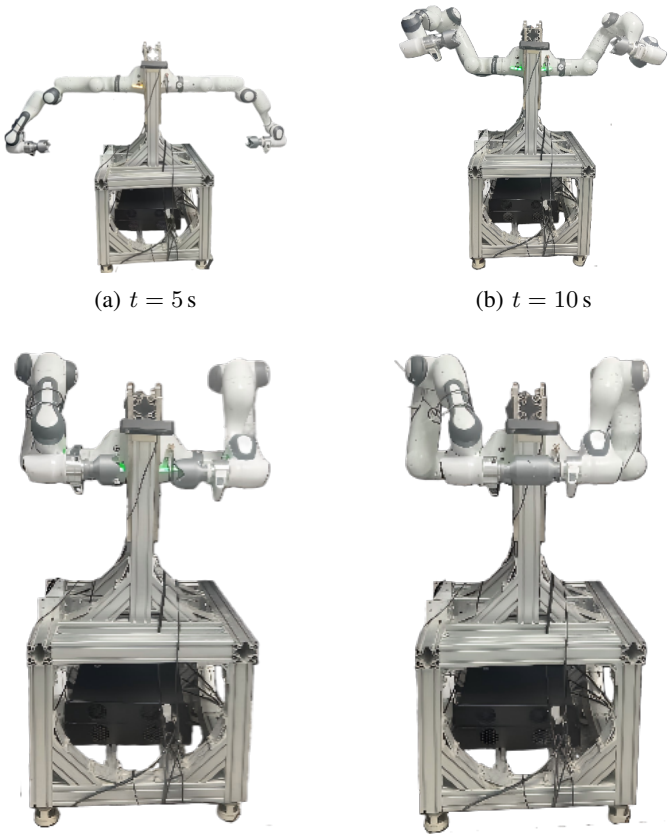
The second stage employs Cartesian space linear interpolation, maintaining constant arm angle, to plan the motion from intermediate to optimal reconfiguration configuration, as shown in Fig. 12 (c) and (d).

The left arm's joint information during the entire dual-arm reconfiguration is shown in Fig. 13. Joint angles change continuously, velocity curves exhibit typical quintic polynomial characteristics, and acceleration curves show no abrupt changes, meeting dynamic continuity requirements.

In summary, this two-stage planning method performs well in configuration transition, orientation alignment, and end-effector path continuity, providing a feasible motion planning scheme for dual-arm reconfiguration tasks.

V. Conclusion

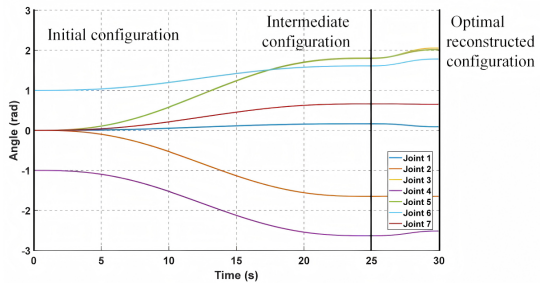
This paper addresses the issues of limited workspace and insufficient task adaptability in traditional fixed-base dual-arm space robots by proposing a solution for cascading dual arms into a hyper-redundant combined arm. Focusing on this scheme, the study investigates two main problems:



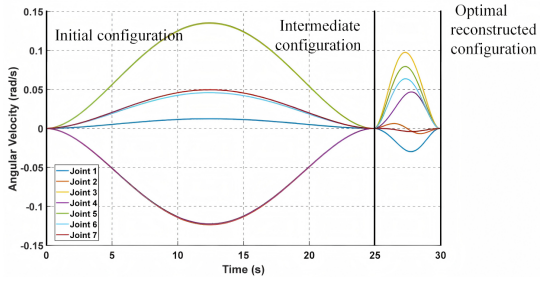
(a) $t = 5$ s (b) $t = 10$ s

(c) $t = 20$ s (d) $t = 30$ s

Fig. 12: Dual-arm reconfiguration motion process



(a) joint angle variation curve



(b) joint angular velocity variation curve

Fig. 13: Joint pose and velocity information during the entire motion process

kinematics during variable configuration and optimization of dual-arm reconfiguration configurations. For the challenging inverse kinematics of offset manipulators, a hybrid solving algorithm combining arm angle method and LM optimization is proposed. By parameterizing with arm angle to obtain analytical initial values and combining numerical optimization, it overcomes defects of traditional methods, achieving a 96% success rate and 12% efficiency improvement over pure numerical methods. For dual-arm reconfiguration, a KD-Tree-based bidirectional nearest neighbor search efficiently defines the cooperative workspace; PSO obtains the optimal reconfiguration configuration; and a reverse segmented trajectory planning strategy realizes smooth transition from initial to optimal configuration. Validation on the Franka dual-arm experimental platform demonstrates stable and continuous motion of the proposed algorithms and planning scheme, providing theoretical and experimental support for the development of reconfigurable space robotic systems.

ACKNOWLEDGMENT

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REFERENCES

[1] J. C. Parrish, "The ranger telerobotic shuttle experiment: An on-orbit satellite servicer," in *Proc. 5th Int. Symp. Artificial Intelligence, Robotics and Automation in Space (ISAIRAS '99)*, 1999, pp. 225–232.

[2] P. Schmaus, D. Leidner, T. Krueger, A. Schiele, B. Pleintinger, R. Bayer, and N. Y. Lii, "Preliminary insights from the METERON SUPVIS Justin space-robotics experiment," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3836–3843, Oct. 2018.

[3] D. Sanchez, W. Wan, and K. Harada, "Four-arm collaboration: Two dual-arm robots work together to manipulate tethered tools," *IEEE/ASME Trans. Mechatronics*, vol. 27, no. 5, pp. 3286–3296, Oct. 2022.

[4] Z. Jiang, X. Cao, X. Huang, H. Li, and M. Ceccarelli, "Progress and development trend of space intelligent robot technology," *Space-Sci. & Technol.*, vol. 2022, Jan. 2022, Art. no. 9832053.

[5] L.-F. Jia, S.-L. Qiao, Y.-F. Tao, J.-G. Zheng, Y.-X. Guo, J. Chen, and Y.-P. Huang, "Research progress on inverse kinematics solution methods of redundant manipulators," *Control and Decision*, vol. 38, no. 12, pp. 3297–3316, 2023.

[6] N. A. Mai and D. X. Bien, "Voice recognition and inverse kinematics control for a redundant manipulator based on a multilayer artificial intelligence network," *J. Robot.*, vol. 2021, Jun. 2021, Art. no. 5805232.

[7] D. Cagigas-Muniz, "Artificial neural networks for inverse kinematics problem in articulated robots," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 107175.

[8] H. Luo and T. L. Lam, "Auto-optimizing connection planning method for chain-type modular self-reconfiguration robots," *IEEE Trans. Robot.*, vol. 39, no. 2, pp. 1353–1372, Apr. 2023.

[9] H. Sawada and S. Matunaga, "Dynamics and control of space robot brachiating motion using handrails," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2003, pp. 3065–3070.

[10] M. Deremetz, M. Debroise, R. Boitte, M. De Stefano, H. Mishra, B. Brunner, G. Grunwald, M. A. Roa, M. Reiner, M. Zavodnik, M. Komarek, J. D'amico, F. Cavenago, J. Gancet, P. Letier, M. Ilzkovitz, L. Gerdes, and M. Zwick, "MIRROR: A modular and relocatable multi-arm robot demonstrator for on-orbit large telescope assembly," presented at the Int. Astronautical Congress (IAC), 2023.

[11] B. Ma, Z. Jiang, Y. Liu, and Z. Xie, "Advances in space robots for on-orbit servicing: A comprehensive review," *Adv. Intell. Syst.*, vol. 5, no. 8, Aug. 2023.

Grasping Trajectory Planning of Industrial Robot Based on Dynamic Movement Primitives

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***Abstract*—Thousands of bent pipes that serve as conduits for water, oil, gas and electricity in planes and aeroengines need to be measured and screened. In order to free workers from repeated and heavy manual labor, the ideal bending measurement process can be completed by industrial robots. Thousands of bending pipes mean thousands of trajectory planning process for robot, which is not only computationally large, but also detrimental to system security. How to find the common characteristics of bending trajectories and generate different obstacle avoidance trajectories that meet the measurement requirements are the key technical problems to be solved. Even though each bend pipe has differences in size and shape and the measurement trajectories are different, the measurement trajectories are similar. A trajectory planning method based on dynamic motion primitives(DMP) is proposed to solve the trajectory planning problem of robotized pipe bending measurement system. The grasping position (the starting point of trajectory planning) and the measuring position (the end point of trajectory planning) of the bending measurement process are different, but the trajectory process and obstacle avoidance characteristics are similar. The DMP method can take into account the difference and shape similarity of the trajectory in the same time, and can form the measurement trajectory for different bending pipes well. In this paper, the traditional DMP theory is described and verified theoretically, and then it is extended to multi-dimensional position, multi-dimensional direction and space obstacle avoidance. Simulation results and experiments verify the effectiveness and efficiency of the proposed method in trajectory planning.**

Keywords—trajectory planning, DMP, industrial robot, grasping

I. INTRODUCTION

There are thousands of curved tubes inside aeroengines and aircraft, known as the arteries of the aircraft, which are the channels for gases, liquids and electrical circuits. The shape of each tube is different from each other. If the actual shape and curvature of the bend is very different from the design shape, it will form a large assembly stress, affecting the safety of the aircraft and the engine. Given the stringent quality requirements in the aerospace field, it is imperative to measure and screened all bent tube structures, resulting in a substantial workload. In order to free workers from repeated and heavy manual labor, the ideal bending measurement process can be completed by industrial robots.

However, thousands of bending pipes mean thousands of planning process for robot, which is not only computationally large, but also detrimental to ensuring system security.

The entire process of measuring bent tubes using a robot bent tube measurement system involves various steps such as measurement and display of the bent tube, quick change of clamping tools, autonomous clamping, obstacle avoidance motion planning, and automatic measurement. And even though each bend pipe has differences in size and shape and the measurement trajectories are different, the measurement trajectories are similar.

The automated pipe bending measurement process requires the robot to automatically clamp the pipe on the feeding platform according to the information transmitted by the vision system, and clamp the pipe to avoid obstacles in the environment and reach the designated position. With the development of trajectory planning technology, there are many ways to solve this problem of robot motion planning with obstacle avoidance so far[1]. In general, the obstacle avoidance motion planning methods for manipulators can be divided into three categories: heuristic algorithm[2], random algorithm[3] and optimization method[4]. Heuristic algorithm is an ergodic search method with certain rules. But the traversal priority policy will be the biggest factor affecting the calculation failure. Because it is impossible for a heuristic algorithm to take into account all possibilities equally, whether it is depth-first traversal or breadth-first traversal, as long as there is a priority strategy, the strategy will tilt towards the priority direction, making the algorithm miss the global optimal and fall into the local optimal possibility. Random algorithm is the most commonly used planning method for robotic arms, and its basic theory is real-time random extended tree (RRT). This approach combines planning and obstacle avoidance perfectly. Combined with the strategies to eliminate local optimal, it can meet most robotic arm planning requirements. The biggest problem of this algorithm to be used in the measurement of bending pipe is its time consumption and randomness. When complex obstacles narrow the feasible path, RRT consumes a lot of computational time. But since RRT

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doesn't learn intelligently about these obstacles or gain experience from previous planning, it recalculates a different trajectory each time. The essence of the optimization method[5] is to transform the trajectory planning problem into a mathematical multi-variable and multi-objective optimization problem[6]. The optimization methods include genetic algorithm, ant colony, particle swarm and so on. However, the optimization algorithm is more suitable for the reuse of trajectories in extreme environments.

When measuring thousands of bends, making thousands of motion plans with those traditional methods from scratch is not an economical and reliable method. In fact, for the planning problem of bending measurement, we need a planning method that can learn the trajectory characteristics from the preliminary planning, so as to reduce the calculation amount and improve the robustness of the trajectory in the later planning. By analyzing the workflow of the bent tube measuring system, it is observed that the trajectories of the manipulators are generally similar, except for the positional differences of the clamping positions and measure positions. Therefore, it is necessary to find a planning method that can modify the starting pose and the target pose, and successfully reproduce the obstacle avoidance trajectory. An improved DMP method is proposed in this paper, which can avoid obstacles in joint control planning and Cartesian space, and realize the recurrence of similar trajectories by reinforcement learning.

II. Dynamic Movement Primitive Algorithm Theory

A. Basic Concepts of Dynamic Movement Primitives

The dynamic movement primitives (DMPs) algorithm is deduced with a stable second-order spring damping system. Additionally, a nonlinear term is incorporated to regulate the system's motion and achieve the intended trajectory.

The formula is as follows:

$$\tau^2 \ddot{y} = \alpha_y(\beta_y(g-y) - \tau\dot{y}) + f(t) \tag{1}$$

$$\tau\dot{x} = -\alpha_x x \tag{2}$$

In which, "y" represents the state of motion of the system, while "$\dot{y}$" and "$\ddot{y}$" represent the first and second derivatives of "y" respectively, which correspond to the speed and acceleration. The target state of the system, referred to as the attractor, is denoted by "g" and the system ultimately converges to this state. The variable "τ" represents a time constant, and "α_y" and "β_y" are constants. By varying the value of "g" different trajectories corresponding to different target states can be obtained. Similarly, by adjusting the value of "τ" trajectories with different speeds of motion can be achieved. Furthermore, by altering the relationship between "α_y" and "β_y" to "$\beta_y=\alpha_y/4$" the system can attain a critical damping state, resulting in a monotonous convergence to "g".

The forced term formula is as follows:

$$f(t) = \frac{\sum_{i=1}^{N}\psi_i(t)\omega_i}{\sum_{i=1}^{N}\psi_i(t)} \tag{3}$$

"f" directly affects the acceleration term of the system, which is equivalent to the action of an external force and is mainly used to change the shape of the trajectory. "ψ_i" is a Gaussian function, and "N" represents the number of basis functions, which "ω_i" is the corresponding weight of each basis function. Different complex trajectories can be obtained by giving different numbers of basis functions or weights corresponding to different basis functions. The forced term in Formula (3) is a time-dependent linear time-variant dynamic system, which cannot be directly coupled with other dynamic systems. Therefore, it is necessary to reconstruct the nonlinear term according to different usages.

B. Cartesian Space Dynamic Movement Primitives

Generally, dynamic movement primitives can be categorized into two types: discrete DMPs and rhythmic DMPs. Rhythmic DMPs are employed to address trajectory planning issues that involve periodic movements, typically in joint space. On the other hand, discrete DMPs are utilized for trajectory planning in Cartesian space.

Based on standard DMP[7], an improved DMP method is proposed as below:

$$\tau^2 \ddot{p} = K(g_p - p) - \tau D\dot{p} - K(g_p - p_0)x + Kf_p(x) \tag{4}$$

$$\tau\dot{x} = -\alpha_x x \tag{5}$$

The variables "g_p", "p", and "p_0", represent the multi-dimensional target position, current position, and initial position vector, respectively. "$\dot{p}$" and "$\ddot{p}$" refer to the velocity vector and acceleration vector, respectively. The elastic coefficient matrix and damping coefficient matrix are represented by "K" and "D" respectively. The relationship between the elastic coefficient "K" and the damping coefficient "D" is expressed as "$4K=D^2$", that make sure the system could achieve a critical damping. Formula (5) represents a canonical system, in which the variable "x" serving as a phase variable. Typically, when "$x=1$", it indicates that the system is in its initial state, and over time, the system will gradually converge to the target state of "$x=0$". The function "$f(x)$" is a non-linear function that is constructed to express of the Cartesian space trajectory. The forced function "$f_p(x)$" is defined as follows:

$$f_p(x) = \frac{\sum_{i=1}^{N}\psi_i(x){\omega_i}^p}{\sum_{i=1}^{N}\psi_i(x)}x \tag{6}$$

$\omega_i^p \in R^3$is the position weight coefficient vector.

To plan the manipulator trajectory effectively, it is necessary to consider both the position and orientation changes in Cartesian space[8]. The orientation is represented by a unit quaternion. An improved DMP method based on quaternion is proposed in this paper which not only could express the posture of the manipulator but also address the zero-crossing issue in orientation problems. Therefore, the enhanced formula for orientation DMPs is written as follows:

$$\begin{aligned}\tau^2\dot{\omega} = {} & K2\log(g_o * \bar{q}) - \tau D\omega - \\ & K2\log(g_o * \bar{q}_0)x + Kf_o(x)\end{aligned} \tag{7}$$

$$\dot{q} = \frac{1}{2}\omega * q \tag{8}$$

$$\tau\dot{x} = -\alpha_x x \tag{9}$$

The system's target orientation, current orientation, and initial orientation are represented by "g_0", "q" and"q_0" respectively. The elastic coefficient matrix and damping coefficient matrix are represented by "K" and "D" respectively. Coercive terms "$f_o(x)$" are defined as follows:

$$f_o(x) = \frac{\sum_{i=1}^{N}\psi_i(x)\omega_i{}^o}{\sum_{i=1}^{N}\psi_i(x)}x \tag{10}$$

$\omega_i o \in R^3$ is the weight coefficient vector for orientation, with all other parameters being the same as those of position DMP. Orientation "q" is represented by the unit quaternion.

$$q=v+\boldsymbol{u}(v\in R,\boldsymbol{u}\in R^3).$$

where is the conjugate vector of "q", $\bar{q}=v-\boldsymbol{u}$. The quaternion product is:

$$q_1 * q_2 = (v_1\boldsymbol{u}_2 + v_2\boldsymbol{u}_1 + \boldsymbol{u}_1\times\boldsymbol{u}_2) + (v_1v_2 - \boldsymbol{u}_1{}^T\boldsymbol{u}_2) \tag{11}$$

$$\log(q) = \log(v+\boldsymbol{u}) = \begin{cases} \cos^{-1}v\dfrac{\boldsymbol{u}}{\|\boldsymbol{u}\|}, & \boldsymbol{u}\neq 0, \|v\|\le 1 \\ [0 \quad 0 \quad 0]^T \end{cases} \tag{12}$$

The relationship between quaternion logarithm and angular velocity is as follows[9]: $\omega=2\log(g_o\times\bar{q})$.

C. Space Obstacle Avoidance Problem

Obstacle avoidance is a crucial problem in comprehensive trajectory planning methods for manipulators[10]. As the manipulator progresses along its path, it is important to account for the deviation angle, denoted as "φ" between the vector "(O-P)" connecting the obstacle position and the current end position of the manipulator, and the current velocity vector "V" of the manipulator. To avoid obstacles, the manipulator had better incorporate a deflection acceleration into its original dynamic movement primitives[11]. This is achieved by introducing a virtual contact force to generate the necessary deflection acceleration[12]. The relationship between these factors is illustrated in the accompanying figure.

The virtual contact force is defined as the dynamic relationship between the contact force-deflection angle and deflection acceleration. Its size is defined as follows:

$$\|f(t)\| = a - bd(t), d(t) \le d_{\max} \tag{13}$$

In which, a and b are two different constants. $d(t)$ indicates the distance between the obstacle and the end of the manipulator. And $f(t)$ satisfy a-$bd_{\max}$=0.

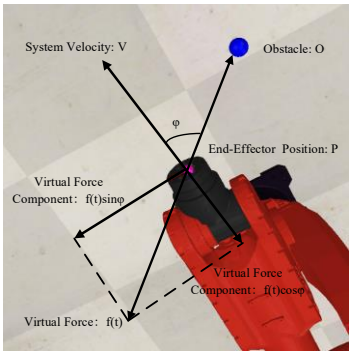


Fig. 1. The relationship of the position and virtual force between the mechanical arm and the obstacle

The deflection angle between the obstacle and the end velocity of the manipulator is defined as follows:

$$\varphi = \cos^{-1}\frac{(o-p)^T v}{\|o-p\|\|v\|} \tag{14}$$

The traditional method has the defect of avoiding obstacles prematurely. In this paper, comprehensive effect of deflection angle and distance are both considered. Deflection acceleration is proposed and defined as follows:

$$P_a(p,v) = k\|f(t)\|\cos\varphi Rv \tag{15}$$

Where R represents the rotation matrix obtained by rotating $\pi/2$ around the vector of (O-P)×V. If there are multiple obstacles, the deflection acceleration generated by each obstacle can also be superimposed:

$$P_a(p,v) = k\sum_{i=1}^{n}\|f_i\|\cos\varphi_i R_i v \tag{16}$$

Where, it represents the number of obstacles (=1,2,3...n). To sum up, the DMPs formula of multi-dimensional position considering obstacle avoidance factors is:

$$\tau^2\ddot{p} = K(g_p - p) - \tau D\dot{p} - K(g_p - p_0)x + Kf_p(x) + \begin{cases} 0, & \text{without obstacle} \\ \tau p_a(p,v) & \text{with obstacle} \end{cases} \tag{17}$$

III. Simulation Verification

Following an analysis of dynamic movement primitives (DMPs) and their associated models, we simulated and validated trajectory planning for multi-dimensional position, orientation, and obstacle avoidance in space. The CoppeliaSim simulation platform and Python software were used for joint simulation, with the T50A-21 mechanical arm serving as the chosen simulation model.

In order to plan the trajectory of dynamic motion elements involving orientation, a set of position and orientation matrices are employed to depict the state of each point within the trajectory. A python program is developed to simulate and validate the posture component of the dynamic movement primitives. The program segregates the position and posture aspects, with the position trajectory being replicated and generalized through the utilization of the multi-dimensional position DMPs algorithm. As for the orientation component, the

orientation rotation matrix is converted into quaternion representation, and the orientation DMPs algorithm is employed to replicate and generalize the orientation. The obtained outcomes are presented as follows:

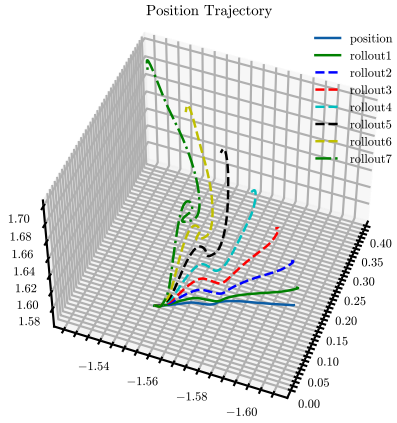


Fig. 2. Trajectory in Cartesian space

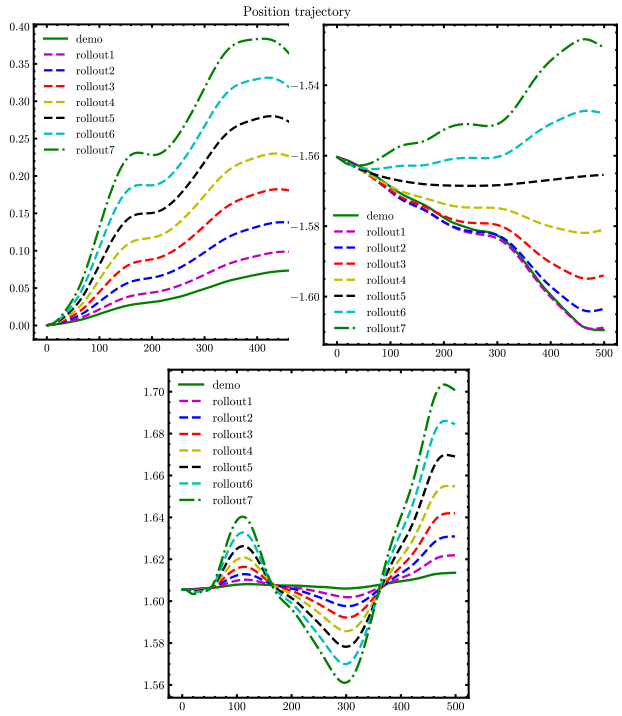


Fig. 3. Position trajectory of x、y、z in Cartesian space

Figure 2 illustrates the outcome of coordinate reproduction and generalization in the X, Y, and Z directions. The combination of the coordinates in the three directions depicted in Figure 3. On the other hand, Figure 4 demonstrates the reproduction and generalization of the real part and three imaginary parts of the orientation quaternion.

In this study, a set of coordinate systems is employed to depict the orientation in the CoppeliaSim simulation platform, enabling a visual representation of the orientation changes. The obtained simulation outcomes are presented below:

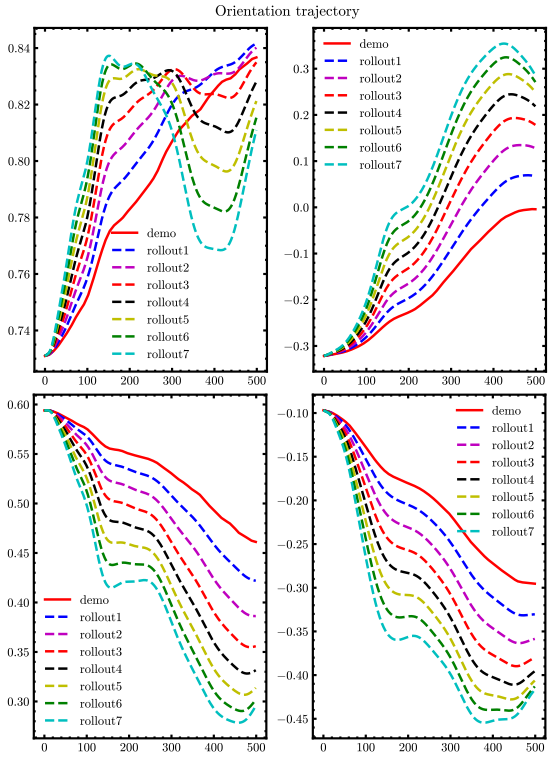


Fig. 4. Results of orientation quaternion

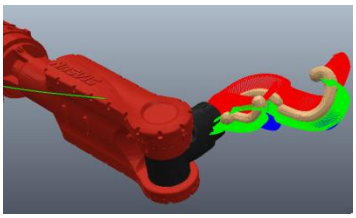


Fig. 5. End orientation trajectory of robotic arm in CoppeliaSim

Figure 5 displays three position trajectories that were chosen from Figure 3. The simulation and verification of the position Dynamic Movement Primitives (DMPs) have been previously addressed in a separate article, and will not be reiterated here. To provide a more visually comprehensible representation of posture changes, the same position trajectory is simulated with various sets of postures. The ensuing simulation outcomes are presented below. Figure 6 is the simulation result of the same position trajectory combined with two different series of orientation coordinate systems. Posture has a similar trend.

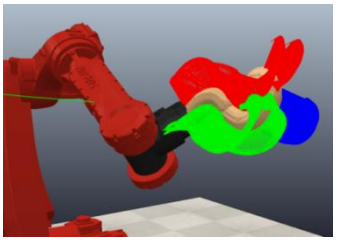


Fig. 6. Simulation Results of the Same Position Trajectory and Different Orientation Trajectories of the mechanical arm in CoppeliaSim

In order to address the issue of avoiding obstacles in a given space, we have chosen to represent the obstacles as ellipsoid envelope balls within the simulation environment. The trajectory of the mechanical arm's end point is assumed to follow a straight path in three-dimensional space, and an obstacle is positioned along this trajectory.

To implement the obstacle avoidance functionality within the dynamic movement primitives, we have developed a Python code. The resulting trajectory, as depicted in the accompanying figure, demonstrates the successful avoidance of the obstacle. The simulation results in CoppeliaSim simulation platform are shown as follows:

Fig. 7. Obstacle avoidance simulation results

It can be seen from the simulation results that the space obstacle avoidance theory of dynamic movement primitives is feasible.

IV. Bend Measurement Experiment

After the simulation experiment is completed, experimental verification is carried out on a real robotic arm. In this section, an experiment for robot bending measurement is designed, and different robotic arm clamping starting points and measurement endpoints are obtained based on different bending characteristics. Different trajectories are planned using dynamic motion primitive methods. Verify the trajectory operation of dynamic motion primitives in terms of multi-dimensional position, posture, obstacle avoidance, etc. The robot bending measurement system is as follows:

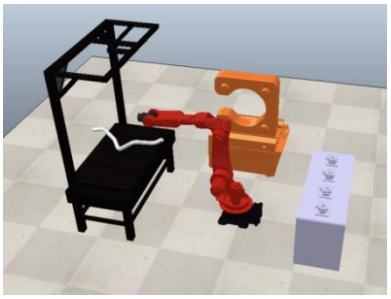
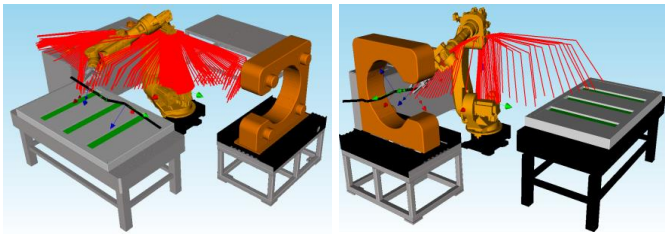


Fig. 8. Bend measurement system

The entire process of bending measurement includes several workflows: bending measurement and display, quick change of clamping tools, autonomous clamping, obstacle avoidance motion planning, and automatic measurement. Mainly conducting obstacle avoidance experiments for robots from gripping bent pipes on the loading platform to measuring bent pipes on the measuring platform, as well as the intermediate process.

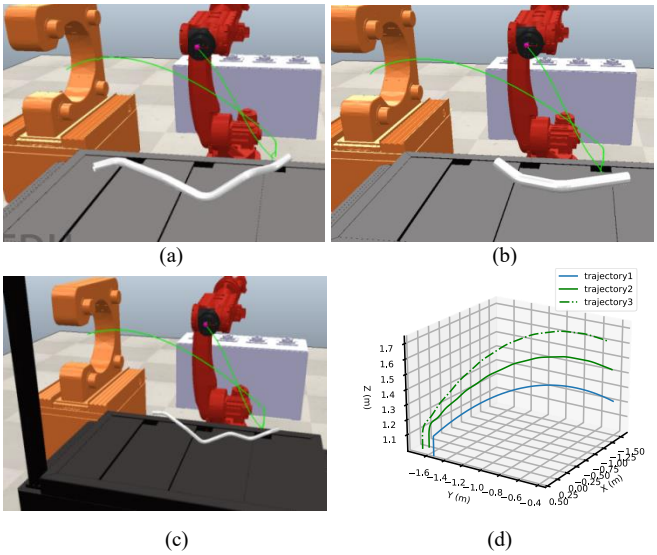
The bending measurement process for pipes is shown in Figure 9.



(a) gripping bent pipes (b) measuring bent pipes

Fig. 9. Gripping and measuring bent pipes on the measuring platform

Plan the trajectory of pipe bending measurement in the simulation platform, and change the initial and end points of the trajectory for different shapes of pipes. The simulation curve are shown in Figure10(a)-(c). The trajectory curve of the robotic arm from the initial point to the end point is shown in Figure 10(d). The bending measurement experiment is show in Figure 11.



(a) (b)

(c) (d)

Fig. 10. Measurement trajectories of different bent pipes

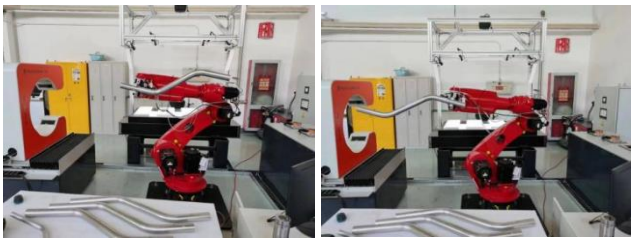


Fig. 11. Measurement platform for bending measurement

The experimental results prove that the trajectories obtained during the measurement process of each bend, except for differences in the initial and end points, have similar overall motion trends. The feasibility of the dynamic movement primitives method has been well demonstrated.

V. Conclusion

This paper focuses on the trajectory planning of a robot bent tube measuring system, specifically from the process of grabbing the bent tube to the actual measuring. The authors analyze the working characteristics of the robot bent tube measuring system and observe that the entire trajectory of the system consists of a series of similar trajectories from the grabbing point to the measuring point. This holds true regardless of the length and shape of the bent tube, with the only difference being the starting and target grabbing positions during the measuring process. Based on this observation, the paper proposes a trajectory planning method that utilizes dynamic movement primitives. By learning the manual teaching trajectory, the mechanical arm of the system can replicate the teaching trajectory using the principles of dynamic movement primitives. Furthermore, the system can generalize the trajectory based on different initial and target positions. This approach significantly enhances the efficiency of the robot bent tube measurement system and reduces the workload associated with trajectory planning.

This paper presents a simulation and verification of trajectory planning for a mechanical arm using Dynamic Movement Primitives (DMPs). The study examines the position, orientation, and obstacle avoidance capabilities of the arm, and the simulation results demonstrate the successful validation of trajectory planning based on the DMPs theory. Specifically, the obstacle avoidance functionality is verified for a single obstacle, and future research aims to extend this verification to multiple obstacles. The primary emphasis of this paper is on the obstacle avoidance behavior of the end effector, while future work will concentrate on investigating obstacle avoidance for the entire mechanical arm.

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References

[1] Huang, Z.Z., Wu, H. “Overview of trajectory planning methods for robot systems”, 2021 IEEE International Conference on Power, Intelligent Computing and Systems (ICPICS), Shenyang China, July 29-31, pp. 1-10, 2021.

[2] Tarek, A.H., Djalal, H., Sofiane, A. “A deep reinforcement learning-based cooperative approach for multi-intersection traffic signal control”, Intelligence, Vol. 114, 105019, pp. 1-10, 2022.

[3] Liu, Y.W., Zhang, X., Qu, T.G., et. al. “Intelligent robot motion trajectory planning based on machine vision”, International Journal of System Assurance Engineering and Management, Vol. 14, pp.776-785, 2023.

[4] Qi, R.L., Tang, Y.G., Zhang, K. “On-line self-calibration method for unattended manipulators based on Gaussian motion model and visual system”, Industrial Robot: the international journal of robotics research and application, Vol. 48, No. 2, pp.300-312, 2021.

[5] Du, Y.X., Chen, Y.H., “Time Optimal Trajectory Planning Algorithm for Robotic Manipulator Based on Locally Chaotic Particle Swarm Optimization”, Internet of Things and Automatic Control, Vol. 31, Vo. 5, PP. 906-914, 2022.

[6] Wang, J.F, Wang, Y. N. , Chen, W. R. , et. Al. “6D trajectory planning based on improved dynamic movement primitivess Control Theory and Applications”, vol. 39, no. 5, pp. 809-818, May. 2015.

[7] Chi M , Yao Y , Liu Y ,et. al. “Learning, Generalization, and Obstacle Avoidance with Dynamic Movement Primitives and Dynamic Potential Fields”, Applied Sciences, vol. 9, no. 8, pp. 1-10, 2015.

[8] N. Wan, D. Xu and H. Ye, "Improved cubic B-spline curve method for path optimization of manipulator obstacle avoidance," 2018 Chinese Automation Congress (CAC), Xi'an, China, 2018, pp. 1471-1476, doi: 10.1109/CAC.2018.8623056.

[9] M. J. A. Zeestraten, I. Havoutis, J. Silvério, S. Calinon and D. G. Caldwell, "An Approach for Imitation Learning on Riemannian Manifolds," in IEEE Robotics and Automation Letters, vol. 2, no. 3, pp. 1240-1247, July 2017, doi: 10.1109/LRA.2017.2657001.

[10] Pham Q C,Nakamura Y. “Affine trajectory deformation for redundant mechanical arms”, Robotics: Science and Systems , pp. 329-336, 2012.

[11] Q. -C. Pham and Y. Nakamura, "A New Trajectory Deformation Algorithm Based on Affine Transformations," in IEEE Transactions on Robotics, vol. 31, no. 4, pp. 1054-1063, Aug. 2015, doi: 10.1109/TRO.2015.2450413.

[12] Ginesi M, Meli D, Roberti A, et. al. “Dynamic Movement Primitives: Volumetric Obstacle Avoidance Using Dynamic Potential Functions”,vol. 101, no. 4, 2021.

Trajectory Tracking Control of Wheeled Mobile Robots

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***Abstract*—In order to solve the trajectory tracking problem of mobile robot with centroid deviation under nonholonomic constraints, a trajectory tracking control method for mobile robot with improved sliding mode variable structure is proposed, and the tracking controller with global asymptotic stability is also obtained by this method. This study is based on the motion model of three-wheeled mobile robot, and the tracking controller design is divided into two parts. In the first part, using X direction and forward direction with virtual feedback variable, the stability of the control system is analyzed by Lyapunov method, and the switching function is designed. The second part, the design of sliding mode controller, which adopts exponential approach law and uses S type exponential function instead of symbol function. This effectively improves the simulation efficiency , and reduces the jitter of the sliding mode variable structure control. The control law of linear velocity and angular velocity is designed to asymptotically stabilize the tracking error of mobile robot and realize the global asymptotic tracking of the reference trajectory. The experimental results show that the control law can be stable in a very short time, and the tracking plane coordinate error and forward angle error can converge to zero in a very short time. Moreover, the mobile robot can track the desired trajectory effectively, and its control effect is obviously improved, and the method has good anti-jamming performance.**

Keywords—three-wheeled mobile robot, sliding mode variable structure, trajectory tracking, globally asymptotically stable, Lyapunov function.

I. Introduction

With the rapid development and widespread application of mobile robots, the problem of trajectory tracking control has attracted significant attention [1]–[7]. The primary objective of trajectory tracking controller design is to ensure that, from arbitrary initial states and under limited external disturbances within a finite time, the robot can accurately follow a desired trajectory by properly regulating its linear and angular velocities.

In the study of trajectory tracking control under nonholonomic constraints, achieving faster convergence is often a key concern, as the convergence rate directly affects tracking accuracy. To address this issue, various control strategies have been proposed. For instance, a novel trajectory tracking approach based on the combination of sliding mode control and Lyapunov theory was presented in [8]. In [9], a sliding mode control law for linear velocity was developed using a backstepping design under the condition that the heading angle error approaches zero.

From the perspective of convergence performance [10]–[12], finite-time control systems are typically governed by fractional-order differential equations rather than conventional ordinary differential equations. As a result, the design and stability analysis of finite-time controllers are more challenging and are often limited to low-order systems. Moreover, studies on mobile robots with center-of-mass offset remain relatively scarce. To address this gap, [13] proposed a trajectory tracking controller based on sliding mode control considering center-of-mass offset.

Motivated by the above studies, this paper develops an improved sliding mode-based trajectory tracking control method for mobile robots with center-of-mass offset. By taking both convergence speed and tracking accuracy into account, the proposed method introduces an enhanced sliding mode control strategy to derive a novel control law. The stability of the tracking controller is analyzed using Lyapunov stability theory, and the global asymptotic stability of the system is rigorously proven.

II. Dynamic Models

A. Mobile Robot Model

The mobile robot studied in this work adopts a typical three-wheeled configuration [13]. In order to accurately track a desired trajectory, the robot must satisfy nonholonomic constraints. Figure 1 illustrates a representative three-wheeled mobile robot, which consists of two large coaxial rear wheels serving as driving wheels and a smaller front wheel acting as a passive caster.Let C denote the position of the robot’s center of mass, and R the midpoint of the coaxial driving wheels, which also corresponds to the geometric center of the robot. The distance between the center of mass and the geometric center is denoted as d. In the 2D X-Y Cartesian coordinate system, the robot’s pose is fully described by $\boldsymbol{p}=(\boldsymbol{x},\boldsymbol{y},\boldsymbol{\theta})^{\boldsymbol{T}}$, where $\boldsymbol{x}$ $\boldsymbol{y}$ represents the global coordinates of point C, and $\boldsymbol{\theta}$ is the angle between the robot’s heading direction and the X-axis. The control input vector is denoted as $\boldsymbol{q}=(\boldsymbol{v},\boldsymbol{w})^{\boldsymbol{T}}$, where $\boldsymbol{v}$ and $\boldsymbol{w}$

represent the linear and angular velocities of the mobile robot, respectively. In practical applications, to prevent wheel slippage due to excessive driving wheel speeds, both the linear velocity and angular velocity of the robot must be bounded. Under these conditions, the nonholonomic constraints of the wheeled mobile robot can be expressed as follows:

$$-\dot{\boldsymbol{x}}\sin\boldsymbol{\theta}+\dot{\boldsymbol{y}}\cos\boldsymbol{\theta}-d\dot{\boldsymbol{\theta}}=0 \tag{1}$$

Assuming that the driving wheels of the mobile robot satisfy the pure rolling condition without slipping, the kinematic model of the mobile robot can be expressed as：

$$\dot{\boldsymbol{p}}=\begin{pmatrix}\dot{\boldsymbol{x}}\\ \dot{\boldsymbol{y}}\\ \dot{\boldsymbol{\theta}}\end{pmatrix}=\begin{pmatrix}\cos\boldsymbol{\theta} & -d\sin\boldsymbol{\theta}\\ \sin\boldsymbol{\theta} & d\cos\boldsymbol{\theta}\\ 0 & 1\end{pmatrix}\boldsymbol{q} \tag{2}$$

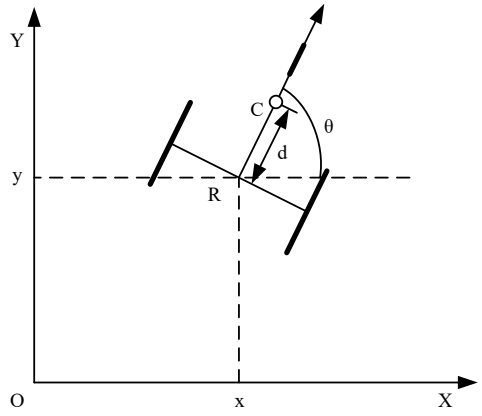

Fig. 1. Three-wheeled mobile robot

B. Mobile Robot Pose Error Model

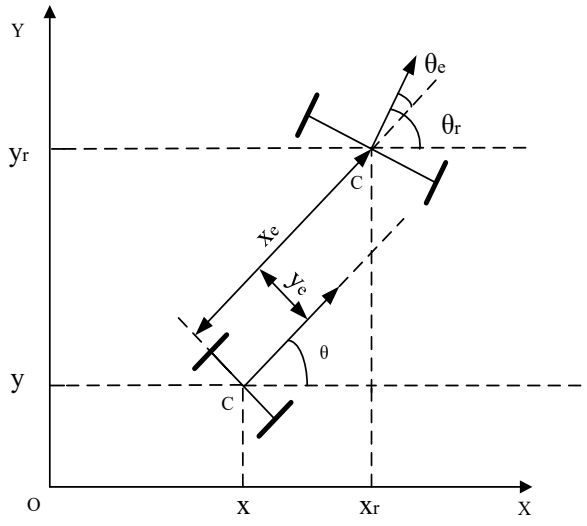

Fig. 2. The description of posture error

In mobile robot trajectory tracking, the objective is to design control laws for v and w to ensure that the robot can accurately follow the reference trajectory. Here, the pose $\boldsymbol{p}_r=(\boldsymbol{x}_r,\boldsymbol{y}_r,\boldsymbol{\theta}_r)^T$ represents the desired reference pose, and $q_r=(v_r,w_r)^T$ denote the corresponding reference linear and angular velocities.As illustrated in Figure 2, the wheeled mobile robot moves from its actual pose $\boldsymbol{p}=(\boldsymbol{x},\boldsymbol{y},\boldsymbol{\theta})^T$ to the reference pose $\boldsymbol{p}_r=(\boldsymbol{x}_r,\boldsymbol{y}_r,\boldsymbol{\theta}_r)^T$, and the corresponding pose error vector $\boldsymbol{p}_e=(\boldsymbol{x}_e,\boldsymbol{y}_e,\boldsymbol{\theta}_e)^T$ is defined to quantify the deviation between the actual and desired states.

Based on the coordinate transformation, the pose error dynamics of the mobile robot can be expressed as:

$$\boldsymbol{p}_e=\begin{pmatrix}\boldsymbol{x}_e\\ \boldsymbol{y}_e\\ \boldsymbol{\theta}_e\end{pmatrix}=\begin{pmatrix}\cos\boldsymbol{\theta} & \sin\boldsymbol{\theta} & 0\\ -\sin\boldsymbol{\theta} & \cos\boldsymbol{\theta} & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\boldsymbol{x}_r-\boldsymbol{x}\\ \boldsymbol{y}_r-\boldsymbol{y}\\ \boldsymbol{\theta}_r-\boldsymbol{\theta}\end{pmatrix} \tag{3}$$

For the three-wheeled mobile robot, by differentiating Eq. (3) and combining it with Eq. (2), the differential equation describing the trajectory tracking error of the mobile robot can be obtained as follows:

$$\dot{\boldsymbol{p}}_e=\begin{pmatrix}\dot{\boldsymbol{x}}_e\\ \dot{\boldsymbol{y}}_e\\ \dot{\boldsymbol{\theta}}_e\end{pmatrix}=\begin{pmatrix}\boldsymbol{y}_e\boldsymbol{w}-\boldsymbol{v}+v_r\cos\boldsymbol{\theta}_e-dw_r\sin\boldsymbol{\theta}_e\\ -\boldsymbol{x}_e\boldsymbol{w}-d\boldsymbol{w}+v_r\sin\boldsymbol{\theta}_e+dw_r\cos\boldsymbol{\theta}_e\\ w_r-\boldsymbol{w}\end{pmatrix} \tag{4}$$

III. Design of Trajectory Tracking Controller for Mobile Robots

The kinematic trajectory tracking of a mobile robot aims to determine appropriate bounded control inputs $\boldsymbol{q}=(\boldsymbol{v},\boldsymbol{w})^T$. For any initial pose error, under the designed control inputs $\boldsymbol{q}$, the pose error $\boldsymbol{p}_e=(\boldsymbol{x}_e,\boldsymbol{y}_e,\boldsymbol{\theta}_e)^T$ dynamics described by Eq. (4) remain bounded and satisfy the following equation:

$$\lim_{t\to\infty}\left\|\begin{pmatrix}\boldsymbol{x}_e & \boldsymbol{y}_e & \boldsymbol{\theta}_e\end{pmatrix}\right\|^T=0 \tag{5}$$

When the unknown parameters d in Eqs. (2) and (4) are set to zero, the three-wheeled mobile robot model simplifies to the two-wheeled mobile robot model described in [8], where the caster wheel serves only as a passive support during motion. Subsequently, the same control design approach can be applied to develop the trajectory tracking controllers for both types of mobile robots.

A. Design of the Switching Function

Lemma [14]: Let $x\in R$ and $|x|<\infty$, $\phi(x)=x\sin(\arctan x)\geq 0$.

To design the switching function for the variable structure control based on the backstepping method, the Lyapunov function is first defined as: $\boldsymbol{V}_y=\frac{1}{2}\boldsymbol{y}_e^2$, Assume $\boldsymbol{x}_e=c\boldsymbol{f}(k\boldsymbol{w})\boldsymbol{y}_e$, $\boldsymbol{f}(k\boldsymbol{w})=\frac{1-\exp(-2k\boldsymbol{w})}{1+\exp(-2k\boldsymbol{w})}$ is a hyperbolic tangent function. $c\boldsymbol{f}(k\boldsymbol{w})\boldsymbol{y}_e$ is a virtual feedback variable，

c,k >0, Based on the properties of the hyperbolic tangent function, it can be seen that: $\boldsymbol{wf}(k\boldsymbol{w}) \geq 0$. Assume $\boldsymbol{\theta}_e = -\arctan(\boldsymbol{v}_r \boldsymbol{y}_e)$, $\dot{\boldsymbol{\theta}}_e = \frac{\partial \boldsymbol{\theta}_e}{\partial v_r} + \frac{\partial \boldsymbol{\theta}_e}{\partial \boldsymbol{y}_e} = -\frac{v_r + \boldsymbol{y}_e}{1 + (v_r \boldsymbol{y}_e)^2} < 0$

For the three-wheeled mobile robot illustrated in Figure 1, and based on the above equations and the lemma, the following expression can be derived:

$$\begin{aligned}
\dot{\boldsymbol{V}}_y &= \boldsymbol{y}_e \dot{\boldsymbol{y}}_e \\
&= \boldsymbol{y}_e(-\boldsymbol{x}_e \boldsymbol{w} - d\boldsymbol{w} + v_r \sin\boldsymbol{\theta}_e + dw_r \cos\boldsymbol{\theta}_e) \\
&= -\boldsymbol{y}_e \boldsymbol{x}_e \boldsymbol{w} - \boldsymbol{y}_e d\boldsymbol{w} + v_r \boldsymbol{y}_e \sin(-\arctan(v_r \boldsymbol{y}_e)) + \\
&\quad \boldsymbol{y}_e dw_r \cos(-\arctan(v_r \boldsymbol{y}_e)) \\
&\leq -\boldsymbol{y}_e \boldsymbol{x}_e \boldsymbol{w} - \boldsymbol{y}_e d\boldsymbol{w} + \boldsymbol{y}_e dw_r \\
&= -\boldsymbol{y}_e^2 c\boldsymbol{f}(k\boldsymbol{w})\boldsymbol{w} + \boldsymbol{y}_e d\dot{\boldsymbol{\theta}}_e \\
&< 0
\end{aligned}$$

From the above, the following conclusion can be drawn: the system will converge as long $\boldsymbol{x}_e$ as it satisfies $c\boldsymbol{f}(k\boldsymbol{w})\boldsymbol{y}_e$, $\boldsymbol{\theta}_e$ satisfies $-\arctan(v_r \boldsymbol{y}_e)$ .Then, the system state variables $\boldsymbol{y}_e$ will converge to zero. Based on this conclusion, the switching function is designed as follows:

$$\boldsymbol{s} = \begin{bmatrix} \boldsymbol{s}_1 \\ \boldsymbol{s}_2 \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}_e - c\boldsymbol{f}(k\boldsymbol{w})\boldsymbol{y}_e \\ \boldsymbol{\theta}_e + \arctan(v_r \boldsymbol{y}_e) \end{bmatrix} \tag{6}$$

By designing the sliding mode controller, $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$ can be driven to approach zero. This implies that the tracking errors $\boldsymbol{x}_e$ converge $c\boldsymbol{f}(k\boldsymbol{w})\boldsymbol{y}_e$, $\boldsymbol{\theta}_e$ ultimately leading to convergence to zero. Consequently, all system state variables $\boldsymbol{y}_e$ converge to zero.

B. Design of the Sliding Mode Controller

To reduce chattering, an exponential reaching law $\dot{\boldsymbol{s}} = -\varepsilon \operatorname{sgn}\boldsymbol{s} - h\boldsymbol{s}$ is adopted. ε, h >0.where ε is a positive real number.By $\frac{1-e^{-s}}{1+e^{-s}}$ replacing the sign function with a continuous-type function, the following equation can be obtained:

$$\dot{\boldsymbol{s}}_i = -\varepsilon_i \frac{1-e^{-s_i}}{1+e^{-s_i}} - h_i \boldsymbol{s}_i, i = 1,2 \tag{7}$$

Here, ε_i is a small positive number and h_i is a real constant. Assume $\alpha = \arctan(v_r \boldsymbol{y}_e)$, From Eqs. (4) and (6), we obtain:

$$\begin{aligned}
\dot{\boldsymbol{s}} &= \begin{bmatrix} \dot{\boldsymbol{s}}_1 \\ \dot{\boldsymbol{s}}_2 \end{bmatrix} = \begin{bmatrix} -\varepsilon_1 \frac{1-e^{-s_1}}{1+e^{-s_2}} - h_1 \boldsymbol{s}_1 \\ -\varepsilon_2 \frac{1-e^{-s_2}}{1+e^{-s_2}} - h_2 \boldsymbol{s}_2 \end{bmatrix} \\
&= \begin{bmatrix} \dot{\boldsymbol{x}}_e - ck\boldsymbol{f}'(k\boldsymbol{w})\dot{\boldsymbol{w}}\boldsymbol{y}_e - c\boldsymbol{f}(k\boldsymbol{w})\dot{\boldsymbol{y}}_e \\ \dot{\boldsymbol{\theta}}_e + \frac{\partial \alpha}{\partial v_r}\dot{v}_r + \frac{\partial \alpha}{\partial \boldsymbol{y}_e}\dot{\boldsymbol{y}}_e \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{y}_e \boldsymbol{w} - \boldsymbol{v} + v_r \cos\boldsymbol{\theta}_e - dw_r \sin\boldsymbol{\theta}_e - ck\boldsymbol{f}'(k\boldsymbol{w})\dot{\boldsymbol{w}}\boldsymbol{y}_e - c\boldsymbol{f}(k\boldsymbol{w})\dot{\boldsymbol{y}}_e \\ w_r - \boldsymbol{w} + \frac{\partial \alpha}{\partial v_r}\dot{v}_r + \frac{\partial \alpha}{\partial \boldsymbol{y}_e}\dot{\boldsymbol{y}}_e \end{bmatrix}
\end{aligned}$$

From the above equation, the kinematic trajectory tracking control law for the three-wheeled mobile robot can be derived:

$$\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix} = \begin{bmatrix} \boldsymbol{y}_e \boldsymbol{w} + v_r \cos\boldsymbol{\theta}_e - dw_r \sin\boldsymbol{\theta}_e - ck\boldsymbol{f}'(k\boldsymbol{w})\dot{\boldsymbol{w}}\boldsymbol{y}_e - c\boldsymbol{f}(k\boldsymbol{w})\dot{\boldsymbol{y}}_e + \varepsilon_1 \frac{1-e^{-s_1}}{1+e^{-s_1}} + h_1 \boldsymbol{s}_1 \\ \dfrac{w_r + \frac{\partial \alpha}{\partial v_r}\dot{v}_r + \frac{\partial \alpha}{\partial \boldsymbol{y}_e}(v_r \sin\boldsymbol{\theta}_e + dw_r \cos\boldsymbol{\theta}_e) + \varepsilon_2 \frac{1-e^{-s_2}}{1+e^{-s_2}} + h_2 \boldsymbol{s}_2}{1 + \frac{\partial \alpha}{\partial \boldsymbol{y}_e}(\boldsymbol{x}_e + d)} \end{bmatrix} \tag{8}$$

When $d = 0$ is chosen, the kinematic trajectory tracking control law of the mobile robot becomes:

$$\begin{bmatrix} \boldsymbol{v}_0 \\ \boldsymbol{w}_0 \end{bmatrix} = \begin{bmatrix} \boldsymbol{y}_e \boldsymbol{w} + v_r \cos\boldsymbol{\theta}_e - ck\boldsymbol{f}'(k\boldsymbol{w})\dot{\boldsymbol{w}}\boldsymbol{y}_e - c\boldsymbol{f}(k\boldsymbol{w})\dot{\boldsymbol{y}}_e + \varepsilon_1 \frac{1-e^{-s_1}}{1+e^{-s_1}} + h_1 \boldsymbol{s}_1 \\ \dfrac{w_r + \frac{\partial \alpha}{\partial v_r}\dot{v}_r + \frac{\partial \alpha}{\partial \boldsymbol{y}_e}(v_r \sin\boldsymbol{\theta}_e) + \varepsilon_2 \frac{1-e^{-s_2}}{1+e^{-s_2}} + h_2 \boldsymbol{s}_2}{1 + \frac{\partial \alpha}{\partial \boldsymbol{y}_e}\boldsymbol{x}_e} \end{bmatrix} \tag{9}$$

Here, $\frac{\partial \alpha}{\partial v_r} = \frac{\boldsymbol{y}_e}{1 + (v_r \boldsymbol{y}_e)^2}$, $\frac{\partial \alpha}{\partial \boldsymbol{y}_e} = \frac{v_r}{1 + (v_r \boldsymbol{y}_e)^2}$.

IV. Experimental Results

In order to illustrate the effectiveness of the method, we compared it with previous methods [16] in three cases. The first case uses only kinematic models and kinematic controllers. In the second case, the dynamic controller and dynamic model are added. In the third case, an adaptive dynamics compensation controller was used in the case of the second addition of the dynamic model.

The robot control law is given by Eq. 4 and 10, and the robust adaptive update law is given by Eq. 13. Then apply the robot control structure shown in Fig. 2, and simulate the application through MATLAB/Simulink.

In all simulations, the common parameters are as follows: fixed sample time of 0.1s; saturation parameters $l_u = 1$, $l_w = 1$.

$$\gamma = diag(1.7, 1.1, 0.5, 0.3, 0.01, 0.5)$$

$$\Gamma = diag(0.0005, 0.001, 0.001, 0.00006, 0.001, 0.001)$$

The robot starts at position $(0.2, 0.0)$ m with orientation 0 degrees, and should track a 8-shaped reference trajectory starting at $(0.0, 0.0)$ m.

Simulation results for kinematic controller and kinematic model:

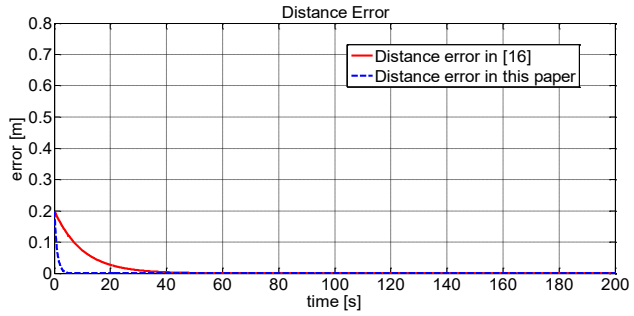

Fig. 3. Shows the actual tracking trajectory of the robot.

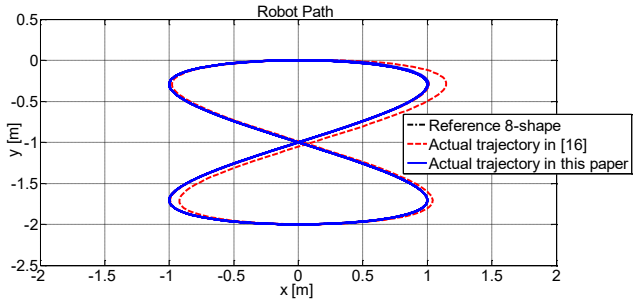

Fig. 4. Represent the tracking distance error

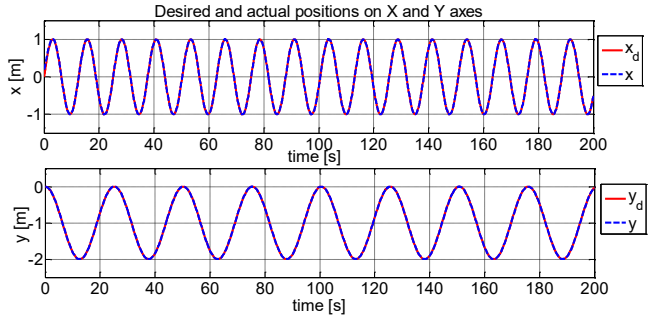

Fig. 5. Represent the velocities tracking of the robot

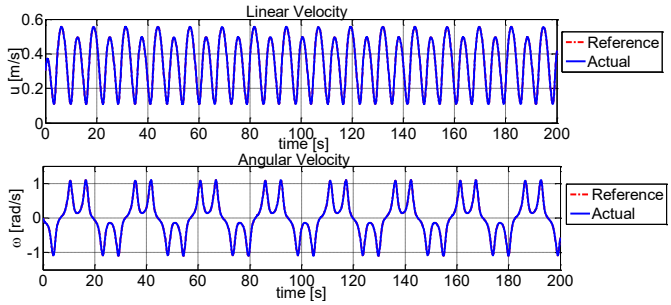

Fig. 6. Represent robot positions tracking on the X and Y axes

The simulation results are shown in Fig. 3 and Fig. 4. The robot path tracking and distance error evolved into perfect tracking. Fig. 5and Fig. 6 show the robot's X-axis and Y-axis position and velocity tracking, respectively. Since the robot dynamics model is not considered, the difference between the reference speed and the actual speed is not seen in Fig. 4b, and the distance error evolution will remain at zero. In addition, the proposed method and the reference [16], the robot path tracking speed Fast, the distance error tends to zero only about 3s.

V. Conclusion

This paper proposes a control law for mobile robots with a shifted center of mass, incorporating virtual feedback metrics for both the X-axis position and the heading angle. Specifically, both the linear and angular velocity controllers are designed based on the sliding mode variable structure method. The proposed control algorithm is characterized by rapid convergence, strong robustness, and excellent stability, enabling the mobile robot to quickly and accurately track a preset figure-eight reference trajectory. Simulation results validate the effectiveness and relevance of this control strategy.

References

[1] Basil N, Marhoon H M, Sahib D F, et al. Accelerated black hole optimization algorithm with enhanced FOPID controller for omni-wheel drive mobile robot system[J]. Neural Computing and Applications, 2025, 37(21): 16983-17014.

[2] Bingol M C. A Safe Navigation Algorithm for Differential-Drive Mobile Robots by Using Fuzzy Logic Reward Function-Based Deep Reinforcement Learning[J]. Electronics, 2025, 14(8): 1593.

[3] Rout R, Bera T K, Chatti N. Fault tolerant self-reconfigurable waypoint guidance for mobile robots under actuator faults[J]. Journal of the Franklin Institute, 2025, 362(10): 107735.

[4] Lee W, Joo H, Kang Y K, et al. A review on the force sensing and force feedback-based navigation of mobile robots[J]. International Journal of Precision Engineering and Manufacturing, 2025, 26(5): 1291-1311.

[5] Hoseinnezhad R. A comprehensive review of deep learning techniques in mobile robot path planning: Categorization and analysis[J]. Applied Sciences, 2025, 15(4): 2179.

[6] Tran Q K, Ryoo Y J. Multi-Sensor Fusion Framework for Reliable Localization and Trajectory Tracking of Mobile Robot by Integrating UWB, Odometry, and AHRS[J]. Biomimetics, 2025, 10(7): 478.

[7] Arega T B, Tesfa Y M, Abdissa C M. Three-wheeled mobile robot trajectory tracking control using nonlinear PID controller based neural network combined with backstepping controller[J]. IEEE Access, 2025.

[8] Louda S, Karkar N, Seghir F, et al. Fuzzy Dynamic Feedback Linearization for Efficient Mobile Robot Trajectory Tracking and Obstacle Avoidance in Autonomous Navigation[J]. International Journal of Robotics & Control Systems, 2025, 5(2).

[9] Dang S T, Dinh X M, Than Ngoc T A, et al. Adaptive Sliding Mode Control for Robust Trajectory Tracking of Car-Like Mobile Robots in Autonomous Navigation Systems[J]. Journal of Applied and Computational Mechanics, 2025: e19517.

[10] Schrick M, Hinckeldeyn J, Thiel M, et al. A microservice based control architecture for mobile robots in safety-critical applications[J]. Robotics and Autonomous Systems, 2025, 183: 104795.

[11] Kang Y, Hao Y, Wang Q, et al. Formation control of networked mobile robots with weighted directed topology using the Udwadia–Kalaba approach[J]. Nonlinear Dynamics, 2025, 113(6): 5423-5438.

[12] Shahab M, Nasir A, Alyazidi N M. Formation control of wheeled mobile robots with fault-tolerance capabilities[J]. Robotics, 2025, 14(5): 59.

[13] Damjanović D, Biočić P, Prakljačić S, et al. A comprehensive survey on SLAM and machine learning approaches for indoor autonomous navigation of mobile robots[J]. Machine vision and applications, 2025, 36(3): 55.

[14] Salimpour S, Peña-Queralta J, Paez-Granados D, et al. Sim-to-real transfer for mobile robots with reinforcement learning: from nvidia isaac sim to gazebo and real ros 2 robots[J]. arXiv preprint arXiv:2501.02902, 2025.

[15] Shen Y, Shen Y, Zhang Y, et al. Research Progress on Path Planning and Tracking Control Methods for Orchard Mobile Robots in Complex Scenarios[J]. Agriculture, 2025, 15(18): 1917.

[16] Martins, Felipe N., Mário Sarcinelli-Filho, and Ricardo Carelli. "A velocity-based dynamic model and its properties for differential drive mobile robots." Journal of Intelligent & Robotic Systems 85.2 (2017): 277-292.

[17] De La Cruz, Celso, and Ricardo Carelli. "Dynamic model based formation control and obstacle avoidance of multi-robot systems." Robotica 26.3 (2008): 345-356.

Robotic Force Control Approach Based on Multi-parameter Optimization in Grinding Process

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***Abstract*—Aiming to resolve the issues of poor force control stability and inconsistent surface quality in the grinding of complex curved components, a control method based on multivariate process parameter rectification has been proposed. This method integrates a particle swarm optimization-based variable impedance strategy with a random forest regression model. The key process parameters are identified through the traditional impedance control framework in conjunction with Preston's theory, and their influence is quantitatively analyzed using machine learning regression techniques. An adaptive control mechanism is established by incorporating feedback from a six-dimensional force sensor. Simulation and experimental results indicate that the most influential theoretical parameters affecting surface quality are the contact force error, feed rate, and the damping coefficient of the robotic system. Although hardware-related parameters such as abrasive belt particle size, belt speed, and target contact force exhibit certain effects, their overall impact remains limited. By employing the proposed force control strategy, the surface roughness is effectively reduced from approximately 0.6 μm to 0.3 μm, resulting in a notable improvement in process stability and consistency. The engineering applicability of the method under complex working conditions has been verified.**

Keywords—Robotic grinding; Force control; Process parameter optimization; Impedance control

I. INTRODUCTION

As the demand for precision machining in advanced equipment manufacturing continues to increase, robotic grinding has become a crucial technique for fabricating complex surfaces, such as aero-engine blades and high-precision molds. However, the intricate coupling mechanisms among multiple process parameters present significant challenges in maintaining consistent quality and process stability in robotic grinding. The strongly nonlinear interactions among key physical parameters—including contact force, belt line speed, feed rate, and grit size—result in limited predictive accuracy for material removal and surface quality, while also increasing the risk of belt wear and surface damage[1]. In engineering applications, uneven machining allowances and the accumulation of contact force errors due to workpiece positioning errors are common, further amplifying the sensitivity of the process parameters. This makes it difficult to ensure actual machining quality with the parameter optimization scheme of offline programming. Currently, the process stability of the contact operation largely relies on the indirect feedback of contact force information. Although the adjustment of each process parameter affects the material removal rate to varying degrees, it is possible to optimize the corresponding parameters using the contact force information collected by the force sensor. Therefore, a relatively stable robot constant force control method can compensate for the accuracy deviation between the mathematical model of the workpiece and the actual workpiece, while also reducing the impact of process parameter selection errors on the consistency of surface quality.

In response to the increasing demand for high-precision, high-quality, and stable grinding of high-end components, researchers have extensively investigated methods to achieve stable constant-force control in industrial robots. Building upon traditional force control approaches, advanced strategies have been proposed to enhance control stability, which are generally categorized into two primary approaches: (1) Impedance control: This method seeks to adjust the robot's dynamic behavior to ensure predictable and controlled responses to external forces. Xu and colleagues developed a compliant deburring robot control strategy based on force impedance, enabling real-time adjustment of position and velocity through normal and tangential forces[2]. Li et al. devised a closed-loop force control system for manipulators, successfully facilitating constant-force tracking over complex surfaces[3]. Zhou et al introduced an adaptive impedance-based hybrid control strategy for robotic grinding, which mitigates contact shocks and enhances both positional and force-tracking accuracy[4]. (2) Hybrid force/position control: This approach involves dividing the task space into position and force control subspaces, which are controlled separately. Zhang proposes a force/position hybrid control scheme for grinding machine based on speed servo. By converting force into speed to control the speed servo robot, it realizes direct control of force and has faster response speed[5]. Zhang and Sun independently proposed fuzzy PID-based hybrid force/position control strategies, with the former combining force and torque to mitigate sharp-edge phenomena[6-7]. In recent years, machine learning has attracted increasing attention and provided new perspectives for addressing the aforementioned challenges. Numerous scholars have explored machine learning methods to address the constant force tracking control problem for robots under uncertain working conditions. Zhang et al. introduced a deep

reinforcement learning-based method to optimize controller parameters online, compensating for end-effector deformation during grinding and enhancing force stability[8]. Weng et al. analyzed the strengths and limitations of traditional techniques, neural networks, and deep reinforcement learning, discussing trends in hybrid control systems[9]. Carlucho I developed an intelligent control framework integrating deep reinforcement learning with adaptive multi-PID strategies to manage systems characterized by complex dynamics[10]. Zhang et al. proposed a PPO-based reinforcement learning algorithm for constant-force grinding and developed a lightweight neural network architecture to enhance control responsiveness[11].

In summary, at the current stage, research efforts remain predominantly concentrated on the analysis of single-parameter influence laws, while the dynamic mechanisms under the synergistic effect of multiple parameters have not been sufficiently investigated. As a result, the determination of the process window continues to rely heavily on operator experience and trial-and-error methods, which proves inadequate for addressing the processing requirements associated with abrupt changes in geometric features of complex surfaces and the machining of novel composite materials. To this end, traditional impedance control methods have been utilized as the foundational force-position control mechanism in decoupled robots, and intelligent control algorithms together with machine learning prediction methods have been integrated to enable dynamic regulation and optimization of multivariate grinding process parameters. A particle swarm variable impedance random forest-based robot grinding force control method, grounded in the rectification of multivariate process parameters, has been proposed as a key technology for addressing the aforementioned engineering challenges.

II. Steady-state Error Analysis of Traditional Impedance Control

A. Workpiece and Belt Sander Contact Force Model

A model of the contact between the robot clamped workpiece and the belt sander is shown in Figure 1. In this model, the robot clamping the workpiece can be considered as a mass-spring-damping system, and the abrasive head of the belt sander can be simplified as a spring-damping system. Where x_r is the reference position planned by the trajectory, x_e is the environmental position, x is the actual position of the end of the workpiece, k_e and b_e are the environmental stiffness and environmental damping coefficients, m_d , b_d , and k_d are the inertia coefficients, damping coefficients and stiffness coefficients, respectively, in the impedance model of the robot, and f_e is the actual contact force between the robot's end blade and the abrasive head [12].

The industrial robot controller controls the robot to reach a specified position, which is x=x_d , where x_d is the desired trajectory of the robot motion. In the impedance model f_d is the desired contact force, then the impedance control model expression for the workpiece and the belt sander in a certain direction is:

$$m_d(\ddot{x} - \ddot{x}_r) + b_d(\dot{x} - \dot{x}_r) + k_d(x - x_r) = f_d - f_e \quad (1)$$

The expression for the contact force for the grinding head of a belt sander is:

$$f_e = b_e\dot{x} + k_e(x - x_e) \quad (2)$$

B. Analysis of Steady-State Error

The impedance control strategy integrates an inner-loop position controller with an outer-loop force controller, illustrated in Fig 2.

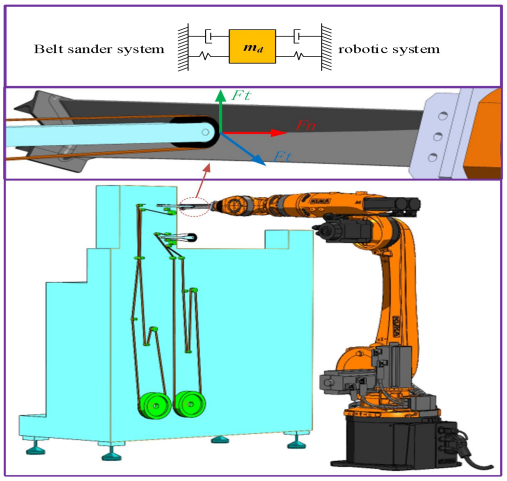


Fig. 1. Workpiece-belt sander interaction model

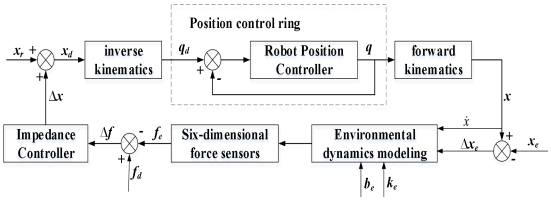


Fig. 2. Schematic diagram of traditional impedance control strategy

The transfer function of the impedance-based control system can be readily obtained from equation (1) as follows:

$$G(s) = 1/(m_d s^2 + b_d s + k_d) \quad (3)$$

Consequently, the relationship between the trajectory correction ΔX and the contact force deviation ΔF is expressed as:

$$\frac{\Delta X}{\Delta F} = \frac{1}{(m_d s^2 + b_d s + k_d)} \quad (4)$$

By substituting $\Delta F = F_d - F_e$ and $\Delta X = X(s) - X_r(s)$ into equation (4), the following expression is derived:

$$(m_d s^2 + b_d s + k_d)\big(X(s) - X_r(s)\big) = F_d(s) - F_e(s) \quad (4)$$

By transforming equation (2), the expression for the actual robot position is obtained:

$$X(s) = \frac{F_e + k_e X_e}{b_e s + k_e} \tag{5}$$

Equation (6) is inserted into (5) and simplified to obtain the following result.

$$\Delta F = \frac{m_d s^2 + b_d s + k_d}{m_d s^2 + (b_d + b_e)s + k_d + k_e} \times$$

$$\left(F_d + k_e X_e(s) - k_e X_r(s) - b_e s X_r(s)\right) \tag{6}$$

According to the final value theorem of Laplace transform, the steady-state error of the contact force deviation is:

$$\Delta F_{ess} = \lim_{t\to\infty} \Delta F(t) = \lim_{s\to 0} s\Delta F(s) =$$

$$\frac{(m_d s^2 + b_d s + k_d)k_e}{m_d s^2 + (b_d + b_e)s + k_d + k_e}\left(\frac{s}{k_e} F_d + s X_e - s X_r - \frac{b_e s^2}{k_e} X_r\right) \tag{8}$$

When the system input is a step input, $F_d = \frac{f_d}{s}$, $X_e(s) = \frac{x_e}{s}$, $X_r(s) = \frac{x_r}{s}$, then:

$$\Delta F_{ess} = \lim_{s\to 0} s\Delta F(s) = \frac{k_d k_e}{k_d + k_e}\left(\frac{f_d}{k_e} + x_e - x_r\right) \tag{7}$$

From equation (9), in order to satisfy the control requirement of zero steady state error of the system, two strategies can be adopted: one is to set the desired stiffness coefficient of the impedance controller to zero (k_d=0); and the other is to adjust the reference trajectory of the robot to satisfy x_r=f_d/k_e+x_e. In the first strategy, the deviation of the system contact force (Δf_{ess}=0) is achieved when the desired contact force is set to f_d=0. Under this condition, the position following error e=x−x_r=0 can also be satisfied, even if the stiffness of the machined part is an arbitrary non-zero value.

When the desired contact force $f_d \neq 0$ and the end position $x \neq x_e$, the contact force error between the robot end and the workpiece must be minimized in the final steady state. To achieve this, the desired stiffness coefficient k_d is set to zero, leading to the following equation relationship.

$$\Delta f = f_d - f_e = m_d \ddot{e} + b_d \dot{e} \tag{8}$$

From equation (10), under any zero-stiffness condition, a suitable set of impedance parameters m_d, b_d can be found to ensure the tracking error e=0.

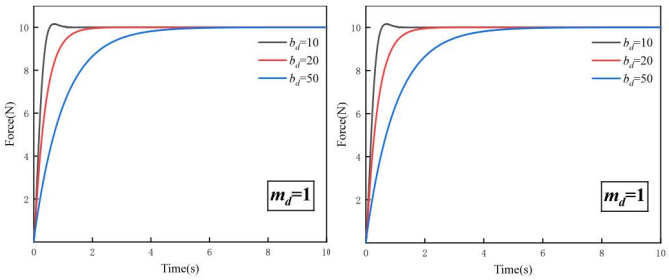
In this study, the reduction of the coupling relationship among various process parameters is investigated with respect to its effect on the surface material removal rate of the workpiece. To this end, the first method of variable impedance control, in which the stiffness coefficient is set to zero, is employed as the basis for adjusting the coupling between the actual position of the robot's end effector and the contact force.

III. Particle Swarm Variable Impedance Random Forest Robot Grinding Force Control Methods

A. Particle Swarm Variable Impedance Parameter Control Method

When the robot end grinding tool and the processed parts contact, due to the inertia of the robot's motion will certainly produce impact force, and the stability of the contact has a greater impact. Therefore, the control characteristics of different impedance parameters are analyzed (set the stiffness coefficient as 0), as shown in Fig. 3.

From Fig. 3, it can be observed that an increase in expected inertia leads to higher acceleration of the robot end during contact with the workpiece. This results in increased overshoot of the contact force, enhanced vibration, and longer regulation time, which may cause system disturbances. Therefore, the inertia coefficient is not suitable for real-time adjustment in the impedance controller. Selecting an appropriate constant value for the expected inertia parameter is a crucial prerequisite for ensuring the stability of the control system. On the contrary, when the desired damping parameter is increased, it only leads to an increase in the regulation time for the system to reach the steady state,and does not have a serious impact on the stability of the system. Therefore, in the next study, only the desired damping coefficient is adjusted in real time in the impedance controller.



(a) Contact force response curves with different inertia coefficients

(b) Contact force response curves with different damping coefficients

Fig. 3. Contact force response curves with different impedance parameters

The particle swarm optimization (PSO) algorithm was proposed by American scholars Russell Eberhardt and James Kennedy in 1995. The PSO algorithm sets up a random initial position of particles, and then finds an approximate optimal solution by iterating through a loop with different set number of iterations. In each update of the system, each random particle continuously updates itself through two judgment parameters pbest (the best position passed by a single particle) and gbest (the best position passed by the whole particle swarm). After reaching the set number of cycles of the system, by constantly searching for the values of the two parameters mentioned above, the particles update their speed and position by the following formula[13].

$$v_i = v_i + c_1 \times rand() \times (pbest_i - x_i)$$

$$+ c_2 \times rand() \times (gbest_i - x_i) \tag{11}$$

$$x_i = x_i + v_i \tag{12}$$

Where, i=1,2,...,N,N is the total number of particles in this swarm, v_i is the velocity of the particles, x_t is the current position of the particles, *rand*() is a random number between, c_1 and c_2 are the learning factors.

In order to realize the automatic dynamic adjustment of the damping coefficient in the impedance controller of the robot, the PSO method is used to adjust the real-time parameter tuning, so that the damping parameter corresponding to the current moment that makes the amount of abrasive removal converge to the ideal value can be found. The flow chart of the control strategy is shown in Fig. 4.

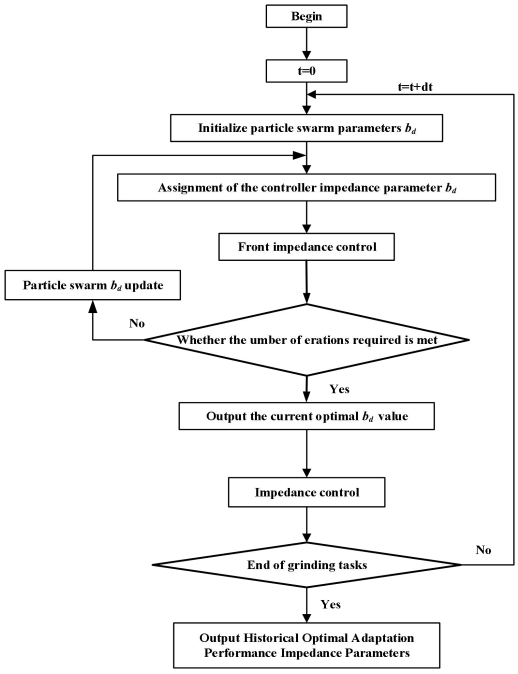


Fig. 4. Flow chart of particle swarm variable impedance parameter control strategy

B. *Analysis of the Importance of Features Affecting the Robotic Grinding Process*

Based on the impedance model in equation (1), setting k_d=0 and combining with equation (10) yields the following:

$$\ddot{x}(t) = \ddot{x}_r(t) + \frac{1}{m_d}\left[\Delta f(t) - b_d\left(\dot{x}(t-1) - \dot{x}_r(t)\right)\right] \quad (13)$$

$$\dot{x}(t) = \dot{x}(t-1) + \ddot{x}(t)T \quad (14)$$

$$x(t) = x(t-1) + \dot{x}(t)T \quad (15)$$

By substituting equations (13) and (14) into (15), the following expression is derived:

$$\Delta x = \dot{x}(t-1)T + \{\ddot{x}_r(t) + \frac{1}{m_d}[\Delta f - b_a(\dot{x}(t-1)$$

$$\dot{x}_r)]\}T^2 \quad (16)$$

In equation (16), Δx can be taken as the theoretical actual grinding removal amount, in which the desired acceleration, the desired grinding speed and the desired inertia coefficient in the impedance controller are the constant values set by the system. Through the above equation, it can be reflected intuitively that the factors that can affect the actual removal amount are: the expected damping coefficient in the impedance controller, the actual contact force error, the actual grinding speed and the sampling time of the robot control system.

The particle swarm variable impedance parameter control strategy is employed to simulate the actual surface grinding process. The control system is initially adjusted from Δt=0.02 s to other sampling intervals such as Δt=0.01 s and Δt=0.03 s. The simulation is then repeated to record parameters including the contact force error, the robot's damping coefficient, and end-effector polishing variables, which are logged in an Excel spreadsheet. The simulation training set of speed-related influence parameters. Random forest regression analysis program was written in PyCharm platform, and the training data were submitted to it for classification analysis, and the results of feature importance analysis are shown in Fig. 5.

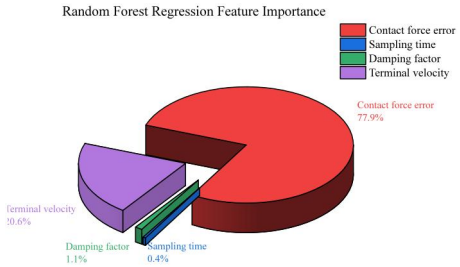


Fig. 5. Importance analysis of grinding process parameters characteristics

C. *Evaluation of Regression Prediction Models*

The random forest regression algorithmic formulation is based on a decision tree regression model, where the prediction function for each decision tree can be expressed as:

$$f_k(x) = \sum_{j=1}^{J_k} c_{kj} I\left(x \in R_{kj}\right) \quad (17)$$

where k denotes the k decision tree, x denotes the input sample, J_k denotes the number of leaf nodes of the k decision tree, c_{kj} denotes the predicted value of the j leaf node of the k decision tree, and R_{kj} denotes the set of samples of the j leaf node of the k decision tree.

The prediction function of multiple decision trees can be expressed as:

$$f(x) = \frac{1}{K}\sum_{k=1}^{K} f_k(x) \quad (18)$$

Where K denotes the number of decision trees.

Common evaluation metrics for random forest regression include mean square error (MSE) and R-squared (R^2). A lower MSE value indicates better model fit, while an R^2 value closer to 1 suggests a higher level of explanatory power. Conversely, higher MSE and lower R^2 values indicate poorer model performance.

The formula for its calculation is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (19)$$

Where n denotes the number of samples, y_i denotes the true value of the i sample, and denotes the predicted value of the i sample.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (20)$$

Where $\bar{y}$ denotes the average of the true values of all samples.

The evaluation results of the console results output for model predictions are shown in Fig. 6.

As can be seen from the fitting result data in the figure, the method for fitting the grinding process of the influence of individual process parameters on the actual amount of removal of the model fitting effect reached the expected ideal level.

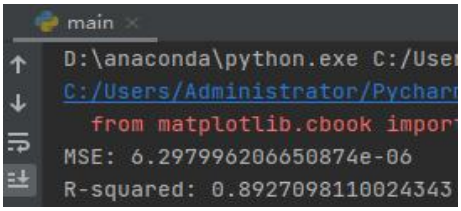


Fig. 6. Output of console fitting results

D. Optimal Grinding Speed Solution Method

Through the analysis of the above equation (14) and Fig. 5, it can be seen that in addition to the contact force error, the grinding speed of the robot also has a more important influence on the overall grinding process. The end speed of the robot is composed of three speeds, as shown in Fig. 7.

Belt sander belt line speed V_s will directly affect the efficiency of grinding and surface quality, high rotational speed per unit time abrasive contact with the workpiece more times, will improve the material removal rate, but may lead to overheating or wear faster. The set motion speed V_t at the end of the robot determines the contact time between the grinding coverage area and the unit point area, and the speed over may lead to uneven grinding, overcutting or undercutting phenomenon. The feed rate V_f of the grinding force has a direct effect on maintaining a constant contact force, especially when the surface of the workpiece is uneven or the hardness of the material varies, and the robot needs to adjust the position of the end at the proper speed to maintain the proper pressure. The first two speeds are constant values set empirically in the grinding control system. Reasonable optimization of the longitudinal feed speed enables a faster response to contact force variations. This adjustment helps maintain a more stable contact force, thereby improving grinding quality and reducing the impact of force fluctuations on surface consistency.

When setting up the grinding trajectory, it is not complicated to plan the desired removal depth at each position. However, if you want to process in real time at the optimal feed grinding speed, you have to solve the nonlinear differential equations for the corresponding moments, which is undoubtedly a difficult and time-consuming process. Using machine learning, the established grinding prediction model was trained using 2000 datasets collected from previous grinding experiments under varying process conditions. Once trained, the model can be applied in real time to determine the optimal grinding speed for test sets corresponding to varying material removal amounts. The model training was considered complete when the performance on the validation set became stable and further increasing the number of trees did not result in a significant improvement in MSE or R^2. The trained model could then be used to provide prediction values as substitutes for repeatedly solving the nonlinear differential equations in real time, as illustrated in Fig. 8.

From Fig. 8, it can be observed that a corresponding actual training speed exists for each material removal amount during the grinding process, indicating that the predicted values are capable of being effectively adapted to variations in the working environment. The predicted value output from the calculation is fed into the speed control interface of the robot control system to automatically adjust the grinding speed of the robot on-the-fly. This is a good solution to the complex nonlinear problem between the grinding process parameters and the real-time optimal grinding speed solution problem in the grinding process.

The PSO algorithm is used to automate the regulation of impedance parameters and the stable control of contact force in the impedance controller, and a classical random forest machine learning method is incorporated to correct the robot's normal grinding and throwing feed rate. So far, a particle swarm variable impedance random forest robot abrasion force control method is proposed to solve the above faced engineering problems. In the following section, the optimization method for the proposed mechanism parameters is validated and analyzed through both simulation and experimental environments. The hardware tool parameters are further corrected via grinding experiments based on Preston's empirical formula, as shown in equation (21)[14].

$$\frac{dH}{dt} = K_p P v_r \quad (21)$$

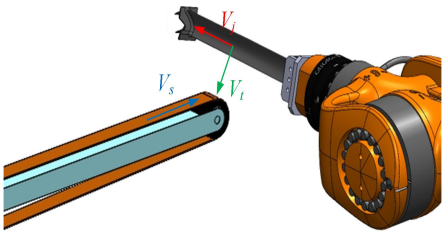


Fig. 7. Schematic diagram of end velocity decomposition

Where dt is the time required for grinding ; dH is the depth of material removal during the time; K_p is the Preston coefficient, which is related to the grain size, shape, hardness of the abrasive, and the material properties of the workpiece to be processed; P is the contact pressure; and v_r is the speed of movement of the abrasive tool relative to the workpiece.

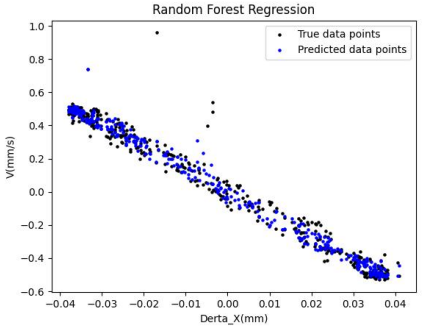


Fig. 8. Comparison of actual grinding speed and predicted grinding speed

According to the analysis of equation (21), several process parameters in the hardware equipment are still identified as influencing factors for the final grinding surface quality. Accordingly, different expected values of contact force, belt speed, and abrasive belt mesh number will be introduced in the experimental phase for further analysis.

IV. Simulation and Experimental Analysis

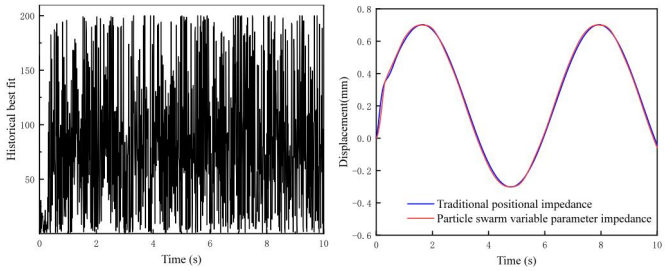
A. Simulation Analysis

A simulation platform was built in Matlab to model the interaction between the robot and external contact forces, through a group of the most curved parts representative of the simulation adjustment that is, when the position of the working condition

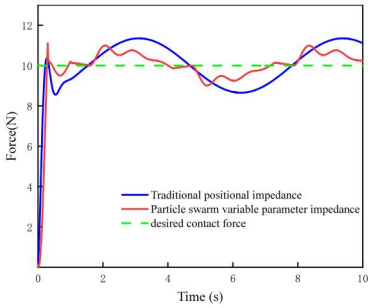
occurs sinusoidal changes in the stiffness remains unchanged, respectively, the particle swarm variable impedance parameter control strategy and the traditional impedance control strategy machining tool displacement and the normal direction of the force on the contact surface and to compare the results of the test.

In the simulation comparison experiment, both control strategies set the initial environment position at x_e=0 mm, the workpiece stiffness is 50 N/mm, the position in the feed direction to do sinusoidal vibration of 0.5 amplitude, 1 rad/s frequency, and the expectation to keep the tracking force f_d is 10N, the traditional impedance control strategy in which the impedance controller parameter m_d, b_d, and k_d are set to 1, 10, and 100, respectively, and the particle swarm variable Impedance parameter control strategy in the impedance controller parameters set m_d, k_d are 1, 0, respectively, the number of particles N= 15, the initial desired damping b_d parameter value is set to 10, the search range is set to [1,200], the limiting speed range is [-2,2], the number of iterations is 200, the inertia weights w= 0.8, the self-learning factor c_1= 0.5, and the swarm learning factor c_2 = 0.2. Based on the above parameter settings, the simulation analysis of the simulated surface condition is shown in Fig. 9.

From the simulation results in Fig. 9, it can be seen that out in the case of sinusoidal changes in the surface position of the workpiece, both control strategies are generally able to track the change rule of the reference position better. The contact force under the traditional impedance control strategy stabilizes within the range of 8.549 N to 11.348 N. In comparison, the particle swarm variable-parameter impedance control strategy results in a fluctuation range of 9.010 N to 11.104 N. After the feed prediction speed is adjusted, the contact force fluctuates between 9.407 N and 10.65 N, representing a 25.19% reduction in fluctuation amplitude compared to the traditional strategy. The effect of model error on contact force caused by inaccurate matching of working condition parameter values is compensated, and the effectiveness of the proposed intelligent algorithm is verified.



(a) Damping parameter adjustment process (b) Comparison of desired positions



(c) Comparison of actual contact forces

Fig. 9. Comparison of surface grinding simulation

B. Analysis of Grinding Experiments

To analyze the influence of hardware parameters on surface quality and verify the proposed control algorithm, the KR22R1610-2 industrial robot manufactured by KUKA (Germany) was selected as the core experimental equipment. This robot is equipped with a smartPAD demonstrator and a KRC4 control cabinet, and communicates via Ethernet using the EtherCAT protocol to ensure high communication efficiency and stability. EtherCAT protocol to build a field bus to ensure the efficiency and stability of the system communication. The sensor is a six-dimensional force transducer model 9105-NETDeltaIP60 produced by ATI of the United States, with a special connection cable and NETBox module for data transmission. The settings and status monitoring of the sensor system can be accessed remotely and monitored in real time through a web browser, which facilitates the dynamic adjustment of parameters during the experiment. The grinding drive system adopts a self-developed belt sander. It is driven by Delta's C2000 inverter, which supports a frequency conversion range of 0.00-60.00 Hz and controls an induction motor with a maximum power of 750 W. This setup enables indirect regulation of the main wheel's rotational speed and the belt's linear speed. The abrasive belt is KK712X aluminum oxide abrasive cloth-based abrasive belt, with a width of 12mm, and has been configured with two grit sizes, P180 and P320. The experimental workpiece is an aero-engine blade, which is made of nickel-based high-temperature alloy

material through the precision casting process, with complex curved surfaces and high hardness characteristics, and is suitable as a typical workpiece for verifying the performance of grinding control. Based on the industrial robot grinding experimental system constructed on the above hardware platform, constant force tracking grinding experiments are carried out to verify the feasibility and stability of the proposed force control algorithm in complex free-form surface machining scenarios. The schematic diagram of the industrial robot grinding system is shown in Fig. 10.

Using offline trajectory planning programming software for aero-engine blade three-dimensional model grinding trajectory planning, in the upper machine grinding control system can realize the contact force three-dimensional space real-time display and data export, the software interface is shown in Fig. 11.

Before the experiment, the robot base coordinate system was calibrated, and the gravity compensation program was run to eliminate the interference of end attachments such as the quick-change head and the clamped workpiece on the contact force measurement acquisition, and the Z-direction grinding attitude was set to be in the same direction as the normal direction of the workpiece surface.

To systematically evaluate the practical effect of impedance control in robotic grinding tasks, an orthogonal experimental scheme, as presented in Table 1, was adopted in this study. This scheme focuses on four variables — control method, desired contact force setting, abrasive belt line speed, and abrasive belt particle size—and analyzes them with respect to contact force fluctuation and surface roughness stability. In the experiment, traditional position impedance control was applied in Experiment 1. Experiments 2 to 5 were conducted using the particle swarm variable impedance control strategy to investigate the stability and processing consistency of the proposed method under complex grinding conditions, as well as the influence of hardware and equipment parameters on grinding surface quality.

According to the process parameters in Table 1, with other set parameters kept constant, the corresponding five grinding experiments were carried out for each of the five regions divided by the blade, as shown in Fig. 12.

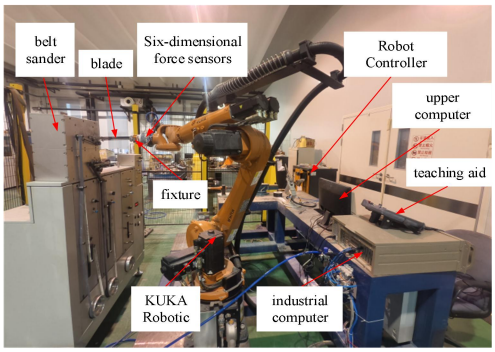
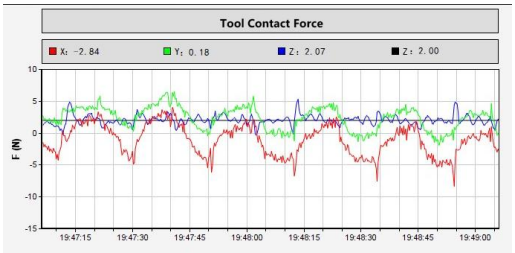
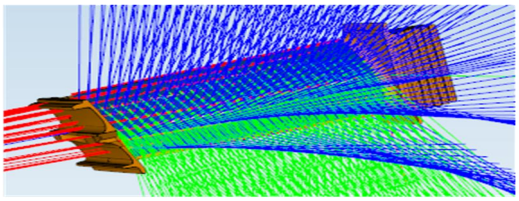


Fig. 10. Automated grinding platform for industrial robots



(a) Force control interface



(b) Abrasive trajectory and attitude planning

Fig. 11. Upper computer interface

At the end of the sanding experiments in different regions of the blade, the actual Z-direction contact force data collected during the sanding process were exported, and the comparative results of the relevant influencing variables were plotted in a single graph for comparative analysis, as shown in Fig. 13.

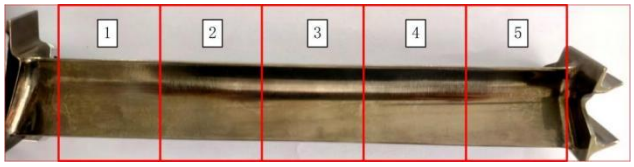
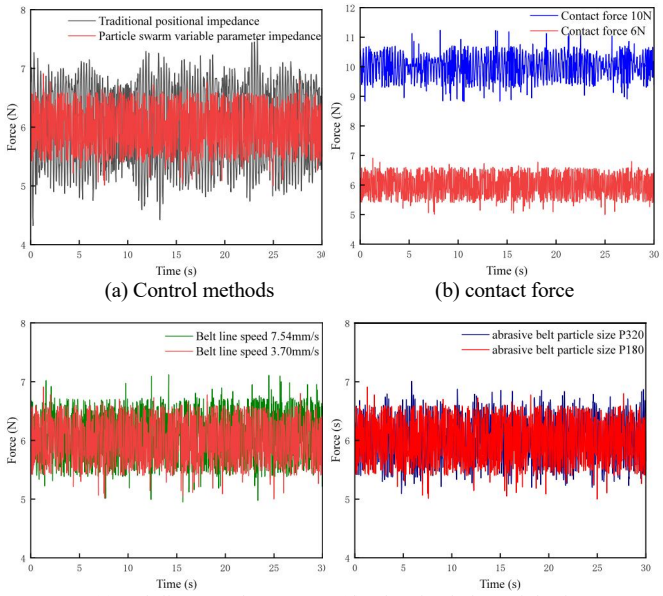


Fig. 12. Aero-engine blades

TABLE I. TABLE OF ORTHOGONAL EXPERIMENTAL DESIGN

No.	Control methods	contact force	Belt line speed	Abrasive belt particle size
1	traditional position impedance control	6 N	3.70 mm/s	P180
2	Particle swarm variable impedance control	6 N	3.70 mm/s	P180
3	Particle swarm variable impedance control	10 N	3.70 mm/s	P180
4	Particle swarm variable impedance control	6 N	7.54 mm/s	P180
5	Particle swarm variable impedance control	6 N	3.70 mm/s	P320

As analyzed in Fig.13, the fluctuation range of the contact force under the particle swarm variable impedance control mode (Experiment 2) is 5.956±0.955 N, which is significantly better than that of the traditional control method (Experiment 1), which is 5.812±1.588 N. The fluctuation of the red curves in Fig. (a) is small, which clearly shows the advantages of the particle swarm variable impedance control in terms of the stability of the contact force and the immunity to disturbances of the system. The fluctuation range of contact force is 10±1.206 N in experiment 3, 6.035±1.084 N in experiment 4, and 6.051 ± 0.958 N in experiment 5. From the experimental results of setting different desired contact force, abrasive belt speed, and Abrasive belt particle size, the three hardware parameters of the fluctuation range of the contact force have less influence on the interference and can still be maintained in a more accurate and reasonable range of deviation, and the particle swarm impedance control shows good environmental adaptation and control robustness in these three groups of comparative experiments. The particle swarm variable impedance control shows good environmental adaptability and control robustness in these three groups of comparative experiments.



(a) Control methods (b) contact force

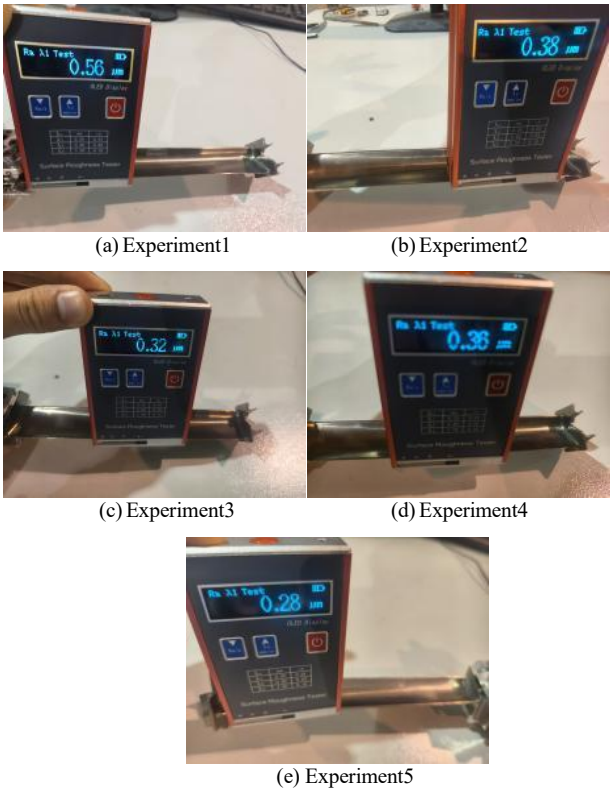
(c) Belt line speed (d) Abrasive belt particle size

Fig. 13. Comparison of Z-direction contact force for orthogonal experiments

After completing the sanding experiment, roughness measurements were taken in the trajectory planning area for the post-sanding position, as shown in Fig. 14.

In the five groups of comparison experiments, ten different locations on the workpiece surface were selected for roughness measurements in the grinding area of each group of experiments, and the roughness distribution of each group of experiments at the corresponding measurement points is shown in Fig. 15.

Based on the comparative analysis of the surface roughness data can further verify the practical effect of the particle swarm variable impedance control method in robotic grinding processing. As shown in the figure, experiment 1 is the traditional control method, and its roughness value fluctuates in a wide range, between 0.57-0.74 μm, with an average value of 0.65 μm, which indicates that the fluctuation of the contact force is significant in the processing, resulting in poor surface quality consistency. In contrast, impedance control was used in Experiments 2 to 5, and the roughness values of each group were significantly lower than those of the traditional control, and the fluctuations were significantly reduced. The mean value of Experiment 2 is 0.35 μm, Experiment 3 is 0.29 μm, Experiment 4 is 0.28 μm, and Experiment 5 is 0.25 μm. Experiment 5 was selected to be the lowest roughness of 0.19 μm after the P320 grit fine grit belt. Roughness comparison results can reflect the hardware equipment parameters expected contact force and belt



(a) Experiment1 (b) Experiment2

(c) Experiment3 (d) Experiment4

(e) Experiment5

Fig. 14. Roughness detection in different areas of the blade

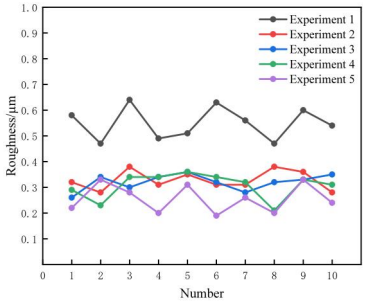


Fig. 15. Comparison of roughness of orthogonal experiments

speed in the actual grinding experimental environment for the final surface quality results of the influence of small, while the selection of a reasonable number of abrasive belt particle size will have a greater enhancement of the quality of the surface

sanding, five groups of comparative experimental results reflect that, in the group of the hardware process parameters in conjunction with a swarm of particles, the variable impedance control method to achieve a more optimal output of the surface quality.

To further verify the disturbance rejection capability of the proposed control strategy, a vibration spectral analysis was conducted based on the measured acceleration signals during the grinding process. The signals were transformed into the frequency domain using the fast Fourier transform (FFT), and the vibration spectra under traditional impedance control and the proposed method were compared, as shown in Fig. 16. The results show that the proposed method exhibits lower amplitude in the dominant vibration frequency range, indicating that it can effectively suppress vibration-induced disturbances and improve process stability.

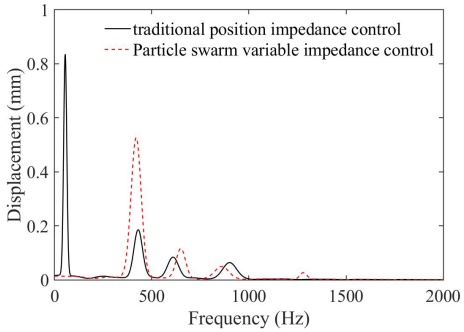


Fig. 16. Frequency domain diagram of grinding process

Comprehensive analysis shows that during the grinding experiments based on the multivariate process parameter rectification of the robot intelligent ability control method, compared with the traditional position impedance control method, it can effectively inhibit the range of fluctuations in the contact force, thus stabilizing the machining process, improving the consistency of surface roughness, and showing good anti-disturbance performance and adaptability, which verifies the feasibility and superiority of the control method in the complex grinding tasks.

V. Conclusion

In this study, a particle swarm variable impedance random forest robotic grinding force control method, based on the parameterization of multivariate grinding process conditions, is proposed to enhance the performance of traditional robotic grinding force control systems under complex working conditions. Through simulation experiments and actual industrial robot grinding experiments, it is verified that the method shows good results in constant force tracking, system response speed and processing quality. The experimental results show that the control algorithm has more stable constant force output capability, faster steady state response characteristics, and more efficient grinding performance under the working conditions of different process parameters, showing good adaptability and robustness. In addition, a certain degree of vibration phenomenon was observed during the blade grinding process, which affected the consistency of the grinding quality and the stability of the system, and future research will further focus on the vibration suppression strategy during the machining process in order to enhance the overall control accuracy and machining quality in the complex surface grinding work.

References

[1] Pokuang Zhou, Yuming Zhou, Qianlong Xie, et al. Adaptive force control for robotic grinding of complex blades[J]. IOP Conference Series: Materials Science and Engineering, 2019, 692(1):012008. DOI:10.1088/1757-899X/692/1/012008

[2] Guanhua Xu, Zhipeng Wang, Jiafu Zhang, et al. Compliance control of deburring robots based on force impedance[C]. 2020 Chinese Automation Congress, Shanghai, China, 2020:79-84. DOI:10.1109/CAC51589.2020.9327372

[3] Shiyi Li, Ming Huang, Zai Shi. Surface tracking system based on closed-loop force control of manipulator[C]. 2020 International Conference on Artificial Intelligence and Electrpmechanical Automation, Tianjin, China, 2020:292-298. DOI:10.1109/AIEA51086.2020.00068

[4] Haibo Zhou, Shitai Ma, Guilian Wang, et al. A hybrid control strategy for grinding and polishing robot based on adaptive impedance control[J]. Advances in Mechanical Engineering, 2021, 13(3):1-16. DOI:10.1177/16878140211004034

[5] Qingwei Zhang, Lili Han, Fang Xu, et al. Research on velocity servo-based hybrid position/force control scheme for a grinding robot[J]. Advanced Materials Research, 2012, 490:589-593. DOI:10.4028/www.scientific.net/amr.490-495.589

[6] Hongyao Zhang, Lun Li, Jibin Zhao, et al. Design and implementation of hybrid force/position control for robot automation grinding aviation blade based on fuzzy PID[J]. International Journal of Advanced Manufacturing Technology, 2020, 107:1741-1754. DOI:10.1007/s00170-020-05061-y

[7] MingJian Sun, Kai Guo, Jie Sun. Research on robot grinding force control method[J]. Intelligent Robotics and Applications, 2021, 13014:821-829. DOI:10.1007/978-3-030-89098-8_77

[8] Tie Zhang, Chao Yuan, Yanbiao Zou. Online optimization method of controller parameters for robot constant force grinding based on deep reinforcement learning rainbow[J]. Journal of Intelligent and Robotic Systems, 2022, 105:85. DOI:10.1007/s10846-022-01688-z

[9] Liguo Weng, Lang Tian, Kai Hu, et al.Overview of robot force control algorithms based on neural network[C]. 2020 Chinese Automation Congress, Shanghai, China, 2020:6800-6803. DOI:10.1109/CAC51589.2020.9327397

[10] Carlucho I, Paula M D, Acosta G G. An adaptive deep reinforcement learning approach for MIMO PID control of mobile robots[J]. ISA Transactions, 2020, 102:280-294. DOI:10.1016/j.isatra.2020.02.017

[11] Tie Zhang, Chao Yuan, Yanbiao Zou. Research on the algorithm of constant force grinding controller based on reinforcement learning PPO[J]. International Journal of Advanced Manufacturing Technology, 2023, 126:2975-2988. DOI:10.1007/s00170-023-11129-2

[12] Luocheng Zheng, Pengfei Rao, Yingke Li, et al. Admittance control based humanoid robot standing balance control[C]. 2019 International Conference on Advanced Robotics and its Social Impacts, Beijing, China, 2019:167-173. DOI:10.1109/ARSO46408.2019.8948802

[13] Fateh M M, Zirkohi M M. Adaptive impedance control of a hydraulic suspension system using particle swarm optimisation[J]. Vehicle System Dynamics, 2011, 49(12): 1951–1965. DOI:10.1080/00423114.2011.564289

[14] Preston F W. The theory and design of plate glass polishing machiness[J]. Journal of the Society of Glass Technology, 1927, 11:214-256

Control Method of Landing Based on Runway Model

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***Abstract*—Depending on the differential global positioning system (GPS) information, height parameter with the characteristic of real time and high precision is acquired. Since the differential GPS height parameter is not sensitive to the terrain, the landing method use differential GPS do not adapt to the slope runway condition, which will lead to the advance or delay of the landing moment. In order to improve the adaptation to the slope runways, this paper carried out a research on the slope runway landing method. Firstly, two-dimensional (2D) runway model is established based on the slope of runway related to the distance to go and relative height, the precision description is gotten by the characteristic of the slope runway. Secondly, a compensation method is carried out, considering not only the relative height but also the relative descent rate, ensuring the consistency of the calculated descent rate and the real descent rate referring to the runway surface. At last, the test result was proposed based on the runway model, which is proved to be reasonable and effective.**

Keywords—Unmanned aerial vehicle (UAV), slope runway model, differential GPS height, slope compensation

I. INTRODUCTION

Differential global positioning system (GPS)method was a normal way for wheel UAV taking off and landing. High precise height and position information is acquired based on the differential GPS data, realizing precise landing control [1-2]. Since the shortage of insensitive to the terrain changes, problems are arisen encountering to runway with slopes, because the relative height is calculated based on the benchmark on the runway. Slope landing method research is concentrated to solve helicopter landing problem [3-5], the wheeled unmanned aerial vehicle (UAV) research is rare. The generation of deviation was deduced by the calculation method, including the relative height and descent rate. The deviation of descent rate was caused by the referring coordinate system, the calculated one was referring to the geodetic coordinate system, while the real one was depended on the slope inclined coordinate system. When the UAV try to land in the direction of climbing the slope runway, the descent rate was increased for the lifting of slope, which causes the advanced landing, meanwhile causing the jumping disaster after landing. For the sake of landing in the direction of downhill the runway, since the runway surface descending, the descent rate will be lower than differential GPS descent rate, which caused the delay of landing. The delay time mentioned above rises landing risk for the landing of UAVs, since the static unstable of the heading of the UAV.

The differential GPS landing method refers to the benchmarks on the runway [3]. The benchmarks are chosen to determine the height when converting to the flight stage of flare and alignment. The inappropriate benchmark point leads to the incorrect height to convert the flight stages, causing adverse effects for the landing. Above all the chosen of the benchmark points are strongly deepened，when implementing the traditional differential GPS landing method.

To realize the consistency of relative height and relative descent rate referring to the slope runway surface of the different airports. This paper proposed a novel method based on the two-dimensional (2D) slope runway model established by the differential GPS pulling distance test [4], much easier than the three-dimensional (3D) reconstruction method in engineering, in which slope data map by the distance to go and the runway height. This model is uploaded to the flight management system (FMS), in which the relative height to the runway surface is real time calculated related to the distance to go. The model is used as the flight stages exchanging judgment of flare and alignment. The descent rate related to the surface of the runway is modified based on the run way slope, which is used in control law calculation, ensuring the consistency between the calculated descent velocity and the GPS differential descent velocity.

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II. Differential Global Positioning System Differential Landing Method

GPS differential landing method is a common way for wheeled UAV landing [6]. Fig. 1 shows the ideal glide path, which shows the glide path for the climbing slope runway condition.

D_1 point indicts the crossing point of the approaching line and the flat runway.

D_2 point indicts the crossing point of the approaching line and the climbing slope runway.

The G (including G_1, G_1) point indicts the ideal landing point, where G_1 is the ideal landing point on the flat runway, G_2 is the slope runway case.

P indicts the start point of flare.

Q indicts the start point of alignment.

In the ideal condition, the UAV starts to control the descent rate, until touching the ground at the terminal point G, the ideal descent rate will reach 0 m/s, and in addition the relative descent rate referring to the runway can reach the idea limit. In the slope runway condition application, G point is chosen as the benchmark of the landing point.

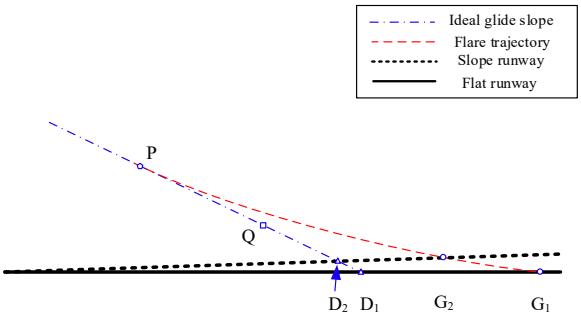

Fig. 1. Schematic diagram of a typical landing trajectory

When the runway is ideal flat, G point is still be chosen as the benchmark of the landing point, the descent rate is as much as the relative runway surface decent rate. While for the sake of climbing slope, the descent rate will increase for the effect of the lifting altimeter of the runway surface, on the contrary the UAV control law is based on the absolute descent rate of the differential GPS.

Without the optimization of the descent rate and height to the terrain, disasters usually happened. The differences of the slope runways will cause the wrong height of the stage of flare and alignment, which disturbs the ordinary landing strategy, especially affect the pitch compensation value. This value is used to overcome the ground effect. This value is highly depend on the precisely measurement of the height, since it is changed accompany with the height. The kinematic of the UAV is expected to meet the descent rate, while the control law use the descent rate of the ground coordination, instead of the runway surface coordination. The relative descent rate to the runway surface is not taken into consideration, which causes the difference between the control instruction and the real descent rate. The increase or decrease of the descent rate relative to the runway surface, will increase the risk of touching the runway in advance or could not reach the surface in delay time, above all, causes the disorder of landing.

This paper carried out a novel method of landing without using the benchmark. To get the height of the runway, differential GPS height of the runway is derived through the distance test of differential GPS. A relative height was derived and used in the FMS software, ensuring the consistence of the height gate of flight stages. The slope information was considered to calculate the descent rate to compensate the GPS differential descent rate, ensuring the compensation between the GPS differential descent rate and the control law used descent rate

III. Runway Slope Adaptive Compensation

A. Establishing Runway Model

To ensure the compensation between the relative height and the descent rate when the flight stage is changed. The absolute altitude and slope parameter at a specific position must be gotten to establish the model [7].

The traditional method of measuring the altitude of the runway is benchmark points use the average GPS method, which costs a lot of time to measure. To establish the entire runway model a large amount of points measurement are needed. Differential GPS data is easy to get form the distance test, which can cover the whole runway and acquire the high-precision of distance to go with the relationship with height. In this paper, the real-time kinematic (RTK) method is used in the measurement of the runway height. The root mean square (RMS)error between the differential RTK and average GPS benchmark points is lower than 0.2m.

The coordinate of the runway is established by the definition of the origin point and the direction of the axis. The origin point is set at the edge of the runway. The direction of landing is the positive direction of X axis. The vertical direction refer to the surface of the runway is defined as Y axis, show as Fig. 2.

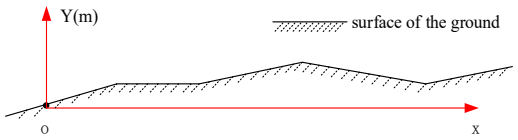

Fig. 2. The definition of the runway coordination

A runway can be described by n feature points. The inclination angles of the n-1 runway segments (from 1 to 2, 2 to 3, …, n-1 to n) are described as (1).

$$\begin{aligned} \gamma_1 &= arc\tan(\frac{H_2 - H_1}{X_2 - X_1}) \\ \gamma_2 &= arc\tan(\frac{H_3 - H_2}{X_3 - X_2}) \\ &\vdots \\ \gamma_{n-1} &= arc\tan(\frac{H_n - H_{n-1}}{X_n - X_{n-1}}) \end{aligned} \tag{1}$$

Where(X_1, H_1) ,(X_2, H_2) ,…,(X_{n-1}, H_{n-1}) (X_{n-1}, H_{n-1}) and (X_n, H_n) denote the coordinates of the 1,2,…,n-1 and n feature points on the runway, γ_1 is deduced as the angel of section 1 and 2. γ_2 is deduced as the angel of section 2 and 3. γ_{n-1} is deduced as the angel of section n-1 and n.

When UAV attends landing process, the slope and runway absolute height is calculated, referring to the X position in the runway coordination. The slope angle γ is calculated as (2) and the absolute altitude of the runway H_r is calculated as (3).

$$\gamma = \begin{cases} 0 & X \le X_1 \\ \gamma_1 & X_1 < X \le X_2 \\ \cdots & \cdots \\ \gamma_{n-1} & X_{n-1} < X \le X_n \\ 0 & X > X_n \end{cases} \tag{2}$$

$$H_r = \begin{cases} H_1 & X \le X_1 \\ H_1 + \dfrac{H_2 - H_1}{X_2 - X_1}(X - X_1) & X_1 < X \le X_2 \\ \cdots & \cdots \\ H_{n-1} + \dfrac{H_n - H_{n-1}}{X_n - X_{n-1}}(X - X_{n-1}) & X_{n-1} < X \le X_n \\ H_n & X > X_n \end{cases} \tag{3}$$

B. Landing Control Method Considering the Runway Model

According to the runway model, the X position can be calculated by the flight parameter of distance to go (DTG).

The runway absolute altitude H_r according to the (3).

The relative height of the UAV H_{rw} can be calculated as (4).

$$H_{rw} = H - H_r \tag{4}$$

The relative height to the surface of the runway was used as the judgment of flight stages changing. The guidance instruction is produced and pitch pre-feedback parameter is calculated at the same moment.

When the main control method will be changed to descent rate instead of the relative height. A compensation of the descent rate must be invited referring to the slope of the runway. The target of the compensation is to control the landing descent rate is in the ideal range when landing. Assuming the slope is r on the real runway surface,based on the relationship of coordination exchanging relationship, the descent rate is calculated as (5).

$$\dot{H}_{rw} = \dot{H}\cos\gamma + V_e \sin\gamma \tag{5}$$

Where $\dot{H}$ is the GPS differential descent rate measure by the global navigation satellite system (GNSS), $\dot{H}_{rw}$ is the descent rate referring to the runway surface, which substituted by $\dot{H}$ into the descent rate control. V_e is defined as the ground speed in even flat.

To avoid the nose wheel landing in advance, the pitch instruction is provided with limited value before UAV touching the ground. When the slope is exist, the limit must be modified with the increase or decrease of the runway slope.

C. Algorithm Test result

This section presents the landing simulation results on uphill and downhill runways. The proposed algorithm incorporates the sloped runway model to realize parameter design and compensation for the entire landing process.

As illustrated in Fig.3 and Fig.4, vertical speed refer to the 0m point is the traditional method of landing using the differential GPS descent rate, while the vertical speed refer to the ground is the modified descent rate. Height and descent rate compensations are respectively implemented for climbing slope (Fig.3) and descending slope landing (Fig.4) landing conditions. The compensation scheme guarantees a desirable vertical descent rate (less than -1 m/s)when touching the ground.

On an downhill runway, a properly increased vertical touchdown speed is adopted to achieve rapid and firm touchdown rather than floating, thus preventing the nose wheel from striking the ground prematurely. On a uphill runway, the decrease of the descent rate is effectively restricted to avoid severe collision between the main landing gear and the runway surface.

The results demonstrated that the proposed compensation strategy can significantly improve landing safety and stability under sloped runway conditions, which validates the effectiveness and practicability of the designed algorithm.

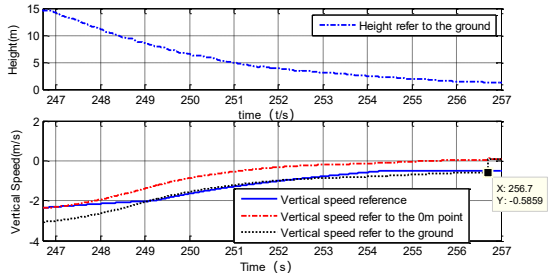

Fig. 3. Climbing slope landing simulation

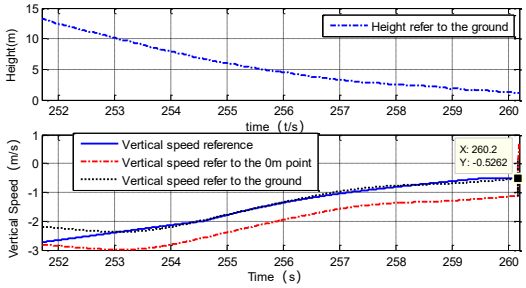

Fig. 4. Descending slope landing simulation

IV. Conclusion

This paper presents a novel method to address the landing problem of wheeled UAVs on sloped runways. A height model is constructed based on real-time GPS differential distance measurements. Meanwhile, compensation strategies for the relative height with respect to the runway surface and the descent rate are designed to improve landing safety on inclined runways, and to maintain consistency between the theoretical descent rate and the actual descent rate relative to the runway.

Algorithm Test result demonstrate the effectiveness and accuracy of the proposed method. It is verified that the presented approach can ensure good consistency between the GNSS-measured relative descent rate and the real relative descent rate relative to the runway surface. In addition, the method has obvious advantages in practical engineering applications and reduces the heavy dependence on high-precision runway benchmarks, which is of great significance for autonomous landing of UAVs under complex runway conditions.

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References

[1] Z. He and X. Wang, “A High-Precision Integrated Navigation Program for Aircraft Landing Phase,” Aero Weaponry, vol. 2(5), pp.12-16, 2013.

[2] X. Zou, P. Cheng and N. Cheng, “A Simulation Model for Airport Runway Capacity Estimation,” 17th International IEEE Conference on Intelligent Transportation Systems, Qingdao, pp. 1462-1467, 2014.

[3] W. Cheng,“ Flight test research of helicopter slope landing,” Flight dynamic, vol. 34(2), pp. 78-85, 2016.

[4] X. F. Wang, G. Lu, Z. Y. Shi, and Y. S. Zhong,“ Robust LQR controller for landing unmanned helicopters on a slope,” Proceedings of the 35th Chinese Control Conference, Chengdu, pp.10639-10644, 2016.

[5] Y. S. Li, G. M. Bai, D. F. Xu and L. S. Geng, “Key Technology of Helicopter’s Slope Landing and Take-off,” Aircraft Design, vol. 41(3),pp.72-76, 2021.

[6] J. Yu and Y. Pu, “Mathematical Model and Analysis of Runway Landing Capacity,” Journal of Southwest Jiao Tong University, vol 17(1), pp. 42-46, 2004.

[7] F. X. Zhang, D. Q. Wang and H. Yuan, “Research on Flight Test Methods for UAV Differential Satellite-Based Takeoff and Landing Guidance Systems,” Advances in Aeronautical Science and Engineering, vol. 14(4), pp. 112-118, 2023.

Image-Guided Robotic Suturing via Real-Time Wound Instance Segmentation and Trajectory Encoding

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***Abstract*—Accurate and real-time wound segmentation is vital for advancing robotic-assisted surgery, where precise boundary detection guides autonomous suturing. This paper proposes a robust instance segmentation pipeline based on three YOLO variants—YOLOv8n-seg, YOLOv11n-seg, and YOLOv12n-seg—specifically tailored for wound boundary extraction under complex clinical conditions. The dataset contains 4,097 annotated images compiled by extending a public Roboflow release with additional wound annotations and 1,575 distractor images. All models were trained under identical settings and evaluated using standard segmentation metrics. Among the evaluated models, YOLOv11n-seg achieved the highest mask mAP@0.5 (0.9634), with 25.77 ms/image inference time and a 116-minute training duration under a consistent experimental protocol. The comparative results quantify precision–recall trade-offs, supporting informed model selection for latency-sensitive surgical applications. Finally, a vision-based post-processing pipeline converts predicted masks into smooth, anatomically aligned suturing trajectories via contour extraction, arc-length resampling, and Gaussian smoothing, enabling a low-latency perception-to-trajectory interface for downstream robotic suturing planning. This work contributes a real-time, image-guided perception module suitable for robot-assisted interventions, bridging medical imaging outputs to actionable geometric trajectories.**

***Index Terms*—Wound segmentation, YOLO, robotic suturing, instance segmentation, medical imaging, real-time inference**

I. Introduction

WOUND segmentation is crucial in computer-assisted diagnosis, surgical planning, and robotic automation. Accurate wound boundary detection supports quantitative clinical assessment and enables downstream tasks such as robotic suturing. Manual or semi-automated methods remain commonplace; however, they are labor-intensive, inconsistent, and unsuitable for real-time application. Because surgeries are dynamic processes requiring low-latency visual perception, robust automated vision systems are increasingly necessary [1]. We position the proposed system as an image-guided perception component for robot-assisted intervention, where segmentation outputs directly support downstream robotic planning and trajectory generation.

Deep-learning approaches, especially deep convolutional neural networks (CNNs), have greatly shifted the landscape of medical image segmentation. Specifically, U-Net [2] and Mask R-CNN networks [3] have demonstrated high accuracy for anatomical boundary segmentation. However, their high computational demands limit use in latency-sensitive applications. Furthermore, these models are rarely integrated into full surgical automation pipelines—such as contour-based trajectory planning. Instance segmentation is particularly advantageous in robotic suturing because it preserves object-level boundary integrity, enabling direct contour extraction and geometric trajectory parameterization without additional region-separation post-processing.

Meanwhile, object detection models such as YOLO (You Only Look Once) have rapidly evolved, combining segmentation accuracy with real-time performance. Recent iterations of YOLO—YOLOv8, YOLOv11, and YOLOv12—integrate advancements (e.g., decoupled heads, attention modules, and lightweight feature fusion), but their potential in surgical settings remains underexplored, particularly for wound segmentation in robotic suturing [4], [5].

To address these gaps, we propose an end-to-end framework for real-time wound instance segmentation and contour-based trajectory extraction using YOLO-based models. The contributions to the field are as follows:

1. *Dataset Enhancement:* Extended a public wound dataset to 4,097 images by adding 1,575 verified distractor (negative) images to reduce false positives under wound-like background patterns.
2. *Controlled Benchmarking:* Evaluated YOLOv8n-seg, YOLOv11n-seg, and YOLOv12n-seg under identical training protocols and hardware to isolate architectural trade-offs in accuracy and latency.
3. *Perception-to-Trajectory Output:* Converted predicted instance masks to evenly spaced suturing waypoints via contour extraction, arc-length resampling, and Gaussian smoothing for downstream motion planning.

The remainder of the paper is organized as follows: Section II presents an overview of prior methods in wound segmentation and robotic suturing, Section III details the

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methodology, including dataset design, model configurations, training setup, and evaluation process; and Section IV presents experimental results including wound segmentation performance and trajectory extraction pipeline results, while Section V concludes with key findings and future directions.

II. Related Work

Wound segmentation has advanced from rule-based image analysis to sophisticated deep learning techniques with real-time and instance-level recognition. This section describes important advances chronologically and by technique.

A. Early Computational Techniques

Early wound segmentation relied on traditional image-processing techniques rather than learning-based models. Common pipelines used intensity thresholding, morphological operations (e.g., opening/closing), and edge detection (e.g., Canny) to approximate wound boundaries. Active-contour methods that use edge maps for boundary attraction improved localization in some cases; however, they often produced fragmented or unstable contours under illumination changes, specular highlights, and low contrast—conditions frequently observed in clinical images [6]. In terms of classical methods, [7] showed that classical segmentation combined with binary closing (Isodata thresholding), could perform well under similar conditions with an IoU 0.717. Nevertheless, these classical approaches often require extensive manual preprocessing to achieve reliable segmentation and do not generalize well to heterogeneous clinical datasets.

More advanced classical pipelines combine denoising with edge detection. For example, hybrid Perona–Malik diffusion and Total Variation regularization can improve edge maps under low-quality imaging, enabling Canny and Sobel detectors to produce usable wound boundaries on noisy diabetic-foot-ulcer images [8]. Nevertheless, although classical methods had sufficient accuracy, they were conditioned on spatial resolution, required tuning of arbitrarily assigned parameters and are not suitable for real-time or robotic applications.

B. Deep Learning in Wound/Tissue Segmentation

Deep learning has radically changed the architecture of wound segmentation for the better with methods like U-Net and its variants dominating the field due to their encoder-decoder structure and skip connections. An ensemble of U-Net and LinkNet models using transfer learning and data augmentation achieved Dice scores of 92.07% and 88.80% on the chronic wound and FUSeg datasets respectively, in a challenge hosted at the MICCAI 2021 conference [9].

In [10], Mask R-CNN enabled non-invasive wound measurement from 2D and 3D data with 87% precision. [11] introduced an ensemble model combining patch-wise and image-wise classifiers for wound type prediction.

C. YOLO and Real-Time Instance Segmentation

Recent YOLO-based models have introduced real-time object detection capabilities with promising results in medical imaging. One study showed how YOLOv5 was used to detect and classify pressure ulcers into clinical stages, achieving strong mAP through architecturally efficient features and data augmentations [12].

While this study focused on classification tasks, instance-level wound segmentation is necessary in addition to classification tasks, and real-world robotic control workflows with surgery. However, systematic benchmarking of newer YOLO segmentation variants on distractor-rich wound datasets remains limited, particularly in pipelines that convert masks to suturing trajectories.

D. Gaps in Literature

Despite notable advances, key gaps persist:

1) Intraoperative limits and poor distractor handling: The majority of studies are completed on static wounds (e.g., foot ulcers) and do not involve consideration of intraoperative complexity. Many models also did not train on negative samples (e.g., healthy skin or background), resulting in a false positive risk because many of the features may have similarities to similar structures, such as folds or facial features [4].

2) No systematic YOLO benchmarking: Few prior works benchmark YOLOv8/11/12 segmentation models on distractor-augmented wound datasets or connect segmentation outputs to trajectory extraction.

3) Segmentation rarely integrated in robotic suturing: Common robotic systems in surgical robotics use control and planning but in full autonomy mode, they often omit deep segmentation as input.

This study addresses these gaps by benchmarking YOLO-based segmentation on a distractor-augmented wound dataset and by extracting smoothed suturing trajectories from predicted masks.

III. Methodology

In this section, the end-to-end methodology that was followed to develop and assess instance segmentation models that identify the boundaries of wounds is described in detail. It includes the process of constructing the dataset, selecting a model architecture, training configuration strategy, evaluation strategy, and post-processing of the boundaries' paths. The pipeline was designed for consistency and reproducibility, with the goal of producing anatomically accurate contours that can be used in downstream tasks such as trajectory planning or semi-autonomous suturing assistance. In Fig. 1, the entire end-to-end methodology pipeline is displayed.

A. Dataset

The dataset contains 4,097 images across three splits: 3,698 for training, 321 for validation, and 78 for testing. Splits were fixed prior to training and maintained consistently across all experiments to avoid data leakage between training, validation, and test sets. This total includes 1,575 distractor (negative) images with wound-like visual patterns (e.g., scars, skin folds, and facial features), used to reduce false positives. Distractor images (negative samples) contained no wound annotations. Each image is annotated with an instance segmentation mask of the wound (e.g., ulcers, incisions, and lesions). Most images contain a single wound instance; however, the instance-segmentation formulation allows multiple wound instances per image when present.

The dataset includes images from a public Roboflow release curated by Noah Oikarinen (Feb. 2024), which originally contained 2,522 annotated images. All additional annotations

and distractor labels were manually verified to ensure consistency with the original labeling protocol. Fig. 2 illustrates (a) a challenging blurred wound image and (b) representative non-wound distractors used during training to reduce false positives in visually similar regions.

B. Image Preprocessing

All images were resized to 640×640 pixels. Pixel intensities were normalized to $[0,1]$ to reduce illumination variability. Annotations were stored as instance masks; during training, masks were rasterized in the format required by the Ultralytics segmentation head.

C. Model Architecture

Three YOLO-based instance segmentation models—YOLOv8n-seg, YOLOv11n-seg, and YOLOv12n-seg—were created using the Ultralytics framework to segment the wound regions for robotic suturing tasks. Each model represents a successive architectural iteration aimed at improving segmentation accuracy and inference efficiency.

YOLOv8n-seg served as the baseline and uses a Cross Stage Partial (CSP) backbone, an anchor-free design, and decoupled heads for classification, localization, and mask prediction. YOLOv11n-seg extends this design with C3k2 blocks, Spatial Pyramid Pooling–Fast (SPPF), and Convolutional Parallel Spatial Attention (C2PSA), improving multi-scale feature learning and spatial sensitivity. YOLOv12n-seg further enhances feature fusion using area-attention mechanisms and Residual Efficient Layer Aggregation Networks (R-ELAN).

This representation of architecture from YOLOv8n-seg to YOLOv12n-seg allowed for a useful comparison of architectural additions, and the impact that these changes had on the performance of the wound segmentation models in visually complex conditions.

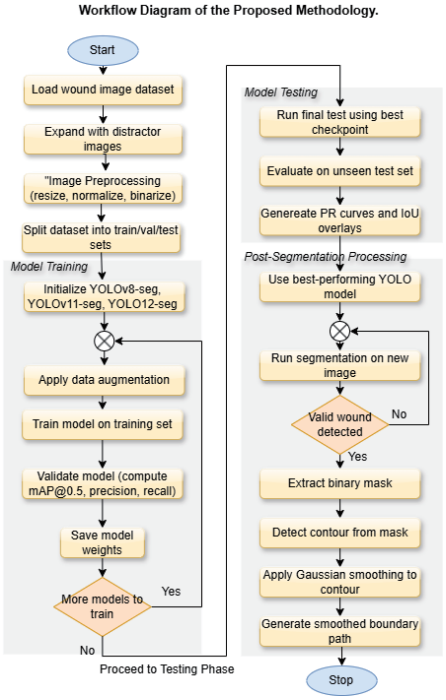

Fig. 1. Workflow of the proposed wound segmentation and post-processing pipeline

Fig. 2. Sample images from the dataset: (a) blurred wound image, (b) non-wound distractor examples.

D. Training Procedure

YOLOv8n-seg, YOLOv11n-seg, and YOLOv12n-seg were all trained with the same parameters so that the architecture could be compared fairly. They were all trained on an NVIDIA GeForce RTX 3050 Laptop GPU (4GB VRAM) for 100 epochs with a batch size of 16, image size of 640×640, and an initial learning rate of 0.01. Training was conducted for a fixed 100 epochs without early stopping; the best validation checkpoint was selected for test evaluation. We fixed random seeds (seed = 42), enabled deterministic behavior where supported, and used automatic mixed precision (AMP) to improve training efficiency. The augmentation pipeline was constant among the models, which consisted of RandAugment, horizontal flipping, and 40% random erasing. The augmentations were applied to each training image to improve generalization capacity against distractors such as skin folds, scars, and textures of backgrounds.

All models were trained on the same training, validation, and test split of the data, and whatever variations in performance obtained from the models can be attributed to the architecture since the hyper-parameters, augmentations, and computing infrastructure was held constant.

E. Model Testing

Each model was tested using the val method found in Ultralytics with split="test" specified, which produces full segmentation metrics on the specified test set. It should be noted that this test data was completely separate from the training and validation sets, and limited to be viewed by the model outside of training or tuning hyperparameters. This is important to be in this format, so that the model generalization performance to a previously unseen dataset can be evaluated fairly. For each architecture (YOLOv8n-seg, YOLOv11n-seg, and YOLOv12n-seg), only the checkpoint with the highest mAP@0.5 from validation was used for testing, which supports a fair and equal comparison across the architectures.

F. Evaluation

Model performance was evaluated using standard instance segmentation metrics, including mask mAP@0.5 (IoU $\geq$ 0.5), precision, and recall, computed against the ground-truth wound masks.

- Intersection over Union (IoU) measures the overlap between the predicted segmentation and the real mask and is calculated as follows in (1).

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (1)$$

- mAP@0.5 (mean Average Precision at IoU $\geqslant$ 0.5) represents the mean precision score averaged over classes at an IoU threshold of 0.5. It indicates whether

the model accurately identifies and segments wound areas with acceptable spatial accuracy.

- Precision is the proportion of predicted wound instances (masks) that match a ground-truth mask at IoU ≥ 0.5.
- Recall is the proportion of ground-truth wound instances that are successfully matched by a predicted mask at IoU ≥ 0.5.

Precision and recall were computed from predicted and ground-truth masks at an IoU threshold of 0.5, as implemented in the Ultralytics evaluation framework. YOLOv8n-seg was treated as the architectural baseline, while the best-performing checkpoint (highest validation mask mAP@0.5) for each model was selected for test evaluation. The mAP metrics and visual overlays of predicted-vs-actual as well as precision–recall curves were computed for qualitative understanding and error assessment.

G. Post-Segmentation Trajectory Extraction for Robotic Suturing

A vision-based post-processing procedure was conducted to assist potential downstream robotic suturing applications by synthesizing wound segmentation masks into anatomically aligned suturing paths. Our post-processing procedure (outlined in the results section) consisted of four stages: (1) The selected YOLOv11n-seg model segmented the wound; (2) Binary masks were extracted from the predicted model outputs; (3) We extracted wound contours from the binary masks (e.g., using connected-component filtering followed by contour tracing); and (4) smooth suturing trajectories were developed using Gaussian filters. The purpose of this procedure was to acknowledge anatomical accuracy while providing continuous geometric information for autonomous or semi-autonomous robotic operations. All aforementioned methods are presented in detail with qualitative and quantitative descriptions in the results section.

IV. Results and Discussion

This section provides a thorough evaluation of the proposed wound segmentation framework, including quantitative metrics and deployment considerations. The evaluation includes training dynamics, computational efficiency, inference latency, and confidence-based performance analysis of three YOLO-based instance segmentation models, enabling selection of the most suitable architecture for real-time robotic suturing applications. Each subsection contains meaningful feedback, supported by resources including metrics, plots, and comparisons to guide use and model deployment decisions.

A. Training and Validation Performance

All three models were trained for 100 epochs under identical conditions, enabling a controlled comparison of architectural differences. Each model demonstrated stable convergence during training, with no obvious signs of divergence.

YOLOv8n-seg achieved peak validation mask mAP@0.5 of 0.9553 at epoch 81 (mAP@0.5:0.95 = 0.7946), indicating strong generalization despite its lightweight architecture. YOLOv11n-seg converged more rapidly, reaching its best mask mAP@0.5 of 0.9634 at epoch 57 (mAP@0.5:0.95 = 0.787), reflecting efficient feature learning and stable optimization. YOLOv12n-seg exhibited a higher initial training loss, likely due to increased architectural complexity, but converged steadily to a peak mask mAP@0.5 of 0.9518 at epoch 93 (mAP@0.5:0.95 = 0.7792). Notably, differences in mAP@0.5 across models were relatively small (<1.2%), indicating comparable segmentation performance across architectures. Although YOLOv12n-seg achieved the highest box precision (0.9498), its slightly lower recall compared to YOLOv11 suggests a more conservative prediction strategy.

The final segmentation metrics for bounding box and mask outputs are summarized in Table I.

B. Computational Cost and Training Efficiency

Beyond being accurate, computational efficiency reduces compute cost and supports deployment on resource-constrained platforms. All models were trained under identical hardware and software conditions to ensure fair comparison of training duration.

Training required 112 min for YOLOv8n-seg and 116 min for YOLOv11n-seg, confirming their efficiency for constrained settings. YOLOv12n-seg required significantly longer (273 min) due to the computational complexity of its deeper architecture and additional attention mechanisms.

The increased inference latency of YOLOv12n-seg yields only marginal accuracy gains over its predecessors. Consequently, YOLOv11n-seg offers the optimal trade-off between segmentation quality and computational efficiency. YOLOv12n-seg remains best suited for non-real-time applications where maximum precision is prioritized over latency and where high-performance compute resources are available.

C. Inference Speed and Real-Time Applicability

Inference latency is critical for deployment in time-sensitive applications such as autonomous robotic suturing. Inference speed was benchmarked on 78 test images using default Ultralytics non-maximum suppression settings and a confidence threshold of 0.25. YOLOv11n-seg achieved the fastest performance at 25.77 ms/image, followed by YOLOv12n-seg at 30.22 ms/image. In contrast, YOLOv8n-seg required 95.68 ms/image under the same hardware configuration (RTX 3050 Laptop GPU, 4GB VRAM).

Overall, YOLOv11n-seg provides the best balance between segmentation accuracy and real-time performance. While YOLOv12n-seg offers slightly higher precision, it incurs additional computational cost. YOLOv8n-seg exhibited slower inference in our setup and may therefore be less suitable for strict latency-constrained applications.

TABLE I. SEGMENTATION PERFORMANCE METRICS (BOX AND MASK).

Model	Box			Mask		
	Precision	*Recall*	*mAP50*	*Precision*	*Recall*	*mAP50*
YOLOv8	0.9312	0.8853	0.9188	0.8945	0.9638	0.9553
YOLOv11	0.9228	0.8945	0.939	0.9128	0.9545	0.9634
YOLOv12	0.9498	0.8899	0.9509	0.889	0.9542	0.9518

D. Test Performance and Confidence-Based Evaluation

YOLOv11n-seg was evaluated on the 78-image test set using confidence-based metrics for both bounding box and

mask outputs. As expected, precision increased at higher confidence thresholds while recall decreased, reflecting the standard sensitivity–specificity trade-off. The optimal mask F1-score was achieved at a confidence threshold of 0.47, indicating stable calibration across operating points. These results further support the model's reliability for precision-critical and latency-sensitive robotic suturing applications.

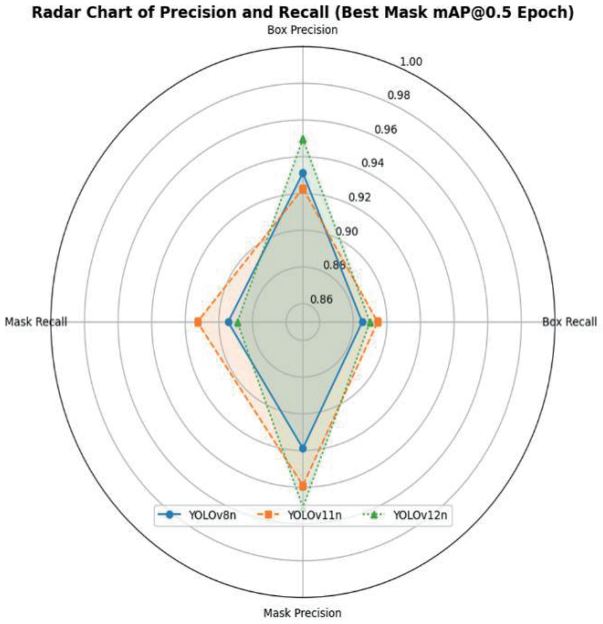

Fig. 3. Radar plot comparing precision and recall metrics for YOLO models at peak mask mAP@0.5 epochs.

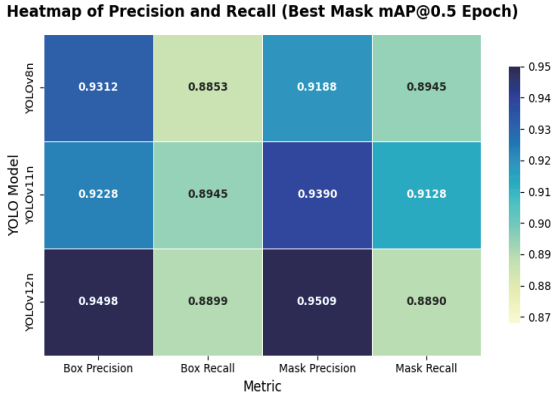

Fig. 4. Heatmap of precision and recall metrics for all YOLO models at best mask mAP@0.5 epoch.

E. Precision–Recall Trade-offs and Model Selection

Precision–recall trade-offs were assessed for all three models using the peak mask mAP@0.5 epochs. Radar plots (Fig. 3) indicated that YOLOv12n-seg exhibited the highest box precision (0.9498), suggesting a more conservative and false-positive minimizing approach to candidate detection. In contrast, YOLOv11n-seg achieved the highest mask precision (0.9128) and the highest box recall (0.8945), indicating stronger overall segmentation reliability and greater sensitivity to potential wound regions. Additionally, YOLOv12n-seg achieved the highest box mAP@0.5 (0.9509), reflecting strong localization performance across detected regions.

These trends were confirmed by a heat map (Fig. 4), and though trade-offs manifest in model performance as high recall, they are especially critical for surgical robotic suturing applications, where avoiding missed wound areas is more beneficial than limiting minor over-segmentation with robotics. Smaller over-segmentation is generally more beneficial to surgical robotics in a physician-led environment than missing wound areas. To summarize the comparative strengths of each architecture across evaluation metrics, Table II highlights the top-performing model for each metric at the best mask mAP@0.5 epoch.

TABLE II. TOP-PERFORMING YOLO MODELS BY METRIC AT BEST MASK MAP@0.5 EPOCH.

Performance Metric	*Top-Performing Model*	*Score*
Box Precision	YOLOv12n	0.9498
Box Recall	YOLOv11n	0.8945
Box mAP@0.5	YOLOv8n	0.9638
Mask Precision	YOLOv12n	0.9509
Mask Recall	YOLOv11n	0.9128
Mask mAP@0.5	YOLOv11n	0.9634

Of the three models, YOLOv11n-seg also had the best inference speed (25.77 ms/image), and training time (116 min) indicating it represents the most balanced trade-off and deployment-ready of the three. YOLOv12n-seg is better suited for scenarios that may require the utmost precision, while YOLOv8n-seg is also deployment applicable and represents the better choice for a more constrained environment.

F. Suturing Trajectory Extraction Pipeline

This pipeline provides anatomically smooth trajectories from segmentation outputs for downstream surgical planning or robotic sutures.

1) Wound Segmentation: YOLOv11n-seg was used to generate predicted segmentation masks from test images using the best performing checkpoint with the highest validation mAP on the segmentation data testing with a curated wound/distractor dataset.

2) Binary Mask Isolation: Only valid wound detector binary masks were retained; anything false positive detections or null output were excluded.

3) Contour Detection: The outer boundary of the wound region was extracted from the binary mask, yielding an ordered set of (x, y) coordinates. The resulting contour was parameterized as an ordered polyline and sampled at fixed arc-length intervals to produce evenly spaced waypoints suitable for downstream motion planning. The generated trajectories are expressed in image pixel coordinates and can be transformed into robot task-space coordinates through external camera calibration and hand–eye calibration procedures.

4) Trajectory Generation via Gaussian Smoothing: a discrete Gaussian filter was applied independently to the x and y portions of the wound contour. The smoothing kernel is defined in (2),

$$G(i) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{i^2}{2\sigma^2}} \quad (2)$$

where i is the index distance and σ is the smoothing parameter. This ensured geometric noise was eliminated and obtained a certain degree of anatomical plausibility. Fig. 5 shows the process from raw segmentation (a), to binary mask (b), to smoothed trajectory (c).

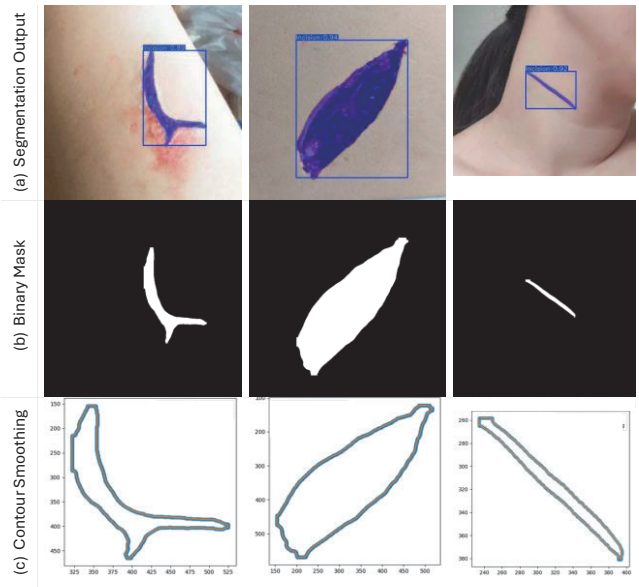

Fig. 5. Suturing trajectory extraction pipeline: (a) YOLO-based segmentation output, (b) binary mask generation, and (c) smoothed wound contour using Gaussian filtering.

V. Conclusion and Future Work

This research outlined and tested an end-to-end pipeline for real-time wound segmentation and robotic trajectory extraction using YOLO-based models. By curating a distractor-augmented dataset and benchmarking three YOLO variants (YOLOv8n-seg, YOLOv11n-seg, and YOLOv12n-seg), YOLOv11n-seg achieved the best balance of segmentation accuracy (mask mAP@0.5 = 0.9634), inference latency (25.77 ms/image), and training time (116 minutes). These results suggest that YOLOv11n-seg can serve as a practical perception module for time-sensitive medical robotics workflows. Moreover, the proposed contour-to-trajectory post-processing reliably converts predicted masks into smooth, anatomically aligned suturing paths, providing a low-latency vision front-end for downstream robotic suturing and a concrete step toward closed-loop, vision-guided execution.

In future work, the following will be pursued:

1. Adapting real-world surgical video data to further test robustness against intraoperative lighting and motion.
2. Incorporate image depth cues and 3D model articulations for improved wound geometry estimation.
3. Extending the framework for multi-class tissue segmentation.
4. Integrating closed-loop robotic execution with respect to visual feedback.

This work establishes a foundation for intelligent vision-guided robotic suturing systems by connecting visual segmentation to autonomous planning.

References

[1] Y. Tian et al., "Toward Autonomous Robotic Micro-Suturing using Optical Coherence Tomography Calibration and Path Planning," in 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France: IEEE, May 2020, pp. 5516–5522. doi: 10.1109/ICRA40945.2020.9196834.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015, vol. 9351, N. Navab, J. Hornegger, W. M. Wells, and A. F. Frangi, Eds., in Lecture Notes in Computer Science, vol. 9351. , Cham: Springer International Publishing, 2015, pp. 234–241. doi: 10.1007/978-3-319-24574-4_28.

[3] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in 2017 IEEE International Conference on Computer Vision (ICCV), Venice: IEEE, Oct. 2017, pp. 2980–2988. doi: 10.1109/ICCV.2017.322.

[4] G. Scebba et al., "Detect-and-segment: A deep learning approach to automate wound image segmentation," Inform. Med. Unlocked, vol. 29, p. 100884, 2022, doi: 10.1016/j.imu.2022.100884.

[5] R. Zhang, D. Tian, D. Xu, W. Qian, and Y. Yao, "A Survey of Wound Image Analysis Using Deep Learning: Classification, Detection, and Segmentation," IEEE Access, vol. 10, pp. 79502–79515, 2022, doi: 10.1109/ACCESS.2022.3194529.

[6] A. O. Khadidos, "Medical Image Segmentation using Edge-Based Active Contours".

[7] B. U. Marcato et al., "Automatic Area Estimation of Mice Wound Images," in Anais do XII Symposium on Knowledge Discovery, Mining and Learning (KDMiLe 2024), Brasil: Sociedade Brasileira de Computação - SBC, Nov. 2024, pp. 1–8. doi: 10.5753/kdmile.2024.241973.

[8] R. H. Mwawado, B. J. Maiseli, and M. A. Dida, "Robust Edge Detection Method for Segmentation of Diabetic Foot Ulcer Images," Eng. Technol. Appl. Sci. Res., vol. 10, no. 4, pp. 6034–6040, Aug. 2020, doi: 10.48084/etasr.3495.

[9] A. Mahbod, G. Schaefer, R. Ecker, and I. Ellinger, "Automatic Foot Ulcer Segmentation Using an Ensemble of Convolutional Neural Networks," 2021, arXiv. doi: 10.48550/ARXIV.2109.01408.

[10] S. Zahia, B. Garcia-Zapirain, and A. Elmaghraby, "Integrating 3D Model Representation for an Accurate Non-Invasive Assessment of Pressure Injuries with Deep Learning," Sensors, vol. 20, no. 10, p. 2933, May 2020, doi: 10.3390/s20102933.

[11] B. Rostami, D. M. Anisuzzaman, C. Wang, S. Gopalakrishnan, J. Niezgoda, and Z. Yu, "Multiclass wound image classification using an ensemble deep CNN-based classifier," Comput. Biol. Med., vol. 134, p. 104536, Jul. 2021, doi: 10.1016/j.compbiomed.2021.104536.

[12] B. Aldughayfiq, F. Ashfaq, N. Z. Jhanjhi, and M. Humayun, "YOLO-Based Deep Learning Model for Pressure Ulcer Detection and Classification," Healthcare, vol. 11, no. 9, p. 1222, Apr. 2023, doi: 10.3390/healthcare11091222.

Optimization Design and Development of Swirler Blades for Aero-Engine Combustors

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***Abstract*—The swirler is the core aerodynamic component of aero-engine combustors, whose blade structure directly determines cold flow field characteristics, fuel-air mixing efficiency and flame stability. Traditional dual-stage swirlers suffer from uneven flow fields, insufficient central recirculation zone stability and weak flow separation suppression, while existing research lacks systematic analysis of maple samara aerodynamic characteristics in staged swirlers. This paper combines bionic structure optimization and CFD simulation, extracts two core aerodynamic features of maple samaras to design three bionic schemes, and sets 10 single-factor control cases with equal-thickness blades as the benchmark, validated via grid independence verification. This study quantifies the regulation effects of bionic schemes, clarifies their adaptation boundary, and establishes a 70% trailing-to-leading edge thickness ratio design criterion. Results show 1.0 mm blades outperform 1.5 mm ones; synchronous thickness gradient optimization increases recirculation zone length by 10.0% with optimal uniformity, while diversion schemes only suit single-stage use and multi-feature coupling causes performance degradation. This work provides a novel theoretical basis for efficient swirler optimization.**

Keywords—combustor swirler, flow field analysis, structural optimization, bionic blade, CFD numerical simulation

I. Introduction

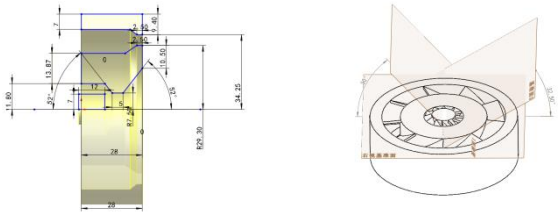
Modern aero-engines impose increasingly stringent low-emission requirements on combustors, and the optimization of the swirler flow field is the core support for low-emission combustion technologies [1].Therefore, the bionic optimization of the swirler blade structure is of great engineering significance. As the core aerodynamic component of an aero-engine combustor, the swirler directly determines the fuel-air mixing efficiency and flame stability by regulating the cold flow field. Conventional equal-thickness swirler blades have the defects of weak flow separation suppression ability and insufficient recirculation zone stability, which can hardly meet the design requirements of advanced aero-engines. Existing studies have verified the application value of bionic structures in fluid machinery optimization, but there is still a lack of systematic analysis on the differential adaptability of maple samara aerodynamic characteristics in dual-stage swirlers and the performance difference between single-feature and multi-feature coupling schemes.

For this reason, this paper takes the dual-stage axial swirler of an aero-engine as the research object, designs three groups of bionic optimization schemes based on the aerodynamic characteristics of maple samaras, and establishes 10 groups of comparison cases through CFD numerical simulation. The influence law of different bionic features on the flow field characteristics of the swirler is analyzed, and the optimal scheme is screened with the range of the central recirculation zone and velocity distribution uniformity as the core indicators, so as to provide a reference for the efficient optimization design of aero-engine combustor swirlers.

II. Bionic Design and Modeling of Swirler Blades

A. Research Object and Benchmark Model

The research object of this paper is a dual-stage counter-rotating axial swirler at the head of a single annular combustor for aero-engines. It adopts the widely used counter-rotating configuration in aero-engines, which can enhance fuel-air mixing through the strong shear effect of airflow. The two-dimensional core dimensions and blade installation datum plane design are shown in (Fig. 1.), and the core structural parameters are listed in TABLE I.



（a）2D core dimension diagram of the dual-stage swirler　（b）Angle design of the blade installation datum plane

Fig. 1. Design drawing of the dual-stage swirler benchmark model

The core design criterion of an axial swirler is the swirl number S, a dimensionless parameter characterizing swirl intensity, which directly determines the shape of the combustor central recirculation zone and flame stability. For axial vane swirlers, the classical calculation formula is:

$$S = \frac{2}{3}tan\,\theta \cdot \frac{R_{outer}^3 - R_{inter}^3}{R_{outer}^2 - R_{inter}^2} \quad (1)$$

In the formula: θ is the blade installation angle,R_{outer} and R_{inter} are the outer diameter and inner diameter of the swirler, respectively.

The blade installation angle is a key parameter for swirl number regulation and directly affects the characteristics of the recirculation zone. A large installation angle tends to form strong swirl and expand the recirculation zone, but may lead to an overly long recirculation zone and vortex core breakdown; an excessively small installation angle weakens the swirl intensity and reduces the flame stability. For axial swirlers in aero-engine combustors, the installation angle is usually selected as 30°–55° to balance the turbulence mixing efficiency and flow loss [11]. In this paper, the installation angle of the main-stage blades is set to 32.5° and the pilot-stage to 50°. Calculated by Eq. (1), the swirl number falls within the classic design range for aero-engine combustors, matching the flow field requirements of low-emission combustion.

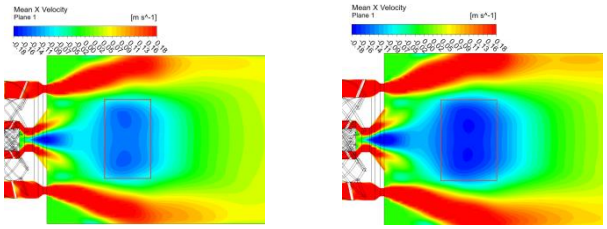
The parameters refer to the classic design criteria of dual-stage swirlers for centrally staged combustors [4, 6]. Benchmark simulation verifies that the cold flow field characteristics of the model are consistent with similar studies [2, 7], so it can be used as the benchmark model for subsequent bionic optimization.

TABLE I. CORE DESIGN PARAMETERS OF THE DUAL-STAGE AXIAL SWIRLER

Design Item	Design Parameter
Number of Main-Stage Blades	***9***
Number of Pilot-Stage Blades	***12***
Main-Stage Installation Angle	***32.5°***
Pilot-Stage Installation Angle	***50°***

B. Determination of Benchmark Thickness Parameters

To determine the benchmark thickness for bionic optimization, single-factor comparison is carried out between 1.0 mm and 1.5 mm equal-thickness blades using the general numerical simulation method verified in Chapter III. The results show that the 1.0 mm blade exhibits significantly better central recirculation zone continuity and flow field uniformity than the 1.5 mm blade (Fig. 2.). Therefore, the conventional 1.0 mm equal-thickness blade is taken as the benchmark for bionic optimization in this paper.



（a）Mean velocity cloud diagram of 1.5 mm blade　（b）Mean velocity cloud diagram of 1.0 mm blade|

Fig. 2. Comparison of axial mean velocity cloud diagrams of cold flow fields for blades with different thicknessesBionic Scheme Design and Modeling

Dong et al. [8] confirmed via Particle Image Velocimetry (PIV) experiments that the thickness gradient structure of maple samaras can form a stable leading-edge vortex with excellent aerodynamic stability, providing key experimental support for the engineering application of this aerodynamic feature. In this paper, 10 mature maple samara samples are measured, and it is further quantified that the blade trailing-edge thickness is about 70% of the leading-edge thickness, with the number of main veins concentrated at 3–5. Meanwhile, Ding et al. [4] have verified the effectiveness of such maple samara bionic structures in dual-stage swirlers, offering an important reference for the design in this paper.

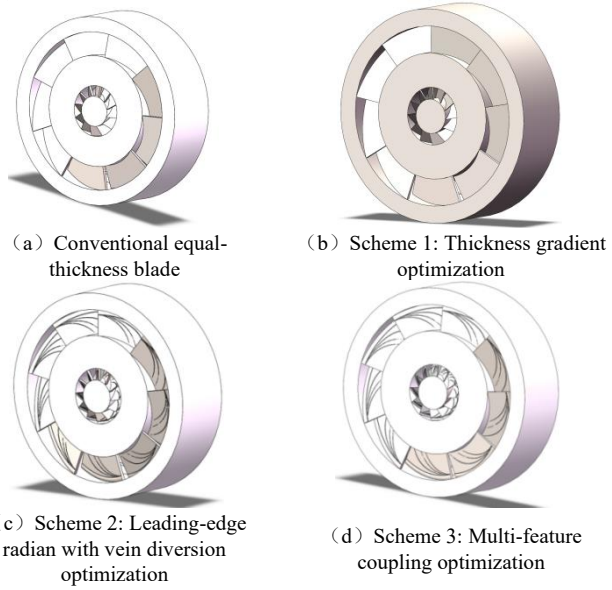
To quantify the core parameters of bionic design, 10 mature maple samara samples are measured, and the key data of blade leading-edge thickness, trailing-edge thickness, and vein number are shown in TABLE II. The measurement results show that the trailing-edge thickness of maple samara blades is about 70% of the leading-edge thickness, and the number of main veins is 3–5, which provides a direct biological prototype for the bionic scheme design in this paper.

TABLE II. MEASURED CORE PARAMETERS OF MAPLE SAMARA SAMPLES

No.	Leading-edge thickness (mm)	Trailing-edge thickness (mm)	Overall blade thickness (mm)	Main vein count
1	***0.48***	***0.34***	***0.40***	***5***
2	***0.52***	***0.36***	***0.42***	***3***
3	***0.50***	***0.35***	***0.38***	***5***
4	***0.47***	***0.33***	***0.39***	***4***
5	***0.55***	***0.38***	***0.45***	***5***
6	***0.45***	***0.32***	***0.37***	***3***
7	***0.53***	***0.37***	***0.43***	***5***
8	***0.49***	***0.34***	***0.41***	***5***
9	***0.51***	***0.36***	***0.44***	***4***
10	***0.46***	***0.32***	***0.36***	***3***
Average	***0.50***	***0.35***	***0.41***	***4.1***

Based on the above measured data, three core aerodynamic features of maple samaras are extracted, and three corresponding bionic optimization schemes are designed:(1) Thickness gradient optimization (trailing-edge thickness = 70% of leading-edge thickness);(2) Leading-edge radian with vein diversion optimization;(3) Multi-feature coupling optimization.

Parametric modeling of conventional blades and three groups of bionic blades is completed in SolidWorks. Sharp corners are rounded to ensure grid quality and calculation convergence. The blade models of different schemes are shown in (Fig. 3.).



（a）Conventional equal-thickness blade

（b）Scheme 1: Thickness gradient optimization

（c）Scheme 2: Leading-edge radian with vein diversion optimization

（d）Scheme 3: Multi-feature coupling optimization

Fig. 3. 3D models of swirlers with different optimization schemes

III. Numerical Simulation and Comparison of Bionic Schemes

A. Cold Flow Field Simulation Solution Settings

All simulations in this paper are cold flow field simulations without combustion chemical reactions. Referring to similar studies [4, 9], the standard k−ε model with standard wall functions is adopted, which has a capture error of less than 5% for the strong swirl flow field of dual-stage swirlers, balancing accuracy and convergence. Boundary conditions refer to aero-engine cold test conditions: the mass flow ratio of the main stage to the pilot stage is set to 7.6:1, the inlet temperature is 300 K, and the outlet is under standard atmospheric pressure. All simulation cases adopt the above unified settings to ensure that comparison results are only caused by blade structural differences, eliminating interference from irrelevant variables.

B. Grid Independence Verification

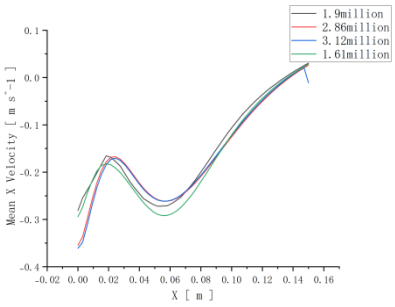


Fig. 4. Mean velocity distribution curves along the central axis with different grid numbers

Poly-Hexcore hybrid meshing is used, with local refinement in regions with intense flow gradients such as blade passages. This meshing strategy effectively captures low-swirl flow field characteristics [5] and ensures the calculation accuracy of recirculation zone details. Four groups of models with different grid numbers are verified, and the results show that 2.86 million grids meet the grid independence requirement (Fig. 4.). The final total grid number is about 2.8 million, with a maximum skewness ≤ 0.7, and the wall y+ satisfies the solution requirements.

C. Design of Simulation Comparison Schemes

To systematically verify the optimization effects of different bionic features, 10 comparison cases are set with conventional equal-thickness blades as the benchmark, following the single-variable principle strictly. Scheme 3 takes Case d (the optimal case of Scheme 1) as the control to explore the regulation effect of multi-feature coupling on the optimal single-feature scheme. Only the bionic structure of the blade is changed in all cases; other geometric parameters, meshing, boundary conditions, and solution settings remain completely consistent. The specific grouping is shown in TABLE III.

TABLE III. Grouping of Simulation Comparison Cases for Bionic Optimization of Dual-stage Swirler

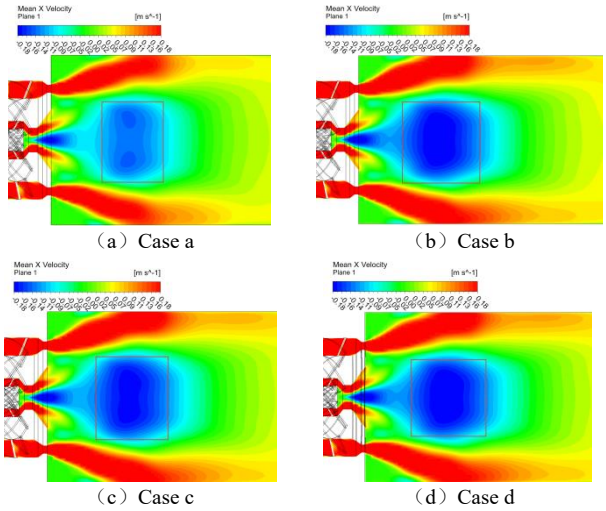
Case	Main-stage blade setting	Pilot-stage blade setting	Core optimization feature
a	*Conventional equal-thickness*	*Conventional equal-thickness*	*Benchmark control group*
b	*Conventional equal-thickness*	*Scheme 1 (thickness gradient)*	*Pilot-stage single-stage thickness gradient*
c	*Scheme 1 (thickness gradient)*	*Conventional equal-thickness*	*Main-stage single-stage thickness gradient*
d	*Scheme 1 (thickness gradient)*	*Scheme 1 (thickness gradient)*	*Dual-stage synchronous thickness gradient*
e	*Scheme 2 (radian + vein diversion)*	*Conventional equal-thickness*	*Main-stage single-stage diversion structure*
f	*Conventional equal-thickness*	*Scheme 2 (radian + vein diversion)*	*Pilot-stage single-stage diversion structure*
g	*Scheme 2 (radian + vein diversion)*	*Scheme 2 (radian + vein diversion)*	*Dual-stage synchronous diversion structure*
h	*Scheme 3 (multi-feature coupling)*	*Scheme 1 (thickness gradient)*	*Main-stage multi-feature coupling*
i	*Scheme 1 (thickness gradient)*	*Scheme 3 (multi-feature coupling)*	*Pilot-stage multi-feature coupling*
j	*Scheme 3 (multi-feature coupling)*	*Scheme 3 (multi-feature coupling)*	*Dual-stage synchronous multi-feature coupling*

D. Analysis of Simulation Results of Different Bionic Schemes

In this paper, the range of the central recirculation zone, axial velocity distribution uniformity, and vortex core stability are taken as the core evaluation indicators to compare and analyze the simulation results of 10 cases. The continuous region with axial velocity ≤ 0 m/s is defined as the effective central recirculation zone, ignoring discrete error points. The end of the recirculation zone is taken as the last coordinate point with velocity ≤ 0, and the next point with velocity > 0 is the starting point of the mainstream region. Accurate quantitative calculation is performed based on the axial velocity coordinates along the center line derived from simulations.

Results of Thickness Gradient Optimization Scheme

Thickness gradient optimization significantly improves the swirler flow field. Compared with the benchmark Case a, single-stage optimization in Cases b and c both expand the central recirculation zone, improve velocity distribution uniformity, and effectively reduce energy loss caused by flow separation on the blade surface. Among them, Case d with dual-stage synchronous thickness gradient optimization shows a significant superposition effect and is the optimal design of Scheme 1. The axial velocity distribution and centerline velocity variation are shown in (Fig. 5. and Fig. 6.).



（a）Case a （b）Case b

（c）Case c （d）Case d

Fig. 5. Comparison of mean velocity cloud diagrams for Scheme 1

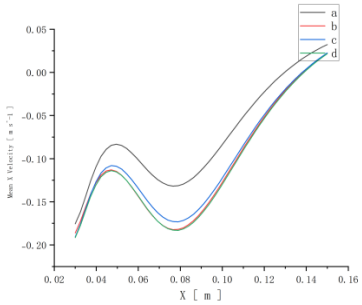


Fig. 6. Global curve of axial mean velocity along the centerline for Scheme 1

TABLE IV. CORE PARAMETERS OF THE RECIRCULATION ZONE FOR BENCHMARK CASE A AND OPTIMIZED CASE D

Case	Start X (m)	End X (m)	Axial length (mm)
Benchmark a	***0.0300***	***0.1280***	***97.96***
Optimized d	***0.0300***	***0.1378***	***107.75***

To accurately quantify the regulation effect of thickness gradient optimization on the central recirculation zone, core boundary parameters of the recirculation zone for benchmark Case a and optimized Case d are extracted based on the centerline axial velocity data from post-processing, according to the definition of the effective recirculation zone in this section. The core parameters are shown in TABLE IV.

The increase ratio of the axial extension length of the central recirculation zone is calculated as:

$$\eta = \frac{L_d - L_a}{L_a} \times 100\% \tag{2}$$

where η is the increase ratio, L_d is the axial length of the recirculation zone for optimized Case d, and L_a is that for benchmark Case a.

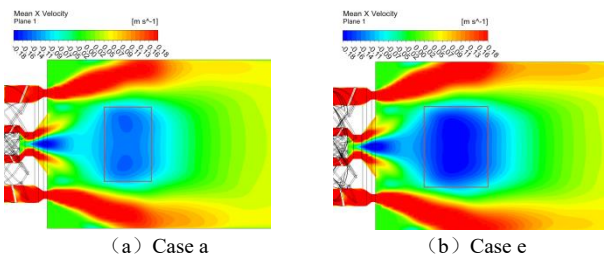
Substituting data from TABLE IV into the formula:

$$\eta = \frac{107.75 - 97.96}{97.96} \times 100\% \approx 10.0\% \tag{3}$$

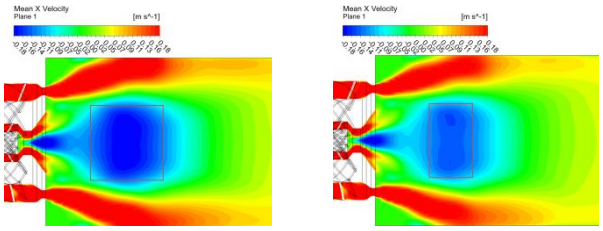
The results show that after dual-stage synchronous thickness gradient optimization, the axial extension length of the swirler central recirculation zone is increased by about 10.0% compared with the benchmark. The coverage of the recirculation zone is significantly expanded, providing more sufficient space for fuel atomization and fuel-air mixing. It can be further verified from the velocity cloud diagrams and centerline curves that the vortex core region in Case d is continuous, complete, and unbroken, without vortex core offset caused by local strong airflow shear. The axial velocity distribution uniformity is the best among all cases in Scheme 1, confirming the optimization effect of the 70% trailing-edge to leading-edge thickness ratio on the dual-stage swirler flow field.

Results of Leading Edge Radian + Vein Diversion Optimization Scheme

Single-stage optimization results of Scheme 2 show that the vortex core area in Cases e and f is significantly larger than that in benchmark Case a, and the centerline velocity of the recirculation zone is obviously improved. Leading-edge radian and vein diversion can effectively suppress flow separation and enhance local swirl stability, which is consistent with the regulation mechanism of blade structure on swirl intensity proposed by Feng et al. [7]. However, in Case g with dual-stage synchronous application, the two-stage enhanced swirl fields cause airflow interference, resulting in vortex core breakdown and recirculation zone fracture, and the optimization effect is significantly reduced, indicating that this scheme is not suitable for synchronous optimization of dual-stage swirlers (Fig. 7. and Fig. 8.).



（a）Case a （b）Case e



（c）Case f　　（d）Case g

Fig. 7. Comparison of mean velocity cloud diagrams for Scheme 2

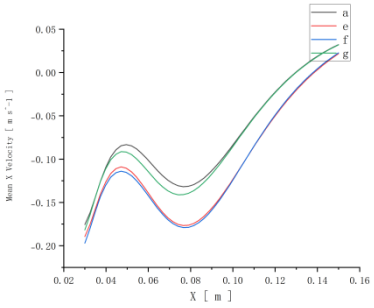
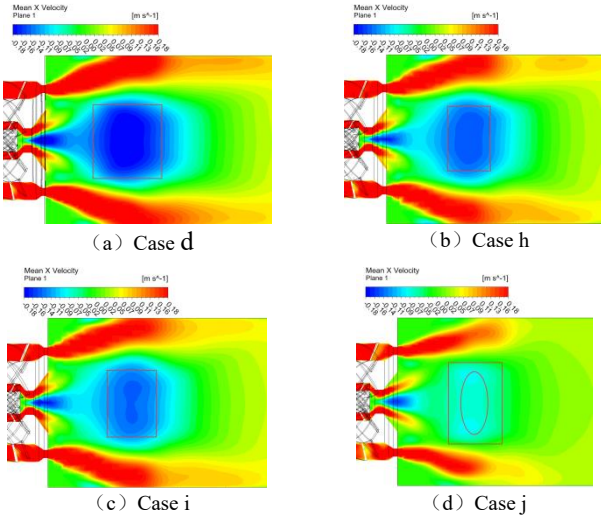


Fig. 8. Global curve of axial mean velocity along the centerline for Scheme 2

Results of Multi-feature Coupling Optimization Scheme

Simulation results of Cases h, i, and j with Case d (the optimal case of Scheme 1) as the control show that from Case d to j, the vortex core area of the central recirculation zone continuously decreases and the core velocity gradually decreases, so the multi-feature coupling optimization fails to achieve the expected effect. The main reason is that the airflow angle regulation effect of the thickness gradient conflicts with the diversion effect of leading-edge radian and vein diversion, destroying the continuity and stability of the flow field. It verifies that the effectiveness of single bionic feature optimization is better than the blind superposition of multiple features (Fig. 9.and Fig. 10.).



（a）Case d　　（b）Case h

（c）Case i　　（d）Case j

Fig. 9. Comparison of mean velocity cloud diagrams for Scheme 3

IV. Results and Discussion

Through the simulation comparison of 10 cases, this paper clarifies the adaptation law and flow field regulation mechanism of different aerodynamic features of maple samaras in dual-stage swirlers. The thickness-gradient structure with a thick leading edge and thin trailing edge forms a reasonable pressure gradient distribution on the blade surface, weakens the adverse influence of adverse pressure gradient on the boundary layer, and suppresses flow separation. This is consistent with the flow field regulation law in centrally staged combustors obtained by Yu et al. [2], further verifying the universality of the optimization scheme. Meanwhile, the airflow accelerates smoothly along the chord direction to avoid flow field disturbance, so dual-stage synchronous application achieves the optimal improvement of recirculation zone stability without additional structural complexity, showing strong engineering adaptability.

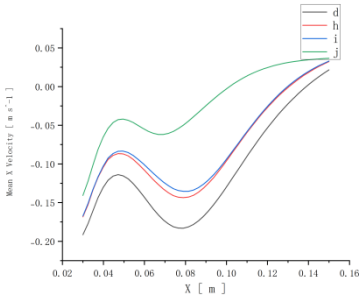


Fig. 10. Global curve of axial mean velocity along the centerline for Scheme 3

The leading-edge radian with vein diversion structure is only suitable for single-stage optimization. In dual-stage application, the two-stage enhanced swirl fields with counter-rotating directions form strong shear interference, directly destroying the continuity of the vortex core. Multi-feature coupling optimization tends to trigger conflicts among flow field regulation mechanisms of different features, failing to form a synergistic effect and leading to performance degradation, which is consistent with the conclusion of swirler structural parameter coupling interference obtained by Tao et al. [9] via neural network analysis.

The main innovations of this paper are as follows:

1) Systematically compare the differential adaptability of three aerodynamic features of maple samaras in the main and pilot stages of dual-stage swirlers, and clarify that the comprehensive effect of single thickness gradient optimization is superior to the blind superposition of multiple features;

2) Quantify the optimization effect of the 70% trailing-edge to leading-edge thickness ratio based on measured data, providing a referable design criterion for the bionic optimization of swirlers;

3) Reveal the adaptation law of bionic features in staged swirlers, providing a theoretical basis for differential optimization design.

V. Conclusion and Future Work

Taking the dual-stage axial swirler of an aero-engine as the research object, this paper designs a bionic optimization scheme based on the aerodynamic characteristics of maple samaras. Through CFD simulation comparisons of cold flow fields in 10 cases, the main conclusions are drawn as follows:

The flow field performance of the 1.0 mm-thickness blade is significantly better than that of the 1.5 mm-thickness blade, which is a reasonable benchmark parameter for bionic optimization.

When the dual-stage swirler synchronously adopts the thickness gradient design with a trailing-edge thickness of 70% of the leading-edge thickness, the flow field optimization effect is optimal. The axial extension length of the central recirculation zone is increased by about 10.0% compared with the benchmark, and the velocity distribution uniformity is the best.

Single-stage optimization with leading-edge radian and vein diversion can enhance local swirl stability, but dual-stage synchronous application easily induces vortex core breakdown.

Multi-feature coupling optimization tends to cause parameter conflicts and reduce the stability of the recirculation zone; single-feature optimization is better than the blind superposition of multiple features.

Future research will be deepened in two aspects: first, carry out thermal combustion coupling simulation to verify the actual influence of the optimized flow field on combustion performance; second, adopt the PIV test method for single-head combustors proposed by Zhang et al. [10] to cross-validate the simulation results of this paper and improve engineering applicability.

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REFERENCES

[1] Y. Liu, X. Sun, V. Sethi, and R. Niti, “Review of modern low emissions combustion technologies for aero gas turbine engines,” Progress in Aerospace Sciences, vol. 94, pp. 12–45, 2017.

[2] H. Yu, J. Suo, P. Zhu, and L. Liang, “Investigation on flow and emission characteristics of centrally staged lean direct injection (LDI) combustor,” Journal of Northwestern Polytechnical University, vol. 36, no. 5, pp. 816–823, 2018.

[3] M. Li, M. Cheng, H. Lin, and H. He, “Influence of main-stage swirl number on flow field of centrally staged combustor,” Aeroengine, vol. 44, no. 6, pp. 50–53, 2018.

[4] M. Ding, L. Dong, R. Ban, and Z. Li, “Numerical simulation on influence of bionic blades of dual-stage swirler on flow field morphology,” Journal of Propulsion Technology, vol. 46, no. 4, pp. 128–139, 2025.

[5] X. Ge, S. Shang, W. Hu, and D. Jin, “Wall-modeled large eddy simulation of flow field characteristics in low-swirl centrally staged combustor,” Aeroengine, pp. 1–7, 2025.

[6] D. Jin, X. Hui, J. Wu, and Y. Liu, “Influence law of dual-stage axial swirler design parameters on combustor ignition performance,” Journal of Aerospace Power, vol. 37, no. 11, pp. 2544–2552, 2022.

[7] Z. Feng, J. Sun, Y. Song, and Z. Liu, “Numerical study on influence of swirler vane installation angle on low-emission combustor performance,” Journal of Engineering for Thermal Energy and Power, vol. 39, no. 11, pp. 159–169, 2024.

[8] L. Dong, G. Wen, Z. Lei, and H. Li, “PIV experimental study on vortex system during autorotation of samara,” Journal of Experiments in Fluid Mechanics, vol. 35, no. 5, pp. 54–60, 2021.

[9] Y. Tao, J. Feng, L. Jiang, and Y. Zhang, “Influence analysis of structural parameters of dual axial swirler based on neural network,” Journal of Propulsion Technology, vol. 45, no. 10, pp. 174–183, 2024.

[10] Y. Zhang, Q. Luo, Z. Ma, and Y. Zeng, “PIV experimental measurement of flow field in single-head combustor,” Journal of Nanjing University of Aeronautics and Astronautics, vol. 50, no. 4, pp. 484–493, 2018.

[11] Z. Deng, Y. Liu, S. Wang, and P. Zhang, “Investigation of swirler configuration impacts on combustion dynamics and NO_x emissions of NH_3/air premixed flames with large eddy simulation,” International Journal of Hydrogen Energy, vol. 77, pp. 220–229, 2024.

Production Process Optimization in Plastic-Part Manufacturing Industry: Design and Analysis of Experiments

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***Abstract*—Smart devices require high-quality plastic components with strict dimensional accuracy to support reliable performance and large-scale production. Production of these components frequently faces serious quality problems related to thickness variation that does not meet the specified standard of 1.75 millimeters. This problem primarily originates from inappropriate settings of injection molding parameters, which affect material flow, cooling behavior, and dimensional stability of plastic parts. In particular, mold temperature, melt temperature, cooling time, and injection pressure represent critical parameters that strongly influence thickness consistency. Therefore, this study aims to determine suitable levels of these parameters through application of a general full-factorial design. To evaluate main effects and interaction effects among the injection molding parameters, the experiment design is planned and consists of the total of 243 experiments due to three replications and treatment combinations. The results indicated that all the parameters statistically have main effects and interactions on thickness variation. The optimization analysis identifies the most optimal production parameters, with mold temperature at 60 degrees Celsius, melt temperature at 230 degrees Celsius, cooling time at 14 second, and injection pressure at 100 megapascal. When the optimal parameter levels were applied to the production process, the proportion of nonconforming products decreases substantially. In particular, thickness variation, which represents the dominant defect type, is completely eliminated, while the total defect rate decreases from 8.5% to 2%. and this result confirms the effectiveness of the proposed experiment and process optimization for improving quality in injection-molded smart device components.**

Keywords—Design of Experiments, Full-factorial design, Injection Molding, Process Optimization, Smart Device, Thickness Variation

I. INTRODUCTION

The rapid expansion of smart devices has transformed modern society through advanced connectivity, automation, and data-driven services across daily life and industrial applications. Smart devices now play an essential role in communication systems, healthcare supervision, home automation, and industrial control, which has accelerated global adoption. As the number of smart devices continues to increase, demand for reliable and high-quality components required for production of these devices also rises significantly. This growth places increasing pressure on manufacturers to ensure consistent quality, dimensional accuracy, and production efficiency of components used in smart devices in order to meet market expectations and support large-scale deployment.

The case study is a manufacturing company that produces and supplies injection-molded plastic components for smart devices, including housings for IoT sensors, smart plugs, and small electronic control units. The company currently faces significant quality problems related to nonconforming plastic parts, which account for approximately 8.5% of total monthly production output. Among all defect types, nonuniform thickness represents the most critical issue, contributing to nearly 45% of total defects, with measured thickness deviations ranging from ±0.25 mm to ±0.40 mm from the specified nominal value. Other quality problems include surface sink marks, which represent about 20% of defects, part warpage after mold ejection at approximately 15%, dimensional deviation at screw bosses and connection points at around 12%, and inconsistent surface finish at roughly 8%. These quantitative results indicate that thickness variation constitutes the dominant quality problem and exerts the greatest impact on rejection and rework rates. The average thickness is lower than the specified standard of 1.75 millimeters (mm). Accordingly, this study places primary emphasis on mitigation of nonuniform thickness, since this defect type accounts for the largest proportion of quality losses in the case study.

The observed defect distribution underscores the need for a systematic approach to examine the effects of production parameters and their interactions in order to improve thickness consistency and overall product quality for plastic components used in smart devices. Preliminary investigation of the production process indicates that the identified quality problem primarily originates from the plastic injection molding stage. Analysis of process records and on-site observations reveals that inappropriate settings of injection molding parameters play a major role in thickness variation. In particular, mold temperature, melt temperature, cooling time, and injection pressure show insufficient control and lack of systematic adjustment.

Based on a review of literature related to determination of optimal machine parameter settings for reduction of nonconforming products, many recent studies apply Design of Experiments (DOE) as a primary analytical approach. Prior

research demonstrates that DOE provides a statistical framework to evaluate both main effects and interaction effects of process parameters and to identify optimal operating conditions [1]. For example, Werdine et al. [2] applied a general full-factorial design to set up the manufacturing parameters and solve defect issues in self-compacting concrete mixed with tire rubber waste. Gul and Iqbal [3] applied a general full-factorial design to minimize defects in microcapsule formation by optimizing solvent type, emulsifier, and processing temperature during the encapsulation of coconut oil. Vidakis et al. [4] implemented a general full-factorial design to identify optimal conditions for reducing defects in 3D-printed PLA parts by analyzing the effects of infill density, printing speed, and layer thickness. Belkadi et al. [5] applied a general full-factorial design to reduce defects in laser powder bed fusion by adjusting laser power, scanning speed, and hatching distance to improve porosity and surface quality. Zahi et al. [6] applied a general full-factorial design to reduce surface defects in 3D printing by varying layer thickness, printing speed, and extrusion temperature. Kurniawansyah et al. [7] used 2^2 full-factorial design to evaluate the effects and interactions of Poloxamer 407 and HPMC concentrations to optimize chemical stability and reduce formulation defects in chloramphenicol in-situ gel formulations. Previous studies have demonstrated that the general full-factorial design has been successfully applied across various industries to improve product quality and effectively reduce the number of nonconforming products as well as analyzing main effects and interaction effects of process parameters.

Therefore, this study aims to employ a general full-factorial design as the primary experimental approach to analyze the influence of critical injection molding process parameters, including mold temperature, melt temperature, cooling time, and injection pressure, on product quality and to identify optimal machine settings that reduce defects and enhance the overall efficiency of the injection molding process. The structure of paper is organized as follows. Section II provides the research methodology. Section III presents the results of the case study and interprets the findings. Finally, Section IV concludes the paper and outlines future works.

II. Materials and Methods

A. Investigation of Production Process

This study aims to improve the plastic-component production process for smart device housings. The production line consists of standardized sequence of operations that ensures dimensional accuracy, surface quality, and repeatability under mass production conditions. The process begins with raw material preparation, where thermoplastic pellets undergo a drying procedure according to supplier specifications in order to prevent moisture-related defects. After preparation, the material enters the injection molding machine through a hopper and moves toward the injection unit.

The subsequent stage involves plastic melting and homogenization inside the barrel, where screw rotation generates heat and pressure until the polymer reaches a molten state suitable for mold cavity filling. After material preparation, the injection molding machine performs the cavity fill phase, during which molten plastic flows into the mold cavity through the runner and gate system. This phase requires precise control of machine parameters to achieve uniform material flow and complete cavity formation. In this study, critical injection molding parameters are defined and adjusted at this stage, which include mold temperature, melt temperature, cooling time, and injection pressure, since these parameters directly affect material flow characteristics, solidification behavior, and final part thickness.

After cavity fill completion, the process proceeds to the solidification stage, where the molded part remains inside the closed mold for specified time period to allow dimensional stabilization. Thermal conditions during this stage influence shrinkage behavior and thickness consistency across the part. Once the solidification period ends, the mold opens and the part exits the mold through the action of ejector pins. The final stage consists of visual inspection and dimensional measurement, where thickness measurements occur at predefined locations to verify compliance with product specifications.

B. Problem Analysis

Problem analysis begins with examination of quality inspection data and production records, which indicates that nonuniform thickness represents the defect with the highest occurrence rate in the plastic component production process. This defect contributes the largest share of rejected and reworked parts and exerts direct impact on production efficiency, material usage, and overall manufacturing cost. Based on this evidence, nonuniform thickness is identified as the primary quality problem that requires systematic investigation. To analyze the root causes of nonuniform thickness, this study employs the cause-and-effect diagram based on the brainstorming among the factory manager and engineers. The diagram enables comprehensive identification of both primary factors and their causes associated with thickness variation in injection-molded plastic components, as presented in Fig. 1.

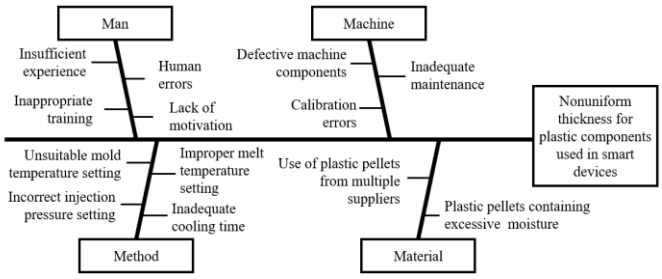

Fig. 1. Cause and effect diagram for analyzing the problem.

The problem involves the four factors, namely man, machine, method, and material. Each factor consists of various causes as follows:

- Man: This factor consists of many causes that primarily originates from human factors. One major cause involves insufficient operator experience in the manufacturing process. Some operators receive training that does not fully align with their actual job responsibilities, which limits their ability to understand process behavior. Human error also represents a significant contributing factor. Examples include incorrect data input during machine setup, selection of incorrect components

or materials during production preparation, and omission of required operational steps due to oversight or fatigue. Furthermore, lack of work motivation can reduce operator attentiveness and adherence to standard operating procedures, which increases the likelihood of mistakes.

- Machine: This factor arises from various causes associated with equipment performance and reliability. One major cause involves inadequate maintenance of injection molding machines. Calibration error represents another important factor. Inaccurate calibration of machine controllers and sensors may cause deviation between set values and actual operating conditions. Furthermore, machine components, such as screws, barrels, and hydraulic elements, can reduce control accuracy and introduce fluctuation in pressure and temperature during production.
- Method: This factor arises from causes related to process configuration and operational procedures. One major cause involves inappropriate selection of injection molding parameter settings, which directly influences process stability and product quality. In particular, unsuitable mold temperature settings may lead to uneven heat transfer and nonuniform solidification. Improper melt temperature can affect material viscosity and flow behavior within the mold cavity. Inadequate cooling time may prevent sufficient dimensional stabilization before part ejection, which increases thickness variation. In addition, incorrect injection pressure settings can result in inconsistent cavity filling and material packing. Lack of a systematic and standardized parameter-setting procedure further amplifies these issues and contributes to nonuniform thickness during production.
- Material: This factor originates from causes related to the quality and consistency of raw plastic materials. One major cause involves use of plastic pellets with insufficient quality, particularly pellets that contain excessive moisture. Another cause arises from simultaneous use of plastic pellets sourced from multiple suppliers, which may introduce variability in material properties such as viscosity and thermal response.

After identification and analysis of all potential causes of the problem, the brainstorming was conducted to identify the most important causes for addressing nonuniform thickness. The management panel indicates that improvement efforts should focus on causes related to machine parameter settings, since these causes not only have a direct impact on thickness variation but also act as underlying drivers that influence other sources of variation.

C. Determination of Process Parameters

Based on the problem analysis and managerial consensus, this study selects four critical injection molding process parameters for experimental investigation. These parameters have direct influence on the thickness of product. The selected parameters include mold temperature, melt temperature, cooling time, and injection pressure.

- Mold temperature controls heat transfer between the molten plastic and the mold surface and affects cooling uniformity and shrinkage behavior. In this study, mold temperature is set at 40 degrees Celsius (°C), 50 °C, and 60 °C to represent practical industrial conditions.
- Melt temperature determines material viscosity and flow characteristics before cavity filling. Melt temperature levels are defined at 220 °C, 230 °C, and 240 °C, which cover the recommended processing range for the selected plastic material.
- Cooling time represents the duration required for part solidification inside the mold prior to ejection. Cooling time is assigned values of 10, 14 and 18 seconds (Sec) to examine its influence on thickness stability and dimensional consistency.
- Injection pressure provides the driving force for cavity filling and material packing. Injection pressure levels are specified at 70, 85, and 100 megapascal (MPa), which reflect common operating settings in industrial injection molding machines.

All the levels of each parameter are presented in Table I.

TABLE I. PARAMETER LEVELS

Parameter	Code	Low	Medium	High
Mold temperature (°C)	A	40	50	60
Melt temperature (°C)	B	220	230	240
Cooling time (Sec)	C	10	14	18
Injection pressure (MPa)	D	70	85	100

D. Experiment Design

This study adopts a general full-factorial experimental design to systematically evaluate the effects of process parameters. The design includes four controllable factors, namely mold temperature, melt temperature, cooling time, and injection pressure. All possible combinations of factor levels are included in the experimental plan, which results in a complete factorial structure. To improve reliability of the experimental results and to provide an adequate estimate of experimental error, each experimental condition is replicated three times under the same operating settings. Therefore, the total number of experiments is 243 times. The order of experimental runs is randomized to reduce the influence of uncontrolled external factors and time-related variation. The data collection plan is summarized as shown in Table II. Thickness measurements are collected for each run at predefined locations on the molded parts and used as the response variable for subsequent statistical analysis.

TABLE II. DATA COLLECTION PLAN

StdOrder	RunOrder	A	B	C	D	Thickness
153	1	60	230	18	100	
172	2	40	230	10	70	
201	3	50	230	10	100	
:	:	:	:	:	:	
80	243	60	240	18	85	

E. Data Analysis

The thickness data collected from the molded parts are compiled for statistical analysis. Data analysis is performed using a statistical software package to evaluate main effects and interaction effects of the selected process parameters. Analysis of variance serves as the primary analytical technique, with statistical significance assessed through p-values. A

significance level 0.05 is adopted for hypothesis testing. When the p-value associated with the parameter or interaction term is less than 0.05, the null hypothesis is rejected, which indicates that the parameter exerts a statistically significant influence on the response variable [8]. This analytical procedure enables objective identification of critical parameters that affect thickness variation in the injection molding process. Hence, the model for process optimization can be written as follows:

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + \delta_l + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\tau\delta)_{il} + (\beta\gamma)_{jk} + (\beta\delta)_{jl} + (\gamma\delta)_{kl} + (\tau\beta\gamma)_{ijk} + (\tau\beta\delta)_{ijl} + (\tau\gamma\delta)_{ikl} + (\beta\gamma\delta)_{jkl} + (\tau\beta\gamma\delta)_{ijkl} + \varepsilon_{ijkl} \quad (1)$$

where y_{ijkl} is the $ijkl$th thickness observation; μ denotes the overall mean effect; τ_i, β_j, γ_k, and δ_l mean the effects of each parameter; $(\tau\beta)_{ij}$, $(\tau\gamma)_{ik}$, $(\tau\delta)_{il}$, $(\beta\gamma)_{jk}$, $(\beta\delta)_{jl}$, $(\gamma\delta)_{kl}$, $(\tau\beta\gamma)_{ijk}$, $(\tau\beta\delta)_{ijl}$, $(\tau\gamma\delta)_{ikl}$, $(\beta\gamma\delta)_{jkl}$, and $(\tau\beta\gamma\delta)_{ijkl}$ represent the effects of the interactions between two, three, and four parameters; ε_{ijkl} is a random error component; i, j, k, l = 1, 2, and 3.

The model adequacy checking is operated to ensure validity and reliability of the results. Residual analysis serves as the primary diagnostic procedure for verification of model assumptions. The residuals obtained from the fitted factorial regression model are examined through graphical and statistical evaluation, namely normal probability plots, residuals versus fitted values, and residuals versus run order. These diagnostic checks assess normality, constant variance, independence, and absence of systematic patterns in the residuals [9]. Finally, the Response Optimizer function is used to support the selection of factor settings that achieve a specified quality objective by combining the estimated effects from the factorial model with an optimization algorithm. In this study, the objective focuses on thickness control, and the optimization criterion specifies a target thickness value of 1.75 mm. Accordingly, all process parameters are set to minimize deviation from the target thickness in the production of smart device components [10].

F. Confirmation of Parameter Levels

After determination of suitable parameter levels, the optimized injection molding settings are applied to the production process for result confirmation. The process operates under the selected mold temperature, melt temperature, cooling time, and injection pressure. After process stabilization, as least 30 plastic components are randomly sampled from the production line and their thickness values are measured using calibrated equipment. The Z-test is conducted to statistically verify whether the process, under the optimized parameter settings, achieves the desired thickness performance. The hypotheses are defined as follows: H_0: $\mu = 1.75$ mm and H_1: $\mu \neq 1.75$ mm. The test decision is based on the p-value at a specified significance level. When the p-value is less than the significance level, the null hypothesis is rejected, which indicates that the process mean thickness differs significantly from the target value. If the p-value exceeds the significance level, the null hypothesis is not rejected, which confirms that the optimized parameter settings achieve thickness performance consistent with the target specification.

III. Results and Discussion

This section presents and discusses the results obtained from the general factorial experimental design applied to the injection molding process. The analysis focuses on evaluation of the effects of mold temperature, melt temperature, cooling time, and injection pressure on thickness variation of plastic components for smart devices. Statistical analysis of the experimental data provides insight into significant main effects and interaction effects among process parameters. The collective thickness data were analyzed to check the main effects and interactions through analysis of variance as presented in Table III.

TABLE III. Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	80	1.221	0.015	1462.630	0.000
Linear	8	0.874	0.109	10474.010	0.000
A	2	0.398	0.199	19051.880	0.000
B	2	0.261	0.131	12523.460	0.000
C	2	0.149	0.074	7122.900	0.000
D	2	0.067	0.033	3197.810	0.000
2-Way Interactions	24	0.289	0.012	1155.530	0.000
A*B	4	0.100	0.025	2406.040	0.000
A*C	4	0.065	0.016	1566.140	0.000
A*D	4	0.044	0.011	1056.410	0.000
B*C	4	0.044	0.011	1057.480	0.000
B*D	4	0.023	0.006	562.320	0.000
C*D	4	0.012	0.003	284.760	0.000
3-Way Interactions	32	0.056	0.002	167.140	0.000
A*B*C	8	0.022	0.003	260.430	0.000
A*B*D	8	0.014	0.002	172.840	0.000
A*C*D	8	0.011	0.001	132.380	0.000
B*C*D	8	0.009	0.001	102.930	0.000
4-Way Interactions	16	0.001	0.000	8.580	0.000
A*B*C*D	16	0.001	0.000	8.580	0.000
Error	162	0.002	0.000		
Total	242	1.223			
	S = 0.003 R-sq = 99.86% R-sq(adj) = 99.79%				

The result of analysis of variance shows that the p-values associated with the main effects (A, B, C, and D) indicate that all factors exhibit statistically significant main effects. The main effects of each factor on thickness of plastic component for smart devices, as shown in Fig. 2. The results indicate that higher mold temperature, melt temperature, and injection pressure contribute to increased thickness, while longer cooling time reduces thickness. Similarly, the two-, three-, and four-way interaction effects are all less than the significance level of 0.05. Similarly, the two-, three-, and four-way interaction

effects are all less than the significance level of 0.05. The interaction effects demonstrate that thickness variation is not governed by individual parameters alone, but rather by combined interactions among the process parameters. The interaction plot shows that parameter influences on thickness are interdependent, as shown in Fig. 3. The results reveal that higher injection pressure improves thickness uniformity more effectively at elevated mold temperatures due to enhanced material flow and reduced premature solidification. The effect of melt temperature depends on cooling time. At short cooling time, higher melt temperature improves flow but causes insufficient solidification and increases variation. At long cooling time, it produces more stable thickness. Higher injection pressure improves thickness consistency more effectively at high mold temperature. This result shows that process parameters must be optimized together. Based on the above results, the process optimization model for this case can be expressed as follows: $y_{ijkl} = 1.700 - 0.049\tau_1 + 0.001\tau_2 + \ldots + 0.001\tau_3\beta_3\gamma_3\delta_2 + 0.006\tau_3\beta_3\gamma_3\delta_3$.

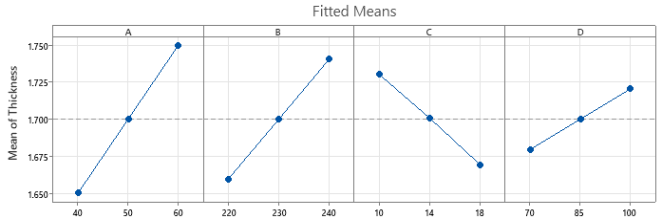

Fig. 2. Main effects plot of thickness for plastic components used in smart devices.

The model adequacy checking was conducted based on residuals, as illustrated in Fig. 4. The normal probability plot of residuals shows that the data points closely follow a straight line, which indicates that the residuals conform well to the normal distribution. The residuals versus fitted values plot displays random scatter around the zero line without any discernible pattern or funnel-shaped structure. This behavior indicates constant variance across the range of fitted thickness values and suggests absence of heteroscedasticity. In addition, the residuals versus observation order plot shows no systematic trend, clustering, or cyclic pattern, which supports the assumption of independence among observations throughout the experimental runs. This analysis confirms that the assumptions of normality, constant variance, and independence are satisfied. These results demonstrate that the developed factorial model provides an adequate representation of the relationship between injection molding parameters and thickness response, and that the model is suitable for subsequent optimization and confirmation analysis.

Next, the Response Optimizer function was applied to determine optimal levels of mold temperature, melt temperature, cooling time, and injection pressure, as presented in Fig. 5. The result indicates that the optimal levels of each parameter are mold temperature at 60 °C, melt temperature at 230 °C, cooling time at 14 Sec, and injection pressure of 100 MPa, respectively. After determination of the optimal parameter levels, the production process is set according to the optimal levels of the parameters. The injection molding process then operates under these settings to produce plastic components under stable conditions. Subsequently, 30 components are randomly sampled and measured their thickness. The statistical hypothesis test was performed based on Z-test with the predefined hypotheses. The result of statistical hypothesis test is illustrated in Table IV.

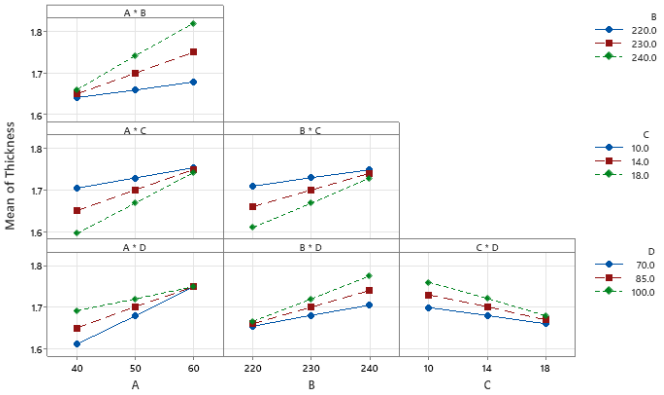

Fig. 3. Interaction effects plot of thickness for plastic components used in smart devices.

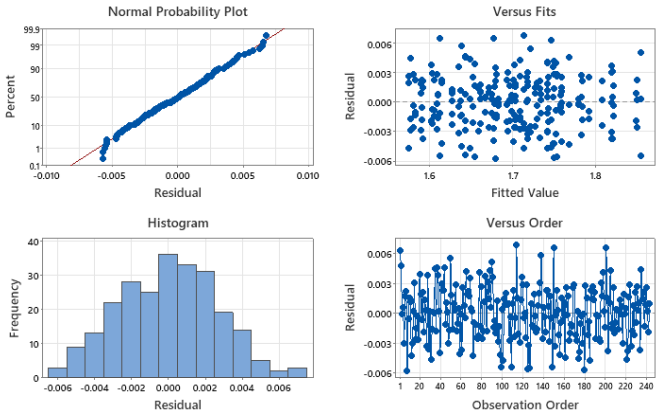

Fig. 4. Residual plots of model adequacy checking for plastic components used in smart devices.

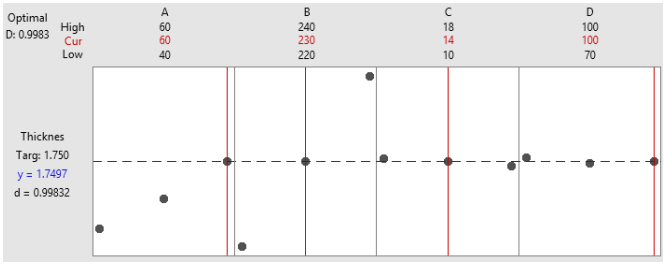

Fig. 5. Optimal parameter levels for the production of plastic components for smart devices.

It was found that the sample mean thickness ($\bar{x}$) obtained from the 30 samples equals 1.751 mm, with standard deviation (SD) of 0.022 mm. When the hypothesis test is conducted at significance level of 0.05, the result indicates that the sample mean thickness does not differ significantly from the standard value of 1.750 mm. This conclusion is supported by the p-value of 0.846, which exceeds the significance level, and therefore the null hypothesis is not rejected. These results confirm that the identified optimal parameter levels enable production of

smart device housings with thickness values that are statistically consistent with the specified standard.

TABLE IV. STATISTICAL HYPOTHESIS TEST RESULT

Sample	$\bar{x}$	SD	Z-Value	P-Value
30	1.751	0.022	0.190	0.846

The practical effectiveness of the improvement was evaluated through the implementation of the identified optimal parameter levels in actual production over a period of three months. During this period, production performance data were systematically monitored to assess long-term process stability and quality improvement. The results demonstrate that defects related to nonconforming thickness are completely eliminated. In addition, other types of defects are also reduced as a result of the overall process improvement, which addresses underlying causes contributing to various defect types. Hence, the overall defect rate decreases from 8.5% to 2% of total production output, which represents a reduction of 76.5%. The remaining defects may be attributed to the other factors that influence the production process but are not addressed in this study. The elimination of thickness variation leads to substantial cost savings; for an annual production of 500,000 units per year with a unit cost of 150 THB, the reduction in defects corresponds to savings of 4,875,000 THB per year.

IV. CONCLUSION

This study aims to reduce thickness variation in injection-molded plastic components used for smart device housings through the design and analysis of experiments. The current study focuses on identification of critical process parameters and determination of suitable operating conditions that support thickness accuracy and process stability in smart manufacturing environments.

The general full-factorial design was applied to examine the effects of mold temperature, melt temperature, cooling time, and injection pressure on product thickness. The analysis of variance confirms that the main and interaction effects among the process parameters exert significant influence on thickness variation. Model adequacy checking based on residual analysis verifies validity of the developed factorial model. The Response Optimizer function is then employed to determine optimal parameter levels that achieve the target thickness of 1.75 mm. The optimization analysis yields the most optimal combination of process parameters, with the mold temperature of 60 °C, the melt temperature of 230 °C, the cooling time of 14 Sec, and the injection pressure of 100 MPa, which satisfies the target thickness requirement. The optimized parameter settings were implemented in the production process and validated through confirmation experiments and statistical hypothesis testing. The results demonstrate that the optimized process achieves thickness values that are statistically consistent with the specified standard. Long-term application of the optimized settings in actual production further leads to a substantial reduction in overall defect rate, with nonuniform thickness reduced to a negligible level. This study contributes to both theory and practice through the integration of statistical experimentation with real industrial implementation. The results expand the theoretical understanding of experimental design for the optimization of production processes, which confirms its effectiveness and practical applicability for quality improvement in injection molding processes for smart device components.

Future research may integrate DOE with machine learning models to enhance process optimization capabilities. For example, key process parameters and corresponding quality outcomes can serve as inputs to train a machine learning model that captures the relationship between inputs and outputs. The trained model can then predict optimal parameter settings and reduce defects. Furthermore, extension of the proposed framework to other materials and products may provide broader validation and strengthen its applicability across diverse smart manufacturing contexts.

REFERENCES

[1] Z. Wang, Y. Feng, Y. Yang, J. Wang, S. Xu, and J. Qin, "Multi-objective optimization of rectangular cooling channel design using Design of Experiments (DOE)," *Applied Thermal Engineering*, vol. 242, 2024, Art. no. 122507.

[2] D. Werdine, G. A. Oliver, F.A. de Almeida, M. de Lourdes Noronha, and G. F. Gomes, "Analysis of the properties of the self-compacting concrete mixed with tire rubber waste based on design of experiments," Structure, vol. 33, pp. 3461-3474, 2021.

[3] Z. Gul and K. Iqbal, "Encapsulation of coconut oil using ethyl cellulose as coating shell for thermoregulating response of cotton," *Journal of Natural Fibers*, vol. 19, no. 16, pp. 14422-14434, 2022.

[4] N. Vidakis, M. Petousis, and J. D. Kechagias, "Parameter effects and process modelling of polyamide 12 3D-printed parts strength and toughness," *Materials and Manufacturing Processes*, vol. 37, no. 11, pp. 1358-1369, 2022.

[5] A. A. Belkadi, O. Kessal, T. Chiker, Y. Achour, A. Rouabhi, O. Messaoudi, and M. L. K. Khouadjia, "Full factorial design of mechanical and physical properties of eco-mortars containing waste marble powder," *Arabian Journal for Science and Engineering*, vol. 48, no. 4, pp. 4325-4338, 2023.

[6] S. E. Zahi, N. Ouazene, C. Tchekiken, W. F. Z. Elaglia, Y. Kerchich, and N. Belhaneche-Bensemra, "Optimization of azithromycin degradation by Fenton oxidation: a comparative study of general full factorial design 3^3 and Box-Behnken design," *Reaction Kinetics, Mechanisms and Catalysis*, vol. 136, no. 4, pp. 2147-2165, 2023.

[7] I. S. Kurniawansyah, T. Rusdiana, I. F. D. Arya, H. Ramoko, and H. A. Wahab, "Optimizing chemically stable chloramphenicol in-situ gel formulations using poloxamer 407 and HPMC through full-factorial design," *Scientific Reports*, vol. 14, 2024, Art. no. 25344.

[8] D. C. Montgomery, Design and Analysis of Experiments. Hoboken, NJ: John Wiley & Sons, 2017.

[9] S. Y. Win and M. Thianwiboon, "Parametric optimization of NACA 4412 airfoil in ground effect using full factorial design of experiment," Engineering Journal, vol. 25, no. 12, pp. 9-19, 2021.

[10] N. Koohathongsumrit and W. Meethom, "An integrated approach of fuzzy risk assessment model and data envelopment analysis for route selection in multimodal transportation networks," *Expert Systems with Applications*, vol. 171, 2021, Art. no. 114342.

A Dual-Layer Fault-Tolerant Strategy for Steer-by-Wire Systems with Heterogeneous Redundancy and Differential Steering

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***Abstract*—Steer-by-Wire (SBW) systems eliminate the mechanical steering column, resulting in an ASIL-D functional safety level. Failure of the steering actuator can cause complete loss of steering capability. This paper proposes a multi-level fault-tolerant control strategy for a dual-redundant SBW actuator architecture, integrating dual-channel heterogeneous redundancy with Differential Drive Assisted Steering (DDAS) backup. The hierarchical framework consists of an upper-layer Linear Time-Varying Model Predictive Control (LTV-MPC) that computes the desired front-wheel steering angle δ_{fd} and additional yaw moment ΔM_z, and a lower-layer controller that activates sliding-mode control (SMC) to generate the differential torque τ_{dif} when both channels fail completely. Four-wheel hub-motor torque is optimally allocated via quadratic programming (QP) that minimizes tire workload. Joint CarSim/Simulink simulations under a 60 km/h double-lane-change (DLC) maneuver with sudden dual-channel failure demonstrate that the proposed strategy limits the maximum lateral position deviation to 0.0937 m, keeps vehicle speed variation below 1 km/h, and achieves an average computation time of only 4.501 ms, satisfying real-time embedded requirements. The method significantly enhances the fail-operational capability of SBW systems.**

Keywords—Steer-by-Wire; fault-tolerant control; differential steering; model predictive control; sliding-mode control; torque allocation

I. INTRODUCTION

Conventional vehicle steering systems rely on a mechanical column for backup, whereas SBW systems depend entirely on electronic control. Any single-point failure can result in total loss of steering functionality. According to ISO 26262 [1], SBW systems must achieve ASIL-D safety integrity[2], which requires high reliability and fault tolerance. Most existing solutions employ hardware redundancy, such as dual-channel actuators and backup control units, to mitigate failures, but completely eliminating risk remains a significant challenge.

Most existing solutions employ hardware redundancy, such as dual-channel actuators and backup control units, to mitigate failures [3, 4]. However, completely eliminating risk remains a significant challenge because of component aging, common-cause failures, and the absence of mechanical fallback in SBW systems. Any actuator, sensor, or control unit failure can compromise vehicle steering and stability. Conventional fault-tolerant approaches often rely solely on hardware redundancy and binary fault judgments (fully operational or fully failed), without fully exploiting partial actuator capabilities or the vehicle's over-actuation potential from distributed drive systems [5].

Existing SBW fault-tolerant strategies fall into two categories. Hardware redundancy approaches [2, 6] employ dual-motor or dual-winding actuators to mitigate single-point failures, but increase cost and remain vulnerable to common-cause faults such as power supply anomalies [3]. Control-based solutions leverage MPC for constrained trajectory tracking [4, 7, 8] and SMC for robust yaw stabilization [9-11]; however, these methods typically assume binary fault states and do not exploit progressive degradation assessment. For complete steering loss, Differential Drive Assisted Steering (DDAS) uses left-right wheel torque differences to generate steering moments via the kingpin offset. Kirli et al. [12] showed that DDAS combined with Vehicle Stability Control requires integrated design to avoid stability deterioration, but their single-layer MPC framework lacks hardware redundancy and progressive fault handling. A recent review [13] confirms that integrated strategies combining multi-level hardware redundancy with DDAS backup remain unexplored.

To address this gap, this paper proposes a hierarchical fault-tolerant framework with four distinguishing features: (i) a fault factor lambda in [0,1] for progressive degradation assessment instead of binary fault judgment, (ii) a dual-layer architecture using LTV-MPC for trajectory tracking/channel arbitration and SMC-based DDAS for ultimate backup, (iii) tire-workload-minimization torque allocation preserving stability margins, and (iv) validated 4.501 ms real-time performance. Unlike [12], the proposed method integrates heterogeneous dual-channel hardware redundancy as the primary defense with DDAS as a last resort, forming a multi-layer safety architecture.

This paper builds upon a previously designed dual-channel heterogeneous redundant SBW actuator architecture and proposes a hierarchical fault-tolerant control framework. The upper layer is responsible for trajectory tracking, channel arbitration, and control reconfiguration, calculating the desired

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front-wheel steering angle and additional yaw moment to maintain vehicle stability. The lower layer activates Differential Drive Assisted Steering (DDAS) as a backup when both primary channels fail completely. By leveraging the over-actuated characteristics of four in-wheel motors and the scrub radius of the kingpin, the proposed strategy physically restores steering capability without relying on mechanical backup [14].

The hierarchical structure ensures layered safety: the upper layer handles normal operation and partial fault scenarios, while the lower layer provides a final line of defense during extreme failures. This design maintains vehicle controllability under dual-channel failures and demonstrates how software-based fault management, combined with mechanical over-actuation, can meet stringent ASIL-D requirements for SBW systems. The proposed approach addresses a critical gap in the literature by enabling safe and reliable operation under "no mechanical fallback" conditions, which is essential for future autonomous and advanced driver-assistance vehicles.

The remainder of this paper is organized as follows. Section II presents the framework and fault factor design. Section III develops the upper-layer LTV-MPC controller. Section IV details the lower-layer SMC-based differential steering and QP torque allocation. Section V provides simulation verification. Section VI concludes the paper.

II. Overall Framework of the Fault-Tolerant Control Strategy

The fault-tolerant control strategy for the SBW system proposed in this paper adopts a two-layer architecture, with the control logic shown in Figure 1. A complete fault-tolerant chain is constructed, spanning from fault detection and channel arbitration to control reconfiguration [4, 6].

The upper-layer controller calculates the desired front-wheel steering angle δ_{fd} and additional yaw moment ΔM_z required to maintain vehicle stability based on the failure degree of the steering actuator and the driver's desired trajectory, using model predictive control theory [7]. It redistributes the basic steering torque and makes decisions for channel switching. When the main control channel fails, the system seamlessly transfers control authority to the backup channel via a cross-channel communication mechanism. If both channels fail, an emergency fault-tolerant mode based on differential steering is activated [12, 15]. The lower-layer controller uses the vehicle's centroid sideslip angle and yaw rate as control parameters. Employing sliding mode control theory, it calculates the required additional yaw moment and coordinates torque allocation through four-wheel hub motors, achieving yaw stability control in the event of steering failure.

To overcome the binary fault judgment limitations of conventional methods, this paper introduces a fault factor $\lambda \in [0,1]$ to quantify the failure degree of the SBW actuators, defined as the ratio of the actual torque output of the steering motor to its desired torque output. Formally, lambda = T_actual / T_desired, where T_actual is the measured torque feedback and T_desired is the commanded torque. The fault response threshold is lambda < 0.9. The system fault identification module continuously calculates the fault factor by comparing the control torque output from each channel's microcontroller with the actual torque feedback from the motor position sensors. When the fault factor falls below a threshold (typically set at 0.9), the corresponding fault tolerant control strategy is triggered: a single-channel failure initiates a main-to-backup switch, while a dual-channel failure activates the differential steering backup [3].

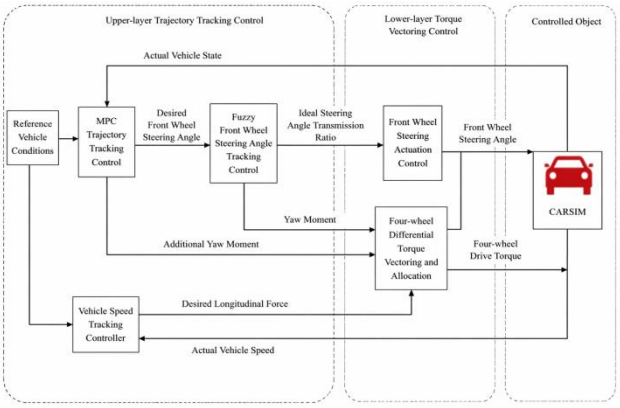

Fig. 1. Overall framework of the SBW fault-tolerant control strategy

III. Upper-Layer Trajectory-Tracking Controller Design

A. Three-Degree-of-Freedom Nonlinear Vehicle Dynamics Model

To capture planar motion while maintaining computational efficiency, a 3-DOF vehicle model is adopted, which is widely used in vehicle dynamics analysis [16, 17]. The following assumptions are adopted: vertical, pitch, and roll motions are neglected; the road is flat; aerodynamic drag and suspension effects are ignored.

Using Newton-Euler equations, the dynamics in the body-fixed frame are:

$$\begin{aligned} m(\dot{v}_x - v_y\gamma) &= F_{x1} + F_{x2} + F_{x3} + F_{x4} \\ m(\dot{v}_y + v_x\gamma) &= F_{y1} + F_{y2} + F_{y3} + F_{y4} \\ I_z\dot{\gamma} &= l_f(F_{y1} + F_{y2}) - l_r(F_{y3} + F_{y4}) + M_z \end{aligned} \tag{1}$$

where m is the vehicle mass, v_x and v_y are the longitudinal and lateral velocities, γ is the yaw rate, F_{xi} and F_{yi} $(i = 1,2,3,4)$ are the longitudinal and lateral tire forces, l_f and l_r are the distances from the center of gravity to the front and rear axles, and M_z is the additional yaw moment from direct yaw control (DYC).The subscript i = 1, 2, 3, 4 denotes the front-left, front-right, rear-left, and rear-right wheels, respectively.

Tire forces are computed using the Magic Formula tire model, which accurately captures nonlinear tire characteristics. For combined-slip conditions, the friction-ellipse principle is applied to correct longitudinal and lateral forces, ensuring commands remain within the tire's physical adhesion limit[18].

The 3-DOF model neglects vertical, pitch, and roll dynamics, so load transfer in aggressive cornering is not captured. The steady-state Magic Formula omits transient tire dynamics. These assumptions are acceptable for the simulated DLC maneuver ($m\mu = 0.8$), but near the friction ellipse limit, model fidelity

degrades. The control architecture can accommodate higher-fidelity models through parameter updates alone.

B. Linearization of Prediction Model and MPC Optimization

To facilitate the real-time computation of the MPC controller, the system state vector is defined as $\xi = [\dot{x}, \dot{y}, \dot{\psi}, \psi, X, Y]^T$, the control input vector as $u = [\delta_f, \Delta M_z]^T$, and the output vector as $\eta = [\dot{\psi}, \psi, Y]^T$. The nonlinear system described by Eqs. (1) can be represented as $\dot{\tilde{\xi}} = A_t \tilde{\xi} + B_t \tilde{u}$. At each control step k, the previous system state ξ_r and control input u_r, are taken as reference points, and a first-order Taylor series expansion is applied to the nonlinear equations while neglecting higher-order terms, yielding the linearized state-space equation:

$$\dot{\tilde{\xi}} = A_t \tilde{\xi} + B_t \tilde{u} \tag{2}$$

Here, $\tilde{\xi} = \xi - \xi_r$ and $\tilde{u} = u - u_r$; the Jacobian matrices are defined as $A_t = \frac{\partial f}{\partial \xi}|_{\xi_r, u_r}$, $B_t = \frac{\partial f}{\partial u}|_{\xi_r, u_r}$.

For implementation in a digital controller, the continuous system is discretized using the forward Euler method. Let the sampling period be T_s; then the discretized state-space equation becomes:

$$\tilde{\xi}(k+1) = A_k \tilde{\xi}(k) + B_k \tilde{u}(k) \tag{3}$$

Here, $A_k = I + T_s A_t$, $B_k = T_s B_t$, and I is the identity matrix.

To incorporate the control increment $\Delta u(k)$ into the state equation and limit actuator rate changes, an augmented state vector is defined as $x(k) = [\tilde{\xi}(k)^T, u(k-1)^T]^T$. The new state-space equation can then be reformulated as:

$$\begin{cases} x(k+1) = \tilde{A} x(k) + \tilde{B} \Delta u(k) \\ \eta(k) = \tilde{C} x(k) \end{cases} \tag{4}$$

Here, $\tilde{A} = \begin{bmatrix} A_k & B_k \\ 0 & I \end{bmatrix}$, $\tilde{B} = \begin{bmatrix} B_k \\ I \end{bmatrix}$, $\tilde{C}$ is the output matrix.

The core of the MPC controller is to solve an optimization problem over a finite prediction horizon to obtain the optimal control sequence [19]. For fault-tolerant control of the SBW system after actuator failure, the design of the objective function must consider three aspects: first, tracking accuracy, i.e., minimizing the deviation between the actual output and the reference trajectory; second, control smoothness, i.e., preventing abrupt fluctuations in the front-wheel steering angle and yaw moment; third, vehicle stability constraints. The optimization objective function J is defined as follows:

$$J(x(t), \Delta U(t)) = \sum_{i=1}^{N_p} \|\eta(k+i|k) - \eta_{ref}(k+i|k)\|_Q^2 + \sum_{i=0}^{N_c-1} \|\Delta u(k+i|k)\|_R^2 + \rho \varepsilon^2 \tag{5}$$

In the equation, N_p denotes the prediction horizon, N_c denotes the control horizon, and η_{ref} is the reference output vector, which mainly includes the desired yaw rate, desired heading angle, and desired lateral displacement. Q and R are the weighting matrices for the output error and control increment, respectively, used to balance tracking performance and control effort. ε is a slack variable, and ρ is its weighting factor, ensuring that the optimization problem remains feasible under strict constraints. (The slack variable ε in Eq.(5) is distinct from the SMC switching gain ϵ in Section IV.)

When solving the above quadratic programming (QP) problem, the system's physical constraints must be strictly enforced, including control input limits, control increment limits, and soft constraints:

$$\begin{cases} u_{min} \le u(k+i) \le u_{max} \\ \Delta u_{min} \le \Delta u(k+i) \le \Delta u_{max} \\ y_{min} - \varepsilon \le \eta(k+i) \le y_{max} + \varepsilon \end{cases} \tag{6}$$

In addition, to prevent the vehicle instability, stability constraints must be imposed on the vehicle center-of-gravity sideslip angle β and the lateral acceleration a_y: $|\beta| \le \beta_{th}$, $|a_y| [\text{cite}_s\text{tart}] \le \mu g$. By solving the above QP problem, the optimal control increment sequence over the control horizon is obtained: $\Delta U^* = [\Delta u^*(k), \ldots, \Delta u^*(k+N_c-1)]^T$. According to the receding horizon principle of MPC, only the first element $\Delta u^*(k)$ of the sequence is applied to the system, yielding the current desired front-wheel steering angle δ_{fd} and the additional yaw moment ΔM_z.

IV. Lower-Layer Differential Steering and Torque-Allocation Controller Design

When both SBW channels fail completely, the front-wheel angle δ_{fd} is dominated by the aligning torque from the road. The principle of differential steering is illustrated in Fig. 2, where steering motion is generated through longitudinal force differences between left and right wheels [5, 20]. The left and right wheels of the front axle are coupled through the steering trapezoid mechanism, achieving geometric constraint of the kingpin steering angles. The longitudinal tire force generated by road interaction has an offset distance relative to the kingpin axis, thereby producing an aligning torque at the kingpin.

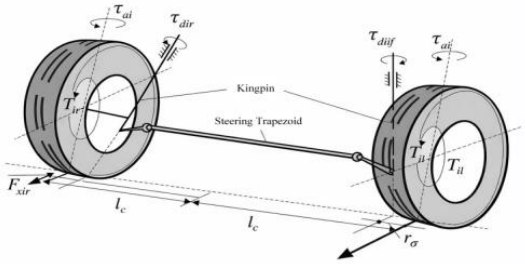

Fig. 2. Principle of differential steering

A. Sliding-Mode Differential Steering Control Law

The physical principle of differential steering is based on the kingpin offset in the vehicle suspension geometry. When a longitudinal force difference ΔF_x arises between the front-left and front-right wheels, the kingpin offset r_σ generates a moment around the kingpin axis, causing the wheels to steer. According to the differential steering mechanism, the resulting

differential torque τ_{dif} induced by the trapezoidal steering constraint can be expressed as:

$$\tau_{dif} = \tau_{dir_r} - \tau_{dir_l} = (F_{xr} - F_{xl})\, r_\sigma \cos\gamma \cos\sigma \quad (7)$$

Here, τ_{dir_r}, and τ_{dir_l} denote the components of the moments about the kingpin generated by the front-right and front-left wheels, respectively; r_σ is the kingpin offset (scrub radius), typically 60–80 mm for passenger vehicles; γ and σ are the kingpin inclination and caster angles, respectively. Under actuator failure conditions, the dynamics of the steering system can be described by:

$$J_{eff}\ddot{\delta}_f + b_{eff}\dot{\delta}_f = \tau_{dif} - T_i - \tau_f \quad (8)$$

Here, J_{eff} denotes the equivalent rotational inertia of the steering system, b_{eff} is the equivalent damping coefficient, τ_f represents the system friction torque, and T_i is the sum of the tire self-aligning torques. The self-aligning torque is the primary factor causing passive steering in the vehicle and represents the main resistance that differential steering must overcome.

To enable the actual front-wheel steering angle δ_f to track the desired angle δ_{fd} quickly and accurately, while compensating for the nonlinear disturbance caused by the self-aligning torque T_i, a sliding-mode variable-structure controller is designed. First, the steering angle tracking error sliding surface s is defined as:

$$s = \delta_{fd} - \delta_f \quad (9)$$

An exponential reaching law is employed to ensure convergence [3]:

$$\dot{s} = -\epsilon \mathrm{sgn}(s) - ks, (\epsilon > 0, k > 0) \quad (10)$$

Stability is verified via Lyapunov's direct method [11]. Define the Lyapunov function candidate:

$$V = \frac{1}{2}s^2 \quad (11)$$

Taking the time derivative and substituting the reaching law $\dot{s} = -\varepsilon\, \mathrm{sgn}(s) - k \cdot s$:

$$\dot{V} = s\dot{s} = s(-\varepsilon\, \mathrm{sgn}(s) - k \cdot s) = -\varepsilon|s| - ks^2 < 0 \ (\forall\, s \neq 0) \quad (12)$$

Proving global asymptotic stability of s = 0.By combining Eqs. (8) and (9), the derivative of the sliding surface is taken and substituted into the system dynamics, yielding the differential torque control law required for differential steering:

$$\tau_{dif} = b_{eff}(\dot{\delta}_{fd} + \epsilon \mathrm{sgn}(s) + ks) + T_i + J_{eff}\ddot{\delta}_{fd} \quad (13)$$

To mitigate the inherent chattering phenomenon of sliding-mode control, a saturation function $\mathrm{sat}(s)$ is used to replace the sign function $\mathrm{sgn}(s)$. The resulting differential torque control law is thus given by:

$$\tau_{dif} = b_{eff}(\dot{\delta}_{fd} + \epsilon \mathrm{sat}(s) + ks) + T_i \quad (14)$$

This control law shows that the required differential torque is composed of three components: a term to counteract system damping, a robustness term from the sliding-mode controller, and a compensation term to overcome the self-aligning torque.

During aggressive DDAS, front tires may approach adhesion limits. Tire force saturation can abruptly degrade lateral capacity [21, 22]. The QP allocation in Section IV-B mitigates this by minimizing squared tire utilization, keeping forces within the friction ellipse. Under $m\mu = 0.8$, utilization remains below saturation; for low-$m\mu$ surfaces, tighter inequality constraints would be needed.

B. Tire-Workload-Minimization Torque Allocation

The upper-layer commands δ_d and M_z are translated into four-wheel longitudinal forces F_{xi}. The optimization objective minimizes the sum of squared tire workload, which is a commonly used criterion in torque allocation and control distribution problems [14]: $\min J = \sum_{i=1}^{4} \left(\frac{F_{xi}^2 + F_{yi}^2}{(\mu F_{zi})^2} \right)$, subject to equality constraints (total longitudinal force, yaw moment, differential force) and inequality constraints (motor torque limits, road adhesion). The QP solver yields the optimal motor torque commands.

V. Simulation Verification and Analysis

To validate the effectiveness of the proposed hierarchical fault-tolerant control strategy for the SBW system under extreme failure conditions, this section employs a co-simulation platform combining CarSim and Simulink to perform an in-depth analysis of the system response following a complete dual-channel failure. In the simulation, the test vehicle travels at an initial speed of 60 km/h on a double-lane change (DLC) road surface with a high friction coefficient (μ=0.8), and at $t = 7$s, a complete failure of both steering actuator channels is randomly introduced. At this moment, the vehicle's mechanical and electric steering capabilities are entirely lost, and the system switches to the differential steering backup mode.

From the overall kinematic response, as shown in Figure 3, before the fault occurs at 7 s, the vehicle accurately executes steering maneuvers under the SBW system. At the instant of failure, the uncontrolled front-wheel steering angles rapidly decay toward zero due to the road centering torque T_i, causing the vehicle to completely lose path-tracking capability, and the lateral deviation increases sharply within a short period, preventing successful obstacle avoidance. After the fault-tolerant control strategy is engaged, the differential torque τ_{dif} generated by the left-right driving force difference on the front axle effectively counteracts the disturbance from the centering torque. As illustrated in Figure 3(f), the reconstructed equivalent front-wheel steering angle closely tracks the desired steering waveform. Although minor fluctuations occur at the switching instant due to the physical characteristics of differential steering, the fast response of the sliding-mode variable-structure controller ensures that both the phase and amplitude of the waveform remain highly consistent with the reference.

As a result, the vehicle's yaw rate, heading angle, and lateral deviation are all constrained within minimal error ranges, with the maximum lateral position deviation reaching only 0.0937 m.

Quantitatively, at the moment of dual-channel failure (t = 7 s), the front-wheel steering angles decay rapidly under the road centering torque, causing lateral deviation to grow immediately. The SMC-based DDAS controller activates, generating differential torque τ_{dif} that counteracts the centering torque and restores steering authority. The peak lateral deviation of 0.0937 m is reached at approximately 1.2 s post-failure, after which the vehicle converges back toward the reference trajectory; yaw rate tracking recovers to within 5% of the reference within 0.8 s. The reconstructed equivalent front-wheel steering angle (Figure 3(f)) tracks the desired waveform with less than 30 ms phase lag, confirming sufficient bandwidth for dynamic maneuvers. Longitudinal speed variation remains within 1 km/h throughout, indicating effective decoupling of longitudinal and lateral dynamics by the QP torque allocation.

This fully validates the critical role of differential steering as a "last line of defense" in maintaining vehicle operability under complete actuator failure.

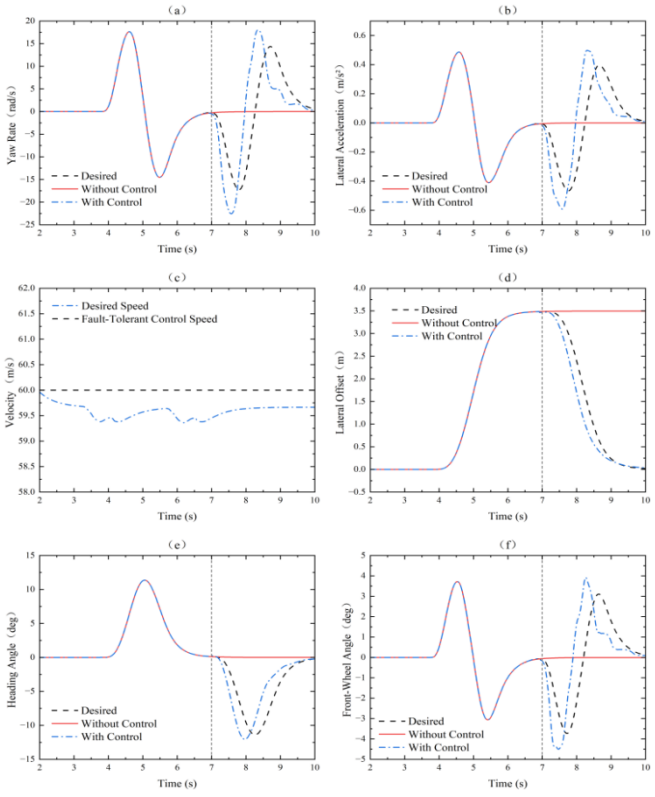

Fig. 3. Simulation results under 60 km/h double-lane-change condition

The optimized allocation of lower-layer torques forms the physical foundation for achieving the fault-tolerant performance described above. As illustrated by the motor torque distribution curves in Figure 4, immediately after the failure at 7 s, the output torques of the front-left and front-right wheel motors transition from a synchronized state to a highly asynchronous and opposite-direction output. The front-left wheel motor significantly increases its driving force, while the front-right wheel motor generates counteracting braking torque. By leveraging the kingpin offset distance r_σ, the longitudinal force difference is converted into a steering torque around the kingpin, enabling wheel deflection even when the mechanical linkage is disconnected.

Simultaneously, the lower-layer allocation algorithm actively engages the rear-axle motors to contribute to compensating the yaw moment, based on the principle of minimizing tire load ratios. This four-wheel coordination mechanism not only shares the longitudinal force burden of the front wheels—preventing tire side-slip risk due to overload from differential steering demands—but also compensates for energy losses caused by increased lateral forces during steering. Consequently, the vehicle speed fluctuations, as shown in Figure 3(c), are constrained within 1 km/h.

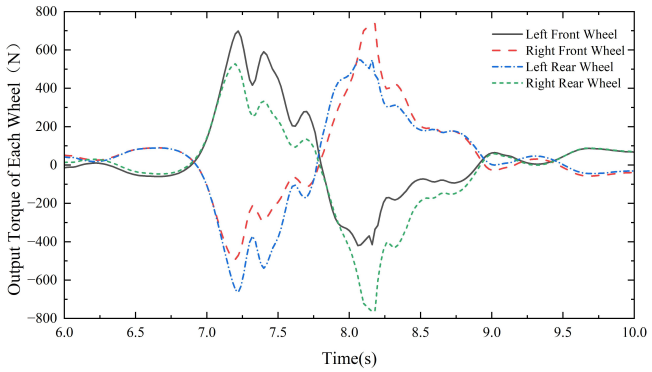

Fig. 4. Motor torque distribution curves

VI. Conclusion

This paper presents a hierarchical fault-tolerant control strategy for SBW systems that integrates dual-channel heterogeneous redundancy with DDAS backup. The upper LTV-MPC layer ensures accurate trajectory tracking under constraints, while the lower SMC + QP layer physically restores steering via differential torque and optimally allocates hub-motor torque to maintain yaw stability. CarSim/Simulink simulations under a 60 km/h double-lane-change scenario with sudden dual-channel failure validate that the maximum lateral deviation is controlled within 0.1 m and real-time performance is satisfied. The proposed method provides an effective "last-resort" backup for SBW systems, enhancing fail-operational capability. Future work includes: (i) HIL validation on a real-time embedded platform to verify 4.501 ms computation under hardware constraints and sensor noise; (ii) real-vehicle experiments on a four-wheel independent-drive EV prototype, including low-friction surfaces ($m\mu$ <= 0.3); (iii) extending the fault factor model to sensor and communication faults for unified fault assessment.

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References

[1] I. O. f. S. (ISO), "ISO 26262: 2018 Road vehicles-functional safety [S]," International Organization for Standardization, 2018.

[2] C. Huang and L. Li, "Architectural design and analysis of a steer-by-wire system in view of functional safety concept," *RELIABILITY ENGINEERING & SYSTEM SAFETY,* vol. 198, JUN 2020, Art no. 106822, doi: 10.1016/j.ress.2020.106822.

[3] C. Huang, F. Naghdy, H. P. Du, and H. L. Huang, "Fault tolerant steer-by-wire systems: An overview," *ANNUAL REVIEWS IN CONTROL,* vol. 47, pp. 98–111, 2019, doi: 10.1016/j.arcontrol.2019.04.001.

[4] H. Zhang, W. T. Jiang, W. Z. Zhao, and Y. H. Li, "Tracking and Fault-Tolerant Controller Design for Uncertain Steer-by-Wire Systems Using Model Predictive Control," *CHINESE JOURNAL OF MECHANICAL ENGINEERING,* vol. 37, no. 1, NOV 25 2024, Art no. 141, doi: 10.1186/s10033-024-01110-9.

[5] J. Tian, J. Tong, and S. Luo, "Differential Steering Control of Four-Wheel Independent-Drive Electric Vehicles," *ENERGIES,* vol. 11, no. 11, NOV 2018, Art no. 2892, doi: 10.3390/en11112892.

[6] G. B. Shi, P. F. Qiao, D. G. Sang, S. Wang, and M. H. Song, "Synchronous and fault-tolerance control for dual-motor steer-by-wire system of commercial vehicle," *PROCEEDINGS OF THE INSTITUTION OF MECHANICAL ENGINEERS PART D-JOURNAL OF AUTOMOBILE ENGINEERING,* vol. 238, no. 7, pp. 1964–1980, JUN 2024, doi: 10.1177/09544070231154961.

[7] B. T. Ren, H. Chen, H. Y. Zhao, and L. Yuan, "MPC-based yaw stability control in in-wheel-motored EV via active front steering and motor torque distribution," *MECHATRONICS,* vol. 38, pp. 103–114, SEP 2016, doi: 10.1016/j.mechatronics.2015.10.002.

[8] P. Stano *et al.*, "Model predictive path tracking control for automated road vehicles: A review," *ANNUAL REVIEWS IN CONTROL,* vol. 55, pp. 194–236, 2023, doi: 10.1016/j.arcontrol.2022.11.001.

[9] "<978-3-319-66266-4.pdf>."

[10] S. H. Ding, L. Liu, and W. X. Zheng, "Sliding Mode Direct Yaw-Moment Control Design for In-Wheel Electric Vehicles," *IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS,* vol. 64, no. 8, pp. 6752–6762, AUG 2017, doi: 10.1109/TIE.2017.2682024.

[11] G. Weibing and J. C. Hung, "Variable structure control of nonlinear systems: a new approach," *IEEE Transactions on Industrial Electronics,* vol. 40, no. 1, pp. 45–55, 1993, doi: 10.1109/41.184820.

[12] A. Kirli, Y. S. Chen, C. E. Okwudire, and A. G. Ulsoy, "Torque-Vectoring-Based Backup Steering Strategy for Steer-by-Wire Autonomous Vehicles With Vehicle Stability Control," *IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY,* vol. 68, no. 8, pp. 7319–7328, AUG 2019, doi: 10.1109/TVT.2019.2921016.

[13] L. Zhang, Z. Q. Zhang, Z. P. Wang, J. J. Deng, and D. G. Dorrell, "Chassis Coordinated Control for Full X-by-Wire Vehicles-A Review," *CHINESE JOURNAL OF MECHANICAL ENGINEERING,* vol. 34, no. 1, MAY 11 2021, Art no. 42, doi: 10.1186/s10033-021-00555-6.

[14] H. Feng *et al.*, "Fault-Tolerant Collaborative Control of Four-Wheel-Drive Electric Vehicle for One or More In-Wheel Motors' Faults," *SENSORS,* vol. 25, no. 5, MAR 2025, Art no. 1540, doi: 10.3390/s25051540.

[15] F. L. J. Silva, L. C. A. Silva, J. Eckert, R. Yamashita, and M. A. M. Lourenço, "Parameter influence analysis in an optimized fuzzy stability control for a four-wheel independent-drive electric vehicle," *CONTROL ENGINEERING PRACTICE,* vol. 120, MAR 2022, Art no. 105000, doi: 10.1016/j.conengprac.2021.105000.

[16] C. E. Beal and C. Boyd, "Coupled lateral-longitudinal vehicle dynamics and control design with three-dimensional state portraits," *VEHICLE SYSTEM DYNAMICS,* vol. 57, no. 2, pp. 286–313, FEB 1 2019, doi: 10.1080/00423114.2018.1467019.

[17] J. J. Hu, J. L. Tao, W. Zhao, and Y. Han, "Modeling and simulation of steering control strategy for dual-motor coupling drive tracked vehicle," *JOURNAL OF THE BRAZILIAN SOCIETY OF MECHANICAL SCIENCES AND ENGINEERING,* vol. 41, no. 4, APR 2019, Art no. 190, doi: 10.1007/s40430-019-1692-0.

[18] K. Berntorp, B. Olofsson, B. Bernhardsson, K. Lundahl, and L. Nielsen, "Models and methodology for optimal vehicle maneuvers applied to a hairpin turn," in *2013 American Control Conference*, 17–19 June 2013 2013, pp. 2139–2146, doi: 10.1109/ACC.2013.6580152.

[19] M. Morari, C. E. Garcia, and D. M. Prett, "Model predictive control: Theory and practice," *IFAC Proceedings Volumes,* vol. 21, no. 4, pp. 1–12, 1988/06/01/ 1988, doi: https://doi.org/10.1016/B978-0-08-035735-5.50006-1.

[20] J. Tian and M. F. Yang, "Hierarchical control of differential steering for four-in-wheel-motor electric vehicle," *PLOS ONE,* vol. 18, no. 6, JUN 9 2023, Art no. e0285485, doi: 10.1371/journal.pone.0285485.

[21] R. R. Wang, H. Zhang, J. M. Wang, F. J. Yan, and N. Chen, "Robust lateral motion control of four-wheel independently actuated electric vehicles with tire force saturation consideration," *JOURNAL OF THE FRANKLIN INSTITUTE-ENGINEERING AND APPLIED MATHEMATICS,* vol. 352, no. 2, pp. 645–668, FEB 2015, doi: 10.1016/j.jfranklin.2014.09.019.

[22] C. E. Beal and J. C. Gerdes, "Model Predictive Control for Vehicle Stabilization at the Limits of Handling," *IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY,* vol. 21, no. 4, pp. 1258–1269, JUL 2013, doi: 10.1109/TCST.2012.2200826.

Resonance Control of a Piezoelectric-Driven Nanopositioning Stage Based on Integral Resonant Control and the ITAE Criterion

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***Abstract*—To address the poor dynamic performance and limited control bandwidth of piezoelectric micro-positioning stages caused by inherent low-damping resonance, as well as the limitations of conventional dual-loop control in which the inner and outer loops are designed independently—making pole placement highly dependent on empirical tuning and making it difficult to balance time-domain and frequency-domain performance—a resonance control method for piezoelectric micro-positioning stages combining integral resonant control and the ITAE criterion is proposed. First, a third-order dynamic model of the piezoelectric micro-positioning stage is established through system identification and then reduced to a second-order model. Subsequently, a dual-loop control architecture consisting of an integral resonant damping loop and a tracking loop is constructed. The closed-loop transfer function of the system is derived, and its denominator is matched with the fourth-order standard characteristic polynomial based on the ITAE criterion. In this way, the adjustable parameters of both the damping controller and the tracking controller are analytically solved simultaneously, enabling optimal pole placement. Simulation results show that the proposed method increases the closed-loop bandwidth of the system to 210 rad/s and effectively suppresses the resonance peak below 0 dB. The step response exhibits zero steady-state error, while the overshoot is reduced to 0.25%, indicating a significant improvement in resonance suppression capability, dynamic response speed, and trajectory tracking accuracy of the piezoelectric micro-positioning stage.**

Keywords—piezoelectric micro-positioning stage; integral resonant damping control; ITAE criterion; dual-loop control architecture; resonance suppression

I. INTRODUCTION

Nano-positioning stages combine the high resolution and fast response of piezoelectric ceramics with the frictionless and backlash-free advantages of flexure mechanisms [1,2]. They play an important role in nanolithography, biological micromanipulation, ultra-precision manufacturing, and related fields [3–5]. However, their inherent low-damping characteristics readily produce pronounced resonance peaks. During high-speed motion within the same frequency band, self-excited oscillations may occur, thereby limiting dynamic performance and control bandwidth and restricting the realization of high-speed and high-precision positioning[6,7]. Therefore, the development of effective resonance suppression and high-performance tracking control methods is of great significance for improving their engineering applicability.

To achieve simultaneous enhancement of resonance suppression and trajectory-tracking accuracy, dual-loop control architectures have been widely adopted. In such structures, the inner loop is responsible for damping control, whereas the outer loop is responsible for tracking control. The outer loop usually employs an integral tracking controller, while the inner-loop damping controller may take various forms, including integral resonant control, positive position feedback, positive velocity-position feedback, and positive acceleration-velocity-position feedback[8–10]. These methods share the common principle of improving damping and suppressing resonance by reshaping the system zero-pole distribution. However, conventional approaches typically design the inner- and outer-loop controllers independently or sequentially. As a result, the outer-loop controller may alter the poles configured by the inner loop, causing the actual bandwidth to fall below the designed value and potentially affecting closed-loop stability. In addition, pole selection is often highly dependent on engineering experience and lacks a systematic design criterion, making the resulting control performance sensitive to the designer's subjective judgment. To address this issue, Russell et al.[11] proposed a simultaneous damping and tracking controller design based on the Butterworth pattern. By configuring the closed-loop poles to follow a standard Butterworth distribution, simultaneous parameter design and bandwidth improvement were achieved. Li et al.[12] further expanded the closed-loop bandwidth by introducing active resonance-frequency shifting. The methods proposed by Tao et al.[13,14] also aim to obtain a flat closed-loop frequency response and maximum bandwidth, and thus belong to frequency-domain shaping strategies. Overall, existing methods have made progress in resonance suppression, but most of them focus primarily on frequency-domain performance optimization and do not fully account for time-domain dynamic

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indices, such as fast transient response, low overshoot, and strong tracking capability. In point-to-point high-speed positioning scenarios that impose stringent requirements on both smoothness and rapidity of transient response, the practical application of these methods still faces certain limitations.

In summary, although considerable progress has been made in resonance suppression, the design process of existing methods often relies on empirical tuning and tends to emphasize frequency-domain optimization, making it difficult to meet the integrated requirements of time-domain dynamic response, such as rapidity and low overshoot, together with high-precision tracking. To solve these problems, this paper proposes a resonance-suppression strategy based on integral resonant control and the ITAE (Integral of Time-weighted Absolute Error) criterion. First, an accurate platform model is obtained through system identification, and an integral resonant system is constructed to improve the damping ratio. More importantly, the ITAE optimal criterion is introduced into the pole-placement process. By matching the ITAE standard characteristic polynomial with the closed-loop characteristic equation of the system, the parameters of the damping controller and tracking controller are solved analytically and synchronously. Thus, resonance can be effectively suppressed while optimal time-domain dynamic response characteristics are obtained, providing theoretical support and a technical solution for high-speed and high-precision nanopositioning applications.

II. Piezoelectric-Driven Nanopositioning Stage and Dynamic Model Identification

The experimental platform used in this study is a PT4V150-400S-S piezoelectric tilt stage manufactured by Shanghai Nano-Motion Technology Co., Ltd. The stage is driven by three piezoelectric ceramic actuators symmetrically arranged at 120° intervals, and multi-degree-of-freedom motion is realized through a flexure amplification mechanism. The maximum stroke of a single drive unit can reach 200μm. The stage is driven by a PCM931S piezoelectric driver, which receives a 0–10 V control signal and amplifies it to a 0–120 V driving voltage applied to the piezoelectric ceramic actuators. Displacement feedback is realized by a strain-gauge sensor integrated into the drive unit. Its output voltage of 0–10 V corresponds to a displacement range of 0–200μm, with a resolution of approximately 7 nm. The experimental system was built based on the Simulink Real-Time real-time simulation environment and a PCI-6221 data-acquisition card, and the control frequency was set to 20 kHz. The composition of the experimental system is shown in Fig. 1. The dynamic model of the platform was identified using the ident system-identification toolbox in MATLAB together with input-output data. The identification target was a third-order transfer-function model with two zeros; hence, the dynamic transfer function of the platform was obtained as follows:

$$G_{\mathrm{P}}(s)=\frac{1098s^2+1.465\mathrm{e}05s+3.566\mathrm{e}09}{s^3+2668s^2+1.956\mathrm{e}06s+4.736\mathrm{e}09} \tag{1}$$

The comparison between the Bode plot of the identified transfer function an d the experimentally obtained frequency response is shown in Fig. 2. As can be observed, a sharp resonance peak appears near the platform resonance region. This peak is the main cause of oscillation during high-speed operation and degrades positioning accuracy.

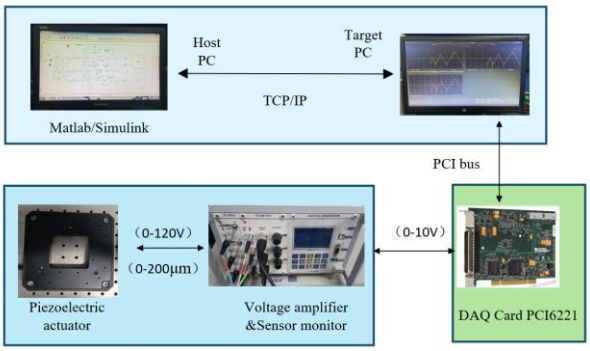

Fig. 1. Experimental system of the piezoelectric-driven nanopositioning stage

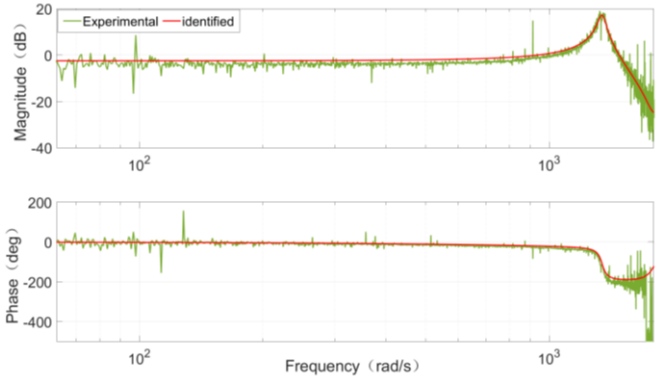

Fig. 2. Comparison between the Bode plot of the identified transfer function and the experimental frequency-response result

III. Integral Resonant Damping Control Based on Model Reduction

This section proposes an integral resonant control method based on model reduction. The identified third-order system is simplified into a second-order model, on the basis of which an integral resonant controller is designed.

A. Reduction Method for the Third-Order Model

Because subsequent integral resonant control is generally designed for a second-order system without zeros, the identified third-order dynamic transfer function must be reduced during controller design. The transfer function shown in Eq. (1) can be rewritten as

$$G_{\mathrm{P}}(s)=\frac{b_2s^2+b_1s+b_0}{(s+c)\left(s^2+2\xi\omega_{\mathrm{n}}s+\omega_{\mathrm{n}}{}^2\right)} \tag{2}$$

During model reduction, the simplified second-order system should remain consistent with the original third-order system in key characteristics, including resonance frequency, damping ratio, and DC gain, in order to ensure equivalence in practical applications and continuity in control performance. To improve controller design accuracy, neither the second-order zero pair nor the first-order inertial component in the original third-order system should be directly neglected; otherwise, the resonance suppression effect may be affected. Therefore, a second-order zero polynomial is connected in series before the original third-order model as the denominator, and a first-order pole is used as the numerator while keeping the overall transfer-

function gain unchanged. The specific transformation process is as follows:

$$G_1(s) = G_P(s)\frac{(s+c)}{b_2 s^2 + b_1 s + b_0}\sigma^2 = \frac{\sigma^2}{s^2 + 2\xi\omega_n s + {\omega_n}^2} \tag{3}$$

where the reduced transfer function represents the second-order dominant vibration mode of the platform. This simplification $\sigma^2 = b_0/c$ from a third-order to a second-order model essentially introduces two zeros and one pole into the controller structure. In this way, the dynamic characteristics of the system in the resonance frequency band are retained, while the key components in the platform dynamic response are also fully reflected, thereby effectively ensuring the accuracy of controller design and the practical control effect.

TABLE I. PARAMETERS OF THE SIMPLIFIED SECOND-ORDER TRANSFER FUNCTION

c	σ	ξ	ω_n rad/s
2613	1168.21	0.02035	1349.64

B. Integral Resonant Damping Control

To address the low-damping resonance characteristics of the piezoelectric nanopositioning stage, an integral resonant control strategy is adopted in this paper. By introducing zeros to reconstruct the open-loop zero-pole distribution and by tuning the integral gain through the root-locus method to achieve maximum damping, the resonance peak can be effectively suppressed and system damping can be improved. Consequently, the control bandwidth is expanded, and the stability and accuracy of trajectory tracking are enhanced. The overall control block diagram is shown in Fig. 3, where d is a constant gain, k is the integral controller gain, and kt is the tracking controller gain.

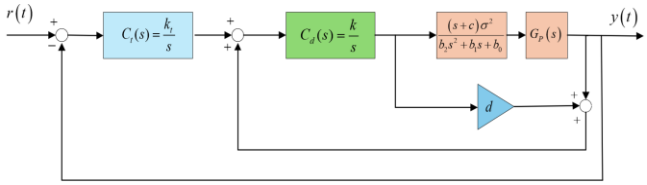

Fig. 3. Block diagram of integral resonant damping control with closed-loop tracking control

In traditional integral resonant damping controller design, a cascaded design procedure is typically adopted: the parameters of the damping controller are first determined according to the plant model to suppress resonance, and the outer-loop tracking controller, such as a PI controller, is then designed independently to improve tracking accuracy. However, this approach usually relies on empirical formulas and trial-and-error tuning and does not consider the dynamic coupling between the inner and outer loops. As a result, the actual damping effect and tracking performance are difficult to optimize simultaneously. To achieve synchronous design of the damping controller and tracking controller, this paper derives the system closed-loop transfer function and matches it with the standard characteristic polynomial under the ITAE criterion, thereby obtaining analytical expressions for the three adjustable parameters kt, k, and d. According to the controller structure in Fig. 3, the closed-loop transfer function can be derived as

$$\begin{aligned} G_{sys}^{num}(s) &= k_t k\sigma^2 \\ G_{sys}^{den}(s) &= s^4 + (2\xi\omega_n - dk)s^3 \\ &+ (\omega_n^2 - 2\xi\omega_n dk)s^2 + k(-\sigma^2 - d\omega_n^2)s + k_t k\sigma^2 \end{aligned} \tag{4}$$

where the numerator G_{sys}^{num} and denominator G_{sys}^{den} of the closed-loop transfer function are denoted separately. The fourth-order characteristic polynomial optimized under the ITAE criterion can be expressed as follows; the detailed analysis is provided in Section 2.3:

$$P(s) = s^4 + 2.1\omega_d s^3 + 3.4\omega_d^2 s^2 + 2.7\omega_d^3 s + \omega_d^4 \tag{5}$$

By matching the denominator in Eq. (4) with the ITAE characteristic polynomial in Eq. (5), the characteristic frequency, damping-controller gain, feedthrough coefficient, and tracking-controller gain can be directly obtained as

$$\omega_d = \frac{\omega_n\left(4.2\xi \pm \sqrt{13.6 - 36.76\xi^2}\right)}{6.8} \tag{6}$$

$$k = \frac{2.1\omega_d\omega_n^2 - 2\xi\omega_n^3 - 2.7\omega_d^3}{\sigma^2} \tag{7}$$

$$d = \frac{2\xi\omega_n - 2.1\omega_d}{k} \tag{8}$$

$$k_t = \frac{\omega_d^4}{k\sigma^2} \tag{9}$$

Here, ω_d is the desired characteristic frequency, k is the integral resonant damping gain, d is the feedthrough coefficient, and k_t is the tracking-controller gain. The derivation of Eqs. (6)–(9) is given in Appendix A.

C. Dynamic-Response Pole Placement Based on the ITAE Criterion

The ITAE (Integral of Time-weighted Absolute Error) criterion is a comprehensive performance index that balances response speed, steady-state error, and overshoot suppression. Its objective function is

$$J_{ITAE} = \int_0^\infty t|e(t)|\,dt \tag{10}$$

where e(t) is the system error and t is the weighting factor. Owing to the multiplicative effect of time t, errors that occur near the later steady-state stage are significantly amplified and penalized. To minimize the ITAE index, the system naturally avoids overshoot oscillations that require a long time to decay, thereby tending to produce a step response with small overshoot and weak oscillation. In addition, to satisfy the requirement of suppressing time-accumulated steady-state error, the system

must be able to eliminate the error rapidly and converge accurately to the steady-state value. For a piezoelectric nanopositioning system, this directly improves both dynamic response speed and steady-state positioning accuracy. The desired characteristic values under the ITAE criterion can be obtained by solving the characteristic polynomial corresponding to the specific system order. The first- to sixth-order characteristic polynomials are shown in Table 2 [15].

TABLE II. CHARACTERISTIC POLYNOMIALS OF DIFFERENT ORDERS UNDER THE ITAE CRITERION

order	characteristic polynomial
2	$s^2+1.4\ \omega_d s+\omega_d^2$
3	$s^3+1.75\ \omega_d s^2+2.15\ \omega_d^2 s+\omega_d^3$
4	$s^4+2.1\ \omega_d s^3+3.4\ \omega_d^2 s^2+2.7\ \omega_d^3 s+\omega_d^4$
5	$s^5+2.8\ \omega_d s^4+5.0\ \omega_d^2 s^3+5.5\ \omega_d^3 s^2+3.4\ \omega_d^4 s+\omega_d^5$
6	$s^6+3.25\ \omega_d s^5+6.6\ \omega_d^2 s^4+8.6\ \omega_d^3 s^3+7.45\ \omega_d^4 s^2+3.95\ \omega_d^5 s+\omega_d^6$

According to Eq. (6), the desired natural frequency of the characteristic polynomial is calculated as ωd = 748.5 rad/s. Substituting this value into Eq. (5) and solving for the roots of the polynomial yields a set of closed-loop poles that can realize optimal ITAE performance: -317.35 ± 945.35i and -468.57 ± 309.98i.

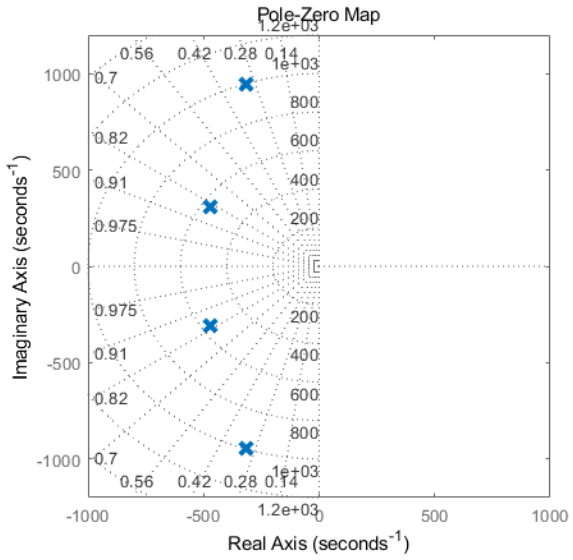

Fig. 4. Ideal closed-loop pole distribution under the ITAE characteristic polynomial

The distribution of these poles in the complex plane is shown in Fig. 4. It can be seen that the ITAE-optimal poles exhibit a unique and highly asymmetric distribution. The conjugate pole pair close to the real axis has a relatively large damping ratio and is mainly responsible for suppressing overshoot and governing the smooth regulation stage of the system response. The other conjugate pole pair, which is farther from the real axis, has a relatively higher undamped natural frequency and is used to improve the initial response speed of the system. This coordinated configuration, in which high damping ensures smoothness and high frequency ensures rapidity, is the fundamental reason why the ITAE criterion can mathematically ensure that the step response is both fast and smooth.

IV. SIMULATION ANALYSIS

A. *Comparison of Closed-Loop Frequency-Response Characteristics*

This section compares the closed-loop frequency-response characteristics under integral resonant control designed according to the ITAE criterion and under a conventional PI controller. The comparison focuses on resonance-peak suppression, bandwidth variation, and the consistency between frequency-domain and time-domain performance.

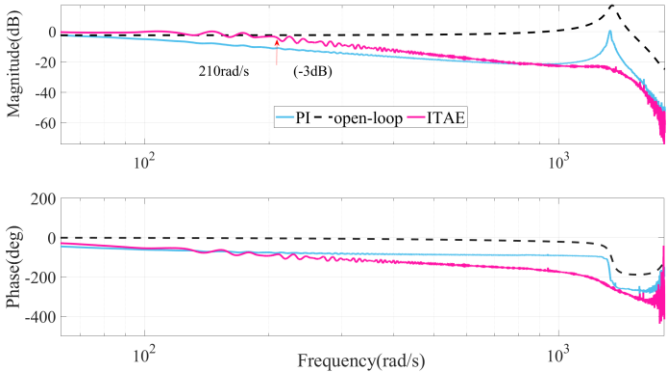

Fig. 5. Comparison of closed-loop frequency responses

As shown in Fig. 5, the open-loop system exhibits a resonance peak of nearly 17 dB around 1342 rad/s. The PI controller has a limited suppression effect on the resonance peak and cannot effectively expand the closed-loop bandwidth. In contrast, the ITAE-based controller suppresses the resonance peak below 0 dB and increases the bandwidth to 210 rad/s, indicating that the proposed pole-placement method improves both damping and response speed.

The reason for this improvement is that the proposed design does not tune the damping loop and the tracking loop independently. Instead, it matches the denominator of the whole closed-loop system to the ITAE polynomial, so the dominant poles are placed in a region that simultaneously increases damping and preserves sufficient response speed. Therefore, the resonance attenuation is not obtained at the expense of an excessively slow tracking response.

B. *Step-Signal Trajectory-Tracking Simulation*

The step-response simulation verifies the time-domain dynamic performance by observing overshoot, rise time, and settling time. As shown in Fig. 6, the step-response curves corresponding to different control methods differ significantly. The PI controller produces obvious oscillations and has a relatively slow rise process because the resonance mode is not sufficiently damped. By contrast, the ITAE-based controller achieves a fast and smooth response with zero steady-state error, and the overshoot is reduced to only 0.25% in the simulation.

The rise time and settling time are 0.0144 s and 0.03165 s, respectively.

This result is consistent with the frequency-response analysis. The reduced resonance peak weakens the residual vibration after the step command, while the expanded bandwidth shortens the time required for the stage to reach the target displacement. These results indicate that the ITAE-based controller is more suitable for high-speed nanopositioning tasks that require both fast response and small overshoot.

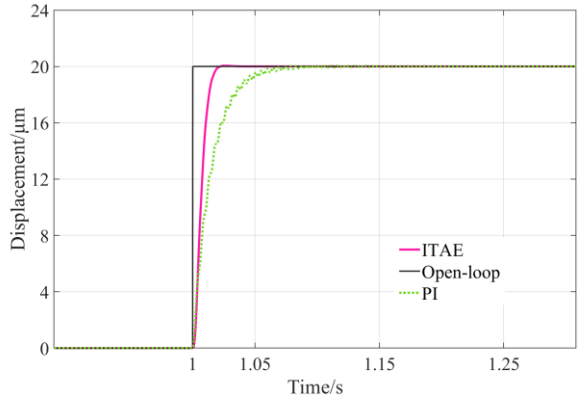

Fig. 6. Step-signal tracking simulation

V. Conclusion

To solve the low-damping resonance problem of piezoelectric micro-positioning stages and overcome the design deficiencies of traditional control methods, this paper proposes an integral resonant control strategy based on the ITAE criterion, and completes model reduction, synchronous controller-parameter design, and multiple groups of simulation experiments. The synchronous controller-parameter design method based on the ITAE criterion breaks through the limitations of the traditional cascaded design of dual-loop control and eliminates performance losses caused by dynamic coupling between the inner and outer loops. Through optimal pole placement, it realizes coordinated optimization of frequency-domain and time-domain performance, effectively suppressing system resonance while ensuring rapid and smooth response. Compared with traditional PI control, the proposed method demonstrates significant advantages in resonance suppression, bandwidth expansion, dynamic response, and trajectory tracking. It can effectively suppress the resonance peak, increase the closed-loop bandwidth of the system, and realize a fast and smooth step response without steady-state error. Meanwhile, it substantially reduces trajectory-tracking errors under high-speed scanning conditions. The proposed control method has a clear design procedure and reproducible parameter-solving process, avoiding the dependence of traditional pole placement on engineering experience. It therefore provides a general and efficient technical solution for resonance control of piezoelectric micro/nano-positioning stages.

Appendix A: Coefficient Matching and Parameter Derivation

For completeness, the coefficient-matching process omitted from the main text is summarized in this appendix. By comparing the denominator of Eq. (4) with the ITAE characteristic polynomial in Eq. (5), the following coefficient equations are obtained:

$$\begin{aligned} 2.1\omega_{\mathrm{d}} &= 2\xi\omega_{\mathrm{n}} - dk \\ 3.4\omega_{\mathrm{d}}^{\,2} &= \omega_{\mathrm{n}}^{\,2} - 2\xi\omega_{\mathrm{n}}dk \\ 2.7\omega_{\mathrm{d}}^{\,3} &= k\left(-d\omega_{\mathrm{n}}^{\,2} - \sigma^2\right) \\ \omega_{\mathrm{d}}^{\,4} &= k_{\mathrm{t}}k\sigma^2 \end{aligned} \tag{A1}$$

From the first, second, and third equations in Eq. (A1), three equivalent expressions of dk can be written as

$$dk = 2\xi\omega_{\mathrm{n}} - 2.1\omega_{\mathrm{d}} \tag{A2}$$

$$dk = \frac{\omega_{\mathrm{n}}^{\,2} - 3.4\omega_{\mathrm{d}}^{\,2}}{2\xi\omega_{\mathrm{n}}} \tag{A3}$$

$$dk = \frac{-k\sigma^2 - 2.7\omega_{\mathrm{d}}^{\,3}}{\omega_{\mathrm{n}}^{\,2}} \tag{A4}$$

Combining Eqs. (A2) and (A3) gives the quadratic equation for the target characteristic frequency:

$$3.4\omega_{\mathrm{d}}^{\,2} - 4.2\xi\omega_{\mathrm{n}}\omega_{\mathrm{d}} + 4\xi^2\omega_{\mathrm{n}}^{\,2} - \omega_{\mathrm{n}}^{\,2} = 0 \tag{A5}$$

Solving Eq. (A5) yields

$$\omega_{\mathrm{d}} = \frac{\omega_{\mathrm{n}}\left(4.2\xi \pm \sqrt{13.6 - 36.76\xi^2}\right)}{6.8} \tag{A6}$$

Substituting Eq. (A2) and (A4) gives the damping gain, feedthrough coefficient, and tracking gain as follows:

$$k = \frac{2.1\omega_{\mathrm{d}}\omega_{\mathrm{n}}^{\,2} - 2\xi\omega_{\mathrm{n}}^{\,3} - 2.7\omega_{\mathrm{d}}^{\,3}}{\sigma^2} \tag{A7}$$

$$d = \frac{2\xi\omega_{\mathrm{n}} - 2.1\omega_{\mathrm{d}}}{k} \tag{A8}$$

$$k_{\mathrm{t}} = \frac{\omega_{\mathrm{d}}^{\,4}}{k\sigma^2} \tag{A9}$$

Acknowledgment

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References

[1] LAI Leijie, MEI Junhua, ZHU Zina. Study on Static and Dynamic Performances of Distributed-compliance Bridge-type Displacement Amplification Mechanism[J]. Piezoelectrics & Acoustooptics, 2018, 40(2): 251-256.

[2] QIU Zhicheng, ZHANG Xianmin, WU Zhigen. A novel compliant XY micro-positioning stage using bridge-type displacement amplifier embedded with Scott-Russell mechanism[J/OL]. Precision Engineering, 2022, 73: 284-298.

[3] DING Bingxiao, LAI Leijie, LI Pengzhi, et al. Analysis and optimization of a novel compact compliant 2-DOF positioner for positioning to assess bio-specimen characteristics[J]. Machines, 2024, 12(6): 421.

[4] DU Gang, KANG Xiaoxu, ZENG Jiangtao, et al. Application of Multilayer Piezoelectric Actuator in Lithography Scanners[J]. Laser & Optoelectronics Progress, 2022, 59(09): 403-412.

[5] LI Linlin, LI Hanxiong, GU Guoying, et al. Design, modeling and control of high-bandwidth nano-positioning stages for ultra-precise measurement and manufacturing: a survey[J/OL]. Journal of Micromechanics and Molecular Physics, 2024, 09(02): 1-30.

[6] WANG Wenwen, LAI Leijie. Data-driven iterative feedforward compensation and adaptive vibration suppression of piezoelectric micro-motion stage[J]. Optics and Precision Engineering, 2024, 32(06): 833-842.

[7] NGUYEN H. N., LE T. D., HUYNH D. T., et al. Bandwidth enhancement in damping control for piezoelectric nanopositioning stages with load uncertainty: Design and implementation[J]. Mechanical Systems and Signal Processing, 2023, 185: 109817.

[8] EL-SAYED A. T., SAYED M., TAWFIK M. A. A modified linear integral resonant controller for suppressing jump-phenomenon and hysteresis in micro-cantilever beam structures[J]. Journal of Sound and Vibration, 2020, 485: 115549.

[9] WU Zhiren, XU Suan, FU Yaqiong, et al. Resonant Control of Third-Order Piezoelectric Nano-Positioning Stage Based on Recursive Positive Position Feedback[J]. Chinese Journal of Sensors and Actuators, 2023, 36(5): 738-743.

[10] LI Linlin, LI Chun-Xia, GU Guoying, et al. Positive acceleration, velocity and position feedback based damping control approach for piezo-actuated nanopositioning stages[J]. Mechatronics, 2017, 47: 97-104.

[11] RUSSELL D, SAN-MILLAN A, FELIU V, et al. Butterworth Pattern-based Simultaneous Damping and Tracking Controller Designs for Nanopositioning Systems[J]. Frontiers in Mechanical Engineering, 2016, 2(2): 21-31.

[12] LI Linlin, APHALE S. S., ZHU Limin. High-bandwidth nanopositioning via active control of system resonance[J]. Frontiers of Mechanical Engineering, 2021, 16(2): 268-281.

[13] TAO Yidan, LI Linlin, LI Han-Xiong, et al. High-bandwidth tracking control of piezoactuated nanopositioning stages via active modal control[J]. IEEE Transactions on Automation Science and Engineering, 2022, 19(4): 2998-3006.

[14] WANG Xiangyuan, LI Linlin, ZHU Zhiwei, et al. Experimental validation of the simultaneous damping and tracking controller design strategy for high-bandwidth nanopositioning – a PAVPF approach[J]. Sensors and Actuators A: Physical, 2022, 338: 113467.

[15] WILLIAMS II Robert L., LAWRENCE Douglas A. Linear State-Space Control Systems[M]. Xi'an: Xi'an Jiaotong University Press, 2014.

Design and Implementation of an Automated Control System for a Gantry Guide Rail Grinder

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***Abstract*—To meet the high-precision and high-efficiency requirements of large-scale guide rail grinding, this paper presents an integrated engineering solution for automating an MK5220 gantry guide rail grinder based on the FANUC Power Mate iD platform. The proposed system's contribution lies in its reliable and coordinated architecture, combining PLC logic, CNC control, and servo drives to achieve stable multi-axis synchronous motion and automated grinding cycles in a heavy-duty industrial environment. By integrating servo-driven feed axes, hydraulic worktable motion, and auxiliary subsystems through unified logic control, the system enables automatic mode switching, closed-loop feedback, and fault protection. Experimental validation on an industrial prototype demonstrates reliable motion accuracy, improved machining stability, and consistent surface quality, confirming the engineering applicability of the proposed solution for heavy-duty precision grinding equipment.The proposed system integrates servo-driven feed axes, hydraulic worktable motion, and auxiliary subsystems through unified logic control, enabling automatic mode switching, closed-loop feedback, and fault protection. Experimental validation on an industrial prototype demonstrates reliable motion accuracy, improved machining stability, and consistent surface quality, confirming the engineering applicability of the proposed solution for heavy-duty precision grinding equipment.**

Keywords—Gantry guide rail grinder, multi-axis coordination, motion control, Servo-driven Feed Mechanism

I. INTRODUCTION

Gantry guide rail grinders play a critical role in the finishing of large machine tool components, such as beds and columns, where high dimensional accuracy and surface quality must be maintained over long travel distances. Due to the large size and heavy mass of gantry-type structures, their machining processes are characterized by high inertia, complex multi-axis coordination, and strict stability requirements. Traditional manual or semi-automatic grinding methods rely heavily on operator experience, often resulting in low efficiency and inconsistent machining quality.

With the development of industrial automation technologies, PLC-based control systems have been widely adopted in gantry automation equipment to improve reliability and operational flexibility[1]. In grinding machine applications, PLC-controlled automation has been shown to effectively enhance production efficiency and reduce dependence on manual operation, especially under heavy-duty working conditions[2]. These studies indicate that PLC-based solutions are particularly suitable for complex industrial grinding equipment.

From the perspective of control system architecture, modular CNC controller designs offer an effective approach for integrating motion control and logic coordination in multi-axis machine tools[3]. Such architectures are well suited for gantry-type CNC machines, where multiple feed axes, auxiliary subsystems, and safety interlocks must operate in a coordinated manner. However, motion accuracy and long-term stability remain key challenges for large-scale gantry CNC equipment.

Previous research has shown that geometric errors and synchronization issues can significantly affect the performance of gantry-type CNC machines, highlighting the importance of closed-loop feedback and coordinated control strategies[4]. In recent years, adaptive PLC orchestration and integrated control logic have been emphasized as effective means to improve system robustness and reduce operational uncertainty in industrial manufacturing systems[5].

For CNC machine tools, PLC-based logic control remains an essential component for managing auxiliary subsystems, safety mechanisms, and operational sequences[6]. Established studies on CNC control system development further indicate that engineering-oriented control designs focusing on reliability and

maintainability are more suitable for industrial applications than overly complex algorithmic solutions[7]. Nevertheless, relatively few studies report comprehensive engineering implementations that integrate PLC logic control, CNC motion control, hydraulic regulation, and safety management into a unified system for gantry grinding equipment.

In this context, the present work focuses on the engineering design and implementation of an automated control system for a gantry guide rail grinder based on a FANUC PLC-CNC platform. While existing studies often focus on individual algorithms or components, relatively few report a comprehensive, industry-ready engineering implementation. This paper addresses this gap by detailing the complete design, integration, and validation of an automated control system for a large-scale gantry grinder. The novelty of this work lies not in the invention of new control theories, but in the systematic engineering application of proven technologies to achieve high performance, reliability, and operational safety in a demanding manufacturing context. The proposed system integrates multi-axis servo control, hydraulic worktable motion, and auxiliary subsystem management within a PLC-centered architecture, and its feasibility is validated through industrial application on a gantry guide rail grinder.

II. SYSTEM ARCHITECTURE AND SPECIFICATIONS

System Architecture and Specifications The MK5220 gantry guide rail grinder consists of a rigid gantry frame, a hydraulically driven reciprocating worktable, and peripheral and universal grinding heads. To support complex machining tasks, a five-axis motion configuration is employed, including V/W feed axes, Y-axis horizontal motion, and Z/A positioning and rotary axes.Table Ⅰ shows the 5-axis motion frame.

TABLE I. 5-AXIS MOTION FRAME

5-axis motion frame	
AXIS	***use***
V-axis & W-axis	feed movements for peripheral and general-purpose grinding heads, ensuring precise depth control.
Y-axis	Horizontal movement of the peripheral grinding head along the gantry
Z-axis & A-axis	Horizontal and rotational positioning of universal heads to accommodate complex rail geometries.

The control system is centered on the FANUC Power Mate iD platform, which integrates CNC interpolation and PLC logic control. Servo drives are used for all feed and positioning axes, while the worktable motion is regulated through PLC-controlled hydraulic units. This layered architecture ensures clear functional division between motion execution and logic coordination, while maintaining high dynamic response and positioning accuracy.Mechanical layout and axis definition of the MK5220 gantry grinding machine is shown in Figure 1.

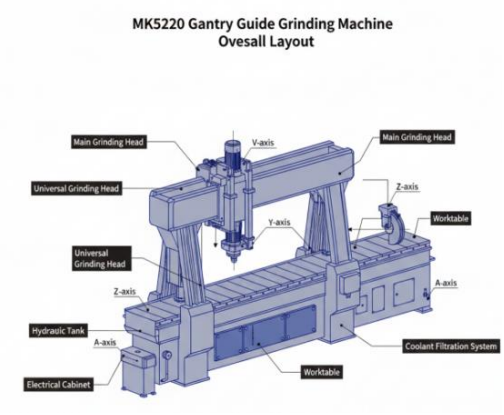

Fig. 1. Mechanical layout and axis definition of the MK5220 gantry grinding machine.

III. HARDWARE CONFIGURATION AND CONTROL LOGIC

The reliability of a control system depends on the sound selection of its hardware components.Multi-axis coordinated motion is realized through CNC interpolation combined with PLC-based sequence control. Closed-loop feedback from encoders is continuously monitored to ensure accurate positioning and synchronized motion. Automatic grinding cycles, including wheel dressing and feed compensation, are implemented to reduce manual intervention and maintain consistent machining conditions.

A. Control Units and Servo Systems

The FANUC Power Mate iD controller was selected for its exceptional reliability in industrial environments. It features integrated PLC functionality to handle complex logic sequences. The drive layer utilizes Beta-series servo motors to ensure high torque and nanometer-level positioning accuracy.

The required torque for the vertical axis (V/W-axis) servo motors is determined based on the axial load and transmission efficiency, expressed as:

$$T_m = 9550 \cdot \frac{P}{n} \tag{1}$$

where T_m is the required motor torque (N • m), F is the maximum axial cutting force (N), L is the lead of the ball screw (mm), η is the mechanical efficiency, and i is the reduction ratio.

Based on the specific parameters of the corresponding motor model, plug these values into the formula to obtain the result:

$$T_{mv} = 9550 \cdot \frac{4}{3000} \approx 12.73\text{N}\cdot\text{m} \tag{2}$$

$$T_{mw} = 9550 \cdot \frac{7}{3000} \approx 22.28\text{N}\cdot\text{m} \tag{3}$$

B. Auxiliary system integration

In addition to motion control, the electrical system integrates three major subsystems: hydraulic system, cooling and filtration system, and safety interlock system. Table Ⅱ illustrates these subsystems and their functions.

TABLE II. SUBSYSTEMS AND THEIR FUNCTIONS

subsystems and their functions	
Subsystems	***functions***
hydraulic system	Controlled by PLC to ensure stable reciprocating motion and lubrication of the worktable.
cooling and filtration system	A closed-loop feedback mechanism is employed to maintain constant temperature during the grinding process, preventing thermal deformation of the workpiece.
safety interlock system	Redundant hardware circuits and software constraints are programmed to prevent collisions within the multi-axis workspace.

C. Automated Dressing and Error Compensation Scheme

To maintain grinding wheel cutting ability and machining accuracy, an automatic dressing cycle is essential because abrasive grains wear down during use, exposing fresh, sharp grains. The proposed system integrates a coordinated correction scheme consisting of three key elements: deviation measurement units, correlation adjustment logic, and closed-loop compensation.

For precise control of the dressing process, the system utilizes dedicated feedback units. Linear scales are installed on the V, W, Y, and Z axes to provide high-resolution position feedback for closed-loop control of the grinding heads during dressing compensation. Furthermore, the dresser units themselves are equipped with limit switches to define safe travel ranges and home positions, ensuring accurate and repeatable dressing cycles.

The dressing and compensation process is governed by a coordinated logic implemented in the PLC. The system supports two operational modes: manual and automatic. In manual mode, operators can initiate dresser movements via the control panel or a handheld unit, with safety ensured by limit switches and software interlocks that prevent simultaneous left/right commands. In automatic mode, when an M-code is invoked by the CNC program, the PLC takes over, sequencing the dresser movements based on predefined parameters. Upon completion of the dressing cycle, the system automatically calculates the reduced wheel diameter. This cumulative reduction is then fed back to compensate the Z-axis grinding depth, ensuring that the programmed depth is accurately achieved on the workpiece.

IV. MULTI-AXIS COORDINATION AND AUTOMATED WORKFLOW

A. Multi-axis Coordinated Motion

Multi-axis synchronization was achieved by leveraging the inherent capabilities of the FANUC PMC (Programmable Machine Controller), which coordinates the axes based on integrated PLC logic. By leveraging the capabilities of the FANUC PMC (Programmable Machine Controller), the system achieves a seamless transition between rapid positioning and precise feed movements.

To ensure the precision of the gantry movement, a closed-loop feedback mechanism is implemented. The real-time positioning error δ is calculated as:

$$\Delta\delta = P_{feedback} - p_{command} \tag{4}$$

where $P_{feedback}$ is the actual position signal obtained from the linear scale, and $P_{command}$ is the target coordinate issued by the FANUC controller.

A detailed diagram of the hardware connections and electrical control architecture based on the FANUC system is shown in Figure 2.

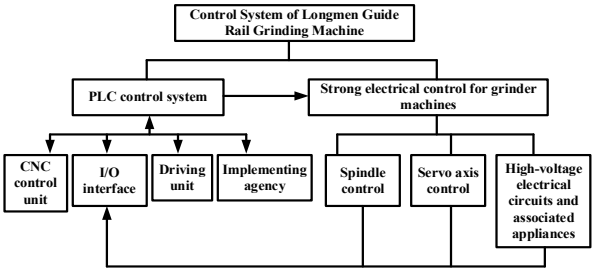

Fig. 2. Overall hardware architecture and electrical control diagram of the FANUC-based system.

B. Automated Grinding Cycle

To minimize manual intervention, automatic grinding wheel dressing is employed to monitor the wheel's condition. Based on preset parameters, it automatically activates correction programs and step-feed controls for incremental adjustments to the Z-axis and W-axis, ensuring the grinding force remains within an optimal range to preserve surface integrity.

The system automatically compensates for the reduction in grinding wheel diameter after each dressing cycle to maintain machining consistency:

$$Z_{actual} = Z_{programmed} + \sum_{j=1}^{m} \Delta d_j \tag{5}$$

where Z_{actual} represents the final compensated grinding depth, and Δd_j is the cumulative dressing amount across m cycles.

V. EXPERIMENTAL VALIDATION AND ENGINEERING PERFORMANCE ANALYSIS

A. System Integration and Control Workflow

The proposed control system for the gantry guide rail grinder is implemented based on a PLC-centered architecture, integrating the CNC system, servo drives, hydraulic units, and auxiliary subsystems into a unified automation platform. To achieve precise coordination between the CNC controller and various actuators, a robust electrical architecture was established.

The FANUC PLC acts as the core controller, responsible for logic coordination, signal processing, and inter-module communication, while the CNC system focuses on motion command generation and axis-level interpolation.

In the overall system architecture, the CNC system outputs motion commands and machining states to the PLC through

dedicated I/O and communication interfaces. The PLC further coordinates the servo drive units to control multi-axis motions, including the worktable feed axes and grinding head axes, while simultaneously managing the hydraulic system, lubrication system, cooling system, and safety-related devices. This layered structure ensures clear functional division and improves system reliability and maintainability.

To meet different operational requirements during machining and commissioning, the control system is designed with three operating modes: automatic mode, manual mode, and debug mode. In automatic mode, the PLC executes predefined machining sequences in coordination with CNC programs, enabling fully automated grinding operations. In manual mode, operators can directly control individual axes and auxiliary devices through the operation panel or hand-held unit, which is mainly used for setup, alignment, and maintenance. The debug mode is designed for system testing and parameter adjustment, allowing controlled single-step execution and signal verification without activating the full machining cycle. Mode switching is realized through interlocked logic conditions to prevent misoperation and ensure operational safety.

The system workflow can be summarized as follows. After power-on and system initialization, the PLC performs self-checking procedures and verifies the status of key components such as servo readiness, hydraulic pressure, and safety interlocks. Once the operating mode is selected, corresponding control logic is activated. During machining, real-time feedback signals from encoders, limit switches, and auxiliary sensors are continuously monitored. In the event of abnormal conditions, the PLC immediately interrupts the current operation and triggers the alarm handling mechanism. This workflow ensures stable system operation and reliable execution of grinding tasks.

Figure 3 demonstrates the integrated control performance in automated grinding cycles.

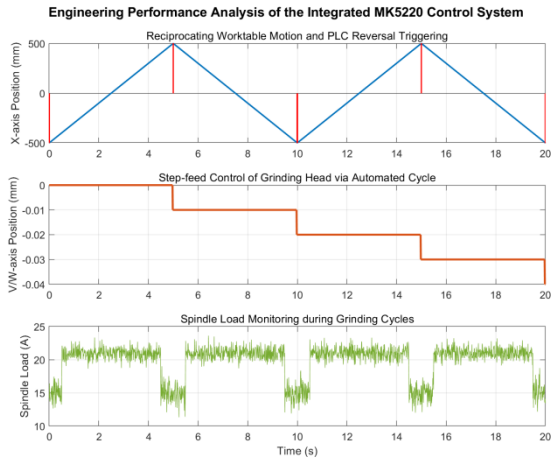

Fig. 3. Automated grinding cycle in operation on the MK5220 prototype, showing coordinated multi-axis movement during a typical machining task.

B. *Motion Accuracy and Stability Validation*

Motion accuracy and operational stability are critical performance indicators for large-scale gantry guide rail grinders. In the proposed system, the worktable feed motion is controlled through a combination of CNC interpolation commands and PLC-coordinated hydraulic regulation, enabling smooth reciprocating movement over long travel distances. The use of closed-loop feedback from position encoders and limit switches allows the system to maintain consistent positioning behavior during repeated operations.

For multi-axis coordinated motion, the servo-controlled grinding head axes operate under closed-loop control with real-time feedback, ensuring synchronized movement and stable dynamic response. The PLC continuously monitors axis states, including servo readiness, motion completion signals, and inter-axis interference conditions. This coordinated control strategy effectively reduces the risk of motion conflicts and enhances overall system stability during complex grinding processes.

In addition, the closed-loop feedback mechanism plays an important role in fault detection and system protection. Feedback signals from encoders, pressure switches, and thermal protection devices are integrated into the PLC logic. When abnormal conditions such as overtravel, overload, or hydraulic pressure loss are detected, the system immediately issues corresponding alarms and executes protective actions, such as stopping axis motion or shutting down related subsystems. This mechanism not only improves operational safety but also contributes to maintaining long-term motion accuracy by preventing mechanical damage and cumulative errors.

C. *Engineering Application Discussion*

Compared with traditional manual or semi-automatic grinding operations, the proposed control system significantly enhances automation level and operational consistency in large guide rail machining. Automated coordination between CNC commands and PLC logic reduces dependence on operator experience, minimizes manual intervention, and improves machining repeatability. This is particularly important for large-scale guide rails, where long travel distances and heavy loads make manual control inefficient and error-prone.

From an engineering application perspective, the system demonstrates strong adaptability to heavy-duty gantry grinding equipment. The integration of servo control, hydraulic regulation, and auxiliary subsystem management enables stable operation under high load and long-duration machining conditions. The modular design of the control logic also facilitates maintenance and troubleshooting, which is beneficial for practical industrial deployment.

Furthermore, the proposed control architecture exhibits good scalability and extensibility. By adjusting PLC logic parameters and reconfiguring axis definitions, the system can be adapted to gantry grinders of different specifications or expanded to support additional axes and functional modules. This flexibility provides a practical foundation for applying the proposed control strategy to other types of large-scale automated grinding equipment.

VI. Experimental Analysis

The developed control system was tested on the MK5220 prototype. The evaluation focused on two key metrics: motion synchronization and surface roughness.

To further quantify the operational stability of the proposed control architecture, we conducted a comparative analysis of positioning accuracy distribution. As shown in Figure 4, the FANUC-based automated system demonstrates a narrower and more concentrated error distribution compared to traditional manual grinding methods. The conventional approach exhibits a more dispersed error distribution due to inertia and human operational variations. The integrated closed-loop feedback and PLC-CNC collaborative control effectively confines positioning deviations within a narrow range of ±0.005 mm. This concentration confirms the system's excellent stability and suitability for heavy-duty precision machining applications.

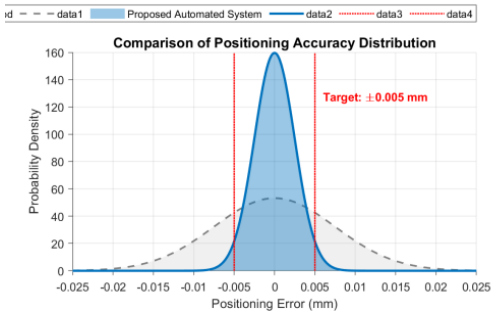

Fig. 4. Comparison of positioning error distribution between the proposed automated control system and a traditional manual/semi-automatic grinding operation

For comparison, positioning data was also collected during operation under a manual control mode on the same machine, where an experienced operator manually guided the axes. The resulting error distribution (labeled as 'Traditional' in Fig. 4) shows significantly higher dispersion, primarily due to inconsistencies in human reaction and control. This comparison validates the performance improvement achieved by the proposed automated system, which confines positioning deviations within a narrow range of ±0.005 mm.

To evaluate the repeatability of the positioning system, a series of 30 repeated positioning tests were conducted on the Y-axis over a travel distance of 500 mm. The actual position was measured using the integrated linear scale. The standard deviation S of the positioning error was calculated using Equation (6):

$$S=\sqrt{\frac{1}{N-1}\cdot\sum_{i=1}^{n}(x_{actual}-\bar{x})^2} \tag{6}$$

Where x_actual denotes the measured position and $\bar{x}$ represents the mean value. Based on the experimental sample data, the calculation result for the standard deviation is 0.0027 mm. This result demonstrates high repeatability under the closed-loop feedback mechanism, far exceeding the requirements of standard industrial guides.

Based on the experimental sample data，The calculation result for the standard deviation is 0.0027mm.

The surface roughness (Ra) of the machined guide remains consistently below 0.4 μ m, indicating the FANUC-based control system exhibits exceptional stability.As shown in Figure 5, the measurement sampling length for the guide rail surface profile is 4.0 mm. After processing with MATLAB, the experimental results indicate that the surface roughness (Ra) of approximately 0.38 μ m meets the high-precision requirements for large gantry components.

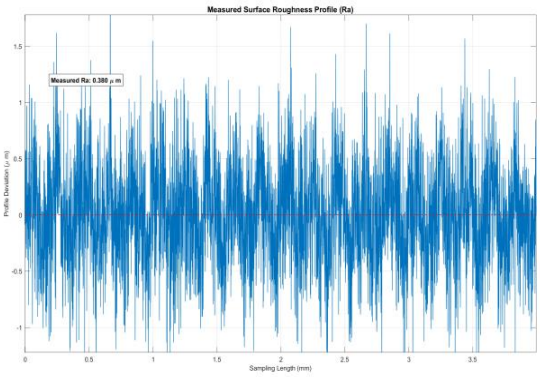

Fig. 5. Measured surface profile of the machined guide rail, processed with MATLAB, showing a roughness Ra of approximately 0.38 μ m.

VII. Conclusion

This paper presented a successfully engineered and implemented control system for the MK5220 gantry guide rail grinder. By integrating standard industrial components—a FANUC PLC-CNC platform, servo drives, and hydraulic systems—within a robust electrical architecture, a highly reliable and automated machining solution was achieved. The key contribution is the demonstration of a practical, scalable, and maintainable system that significantly enhances the automation level and machining consistency of large-scale grinding equipment, moving beyond theoretical discussions to provide a validated engineering blueprint for similar industrial applications. Future work will focus on integrating AI-based predictive maintenance and real-time vibration compensation to further elevate the intelligent capabilities of the grinding process.

References

[1] S. Krishnan and A. V. U., “XYZ gantry: A PLC-based automated gantry system for precision handling,” International Journal of Engineering Research & Technology, vol. 14, no. 4, pp. 443–448, 2025.

[2] A. K. Yadav and R. Singh, “Industrial grinding automation using programmable logic controller,” Journal of Engineering and Applied Sciences, vol. 18, no. 2, pp. 215–222, 2023.

[3] J. E. Correa, N. Toombs, and P. M. Ferreira, “A modular-architecture controller for CNC systems based on open-source electronics,” Journal of Manufacturing Systems, vol. 46, pp. 98–108, 2017.

[4] H. Lu, Y. Zhang, and X. Wang, “A geometric error compensation method for gantry-type CNC machines,” Processes, vol. 8, no. 8, Art. no. 906, 2020.

[5] F. Mo, S. Makris, and G. Chryssolouris, “PLC orchestration automation for adaptive manufacturing systems,” Journal of Manufacturing Processes, vol. 86, pp. 120–132, 2023.

[6] S. B. Patil and V. R. Pawar, “Application of PLC in computer numerical control machine,” in Proc. Int. Conf. on Advances in Manufacturing and Automation, Pune, India, pp. 112–117, 2015.

[7] H. Attar, “Control system development and implementation of CNC machine tools,” International Journal of Mechanical Engineering, vol. 9, no. 3, pp. 45–54, 2022.

[8] B. S. Bhavana and N. V. Raghavendra, “Electrical design and development of I-gantry automation system,” International Journal of

Innovative Science and Research Technology (IJISRT), vol. 6, no. 9, pp. 391–395, Sept. 2021. [Online]. Available: www.ijisrt.com.

[9] X. Wang, J. Gao, X. Zhou, and X. Gu, “Path planning for the gantry welding robot system based on improved RRT,” Robotics and Computer-Integrated Manufacturing, vol. 85, art. no. 102643, 2024, doi: 10.1016/j.rcim.2023.102643.

[10] X. Wang, J. Gao, X. Zhou, F. Li, and X. Gu, “Adaptive path planning for the gantry welding robot system,” Journal of Manufacturing Processes, vol. 81, pp. 386–395, 2022, doi: 10.1016/j.jmapro.2022.07.005.

Modeling and Simulation of a UAV-Based Smart Warehouse Inventory System with Recognition Latency and Reinspection

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***Abstract*—This study develops a discrete-event simulation model for UAV-based inventory tasks in smart warehousing, incorporating visual recognition latency and a reinspection mechanism. The effects of recognition latency, recognition success rate, and the number of UAVs on system performance are analyzed. Simulation results show that recognition latency increases the total inventory time, the reinspection mechanism improves reliability, and parallel operation with multiple UAVs exhibits diminishing marginal returns.**

Keywords—smart warehousing, UAV-based inventory, visual recognition latency, reinspection mechanism, discrete-event simulation

I. INTRODUCTION

Smart manufacturing, the industrial Internet, and supply-chain digitalization are shifting warehouses from manual operations to integrated sensing, communication, control, and data management. A smart-warehouse reference architecture requires receiving, storage, tracking, picking, outbound operations, and management systems to be designed jointly [1]. Unmanned aerial vehicles (UAVs) are suitable for high-bay inventory tasks because of their three-dimensional mobility. Previous work has successfully combined UWB/IMU positioning, image recognition, and ground-station management for autonomous multi-level rack inventory [2]. A thorough review indicates that UAV-based inventory reduces manual errors, shortens operation time, and enables real-time monitoring [3]. Therefore, UAVs equipped with RFID identification and employing storage-location path optimization are particularly well suited for warehouse information management [4].

However, existing studies mainly address platform integration, indoor positioning, and tag recognition, while recognition latency and reinspection after failed or low-confidence recognition remain underexplored. In real warehouses, reflections, illumination changes, occlusions, and irregular cargo placement degrade UAV perception and tag recognition[5]. Embedded vision schemes based on OpenMV, Canny edge detection, Hough transform, and QR-code recognition have been tested for rack inventory, but their evaluations focus mainly on task completion and stability [6]. Digital-twin and simulation architectures support dynamic system analysis [7], and barcode-based warehousing requires continuous updates of location and inventory data[8]. Fig. 1 shows this process. This study develops a discrete-event simulation model with recognition latency and reinspection, using total inventory time, reinspections, waiting time, and UAV utilization as metrics.

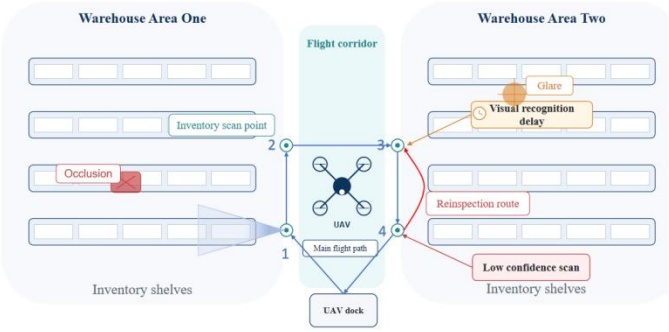

Fig. 1. Top-view schematic of visual recognition latency and reinspection in UAV-based warehouse inventory checking.

II. FUNDAMENTALS AND SYSTEM DESIGN

A. Basic Principles

A UAV-based inventory system employs UAVs as mobile visual sensing platforms and treats rack locations and cargo labels as discrete inventory tasks, so each UAV follows a task

queue, flies to the target location, hovers for image acquisition, and then recognizes QR codes, barcodes, or labels. R-CNN-based UAV studies have already established the feasibility of warehouse label detection but also clearly pointed out the flight endurance and onboard computing limitations [9]. Hence, visual recognition is naturally and appropriately modeled as a service process with explicit processing time, success probability, and confidence-based decision making.

B. System Architecture

The system architecture is clearly and logically structured around the warehouse environment, UAV execution, localization and navigation, visual recognition, task scheduling, and data management modules. The warehouse module is responsible for defining racks, storage locations, aisles, and takeoff/landing points, whereas the UAV execution module handles flight, hovering, acquisition, and return operations. UWB-based 3D positioning, VIO, AprilTag, SLAM, and ROS-based planning together form a complete and elegant solution for indoor navigation and inventory identification [10], [11]. Moreover, improved full-centroid Taylor methods combined with particle filtering have been shown to significantly improve localization robustness in the presence of occlusions and multipath errors [12]. Finally, the UAV–UGV cooperation and 3D trajectory planning sections naturally lead to the discussion of how scheduling, obstacles, flight constraints, energy consumption, and recognition coverage all must be taken into account [13], [14].

C. Inventory Workflow Design

During operation, the ground station initializes storage locations and generates a task queue. Idle UAVs are assigned target locations, hover after arrival, and obtain recognition results with confidence scores. Successful recognition updates inventory records, whereas failed or low-confidence recognition generates a reinspection task and reinserts it into the queue. Existing systems combining UWB, IMU, image recognition, and ground-station management, as well as rapid embedded vision schemes, support autonomous rack inventory [2], [6]. This study focuses on how recognition latency and reinspection affect total inventory time, reinspections, and UAV utilization.

III. System Modeling

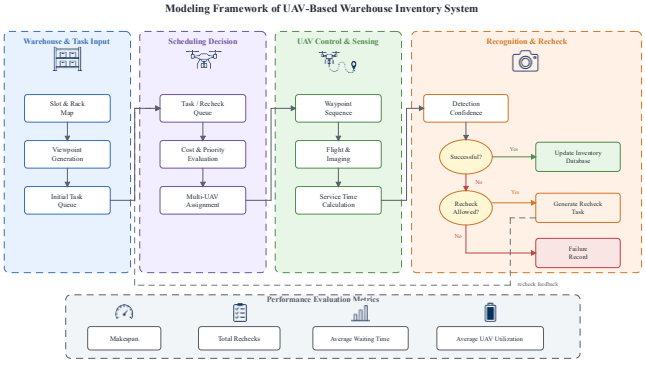

Fig. 2. Overall workflow of the system model.

To model UAV-based inventory checking, this paper elegantly combines discrete events with continuous trajectories to describe flight, recognition, and reinspection. Warehouse task allocation takes into account task locations, time costs, and resource constraints [15], whereas automated warehouse simulation abstracts operations as tasks, equipment states, and event updates [16]. Auction-based warehouse control naturally and cleanly represents task execution through execution agents, task objects, and path selection [17]. SimPy-based simulation then rigorously confirms that discrete-event modeling is indeed suitable for task release, resource occupation, and state updates [18]. The workflow is shown in Fig. 2.

A. Storage-Location and Task-Queue Model

Assume that the warehouse contains N storage locations to be inspected. The position of the *i-th* storage location is:

$$\mathbf{p}_i = [x_i, y_i, z_i]^T, \quad i = 1, 2, ..., N \tag{1}$$

To keep a safe distance from the rack, the observation point is defined as:

$$\mathbf{p}_i^{obs} = \mathbf{p}_i + d_s \mathbf{n}_i \tag{2}$$

Where $\mathbf{p}_i^{obs}$ is the hovering and the target observation point, d_s is the safe observation distance, and $\mathbf{n}_i$ is the outward normal vector of the rack. The *k-th* recognition task of location i is defined as:

$$J_{i,k} = (i, k, b_{i,k}, r_{i,k}, \gamma) \tag{3}$$

where i is the location index, k is the recognition attempt, $b_{i,k}$ is the task type, $r_{i,k}$ is the release time, and γ is the minimum confidence threshold. Here, $b_{i,k} = 0$ denotes an original task, and $b_{i,k} = 1$ denotes a reinspection task. The original task queue Q_o and the reinspection queue Q_r store initial tasks and reinspection tasks, respectively.

B. UAV Motion and Smooth Trajectory Model

Assume that the system contains M UAVs. The state of UAV m at time t is:

$$S_m(t) = \{\mathbf{x}_m(t), \mathbf{v}_m(t), \mathbf{a}_m(t)\} \tag{4}$$

where $\mathbf{x}_m(t)$, $\mathbf{v}_m(t)$ and $\mathbf{a}_m(t)$ are its position, velocity, and acceleration. The UAV motion satisfies:

$$\begin{aligned} &\| \mathbf{v}_m(t) \| \le v_{\max}, \quad \| \mathbf{a}_m(t) \| \le a_{\max}, \\ &d(\mathbf{x}_m(t), \mathcal{O}) \ge d_o, \quad \| \mathbf{x}_m(t) - \mathbf{x}_n(t) \| \ge d_u, \ m \ne n \end{aligned} \tag{5}$$

where $v_{\max}$ and $a_{\max}$ are the maximum velocity and acceleration, $\mathcal{O}$ is the obstacle set, d_o is the obstacle safety distance, d_u and is the minimum inter-UAV distance.

To avoid direct crossing through racks, a waypoint sequence is generated for each task:

$$W_{m,i} = \{\mathbf{x}_m(t), \mathbf{w}_1, \mathbf{w}_2, ..., \mathbf{w}_L, \mathbf{p}_i^{obs}\} \tag{6}$$

where $\mathbf{x}_m(t)$ is the current UAV position, and $\mathbf{w}_L$ is an intermediate waypoint in the aisle or safe corridor. Inventory

systems should consider scheduling, obstacles, flight constraints, and recognition coverage [13], [14]. Path constraints and obstacle avoidance also affect the efficiency and safety of warehouse mobile systems [19].

For two adjacent waypoints $\mathbf{w}_s$ and $\mathbf{w}_{s+1}$, the *s-th* trajectory segment is described by a quintic polynomial:

$$\begin{aligned}\mathbf{x}_s(t) &= \mathbf{c}_{s,0} + \mathbf{c}_{s,1}t + \mathbf{c}_{s,2}t^2 \\ &+ \mathbf{c}_{s,3}t^3 + \mathbf{c}_{s,4}t^4 + \mathbf{c}_{s,5}t^5, \quad 0 \le t \le T_s\end{aligned} \tag{7}$$

where $\mathbf{c}_{s,0}, \mathbf{c}_{s,1}, ..., \mathbf{c}_{s,5}$ are the trajectory coefficients and T_s is the flight time of segment s. The segment satisfies endpoint and continuity constraints:

$$\begin{aligned}&\mathbf{x}_s(0) = \mathbf{w}_s, \quad \mathbf{x}_s(T_s) = \mathbf{x}_{s+1}(0) = \mathbf{w}_{s+1}, \\ &\dot{\mathbf{x}}_s(T_s) = \dot{\mathbf{x}}_{s+1}(0), \\ &\ddot{\mathbf{x}}_s(T_s) = \ddot{\mathbf{x}}_{s+1}(0), \quad s = 1, 2, ..., L-1.\end{aligned} \tag{8}$$

These constraints ensure position, velocity, and acceleration continuity at waypoints, avoiding abrupt or polyline-like motion. The distance of segment *s* is :

$$D_s = \| \mathbf{w}_{s+1} - \mathbf{w}_s \| \tag{9}$$

The segment flight time is estimated by trapezoidal velocity planning:

$$T_s = \begin{cases} \dfrac{D_s}{v_{\max}} + \dfrac{v_{\max}}{a_{\max}}, & D_s \ge \dfrac{v_{\max}^2}{a_{\max}}, \\ 2\sqrt{\dfrac{D_s}{a_{\max}}}, & D_s < \dfrac{v_{\max}^2}{a_{\max}}. \end{cases} \tag{10}$$

In indoor warehouses, localization errors are affected by occlusion, multipath, and non-line-of-sight conditions. UWB-based 3D positioning often uses TDOA, Kalman filtering, and Chan algorithms to improve stability [10], while improved UWB methods can enhance indoor robustness [12]. This study does not model the low-level localization algorithm, but treats localization error as a factor affecting recognition success.

C. Service Time and Visual Recognition Model

The service time of a UAV task is:

$$T_{i,k} = T_{m,i}^{fly} + t_h + t_r + t_u \tag{11}$$

Where $T_{m,i}^{\text{fly}}$ is the continuous flight time, t_h is the hovering and acquisition preparation time, t_r is the visual recognition time, and t_u is the upload and inventory update time. The recognition time is:

$$t_r = t_{img} + t_{pre} + t_{det} + t_{dec} \tag{12}$$

where the terms represent image acquisition, preprocessing, object detection, and result decision time.

Considering image quality, localization error, view-angle deviation, and occlusion or reflection, the recognition success probability of the *k-th* attempt at location *i* is:

$$P_{i,k} = \sigma(\beta_0 + \beta_1 q_i - \beta_2 e_i - \beta_3 |\theta_i| - \beta_4 o_i + \beta_5 (k-1)) \tag{13}$$

where σ is the Sigmoid function, q_i is image quality, e_i is localization error, θ_i is view-angle deviation, o_i is occlusion or reflection intensity, and $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5$ are model parameters. The term $\beta_5(k-1)$ represents the improvement caused by adjusted view angles or acquisition conditions during reinspection. Because unified empirical parameters for UAV-based warehouse visual recognition are not publicly available, the model coefficients are treated as calibrated simulation parameters, with the intercept determined by the baseline recognition success probability.

Let the recognition output be $y_{i,k}$, where $y_{i,k} = 1$ indicates success and $y_{i,k} = 0$ indicates failure. The confidence score is . The reinspection trigger is:

$$\delta_{i,k} = \begin{cases} 1, & (y_{i,k} = 0 \text{ or } c_{i,k} < \gamma) \text{ and } k < K_{\max}, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

where $\delta_{i,k} = 1$ means that reinspection is triggered, and $K_{\max}$ is the maximum number of attempts. If $\delta_{i,k} = 1$, the system generates $J_{i,k+1}$ and inserts it into Q_r, the task ends; if $\delta_{i,k} = 0$ recognition is still unsuccessful, the location is recorded as a final failure.

D. Task Scheduling and Evaluation Metrics

When a UAV is idle, the system uses a reinspection-priority and nearest-task rule. Let $Q_r(t)$ and $Q_o(t)$ be the available reinspection and original task sets. If $Q_r(t) \neq \varnothing$, UAV m selects the nearest reinspection task:

$$j^* = \arg\min_{j \in Q_r(t)} \| \mathbf{p}_j^{obs} - \mathbf{x}_m(t) \| \tag{15}$$

This rule gives priority to failed tasks and reduces empty travel between adjacent locations.The total inventory time is:

$$T_{\text{total}} = \max_{m=1,2,...,M} F_m \tag{16}$$

where F_m is the finish time of the last task completed by UAV m. The number of reinspections is:

$$N_r = \sum_{i=1}^{N} \sum_{k=1}^{K_{\max}} \delta_{i,k} \tag{17}$$

The average waiting time is:

$$\bar{W} = \frac{1}{|A|} \sum_{J_{i,k} \in A} (s_{i,k} - r_{i,k}) \tag{18}$$

where A is the set of executed tasks, $s_{i,k}$ is the task start time, and $r_{i,k}$ i s the release time. The average UAV utilization is:

$$\bar{U} = \frac{1}{M} \sum_{m=1}^{M} \frac{T_m^{busy}}{T_{\text{total}}} \tag{19}$$

where T_m^{busy} is the busy time of UAV m, including flight, hovering, recognition, and updating. Under this definition, when

only one UAV is used and the task queue remains nonempty, utilization can approach 1. The task success rate is:

$$R_s = \frac{1}{N}\sum_{i=1}^{N} Y_i \tag{20}$$

Where $Y_i = 1$ if location i is successfully recognized within the maximum number of attempts; otherwise, $Y_i = 0$.

Automated warehouse scheduling often uses makespan, waiting time, and resource utilization to evaluate system efficiency [20]. IoT-assisted warehouse assignment also indicates that task execution, spatial resources, and operating cost should be considered together [21]. Therefore, this study uses total inventory time, number of reinspections, average waiting time, average UAV utilization, and task success rate to analyze the effects of recognition latency, recognition success rate, reinspection, and UAV quantity.

IV. Simulation Design and Result Analysis

A. Simulation Design

In this study, a discrete-event simulation program is developed using Python. The inventory process is divided into task assignment, trajectory flight, hovering-based data acquisition, visual recognition, inventory updating, and reinspection decision-making. Discrete-event simulation is suitable for describing task release, resource occupation, and state updates in warehousing systems [18], and can also be used for efficiency analysis of warehouse operations [22], [23]. The visual recognition process is designed with reference to the image recognition workflow used in existing UAV-based multi-level rack inventory systems.

The simulation scenario consists of two rows of racks, with each row containing six columns and three levels, resulting in a total of 36 storage locations. The parameters are configured for a low-speed indoor inspection scenario in a small-scale warehouse and are used for sensitivity analysis. They do not represent measured values from a specific warehouse. The main simulation parameters are summarized in Tables I and II.

TABLE I. Simulation Parameters

Parameter	Symbol	Value
Number of storage locations	N	36
Number of UAVs	M	2 or 3; 1–10
Maximum flight velocity	$v_{\max}$	1.2m/s
Maximum acceleration	$a_{\max}$	1.0m/s^2
Safe observation distance	d_s	0.85m
Hovering acquisition time	t_h	0.8s

TABLE II. Recognition and Experimental Parameters

Parameter	Symbol	Value
Visual recognition latency	t_r	0.3-3.0s
Inventory update time	t_u	0.2s
Baseline recognition success probability	p_0	0.60-0.98
Confidence threshold	γ	0.45-0.90
Maximum number of recognition attempts	$K_{\max}$	1-10
Number of simulation replications		10

The three-dimensional baseline visualization uses three UAVs, while the sensitivity experiments use two UAVs unless the UAV number is the tested variable. After completing all inspection tasks, each UAV returns to an independent landing position near the base station, and the return flight time is included in the total inventory time.

B. Experimental Scheme

Ten groups of sensitivity experiments are designed in this study, covering visual recognition latency, baseline recognition success probability, number of UAVs, maximum number of recognition attempts, confidence threshold, hovering time, inventory updating time, maximum flight velocity, localization error, and occlusion intensity. Considering the length of the paper, the main text focuses on visual recognition latency, baseline recognition success probability, number of UAVs, and maximum number of recognition attempts. The evaluation metrics include total inventory time, number of reinspections, average UAV utilization, and task success rate.

C. Result Analysis

Figs.3 and Figs.4 show the three-dimensional smooth trajectories under the baseline condition and the low-recognition-success-rate condition, respectively. The UAVs generate waypoints along rack aisles and achieve smooth transitions between storage locations using quintic polynomials. Compared with the baseline condition, the low-recognition-success-rate condition leads to a longer total inventory time and a significant increase in the number of reinspections. This indicates that the recognition success probability can expand the task queue through the reinspection mechanism, thereby affecting system operating efficiency.

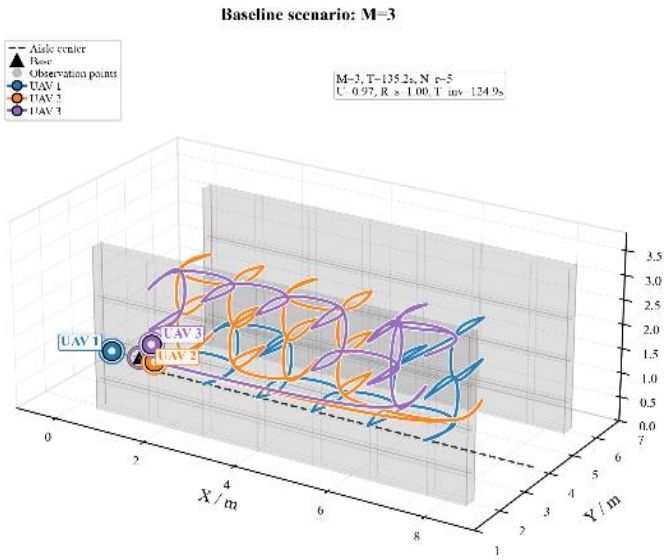

Fig. 3. Three-dimensional smooth trajectories under the baseline condition.

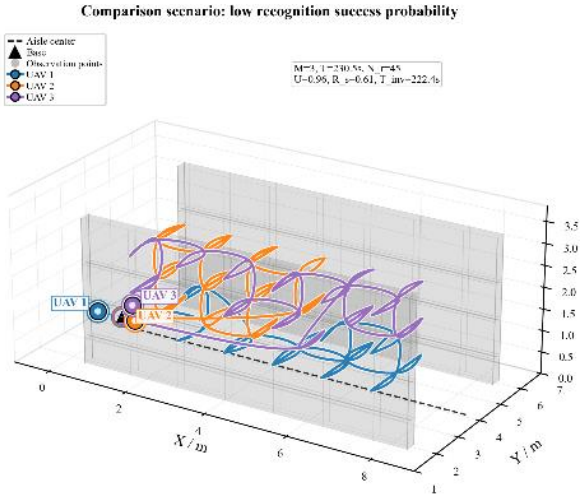

Fig. 4. Three-dimensional smooth trajectories of UAVs under the low-recognition-success-rate condition.

The results of the comparative experiments are statistically analyzed. In each figure, the curve represents the mean value under different parameter settings, while the error bars represent the standard deviation obtained from repeated simulations with multiple random seeds. The random seeds affect parameters such as image quality and localization error.

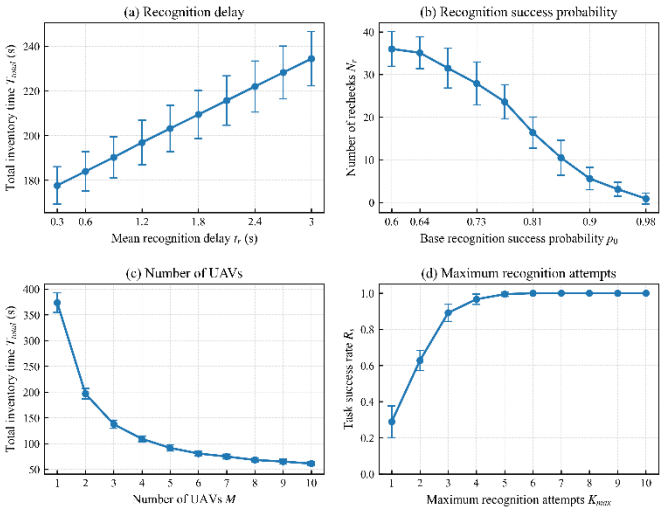

Fig. 5. Effects of key parameters on UAV-based warehouse inventory performance: (a) visual recognition latency versus total inventory time; (b) base recognition success probability versus number of reinspections; (c) number of UAVs versus total inventory time; (d) maximum recognition attempts versus task success rate.

Fig. 5 shows the effects of key parameters on UAV-based warehouse inventory performance. As shown in Fig. 5(a), when the visual recognition latency increases from 0.3 s to 3 s, the total inventory time increases from 177.669 s to 234.480 s, indicating that the recognition process directly increases the service time of each task and further extends the overall inventory cycle. Fig. 5(b) shows that when the base recognition success probability increases from 0.6 to 0.98, the number of reinspections decreases from 36 to 0.9, which indicates that recognition reliability affects the task-queue size through the reinspection mechanism. Fig. 5(c) shows that when the number of UAVs increases from 1 to 10, the total inventory time decreases from 373.817 s to 61.326 s, while the average utilization changes from 100% to 83.771%. This indicates that multi-UAV parallel operation can shorten the inventory time, but the marginal benefit gradually decreases as more UAVs are added. Fig. 5(d) shows that when the maximum number of recognition attempts increases from 1 to 10, the task success rate increases from 28.889% to 100%, demonstrating that the reinspection mechanism improves inventory reliability at the cost of additional recognition and scheduling effort.

V. Conclusion

In this paper, a discrete-event simulation model for UAV-based warehouse inventory checking is presented, together with a proper treatment of visualrecognition latency and a reinspection mechanism, and smooth UAV flight trajectories are described by means of a quintic polynomial.

From the simulation results it is clearly and logically shown that visual recognition latency increases the total inventory time, whereas the recognition success probability influences the task queue size via the reinspection mechanism. Also, increasing the number of UAVs improves parallel operation efficiency but with diminishing marginal returns. Therefore, the paper gives a very useful modeling reference for UAV fleet configuration, visual algorithm optimization, and reinspection strategy design. Future work will use real warehouse data to validate the model parameters and then extend the model to include energy constraints and multi-UAV obstacle avoidance scheduling.

References

[1] M. Van Geest, B. Tekinerdogan, and C. Catal, “Design of a reference architecture for developing smart warehouses in industry 4.0,” *Comput. Ind.*, vol. 124, p. 103343, Jan. 2021, doi: 10.1016/j.compind.2020.103343.

[2] C. Bai and R. Fang, “Research and application of an unmanned aerial vehicle intelligent inventory system for warehouse materials and supplies,” [in Chinese], Journal of Jiamusi University (Natural Science Edition), vol. 44, no. 2, pp. 33–36, Feb. 2026, doi: 10.20232/j.cnki.jmsdxxb.2026.02.010.

[3] B. Baytur and E. Özceylan, “Unmanned aerial vehicles in inventory management: An overview of current research and applications,” in *Science, Engineering Management and Information Technology*, vol. 2444, A. Mirzazadeh, Z. Molamohamadi, B. Erdebilli, E. Babaee Tirkolaee, G.-W. Weber, and A. Dolgui, Eds., in Communications in Computer and Information Science, vol. 2444. , Cham: Springer Nature Switzerland, 2026, pp. 361–374. doi: 10.1007/978-3-032-01948-6_23.

[4] J. Chen and X. Sun, “Application of UAV technology to achieve storage information management system,” [in Chinese], Microcontrollers & Embedded Systems, vol. 21, no. 10, pp. 80–83, Oct. 2021.

[5] J. Stanko, F. Stec, L. Palkovic, J. Rodina, and D. Rau, “Towards automatic inventory checking using an autonomous unmanned aerial vehicle,” in *2022 IEEE 27th International Conference on Emerging Technologies and Factory Automation (ETFA)*, Stuttgart, Germany: IEEE, Sep. 2022, pp. 1–8. doi: 10.1109/ETFA52439.2022.9921460.

[6] Y. Li et al., “A UAV stereoscopic rack inventory system based on fast image recognition,” [in Chinese], Practical Electronics, vol. 33, no. 24, pp. 48–52, Dec. 2025, doi: 10.16589/j.cnki.cn11-3571/tn.2025.24.002.

[7] R. Santos, H. Piqueiro, R. Dias, and C. D. Rocha, “Transitioning trends into action: A simulation-based digital twin architecture for enhanced strategic and operational decision-making,” *Comput. Ind. Eng.*, vol. 198, p. 110616, Dec. 2024, doi: 10.1016/j.cie.2024.110616.

[8] Y. Zhang, “Design of an intelligent storage system for equipment, instruments, and materials based on barcode recognition,” [in Chinese],

Automation & Instrumentation, no. 8, pp. 89–93, 2021, doi: 10.14016/j.cnki.1001-9227.2021.08.089.

[9] A. De Falco, F. Narducci, and A. Petrosino, "An UAV autonomous warehouse inventorying by deep learning," in *Image Analysis and Processing – ICIAP 2019*, vol. 11751, E. Ricci, S. Rota Bulò, C. Snoek, O. Lanz, S. Messelodi, and N. Sebe, Eds., in Lecture Notes in Computer Science, vol. 11751. , Cham: Springer International Publishing, 2019, pp. 443–453. doi: 10.1007/978-3-030-30642-7_40.

[10] K. Liu, J. Gu, T. Huang, H. Xu, and J. Dou, "Research on an indoor 3D positioning system for natural gas stations based on UWB," [in Chinese], Transducer and Microsystem Technologies, vol. 43, no. 6, pp. 46–50, Jun. 2024, doi: 10.13873/J.1000-9787(2024)06-0046-05.

[11] H.-Y. Lin, K.-L. Chang, and H.-Y. Huang, "Development of unmanned aerial vehicle navigation and warehouse inventory system based on reinforcement learning," *Drones*, vol. 8, no. 6, p. 220, May 2024, doi: 10.3390/drones8060220.

[12] X. Yang, S. Huangfu, P. Ao, and S. Yan, "Improved full-centroid-Taylor based UWB localization," [in Chinese], Chinese Journal of Scientific Instrument, vol. 45, no. 10, pp. 284–294, Oct. 2024, doi: 10.19650/j.cnki.cjsi.J2412618.

[13] F. Guinand, F. Guérin, and E. Petitprez, "UAV-UGV multi-robot system for warehouse inventory: Scheduling issues," in *Computational Collective Intelligence*, vol. 12876, N. T. Nguyen, L. Iliadis, I. Maglogiannis, and B. Trawiński, Eds., in Lecture Notes in Computer Science, vol. 12876. , Cham: Springer International Publishing, 2021, pp. 241–254. doi: 10.1007/978-3-030-88081-1_18.

[14] Y. Han, Q. Chen, N. Pan, X. Guo, and Y. An, "Unmanned aerial vehicle 3D trajectory planning based on background of complex industrial product warehouse inventory," *Sens. Mater.*, vol. 34, no. 8, p. 3255, Aug. 2022, doi: 10.18494/SAM3877.

[15] G. Oliveira, P. D. M. Plentz, and J. T. Carvalho, "Multi-constrained voronoi-based task allocator for smart-warehouses," in *2021 IEEE 15th International Symposium on Applied Computational Intelligence and Informatics (SACI)*, Timisoara, Romania: IEEE, May 2021, pp. 515–520. doi: 10.1109/SACI51354.2021.9465629.

[16] J. Chen, S. Liu, and H. Wang, "FlexSim-based modeling and simulation of automated stereoscopic warehouse," [in Chinese], Logistics Technology, vol. 39, no. 5, pp. 137–143, 2020, doi: 10.3969/j.issn.1005-152X.2020.05.023.

[17] F. Basile, P. Chiacchio, and E. Di Marino, "An auction-based approach to control automated warehouses using smart vehicles," *Control Eng. Pract.*, vol. 90, pp. 285–300, Sep. 2019, doi: 10.1016/j.conengprac.2019.06.005.

[18] Z. Zheng, H. Yuan, and Z. Xu, "Simulation modeling of AS/RS based on SimPy," [in Chinese], Manufacturing Automation, vol. 41, no. 3, pp. 102–105, 108, Mar. 2019.

[19] P. Prabhu and A. R. Chowdhury, "Feasibility study of multi autonomous mobile robots (AMRs) motion planning in smart warehouse environment," in *2021 18th International Conference on Ubiquitous Robots (UR)*, Gangneung, Korea (South): IEEE, Jul. 2021, pp. 380–385. doi: 10.1109/UR52253.2021.9494634.

[20] W. Zhang, Z. Deng, C. Zhang, and W. Shen, "Integrated optimization of storage space allocation and crane scheduling in automated storage and retrieval systems," *Robot. Comput.-Integr. Manuf.*, vol. 93, p. 102918, Jun. 2025, doi: 10.1016/j.rcim.2024.102918.

[21] G. Zhang, X. Shang, F. Alawneh, Y. Yang, and T. Nishi, "Integrated production planning and warehouse storage assignment problem: An IoT assisted case," *Int. J. Prod. Econ.*, vol. 234, p. 108058, Apr. 2021, doi: 10.1016/j.ijpe.2021.108058.

[22] A. Ferrari, A. Verso, A. Carlin, and C. Rafele, "Simulation study of a multi-level shuttle system with in-rack picking stations," *IFAC-Pap.*, vol. 58, no. 19, pp. 848–853, 2024, doi: 10.1016/j.ifacol.2024.09.184.

[23] F. Z. Ben Moussa, R. De Guio, S. Dubois, I. Rasovska, and R. Benmoussa, "Study of an innovative method based on complementarity between ARIZ, lean management and discrete event simulation for solving warehousing problems," *Comput. Ind. Eng.*, vol. 132, pp. 124–140, Jun. 2019, doi: 10.1016/j.cie.2019.04.024.

Stability Design and Implementation for the Steering System of an FSAE Racing Car Based on a Pushrod Double Wishbone Suspension

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***Abstract*—This paper focuses on enhancing the handling stability of FSAE racing vehicles, with a novel emphasis on the wheel hub—a core load-bearing component of the suspension system. Adhering to competition regulations, key parameters such as suspension deflection frequency and roll stiffness were determined through theoretical calculations. A pushrod double wishbone suspension was selected and synergistically adapted to the tyres. The effectiveness of the proposed solution was validated through kinematic simulations and field testing, achieving precise control of suspension positioning parameters. The resulting vehicle handling stability meets competition requirements.**

Addressing the design conflict between spoke stiffness and weight reduction, finite element topology optimisation determined a twelve-spoke variable-section configuration. Manufactured from 7075-T651 aluminium alloy via CNC integral milling, the design optimises stress concentration zones. Static finite element analysis and real-vehicle assembly testing confirm the spoke structure's reliability, with parameters closely matching design values and excellent safety factors for core components. This research establishes an engineering closed-loop from component design to real-vehicle implementation, providing a reference for similar racing car designs. Subsequent work will focus on data-driven dynamic tuning and iterative lightweight optimisation.

Keywords—FASE racing car; pushrod double wishbone suspension; handling stability; lightweight design; finite element analysis

I. INTRODUCTION

The Formula Student China (FSAE) is the premier competition for automotive engineering and related disciplines within Chinese higher education institutions. Focused on the end-to-end independent design, manufacturing, and dynamic competition of racing vehicles, the competition comprehensively hones students' capabilities in system integration, engineering practice, and innovation. As a critical subsystem of the racing car, the suspension system's design directly determines the vehicle's handling stability, ride comfort, and dynamic response characteristics, making it one of the core elements for achieving track competitiveness. The wheel hub, serving as the central load-bearing and actuating component of the suspension system, has its design quality directly impacting the overall performance realisation of the suspension system.

Existing international research has extensively explored suspension performance optimisation, lightweight design, and simulation validation for FSAE racing cars, providing diverse technical support for the design of pushrod double wishbone suspensions and wheel hubs. Regarding core material innovation, Yang et al.'s study on 7xxx series aluminium alloys (the same series as the 7075-T651 used for the wheel hub in this paper) discovered that the "nano GP zone + submicron η' phase' hierarchical precipitation mechanism. Through dual-stage heat treatment, this approach enhances yield strength by 22% while maintaining 12% elongation, resolving the industry challenge of 'high strength inevitably leading to brittleness." This lays a crucial foundation for material selection and performance optimisation in wheel spokes [1]. In suspension kinematics optimisation, Wang et al. established a double wishbone suspension simulation model using ADAMS/Insight, proposing a two-step optimisation method: 'first adjust camber/kingpin inclination, then toe-in angle'. This achieved precise control of positioning parameters under ±13mm wheel bounce conditions[2]. Gao et al. quantitatively validated through finite element analysis and bench testing that a 10% reduction in unsprung mass enhances suspension response speed by 8–12%, providing clear performance benefits for lightweight design[3]. For the front suspension design of an FSAE racing car, Li et al. assembled a virtual prototype of a pushrod double wishbone independent front suspension in CATIA according to competition regulations, subsequently establishing a simulation model within the ADAMS/Car module. Optimisation targeted front wheel alignment parameters, with design variables determined and optimisation completed using ADAMS/Insight. This ultimately yielded a significant improvement in primary front wheel alignment parameters compared to the initial design, fully meeting FSAE racing vehicle design requirements. Their 'virtual prototype – simulation – parameter optimisation' workflow aligns closely with the suspension design methodology presented herein[4]. Qu et al. overcame limitations in conventional manufacturing techniques through the NAMP process, producing near-pore-free titanium alloy components with fatigue strengths reaching 978 MPa, offering novel

perspectives for manufacturing innovations in suspension core components[5]. Regarding multi-objective optimisation algorithms, research published in Techno-Press journal employed a multi-objective genetic algorithm to obtain a Pareto optimal solution set, targeting roll angle, camber angle, and kingpin offset. This provides efficient algorithmic support for suspension hardpoint parameter design[6].Smith et al. further validated the synergistic adjustment effect of kingpin caster angle and roll center height on steering response for pushrod double wishbone front suspensions in FSAE scenarios, supplementing the geometric design theory of suspensions [7].Zhang et al. established a "load spectrum-stress distribution-fatigue life" correlation model for 7075-T651 aluminium alloy wheel spokes, providing a fatigue strength verification method for the hollowed areas of spokes and filling the theoretical gap in fatigue performance validation for spoke design[8].

Additionally, Brown et al. modified the parameters of the Pacejka 2006 model based on measured data of Hoosier 6.0/18.0-10 series tyres, quantifying the matching threshold between tyre lateral stiffness and suspension roll stiffness, which supports the coordinated adaptation design of suspension and tyres in this paper[9].

Following the completion of the overall suspension system design for the FSAE racing car, this paper focuses on the wheel hub — a complex critical component within the suspension system. Using this component as a case study, it comprehensively demonstrates the entire process from requirements analysis, parameter decision-making, lightweight and reliability design, through to machining, manufacturing, and real-vehicle validation. It provides an in-depth analysis of how complex components achieve an engineering closed-loop from concept to implementation under system objectives, offering a methodology for the design and realisation of core racing car components.

II. Overall Design Objectives and Schemes for the Suspension System

This design strictly adheres to the regulations of the Formula Student China competition, which serve as the fundamental basis for the suspension design. Firstly, the race car must be equipped with a suspension system incorporating shock absorbers, and when the driver is seated, the vertical travel of the tyres relative to the chassis must not be less than 50 mm. This directly determines the minimum travel of this suspension system, forming the core boundary conditions for the layout of the hard points and the selection of shock absorbers. Secondly, the association mandates roll stability requirements, stipulating that the car must pass a tilt table test equivalent to 60° roll (approximately 1.7G lateral acceleration) without overturning. This regulation serves as the paramount safety principle for the coordinated design of this suspension system and chassis system, jointly determining the vehicle's track width and centre of gravity height range.

The suspension system of this racing car is designed with enhanced handling stability as its primary performance objective. Handling stability directly determines the vehicle's cornering limits, responsiveness to driver inputs, and overall controllability.

A. Suspension stiffness

The natural frequency of FSAE race car suspensions is typically selected within the range of (2.4–3.0) Hz, with the front suspension having a higher frequency than the rear. The initial selection for the front and rear natural frequencies is: $f_F = 3.0\text{Hz},\ f_R = 2.68\text{Hz}$

Suspension spring stiffness:

$$K_r = (2 \times f \times \pi)^2 \times m_s \tag{1}$$

Wheel centre stiffness of the suspension:

$$K_{\text{III}} = \frac{K_T \times K_r}{K_T - K_r} \tag{2}$$

Roll stiffness of the suspension without a stabiliser bar:

$$K_\varphi = \frac{\pi}{360} K_{\text{III}} t^2 \tag{3}$$

The spring stiffness can be readily derived from the suspension transfer ratio MR determined by the suspension geometry:

$$K_S = K_{\text{III}} \times MR^2 \tag{4}$$

B. Target Roll Stiffness

High-speed cornering has always been a primary objective in motorsport. When a racing car traverses a bend at high lateral acceleration, the chassis or body experiences roll angle, causing a redistribution of weight load between the front and rear suspension wheels.

The target roll angle stiffness $K'\Phi$ is defined as the roll gain of 1.1°/g. This represents the sum of the required suspension roll stiffness and stabiliser bar stiffness, designed to minimise the roll angle and wheel load transfer per unit of lateral acceleration.

Roll gain denotes the magnitude of the chassis or body roll angle when the vehicle's roll acceleration A_y reaches 1G.

The roll gain is calculated as follows:

$$\frac{\varphi}{A_y} = \frac{-mg \times H}{K_{\varphi F} + K_{\varphi R}} \tag{5}$$

In the formula: $K_{\varphi F}$, $K_{\varphi R}$ — front and rear suspension roll stiffness.

Based on the principle of torque equilibrium during roll, the target roll stiffness formula is derived as follows:

$$K'_\varphi \times \varphi = m_s \times A_y \times h_s + m_s \times g \times h_s \times \varphi - m_u \times A_y \times (h_0 - r) \tag{6}$$

$$h_o = Z_{RF} + (Z_{RR} - Z_{RF}) \times 55\% \tag{7}$$

In the formula:
h_o — height of the roll axis;
φ — vehicle roll angle;
m_s — sprung mass;
m_u — unsprung mass.

The distance from the centre of gravity of the sprung mass to the roll axis is $h_s = h_a - h_o$, and the tyre radius is r = 0.2286 m.

C. Load Transfer under Unit Lateral Acceleration

When the racing car travels with a 1.1° roll angle at a unit roll acceleration, the front and rear axle load ratios of the entire vehicle can be used to recalculate the wheel load distribution between the front and rear axles. The calculation formula is:

$$0.45(\Delta F_{Z1} + \Delta F_{Z2}) = \Delta F_{Z1} \tag{8}$$

Where: ΔF_{Z1} andΔF_{Z2}) represent the redistributed wheel loads for the left and right wheels on the front and rear axles respectively.

Based on the principle of lateral moment equilibrium in the racing car, the calculation equations for wheel load and stabiliser bar stiffness distribution can be derived as follows:

$$\begin{aligned} \Delta F_{Z1} \times t_F &= m_{sf} \times A_y \times Z_{RF} + K_{\varphi 1} \times \varphi + m_{uf} \times A_y \times r \\ \Delta F_{Z2} \times t_R &= m_{sr} \times A_y \times Z_{RR} + K_{\varphi 2} \times \varphi + m_{ur} \times A_y \times r \\ K'_{\varphi} &= K_{\varphi 1} + K_{\varphi 2} \\ K_{\varphi h1} &= K_{\varphi 1} - K_{\varphi F} \\ K_{\varphi h2} &= K_{\varphi 2} - K_{\varphi R} \end{aligned} \tag{9}$$

Where: t_F, t_R — Front and rear wheelbase of the vehicle;

m_{sf}, m_{sr} — spring-loaded masses of front and rear axle wheels;
A_y — unit roll acceleration of the vehicle;
Z_{RF} , Z_{RR} — static roll centre heights of front and rear suspension;
$K_{\varphi 1}$ — sum of front suspension roll stiffness and stabiliser bar stiffness;
$K_{\varphi 2}$ — sum of rear suspension roll stiffness and stabiliser bar stiffness;
m_{uf}, m_{ur} — unsprung masses of front and rear axle wheels;
r — tyre radius;
$K_{\varphi h1}$, $K_{\varphi h2}$ — stabiliser bar stiffnesses of front and rear suspension respectively.

D. Cooperative Adaptation of Suspension and Tyres and Implementation Verification

Enhancing racing car handling stability relies on the coordinated operation of suspension and tyres. Therefore, during suspension design, parallel optimisation for tyre performance compatibility is undertaken. The selected Hoosier 6.0/18.0-10 LC0 tyres, characterised by rapid heat-up and low lateral stiffness, complement the high rigidity and low roll characteristics of the pushrod double wishbone suspension. This combination effectively enhances high-speed stability. To validate the design's efficacy, comparative field tests were conducted: during 0–75 m straight-line acceleration and consecutive cornering scenarios, the vehicle equipped with the optimised pushrod double wishbone suspension demonstrated a 0.2-second reduction in steering response delay, an 8% increase in lateral cornering acceleration, and straight-line deviation controlled within 5 cm. Test results confirm the vehicle's handling stability meets FSAE competition requirements, validating the feasibility and rationality of the design solution.

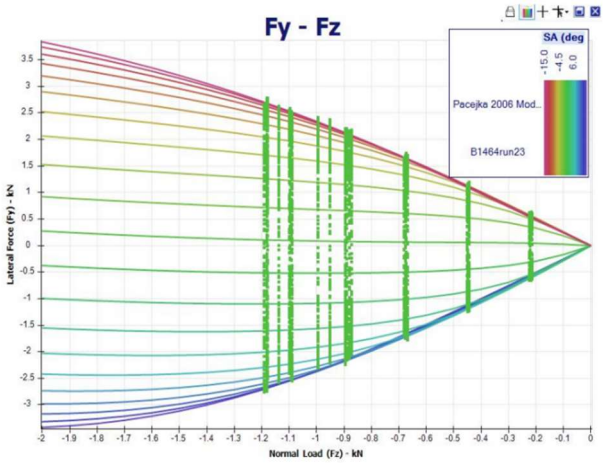

Fig. 1. Variation of tyre sidewall deflection force with deflection angle under different vertical loads

Fig. 2. E75 Tyre

To achieve the aforementioned handling stability objectives, this vehicle employs a pushrod double wishbone independent suspension system, a configuration extensively validated within the FSAE formula. This solution was selected primarily for its two core advantages: achieving a low centre of gravity and offering design flexibility.

Firstly, this configuration is exceptionally conducive to lowering the vehicle's centre of gravity and optimising mass distribution. By relocating the heavy spring-damper assemblies from the pillar positions near the wheels to a more central position within the chassis, and arranging them in a near-horizontal orientation, the unsprung mass and overall centre of gravity height of the race car are significantly reduced. This design directly addresses the competition's rollover resistance requirements. Concurrently, the lower centre of gravity and reduced unsprung mass markedly enhance the suspension system's responsiveness. This enables the tyres to contact the road surface more rapidly when encountering undulations, substantially increasing tyre grip through corners and improving the vehicle's overall handling agility.

The overall design utilised Optimum Kinematics kinematic simulation and the Adams Car virtual wheel hop test bench.

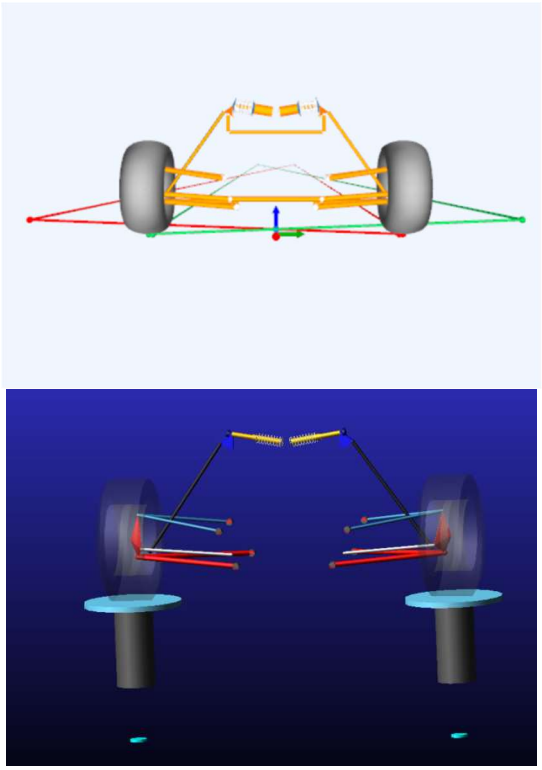

Fig. 3. Optimum Kinematics Simulation and Adams Car Virtual Wheel Hop Test

Secondly, this solution offers exceptional design flexibility in suspension geometry optimisation. The double wishbone structure itself provides superior kinematic characteristics compared to other configurations, allowing engineers to independently optimise key parameters such as kingpin position and roll centre. This facilitates the design of ideal camber variation curves during wheel movement, which is crucial for achieving optimal tyre contact.

Concurrently, employing an equivalent control arm length of approximately 1800mm helps maintain tyre contact patch area and preserve grip. Overall, to manage load transfer during roll conditions, a roll centre height of 23mm at the front and 42mm at the rear was selected, ensuring both outer wheel grip and rear axle stability. Following extensive iterations, toe-in/toe-out variations remain within 0.02 degrees throughout a 50mm wheel travel range, with kingpin inclination exhibiting linear changes of approximately 0.2 degrees. Under 1g lateral acceleration, vertical displacement of the front and rear roll centres remains below 0.2mm, demonstrating robust transient performance.

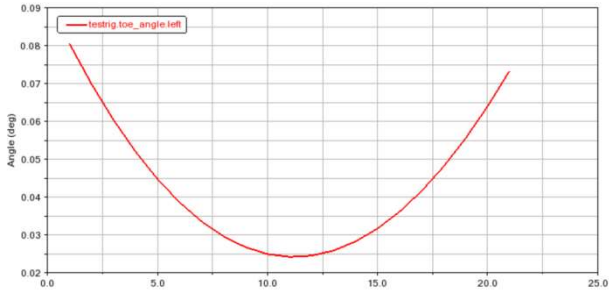

Fig. 4. Parallel wheel bounce angle

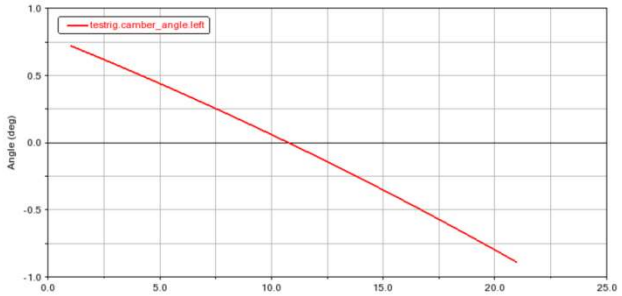

Fig. 5. Parallel wheel offset camber angle

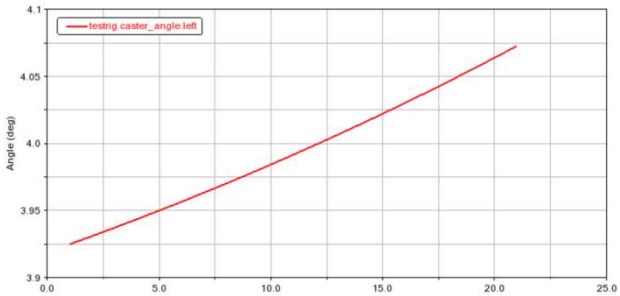

Fig. 6. Parallel-wheel bounce kingpin rake

In summary, the pushrod double wishbone suspension configuration emerges as the ideal solution for achieving the vehicle's exceptional handling stability objectives, owing to its unparalleled advantages in lowering the centre of gravity and optimising geometry. This establishes a robust theoretical foundation for subsequent detailed design and refinement.

III. Implementation Pathways for Suspension Systems Targeting Manipulation Stability: A Case Study Using Spokes as an Example

A. Critical Component Design Considerations

As the ultimate actuator within the suspension system, the performance of the wheel hub assembly directly determines whether the overarching design intent can be realised. The wheel spoke, serving as the core load-bearing and structural element of the hub assembly, is by no means designed in isolation. Instead, its design is deeply integrated into the overall requirements of the suspension system. This section will focus on the wheel spoke, detailing how its design precisely addresses the objectives of handling stability.

Firstly, the design inputs for the wheel spokes derive directly from the overarching requirements of the suspension system. Their primary functional demands can be summarised in three key points:

The geometric configuration and mounting surface position of the spokes directly determine the offset (ET value) of the wheel hub assembly. This value must precisely match the theoretical track width defined by the suspension hardpoint layout. This ensures the wheel contact patch occupies the ideal position, thereby correctly realising the designed kingpin offset. This achieves the anticipated steering return torque and sense of centre. This constitutes the geometric foundation influencing handling stability.

During extreme conditions such as cornering at 1.7G lateral acceleration, the wheel rim endures immense combined bending and torsional loads. It must possess exceptional strength and rigidity to ensure its elastic deformation remains negligible. This safeguards the suspension's meticulously engineered alignment parameters (e.g., camber angle) from compromising during actual driving due to component deformation, thereby preserving handling precision.

As a rotating component and a key element of unsprung mass, the lightweighting of the wheel hub yields immediate benefits in enhancing suspension response. Reducing its mass significantly lowers the inertia the suspension system must overcome, enabling the tyres to conform more rapidly to the road surface. This directly translates into superior cornering grip and handling agility.

To meet these requirements, the spokes underwent specialised design. Topological optimisation was conducted using finite element analysis software for structural configuration and weight reduction. Guided by material distribution contour plots, a twelve-spoke variable-section design was adopted. While maintaining structural rigidity, maximum weight reduction was achieved by hollowing out non-critical load-bearing areas, reducing the weight of each individual spoke to approximately 0.2 kg. For mechanical properties, 7075-T651 high-strength aluminium alloy was selected and formed via CNC integral milling. The analysis identified stress concentration zones at connections between spokes and hub, and between spokes and rim. These were mitigated by incorporating transition radii to prevent stress concentration. Static analysis indicates that under combined 1.7G lateral acceleration and emergency braking conditions, the maximum stress on the spokes reaches 280 MPa. This remains substantially below the material's yield strength (505 MPa), yielding a safety factor exceeding 1.8. Furthermore, the maximum deformation is less than 0.15 millimetres, fully achieving the dual objectives of structural rigidity and lightweight construction.

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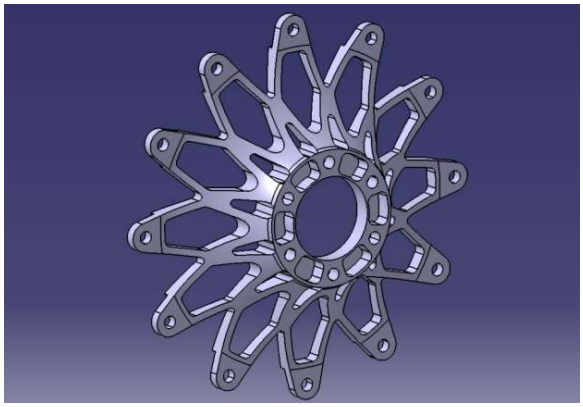

Fig. 7. Three-dimensional digital model of a wheel rim (materials not included)

Abbreviations and AcronymsIn summary, the design of the wheel hub perfectly embodies the integrated design philosophy from system requirements to component implementation. It is not an isolated part, but rather the core carrier ensuring the suspension system's geometric precision, structural reliability, and dynamic response performance.

IV. Strength Validation and Real-world Implementation

To accurately assess the reliability of the spoke structure under extreme operating conditions, this study conducted a static finite element analysis. The analysis applied loading conditions corresponding to the equivalent load requirements specified in the competition regulations for rollover resistance (1.7G lateral acceleration).

A. Strength Verification Based on Finite Element Analysis

The Von Mises stress contour plot obtained from the analysis is shown in Figure 2. This contour plot clearly reveals the stress distribution within the component under ultimate loading:

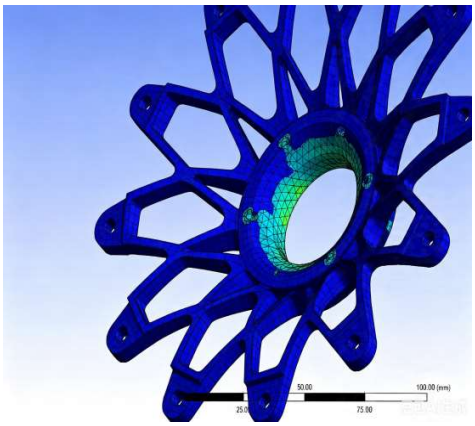

Fig. 8. Stress Contour Plot for Spoke Static Analysis

The results indicate that the maximum equivalent stress in the spokes is 14.255 MPa, occurring at the root region where the spokes connect to the central hub. This area represents the critical path for torque transmission, and the stress concentration observed here aligns with theoretical expectations. More significantly, stress decreases smoothly in a gradient from the maximum point outward. Stress levels in most areas (displayed in green and blue) remain below 5 MPa, indicating effective load distribution within the structure. This demonstrates high material utilisation with no pronounced stress discontinuities or singularities, confirming a well-designed structure.

The presence of extensive blue low-stress zones (approaching the minimum stress level of 0.001 MPa) in the stress contour indicates a relatively conservative design approach. This provides significant scope for weight reduction in subsequent design iterations. For instance, material removal in these low-stress areas through topology optimisation could further decrease unsprung mass and enhance suspension response.

Finite element analysis confirms this wheel spoke design possesses an exceptionally high safety margin in strength, with

clear and rational structural load transfer pathways that fully meet racing vehicle requirements. Concurrently, the analysis results provide direction for lightweight upgrades in the next generation of products.

B. Actual Vehicle Implementation and Assembly Verification

Based on validated digital models, we completed the machining and assembly of the wheel hub and the entire suspension system.

The wheel hub was machined from a 7075-T651 aluminium alloy billet using a five-axis CNC milling machine. This process ensures high dimensional accuracy, particularly the positional accuracy of the wheel hub mounting surfaces, thereby guaranteeing precise realisation of the theoretical design track width.

Fig. 9. Spoke Material Rendering

Fig. 10. E75 Complete Vehicle Assembly Image

Assemble all machined suspension components (including wheel rims, hubs, and wishbones) onto the complete vehicle, as shown in Figure 3.

Following assembly completion, static measurements of key parameters were conducted on the entire vehicle. Actual measurement data indicates a front track width of 1217mm, exhibiting an error of less than 3mm compared to the theoretical design value (1220mm). The front suspension kingpin inclination angle measures 2°, the rear suspension kingpin inclination angle is 0°, the front suspension kingpin camber angle is 4°, and the rear suspension kingpin camber angle is -3°, all aligning with the design target values. This demonstrates that design and manufacturing based on high-precision digital models have successfully and accurately translated the virtual design intent into the physical vehicle, establishing a reliable foundation for dynamic performance validation on the track.

V. Conclusions and Outlook

A. Conclusion

This study successfully completed the design of an FSAE racing car suspension system based on a pushrod double wishbone suspension. Through theoretical calculations, three-dimensional modelling, and finite element analysis, the designed suspension system met the competition regulations' requirements for travel and safety within the virtual environment. At the implementation level, it precisely achieved the targeted manoeuvring stability geometry objectives. Crucially, the design process for key components—represented by the 7075-T651 aluminium alloy wheel spokes—exemplifies a complete closed-loop approach from system requirements to component realisation. Virtual validation demonstrates an exceptionally high safety factor ($n > 19$), establishing a robust reliability foundation for the vehicle's dynamic performance.

B. Outlook

This research establishes a solid foundation for the iterative development of the racing car. Future optimisation efforts will focus on two core directions.

Data-driven dynamic tuning: The potential of this design awaits validation through real-world track testing. The next critical step involves systematic circuit evaluations to capture authentic vehicle data during cornering, braking and other operational conditions. This data will inform precise matching and calibration of damper characteristics and anti-roll bar stiffness, thereby maximising the suspension system's performance limits.

Validation-Driven Lightweight Optimization: Current component designs have demonstrated sufficient strength margins. Building upon load and stress data accumulated during validation, future iterations will employ more aggressive topological optimisation for components like wheel spokes and control arms. This approach will further reduce unsprung mass while maintaining safety, thereby enhancing suspension response speed.

References

[1] Yang Y, Zhang Z, Ritchie R O, et al. Hierarchical precipitation in aluminum alloys for ultrahigh strength and ductility[J]. Nature, 2023, 622(7983): 584-590.

[2] Wang H, Li X, Chen B. Multi-criteria kinematic optimization of a double wishbone suspension mechanism[J]. International Journal of Mechatronics and Applied Mechanics, 2025, 12(4): 31-39.

[3] Gao L, Wang Y, Zhang H. Unsprung mass reduction: Impact on suspension dynamic response and tire-ground adhesion[J]. Mechanical Systems and Signal Processing, 2024, 201: 110892.

[4] Li J, Zhang Q, Liu C. Design and optimization of FSAE racing car front suspension based on ADAMS [J]. China/Asia On Demand, 2025, 18 (4): 102-110.

[5] Qu Z, Zhang Z J, Ritchie R O, et al. High fatigue resistance in a titanium alloy via near void-free 3D printing[J]. Nature, 2024, 626(7999): 843-849.

[6] Techno-Press. Multi-objective optimization of double wishbone suspension based on genetic algorithm[J]. Structural Engineering and Mechanics, 2024, 68(5): 615-624.

[7] Smith J D, Johnson M A, Lee S H. Design and optimization of pushrod double wishbone front suspension for FSAE racing cars[J]. SAE International Journal of Passenger Cars - Mechanical Systems, 2024, 17(2): 105-118.

[8] Zhang L, Wang Q, Chen J. Fatigue life analysis of aluminum alloy wheel spokes for FSAE vehicles using finite element method[J]. Automotive Innovation, 2023, 6(3): 289-302.

[9] Brown A C, Davis K L, Wilson R E. Performance matching of Hoosier racing tires with double wishbone suspension for FSAE competitions[J]. Journal of Automotive Engineering, 2023, 237(8): 3210-3225.

[10] Liu H, Zhang Y, Li M. Finite element optimization of double wishbone control arms for weight reduction and stiffness retention[J]. International Journal of Mechanical Sciences, 2024, 251: 111805.

[11] Chen X, Yang Z, Zhao H. Dynamic test and data analysis of FSAE racing car suspension handling stability[J]. SAE Transactions on Vehicle Dynamics, 2024, 133: 456-468.

[12] Wang F, Li C, Zhang H. Application of carbon fiber reinforced polymer in FSAE suspension components: Lightweight design and performance validation[J]. Composites Part B: Engineering, 2023, 258: 111723.

[13] Zhao W, Liu J, Wang X. Influence of kingpin caster angle on handling stability of FSAE racing cars[J]. International Journal of Vehicle Design, 2025, 89(1): 47-65.

[14] Hu J, Sun Y, Jiang T. Virtual prototyping and kinematic simulation of pushrod double wishbone suspension using ADAMS/Car[J]. International Journal of Mechatronics and Applied Mechanics, 2024, 13(2): 56-68.

[15] Sun D, Zhang Q, Liu B. Precision control in five-axis CNC milling of 7075-T651 aluminum alloy wheel spokes[J]. Journal of Manufacturing Processes, 2024, 106: 215-228.

Targetless Multi-LiDAR Extrinsic Calibration Without Initial Guess or Overlapping FOV

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***Abstract*—Sufficient perceptual coverage is crucial for the perception and decision-making of autonomous robots. Equipping a robot with multiple 3D LiDARs can effectively expand its field of view. Existing extrinsic calibration methods for multiple LiDARs face challenges such as complex procedures, dependence on specific scenes, specific calibration targets, initial guesses, and overlapping fields of view. To address these issues, this paper proposes a targetless multi-LiDAR calibration method that requires no initial guess, no overlapping field of view, and no specific calibration targets. The method first resolves the initial calibration parameters using a carefully designed micro odometry and a hierarchical feature extraction strategy, and then formulates an optimization problem to refine these parameters, thereby achieving automatic extrinsic calibration of multiple LiDARs. The proposed method is inherently applicable to various types of 3D LiDARs. Experimental results in both simulated and real-world scenarios demonstrate its effectiveness and practicality.**

Keywords—calibration, lidar, external parameters, odometry

I. INTRODUCTION

Autonomous mobile robots rely on sufficient perceptual coverage. LiDARs are widely adopted on mobile and manipulation platforms for their long range, large field of view (FOV), and high accuracy. However, mechanical design constraints, mounting positions, and FOV limitations often prevent a single LiDAR from meeting omnidirectional perception requirements. Therefore, multiple LiDARs are typically employed to complement blind areas through heterogeneous FOV complementarity. To achieve accurate cross-LiDAR data fusion, calibrating the extrinsic parameters (6-DoF relative pose) between multiple LiDARs is essential [1].

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As shown in Fig.1, a quadruped robot is equipped with three LiDARs to achieve sufficient perceptual coverage. This design

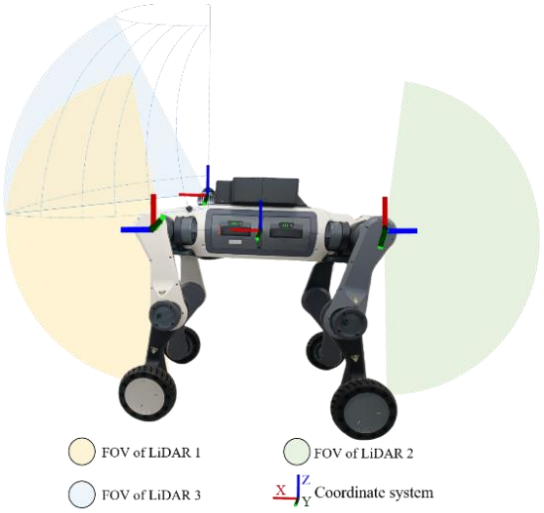


Fig. 1. Quadruped robot designed for autonomous navigation tasks equipped with three LiDARs.

enables the robot to comprehensively perceive surrounding obstacles during autonomous navigation, thereby avoiding potential collision damage. The prerequisite for cooperative operation of multiple LiDARs is the accurate calibration of their extrinsic parameters.

Extrinsic calibration methods for multi-LiDAR systems can be broadly categorized into three paradigms: target-based methods, targetless methods, and online extrinsic calibration methods [2]–[4]. Target-based methods rely on known calibration objects within overlapping FOVs. They extract geometric features and minimize reprojection errors. However,

they require manual setup and are hard to automate. Targetless methods use environmental features or sensor motion to build correspondences. They depend on overlapping FOVs and reliable feature detection. Online methods integrate with SLAM, treating extrinsics as state variables. Without accurate initial values, drift can accumulate and degrade accuracy of the whole system. Motion-based methods avoid overlapping FOV and initial value needs, but ground vehicles' near-planar motion often fails to provide full 6-DoF constraints.

To overcome these limitations, we propose a targetless multi-LiDAR calibration method that requires no initial guess, overlapping FOV, or special targets. It computes initial calibration parameters via a micro odometry and hierarchical feature extraction, then refines them with two stage optimization. Moreover, the method works without point-level feature extraction, making it suitable for all LiDAR types. More specifically, our contributions are as follows:

- A targetless multi-LiDAR calibration algorithm without initial guess or overlapping FOV. It builds a local map using tight-coupled micro odometry, obtains an initial closed-form solution via hierarchical feature extraction, and refines it through two-stage optimization.
- A mechanism for constructing the complete initial pose based on planner structures. Using a hierarchical feature extraction strategy. PCA on the dominant point cloud yields the principal eigenvector, which together with the plane normal forms a 3-DoF rotation matrix. The subordinate point cloud's centroid then regularizes the axes to obtain the full 6-DoF initial pose.
- A micro odomtry compatible with various 3D Lidars. A tightly coupled error-state Kalman filter and point-to-plane ICP registration combined with point-level motion distortion removal enable high-accuracy local mapping. As no point-level features are required, the method applies to all LiDAR types.

II. Related Work

Multi-LiDAR extrinsic calibration is critical for perceptual coverage and data fusion, and has thus attracted considerable research interest. Existing methods fall into three categories: target-based, targetless, and online calibration.

A. Target-Based Calibration

Target-based methods rely on artificial calibration targets and overlapping sensor fields of view. They exploit the known target geometry and common viewing region to establish feature correspondences and build constraint relationships. For instance, Gao et al. placed multiple reflective tapes on a vertical rod to determine feature correspondences [5]. Pereira et al. placed a sphere within the common FOV, used point cloud shape fitting to estimate its centroid, and employed the centroid as a registration feature to construct constraints [6]. Pusztai et al. used an ordinary rectangular box as a calibration object and performed iterative optimization for extrinsic calibration [7]. a four-hole board where hole detection and homography yield 3D circle centers, with two laser beams per hole segmenting correspondences [8].

B. Targetless Calibration

These methods exploit natural environment features to avoid dedicated targets. Representative works: ground plane alignment with genetic algorithm for yaw correction [9]; three independent planes for closed-form initialization and nonlinear refinement [10]; hand-eye calibration from sensor motion followed by point-to-plane refinement [11]; RANSAC plane fitting and polar-coordinate intersection points for nonlinear optimization [12]; FGRSC (filtering grid and grid RANSAC) using curvature and planar probability estimation [13]; CROON with coarse-to-fine registration using road scene features [14]; IMU signal matching for initial guess, refined by ICP and feature matching, with observability-based subset selection [15].

C. Online Extrinsic Calibration

Online calibration integrates with SLAM (filtering or optimization), treating extrinsics as state variables for real-time estimation and refinement during robot operation. Lin et al. proposed a decentralized EKF-based framework for simultaneous calibration, localization, and mapping, sharing computation and eliminating single points of failure; extrinsics are state variables in the Kalman filter [16]. Jiao et al. developed a system: extract edge/plane features, motion/extrinsic initialization, sliding-window multi-LiDAR odometry for pose estimation, online extrinsic refinement, and global map optimization [17]. Liu et al. created FOV overlap through motion to generate common features, built a factor graph of LiDAR poses and extrinsics, solved via graph optimization—works with little or no initial overlap, and produces a globally consistent map [18].

III. Methodology

A. System Overview

Fig.2. illustrates the overall framework of the proposed algorithm. Without loss of generality, we describe the method using two LiDARs (denoted as A and B) on a multi-LiDAR robotic system. As most 3D LiDARs are integrated with an IMU, we first present the case with IMUs, and also provide a corresponding scheme for LiDARs without an IMU. The algorithm comprises four modules: data preprocessing, micro LiDAR odometry, full initial pose construction, and two-stage optimization.

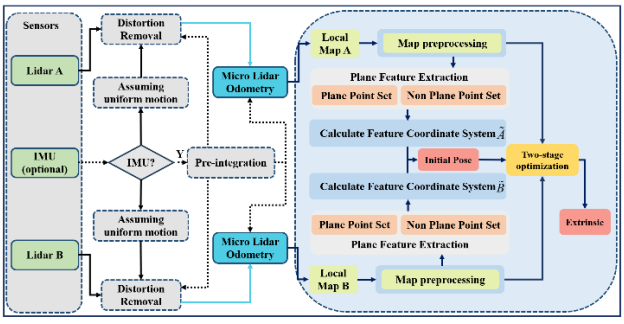


Fig. 2. System overview

Data preprocessing independently acquires raw point clouds and IMU data from each LiDAR, performs static initialization, synchronizes and packs data by frame timestamps, and corrects motion distortion via IMU pre-integration (or

constant-velocity assumption for LiDARs without IMU). Micro odometry then fuses LiDAR and IMU data using a tightly coupled error-state Kalman filter (ESKF), registering motion-distortion-corrected point clouds against an ikd-tree-maintained local map without feature extraction. To eliminate the need for an initial guess, a mechanism constructs a full initial pose from ground geometric structures: PCA on the dominant (planar) point cloud provides the principal direction, combined with the plane normal to form a 3-DoF rotation matrix, while the subordinate cloud centroid regularizes axes to obtain the full 6-DoF transform. Finally, two-stage registration applies NDT for coarse refinement and GICP for fine alignment to balance robustness and accuracy.

B. Micro Lidar Odomtry

Micro lidar odomtry adopts an error-state Kalman filter (ESKF) to achieve data fusion between LiDAR and IMU. The state variable of the ESKF is denoted as x which is defined on a high-dimensional manifold $\mathcal{M}$:

$$x = \left[p, v, R, b_g, b_a, g \right]^T \in \mathcal{M} \tag{1}$$

The error state δx is defined on the tangent space of the manifold in the vicinity of the operating point: $\delta x \in \mathbb{R}^{18}$. The discrete-time motion equations for the nominal state and the error state in the prediction step can be expressed as:

$$x(t+\Delta t) = \begin{cases} p(t) + v\Delta t + \frac{1}{2}\left(R\left(\tilde{a} - b_a\right)\right)\Delta t^2 + \frac{1}{2}\Delta t^2 \\ v(t) + R\left(\tilde{a} - b_a\right)\Delta t + g\Delta t \\ R(t)Exp\left(\left(\tilde{w} - b_g\right)\Delta t\right) \\ b_g(t) \\ b_a(t) \\ g(t) \end{cases} \tag{2}$$

$$\delta x(t+\Delta t) = f\left(\delta x(t)\right) + F\delta x(t) + w, w \in N(0, Q) \tag{3}$$

Where F is given by the linearized form of the motion error:

$$F = \begin{bmatrix} I & I\Delta t & 0 & 0 & 0 & 0 \\ 0 & I & -R\left(\tilde{a} - b_a\right)^{\wedge}\Delta t & 0 & -R\Delta t & I\Delta t \\ 0 & 0 & Exp\left(-\left(\tilde{w} - b_g\right)\Delta t\right) & -I\Delta t & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & 0 & I \end{bmatrix} \tag{4}$$

ESKF prediction predicts both the nominal and error states. The error state is reset to zero after each update, so the motion state uses the nominal state, while the error distribution in prediction is obtained from the error state expression:

$$P_{pred} = FPF^T + Q \tag{5}$$

Where Q is the noise matrix and P is the previous state covariance. For a LiDAR without an IMU, the nominal state prediction assumes constant velocity, and the error state noise matrix is set empirically.

The update step uses motion-distortion compensated points, all projected to the scan end time, so points in a LiDAR frame are effectively sampled simultaneously at that time. Let the k-th point cloud frame be $\left\{ p_i, i = 1, 2...m \right\}$ and the LiDAR pose at acquisition be $T_k^M = \left(R_k, p_k \right)$. Assuming these points project onto a small local planar region on the map, we have:

$$0 = \left(u_i^M\right)^T \left(R_k p_i + p_k - q_i \right) + v, v \sim N\left(0, V_i\right) \tag{6}$$

where u_i^M is the local plane normal and q_k a point on the plane. Point cloud registration iteratively updates the nominal state from the predicted state $x_{k,pred}$. At the j-th iteration, let the error state be δx_j with covariance P_j, the nominal state be x_j. The increment δx_{j+1} is computed, and a tangent space projection transformation is applied to the covariance matrix P_k:

$$J_j = diag\left(I \quad I \quad I - \frac{1}{2}\left(Log\left(R_j^T R_{k,pred} \right) \right)^{\wedge} \quad I \quad I \quad I \right) \tag{7}$$

$$P_j = J_j P_{k,pred} J_j^T \tag{8}$$

The local map is maintained by an ikd-tree. In each iteration, the LiDAR point cloud is projected to the map frame using the pose x_j. For each projected point, the five nearest map points are retrieved via the ikd-tree. A least-squares plane fit checks coplanarity; points below a threshold become valid observations p_i for point-to-plane residuals. The observation equation is:

$$z_{i,j} = \left(u_{i,j}^M\right)^T \left(R_{k,j} p_{i,j} + p_{k,j} - q_{i,j} \right) + v, v \sim N\left(0, V_i\right) \tag{9}$$

Using the $\oplus$ operator from [22], each iteration solves a maximum a posteriori (MAP) estimation problem:

$$\delta x_j = \arg\min_{\delta x} \left(\left\| \delta x_j \right\|_{P_j}^2 + \sum_{i=1}^{m} \left\| z_{i,j} - H_{i,j}\left(x_j \oplus \delta x \right) \right\|_V^2 \right) \tag{10}$$

Where $H_{i,j}$ is the observation Jacobian w.r.t. δx_j at zero:

$$H_i = \begin{bmatrix} \frac{\partial z_i}{\partial p} & 0 & \frac{\partial z_i}{\partial R} & 0 & 0 & 0 \end{bmatrix} \tag{11}$$

$$H_j = \begin{bmatrix} ... & H_i & ... \end{bmatrix}^T \tag{12}$$

$$V = diag\left(... \quad V_i \quad ... \right) \tag{13}$$

Using the SMW identity [22] to handle varying observation dimensions, the ESKF update upon filter convergence is:

$$K_j = \left(P_j^{-1} + H_j^T V^{-1} H_j\right) H_j^T V^{-1} \tag{14}$$

$$\delta x_j = K_j \left(z_j - h\left(x_j\right)\right) \tag{15}$$

$$P_{j+1} = \left(I - \left(P_j^{-1} + H_j^T V^{-1} H_j\right)^{-1} H_j^T V^{-1} H_j\right) P_j \tag{16}$$

After each update, a minimum-variance unbiased estimate of the state is obtained. The k-th point cloud is projected into the local map maintained by the ikd-tree using the updated pose. The micro odometry builds two local maps, Map_A and Map_B, each centered at its LiDAR origin. We indirectly compute the extrinsic parameters between the LiDARs by solving for the relative pose between these two maps.

C. Complete Initial Pose Construction

To avoid dependence on initial calibration values, we construct a complete initial pose from geometric structures. From each local map, we extract generic features via a hierarchical method, build feature coordinate systems, and compute the initial extrinsic parameters from their relative transformation.

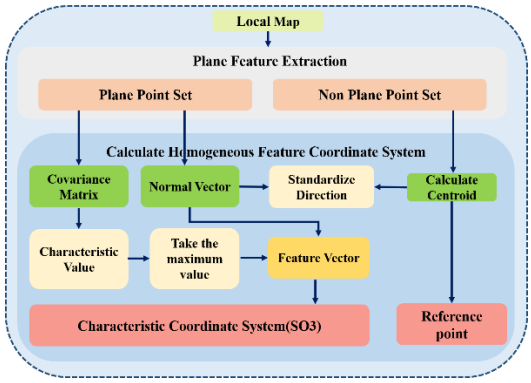


Fig.3. Schematic diagram of feature coordinate system construction.

As shown in Fig. 2, the proposed mechanism has four steps: map preprocessing, extraction of dominant/subordinate feature clouds, feature coordinate system construction, and initial parameter computation. Preprocessing applies a three-stage filter (voxel filtering, statistical outlier removal, Euclidean clustering) to remove dynamic objects, outliers, and non-structural interference. For outlier removal, a K-nearest neighbor method computes each point's mean distance to its K neighbors; assuming a Gaussian distribution, points beyond a threshold are rejected—suppressing sensor noise and sporadic points. Euclidean clustering builds a KD-tree with a set tolerance, extracts the largest connected component (by point count) as the valid scene, and discards distant scattered points.

We segment planar point clouds from local maps using RANSAC. The largest planar component (typically the ground) becomes the dominant feature cloud; the remaining non-planar points form the subordinate cloud. The output plane coefficient (a,b,c,d) satisfy $ax+by+cz+d=0$. For LiDAR A, denote the dominant cloud as P_a, the subordinate cloud as N_a, and the plane normal as $n=(a,b,c)$, normalized to $\hat{n}$.

The construction of the feature coordinate system is depicted in Fig.3. We perform principal component analysis (PCA) on the dominant feature point cloud: we compute its covariance matrix and extract the principal direction. Suppose P_a contains N three-dimensional points:

$$P_a = \{p_1, p_2, ..., p_n\}, p_i = (x_i, y_i, z_i)^T \tag{17}$$

The covariance matrix C of P_a is given by:

$$W = \left[p_1 - \frac{1}{N}\sum_{i=1}^{N} p_i, p_2 - \frac{1}{N}\sum_{i=1}^{N} p_i, ..., p_N - \frac{1}{N}\sum_{i=1}^{N} p_i\right]^T \tag{18}$$

$$C = \frac{1}{N} W W^T \tag{19}$$

We then perform eigendecomposition:

$$C = Q \Lambda Q^T, \Lambda = diag\left(\lambda_1, \lambda_2, \lambda_3\right) \tag{20}$$

$$v_{\max} = Q_{:,2} \tag{21}$$

The covariance matrix is positive semidefinite, so eigenvalues satisfy $0 \le \lambda_1 \le \lambda_2 \le \lambda_3$. The eigenvector $v_{\max}$ of the largest eigenvalue λ_3 defines the principal direction.

To ensure a unique feature coordinate system, we resolve the plane normal's directional ambiguity using the subordinate cloud's centroid. Let the centroid of N_a be p_{nc}:

$$P_{nc} = \frac{1}{N}\left(\sum_{i=1}^{N} x_i, \sum_{i=1}^{N} y_i, \sum_{i=1}^{N} z_i,\right) \tag{22}$$

Construct vector $\hat{p}_{nc}$ from dominant to subordinate centroid. The angle θ between $\hat{n}$ and $\hat{p}_{nc}$ regulates the normal direction:

$$\cos\theta = \frac{\hat{n} \cdot \hat{p}_{nc}}{\|\hat{n}\| \cdot \|\hat{p}_{nc}\|} \tag{23}$$

If the dominant cloud's unit normal and subordinate centroid vector form an obtuse angle, reverse the normal:

$$\hat{n} = -\hat{n} \tag{24}$$

A right-handed feature coordinate system is then built from orthogonal basis vectors $v_{\max}$ and $\hat{n}$:

$$R_a = \left[\pm v_{\max}, \frac{z_a \times x_a}{\|z_a \times x_a\|_a}, \hat{n}\right] \tag{25}$$

Numerical errors in constructing a rotation matrix from point clouds often violate orthogonality. We thus apply the Rodrigues transformation [22] to map the approximate matrix to rotation vector space and resynthesize a strictly orthogonal matrix. This ensures SO(3) compliance and prevents cumulative pose/transform errors.

Applying the above procedure to local maps from the two LiDARs yields orthogonal feature frames R_a and R_b , The complete initial extrinsic parameters can be expressed as:

$$T_{AB} = \left(R_a R_b^{-1}, p_{nc,b} - R_a R_b^{-1} p_{nc,a} \right) \tag{26}$$

D. Two Stage Optimization And Registration

Using the initial extrinsic parameters from Section III-C, two-stage registration applies cascaded registration to the two local maps using the constructed initial estimate. First stage: The initial estimate serves as the starting transformation; Taking advantage of NDT's wide convergence region, a coarse registration is carried out to preliminarily refines the parameters. Second stage: The NDT result is used as the initial guess for fine registration. GICP builds local Gaussian distributions for corresponding points, uses the Mahalanobis distance as the error metric, and adaptively incorporates local geometry. It enforces strong normal constraints while tolerating tangential errors, improving accuracy. This two-stage strategy balances robustness and precision.

IV. Experiment

We evaluated our method in simulation and on a real robot, using both mechanical and hybrid solid-state LiDARs. In simulation, calibration results were compared to ground truth; on the real robot, mechanical design values served as reference.

A. Simulation Experiments

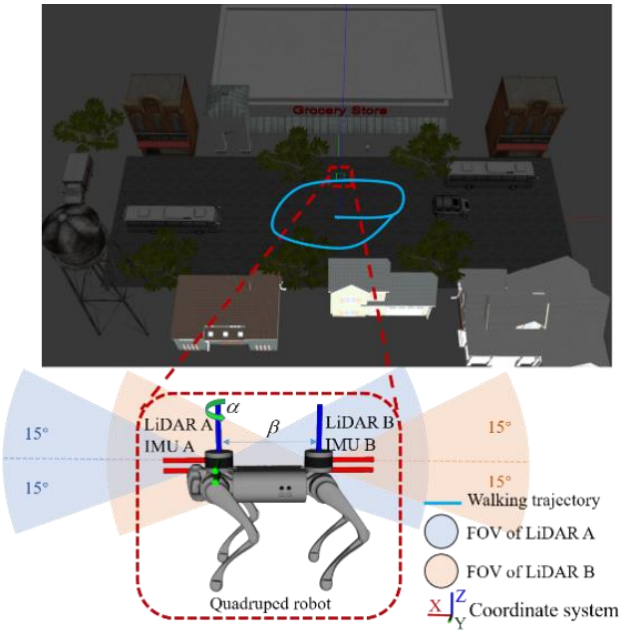
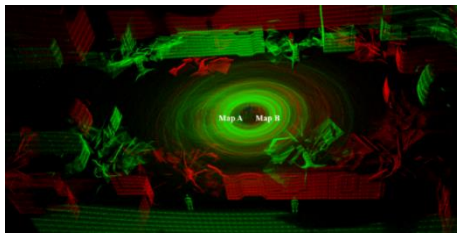


Fig. 4. Experimental setup in the simulation environment.

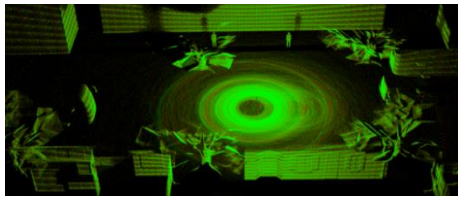
We built a Gazebo simulation environment (Fig. 4) containing a quadruped robot model with 12 joint DoFs for kinematics and dynamics. A virtual model control (VMC) algorithm enabled stable biomimetic walking. Two 16-beam mechanical LiDARs (360°×30° FOV, -15° to +15° vertically) were mounted at the front and rear, each integrated with a 6-axis IMU. The LiDAR measurement frequency was 10 Hz, and the IMU frequency was 100 Hz.

TABLE I. Translation and Rotation Error Analysis of Grand Truth and Calibration Results in Simulation Experiments

Direction	Rotation Error(rad)		Translation Error(m)	
	Mean	*STD*	*Mean*	*STD*
X	0.023	0.0013	0.042	0.0055
Y	0.016	0.0016	0.032	0.0085
Z	0.035	0.0042	0.025	0.0024
roll	0.034	0.0053	0.022	0.0026
pitch	0.021	0.0029	0.037	0.0038
yaw	0.046	0.0055	0.034	0.0054



(a)



(b)

Fig. 5. (a)Two local maps, with red point clouds from LIDAR A and green point clouds from LIDAR B. (b) After calibration and external parameter correction, the two local maps overlap.

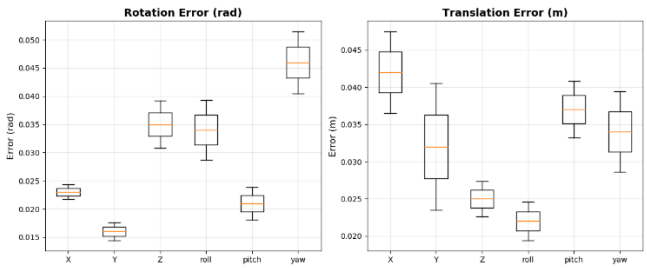


Fig. 6. Performance analysis of our calibration algorithm based on translation and rotation errors.

During the experiments, we varied the relative pose between two LiDARs: for orientation, LiDAR B rotated about the X/Y/Z axes with rotation angles α from 0° to 180° in 30° steps (LiDAR A fixed, translation constant); for translation, LiDAR B moved along X/Y/Z with displacements β from 0 m to 0.5 m in 0.1 m steps (rotation constant). For each configuration, the robot walked a small arbitrary route using a trot gait, allowing both LiDARs to scan the environment for micro odometry to build local maps. In fact, the required motion path merely allows each LiDAR to perceive a local neighborhood. As vast majority of scenes contain a ground plane, our method uses planar features as the primary geometric primitive, ensuring broad applicability. URDF-specified poses served as ground truth. Fig. 4 shows one mounting case; the resulting local maps (red: LiDAR A, green: LiDAR B) are shown in Fig. 5(a). In our experiments, the

rotation error of the calibration was quantified using the L2 norm of the rotation axis-angle vector, whose direction indicates the rotation axis and whose magnitude represents the rotation angle in radians. The translation error was quantified as the L2 norm of the translation error vector. The formulas are given below:

$$\begin{cases} e_r = \left\| \log\left(R R_{gt}^{-1} \right)^{\vee} \right\|_2 \\ e_t = \left\| t_{gt} - t \right\|_2 \end{cases} \tag{27}$$

Where R_{gt} and t_{gt} denote the ground truth rotation and translation, R and t denote the calibration results obtained by our algorithm. According to our experimental design, each translational component was tested six times, and each rotational component was tested five times. We report mean and standard deviation in Table I. Fig. 5(b) shows the calibrated point cloud for the configuration in Fig. 5(a). The fused point cloud after transforming LiDAR B to LiDAR A using calibration results is presented in Fig. 6, where red and green clouds align well. Our method achieves rotation error below 0.046 rad and translation error below 0.042 m, demonstrating its performance.

B. Experiments on Real Robots

Real-world calibration experiments were conducted on the four-wheeled legged robot shown in Figure 1. The robot is equipped with three LiDARs. Specifically, two RoboSense Airy mechanical LiDARs are mounted at the front and rear, respectively, each with a FOV of 360° (horizontal) × 90° (vertical). One Livox Mid-360 hybrid solid-state LiDAR is installed on top, with an FOV of 360° (horizontal) × 59° (vertical). We denote the front Airy as LiDAR 1, the rear Airy as LiDAR 2, and the Mid-360 as LiDAR 3. The robot follows a small arbitrary path that allows each LiDAR's field of view to cover the full 360° region around the robot. The mechanical mounting poses are used as a reference to evaluate the practicality of the proposed calibration algorithm. No initial guess is provided to the calibration algorithm in the experiments. The difference between the calibrated extrinsic parameters and the mounting-based reference is calculated using the error metric defined in Eq. (27). The experimental results are presented in Table II. Considering the presence of mounting errors, the results demonstrate the effectiveness of the proposed calibration algorithm.

TABLE II. TRANSLATION AND ROTATION ERROR ANALYSIS OF EXPERIMENTAL RESULTS ON REAL ROBOTS

LIDARS	Rotation Error(rad)	Translation Error(m)
LIADR 2 to LIDAR 1	0.042	0.036
LIADR 3 to LIDAR 1	0.035	0.045

V. CONCLUSION

We propose a targetless multi-LiDAR extrinsic calibration method requiring no initial guess, overlapping FOV, or specific targets. A micro-odometry module and hierarchical feature extraction provide initial parameters, refined by two-stage registration for offline automatic calibration. The method works on different types of 3D LiDARs. Experiments on simulation and a real quadruped robot show rotation error below 0.046 rad and translation error below 0.042 m, confirming effectiveness. Future work will explore online and adaptive calibration for dynamic environments.

REFERENCES

[1] Lin J, Liu X, Zhang F. A decentralized framework for simultaneous calibration, localization and mapping with multiple LiDARs[C]//2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2020: 4870-4877.

[2] Chang D, Zhang R, Huang S, et al. Versatile multi-lidar accurate self-calibration system based on pose graph optimization[J]. IEEE Robotics and Automation Letters, 2023, 8(8): 4839-4846.

[3] Zhou L, Li Z, Kaess M. Automatic extrinsic calibration of a camera and a 3d lidar using line and plane correspondences[C]//2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2018: 5562-5569.

[4] Xie G, Xu T, Isert C, et al. Online active calibration for a multi-lrf system[C]//2015 IEEE 18th International Conference on Intelligent Transportation Systems. IEEE, 2015: 806-811.

[5] Gao C, Spletzer J R. On-line calibration of multiple lidars on a mobile vehicle platform[C]//2010 IEEE International Conference on Robotics and Automation. IEEE, 2010: 279-284.

[6] Pereira M, Santos V, Dias P. Automatic calibration of multiple LIDAR sensors using a moving sphere as target[C]//Robot 2015: Second Iberian Robotics Conference: Advances in Robotics, Volume 1. Cham: Springer International Publishing, 2015: 477-489.

[7] Pusztai Z, Eichhardt I, Hajder L. Accurate calibration of multi-lidar-multi-camera systems[J]. Sensors, 2018, 18(7): 2139.

[8] Zhang J, Zhang R, Yue Y, et al. SLAT-Calib: Extrinsic calibration between a sparse 3D LiDAR and a limited-FOV low-resolution thermal camera[C]//2019 IEEE international conference on robotics and biomimetics (ROBIO). IEEE, 2019: 648-653.

[9] Maroli J M, Özgüner Ü, Redmill K, et al. Automated rotational calibration of multiple 3D LIDAR units for intelligent vehicles[C]//2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC). IEEE, 2017: 1-6.

[10] Jiao J, Liao Q, Zhu Y, et al. A novel dual-lidar calibration algorithm using planar surfaces[C]//2019 IEEE Intelligent Vehicles Symposium (IV). IEEE, 2019: 1499-1504.

[11] Jiao J, Yu Y, Liao Q, et al. Automatic calibration of multiple 3d lidars in urban environments[C]//2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2019: 15-20.

[12] Kim D H, Kim G W. Automatic multiple LiDAR calibration based on the plane features of structured environments[J]. IEEE Access, 2021, 9: 84387-84402.

[13] Wang S, Zhang H, Wang G, et al. FGRSC: Improved calibration for spinning LiDAR in unprepared scenes[J]. IEEE Sensors Journal, 2022, 22(14): 14250-14262.

[14] Wei P, Yan G, Li Y, et al. Croon: Automatic multi-lidar calibration and refinement method in road scene[C]//2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2022: 12857-12863.

[15] Das S, Boberg B, Fallon M, et al. IMU-based online multi-LiDAR calibration[C]//2024 IEEE Intelligent Vehicles Symposium (IV). IEEE, 2024: 3227-3234.

[16] Lin J, Liu X, Zhang F. A decentralized framework for simultaneous calibration, localization and mapping with multiple LiDARs[C]//2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2020: 4870-4877.

[17] Jiao J, Ye H, Zhu Y, et al. Robust odometry and mapping for multi-lidar systems with online extrinsic calibration[J]. IEEE Transactions on Robotics, 2021, 38(1): 351-371.

[18] Liu X, Zhang F. Extrinsic calibration of multiple lidars of small fov in targetless environments[J]. IEEE Robotics and Automation Letters, 2021, 6(2): 2036-2043.

[19] Xu W, Cai Y, He D, et al. Fast-lio2: Fast direct lidar-inertial odometry[J]. IEEE Transactions on Robotics, 2022, 38(4): 2053-2073.

[20] Magnusson M. The three-dimensional normal-distributions transform: an efficient representation for registration, surface analysis, and loop detection[D]. Örebro universitet, 2009.

[21] Koide K, Yokozuka M, Oishi S, et al. Voxelized GICP for fast and accurate 3D point cloud registration[C]//2021 IEEE international conference on robotics and automation (ICRA). IEEE, 2021: 11054-11059.

[22] Barfoot T D. State estimation for robotics[M]. Cambridge University Press, 2024.

Hybrid Force-Position Control of Linear Drive Systems without Force Sensor

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***Abstract*—This paper addresses the challenge of achieving compliant control in linear drive systems in contact-type operation systems and smart furniture applications, given the absence of force sensors. To tackle this issue, this paper introduces a linear drive compliant control method based on motor current feedback without force sensors. Initially, by filtering the motor current signal, a mapping relationship between motor current and output force is established. Based on this, a compliant control framework utilizing hybrid force-position control is constructed. Furthermore, to tackle the complex and uncertain environment-specific challenges of force-position coordination control, this paper investigates a fuzzy hybrid force-position control method based on a fuzzy adaptive mechanism. Experimental results indicate that, compared to traditional control methods, compliant control without force sensors significantly enhances the work efficiency and safety of linear drives, offering an effective solution for low-cost, high-reliability contact-type operations and smart furniture.**

Keywords—linear drive system, compliant control, fuzzy adaptive control, hybrid force and position control

I. Introduction

Motion control is one of the key components of linear drive systems, primarily used to achieve linear motion in a specific axial direction. Linear drive systems typically consist of key components such as motors, lead screws, torque sensors, and encoders. Therefore, linear drive technology is receiving increasing attention due to its superior performance advantages. Especially in application scenarios requiring high response speed and precise displacement control, linear drive systems demonstrate their unique value.

The earliest force/position hybrid control was proposed by Mason et al., aiming to achieve compliance of the end-effector while controlling both force and position. However, this control method is overly complex and repetitive[1]. Literature [2] combines force/position hybrid control with sliding mode control to address the force and position tracking problem of robots, making force control more stable and robust. Literature [3] proposes adaptive fuzzy force/position hybrid control for force and position control in unknown environments, enabling the system to achieve high control accuracy in positional environments. Literature [4] employs adaptive control to handle model uncertainties and external disturbances, but does not consider the compliance of human-robot interaction. To ensure the compliance of the end-effector of the actuator, literature [5] proposes a velocity-based variable impedance adaptive interaction control method, which adjusts the damping parameters of the admittance controller by tracking the interaction force error. However, both methods utilize force sensors and assume them to be ideal components, which introduces noise into the system and increases its complexity.

Zhang et al. [6] obtained the magnitude of the control torque by measuring the current difference and multiplying it by the torque constant. LEE et al. [7] proposed a collision detection method that does not require any sensors, estimating the external torque caused by collisions by monitoring the motor current. However, existing research mostly focuses on torque control of robot joints or simple collision detection, and targeted research on deeply integrating current-based force estimation methods with force-position hybrid control of linear drive systems is still insufficient. To this end, this paper aims to study the force-position hybrid control technology of linear drive systems without relying on external force sensors.

II. System Modeling and Force Estimation

A. Mathematical Model of Linear Drive System

As the core component of a linear drive system, the establishment of the mathematical model for the sliding table module is crucial for achieving high-precision control. The sliding table module can be regarded as a typical mechanical system. Assuming the mass of the sliding table is m, according to Newton's second law, the dynamic equation of the sliding table can be expressed as:

$$\mathrm{F} = \mathrm{m} \cdot \frac{d^2x}{dt^2} + b \cdot \frac{dx}{dt} + k \cdot x \tag{1}$$

Where x represents the displacement of the sliding platform, $\mathrm{v} = \frac{\mathrm{dx}}{\mathrm{dt}}$ denotes the velocity, $\mathrm{a} = \frac{\mathrm{d^2x}}{\mathrm{dt^2}}$ signifies the acceleration, b is the viscous friction coefficient (related to velocity), and k is the equivalent stiffness coefficient. The right side of the equation sequentially represents the inertial force, viscous friction force, and elastic force. The linear drive system is illustrated in Fig. 1.

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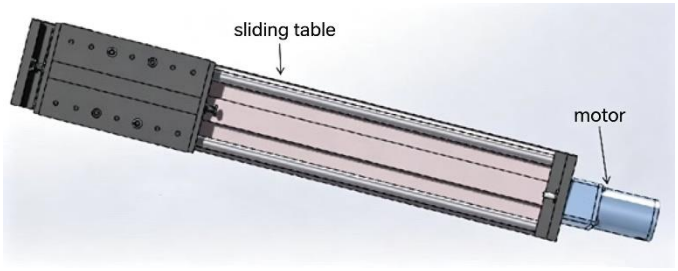

Fig. 1. Linear drive system

DC brush motors are characterized by simple control, good torque-current linearity, and fast response speed, making them highly suitable for force-position hybrid control applications that require precise adjustment of both position and torque. This article selects a 36mm planetary reduction brush motor.

The motor converts electrical energy into mechanical energy through electromagnetic effect, and the torque generation equation is:

$$T_m = K_t \cdot I_a \tag{2}$$

Where T_m is the motor output torque, K_t is the torque constant, and I_a is the armature current.

The back electromotive force equation can be expressed as:

$$E_b = K_e \cdot \omega_m \tag{3}$$

Where E_b represents the back electromotive force, K_e represents the back electromotive force constant, and for an ideal motor, typically $K_t = K_e$, ω_m represents the angular acceleration of the motor.

For the mechanical transmission part, the conversion of position, speed, and acceleration can be expressed as follows:

$$\begin{cases} s = \frac{L}{2\pi} \cdot \theta = \frac{L}{2\pi} \int \omega_m \, dt \\ v = \frac{ds}{dt} = \frac{L}{2\pi} \cdot \omega_m \\ a = \frac{dv}{dt} = \frac{L}{2\pi} \cdot \alpha_m = \frac{L}{2\pi} \cdot \frac{d\omega_m}{dt} \end{cases} \tag{4}$$

Where s represents linear displacement, θ represents motor rotation angle, L represents lead of the screw rod, v represents linear velocity, a represents linear acceleration, and α_m represents motor angular acceleration.

Based on the law of conservation of energy, the relationship between force and torque can be derived:

$$F = \frac{2\pi \cdot \eta}{L} \cdot T_m = \frac{2\pi \cdot \eta \cdot K_t}{L} \cdot I_a \tag{5}$$

Where F represents linear thrust, and η represents transmission efficiency, typically ranging from 0.85 to 0.95.

According to Newton's second law:

$$F = m \cdot a + F_f + F_{ext} \tag{6}$$

Substitute into the thrust formula:

$$\frac{2\pi \cdot \eta \cdot K_t}{L} \cdot I_a = m \cdot a + F_f + F_{ext} \tag{7}$$

The output torque of the motor is used to accelerate the system and overcome resistance:

$$T_m = J_{total} \cdot \alpha_m + T_f + T_{ext} \tag{8}$$

Where J_{total} represents the total moment of inertia of the system.

Replace with a specific expression:

$$K_t \cdot I_a = J_{total} \cdot \frac{2\pi}{L} \cdot a + \frac{F_f \cdot L}{2\pi \cdot \eta} + \frac{F_{ext} \cdot L}{2\pi \cdot \eta} \tag{9}$$

After conversion, we can obtain:

$$I_a = \frac{1}{K_t} \left[J_{total} \cdot \frac{2\pi}{L} \cdot a + \frac{(F_f + F_{ext}) \cdot L}{2\pi \cdot \eta} \right] \tag{10}$$

B. Force Estimation Method and Verification for Linear Drive System

The core concept of the current-based force estimation model lies in indirectly obtaining the contact force during the interaction between the linear drive end effector and the environment by analyzing the current changes of the motor in real time. Its physical basis is that during the motor's motion, there exists a certain linear relationship between its output torque and armature current in a steady state, and this torque can be converted into linear thrust through transmission mechanisms such as ball screws.

Based on the motor's rated voltage $U_N = 24V$, rated current $I_N = 3A$, and rated speed $n_N = 5000rpm$, the input power can be calculated as:

$$P_{in} = U_N \cdot I_N = 24 \times 3 = 72W \tag{11}$$

The efficiency of a brushed DC motor during operation is generally considered to be $\eta_m \approx 0.75$, hence the output mechanical power is:

$$P_{out} = \eta_m \cdot P_{in} = 0.75 \times 72 = 54W \tag{12}$$

Rated torque:

$$T_N = \frac{P_{out}}{\omega_N} \tag{13}$$

$$\omega_N = \frac{2\pi n_N}{60} = \frac{2\pi \times 5000}{60} \approx 523.6 rad/s \tag{14}$$

$$T_N = \frac{54}{523.6} \approx 0.103N \cdot m \tag{15}$$

From $T_N = K_t \cdot I_N$ (neglecting the no-load current), we obtain:

$$K_t = \frac{T_N}{I_N} = \frac{0.103}{3} \approx 0.0343N \cdot m/A = 34.3mN \cdot m/A \quad (16)$$

The reduction ratio is i = 4.25, the efficiency of the reducer is $\eta_g \approx 0.9$, and the relationship between the output shaft torque Tout of the reducer and the motor current I_A is as follows:

$$T_{out} = i \cdot \eta_g \cdot K_t \cdot I_A \approx 0.131 I_A \quad (17)$$

The relationship between the sliding table thrust F and the reducer output torque T_{out} is as follows:

$$F = \frac{2\pi\eta_s}{p} \times 10^3 \cdot T_{out} \approx \frac{822\eta_s}{p} \cdot I_A \quad (18)$$

Where the lead of the screw rod p is 5mm and the efficiency of the screw rod $\eta_s = 0.9$, then:

$$F = 148 I_A \quad (19)$$

It is worth noting that the mapping relationship between current and output force is influenced by various factors. In the low force range, the nonlinear characteristics of the system are particularly evident. This is because static friction, small clearances in transmission components, and the cogging effect of the motor itself all have a significant impact on the measured current values, leading to deviations in force estimation. In the medium to high force range, the system enters a working zone dominated by inertial forces and viscous friction, and its dynamic characteristics become closer to a linear model, resulting in relatively high accuracy in force estimation. The contact force collected in real-time through a tensiometer is correlated with the experimentally measured current values of the motor during operation. Linear fitting is performed on the collected current and contact force values. The relationship between current and contact force is shown in Fig. 2.

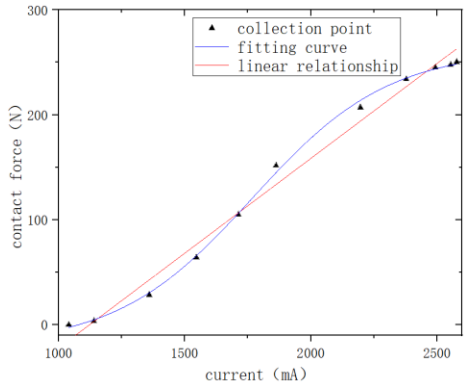

Fig. 2. The relationship between current and force

The linear relationship between the current I and the contact force F can be approximated as follows:

$$F = \frac{I-1035}{6.2} \quad (20)$$

When F = 0, I=1035mA, which represents the required current for the system during no-load operation, including the no-load current of the motor itself, the friction loss of the reducer, and the friction of the slide table guide rail. The slope k=0.1613N/mA=161.3N/A, which is in good agreement with the theoretical value of 148N/A we derived earlier. The error is about 8%. The measured value is larger, indicating that system friction cannot be ignored, which is precisely the difference between the actual system and the ideal model.

III. Force and Position Hybrid Control Method for Linear Drive

In practical applications, facing unknown changes in environmental stiffness and frequent switching between free and constrained spaces, fixed-parameter hybrid force-position controllers often struggle to balance fast response and smooth contact, leading to issues such as large force tracking errors or even system instability. To address the issues of traditional hybrid force-position control, such as the tendency to produce shocks during mode switching and insufficient adaptability to environmental changes, fuzzy logic control is introduced, as shown in Fig. 3. The core idea of this method is to use the system's real-time force tracking error and its change rate as inputs to the fuzzy controller, and the weight distribution coefficient λ of the position loop and force loop as the output.

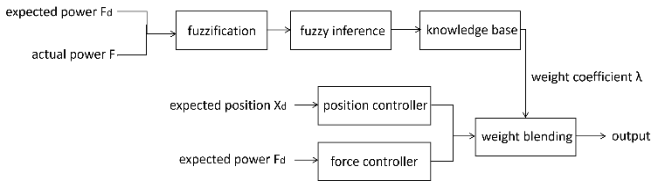

Fig. 3. Fuzzy control method

Combining fuzzy control with force-position hybrid control often achieves better results in practical engineering applications than using a single control strategy. Traditional force-position hybrid control employs a binary selection matrix, where each element is either 0 or 1. Fuzzy logic can soften this approach, enabling a smooth transition. At the moment of contact, instead of a rapid switch, the position loop gradually exits and the force loop gradually intervenes, achieving a compliant switch without impact, thus avoiding the most common switching oscillation problem in force-position hybrid control.

Force and position hybrid control under fuzzy selection matrix:

$$\begin{cases} V_{position} = f_{position}(X_d - X) \\ V_{force} = f_{force}(F_d - F) \\ V_{cmd} = \lambda \cdot V_{position} + (1-\lambda) \cdot V_{force} \end{cases} \quad (21)$$

To ensure the efficiency and accuracy of the system, we have adopted a high-performance motion controller, which boasts a high processing speed and high-precision motion control capabilities. It is capable of processing the control signals required by the system in real time. The mechanical structure design of the experimental platform should have good stability and adjustability. We have adopted a rigid frame structure to ensure the stability of the system during dynamic operation, in order to reduce errors introduced during the transmission process.The experimental platform is shown in Fig. 4.

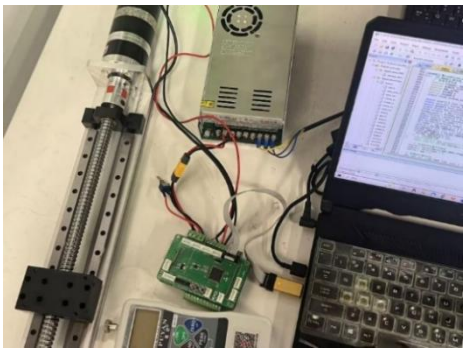

Fig. 4. Force and position hybrid control experimental platform

Perform hybrid force-position control on the experimental platform. Set the expected position of the device to 200mm and the expected current to 1500mA. Apply fixed hybrid force-position control parameters and fuzzy hybrid force-position control parameters for control, respectively. The filtered current curves are shown in Fig. 5.

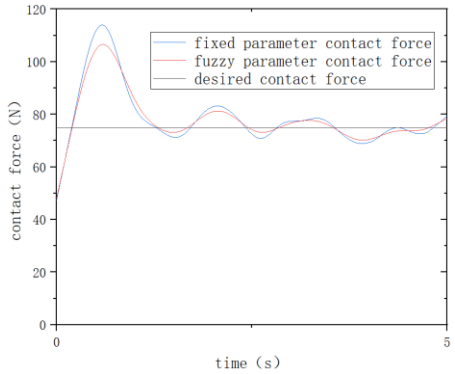

Fig. 5. Force response curve of force position control

Assuming that the linear drive system encounters an obstacle during operation and is required to maintain an output force not exceeding 250N, the actual effects of fuzzy force-position hybrid control and fuzzy impedance control are compared, as shown in Fig. 6.

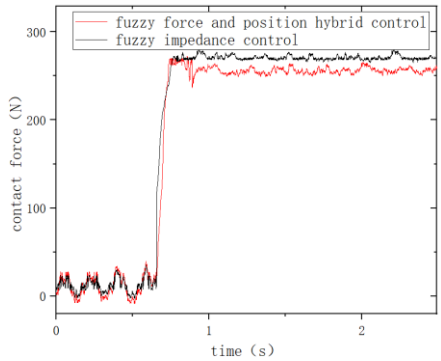

Fig. 6. Comparison Diagram of Two Types of Fuzzy Control

The comparison between fuzzy force position hybrid control and fuzzy impedance control is shown in TABLE I.

TABLE I. CONTROL CONTRAST

dimension	**Comparison between Fuzzy Force Position Hybrid Control and Fuzzy Impedance Control**	
	Fuzzy force position hybrid control	***Fuzzy impedance control***
goal	Simultaneously track Xd and Fd	Maintain an expected dynamic relationship
fuzzy adjustment object	Adjust the weight factor λ	Adjust physical parameters K, B, M
response to environmental changes	Conquer the environment by adjusting weights (while satisfying both X and F)	Adapt to the environment by adjusting parameters

Both control methods can achieve stable force control. The fuzzy force-position hybrid control can accurately track the desired force with minor continuous fluctuations. The fuzzy impedance control is relatively stable but cannot accurately track the desired force.

IV. CONCLUSION

This paper achieves compliant control without force sensors in a linear drive system. By introducing a fuzzy force-position hybrid control strategy, it effectively resolves the contradiction between the difficulty in obtaining force information in low-cost scenarios and the requirement for contact compliance. The research results provide an effective solution for low-cost, high-reliability contact operations and smart furniture, and hold significant theoretical and practical value for the intelligent upgrade of smart furniture products and efficient flexible operations in the field of intelligent manufacturing.

REFERENCES

[1] Mason M T. Compliance and force control for computer controlled manipulators[J]. IEEE Transactions on Systems, Man, and Cybernetics, 1981, 11(6): 418-432.

[2] Raibert M H, Craig J J. Hybrid position/force control of manipulators[J]. Journal of Dynamic Systems, Measurement, and Control, 1981, 102(2): 126-133.

[3] Ravandi A, Khanmirza E, Daneshjou K. Hybrid force/position control of robotic arms manipulating in uncertain environments based on adaptive fuzzy sliding mode control[J]. Applied Soft Computing, 2018, 70: 864-874.

[4] Wang Z, Zou L, Su X, et al. Hybrid force/position control in workspace of robotic manipulator in uncertain environments based on adaptive fuzzy control[J]. Robotics and Autonomous Systems, 2021, 145: 1-15.

[5] Kong L, He W, Yang C, et al. Adaptive fuzzy control for coordinated multiple robots with constraint using impedance learning[J]. IEEE Transactions on Cybernetics, 2019, 49(8): 3052-3063.

[6] Zhang H, Ahmad S, Liu G. Torque estimation for robotic joint with harmonic drive transmission based on position measurements[J]. IEEE Transactions on Robotics, 2015, 31(2): 322-330.

[7] Lee S D, Song J B. Sensorless collision detection based on friction model for a robot manipulator[J]. International Journal of Precision Engineering and Manufacturing, 2016, 17(1): 11-17.

Design of an Intelligent Monitoring and Early Warning System for Ground Anchors used in Transmission Tower Structures

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***Abstract*—Traditional transmission tower ground anchors rely on periodic manual inspections for safety assurance, yet face challenges such as delayed response times, high labor costs, and difficulty in promptly identifying potential hazards. This paper integrates force sensors and guy wire sensors with conventional ground anchors, proposing an optimized intelligent monitoring and early-warning system to enhance the safety alert capabilities of transmission tower anchors and advance grid intelligence. A tension sensor at the anchor pull ring provides real-time force monitoring; a secondary non-load-bearing anchor body at the anchor base connects to guy wire sensors for displacement measurement of ship-shaped anchors; a dedicated secondary anchor release mechanism enables automatic deployment. The monitoring system employs an STM32 microcontroller-based embedded controller transmitting signals via a LoRa module for remote monitoring. Final finite element stress analyses of both the anchor body and main structure confirm the systems compliance with operational requirements.**

Keywords—intelligent ground anchor; structure; monitoring and early warning; sensor; finite element analysis

I. INTRODUCTION

Transmission towers are constantly exposed to the natural environment and are susceptible to various natural disasters such as wind, snow, and earthquakes, making them highly prone to collapse or deformation, thereby endangering grid safety. With the expansion of grid scale and increased load, the requirements for the reliability and durability of transmission towers have become increasingly stringent[1].

During the construction of transmission towers, ground anchors are used for fixation to enhance tower stability and ensure safe operation under various external forces. These anchors serve as attachment points for temporary guy wires; connecting the guy wires to the anchors provides stable lateral tension, maintaining the towers upright position during installation and ensuring both construction safety and tower verticality. This helps resist lateral forces from strong winds and prevents tower tilting or collapse[2]. Additionally, during ice and snow conditions, conductors and ground wires may increase in weight due to ice accumulation; the ground anchors distribute these additional loads through the guy wires, preventing tower overload[3].

The safety monitoring of traditional anchor systems remains highly dependent on manual inspection operations[4]. Staff must frequently visit the work site to assess the condition of ground anchor structures using basic methods such as visual inspection and manual tapping. This approach not only leads to subjective judgment biases but also makes it difficult to promptly detect hidden safety hazards like metal fatigue, foundation settlement, or corrosion deterioration.

Moreover, traditional ground anchor structures predominantly rely on static designs, lacking intelligent monitoring and control capabilities, making them ill-suited for adapting to complex and dynamic external environments[5-6]. Currently, there are several application cases of intelligent

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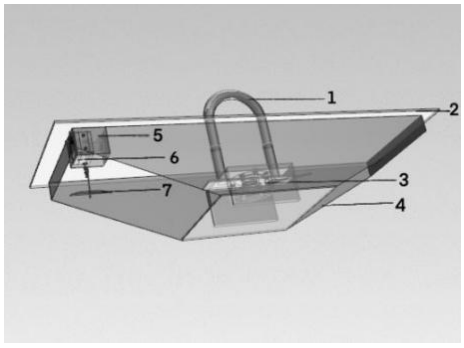
power transmission systems, primarily deployed on high-voltage and ultra-high-voltage transmission lines. Through real-time monitoring and feedback mechanisms, these systems enhance the safety and reliability of transmission towers.[7-9]This paper focuses on the structure of widely used ship-type ground anchors, employing tension and position sensors to monitor anchor status, thereby improving the construction environment for power grids and enhancing the intelligence level of grid development.

The intelligent ground anchor detection and early warning system address the limitations of traditional ground anchors in real-time monitoring and dynamic regulation, and tailored to the construction environment of transmission towers, the system integrates real-time anchoring force sensing, dynamic displacement monitoring, and remote surveillance capabilities. Feasibility of the solution was verified through finite element simulations.

II. Intelligent Anchor Structure Design

The intelligent ground anchor system for transmission towers consists of five components: a pull ring, secondary ground anchors, a secondary anchor deployment device, a top plate, and a cavity, as shown in Figure 1.



1. Ground anchor pull ring 2. Top plate 3. Cavity sensor 4. Cavity 5. Secondary ground anchor launch device 6. Motor 7. Secondary ground anchor

Fig. 1. Schematic diagram of ground anchor structure

A. Improvement of the Loop Structure

The intelligent anchor employs a sealed chamber to ensure the stable and proper operation of smart devices, with the pull ring in the load-bearing structure modified to a movable design. Post-integration of smart functionality, real-time monitoring of the anchors tension and displacement becomes feasible.

Since real-time tension monitoring is required and installing sensors at the pull ring poses significant challenges, the pull ring has been redesigned as a movable structure. The pull ring is cast from Q345 steel, featuring a circular cross-section with a hot-dip galvanized surface and a design safety factor of ≥2.0. Four trapezoidal reinforcement ribs are symmetrically welded around the root of the pull ring using continuous angle welds, distributing 30% of the tension load and reducing stress concentration, as shown in Figure 2. The reinforcement ribs are made of Q345 steel, and the weld joints undergo UT testing in accordance with the GB 50661 standard.

Fig. 2. Ground anchor pull ring

B. Secondary ground anchor

The secondary ground anchor consists of an anchor pile and two anchor arms connected by pins. Its function is to connect the wiring points of the displacement sensor and secure the sensor cables, as shown in Figure 3. Before hoisting, the secondary ground anchor is inserted into the secondary anchor deployment device; once the anchor is raised above ground level, the device is activated to drive the anchor into the ground. When the main body of the anchor is pulled upward by power tower cables, the anchor pile is lifted by the displacement sensor cables, causing the anchor arms to open. The arm structure is then secured by soil pressure, as illustrated in Figure 4.

Fig. 3. Secondary Ground

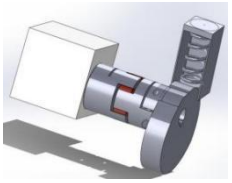


Fig. 4. Secondary Ground Anchor Launching Device

The secondary ground anchor launching device consists of a stepper motor, an electromagnetic clutch, a coupling, and a cam. The stepper motor drives the cam to actuate the spring, while the electromagnetic clutch locks the spring to store energy. Upon deactivation of the electromagnetic clutch, the spring releases, driving the secondary ground anchor into the soil.

C. Roof and Cavity

The deck plate, serving as the main framework of the ship-shaped ground anchor, is typically constructed from robust materials such as cast steel or high-strength alloy steel, enabling it to withstand significant tensile and compressive forces while preventing deformation or damage during operation. Its distinctive ship-shaped design creates a large contact area with the ground or foundation structure, allowing for uniform load distribution and reduced ground pressure, thereby enhancing the stability and reliability of the ground anchor[10-11].Most importantly, the ship-shaped anchor features a relatively large cavity that provides ample space for housing control system components, including sensors, control boards, and batteries, as shown in Figures 5 and 6.

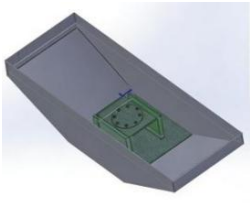


Fig. 5. Ground anchor cavity

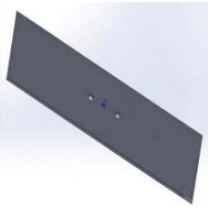


Fig. 6. Ground anchor top plate

III. Design of an Intelligent Anchor Monitoring and Early Warning System

A. System Architecture

The intelligent ground anchor proposed in this paper integrates force sensors and cable-type displacement sensors to enable synchronous acquisition of anchoring force and displacement. An embedded control board developed using the STM32 microcontroller employs the LoRa network to ensure low-latency data transmission in field environments. The overall system architecture is illustrated in Figure 7, comprising a data acquisition layer, a core control layer, and a LoRa remote communication layer.

The system simultaneously collects key parameters of ground anchors in real-time, processes them through edge computing, and uploads the data to the cloud management platform for monitoring.

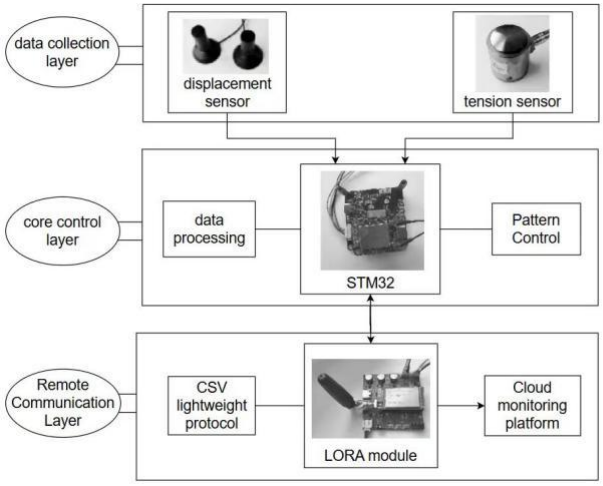


Fig. 7. Overall structure diagram of control system

B. Communication Scheme Design

The intelligent ground anchor communication solution employs a LoRa module that works in tandem with an STM32 controller. The LoRa module serves as the core for remote, low-power wireless transmission. The system collects real-time ground anchor status data using tension and displacement sensors, processes the data via the STM32, packages it using the lightweight CSV protocol, and transmits it via the LoRa module at fixed intervals. Its low-power characteristics enable intermittent operation, making it suitable for off-grid field environments with battery life spanning several months in continuous operation and several days in intermittent mode. The LoRa signal offers strong penetration and long transmission range, ensuring stable coverage across complex terrains surrounding transmission towers. Data is ultimately aggregated through a LoRa gateway and uploaded to a cloud platform, enabling remote real-time monitoring of anchoring force and displacement, overload alerts, and historical data queries.

IV. Control Section Design

A. Sensor Data Acquisition and Conversion

Digital acquisition of sensor analog signals and conversion of physical quantities. Utilizing the built-in ADC in Arduino, the analog voltage is read and the raw data is converted into actual engineering values according to the formula. This linear mapping method ensures that the measured values accurately correspond to the actual physical state[13].

B. Cam Mechanism Control

The cam control function achieves precise spring compression: First, the stepper motor drives the cam to rotate 120° to its maximum lift position; the remaining steps are monitored in real time to confirm completion of the movement; finally, the electromagnetic clutch engages to lock the cam, maintaining the compressed state for 500 ms to ensure full spring energy storage. The AccelStepper library is employed to implement trapezoidal speed curve control, preventing mechanical impact.

C. Communication Protocol Design and Power Consumption Optimization

The system outputs monitoring data in CSV format, with each data frame containing time, displacement, and force values, all rounded to two decimal places. This lightweight protocol is compatible with various serial analysis tools and can be directly integrated with cloud-based monitoring platforms, balancing real-time data availability with bandwidth consumption.

The system supports two operating modes for outdoor power supply environments. Continuous mode enables real-time monitoring, with mode switching achieved through pre-compiled commands to meet varying power consumption requirements across different application scenarios.

The code is as follows:

Enter the sensors simulated voltage value and set the system operating mode

Output the CSV-format data frame sent by Lora

(1) During system parameter initialization,
Set the maximum range of the displacement sensor
Set the maximum range of the tension sensor
-Set the reference voltage V
-Set the sending interval T
(2) Set the work mode to:
Continuous mode: Continuous monitoring
- -Interval mode: Specify the work time interval
(3) Set the main circulation process as follows:
-If the mode is intermittent
-If the current minute setting
-Execute monitoring and sending process
(4) The monitoring and transmission process is as follows:
Call the function `Collect Sensor Data`
Call the function to package CSV data
Send data frames via LORA
- delayed T

V. Strength Verification and Force Analysis of Ground Anchors for Transmission Tower

A. Strength Verification of Each Component of the Intelligent Anchor

Taking the design and verification of a 25-ton traction ground anchor as an example, the anchor parameters are:

TABLE I. List of Ground Anchor Parameters

Roof plate length /m	Bottom plate length /m	width /m	altitude /m
1.1	0.3	0.42	0.16

(1) Compressive strength verification of the main anchor body

The main anchor body is constructed from Q345 steel, with a strength of ≥ 305 MPa, and a maximum design load of F_{max}= *80 kN; cross-sectional area A = 0.451* m^2Actual compressive stress:

$$\sigma=\frac{Fmax}{A}=\frac{80\times10^3N}{0.451m^2}=0.177\ \text{MPa}\leqq \text{fc} \quad (1)$$

(2) Calculation of the tensile strength of the hazardous surface of the pull ring

The yield strength of Q345 steel at its maximum design load is F.$_{max}$= *100 kN. The safety factor is set to 2.0; allowable stress:*

$$[\sigma_1]=\frac{345}{3}=172.5\ \text{M P a}。 \quad (2)$$

Basic properties of the base material: Tensile strength: 470–630 MPa. For the weld area, ideally match the welding material to achieve a tensile strength of ≥ 500 MPa for the weld metal. Actual weld strength may be slightly higher or lower depending on the welding process. A safety factor of 2.0 is applied.

$$[\sigma_2]=\frac{470}{2=235\ \text{MPa},}[\sigma_2]>[\sigma_1] \quad (3)$$

$$[\sigma_{max}]=\frac{F}{A}=\frac{50\times10^3}{\pi\times15^2mm^2}=70.74\ \text{MPa}<[\sigma_1] \quad (4)$$

(3) Calculation of the tensile strength of the hazardous surface at the bottom structure of the anchor body

Permissible stress, maximum load as for the pull ring, hazardous cross-sectional area:

$$A=20\times193=3860mm^2$$

$$[\sigma_{max}]=\frac{F}{A=}\ \frac{50\times10^3N}{3860mm^{2=}\ 12.95\text{MPa}<}[\sigma_1] \quad (5)$$

(4) Calculation of shear strength for the hazardous surface of the anchor bodys bottom structure

The pulling force F = 100 kN. $\sigma_{b]}$ *For a value of 470 MPa, the allowable shear stress [τ] = 0.5*$\sigma_{b=235MPa}$

Shear area A = 2 × (6 × 193.9) mm^2= 2326.8mm^2

$$\tau=\frac{F}{A}=\frac{100\times10^3N}{2326.8mm^2}\approx42.98\text{MPa}<[\tau] \quad (6)$$

The bottom structure of the anchor meets the required strength requirements.

(5) Calculation of shear strength of the hazardous surface of the pull ring

The pulling force F = 100 kN.

σ_b *For a value of 470 MPa, the allowable shear stress [τ] is 0.5.* σ_{b} *= 235 MPa; shear area A = 2 × (3 × 60)* mm^2=360 mm^2

$$\tau=\frac{F}{A}=\frac{100\times10^3N}{360mm^2}\approx278\text{MPa}>[\tau] \quad (7)$$

The shear stress on the hazardous surface of the pull ring exceeds the allowable stress, thus failing to meet the strength requirements. Consequently, the thickness of the end plate of the pull ring is reduced to 5 mm.

$$\tau=\frac{F}{A}=\frac{100\times10^3N}{600mm^{2\approx}167\text{MPa}<[\tau]} \quad (8)$$

Therefore, the pull ring meets the required strength.

(6) Strength verification of the secondary ground anchor launch device

Analyze the resistance of the soil at the end of the secondary ground anchor:

The bearing capacity q of hard soil is approximately 200–500 kPa.

End force F_{tip}= q × cross-sectional area.

Lateral side drag:

Lateral friction force F_{side}= f × lateral area; hard soil: f ≈ 2–5 N/cm²

$$F_{side}=\pi\times d\times L=3.\ 14\times1\text{cm}\times10\ \text{cm}=31.4\ \text{cm}^2 \quad (9)$$

$$F_{total}=F_{tip}+F_{side}\approx150–300\text{N} \quad (10)$$

Therefore, the device is theoretically capable of anchoring secondary anchors into hard soil.

Contact Stress Verification:

$$\sigma_H=\sqrt{\frac{F\times E}{2\pi\times(1-\nu^2)}} \quad (11)$$

Substituting F = 1500 N, E = 206 GPa, and Poissons ratio ν = 0.3,

$$R_{eq}=\frac{r_b\times r_r}{r_b\times r_{r=6.67mm,}}\sigma_{H=}\ 1.21Gpa$$

Permissible contact stress for Q345 steel

modulus of section W：

$$W=\frac{\pi d^3}{32}=6.28\times10^{-7}\text{m}^3 \quad (12)$$

Torsional section modulus W_p：

$$W_p=2\ \text{W}=1.26\times10^{-6}\text{m}^3 \quad (13)$$

For W and W in the above formula:$_P$ Substituting yields

B. Finite Element-Based Stress Analysis of Ground Anchors

To get a better idea of the ground anchor's strength, we ran a finite element analysis on the whole anchor and its parts. As shown in Figure 8, when a 100 kN force is applied, the max stress hits 2.932×10^8 MPa, which is still below the yield limit. So the material and structure are solid. The strain mainly shows up in the pull ring and at the bottom of the anchor, and there are no weak spots, meaning it meets the design goals.

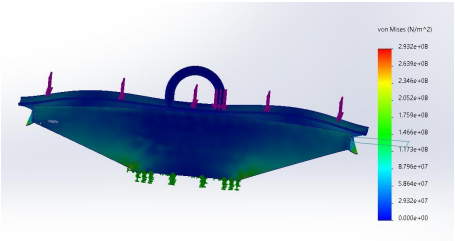


Fig. 8. Stress distribution and strain condition of the main body of the ground anchor

A finite element analysis was conducted on the pull ring under a total tension force of F = 100 kN. The reinforcing ribs and the pull ring share the load collaboratively: the four ribs collectively bear 30 kN, with each rib supporting 7.5 kN. Consequently, the total stress on the pull ring is 70 kN. Figure 9 illustrates the stress and strain distribution under a 70 kN force for Q345 steel with a yield limit of 345 MPa. The maximum stress shown in the figure is 3.237×10^8 Pa, below the yield limit of Q345 steel, confirming the suitability of both material and structure. The reinforcing ribs effectively distribute 30% of the total load, resulting in a relatively uniform stress distribution without significant local stress concentrations. The strain distribution indicates coordinated structural deformation with no hazardous sections identified, and the maximum stress is slightly below the materials yield limit, demonstrating structural safety under static loads and validating the material-structure combination.

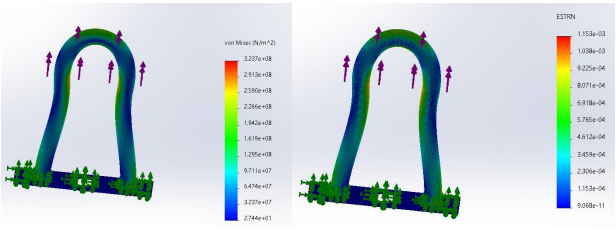


Fig. 9. Stress and strain conditions of the anchor pull ring

VI. Conclusion

Addressing the limitations of traditional transmission tower ground anchors in real-time monitoring, dynamic regulation, and construction safety, this paper proposes an intelligent ground anchor solution for transmission towers that integrates mechanical structure optimization with an intelligent control system. The movable pull ring is fabricated from Q345 steel with added reinforcing ribs and incorporates a spoke-type force sensor to enable precise real-time monitoring of anchoring tension. A non-load-bearing secondary anchor body and a spring-powered release mechanism are incorporated, with a stepper motor driving the cam mechanism to facilitate unmanned automatic deployment and reduce construction risks. The anchor body cavity is optimized to provide a sealed environment for the control system.

The intelligent control system Based on an STM32 microcontroller, it integrates rope displacement sensors and force sensors, with data transmitted remotely via a LORA module. Utilizing low-power algorithms, the system supports both continuous and intermittent operation modes, making it suitable for outdoor environments. The strength of critical components is verified using classical mechanical formulas to ensure structural reliability, while finite element analysis confirms the stress distribution of the anchor under a 100 kN load, guaranteeing structural safety. The practical value of the intelligent anchor lies in its real-time monitoring of anchoring force and displacement, enabling early detection of failure risks and reducing the likelihood of tower collapse accidents. It cuts manual inspection costs by 40%-50% and enhances construction efficiency in complex terrains.

References

[1] Nie Feng. Research on the Stability and Online Monitoring System of Transmission Iron Towers [J]. Shihezi Science and Technology, 2024, (04):69-70.

[2] Mukherjee S,Babu S L G . Analysis of anchor uplift capacity in spatially varying soil using MARS model [J]. Acta Geotechnica, 2024, 19 (8): 5089-5109.

[3] Ma Xinghe, Zhu Xing.Remote Monitoring System for Transmission Tower Tilt [J].Journal of Ordnance Equipment Engineering, 2023, (201):206-211.

[4] Ma Jian, Yang Mangsheng, Wang Henan, et al.Comprehensive Evaluation System for the Safety of Transmission Line Tower Foundations [J].Journal of Safety and Environment, 2020,20(1):25-30.

[5] Xu C, Li Q, Zhou Q, Xu C, Li Q, Zhou Q, et al. Power line-guided automatic electric transmission line inspection system[J]. IEEE Transactions on Instrumentation and Measurement, 2022, 71:1–18.

[6] Zhang Xinchun, Wang Zhangqi, Ou Jian.Research on the optimized design of ground anchor structures for transmission line guy wires [J].Grid and Clean Energy, 2014,30(12):10-14.

[7] Zhao Huanyu.Research and Development of an Intelligent Leveling Instrument Detection Device [J].Measurement Technology, 2015, (11):44-47.

[8] Feng Liang, He Rui, Yang Hong.Experimental Study on the Tensile Resistance Testing of Anchoring Devices in Mobile Anchor Projects [J].Product Reliability Report, 2024, (09):105-107.

[9] Yang Yan, Qin Zuowei, Liao Yuan, et al.Mechanical Property Analysis of the Steel-Concrete Composite Section in Partially Anchored Cable-Stayed Bridge [J].Bridge Construction, 2024,54(04):93-98.

[10] Jiang Wenqiang, An Liqiang, Wang Yedi, et al.Low-temperature tensile and impact tests on angle steel, the primary material for transmission towers [J].Vibration, Testing and Diagnosis, 2017,37(5):1040-1045,1072.

[11] He Jiasheng. Research on State Monitoring and Risk Assessment Models for Transmission Tower [D].Nanning, Guangxi: Guangxi University, 2024.

[12] Wang S, Tan W, Yang T, Wang S, Tan W, Yang T, et al. High-voltage transmission line foreign object and power component defect detection based on improved YOLOv5[J]. Journal of Electrical Engineering & Technology, 2023: 851–866.

Research on Application and Practice of Train Control Center System

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***Abstract*—The Train Control Center (TCC) is the core ground subsystem of the Chinese Train Control System Level 2 (CTCS-2), generating movement authority for trains. While the TCC's architecture and interface protocols are well documented, the existing literature lacks empirical validation from field deployment and systematic fault analysis. TCC is applicable to interlocking stations, block posts and relay stations on lines equipped with the CTCS-2 and Chinese Train Control System Level 3 (CTCS-3) train control systems, and can also be used for CTCS-0 stations connected to CTCS-2 and CTCS-3 lines. On the basis of introducing the structure and functions of the TCC system, this paper expounds the interface modes between TCC and other signal subsystems, and analyzes the TCC fault analysis and handling methods from the perspective of engineering practice. This work provides a reference for both TCC system designers and field maintenance personnel.**

Keywords—train control center, Chinese train control system, balise

I. INTRODUCTION

The Train Control Center (TCC) system controls track circuit coding and active balise information according to the information such as the position of each train under its jurisdiction, interlocking routes and temporary line speed limit status [1], so as to provide movement authority for trains. According to the station type, TCC is divided into Station TCC and Relay Station TCC.

TCC is configured with communication interfaces for the ZPW-2000 series track circuits, Computer Based Interlocking (CBI), Temporary Speed Restriction Server (TSRS) [2], adjacent TCC, Lineside Electronic Unit (LEU), Centralized Signaling Monitoring system (CSM) and Centralized Traffic Control (CTC) equipment. The interface configuration with other external equipment varies with different types of TCC, as shown in Fig. 1.

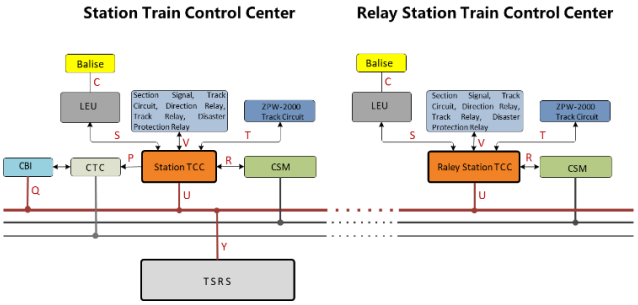

Fig. 1. Interface Configuration of the Train Control Center.

(Notes: P: Interface with CTC; Q: Interface with interlocking slave computer; R: Interface with centralized monitoring system; S: Interface with LEU; T: Interface with track circuit; U: Interface of safety information network between train control centers)

Station TCC is installed at high-speed railway interlocking stations, and directly interfaces with CBI, track circuits, TSRS, adjacent TCC, LEU, CTC and CSM [3], and administers the Relay Station TCC within its jurisdiction. Relay Station TCC is installed at signal relay stations, and directly interfaces with track circuits, TSRS, master station TCC, adjacent TCCs, LEU and CSM [4]. Relay Station TCC must be subordinate to a Station TCC, receives line direction information from the Station TCC, and sends the corresponding track section status information to its subordinate Station TCC.

Each TCC device is assigned a unique independent device number as its digital identifier, which is unique nationwide and also used as the device identifier for secure communication. The TCC software adopts modular, structured and standardized design, enabling the independent setting of configuration data and programs, and ensuring that changes in the main line data will not lead to modifications of the TCC software.

II. STRUCTURE OF THE TRAIN CONTROL CENTER SYSTEM

The TCC is mainly composed of a safety host unit, a communication interface unit, a drive and acquisition unit, an auxiliary maintenance unit and a redundant power supply unit, as shown in Fig. 2.

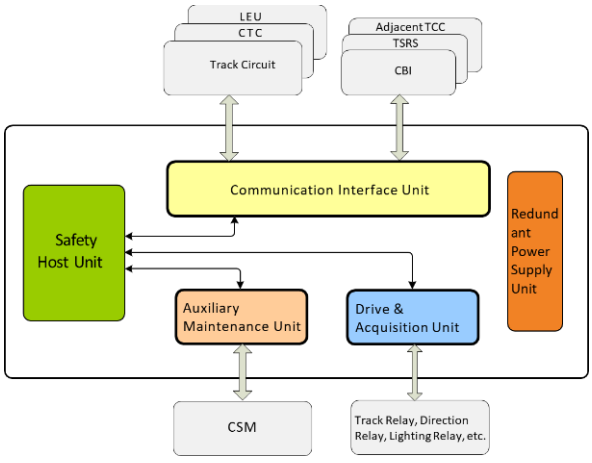

Fig. 2. Composition of TCC Equipment.

The TCC host adopts a 2-out-of-2-by-2 safety computer platform complying with the Fault-Safe principle or other platforms with safety redundant structure as the main logic operation unit [5]. The TCC host unit communicates with the redundant communication channels, communication interface units and drive and acquisition units. The TCC communication interface unit communicates with external equipment such as track circuits, LEU, CBI, TSRS and CTC through redundant communication channels. The TCC drive and acquisition unit adopts a safety redundant drive and acquisition hardware structure to realize the driving of external relays and the acquisition of their status.

The TCC is configured with redundant power supply units to provide reliable power supply for each unit in the TCC. The TCC auxiliary maintenance unit is equipped with a display, keyboard and mouse, all of which are uniformly installed in the TCC cabinet.

A. *Safety Host Unit*

The safety host consists of two parallel systems operating in master-slave mode simultaneously. Each system adopts a Fault-Safe dual-CPU processor, referred to as the logic control unit, which is used to complete the TCC logic operation and the management of the software and hardware of the train control system. The two systems exchange information through a high-speed channel established by a parallel interface (FIFO) to realize synchronization and switching between the dual systems [6]. As the core equipment of the TCC, the safety host is responsible for the logical processing and system management of the TCC.

B. *Maintaining the Integrity of the Specifications*

The drive and acquisition unit adopts an intelligent controller composed of Fault-Safe dual CPUs. Its output circuit is designed in accordance with the Fault-Safe principle, and the input acquisition circuit can detect the fault of the input circuit through an effective self-test function to ensure the security of input information. The output drive adopts dynamic and static signal comparison and verification by dual CPUs to ensure the security of output. The electronic terminal adopts a parallel dual-system structure, so that the fault of a single board will not affect the input and output of the system. The output directly drives safety relays, and the input collects DC 24V signals [7]. The drive and acquisition unit collects the status of external relays and transmits it to the host unit, and drives external relays upon receiving commands from the host unit.

C. *Communication Interface Unit*

The LEU communication interface unit is installed in the main TCC equipment cabinet. The LEU communication interface board is used for secure data transmission between the TCC host and LEU equipment, transmitting message data to the LEU and receiving balise status data transmitted by the LEU at the same time. The CTC interface board is only installed in the main TCC equipment cabinet of stations or block posts configured with CTC station machines [8]. The CTC communication interface board is used for communication between the TCC and CTC equipment, transmitting the status of section track circuits and low-frequency code information to the CTC equipment, as well as the section operation direction information.

D. *Auxiliary Maintenance Unit*

The auxiliary maintenance unit is used to monitor the working status of the TCC, record the communication data of the TCC, and send data to the CSM. The display equipped with the auxiliary maintenance unit serves as the terminal display device for monitoring and maintenance.

E. *Redundant Power Supply Unit*

The redundant power supply unit is configured with independent redundant power modules. The power supply installed in the main equipment cabinet converts the AC 220V power supplied by the power supply panel into two independent channels of 24V logic power and 24V drive-acquisition power, which supply power to the host unit, input/output interface unit, LEU communication interface unit and track circuit communication interface unit respectively. The redundant power supply installed in the LEU cabinet converts the AC 220V power supplied by the power supply panel into one channel of 24V logic power to supply power to the LEU equipment independently.

F. *Power Interface*

The TCC equipment is supplied with two independent channels of AC 220V power by external power supply panel equipment. For TCC configured with remote LEU, an adjustable power supply is also required.

The TCC classifies external equipment into different priority levels according to their importance. The higher the equipment priority, the greater the impact of the equipment interface status on the system integrity. The external device interfaces that affect TCC system switching include: TC, CBI, TSRS, LEU, adjacent station TCC, IO and CTC. The priority classification of the TCC is shown in Fig. 3.

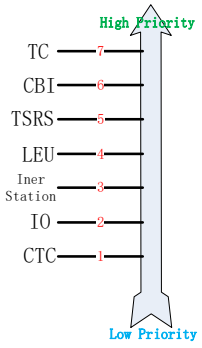

Fig. 3. Priority Classification of the Train Control Center.

III. Functions of the Train Control Center

A. *Temporary Speed Limit Function*

The TCC is in an uninitialized state after startup. When the TCC receives the temporary speed limit initialization command issued by the TSRS, it executes the initialization command and switches to the initialized state. The TCC shall maintain the initialized state in case of communication interruption with the TSRS, and send the initialization state to the TSRS every cycle. When the TCC receives a temporary speed limit command

requiring setting verification from the TSRS, it shall check the validity of the temporary speed limit command according to the current execution status of the temporary speed limit, and feed back the setting verification result of the temporary speed limit to the TSRS. The verification contents are shown in TABLE I.

TABLE I. TEMPORARY SPEED LIMIT VERIFICATION CONTENTS

Content	Result	Status
Whether the kilometer post without coverage mark is within the jurisdiction	Yes	Normal
	No	Abnormal
For main line speed limit: the minimum speed is 45km/h, in steps of 5km/h, up to the maximum line speed; For siding speed limit: the speed is divided into two grades of 45km/h and 80km/h	Yes	Normal
	No	Abnormal
Whether there is regional overlap	Yes	Abnormal
	No	Normal
Whether the number of speed limits set within the one-way jurisdiction of the balise exceeds 3	Yes	Abnormal
	No	Normal
Whether the TSRID number is duplicated with the temporarily speed limit that has been successfully executed	Yes	Abnormal
	No	Normal

B. Balise Message Generation and Transmission Function

The TCC sends the balise permission-to-pass message for large-number turnouts when communication with the CBI is interrupted, and does not send the balise data packet for large-number turnouts during reverse operation. The balise message for large-number turnouts is sent when the following conditions are met; otherwise, the permission-to-pass message is sent:

- The communication between TCC and CBI is normal;
- The initialization of the temporary speed limit is completed;
- The UUS signal is displayed for the lateral receiving signal;
- The length of the route movement authority exceeds the braking distance check range (the braking distance for decelerating to 0km/h at the lateral allowable speed of large-number turnouts);
- There is no temporary speed limit lower than the lateral allowable speed of large-number turnouts within the lateral route range and the braking distance of the departure section (the braking distance for decelerating from the maximum speed at the station exit, not exceeding 305km/h, to 45km/h);
- All block sections (routes) between the block section where the balise for large-number turnouts is located (excluded) and the signal machine protecting the large-number turnouts are idle.

After the TCC sends the balise data packet for large-number turnouts, if it detects that the conditions for sending the data packet are not met, it shall immediately stop sending the data packet. If the unmet conditions are Item 4 or Item 5 above, the TCC shall simultaneously send a corresponding route downgrade command to the CBI.

C. Station Signal Machine Downgrade Function

When the turnouts of the siding route protected by the signal machine are turnouts below No.18 or there is a temporary speed limit lower than 80km/h on the lateral receiving route, the TCC shall control the approach section to send the UU code after the lateral receiving signal is displayed, and send an inbound signal machine downgrade command to the CBI. When there is a temporary speed limit lower than 80km/h in the first block section outside the inbound (receiving route) signal machine (excluding the range between the balise and the signal machine), the code sending of the approach section is downgraded to UU, and an inbound signal machine downgrade command is sent to the CBI. When arranging a siding through route with straight departure conditions, if there is a temporary speed limit lower than 80km/h on the route (including the departure route) or within the braking distance of the departure section (decelerating from 80km/h to 45km/h), the TCC shall send an inbound signal machine downgrade command for displaying UU to the interlocking equipment.

When there is a temporary speed limit lower than 80km/h in the siding area, on the lateral departure route and within the L1 range of the departure section (decelerating from 85km/h to 45km/h), the TCC shall control the corresponding departure track to send the UU code after the lateral departure signal is displayed. If the first departure section sends HU, H, JC, HB or UU codes, the TCC shall still control the corresponding departure track to send the UU code after the lateral departure signal is displayed.

D. Track Section Status Judgment Function

When the TCC does not execute the section occupation logic check (including when the TCC section occupation logic check function is turned off), the collected track section status is directly used as the logical status of the track section, and the judgment logic is shown in TABLE II. An alarm shall be triggered if the inconsistency between the GJ status and the TC status lasts for more than 4.5 seconds due to fault.

TABLE II. TRACK SECTION STATUS JUDGMENT LOGIC

Collected GJ Status	TC Communication Status	Collected Track Section Status
Clear	Occupied	Occupied
Occupied	Clear	Occupied
Occupied	Occupied	Occupied
Clear	Clear	Clear

When the TCC performs the section occupation logic check, it shall first obtain the collected track section status, and judge the logical status of the track section according to the train operation sequence. The basic statuses include fault occupation, normal occupation, shunt loss and idle. The basic judgment principles are as follows:

- Fault occupation: The train does not occupy the block section, but the equipment status of the block section is occupied due to fault;
- Normal occupation: The train occupies the block section, and the equipment status of the block section is occupied;
- Shunt loss: The train occupies the block section, but the equipment status of the block section is idle due to fault;
- Idle: The train does not occupy the block section, and the equipment status of the block section is idle.

E. Track Circuit Coding Function

When the train route signal is not displayed, the TCC shall send the HU code or detection code to the tracks, and send the detection code to the turnout sections. For stations where only EMU are operated, if no active balise is installed at the main line departure signal machine, the detection code shall be sent to the turnout sections when a main line transfer departure route is arranged. For a block section composed of multiple track sections, the normal code shall be sent to the section where the train is located and all block sections in front of the train's running direction, and the detection code shall be sent to all sections behind the train. If the low-frequency code of the boundary track circuit transmitted by the adjacent station TCC is the detection code due to fault, the HU code shall be sent to the boundary section of the local station.

After the train receiving route signal is displayed, the TCC controls the relevant sections of the receiving route to send codes according to the status of the departure signal machine, and the code sending of the receiving route sections is consistent with that of the track sections. After the train departure route signal is displayed, the departure track sends codes according to the status of the departure signal machine and the status of the first departure section, and the code sending of the departure route sections is consistent with that of the first departure section. When a shunting receiving route is arranged or the shunting departure route signal is displayed, the detection code shall be sent to the route sections.

For tracks equipped with protection sections, when the track section in front of the train's running direction is occupied, all track sections behind in the track shall switch to code sending in the opposite direction (except for shunting routes). When the code sending direction of the protection section is consistent with the direction of the protected signal machine, the H code shall be sent to the protection section if the train signal of the signal machine is not displayed; if the signal is displayed, the code sending principle of the protection section is consistent with that of the track.

F. Track Circuit Code Sending Direction Control Function

If a direction switching relay (FQJ) is installed for the station track circuit to realize code sending direction switching, the TCC drives the corresponding FQJ according to the train route to control the consistency between the track circuit direction and the route direction. The FQJ is driven to drop for the forward operation of station track sections, and pick up for the reverse operation. The TCC collects the status of the FQJ, and shall send the HU code or JC code and an alarm message about the fault at the same time if there is an inconsistency between the drive and collection statuses. The default direction of station track sections is the positive line operation direction. After the TCC equipment is started and before the TCC receives route information, the code sending direction of station sections is the default direction, and the FQJ is driven to drop. The FQJ maintains the original code sending direction for the track sections on the station route after the front section is normally occupied or the local section is unlocked without fault.

For a track composed of multiple track sections, when the train occupies the track section in front of its running direction in the track, the track sections behind shall switch to code sending in the opposite direction (except for shunting routes) to realize the double-ended code sending function of the track. When a shunting receiving route is established, if the track section in front of the corresponding track is occupied under the existence of the shunting receiving route, the HU code shall be sent to the track sections behind, and the code sending direction of all track sections in the corresponding track shall remain unchanged to avoid fault.

G. Section Signal Machine Control Function

The TCC shall drive the LJ, UJ, LUJ and HJ relays to realize the lighting of section signal machines. The TCC shall control the section through signal machines to turn off the lights when the current operation direction is reverse or the section direction is unknown due to fault. The TCC shall collect the filament status of the through signal machines, and shall perform logical processing according to TABLE III and output an alarm message about the fault in case of filament breakage of the signal machines. For sections without through signal machines, the red light filament breakage transfer of the inbound signal machines shall not be performed for fault avoidance.

TABLE III. PROCESSING OF SIGNAL MACHINE FILAMENT BREAKAGE FAULT

Signal Machine Drive	**Fault Condition**	**Outdoor Signal Machine Display after Fault**	**Code Sending of Track Section outside the Signal Machine**
H	H	Red light transfer	Detection code
U	U	Lights off	Filament breakage fault has no impact on the track circuit code sending logic
LU	L	Downgraded to display U lamp	
	U	Lights off	
L	L	Lights off	

IV. INTERFACES OF THE TRAIN CONTROL CENTER

The TCC installed at stations and block posts directly interfaces with CBI, CTC, TC, relays, TSRS, adjacent TCC, LEU and CSM; the TCC installed at relay stations directly interfaces with TC, relays, TSRS, adjacent TCC, LEU and CSM. The TCC interface information transmission is shown in Fig. 4.

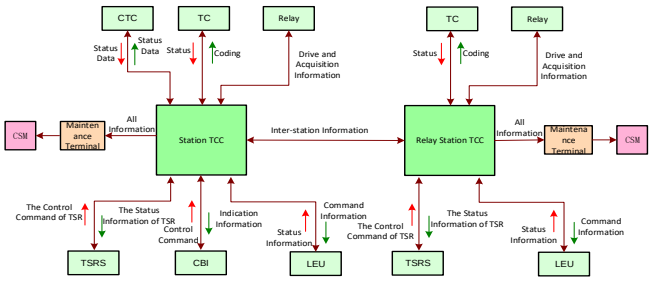

Fig. 4. TCC Interface Information Transmission.

The TCC and the CTC system are connected by RS-422 cable dual-channel cross redundancy, and point-to-point direct connection is realized through the RS-422 serial interface on the CTC interface module. Information is sent at a fixed cycle of 500ms, adopting the RSSP-I safety protocol to prevent communication fault.

The TCC and the TSRS are connected by an Ethernet interface, with both equipment and communication network configured in redundancy. They access the secure data network through the 100M Ethernet port of the interface module. Each TCC communicates with only one TSRS, with information sent at a cycle of 500ms, adopting the RSSP-I safety protocol to avoid fault.

The TCC and the CBI are connected by an Ethernet interface, with both equipment and communication network configured in redundancy. They access the secure data network through the 100M Ethernet port of the interface module. Each TCC communicates with only one CBI, with information sent at a cycle of 250ms, adopting the RSSP-I safety protocol to prevent fault.

Adjacent TCC are connected by an Ethernet interface, with both equipment and communication network configured in redundancy. They access the secure data network through the 100M Ethernet port of the interface module. Each TCC can communicate with a maximum of 8 adjacent TCC, with information sent at a cycle of 500ms, adopting the RSSP-I safety protocol to avoid fault.

The TCC and the ZPW-2000A track circuit system are directly interfaced with the track circuit communication panel via the CAN bus, and the CAN communication bus of the ZPW-2000A is connected through the CAN interface on the TC interface module with a rate of 1Mbps. Each TC interface module is connected to 10 ZPW-2000A communication panels, with master-slave information interaction at a fixed cycle of 250ms. The TCC acts as the master device and the track circuit as the slave device, with a dedicated protocol adopted to ensure the security of interface data and prevent fault.

Each TCC is connected to each LEU by 10Mbps Ethernet dual-channel redundancy, with master-slave information interaction at a cycle of 250ms. The TCC acts as the master device and the LEU as the slave device, adopting the SACEM safety protocol to avoid communication fault.

The TCC is connected to the centralized monitoring network, and can communicate with the CSM station machine, track circuit maintenance machine and signal supervision terminal at the same station. It accesses the CSM network through the 100Mbps Ethernet port of the TCC maintenance terminal. The TCC serves as the server, and the CSM station machine, track circuit maintenance machine and signal supervision terminal act as clients. Information is sent in two ways: at a cycle of 1s and real-time sending upon generation. The network is configured with redundancy to reduce the risk of fault.

V. TCC Fault Analysis

The fault information of TCC equipment shall include two parts: fault description and fault data. The fault description shall contain: time, location/equipment name, software version, fault phenomenon (alarm screenshot) and other contents related to the fault.

A. Fault Analysis of Direction Switching Failure

According to the control principle of the train control section operation direction, the entire process of direction switching can be divided into four stages:

- The CBI of the original receiving station issues a direction switching command to the TCC of the original receiving station;
- After checking that the section conditions meet the direction switching requirements and no fault exists, the TCC of the original receiving station sends a direction switching request to the TCC of the adjacent direction switching response station;
- After checking that the section conditions meet the direction switching requirements and no fault exists, the TCC of the direction switching response station drives the ZGFJ or FGFJ to actuate the FJ, and the FJ drives the FQJ to switch the port direction of the local station to receiving, and sends a direction switching permission to the original receiving station;
- The original receiving station also drives the ZGFJ or FGFJ to actuate the FJ, and the FJ drives the FQJ to switch the port direction of the local station from receiving to departure, completing the change of the section operation direction without fault.

Attention: The relevant relays of the original receiving station and departure station shall complete the corresponding actions within 13 seconds after the ZGFJ or FGFJ is driven respectively; if the time exceeds 13 seconds, the direction switching fault occurs, and the section operation direction of the station port remains unchanged.

B. Fault Analysis of Inconsistency between Drive and Acquisition Statuses

The TCC collects the status of the relay to confirm whether the relay action is in place after outputting the relay drive command. If the TCC collects a relay status inconsistent with the drive status, a fault of inconsistency between drive and acquisition statuses will be reported, and an alarm message about the fault will appear on the TCC maintenance terminal.

If only one of the System A and System B of the TCC maintenance terminal is found to have the fault of inconsistency

between drive and acquisition statuses after inquiry, it is considered that the drive board, acquisition board or acquisition channel of the faulty system has a fault. If both System A and System B are found to have the inconsistency between drive and acquisition statuses, it indicates an abnormal relay collection status caused by fault, and the status of the relevant relays needs to be investigated to eliminate the fault.

VI. Conclusion

The TCC is a key component of the ground subsystem of the CTCS-2 train control system, mainly composed of a safety host unit, a communication interface unit, a drive and acquisition unit, an auxiliary maintenance unit and a redundant power supply unit. It has the functions of temporary speed limit control, balise message generation and transmission, station signal machine downgrade, track section status judgment, track circuit coding, track circuit code sending direction control, section signal machine control and section operation direction control, and interfaces with systems and equipment such as CBI, CTC, TC, relays, TSRS, adjacent TCC, LEU and CSM. The redundant design of the system effectively reduces the risk of single-point fault. Accurate and rapid analysis and handling of TCC fault are the key to ensuring the stable and reliable operation of the system, and also the basic guarantee for the safe and on-time operation of China's high-speed railways.

References

[1] Sheng Luo, Tao He. Modeling and Analysis of Train Control Center Simulation System Based on CPN[C]//2025 International Conference on Electrical Automation and Artificial Intelligence. 2025:616-619.

[2] Yingcai Zhao, Xiaochun Wu. Information Security Risk Assessment Approach for Train Control Center and Interlocking Integrated System Based on FUCOM-CRADIS[C]//Proceedings of the 6th International Conference on Electrical Engineering and Information Technologies for Rail Transportation: Safety Assurance and Operation Maintenance Technology of Rail Transportation System. 2024:549-557.

[3] Lei Yuan, Yinghui Li, Guodong Wei, et al. Research on a Network Diagnosis Method for a Train Control Center and Interlocking Integrated System Based on a Fuzzy Broad Learning System Model[J]. Electronics,2025,14(4). DOI:10.3390/electronics14040691.

[4] Anyi Tang, Gang Niu, Xuanrui Zhu, et al. A Three-Interface Data-based Wireless Communication Timeout Fault Diagnosis Method for China Train Control System[C]//2023 Global Reliability and Prognostics and Health Management Conference (PHM-Hangzhou 2023). 2023:1-4.

[5] Haotong Xu, Kaicheng Li, Lei Yuan, et al. Scenario-based requirements analysis for Chinese new train control system[C]//International Conference on Frontiers of Traffic and Transportation Engineering (FTTE 2022). 2022:123402R-1-123402R-9.

[6] Lijuan Shi, Anping Hang, Zuoning Jia, et al. Risk Analysis of Train Control System Based on Fuzzy Mathematics and Bayesian Network[C]//Proceedings of the 4th International Conference on Electrical and Information Technologies for Rail Transportation: Rail Transportation System Safety and Maintenance Technologies. 2021:607-619.

[7] Qixuan Kang, Chunming Li, Linguo Chai, et al. Modeling and Analysis of Switching Process of Train Control System Based on UML and HTCPN[C]//2023 China Automation Congress (CAC). 2023:4079-4083.

[8] Investigators at Beijing Jiaotong University. The Search-based Mutation Testing of the Chinese Train Control System Level 3 On Board a Train Control System[J]. Intelligent Transportation Systems,2022(Feb.23):3-4.

Research on the Neural Patterns of Altitude-Induced Brain Fatigue Based on fNIRS

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***Abstract*—This study recruited 40 healthy participants and employed functional near-infrared spectroscopy (fNIRS) to collect hemodynamic signals during a VR-simulated altitude fatigue task, aiming to identify differences in brain network and modal controllability characteristics between fatigue and non-fatigue states. The results revealed that the goodness of fit for the linear negative correlation between modal controllability and nodal strength decreased substantially under fatigue conditions. Meanwhile, modal controllability in the right premotor cortex (PMCr) was significantly elevated ($p < 0.001$), whereas that in the prefrontal cortex (PFC), primary motor cortex (M1), and supplementary motor area (SMA) was significantly reduced ($p < 0.01$). Brain network analysis further showed that functional connectivity of the SMA was significantly enhanced under fatigue ($p < 0.001$), with both inter-density and intra-density properties significantly increased, while the clustering coefficient was significantly decreased ($p < 0.001$). These findings demonstrate that fNIRS, combined with brain functional network and modal controllability theory, can effectively characterize the neural signatures of altitude-induced brain fatigue, providing preliminary methodological support for the development of an altitude brain fatigue analysis system.**

Keywords—brain fatigue, fNIRS, physiological pattern analysis

I. Introduction

High-altitude work is an indispensable operational scenario in modern aviation, power grid inspection, and related fields [1]. However, factors like hypoxia and high cognitive workload at altitude can readily induce mental fatigue. This type of fatigue is characterized by strong insidiousness and pronounced cumulative effects, often causing attentional lapses and delayed judgment before being noticed by the workers themselves, thereby becoming a major contributor to safety incidents [2]. Currently, detection of high-altitude mental fatigue still heavily relies on subjective scales and behavioral assessments, lacking objective indicators that directly reflect changes in neural states [3], which makes it difficult to identify early signs of fatigue. Consequently, uncovering objective neural signatures of high-altitude cerebral fatigue has become an urgent imperative for ensuring operational safety.

From the perspectives of brain networks and control theory, efficient brain function depends on dynamic reconfiguration and switching among regions within neural functional networks [4]. By characterizing functional connectivity and nodal attributes between brain regions, functional brain networks can reveal inter-regional information integration patterns [5]. Meanwhile, network control theory further introduces the concept of "modal controllability," which quantifies a brain region's capacity to drive the network toward easily transitioning neural modes—that is, nodes with higher modal controllability can steer brain states into distinct dynamic patterns with lower energetic cost [6]. Therefore, integrating brain functional networks with modal controllability holds promise for uncovering the neural mechanisms of high-altitude cerebral fatigue from a novel dimension of dynamic controllability in network topology.

Among current neuroimaging modalities, functional near-infrared spectroscopy (fNIRS) is particularly well-suited for brain function monitoring in high-altitude, high-pressure environments, owing to its non-invasiveness, portability, and strong resilience to motion artifacts [7]. To date, fNIRS has been successfully applied to brain function studies in everyday scenarios such as emotion assessment and fatigue-related driving [8]. However, existing studies have predominantly focused on functional connectivity analysis or regional activation differences under normal conditions, neither

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incorporating brain control theory nor addressing the unique characteristics of cerebral fatigue under high-altitude hypobaric conditions. This limits our in-depth understanding of the neural mechanisms underlying high-altitude mental fatigue and hinders the development of portable neural warning technologies for real-world applications.

To this end, the aim of this study is to use fNIRS to construct participants' brain functional networks and modal controllability features under simulated high-altitude conditions, thereby revealing the neural patterns of high-altitude mental fatigue, which will provide new perspectives and evidence for the objective monitoring and early warning of mental fatigue in high-altitude workers.

II. Materials and Methods

A. Participants

This study recruited 40 undergraduate and graduate students from Soochow University. All participants were right-handed, in good health, and had no history of psychiatric or cerebrovascular diseases. On the day of the experiment, they were asked to refrain from consuming caffeine or other stimulant beverages. Written informed consent was obtained from all volunteers before the experiment. This study was approved by the Ethics Committee of Soochow University (ECSU) (SUDA20210909H01). All participants provided written informed consent, and all procedures were performed in accordance with the Declaration of Helsinki.

B. Experimental Protocol and Data Acquisition

1) Experimental Procedure

A VR-based high-altitude work simulation task was employed to induce cerebral fatigue. The VR fatigue-induction task was administered at one of three time points: before commencing work in the morning, before dinner in the afternoon, or before rest in the evening.

The experimental procedure is shown in Fig.1. Before and after the VR task, participants completed the Multidimensional Fatigue Inventory (MFI-20) and underwent behavioral tests including the N-back task (N=1) and a mental rotation (MR) task. During both the VR task and the behavioral tests, a functional near-infrared spectroscopy (fNIRS) device was used to record cerebral hemoglobin signals. No rest was given between the VR fatigue induction task and the behavioral tests. The MFI-20, N-back, and MR tasks were primarily used to assess changes in mental fatigue levels before and after the experiment, while the cerebral hemoglobin signals were used to capture physiological information during fatigue development, thereby enabling analysis of the neural patterns of fatigue.

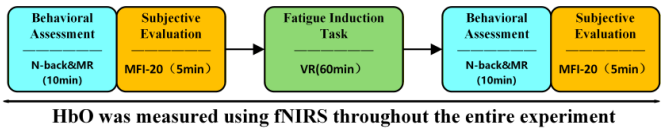

Fig. 1. Experimental procedure

2) NIRS Data Acquisition

Cerebral hemodynamic activity was recorded using a multi-channel fNIRS system (Brainsight C01B, Suzhou, China) at a sampling rate of 14.29 Hz (Fig.2.(a)). Based on the Brodmann cortical parcellation system [9], the optode arrangement comprised 15 light sources and 16 detectors, yielding 50 valid measurement channels. These channels were further grouped into several regions of interest (ROIs) covering four functional brain areas: the prefrontal cortex (PFC), premotor cortex (PMC), supplementary motor cortex (SMA), and primary motor cortex (M1). The correspondence between ROIs and measurement channels is detailed in TABLE I, where l, m, and r denote left, medial, and right, respectively. Channel positioning followed the international 10–20 system [10], and the specific channel layout is illustrated in Fig.2.(b).

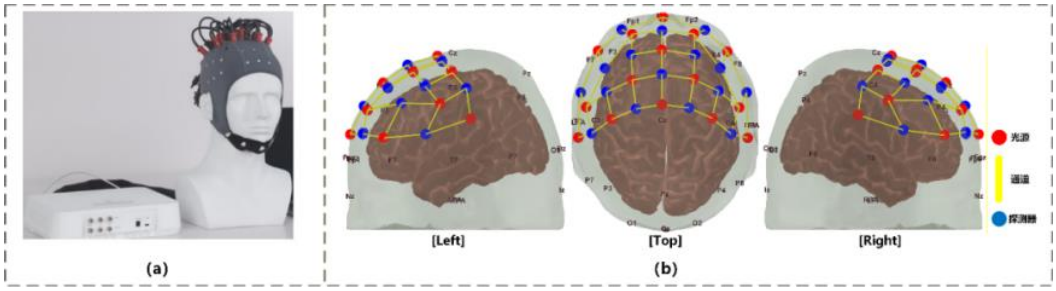

Fig. 2. fNIRS equipment and channels. (a): fNIRS device (b): fNIRS device chann

TABLE I. Corresponding Device Channels for ROI Brain Regions

ROI name	Channels within ROI
PFCl	4, 6, 10, 11, 14
PFCm	1, 2, 3, 8, 9, 15, 16, 17
PFCr	5, 7, 12, 13, 18
PMCl	23, 24, 28, 31, 32
SMA	19, 20, 25, 26, 27, 33, 34

PMCr	21, 22, 29, 30, 35
M1l	37, 38, 43, 45, 46
M1m	39, 40, 41, 47, 48
M1r	36, 42, 44, 49, 50

3) Task Paradigms

The VR fatigue-induction task is illustrated in Fig.3. In the fatigue task, participants first stood on a flat wooden plank to simulate the narrow movement space typical of high-altitude work. The VR headset then displayed a narrow beam platform in various scenarios, with a simulated net held in the right hand and birds continuously moving around. Participants were required to navigate along the simulated beam while capturing the moving birds with the net to earn points. Task difficulty increased progressively over time as scores accumulated.

The procedural diagrams for the N-back and mental rotation (MR) tasks are shown in Fig.4. During the N-back (n=1) behavioral task, a blue square was displayed at the center of the monitor every 3 s, accompanied by a randomly spoken letter. Participants compared whether the current and previous stimuli matched: they pressed key A when the current letter matched the previous one, and key S when the current square position matched the previous one. During the MR behavioral task, normal or mirror-reversed letter R stimuli were displayed on the monitor. Participants judged whether the two stimuli were identical after rotation, pressing key T for a match and key F for a mismatch. Response times and accuracy rates were recorded for each task.

Fig. 3. VR High-altitude simulation task

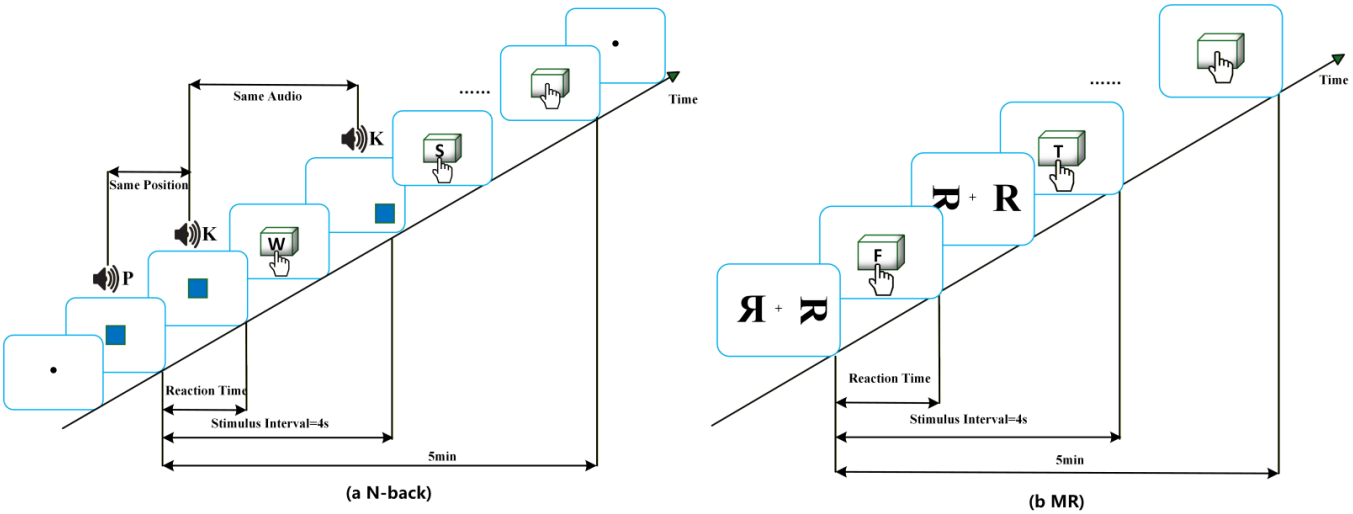

Fig. 4. Experimental paradigms of N-back task and MR test.

C. Data Processing and Analysis

1) Fatigue Level Classification

Based on the MFI-20 scale scores and the behavioral performance on the N-back and MR tasks, seven-dimensional feature vectors were constructed to characterize each participant's pre- and post-task performance. Z-score standardization was applied to unify the feature scales. Subsequently, an exhaustive search combined with K-means clustering was employed to partition the feature samples into non-fatigued and fatigued categories, with the silhouette coefficient [11] computed to evaluate clustering quality. The results indicated that the MR accuracy feature yielded the highest silhouette coefficient (0.5906) with balanced intra-cluster sample sizes. Accordingly, MR accuracy was selected for fatigue classification, with a threshold of 0.772 serving as the fatigue label criterion.

2) Data Preprocessing

For the raw fNIRS physiological data, the mean interpolation method [12] was first used to identify and remove anomalous channel data. Then, a linear fitting function of resting-state blood oxygen was calculated for each channel, and baseline correction was performed by subtracting the fitted values from the raw values to obtain the baseline hemoglobin (HBO) data [13]. On this basis, the temporal derivative distribution repair (TDDR) method [14] was applied to remove abnormal spikes and baseline drifts. Finally, a second-order Butterworth Type II filter was used to bandpass filter the signals to the range of 0.01–2.0 Hz, while eliminating high-frequency noise.

3) Brain Feature Calculation

Based on the brain regions corresponding to each fNIRS channel as specified in Table 1, an entropy-weighted averaging method was employed to compute the hemoglobin concentration for each brain region, thereby mitigating inter-individual variability in oxyhemoglobin levels while assessing overall cortical hemodynamic activity. The formula is expressed as:

$$y = \sum_j X_{ij}\, \omega_j (i = 1,2,3, \dots, N, j = 1,2,3, \dots, M) \quad (1)$$

where X denotes the raw oxyhemoglobin concentration matrix, and ω_j represents the weight of each channel derived from the entropy of the hemoglobin concentration information within that channel.

a) Modal Controllability

Modal controllability was computed based on the adjacency matrix. First, Spearman correlation coefficients across all brain channels were calculated to construct a 9×9 adjacency matrix. Modal controllability was then derived as follows:

$$\emptyset_i = \sum_{j=1}^{N} (1 - \lambda_j^2(A)) v_{ij}^2 \quad (2)$$

where v_{ij} denotes the elements of the eigenvector matrix of the adjacency matrix A, and λ_j represents the j-th eigenvalue of A.

b) Brain Functional Network Feature Extraction

Based on the Spearman correlation adjacency matrix derived from all brain channels, four network properties were computed: mean node degree, clustering coefficient, inter-density, and intra-density.

The mean node degree reflects the overall connectivity of the network, expressed mathematically as:

$$S = \frac{1}{N} \sum_{i=1}^{N} S_i^{\omega} \quad (3)$$

where S_i^{ω} denotes the weighted degree of node i, i.e., the sum of edge weights connected to node i, and N represents the total number of nodes in the network.

The clustering coefficient reflects the degree of node clustering in the network; higher values indicate a more clustered network topology. It is defined as:

$$C = \frac{1}{N} \sum_{U \in G} \frac{1}{\deg(u)(\deg(u)-1)} \sum_{vw} W_{uv}^{\wedge} W_{uw}^{\wedge} W_{vw}^{\wedge\, 1/3} \quad (4)$$

where $\deg(u)$ denotes the degree of node u, and u, v, w form a triplet with three associated edges, where the edge weight $W_{uv}^{\wedge} = W_{uv} / \max(w)$.

Inter-density is defined as the ratio of the actual number of inter-hemispheric connections to the total possible connections, measuring inter-hemispheric connectivity. The formula is:

$$K_{inter} = \frac{1}{N_S^2} \sum_{i,j \in S_{A,U}} a_{i,j} \quad (5)$$

where $S_{A,U}$ denotes the node sets of the affected and unaffected hemispheres, respectively, and $a_{i,j}$ represents the weighted connection strength between nodes.

Intra-density is defined as the ratio of the actual number of intra-hemispheric connections to the total possible connections, measuring the connectivity strength within each hemisphere. The definition is:

$$K_{intra} = \frac{2}{N_S(N_S-1)} \sum_{i \neq j \in S} a_{i,j} \quad (6)$$

4) Statistical Analysis

Statistical analyses were performed using the scipy.stats module in Python (version 3.11). The Shapiro-Wilk test [21] was employed to assess the normality of variable distributions. Paired-sample t-tests and independent-sample t-tests were conducted for within-group and between-group comparisons of each metric between patients and healthy controls, respectively. For data deviating from normality, the Mann-Whitney U test was applied as a non-parametric alternative. All statistical tests were followed by False Discovery Rate (FDR) correction for multiple comparisons. For all analyses, statistical significance was defined as a two-tailed $p < 0.05$.

III. Results

A. Brain Fatigue Classification

The behavioral task scores corresponding to the fatigue classification results are presented in Fig.5. The figure displays the normalized scores for each fatigue level (no-fatigue and

fatigue grades) across the MFI-20 questionnaire, the MR test, and the N-back test. For each level, the mean and standard deviation were computed across all samples.

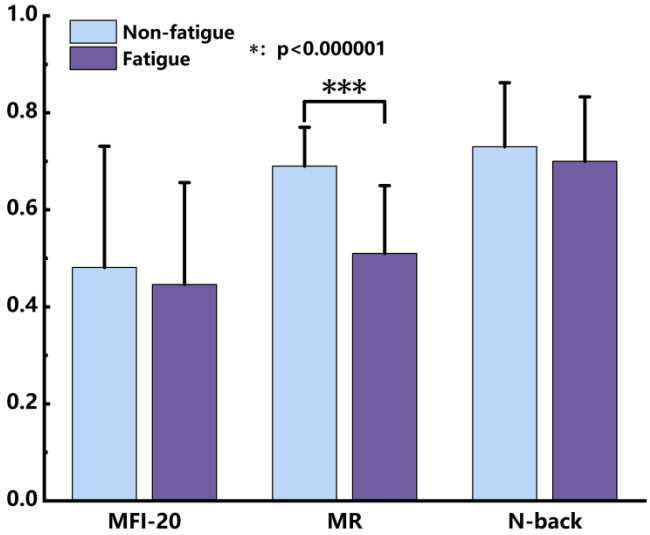

Fig. 5. Comparison of brain fatigue level test performance

Specifically, there were no significant differences in the MFI-20 and N-back scores between the non-fatigue and fatigue samples. However, the overall composite scores of participants decreased under the fatigue condition, reflecting an increased level of fatigue and a prolonged reaction time to task performance. Between the non-fatigue and fatigue conditions, the paired t-test revealed a highly significant decrease in participants' performance scores on the MR behavioral task under fatigue ($t = 7.099$, $p < 0.00001$), indicating increased reaction time and decreased accuracy.

B. Brain Modal Controllability

As shown in Fig.6., both the z-transformed modal controllability and z-transformed nodal strength exhibited a linear negative correlation in both groups. Under fatigue conditions, the goodness of linear fit decreased substantially (R^2: no-fatigue = 0.73, fatigue = 0.29). Furthermore, as illustrated in Figure 6(b), except for the M1r and PMCl brain regions, significant differences in modal controllability were observed between the two groups across almost the entire brain during the task. Specifically, participants in the fatigue state showed a significant increase in modal controllability in the PMCr region, whereas significant decreases were found in the PFC (l, m, r), M1 (l, m), and SMA regions. Detailed statistical results are presented in Table II.

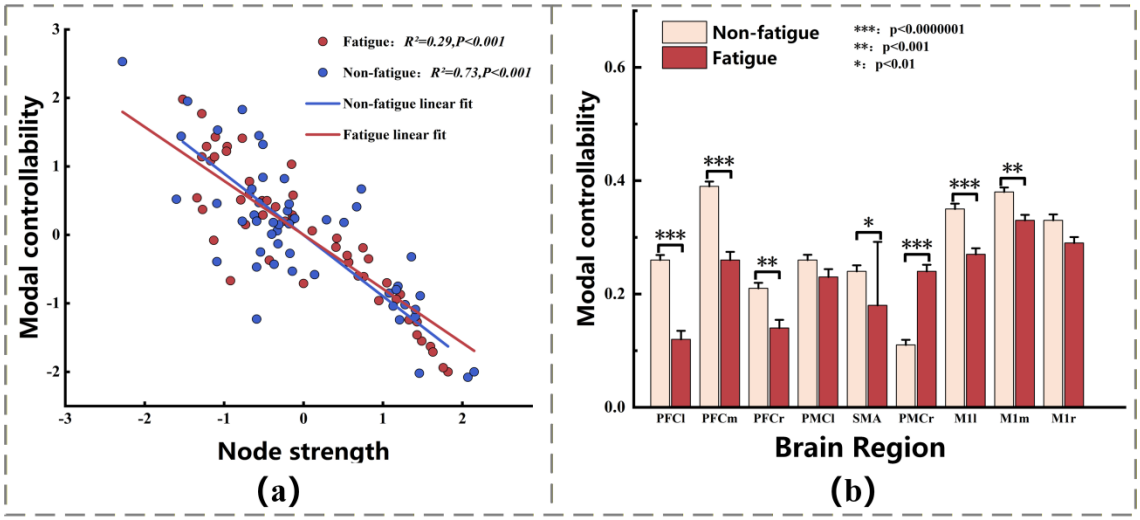

Fig. 6. Comparison of brain modal controllability. (a) Linear fitting between nodal strength and modal controllability. (b) Brain regions with differential modal controllability.

TABLE II. BRAIN REGIONS WITH SIGNIFICANT DIFFERENCES IN MODAL CONTROLLABILITY

Differential Brain Region	t-value	p-value
PFCl	8.059	p<0.0000001
PFCm	7.918	p<0.0000001
PFCr	4.423	p=0.00005
SMA	3.615	p=0.0011
PMCr	-8.575	p <0.0000001
M1l	6.088	p<0.0000001
M1m	3.732	p=0.00081

C. Brain Functional Network

The differential connections between the non-fatigue and fatigue conditions in participants are shown in Fig.7.(a). Under the fatigue condition, the SMA region exhibited the highest differential connectivity with other brain regions. Specifically, the connections of SMA-PFCl, PMCl, PMCr, M1r ($p < 0.00001$), PFCl-PFCm, PMCl ($p < 0.0001$), PFCm-PFCr, M1m ($p < 0.001$), M1l-M1m, M1r ($p < 0.01$), and PMCl-M1r ($p < 0.0001$) were significantly stronger than those in the non-fatigue condition. However, under the non-fatigue condition,

the overall number of differential connections increased, with the M1l and M1m regions showing the highest differential connectivity with other brain regions. Specifically, the connections of M1l-PFCm, PFCr, PMCl, PMCr ($p < 0.00001$), M1m-PFCl, PFCr, PMCl, PMCr, M1m ($p < 0.00001$), PFCm-PMCl, PMCr ($p < 0.0001$), PFCr-PMCl, PMCr ($p < 0.001$), and PMCl-PMCr ($p < 0.0001$) were significantly enhanced compared to the fatigue condition. Meanwhile, as shown in Fig.7.(b), analysis of network coefficients revealed that under the fatigue condition, both inter-density and intra-density network properties increased significantly (inter-density: $t = 6.634$, $p < 0.00001$; intra-density: $t = 7.738$, $p < 0.00001$; after FDR correction), whereas the clustering coefficient decreased significantly ($t = -4.387$, $p = 0.00004$; after FDR correction).

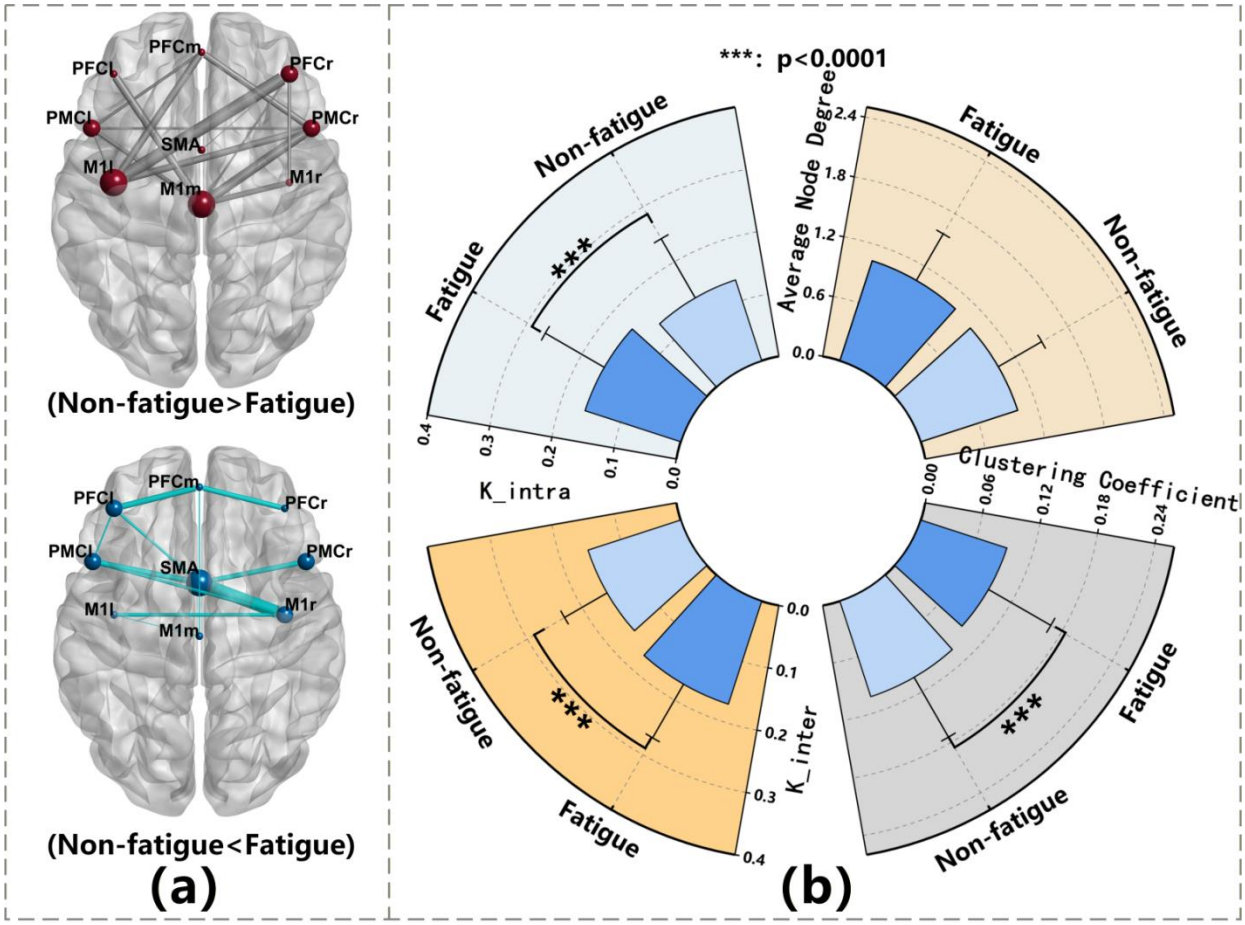

Fig. 7. Brain functional network properties. (a) Differential brain connectivity; (b) Differential brain network metrics

IV. Discussion

The results of this study demonstrate that fNIRS, combined with brain functional network and modal controllability theory, can effectively characterize the neural signatures of altitude-induced brain fatigue.

A. Brain Modal Controllability

From a cognitive perspective, brain regions with higher modal controllability may be crucial for the brain to switch between numerous cognitive functions that require substantial cognitive effort [6]. Previous research [15] has shown a linear trend between modal controllability and nodal strength: the stronger the nodal strength, the lower the modal controllability. This is consistent with the findings of the present study. It indicates that brain regions with stronger connections require lower "control energy" for dynamic regulation, whereas weakly connected regions require higher controllability to achieve flexible switching. In VR high-altitude tasks, this property may directly relate to the allocation of control resources under fatigue. Notably, our study further revealed that under the fatigue condition, the goodness-of-fit (R^2) of this linear relationship decreased substantially. This suggests a possible dissociation of the coupling between brain controllability and brain connectivity, making it difficult for the brain to efficiently allocate control energy based on connectivity strength. In VR high-altitude tasks, such disruption of network properties may directly lead to an imbalance in the allocation of control resources under fatigue, resulting in reduced precision of neural regulation and consequently a loss of flexibility in executive functions. Furthermore, our study found that when participants shifted to the fatigue state, the modal controllability of the right premotor cortex increased significantly, whereas that of the entire frontal lobe cortex, the supplementary motor area, and the left and middle primary motor cortices decreased significantly. From the perspective of cognitive control, the transition from a non-fatigue to a high-altitude fatigue state implies that the brain needs to simultaneously cope with the dual demands of physiological fatigue and high-altitude stimulation. At this point, the brain gradually undergoes a

region-specific reallocation of resources [16]. The significant increase in modal controllability of the right premotor cortex may reflect the brain's compensatory response under fatigue, characterized by high-energy-cost control exertion due to impaired neural regulation stability. The right premotor cortex is closely involved in the planning of motor sequences and sensorimotor integration [17]. In the early stage of the high-altitude task, participants rely on this region for fine, conscious motor control to maintain balance and postural transitions. However, as fatigue accumulates, neural metabolic resources become depleted, and the brain may struggle to maintain efficient automatic processing. Consequently, this region may need to invest greater "control effort" to forcibly sustain motor command output, manifesting as an abnormally elevated modal controllability. In contrast, the significant decrease in modal controllability across the entire frontal lobe cortex suggests that this higher-order cognitive hub, due to resource depletion under fatigue, loses its global coordination ability and cannot effectively recruit core cognitive functions such as attention, working memory, and conflict resolution for compensation [18]. Concurrently, the decreased modal controllability in the supplementary motor area and the left and middle primary motor cortices directly points to weakened control over the motor output stage [19]. The present study interprets this as a possible compensatory failure mechanism: owing to the functional decline of the prefrontal higher-order control center, the brain is unable to effectively transmit commands downward to the motor execution areas. This leads to the left/middle primary motor cortices losing their top-down regulation, thereby reducing the stability of motor command execution. Collectively, these findings reflect the neural manifestation of a weakening control system in the brain under high-altitude fatigue stress.

B. Brain Functional Network Analysis

Functional connectivity networks of brain regions primarily characterize the interactions between different brain regions and the changes in brain networks by measuring the temporal similarity of signals across brain regions [20]. In this study, it was found that under the non-fatigue condition, the left and middle primary motor areas exhibited high differential connectivity with other brain regions, indicating that the motor execution network is in a broadly distributed, preparatory state at baseline. Under the fatigue condition, however, the supplementary motor area showed high differential connectivity with other brain regions, accompanied by a significant increase in both inter-density and intra-density network properties, while the clustering coefficient decreased significantly. This contradictory change in network properties may reflect that under fatigue, although the brain attempts to strengthen intra-modular connectivity, the aggregation efficiency of local networks declines [21], manifesting as a dense but inefficient state of local functional isolation. Compared with the non-fatigue condition, although there are more intra-brain connections among motor-related regions under fatigue, they fail to form efficient local information-transferring clusters, and the interconnectivity among neighboring nodes is impaired, potentially leading to reduced robustness of local information processing. Such structural changes may, to some extent, reflect that when resources are depleted, the brain cannot maintain fine-grained local coordination and only barely preserves signal transmission by blindly increasing connection density, at the cost of network information processing efficiency, which objectively leads to a decline in task performance.

C. Limitations

This study has certain limitations. The current conclusions are based on VR-based altitude tasks with a limited sample size. Future studies should increase the sample size and conduct field-based altitude research to achieve deeper validation.

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REFERENCES

[1] Fan, H., Meng, Y., Zhu, L., Fan, M., Wang, D. M., & Zhao, Y. Q. (2024). A review of methods for assessment of cognitive function in high-altitude hypoxic environments. *Brain and Behavior*, *14*(2), e3418.

[2] Guo, W. Y., Bian, S. Z., Zhang, J. H., Li, Q. N., Yu, J., Chen, J. F., ... & Huang, L. (2017). Physiological and psychological factors associated with onset of high-altitude headache in Chinese men upon acute high-altitude exposure at 3700 m. *Cephalalgia*, *37*(4), 336-347.

[3] Sun, L., An, Y., Yuan, X., Zhang, H., & Cheng, F. (2025). Multidimensional Fatigue Assessment of High-Altitude Electrical Workers: A Comprehensive Analysis Based on Physiological and Psychological Indicators: Electrical Workers: A Comprehensive Analysis. *Applied Psychophysiology and Biofeedback*, 1-27.

[4] Wu, Z., Huang, L., Wang, M., & He, X. (2024). Development of the brain network control theory and its implications. *Psychoradiology*, *4*, kkae028.

[5] Shine, J. M., Bissett, P. G., Bell, P. T., Koyejo, O., Balsters, J. H., Gorgolewski, K. J., ... & Poldrack, R. A. (2016). The dynamics of functional brain networks: integrated network states during cognitive task performance. *Neuron*, *92*(2), 544-554.

[6] Gu, S., Pasqualetti, F., Cieslak, M., Telesford, Q. K., Yu, A. B., Kahn, A. E., ... & Bassett, D. S. (2015). Controllability of structural brain networks. *Nature communications*, *6*(1), 8414.

[7] Wilcox, T., & Biondi, M. (2015). fNIRS in the developmental sciences. *Wiley Interdisciplinary Reviews: Cognitive Science*, *6*(3), 263-283.

[8] Yan, Y., Guo, Y., & Zhou, D. (2025). Mental fatigue causes significant activation of the prefrontal cortex: A systematic review and meta-analysis of fNIRS studies. *Psychophysiology*, *62*(1), e14747.

[9] Amunts, K., & Zilles, K. (2015). Architectonic mapping of the human brain beyond Brodmann. Neuron, 88(6), 1086-1107.

[10] Rich, T. L., Menk, J. S., Rudser, K. D., Chen, M., Meekins, G. D., Peña, E., ... & Gillick, B. T. (2017). Determining electrode placement for transcranial direct current stimulation: a comparison of EEG-versus TMS-guided methods. Clinical EEG and neuroscience, 48(6), 367-375.

[11] Zhang, Y., Liu, N., & Wang, S. (2018). A differential privacy protecting K-means clustering algorithm based on contour coefficients. PloS one, 13(11), e0206832.

[12] Du, P., Bai, X., Tan, K., Xue, Z., Samat, A., Xia, J., ... & Liu, W. (2020). Advances of four machine learning methods for spatial data handling: A review. Journal of Geovisualization and Spatial Analysis, 4(1), 13.

[13] Newville, M., Stensitzki, T., Allen, D. B., Rawlik, M., Ingargiola, A., & Nelson, A. (2016). LMFIT: Non-linear least-square minimization and curve-fitting for Python. Astrophysics Source Code Library, ascl-1606.

[14] Fishburn, F. A., Ludlum, R. S., Vaidya, C. J., & Medvedev, A. V. (2019). Temporal derivative distribution repair (TDDR): a motion correction method for fNIRS. Neuroimage, 184, 171-179.

[15] Wiles, L., Gu, S., Pasqualetti, F., Parvesse, B., Gabrieli, D., Bassett, D. S., & Meaney, D. F. (2017). Autaptic connections shift network excitability and bursting. Scientific Reports, 7(1), 44006.

[16] Shine, J. M., & Poldrack, R. A. (2018). Principles of dynamic network reconfiguration across diverse brain states. NeuroImage, 180, 396-405.

[17] Ebbesen, C. L., & Brecht, M. (2017). Motor cortex—to act or not to act?. Nature Reviews Neuroscience, 18(11), 694-705.

[18] Lynn, C. W., & Bassett, D. S. (2019). The physics of brain network structure, function and control. Nature Reviews Physics, 1(5), 318-332.

[19] Perrey, S. (2013). Promoting motor function by exercising the brain. Brain sciences, 3(1), 101-122.

[20] Craddock, R. C., Tungaraza, R. L., & Milham, M. P. (2015). Connectomics and new approaches for analyzing human brain functional connectivity. Gigascience, 4(1), s13742-015.

[21] Hramov, A. E. E., Frolov, N. S., Maksimenko, V. A., Kurkin, S. A., Kazantsev, V. B., & Pisarchik, A. N. (2021). Functional networks of the brain: from connectivity restoration to dynamic integration. Physics–Uspekhi, 64(6), 584-616.

Experimental Study on Treadmill Walking with a Lower-Limb Exoskeleton Based on Motion Capture and Surface Electromyography

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***Abstract*—To evaluate the assistive effect of a lower-limb exoskeleton during treadmill walking, this study conducted a synchronized analysis based on motion capture and surface electromyography. Ten healthy male participants completed treadmill walking tests at 4.5 km/h under the without-exoskeleton and with-exoskeleton conditions The left knee joint angle was extracted from OptiTrack data, while the RMS values and EMG envelopes of the left rectus femoris and left gluteus maximus were obtained from Delsys surface EMG signals. Gait-cycle boundaries identified from the left knee angle curve were synchronously mapped to the EMG signals, enabling joint kinematic and muscle activation features to be compared within a unified normalized gait-cycle framework. The results showed that wearing the exoskeleton did not substantially disrupt the basic periodic waveform of treadmill walking.The left knee ROM increased from 51.73 ± 4.0° to 57.75 ± 6.2°, and the difference was statistically significant. Meanwhile, the normalized RMS values of the left rectus femoris and left gluteus maximus decreased to 89.49 ± 6.5 % and 90.84 ± 8.8%, respectively. The synchronized EMG envelope analysis further showed that the reduction in muscle activation was mainly concentrated in specific gait phases. These results indicate that the exoskeleton produced phase-dependent modulation of target muscle activation while maintaining the basic gait-cycle structure. The proposed synchronized analysis procedure provides a feasible method for evaluating lower-limb exoskeleton assistance during treadmill walking.**

Keywords—lower-limb exoskeleton, motion capture, surface electromyography, treadmill walking, assistance evaluation

I. INTRODUCTION

Lower-limb exoskeletons are widely used in walking assistance, rehabilitation training, and human–machine integrated systems. Their main purpose is to provide external mechanical assistance while maintaining a relatively natural gait pattern. With the development of exoskeleton structure, actuation, and control strategies, how to objectively and practically evaluate their assistive effects has become an important issue in related research [1].

Existing studies usually evaluate lower-limb exoskeletons using kinematic parameters, kinetic indicators, metabolic cost, or surface electromyography signals [2]-[4]. For treadmill walking, joint kinematics can describe gait rhythm and movement pattern changes, while surface electromyography can reflect the activation level of target muscles. However, these two types of indicators are often analyzed separately. As a result, the temporal correspondence between joint motion changes and muscle activation changes before and after wearing an exoskeleton is not always clear.

Therefore, this study focuses on treadmill walking with a lower-limb exoskeleton and establishes a synchronized analysis procedure based on motion capture and surface electromyography. Ten healthy participants performed treadmill walking at 4.5 km/h under without-exoskeleton and with-exoskeleton conditions. The left knee joint angle was used to divide normalized gait cycles, and the same gait-cycle boundaries were mapped to the EMG signals of the left rectus femoris and left gluteus maximus. By combining left knee kinematics, normalized RMS values, and synchronized EMG envelope features, this study preliminarily evaluates the influence of exoskeleton wearing on gait structure and target muscle activation during treadmill walking.

The main contribution of this study is that kinematic and EMG signals are analyzed within the same normalized gait-cycle framework. This allows the changes in knee joint motion and target muscle activation to be compared more directly, providing a simple and practical method for evaluating the assistive effect of lower-limb exoskeletons under treadmill walking conditions.

II. EXPERIMENTAL DESIGN AND METHODS

A. Subjects and Experimental Equipment

Ten healthy adult male volunteers participated in this study. The participants were 22–25 years old, with heights of 165–180 cm and body weights of 60–90 kg. All participants had no obvious musculoskeletal disorders or lower-limb injuries affecting normal gait, and all completed informed consent and device fitting before the experiment.

The experimental system consisted of a lower-limb exoskeleton, an OptiTrack motion capture system, and a Delsys surface electromyography acquisition system. The

OptiTrack system was used to collect three-dimensional marker trajectories and calculate lower-limb kinematic parameters. The Delsys system was used to collect surface EMG signals from the target muscles. The two systems were connected through a hardware synchronization device to ensure synchronous data acquisition. Reflective markers were placed on key anatomical locations of the lower limbs, and EMG electrodes were attached along the muscle fiber direction of the target muscles EMG electrodes were attached along the muscle fiber direction of the target muscles after skin preparation and signal-quality checking.

B. Experimental Task and Procedure

Two experimental conditions were used in this study: walking without the exoskeleton and walking with the exoskeleton. Each participant completed treadmill walking tests under both conditions. During each trial, the participant first stood still for 1 min, then walked on the treadmill at 4.5 km/h for 3 min, and finally stood still for another 1 min. Before formal data acquisition, participants completed device wearing, signal checking, and short-term familiarization practice. Rest intervals were arranged between different trials to reduce the influence of fatigue.

In subsequent analysis, the stable treadmill walking segmen t in the middle of the recording was selected as the valid analys is interval. This procedure avoided the standing, initiation, ada ptation, and deceleration phases at the beginning and end of the trial, ensuring the comparability of kinematic and EMG indica tors under the two experimental conditions.

Fig. 1. Treadmill walking experiment with a lower-limb exoskeleton.

C. Data Processing and Feature Extraction

In terms of kinematic data processing, the raw marker trajectories exported from OptiTrack missing marker points were first interpolated, and the trajectories were then smoothed For the kinematic analysis of the left knee joint, the left thigh marker (LThigh), left lateral knee marker (LKneeOut), and left lateral ankle marker (LAnkleOut) were selected to construct the geometric angle of the left knee joint. Specifically, with the left lateral knee marker as the vertex, the thigh vector and shank vector were constructed, and the left knee joint angle was calculated from the included angle between the two vectors. In this study, the left knee geometric angle constructed based on the spatial geometric relationship of key markers was used as a simplified kinematic indicator, mainly for comparing treadmill walking trends before and after wearing the exoskeleton. To reduce the influence of high-frequency noise, the left knee joint angle sequence was filtered using a fourth-order Butterworth low-pass filter with a cut-off frequency of 6 Hz. Subsequently, gait cycles were segmented based on the filtered left knee angle curve, and the interval between adjacent valley points was defined as one complete gait cycle[5]. Abnormal cycles with durations shorter than 0.8 s or longer than 1.5 s were excluded. Each valid gait cycle was normalized to 0%–100% of the gait cycle using linear interpolation, with a total of 101 sampling points. Multiple valid cycles were then averaged to obtain the mean normalized gait-cycle curve. In the calculation of the left knee joint angle, with the left lateral knee marker as the vertex, the thigh vector u and the shank vector v were constructed, and the left knee joint angle can be expressed as follows:

$$\theta = \arccos\left(\frac{u \cdot v}{\| u \| \| v \|}\right) \tag{1}$$

where u·v denotes the dot product of the two vectors, and ‖u‖ and ‖v‖ denote the magnitudes of vectors u and v, respectively.

To characterize the overall motion amplitude of the left knee joint within the normalized gait cycle, the range of motion (ROM) was defined as follows:

$$ROM = \theta_{max} - \theta_{min} \tag{2}$$

where θ_{max} and θ_{min} represent the maximum and minimum values of the mean normalized gait-cycle curve, respectively.

In terms of surface electromyography data processing, the left rectus femoris and left gluteus maximus were selected as the target muscle groups in this study. For sEMG signal preprocessing, the raw sEMG signals were first demeaned to eliminate DC offset. Subsequently, a fourth-order Butterworth band-pass filter with a frequency range of 20–450 Hz was applied. This frequency band can preserve the main effective frequency components of surface electromyography signals while reducing the effects of motion artifacts and high-frequency noise. Considering the possible presence of power-line interference in the experimental environment, a 50 Hz notch filter was further applied. The above parameter settings were based on commonly used frequency ranges in surface electromyography signal processing and relevant SENIAM recommendations, and were considered sufficient for analyzing the EMG amplitude of lower-limb muscles during treadmill walking. In this study, the root mean square (RMS) value was used as the main indicator for EMG analysis. Specifically, the RMS value was directly calculated from the entire preprocessed sEMG signal within the stable treadmill walking segment. Considering that the raw amplitude of surface electromyography signals across different participants is easily affected by electrode placement, skin impedance, and individual differences, the RMS value under the without-

exoskeleton condition was used as the baseline for each participant, and the RMS value under the with-exoskeleton condition was normalized accordingly [6]. The expression is as follows:

$$RMS_{norm} = \frac{RMS_{with}}{RMS_{without}} \times 100\% \quad (3)$$

For the EMG signal within a single gait cycle or a stable walking segment, the root mean square (RMS) value is defined as follows:

$$\mathrm{RMS} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^{\,2}} \quad (4)$$

where n denotes the number of signal sampling points.

The stable treadmill walking segment was selected according to the following principle: within the complete 5-min recording, the first 1 min mainly corresponded to standing, initiation, and adaptation, whereas the last 1 min mainly corresponded to deceleration or preparation for the end of the trial. Therefore, the continuous walking phase in the middle of the recording was preferentially selected as the valid analysis interval. Considering the completeness of each trial recording and the stability of gait rhythm, the 70–230 s interval was mainly selected as the stable walking segment in this study. This time window was approximately 160 s in length, covering a sufficient number of consecutive gait cycles while avoiding possible unstable motion states at the beginning and end of the experiment. Therefore, it was suitable for subsequent extraction of kinematic and EMG indicators. Based on the above principle, both kinematic and surface electromyography analyses were performed using the stable treadmill walking segment under each condition to ensure the comparability and reliability of the results. To realize synchronized joint analysis of motion capture and surface electromyography, this study established a kinematic–EMG correspondence analysis procedure under unified gait-cycle boundaries based on the hardware-synchronized acquisition results from OptiTrack and Delsys. First, the left knee joint angle curve was extracted within the stable treadmill walking time window, and the boundaries of consecutive gait cycles were determined based on the valley points of the filtered left knee angle curve. Second, the start and end times of each gait cycle were mapped onto the synchronously acquired sEMG time axis, and the signals of the left rectus femoris and left gluteus maximus were segmented accordingly. Then, the EMG signals within each gait cycle were subjected to full-wave rectification, smoothing, and time normalization to obtain the mean EMG envelope curve over 0%–100% of the gait cycle. Finally, the left knee joint angle curve, target muscle RMS values, and EMG envelope amplitude changes were synchronously compared within the same gait-cycle framework. The core of this procedure is to use the same set of gait-cycle boundaries to constrain both kinematic and EMG analyses, so that changes in joint motion and muscle activation can be matched cycle by cycle.

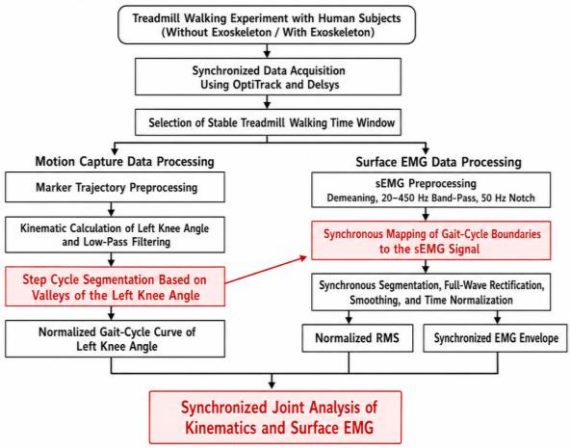

Fig. 2. Workflow of synchronized motion capture and surface electromyography analysis based on unified gait-cycle boundaries.

D. Statistical Analysis

This study adopted a repeated-measures design. For each participant, the left knee ROM, RMS of the left rectus femoris, and RMS of the left gluteus maximus were extracted under the two conditions. The left knee ROM was compared using a paired-sample t-test. For EMG indicators, the normalized RMS value under the with-exoskeleton condition was compared with the 100% baseline using a one-sample t-test. The significance level was set at $p < 0.05$, and all results are expressed as mean ± standard deviation.

III. Results and Analysis

A. Kinematic Results of Treadmill Walking

To analyze the influence of exoskeleton wearing on the knee joint movement pattern during treadmill walking, this study first compared the normalized gait-cycle curves of the left knee joint from ten participants under the without-exoskeleton and with-exoskeleton conditions. Figure 3 presents the mean variation curves of the left knee joint angle under the two conditions. The results showed that the left knee joint angle curves under both conditions exhibited clear periodic variation characteristics, indicating that the participants maintained a basic treadmill walking rhythm in both states.

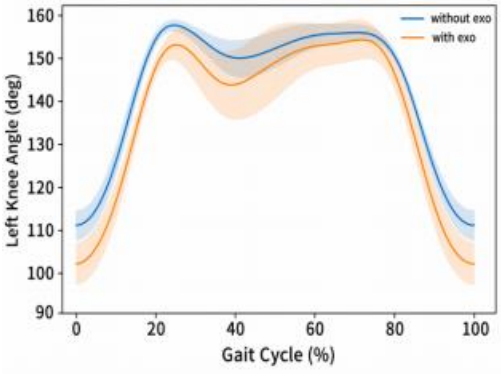

Fig. 3. Normalized mean gait-cycle curves of the left knee joint from ten subjects.

As shown by the mean curves, the overall waveform of the left knee joint angle was consistent between the with-exoskeleton and without-exoskeleton conditions, with generally corresponding peak and valley positions, and both conditions showed stable periodic variation. After wearing the exoskeleton, the curve exhibited a certain degree of local deviation during some gait phases, but no abnormal waveform or obvious abrupt change was observed. This indicates that while providing assistance, the exoskeleton may have affected the motion distribution of the knee joint during the stance and swing phases to some extent. However, this effect was mainly reflected as an adjustment of the movement pattern rather than a disruption of the basic gait structure.

To further quantitatively characterize changes in the overall motion amplitude of the knee joint, the range of motion (ROM) of the left knee joint was calculated in this study. The results showed that the left knee ROM was 51.73 ± 4.0° under the without-exoskeleton condition and 57.75 ± 6.2° under the with-exoskeleton condition. Compared with the without-exoskeleton condition, the left knee ROM increased by 11.63% after wearing the exoskeleton. The paired-sample t-test showed that the difference was statistically significant ($p = 7.91 \times 10^{-3}$).

Combined with Figure 3, the overall waveform of the left knee joint angle was generally consistent between the with-exoskeleton and without-exoskeleton conditions, and the peak and valley positions were approximately aligned. This indicates that wearing the exoskeleton did not obviously disrupt the basic periodic structure of treadmill walking. However, the ROM statistical results showed that the overall motion amplitude of the knee joint increased after wearing the exoskeleton. The quantitative statistical results of the left knee ROM are shown in Table 1.

B. Surface Electromyography Results of Treadmill Walking

Based on the kinematic analysis, this study further compared the changes in surface EMG amplitude of the target muscle groups before and after wearing the exoskeleton to evaluate the influence of the exoskeleton on lower-limb muscle activation levels. Figure 4 presents the statistical results of the normalized RMS values of the left rectus femoris and left gluteus maximus from the ten participants. Taking the RMS value under the without-exoskeleton condition as the 100% baseline, the normalized RMS of the left rectus femoris after wearing the exoskeleton was 89.49 ± 6.5%, representing a decrease of 10.51% from the baseline. The statistical test showed that this difference was significant ($p = 6.33 \times 10^{-4}$). The normalized RMS of the left gluteus maximus was 90.84 ± 8.8%, representing a decrease of 9.16% from the baseline, and the difference was also statistically significant ($p = 9.35 \times 10^{-3}$).

This result indicates that, under the treadmill walking condition in this study, the average EMG activation levels of the two target muscle groups after wearing the exoskeleton were significantly lower than the without-exoskeleton baseline. This suggests that the exoskeleton may reduce the muscle activation demand of the rectus femoris and gluteus maximus during treadmill walking. The quantitative statistical results of the normalized RMS values of the rectus femoris and gluteus maximus are also shown in Table 1.

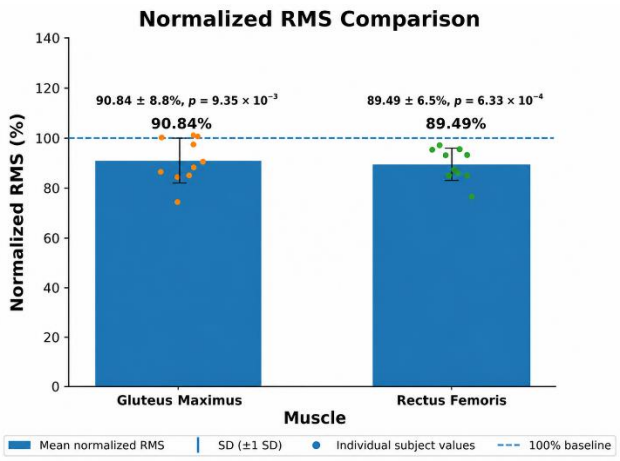

Fig. 4. Comparison of mean normalized RMS values of the main target muscles from ten subjects.

TABLE I. QUANTITATIVE COMPARISON OF LEFT KNEE ROM AND NORMALIZED EMG RMS UNDER WITHOUT-EXOSKELETON AND WITH-EXOSKELETON CONDITIONS

Feature	*n*	*Without exoskeleton*	*With exoskeleton*	*Change*	*Statistical test*	*p-value*
Marker-derived left knee ROM (°)	10	51.73 ± 4.0°	57.75 ± 6.2°	+11.63%	Paired-sample t-test	7.91×10^{-3}
Rectus femoris RMS (% baseline)	10	100.00 ± 0.00	89.49 ± 6.5%	−10.51%	One-sample t-test against 100% baseline	6.33×10^{-4}
Gluteus maximus RMS (% baseline)	10	100.00 ± 0.00	90.84 ± 8.8%	−9.16%	One-sample t-test against 100% baseline	9.35×10^{-3}

Note: The data are expressed as mean ± standard deviation. The normalized RMS values were calculated using the without-exoskeleton condition as the 100% baseline. A value of $p < 0.05$ indicates a statistically significant difference.

C. Synchronized Joint Analysis

To further reveal the specific distribution of the exoskeleton assistive effect within the gait cycle, this study performed a synchronized joint analysis of the left knee joint angle curve, the EMG envelope of the left rectus femoris, and the EMG envelope of the left gluteus maximus under a unified gait-cycle framework. The results are shown in Figure 5.

Figure 5(a) presents the normalized gait-cycle curve of the left knee joint, which was used to characterize changes in the overall gait structure before and after wearing the exoskeleton. The results show that the overall trends of the two curves were consistent, with only slight local phase shifts, indicating that exoskeleton assistance did not obviously disrupt the basic gait pattern. Figure 5(b) presents the synchronized EMG envelope of the left rectus femoris within the normalized gait cycle. The results show that, after wearing the exoskeleton, the rectus femoris envelope was generally lower at approximately 30%–

50% of the gait cycle, and the amplitude of the late peak near approximately 85% of the gait cycle also decreased. This indicates that the influence of the exoskeleton on the rectus femoris was mainly reflected in a reduction in activation amplitude within specific gait intervals, rather than a uniform change throughout the entire gait cycle. Figure 5(c) presents the synchronized EMG envelope of the left gluteus maximus. The shaded areas indicate the gait-cycle intervals in which the EMG envelope amplitude decreased more obviously after wearing the exoskeleton. The results show that the main peaks of the gluteus maximus under both with-exoskeleton and without-exoskeleton conditions were located at approximately 20%–30% of the gait cycle, with generally consistent peak positions. However, the main peak amplitude decreased after wearing the exoskeleton, while the differences in the remaining intervals were relatively small. This suggests that the influence of the exoskeleton on the gluteus maximus was mainly reflected in a reduction in EMG amplitude during the main activation phase.

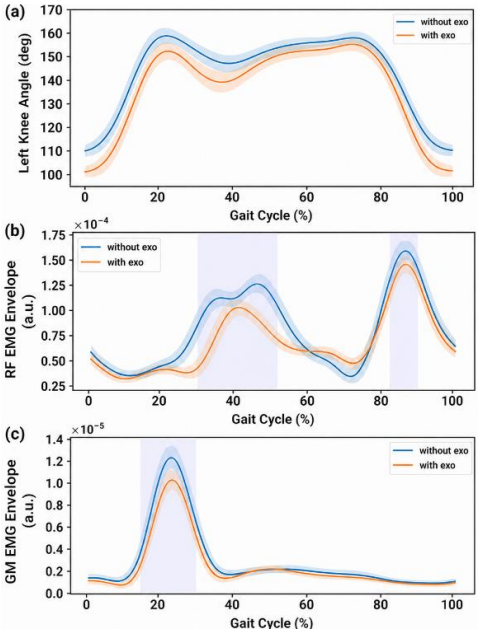

Fig. 5. Joint analysis of mean left knee kinematics and synchronized EMG envelopes from ten subjects.

D. Discussion

Based on the integrated results of kinematics, surface EMG RMS, and synchronized EMG envelopes, the participants still maintained a relatively stable periodic structure of treadmill walking after wearing the exoskeleton, indicating that the exoskeleton did not obviously disrupt the basic gait pattern. Meanwhile, the normalized RMS values of the left rectus femoris and left gluteus maximus were both lower than the without-exoskeleton baseline, and the synchronized envelope results showed that the reduction in EMG amplitude was mainly concentrated in specific gait phases. This suggests that the assistive effect of the exoskeleton may be more reflected in local modulation of the activation timing and amplitude of the target muscle groups, rather than a uniform reduction in muscle activity throughout the entire gait cycle.

It should be noted that this study still has certain limitations. First, the sample size was limited to ten healthy adult male participants, which was relatively small, and female participants, older adults, and individuals with lower-limb motor dysfunction were not included. Therefore, the results mainly reflect the preliminary responses of healthy young males under laboratory treadmill conditions. Second, this study only analyzed treadmill walking at a fixed speed of 4.5 km/h, and did not cover more complex movement tasks such as different walking speeds, overground walking, turning, uphill and downhill walking, or stair ascent and descent. Third, this study mainly analyzed the kinematics of the left knee joint and the surface EMG signals of the left rectus femoris and left gluteus maximus, without further integrating indicators such as hip and ankle kinematics, joint moments, metabolic cost, and subjective comfort for comprehensive evaluation. Future studies should expand the sample size, include more task scenarios, and further verify the stability and applicability of the proposed synchronized evaluation method.

IV. Conclusion

This study investigated the influence of a lower-limb exoskeleton on treadmill walking based on synchronized motion capture and surface electromyography. The gait-cycle boundaries extracted from the left knee joint angle curve were mapped to the EMG signals, enabling kinematic and muscle activation features to be compared within a unified normalized gait-cycle framework.

The results showed that wearing the exoskeleton did not obviously disrupt the basic periodic structure of treadmill walking. The left knee ROM increased from 51.73 ± 4.0° to 57.75 ± 6.2°, indicating that the overall motion amplitude of the knee joint increased after wearing the exoskeleton. Meanwhile, the normalized RMS values of the left rectus femoris and left gluteus maximus decreased to 89.49 ± 6.5% and 90.84 ± 8.8%, respectively, suggesting that the exoskeleton may reduce the average activation level of the target muscles during treadmill walking.

The synchronized EMG envelope results further showed that the reduction in muscle activation was mainly concentrated in specific gait phases rather than uniformly distributed throughout the entire gait cycle. Therefore, the proposed synchronized analysis method can more clearly describe the phase correspondence between joint motion and muscle activation changes. This method provides a feasible analytical approach for evaluating the assistive effect of lower-limb exoskeletons during treadmill walking.

The present study is still a preliminary investigation under laboratory treadmill conditions with a fixed walking speed and a limited sample size. Future studies should include more participants and more diverse locomotion tasks, such as multi-speed walking, overground walking, turning, slopes, and stair ascent and descent, to further verify the applicability of the proposed method.

References

[1] R. Baud, A. R. Manzoori, A. Ijspeert, and M. Bouri, “Review of control strategies for lower-limb exoskeletons to assist gait,” Journal of

NeuroEngineering and Rehabilitation, vol. 18, no. 1, Art. no. 119, 2021, doi: 10.1186/s12984-021-00906-3.

[2] C. Rodrigues-Carvalho, M. Fernández-García, D. Pinto-Fernández, C. Sanz-Morere, F. O. Barroso, S. Borromeo, C. Rodríguez-Sánchez, J. C. Moreno, and A. J. del-Ama, "Benchmarking the effects on human-exoskeleton interaction of trajectory, admittance and EMG-triggered exoskeleton movement control," Sensors, vol. 23, no. 2, Art. no. 791, 2023, doi: 10.3390/s23020791.

[3] Y.-K. Kong, K.-H. Choi, M.-U. Cho, S.-Y. Kim, M.-J. Kim, J.-W. Shim, S.-S. Park, K.-R. Kim, M.-T. Seo, H.-S. Chae, and H.-H. Shim, "Ergonomic assessment of a lower-limb exoskeleton through electromyography and AnyBody Modeling System," International Journal of Environmental Research and Public Health, vol. 19, no. 13, Art. no. 8088, 2022, doi: 10.3390/ijerph19138088.

[4] L. Xu, X. Liu, Y. Chen, L. Yu, Z. Yan, C. Yang, C. Zhou, and W. Yang, "Reducing the muscle activity of walking using a portable hip exoskeleton based on human-in-the-loop optimization," Frontiers in Bioengineering and Biotechnology, vol. 11, Art. no. 1006326, 2023, doi: 10.3389/fbioe.2023.1006326.

[5] N. Zahradka, K. Verma, A. Behboodi, B. Bodt, H. Wright, and S. C. K. Lee, "An evaluation of three kinematic methods for gait event detection compared to the kinetic-based 'gold standard'," Sensors, vol. 20, no. 18, Art. no. 5272, 2020, doi: 10.3390/s20185272.

[6] M. Besomi et al., "Consensus for experimental design in electromyography (CEDE) project: Amplitude normalization matrix," Journal of Electromyography and Kinesiology, vol. 53, Art. no. 102438, 2020, doi: 10.1016/j.jelekin.2020.102438.

Machine Vision-Based Bottle Mouth Defect Inspection System for Industrial Production Line

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***Abstract*—In industrial automated production, product quality inspection serves as a critical link to guarantee production safety and operational reliability. Aiming at the problems of low efficiency and high false detection rate in manual inspection for glass bottle mouth defects during production, this paper proposes a machine vision inspection system based on deep learning. In the experimental phase, a dataset containing typical bottle mouth defects is constructed, and data augmentation strategies are adopted to address the issue of sample imbalance. The test results show that the proposed system outperforms traditional methods in terms of accuracy and real-time performance, and can fully meet the real-time inspection requirements of industrial assembly lines. It provides an efficient and reliable automatic solution for the quality inspection of industrial products.**

Keywords—Machine Vision, Feature Extraction, Bottle Mouth Defect Detection

I. INTRODUCTION

With the development of the information revolution, computer vision and deep learning technologies have achieved remarkable breakthroughs. Automated detection systems based on machine vision have gradually become a research hotspot in the field of industrial quality inspection.

Compared with traditional image processing algorithms such as SVM and HOG, deep learning models feature end-to-end feature learning. They can adaptively extract high-level semantic features of defects, effectively improving detection accuracy and efficiency under complex background conditions[1]. As a representative single-stage object detection algorithm, YOLOv8 has achieved a mean average precision (mAP) of over 95% in the surface defect detection of industrial components by virtue of its high detection speed and lightweight architecture[2].

Nevertheless, the detection of tiny defects on glass bottle mouths in industrial assembly lines still faces three major challenges. First, defect scales vary greatly, with defect sizes ranging from 0.5 mm to 5 mm, making it difficult for existing deep learning models to effectively characterize tiny objects. Second, frequent illumination fluctuations and glass reflection interference in industrial production environments tend to increase the false detection rate. Third, industrial assembly lines operate at high speeds, and current models still have obvious shortcomings in meeting the real-time operation requirements of high-speed production lines.

Image classification algorithms based on computer vision and deep learning can automatically extract features from training data. With the increase of network depth, they are able to learn more discriminative high-level feature representations, thereby greatly improving classification accuracy.

Benefiting from the powerful representation learning capability of deep neural networks, such methods possess excellent robustness against noise and interference in input data, and can adapt to complex practical application scenarios. Nevertheless, the training of deep learning models requires massive labeled data and substantial computing resources, which remains one of the major challenges for their large-scale deployment in industrial applications[3].

With the advent of the Industry 4.0 era, computer vision technology has been increasingly widely applied in the field of intelligent manufacturing. Defect detection systems based on machine vision have been deployed in numerous industries. Featuring high efficiency and high precision, such systems have become an indispensable technical method for industrial quality control and will be continuously applied in various sectors, especially in assembly line operations.

Taking glass bottle mouth defect detection as the research objective, this project addresses the drawbacks of traditional manual inspection, including low efficiency and poor accuracy, which can hardly meet the stringent quality control requirements of modern industrial production. As a vital packaging material for food, pharmaceutical, cosmetic and other industries, glass bottles with mouth defects such as cracks, notches and deformation will directly compromise the sealing performance and safety of products.

Combined with the domestic and international research status of machine vision recognition and classification, this paper proposes a glass bottle mouth defect detection method based on machine vision and deep learning. A complete bottle mouth defect detection system is researched and designed to realize rapid and accurate automatic identification of glass bottle mouth defects in industrial production.

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II. Overall Design of Bottle Mouth Defect Detection System

A. Introduction to Detected Objects of Bottle Mouth Defects

Taking glass bottle mouths in multiple scenarios as the research objects of recognition, as shown in Figure 1, these are images of glass bottle mouths collected under different working conditions. Defects on glass bottle mouths present diverse morphological shapes and dimensional features. It is necessary to conduct accurate defect detection, recognition and classification, and assign defective products to corresponding production procedures and equipment, so as to realize an efficient production process.

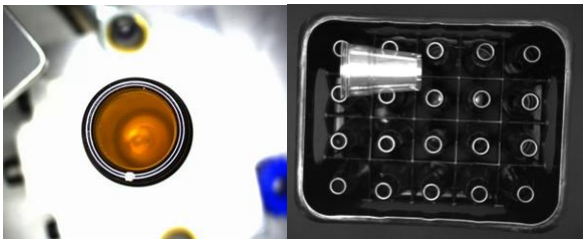

Fig. 1. Physical Diagrams of Detected Objects

B. Technical Index Requirements

To facilitate operators in using the bottle mouth defect detection system, a software interface with favorable human-computer interaction performance needs to be designed. Meanwhile, to assist enterprises in production planning, inventory management and supply chain coordination, as well as further improve production efficiency and resource utilization, the detection capability of the system for various defect types requires continuous optimization and enhancement.

III. System Composition and Principle

The functional requirements, technical indicators of the machine vision-based bottle mouth defect detection system, as well as the overall architecture of the visual recognition system for bottle mouth defect detection, are illustrated in Figure 2. The system consists of two parts: hardware and software. The hardware system includes the glass bottle conveying module, image acquisition module and system control module. The software system comprises the image acquisition module, image processing module, data management module, data storage module and interactive interface module.

The glass bottle conveying module is responsible for transporting glass bottles according to operational tasks and supports adjustable conveyor speed. A camera is installed above the conveyor belt to detect defects on glass bottles during transportation. Bottles to be identified are delivered in an orderly manner by the conveying module. If any bottle mouth defect is detected, the system triggers an alarm and conveys the defective product to the abnormal area.

When glass bottles reach the image acquisition module, the upper computer sends an image capture signal to obtain image data of the targets. After receiving the images, the upper computer invokes the well-trained glass bottle defect detection algorithm to identify bottle mouth defects. Upon completion of detection, the upper computer records and statistically analyzes the inspection results. The conveying module diverts defective glass bottles separately to facilitate subsequent manual processing. The system schematic diagram shown in Figure 3.

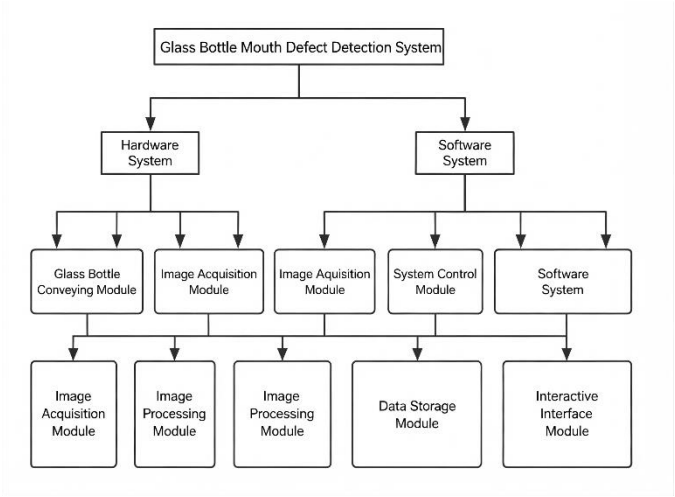

Fig. 2. Overall System Design Framework

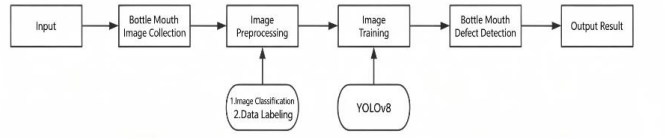

Fig. 3. System Structure Diagram

A. Hardware System Construction

The hardware part of the bottle mouth defect detection system is based on the glass bottle conveying module. According to the requirements of bottle mouth defect detection, an image acquisition module and an image processing module are added to this module. The system control module serves as the control center of the system, coordinating and controlling the operations of each module to complete the detection and classification of bottle mouth defects.

B. Glass Bottle Conveyor Module

To ensure that the glass bottles to be inspected are successfully conveyed to the position directly under the industrial camera, allowing the camera to capture frontal images of the bottle mouths, and continue moving forward in an orderly manner after identification and classification, this paper adopts a speed-multiplying chain conveyor line as the glass bottle conveying device, as shown in Figure 4. The glass bottles to be inspected are placed on the conveyor belt, and the motor drives the speed-multiplying chain to transport the bottles forward. The speed-multiplying chain consists of large-diameter rollers and small-diameter rollers, which achieve the effect of speed-multiplying conveying and sorting through the difference in inner and outer diameters.

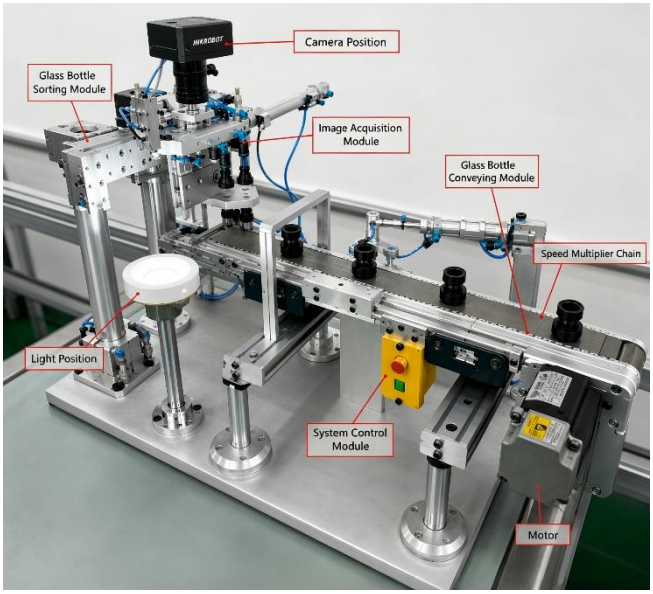

Fig. 4. Glass Bottle Conveying Prototype

In engineering application projects of machine vision, obtaining high-quality, high-resolution images with distinct features is an essential prerequisite for ensuring the accuracy of image processing and model training. The selection and configuration of the light source are decisive factors affecting image quality. Studies have shown that in the inspection of highly reflective glass surfaces, an improper lighting scheme can reduce the image signal-to-noise ratio (SNR) by more than 60%[4].In this system, a high-brightness LED area light source is selected. It has the advantages of long service life, fast response speed, freely selectable colors, and low overall cost, and is widely used in machine vision inspection platforms[16]. The light source is installed 15 cm above the bottle mouth inspection station (with a positioning accuracy of ±2 mm) to achieve feature extraction and capture of the bottle mouth area. This light source is shown in Figure 5.

Fig. 5. LED Light

A camera is a device used to capture images or videos, converting optical information into digital signals that can then be processed and analyzed by a computer. Since the image acquisition task in this system requires high real-time performance and efficiency, and given the special nature of glass materials, defects on glass bottle mouths are generally tiny, imposing high requirements on image quality. It is therefore necessary to strike a balance between speed, accuracy, and cost. For these reasons, this paper adopts a CMOS camera as the image acquisition device for the system.

The camera model selected in this paper is the Hikvision MV-CH250-90GM.Equipped with 90fps frame synchronization technology, this camera can quickly capture every detail of the bottle mouth data. The exposure time range of the MV-CH250-90GM can be as low as less than 10 μs, meeting the industrial standard of exposure time ≤ 500 μs under stable lighting conditions. It is an excellent choice for high-speed inspection tasks on production lines. The appearance of the camera is shown in Figure 6.

Fig. 6. Hikvision MV-CH250-90GM Industrial Camera

The system control module is mainly composed of an upper computer and a lower computer, which realize signal acquisition and motion control functions of the hardware system in this project.The industrial personal computer (IPC) is adopted as the upper computer. It communicates with the industrial camera via Gigabit Ethernet to collect bottle mouth image data in real time. After image acquisition, the visual defect detection and classification algorithm is invoked to process bottle mouth images, complete defect identification, and output detection results.A programmable logic controller (PLC) is used as the lower computer. Its input ports are connected to photoelectric switches to detect the in-place signal of the tooling plate, while the output ports are connected to relays. The relays control the on–off of solenoid valves, so as to drive the cylinders to perform precise actions.Data interaction between the upper and lower computers is realized through Ethernet. The lower computer transmits the in-place signal of glass bottles to the upper computer, and the upper computer feeds back the defect detection completion signal to the lower computer. Relying on real-time communication protocols, all modules are coordinated to ensure the efficient operation of the entire bottle mouth defect detection process.

IV. Software System Design

After the completion of glass bottle mouth defect detection, a recognition completion signal is sent to realize functions such as image acquisition, image processing, and data management. This paper focuses on the development of a glass bottle mouth defect detection system based on machine vision. The system is developed on the PyCharm platform using the Anaconda3 interpreter, with the Python programming language and the Ultralytics open-source vision library. The software interactive interface is designed using the PyQt extension tool.

PyCharm[5] is a professional integrated development environment (IDE) launched by JetBrains and serves as an intelligent Python development engine. With its intelligent features, it has become the preferred choice for large-scale project development. The tool provides code auto-completion, real-time error checking, a graphical debugger, and version control integration, significantly reducing coding complexity.

Its professional edition supports web frameworks such as Django and Flask, as well as scientific toolkits (eg., Jupyter Notebook), while the community edition meets basic Python development requirements. PyCharm is particularly suitable for long-term engineering projects, and its modular design effectively manages code structure and avoids messy script accumulation.

Anaconda3[6] is an out-of-the-box Python distribution optimized for data science and machine learning. Its core advantages are as follows: first, it integrates more than 600 mainstream data science libraries (such as NumPy, Pandas, and Scikit-learn), eliminating the trouble of manual dependency installation; second, Anaconda3 isolates dependencies of different projects through virtual environments, effectively solving version conflicts; third, Anaconda is bundled with Jupyter Notebook (interactive programming) and Spyder (lightweight IDE), forming a complete data analysis workstation with a full ecological toolchain.

Ultralytics[7] is a benchmark library in the field of computer vision. In particular, its YOLOv8 model features high accuracy and real-time performance, and is widely used in real-time detection tasks (e.g., real-time pedestrian detection, real-time vehicle detection). The characteristics of the library include ease of use: model training, validation, and deployment can be completed with only 10 lines of code; multi-framework compatibility supports exporting to ONNX, TensorRT, and other formats for cross-platform deployment; the library provides a full workflow from data annotation tools (such as LabelImg) to model serving, covering the entire model training pipeline.

PyQt[8] introduces the C++ capabilities of the Qt framework to Python, making user operations simpler and more convenient for building professional desktop applications. Its core values are reflected in: providing Qt Designer for drag-and-drop UI generation and automatic conversion to Python code; providing advanced controls such as QChart and QtSQL for database connection; supporting one-time development and deployment to Windows, macOS, and Linux. This system uses PyQt to quickly build visualization tools for YOLO detection results.

V. Study on Bottle Mouth Defect Detection Methods

The YOLOv8 model[9] is designed for tasks such as object detection, image classification, and instance segmentation. Its overall architecture is illustrated in Figure 7. The core function of the backbone is to extract features from input images and convert them into feature representations with rich semantic information. This structure consists of convolution (Conv), C2F, and spatial pyramid pooling fast (SPPF) modules, with the detailed structure shown in Figure 8.The Conv component performs feature extraction through convolution kernels while minimizing the loss of gradient flow information. In the YOLOv8 model, the C2F structure captures richer gradient flow information to enhance feature extraction capability. The SPPF module converts feature map information of different sizes into feature vectors of a unified dimension and ultimately realizes feature fusion[21]. In addition, the backbone integrates multiple convolutional and deconvolutional layers. With residual connections and bottleneck blocks, it reduces network scale and improves overall model performance.

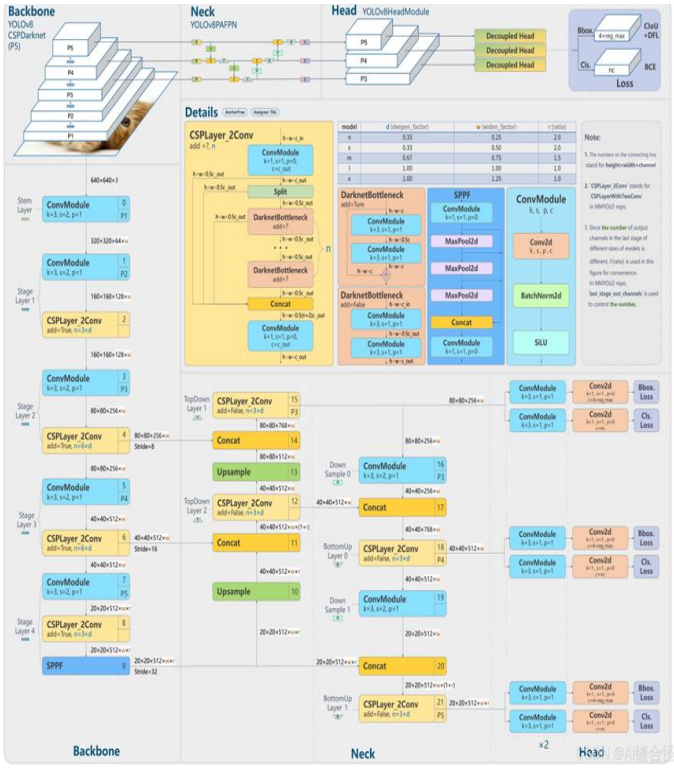

Fig. 7. YOLOv8 Architecture Diagram

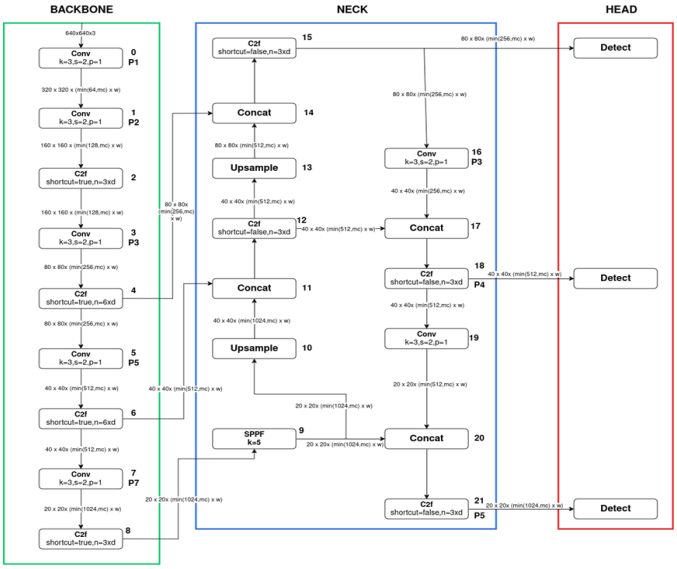

Fig. 8. Backbone, Neck and Head Structure Diagram

The Neck of the model serves as the intermediate layer, which is mainly responsible for multi-scale fusion of features extracted by the Backbone. By integrating feature maps at different scales, this layer strengthens feature representation capability. Feature fusion is realized based on the architectures of the Feature Pyramid Network (FPN) and Path Aggregation Network (PAN). The model ingeniously combines top-down and bottom-up information flows within the network. This

process not only expands the receptive field, but also fuses multi-scale information into feature maps, thereby improving the overall detection performance. The Neck structure can be adjusted and optimized according to actual requirements to adapt to various object detection tasks and datasets.

The Head is the final stage of the model, and its architecture is dynamically adjusted according to task types. Its core function is to extract multi-scale target category and location information using feature maps of different sizes. The Head is mainly composed of a detection head and a classification head. The detection head consists of a series of convolutional and deconvolutional layers and is responsible for generating detection results. YOLOv8 adopts an anchor-free mechanism, which directly regresses the category and confidence of bounding boxes. The classification head classifies each feature map through global average pooling and outputs the probability distribution of each category via dimensionality reduction. This design enables the model to handle multi-class classification tasks more efficiently.

In the field of deep learning, the performance improvement of the YOLOv8 model benefits from its innovative network architecture. The model adopts multi-scale feature fusion technology to integrate information from feature maps of different scales. While maintaining high resolution, it enhances the model's detection capability for small targets. In addition, YOLOv8 introduces an attention mechanism, which adaptively adjusts the importance of different regions in feature maps to further improve recognition accuracy and robustness.

In practical applications, the high speed and accuracy of the YOLOv8 model endow it with great potential in fields such as real-time video analysis, environmental perception for autonomous vehicles, and industrial visual inspection. Its high computational efficiency benefits from lightweight model design, enabling YOLOv8 to run on edge computing devices and meet the requirements of low latency and high efficiency. Meanwhile, the open-source nature of YOLOv8 provides researchers and developers with a comprehensive experimental platform, facilitating further optimization and innovation of detection algorithms. The YOLOv8 architecture is adopted to realize bottle mouth defect detection. YOLOv8 is equipped with Active Learning capability, which enables manual labeling of uncertain samples (such as blurry images or edge cases) to iteratively boost model performance.

In this process, combined with the YOLOv8 Backbone (e.g., CSPDarknet) and Cross Stage Partial (CSP) structure, multi-level features of bottle mouth images are effectively extracted, improving the detection sensitivity for tiny defects such as cracks and scratches.The Bidirectional Feature Pyramid Network (BiFPN) fuses multi-scale feature information and further enhances the recognition ability for micro defects and complex backgrounds. The Decoupled Head strategy separates classification and regression tasks, optimizing the model's prediction accuracy for defect locations and categories.

Mosaic augmentation trains the model by stitching multiple images to simulate complex scenarios, and strengthens the model robustness against bottle mouth defects under varying positions, scales and lighting conditions.

This paper proposes a multi-scale feature fusion extraction method based on YOLOv8 for training and extracting bottle mouth image data. It mainly realizes feature extraction and fusion for defects including cracks, notches, stains and occlusions. The generated feature vectors possess strong representation capability, which can comprehensively and accurately describe the image information of bottle mouths. The overall framework of the bottle mouth defect detection algorithm is shown in Figure 9.

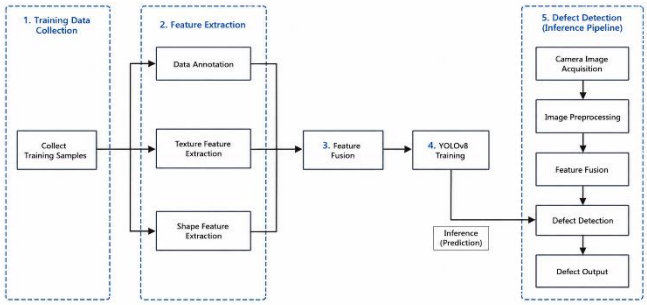

Fig.9. Material Identification and Classification Framework

In the field of deep learning, the convolutional neural network architecture of YOLOv8 is not limited to feature extraction, but also involves network optimization and training strategies. To further improve model performance, YOLOv8 adopts a variety of technologies, including Data Augmentation, Regularization, and Batch Normalization. Data augmentation artificially expands the training dataset and enhances the model's generalization ability. Regularization techniques such as dropout and weight decay help reduce overfitting. Batch normalization ensures stability during network training and accelerates the convergence speed. Figure 10 presents the feature extraction results of this project.

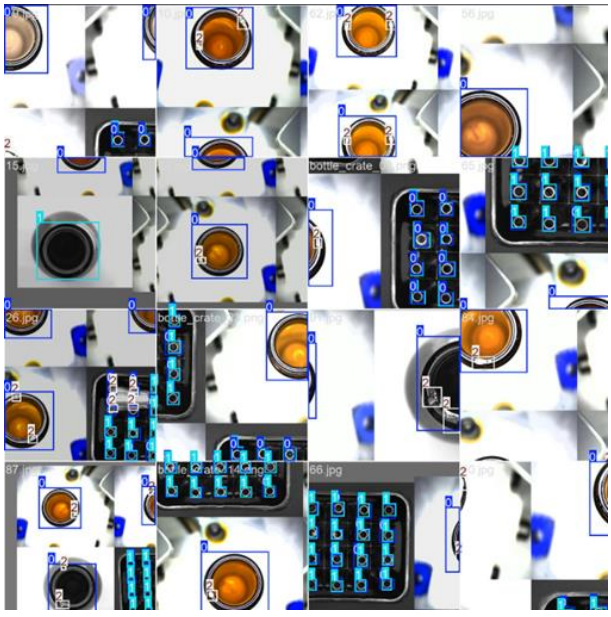

Fig. 10. Training Set Feature Extraction Results

YOLO series models are widely applied due to their capability of real-time multi-object detection. The loss function serves as a key criterion for evaluating model performance. The powerful performance of YOLO models is largely attributed to its specially designed loss functions, including Focal Loss and SIoU Loss.

Figure 11 shows the loss function plot of this project.

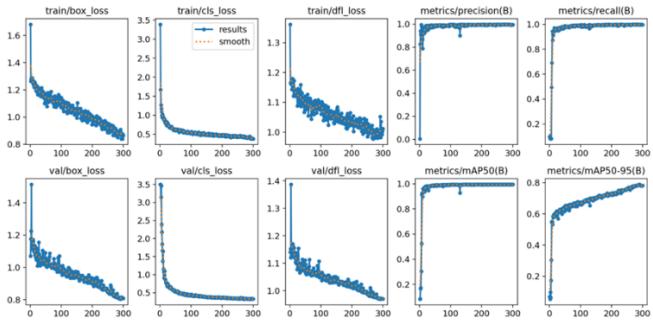

Fig. 11. Loss functions

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REFERENCES

[1] Key Laboratory of Metallurgical Equipment and Control Technology of Ministry of Education, Wuhan University of Science and Technology, "Defect Detection of Glass Bottle Mouths Based on Convolutional Neural Networks," *Manufacturing Automation*, vol. 44, pp. 104–160, June 2022.

[2] Wang. J G, "Review on the Development of Machine Vision Technology in Intelligent Manufacturing," *Acta Automatica Sinica*, vol. 47, no. 3, pp. 521–532, 2021.

[3] Tao. X, "Detection of Power Line Insulator Defects Using Aerial Images Analyzed With Convolutional Neural Networks," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 48, no. 7, pp. 1124–1135, 2018.

[4] Li. K B, "Surface Defect Detection of Automobile Turbine Housing Parts Based on Machine Vision," *Nanjing University of Aeronautics and Astronautics*, 2019.

[5] Hao Z, Liang X, Li G, "Quality control and preprocessing of sequencing reads," *Bio-Protocol*, vol. 12, 2022.

[6] Guo. Y C , Tang. Y J, "Steel Surface Defect Detection Algorithm Based on MHD-YOLO," *Laser & Optoelectronics Progress*, vol. 62, no. 20, pp. 122–131, May 2025.

[7] Fraunhofer Institute, "Real-time 3D Inspection for Automotive Components Based on ToF Camera and YOLOv8," *Journal of Intelligent Manufacturing*, vol. 34, no. 2, pp. 567–580, 2023.

[8] Diwan. T, Anirudh. G, Tembhurne. J V, "Object detection using YOLO: challenges, architectural successors, datasets and applications," *Multimedia Tools and Applications*, vol. 82, pp. 9243–9275, 2025.

[9] Gupta. S, "3D Defect Detection Using Structured Light and Deep Learning," *IEEE Transactions on Industrial Electronics*, vol. 68, no. 5, pp. 4321–4330, 2021.

A Lightweight Structure-Aware CNN-Transformer Network for Robust Handwritten Character Recognition in Robotic Automation Scenarios

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***Abstract*—Handwritten character recognition is a critical visual perception capability for robotic automation systems, with applications in scenarios such as inspection logs, warehouse labels, and maintenance work orders. To address the requirements for high accuracy, lightweight design, and real-time deployment on robot-side systems, this paper proposes the Lightweight Structure-Aware CNN-Transformer Network (LSA-CTNet). It combines a compact convolutional neural network (CNN) to extract local stroke features, a Transformer encoder to model global spatial relationships, and a structure-aware auxiliary branch to enhance the ability to distinguish visually similar characters. Experiments conducted on the EMNIST ByClass dataset demonstrate that LSA-CTNet achieves a Top-1 accuracy of 92.62%, a Top-5 accuracy of 99.91%, and a Character Error Rate (CER) of 7.38%, with 0.08 million parameters and an inference latency of 1.131 ms per image. The results demonstrate that the proposed method significantly improves handwritten character recognition performance while ensuring lightweight architecture and inference efficiency, making it suitable for deployment in robot vision modules.**

Keywords—Handwritten Character Recognition, CNN, Transformer, Structure-Aware Learning, Robotic Automation

I. INTRODUCTION

With advancements in fields such as manufacturing, logistics, inspection, and service robotics, robotic automation systems are gradually evolving from simple execution devices into intelligent systems that integrate perception, understanding, decision-making, and interaction. In practical applications, robots must not only perform tasks such as movement, grasping, and manipulation, but also recognize handwritten text on-site, such as inspection records, warehouse labels, and maintenance work orders. Handwritten character recognition within the robot vision pipeline must ensure high accuracy while also being lightweight and efficiently deployable to meet real-time inference requirements at the edge.

Handwritten characters vary in stroke thickness, slant, connection patterns, and overall shape. Furthermore, there are numerous visually similar categories among digits, uppercase letters, and lowercase letters (such as “0” and “O,” “1” and “I” or “l”), which increases the difficulty of recognition. Traditional convolutional neural networks (CNNs) can extract local stroke features, but due to the limitations of their local receptive fields, they struggle to capture the overall structural relationships of characters. While Transformers possess global modeling capabilities, their computational overhead for large-scale structures is high, making them unsuitable for deployment on robots.

To address the aforementioned issues, this paper proposes a lightweight structure-aware CNN-Transformer network (LSA-CTNet). This model extracts local features using a compact CNN, models the global spatial relationships among characters using a low-cost Transformer encoder, and introduces a structure-aware auxiliary branch to provide additional structural supervision, thereby enhancing the ability to distinguish visually similar characters. Compared to existing methods, LSA-CTNet achieves higher recognition accuracy while maintaining lightweight architecture and inference efficiency.

This paper focuses on the task of isolated handwritten character recognition in robotic automation scenarios. By integrating the model into a visual pipeline comprising “robotic image acquisition—character region extraction—character recognition—structured information generation,” it provides an efficient and lightweight character recognition module for inspection, warehousing, and maintenance tasks. Experiments were conducted using the EMNIST ByClass dataset to evaluate model performance. The results demonstrate that LSA-CTNet outperforms mainstream comparison models in terms of accuracy, character error rate, model parameter count, and inference latency, validating its potential for application in robotics.

II. RELATED WORK

A. Traditional Methods and Convolutional Neural Network Methods

Early handwritten character recognition primarily followed a “preprocessing-feature extraction-classifier discrimination” workflow. Commonly used manual features included structural, statistical, boundary, and directional gradient features, while classifiers included MQDF, MLP, RBF, LVQ, and SVM [1-4]. Although these methods are highly interpretable, their

performance depends on the design of manual features, and they have limited generalization capabilities regarding variations in writing styles, stroke deformations, and visually similar characters. With the advancement of deep learning, neural networks-particularly CNNs-can directly learn local stroke, edge texture, and high-level structural features from images, reducing reliance on manually designed features. However, the local receptive field of CNNs limits their ability to model long-range dependencies between different regions of a character and to capture overall structural information [5-10].

B. Complex Character Recognition and Lightweighting Methods

As recognition tasks have expanded from digital characters to multilingual and complex character sets, the size of character classes, structural complexity, and the number of visually similar classes have increased significantly. Alom et al. [11] systematically evaluated the performance of deep convolutional networks such as VGG, ResNet, and DenseNet in Bengali handwritten character recognition, demonstrating that deep convolutional models are capable of extracting more robust discriminative features. Balaha et al. [12] developed a deep learning system for Arabic handwritten character recognition and analyzed the impact of initialization methods, optimizers, data augmentation, and regularization strategies on model performance. Pomazan et al. [13] further compared the suitability of various CNN architectures for Arabic character recognition tasks. Regarding handwritten Chinese character recognition, Jin Lianwen et al. [14] pointed out that the main challenges include a large number of classes, complex stroke structures, a high number of visually similar characters, and the lack of stroke order information in offline recognition.

Recognizing complex characters typically requires stronger feature representation capabilities, but this also leads to an increase in model size and computational overhead. In robotic automation scenarios, recognition models must not only ensure high accuracy but also adapt to the computational resource constraints of robotic or edge devices. Xiao et al. [15] proposed a fast and compact convolutional neural network from the perspective of model compression and acceleration to reduce the computational and storage overhead of handwritten Chinese character recognition models for large character classes. The aforementioned research indicates that for complex character recognition tasks, models must strike a balance between recognition performance, parameter scale, and inference efficiency. Therefore, constructing recognition models that balance classification capability with lightweight deployment is a critical requirement for handwritten character recognition in robotic automation scenarios.

C. Attention Mechanisms and Complex Scene Recognition Methods

In recent years, research on handwritten character recognition has gradually expanded from single-classification models to areas such as transfer learning, classifier ensemble, few-shot learning, attention mechanisms, and object detection frameworks. Maitra et al. [16] utilized a CNN trained on Bengali character data as a general-purpose feature extractor and combined it with an SVM to achieve digit recognition across multiple writing systems, demonstrating the cross-script transferability of CNN features. Bora et al. [17] combined a CNN with an ECOC classifier to improve the recognition performance of the traditional Softmax classification layer. Li Yuanmu and Wang Zhanqing [18] proposed a matching network based on deep Gaussian processes to address the problem of few-shot handwritten character recognition. Zhuo Tiantian and Sang Qingbing [19] introduced attention mechanisms and composite convolutions into the CRNN+CTC framework for offline English handwritten character recognition. Albahli et al. [20] applied Faster-RCNN to the handwritten character recognition task, improving recognition performance through joint modeling of region detection and classification.

While existing research has made progress in traditional features, CNNs, complex character recognition, lightweight models, and attention mechanisms, handwritten character recognition for robotic automation systems still requires simultaneous consideration of local stroke features, global structural relationships, the differentiation of visually similar characters, and lightweight model deployment. Based on this, this paper proposes LSA-CTNet, which combines lightweight CNNs, Transformer-based global relationship modeling, and structure-aware auxiliary learning to enhance the overall performance of handwritten character recognition models.

III. PROPOSED METHOD

A. Overall Framework

To balance local feature extraction, global structural modeling, and lightweight deployment, this paper proposes the Lightweight Structure-Aware CNN-Transformer Network (LSA-CTNet). The model consists of a lightweight CNN feature extraction module, a Transformer global relationship modeling module, and a structure-aware auxiliary branch. Its overall architecture is shown in Fig. 1.

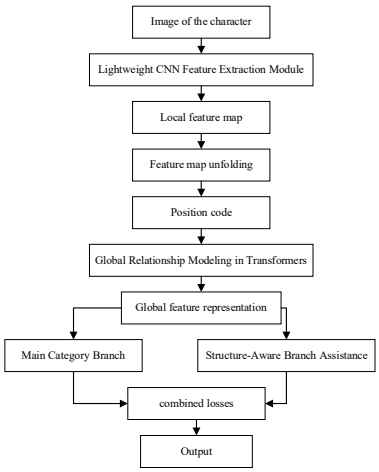

Fig. 1. Overall Network Architecture of LSA-CTNet

First, the model uses a lightweight CNN module to extract local stroke, edge texture, and inflection structure features from character images, yielding compact convolutional feature maps. Subsequently, the convolutional feature maps are unrolled into a token sequence based on spatial position, and learnable positional encoding is incorporated to preserve spatial information between different local regions. Next, the

Transformer encoder models global relationships within the token sequence using a multi-head self-attention mechanism, thereby capturing long-range dependencies between different regions of the characters. Finally, the model outputs character recognition results for 62 classes via the main classification branch and utilizes a structure-aware auxiliary branch during training to predict character structural attributes, thereby enhancing the model's ability to distinguish between visually similar characters and overall morphological differences.

The overall process of the model can be represented as follows:

$$F_C = f_{cnn}(x) \tag{1}$$

$$F_t = f_{trans}(F_C) \tag{2}$$

$$y = f_{cls}(F_t), \quad \hat{s} = f_{aux}(F_t) \tag{3}$$

Here, x denotes the input character image; F_C denotes the local features extracted by the CNN; F_t denotes the global features encoded by the Transformer; y denotes the character class prediction result; and $\hat{s}$ denotes the structural attribute prediction result.

To further illustrate the network architecture of LSA-CTNet, Table I. presents the model's key structural parameters.

TABLE I. LSA-CTNET NETWORK ARCHITECTURE PARAMETERS

Model Layer	Model Structure	Core size/parameters
CNN-1	CNN+BN+ReLU	3×3,16,s=1
Downsample-1	MaxPool	2×2,s=2
CNN-2	CNN+BN+ReLU	3×3,32,s=1
Downsample-2	MaxPool	2×2,s=2
LightConv-1	DWConv+PWConv+BN+ReLU	3×3DWConv,1×1 PWConv,48
LightConv-2	DWConv+PWConv+BN+ReLU	3×3DWConv,1×1 PWConv,64
Tokenization	Flatten	64tokens
Position Encoding	Trainable position encoding	64×64
Transformer Encoder	MHSA+FFN	2layers,4heads,dim=64
Global Pooling	Average Token Pooling	-
Main Classifier	FC+Softmax	-
Auxiliary Branch	FC+ Multi-task output	-

B. Lightweight CNN Feature Extraction Module

The local strokes, edge orientations, structural transitions, and connection patterns of handwritten characters serve as crucial criteria for distinguishing between different categories. Convolutional Neural Networks (CNNs), characterized by their local receptive fields and weight-sharing mechanisms, are capable of effectively extracting local structural information from character images. Consequently, this paper employs a lightweight CNN as the front-end feature extraction module to derive robust representations of local strokes with a reduced parameter count.

As shown in Fig. 2, the lightweight CNN module consists of standard convolutional layers, downsampling layers, and depthwise separable convolutional layers. The model first employs two sets of Conv + BN + ReLU units to extract low-level edge features and local stroke features, while progressively reducing the spatial dimensions of the feature maps through max-pooling operations. Subsequently, the model utilizes depthwise separable convolutions to further extract high-level local structural features. Depthwise separable convolution decomposes standard convolution into depthwise convolution and pointwise convolution, enabling local feature transformation with fewer parameters and lower computational complexity, thereby enhancing the lightweight nature of the model.

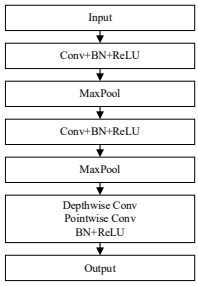

Fig. 2. Structure of a Lightweight CNN Local Feature Extraction Module

The basic form of a standard convolutional unit can be expressed as:

$$F_i = \sigma(BN(Conv(F_{i-1}))) \tag{4}$$

Here, F_i denotes the output features of the i layer, $Conv(\cdot)$ denotes the convolution operation, $BN(\cdot)$ denotes batch normalization, and $\sigma(\cdot)$ denotes the activation function.

A depthwise separable convolution unit can be represented as:

$$F_i = \sigma(BN(PWConv(DWConv(F_{i-1})))) \tag{5}$$

Specifically, $DWConv(\cdot)$ denotes depthwise convolution, which is used to extract spatial local features within each channel; $PWConv(\cdot)$ denotes 1×1 pointwise convolution, which serves to facilitate information fusion across channels. Through this architecture, the model is able to preserve its capacity to represent local stroke combination features while simultaneously reducing the number of parameters.

Ultimately, the output dimensions of the lightweight CNN module are:

$$F_C \in R^{C \times H' \times W'} \tag{6}$$

Here, H', W', and C denote the height, width, and number of channels of the convolutional feature map, respectively. This feature map not only preserves the local structural information within the character image but also significantly compresses its spatial dimensions, thereby providing a compact input representation for subsequent global relationship modeling via the Transformer. Compared to directly processing raw images with a Transformer, utilizing a CNN to first extract local features reduces the number of tokens, thereby lowering the computational overhead of self-attention.

C. Transformer Global Relation Modeling Module

Although CNNs are effective at extracting local stroke and edge texture features, their local receptive field mechanism limits their capacity to model long-range spatial dependencies. For the task of handwritten character recognition, the relative positions between different regions of a character—as well as its overall layout—play an equally significant discriminative role. This is particularly true in the recognition of visually similar characters, where relying solely on local texture features may prove insufficient to reliably distinguish between visually analogous categories. To address this, this paper introduces a Transformer encoder following a lightweight CNN to enhance the model's ability to capture global structural relationships.

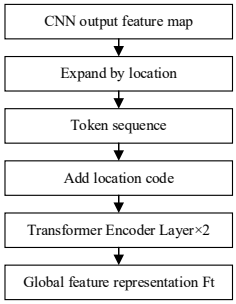

Fig. 3. The Process of Mapping Convolutional Feature Maps to Transformer Sequence Models

As shown in Fig. 3, the feature maps output by the CNN are flattened into a sequence of tokens based on their spatial positions. Since the spatial dimensions of the feature maps are [8, 8], this flattening process yields a token sequence of length 64; each token corresponds to a specific spatial location within the feature map and possesses a feature dimension of 64. This process can be expressed as follows:

$$Z = [z1, z2, \cdots, z_N], \quad N = H^{'} \times W^{'} \tag{7}$$

Here, Z denotes the flattened token sequence, and z_i represents the local feature vector corresponding to the i-th spatial position.

To preserve the spatial positional information of the tokens, this paper incorporates learnable positional encodings into the token sequence:

$$Z_0 = Z + E_{pos} \tag{8}$$

Here, $E_{pos} \in R^{N \times C}$ denotes the positional encoding, and Z_0 represents the Transformer input sequence after the addition of positional information.

The Transformer encoder computes the correlations between different tokens using a multi-head self-attention mechanism. The computation process for single-head self-attention can be expressed as:

$$Attention(Q,K,V) = Softmax(\frac{QK^T}{\sqrt{d}})V \tag{9}$$

Here, Q, K, and V represent the query, key, and value matrices, respectively, and d denotes the feature dimension. Multi-head self-attention further maps features into multiple subspaces to model relationships; its computational form is as follows:

$$MHSA(Z) = Concat(head_1, head_2, \ldots, head_h)W^O \tag{10}$$

Here, h denotes the number of attention heads—set to 4 in this paper—and W^O represents the output projection matrix. The Transformer encoder consists of multi-head self-attention layers and feed-forward networks, utilizing residual connections and normalization operations to enhance training stability. Following Transformer encoding, the model acquires feature representations that capture global spatial relationships, F_t thereby strengthening its capacity to model the holistic structure of characters and distinguish between visually similar categories.

D. Structure-Aware Auxiliary Learning Branch

In the task of mixed handwritten character recognition, categories such as digits, uppercase letters, and lowercase letters exhibit certain visual similarities; consequently, some characters are prone to confusion due to resemblances in local strokes or overall contours. If training relies solely on the primary classification loss, the model may focus predominantly on local textural differences while failing to adequately learn the global structural attributes of the characters. To enhance the model's capacity to represent character structural information, this paper designs a structure-aware auxiliary learning branch that introduces additional structural supervision during the training phase.

As shown in Fig. 4, the structure-aware auxiliary branch shares the global feature representations extracted by the CNN-Transformer with the main classification branch. The main classification branch is responsible for predicting 62 character class labels, while the structure-aware auxiliary branch is dedicated to predicting structural attributes-such as character type, closed structures, and overall morphology. This branch participates in joint optimization only during the training phase, serving to guide the model in learning more robust character structure representations; during the inference phase, the auxiliary branch can be removed based on deployment requirements-retaining only the backbone network and the main classification branch-thereby avoiding any additional inference overhead.

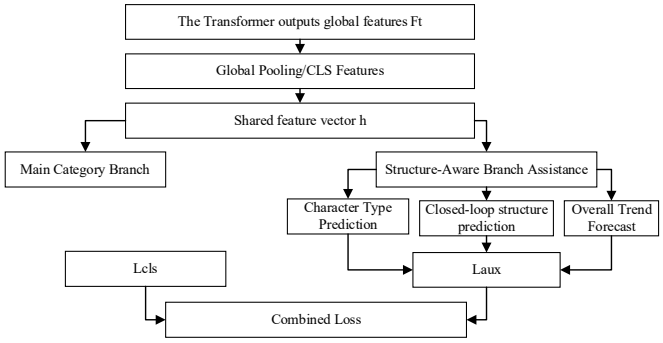

Fig. 4. Structure-Aware Learning Branches and Joint Loss Design

Structure-aware auxiliary labels are derived by mapping the original character class labels and do not require additional

manual annotation. This paper constructs auxiliary supervision signals from three perspectives-character type, closed structure, and overall morphology-with the classification scheme presented in Table II.

TABLE II. STRUCTURE-AWARE ASSISTED LABELING METHOD

Structural Properties	Number of classes	Classification method	Sample characters
Character type	3	Numbers/Uppercase letters/Lowercase letters	0-9/a-z/A-Z
Closed structure	3	Closed/Semi-closed/Open	0,8,O/C,G/1,l/L
Overall shape	4	Vertical/Curved/Crossed/Mixed	1,l,I/S,C/X,x/R,K

Here, the character type serves to distinguish the broad category to which a character belongs; the closed structure describes the presence-or approximate presence-of enclosed regions within the character's contours; and the overall morphology characterizes the character's primary geometric structure. Through this auxiliary supervision, the model is able to learn more structural character representations-beyond the scope of the primary classification task-thereby enhancing its ability to distinguish between visually similar characters.

The structure-aware auxiliary branch can be expressed as:

$$\hat{s} = f_{aux}(F_t) \tag{11}$$

Here, F_t denotes the global feature representation output by the Transformer; $f_{aux}(\cdot)$ denotes the structure-aware auxiliary branch; and $\hat{s}$ denotes the structural attribute results predicted by the model. If multiple structural attributes are considered simultaneously, the auxiliary branch can be further expressed as:

$$\hat{s} = \{\hat{s}^{typc}, \hat{s}^{close}, \hat{s}^{shape}\} \tag{12}$$

Among these $\hat{s}^{typc}$, $\hat{s}^{close}$ and $\hat{s}^{shape}$ respectively represent the prediction results for character type, closed structure, and overall morphology.

E. Loss Function Design

This paper employs a model jointly optimized using a primary classification loss and a structure-assisted loss. The primary classification task serves to constrain the model to output the correct labels for the 62 character classes, while the structure-assisted task guides the model in learning character structural attributes. The primary classification loss utilizes the cross-entropy loss function, defined as follows.

$$L_{cls} = -\sum_{i=1}^{K} y_i \log(y_i) \tag{13}$$

Here, K denotes the number of character classes; in this paper $K = 62$, y_i the true class labels and y_i the model's predicted probabilities.

The structure-assisted task also employs a cross-entropy loss. For a single structural attribute, the loss is defined as:

$$L_{aux} = -\sum_{j=1}^{M} s_j \log(\hat{s}_j) \tag{14}$$

$$L_{aux} = -\sum_{j=1}^{M} s_j \log(\hat{s}_j) \tag{15}$$

Here, M denotes the number of structural attribute categories; s_j represents the true structural label; and $\hat{s}_j$ represents the prediction probability from the auxiliary branch.

Since the structural auxiliary branch in this paper encompasses three auxiliary tasks—character type, closure structure, and overall morphology—the structural auxiliary loss can be further expressed as:

$$L_{aux} = L_{type} + L_{close} + L_{shape} \tag{16}$$

Here, L_{type}, L_{close} and L_{shape} denote the character type prediction loss, the closed-structure prediction loss, and the overall morphology prediction loss, respectively.

The total model loss is defined as:

$$L = L_{cls} + \lambda L_{aux} \tag{17}$$

Here, λ the auxiliary loss weight serves to balance the influence of the primary classification task and the structural auxiliary task on model training. Through joint optimization, the model is able not only to learn discriminative features for character classification but also to explicitly focus on character structural attributes, thereby enhancing its ability to distinguish between visually similar characters.

IV. EXPERIMENTS AND RESULTS

A. Datasets, Experimental Settings, and Evaluation Metrics

This paper employs the open-source EMNIST ByClass dataset to experimentally validate the proposed LSA-CTNet model. The EMNIST ByClass dataset comprises various handwritten character categories-including digits, uppercase English letters, and lowercase English letters-totaling 62 distinct classes. Compared to handwritten recognition datasets containing only digits, EMNIST ByClass features a larger number of categories; furthermore, certain classes exhibit strong visual similarities-such as "0" and "O," or "1," "I," and "l"-thereby enabling an effective assessment of the model's capability to recognize mixed handwritten characters and visually similar classes. Sample instances from the dataset are illustrated in Fig. 5.

Fig. 5. Samples of all letters and digits in the EMNIST dataset

It should be noted that EMNIST ByClass mainly contains clean and cropped isolated character images. Therefore, this study focuses on controlled handwritten character recognition. Real-world degradations such as blur, occlusion, illumination variation, and background interference are not further evaluated due to the lack of real robotic acquisition equipment and real-scene handwritten character samples. Future work will focus on collecting real-scene robotic handwritten character samples to further verify the robustness of the proposed model.

In the experiment, all character images were uniformly resized to 32×32, and their pixel values were normalized. The model was trained using a combination of the main classification loss and a structural auxiliary loss. The Adam optimizer was selected, with a batch size of 1024, an initial learning rate of 0.001, and a maximum of 100 training epochs. To ensure experimental fairness, all comparison models were trained and tested using the same training and test set splits and the same input dimensions. Inference latency was tested on the same hardware platform and with the same input dimensions, and the average inference time per image was recorded. The experimental platform consists of an NVIDIA GeForce RTX 3090 GPU (24 GB), with the deep learning framework PyTorch 2.0.0, CUDA version 11.8, and Python version 3.8.

To comprehensively evaluate model performance, this study established evaluation metrics across two dimensions: recognition accuracy and deployment efficiency. Recognition accuracy was assessed using Top-1 accuracy, Top-5 accuracy, and the Character Error Rate (CER); deployment efficiency was evaluated based on the number of model parameters, floating-point operations (FLOPs), and the inference latency per image.

Top-1 accuracy serves to quantify the proportion of instances where the model's predicted class matches the true class, and is defined as follows:

$$Accuracy = \frac{N_{correct}}{N_{total}} \tag{18}$$

The Character Error Rate (CER) is used to measure the proportion of errors made by a model during recognition; it is defined as:

$$CER = \frac{N_{error}}{N_{total}} \tag{19}$$

Here, $N_{correct}$ denotes the number of misclassified samples. Top-5 accuracy is used to measure whether the true class appears among the top five classes with the highest predicted probabilities. The number of parameters, floating-point operations, and inference latency are used to measure the model's storage overhead, computational complexity, and real-time deployment performance, respectively.

B. Comparison of Model Settings and Analysis of Recognition Performance

To validate the effectiveness of the proposed method, LeNet-5, ResNet18, MobileNetV3, CRNN, and ViT-Tiny were selected as comparative models. Each model possesses distinct structural characteristics, enabling a comparative analysis of the proposed method from various perspectives-including shallow convolutional networks, deep residual networks, lightweight networks, sequence modeling networks, and Transformer networks. The configurations for the comparative models are presented in Table III.

TABLE III. COMPARISON OF MODEL DESCRIPTIONS

Model	Structural Features	Purpose of the Comparison
LeNet-5	Shallow Convolutional Neural Networks	Comparing the basic CNN
ResNet18	Residual Convolutional Network	Comparing the representational power of deep CNNs
MobileNetV3	Lightweight Convolutional Networks	Comparing the Efficiency of Edge Deployment
CRNN	CNN + RNN	Comparing the Performance of Sequence Modeling Structures
ViT-Tiny	Lightweight Transformer	Comparing the performance of pure Transformer models

A comparison of the recognition performance of different models on the EMNIST ByClass test set is presented in Table IV.

TABLE IV. COMPARISON OF RECOGNITION PERFORMANCE ACROSS DIFFERENT MODELS

Model	Top-1 accuracy (%)	Top-5 accuracy (%)	CER/%
LeNet-5	86.22	99.59	13.78
ResNet18	87.47	99.78	12.53
MobileNetV3	86.50	99.64	13.50
CRNN	87.25	99.80	12.75
ViT-Tiny	87.22	99.80	12.78
LSA-CTNet	92.62	99.91	7.38

As shown in Table IV, there are differences in recognition performance among various models on the EMNIST ByClass dataset. As a shallow convolutional neural network, LeNet-5 achieved a Top-1 accuracy of 86.22% and a Character Error Rate (CER) of 13.78%, indicating its limited ability to distinguish between complex and visually similar characters. ResNet18, leveraging its residual structure, achieves a Top-1 score of 87.47%, demonstrating that increasing network depth enhances feature representation. MobileNetV3, designed for lightweight deployment, achieves a Top-1 score of 86.50%, slightly lower than ResNet18, suggesting that lightweight design may compromise certain aspects of feature representation. CRNN and ViT-Tiny represent sequence modeling and self-attention structures, respectively, with Top-1 scores of 87.25% and 87.22%. While they show some improvement in candidate category coverage, the models still rely on local stroke and two-dimensional structural features.

In contrast, LSA-CTNet achieved a Top-1 accuracy of 92.62% and a CER of 7.38%, significantly outperforming other models, with a Top-5 accuracy of 99.91%. These results demonstrate that the synergistic interaction between the lightweight CNN for extracting local features, the Transformer encoder for modeling global relationships, and the structure-aware auxiliary branch effectively enhances the recognition

capability for mixed handwritten characters and visually similar characters.

C. Analysis of Model Complexity and Deployment Efficiency

Given that robotic automation systems typically need to operate on platforms with limited computational resources, this paper further compares the parameter count, floating-point operations, and inference latency of different models. The experimental results are presented in Table V.

TABLE V. COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY

Model	Parameter/M	FLOPs/M	Inference latency (ms)
LeNet-5	0.07	0.42	0.199
ResNet18	11.2	457.76	1.496
MobileNetV3	1.58	5.41	3.117
CRNN	3.35	124.58	0.66
ViT-Tiny	1.8	87.36	1.749
LSA-CTNet	0.08	15.08	1.131

As shown in Table V, there are significant differences among the models in terms of the number of parameters, computational complexity, and inference latency. LeNet-5 has the lowest number of parameters and FLOPs and the fastest inference speed, but its Top-1 accuracy is relatively low, indicating that shallow networks struggle to fully capture the features of 62 handwritten character classes. ResNet18 has 11.2 million parameters and 457.76 million FLOPs, resulting in high computational overhead, which is not conducive to edge deployment. MobileNetV3 has lower FLOPs, but its inference latency is 3.117 ms, indicating that actual speed is influenced by both network architecture and hardware implementation. Both CRNN and ViT-Tiny have higher parameter counts and computational demands than LSA-CTNet; while ViT-Tiny can model global relationships, its computational burden remains high.

In comparison, LSA-CTNet has only 0.08 million parameters, 15.08 million FLOPs, and an inference latency of 1.131 ms, while achieving a Top-1 accuracy of 92.62% and a CER of 7.38%. The results demonstrate that LSA-CTNet achieves optimal recognition performance while ensuring lightweight architecture and efficient inference, making it suitable for deployment in robotic automation scenarios.

It should be noted that the inference latency in this study is measured on a unified GPU-based platform rather than on actual robotic embedded hardware. Future work will further evaluate the proposed model on robotic embedded hardware and robotic edge devices to verify its practical deployment performance.

D. Ablation Study Analysis

To further validate the contribution of each constituent module within LSA-CTNet to recognition performance, this paper conducts ablation experiments on the EMNIST ByClass dataset. These ablation experiments are performed under identical data splits, input dimensions, and training strategies—varying only the model architecture or auxiliary supervision settings—to ensure the comparability of the results across different experiments. This study focuses specifically on analyzing the impact of the lightweight CNN feature extraction module, the Transformer-based global relation modeling module, positional encoding, and the structure-aware auxiliary learning branch on overall model performance.

TABLE VI. ABLATION TEST RESULTS FOR DIFFERENT MODULES

Model	Top-1 accuracy (%)	Top-5 accuracy (%)	CER/%
Light-CNN	86.71	99.66	13.29
Light-CNN+Transformer	87.85	99.70	12.15
LSA-CTNet w/o PE	88.30	99.75	11.70
LSA-CTNet w/o Aux	90.10	99.73	9.90
LSA-CTNet	92.62	99.91	7.38

As shown in the Table VI, when using only Light-CNN, the Top-1 accuracy is 86.71% and the CER is 13.29%, indicating that the convolutional modules can extract local stroke and edge features but have limited ability to model global spatial relationships, making it difficult to distinguish between visually similar characters. After incorporating the Transformer, the Top-1 score improved to 87.85% and the CER decreased to 12.15%, demonstrating that the Transformer encoder can effectively model global character relationships and enhance class discrimination capabilities.

When position encoding (PE) is removed (w/o PE), the Top-1 accuracy is 88.30% and the CER is 11.70%, which is slightly lower than that of the full model, indicating that position encoding provides spatial-temporal information to the Transformer, enhancing its understanding of global structure. When auxiliary branches are removed (w/o Aux), the Top-1 score is 90.10% and the CER is 9.90%, indicating that auxiliary branches provide additional structural supervision during training, enhancing the model's ability to distinguish between visually similar characters and overall character shapes.

The full LSA-CTNet achieved the best performance across all metrics, with a Top-1 accuracy of 92.62%, a Top-5 accuracy of 99.91%, and a CER of 7.38%. This demonstrates that the synergy between the lightweight CNN, Transformer, positional encoding, and auxiliary branches effectively enhances the performance of complex handwritten character recognition. Performance fluctuations remain within a reasonable range, consistent with training randomness and module interaction effects.

E. Training Process and Convergence Analysis

To further assess the training stability of the proposed model, this paper plots the loss and accuracy curves of LSA-CTNet on the training and validation sets, as shown in Fig. 6

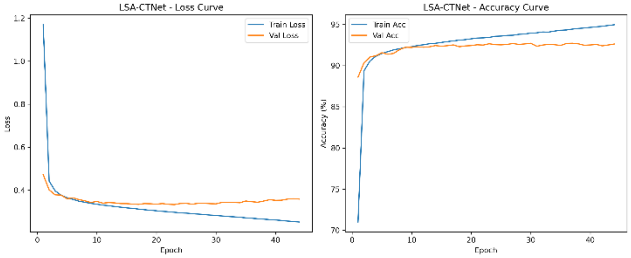

Fig. 6. Training Process Curves of LSA-CTNet on the EMNIST ByClass Dataset

As shown in Fig. 6, the training process of LSA-CTNet on the EMNIST ByClass dataset was generally stable. The loss decreased rapidly during the early stages of training, indicating that the network was able to quickly learn local stroke and edge features. As the number of iterations increased, the training and validation losses gradually leveled off, and the improvement in validation accuracy slowed, suggesting that the model had largely completed feature learning and possessed good generalization capabilities.

To mitigate the risk of overfitting, an early stopping strategy was implemented during training: the process was terminated prematurely if the optimal accuracy on the validation set did not improve over seven consecutive training iterations, and the model parameters with the best performance were retained. Analysis of the loss curve, accuracy curve, and early stopping mechanism indicates that LSA-CTNet achieves stable convergence with relatively low model complexity, while delivering a balanced combination of recognition accuracy and generalization ability.

Based on the experimental results described above, LSA-CTNet outperforms LeNet-5, ResNet-18, CRNN, and ViT-Tiny in terms of Top-1 accuracy, CER, number of parameters, and inference latency. Ablation studies further validate the role of lightweight CNNs, Transformers, positional encoding, and structure-aware auxiliary branches in enhancing performance, demonstrating that the model holds significant practical value for complex handwritten character recognition and deployment on robots.

V. Discussion

This paper proposes LSA-CTNet, a lightweight structure-aware CNN-Transformer network designed for the task of isolated handwritten character recognition in robotic automation scenarios. By combining a lightweight CNN with the global modeling capabilities of a Transformer and structure-aware learning, the model effectively improves recognition performance for mixed handwritten characters and visually similar characters. The model can be integrated into robotic vision pipelines to enable efficient recognition of information such as inspection records, warehouse labels, and maintenance work orders.

The experiment was validated on the EMNIST ByClass dataset, where LSA-CTNet achieved a Top-1 accuracy of 92.62%, a Top-5 accuracy of 99.91%, and a CER of 7.38%, with only 0.08 million parameters, 15.08 million FLOPs, and an inference latency of 1.131 ms per image. Compared with LeNet-5, ResNet-18, MobileNet-V3, CRNN, and ViT-Tiny, the model achieves a more balanced performance in terms of recognition accuracy, model size, and inference efficiency. Ablation studies indicate that lightweight CNNs, Transformers, positional encoding, and auxiliary branches all contribute positively to performance improvements.

Future research will incorporate real-world robotic scenario data to analyze the impact of lighting, viewing angles, motion blur, and detection errors on recognition performance, and will explore model pruning, quantization, and embedded acceleration methods to further improve deployment efficiency on the robot side.

References

[1] Liu C L, Sako H, Fujisawa H. Performance evaluation of pattern classifiers for handwritten character recognition[J]. International Journal on Document Analysis and Recognition, 2002, 4(3): 191-204.

[2] Vamvakas G, Gatos B, Perantonis S J. Handwritten character recognition through two-stage foreground sub-sampling[J]. Pattern Recognition, 2010, 43(8): 2807-2816.

[3] Pal A, Singh D. Handwritten English character recognition using neural network[J]. International Journal of Computer Science & Communication, 2010, 1(2): 141-144.

[4] Zhang Kaibing. A Novel Method for Handwritten Digit Recognition Based on Multi-Feature Combination and Multi-Neural Network Classifier Ensemble [J]. Journal of Xihua University (Natural Science Edition), 2011, 30(1): 65-68.

[5] Pan Yuanyuan, Zhang Li, Duan Lingling, et al. Research on Handwritten Digit Recognition Using Discrete Hopfield Neural Networks [J]. Complex Systems and Complexity Science., 2019, 15(1): 75-79.

[6] Lu Jian, Lin Jiajun. Handwritten Character Recognition Based on CUDA and Deep Belief Networks [J]. Journal of East China University of Science and Technology (Natural Science Edition), 2015 (2): 210-215.

[7] Yuan A, Bai G, Jiao L, et al. Offline handwritten English character recognition based on convolutional neural network[C]//2012 10th IAPR International Workshop on Document Analysis Systems. IEEE, 2012: 125-129.

[8] Rahman M M, Akhand M A H, Islam S, et al. Bangla handwritten character recognition using convolutional neural network[J]. International Journal of Image, Graphics and Signal Processing, 2015, 7(8): 42-49.

[9] Khandokar I, Hasan M, Ernawan F, et al. Handwritten character recognition using convolutional neural network[C]//Journal of Physics: Conference Series. IOP Publishing, 2021, 1918(4): 042152.

[10] Lü Hong. Design of a Handwritten Digit Recognition System Based on Convolutional Neural Networks [J]. Intelligent Computer and Applications., 2019, 9(2): 54-56.

[11] Alom M Z, Sidike P, Hasan M, et al. Handwritten Bangla Character Recognition Using the State-of-the-Art Deep Convolutional Neural Networks[J]. Computational intelligence and neuroscience, 2018, 2018(1): 6747098.

[12] Balaha H M, Ali H A, Saraya M, et al. A new Arabic handwritten character recognition deep learning system (AHCR-DLS)[J]. Neural Computing and Applications, 2021, 33(11): 6325-6367.

[13] Pomazan V, Tvoroshenko I, Gorokhovatskyi V. Handwritten character recognition models based on convolutional neural networks[J]. 2023.

[14] Jin Lianwen, Zhong Zhuoyao, Yang Zhao, et al. A Survey on the Application of Deep Learning in Handwritten Chinese Character Recognition [J]. Acta Automatica Sinica., 2016, 42(8): 1125-1141.

[15] Xiao X, Jin L, Yang Y, et al. Building fast and compact convolutional neural networks for offline handwritten Chinese character recognition[J]. Pattern Recognition, 2017, 72: 72-81.

[16] Maitra D S, Bhattacharya U, Parui S K. CNN based common approach to handwritten character recognition of multiple scripts[C]//2015 13th International conference on document analysis and recognition (ICDAR). IEEE, 2015: 1021-1025.

[17] Bora M B, Daimary D, Amitab K, et al. Handwritten character recognition from images using CNN-ECOC[J]. Procedia Computer Science, 2020, 167: 2403-2409.

[18] Li Yuanmu, Wang Zhanqing. An Improved Small-Batch Handwritten Character Recognition Algorithm [J]. Small and Micro Computer Systems., 2020, 41(7): 1541-1546.

[19] Zhuo Tiantian, Sang Qingbing. Application of Attention Mechanisms and Compound Convolution in Handwritten Recognition [J]. Computer Science and Exploration., 2022, 16(4): 888-897.

[20] Albahli S, Nawaz M, Javed A, et al. An improved faster-RCNN model for handwritten character recognition[J]. Arabian Journal for Science and Engineering, 2021, 46(9): 8509-8523.

Intelligent Classroom Behavior Recognition and Analysis System Based on YOLOv8

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***Abstract*—With the advancement of educational digitalization, intelligent quantitative analysis of students' classroom behaviors is vital to smart education. Manual observation is inefficient and subjective, while traditional algorithms including Haar+AdaBoost and YOLOv3 suffer from low accuracy, unsatisfactory real-time performance and poor portability in complex classroom settings. Different from previous work using off-the-shelf detectors, this paper presents a classroom-oriented behavior analysis system based on optimized YOLOv8n.We build a multi-scenario dataset with over 5,000 annotated images of six student behaviors, covering various lighting conditions, crowd densities and camera angles to narrow the domain gap for classroom detection. A hybrid data augmentation strategy integrating HSV perturbation, geometric transformation and Mosaic augmentation is adopted to handle object occlusion, small distant targets and varying illumination. A lightweight end-to-end platform based on Streamlit is also developed for inference, data statistics and visualization, which supports edge deployment. Experimental results demonstrate that the optimized YOLOv8n achieves excellent overall performance, with mAP@0.5 of 57.0%, an inference speed of 280–350 FPS, 3.2 million parameters and a model size of 6.5 MB. It outperforms YOLOv3, YOLOv5s and YOLOv7-tiny in all key indicators. In resource-constrained classroom environments, the lightweight single-stream architecture of YOLOv8n also delivers better performance than multi-structure models like YOLOv7-tiny.**

Keywords—YOLOv8, Behavior Detection, Smart Education

I. Introduction

Classrooms serve as the core venue for teaching and teacher-student interaction. Students' behaviors directly reflect their learning engagement and provide important references for teaching improvement. Traditional manual observation is labor-intensive, subjective and hard to quantify [1] Early algorithms such as Haar+AdaBoost and YOLOv3 suffer from low accuracy, poor real-time performance, large model size and poor deployability [7] Although deep learning techniques including CNN, LSTM and attention mechanisms have been adopted for classroom monitoring, most existing studies merely apply general detection models [9][10] They fail to address practical challenges such as dense small targets, occlusion, varying illumination and limited computing resources. Moreover, the absence of standardized comparative experiments leads to insufficient basis for model selection.

This paper proposes a classroom-oriented behavior analysis system based on optimized YOLOv8n. The main innovations are as follows. First, a multi-scenario dataset containing over 5,000 images of six typical classroom behaviors is established. It covers diverse lighting conditions, camera views and student densities to reduce the domain gap between general models and classroom detection tasks. Second, a dedicated data augmentation scheme combining HSV perturbation, geometric transformation and Mosaic augmentation is designed to boost detection performance in complex scenarios. Third, comparative experiments on four generations of YOLO models under identical settings verify that the lightweight single-stream YOLOv8n achieves higher accuracy with fewer parameters than multi-structure models like YOLOv7-tiny, making it more suitable for classroom environments with limited computing power. Fourth, a lightweight visual analysis platform based on Streamlit is developed, which can run on ordinary CPU devices. This paper also analyzes five major deployment challenges including device adaptation, multi-stream inference, privacy protection, behavior refinement and model maintenance, providing guidance for practical applications.

II. Algorithm Design and Improvement

This study employs YOLOv8n as the baseline detector for end-to-end classroom behavior detection. An input image is first

processed by data augmentation, then a CSPDarknet backbone extracts multi-level features. A bidirectional FPN+PAN neck fuses P3, P4, and P5 through complementary semantic and spatial information exchange, yielding detection scales N3 (80×80), N4 (40×40), and N5 (20×20). At each scale, a decoupled head performs classification and bounding box regression in an Anchor-Free manner. The model is optimized by CIoU, DFL, and BCE losses during training; at inference, NMS removes redundant predictions, producing detections across six behavior categories: hand-raising, reading, writing, using phone, bowing the head, and leaning over the table. The model contains 3.01M parameters and achieves 355 FPS on an NVIDIA RTX 5060. The pipeline of the proposed algorithm is shown in (Fig. 1.).

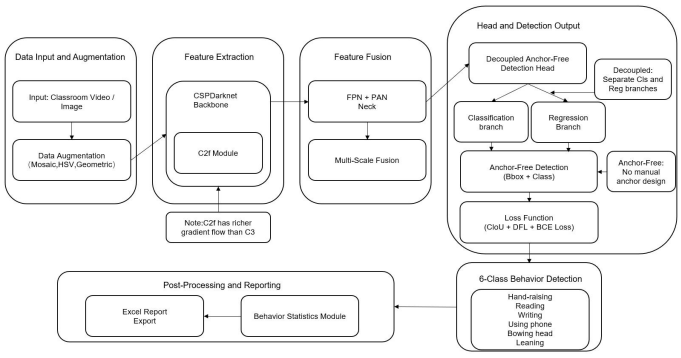


Fig. 1. Pipeline of Improved YOLO-Based Classroom Behavior Recognition

Three complementary augmentation operations are applied during training. Mosaic augmentation composites four random training samples into one image, forcing the model to handle diverse scales and occlusion patterns prevalent in classroom scenes. HSV color jitter performs random perturbation on hue, saturation and brightness within reasonable value ranges, to simulate complex classroom illumination variations such as curtain adjustment and window backlighting. Geometric transformations adopt random rotation, scaling and translation within preset ranges, together with horizontal flipping at a probability of 50%, to capture viewpoint variations from different camera angles and classroom layouts. All augmentation strategies are only applied in the training phase and bring no additional computational overhead during inference.

The backbone adopts CSPDarknet with the C2f (CSP Bottleneck with 2 convolutions + Faster) module as its core building block. Compared to the C3 module in YOLOv5, C2f splits the input feature along the channel dimension into two branches — one undergoes transformation through multiple Bottleneck units while the other passes through directly — then concatenates and fuses them via 1×1 convolution. This cross-layer connection enriches gradient flow, facilitating fine-grained discrimination between visually similar categories such as bowing the head versus leaning over the table. Four feature scales are produced: P2 (4× downsampling, for tiny targets such as phones), P3 (8×, medium-scale behaviors), P4 (16×, larger targets), and P5 (32×, global context). An SPPF module after P5 performs three consecutive 5×5 max-pooling operations, expanding the receptive field to 17×17 without additional parameters.

The neck employs a bidirectional FPN+PAN architecture. The top-down FPN pathway upsamples P5 progressively, concatenating with P4 and P3 before C2f refinement, injecting high-level semantics into shallower layers so that high-resolution maps acquire categorical discriminability. The bottom-up PAN pathway downsamples the P3 fused feature progressively, concatenating with P4 and P5 fused features to transmit precise spatial localization back to deeper layers. The three output scales N3, N4, and N5 correspond to small, medium, and large targets respectively, each possessing both strong semantic understanding and accurate spatial localization — FPN provides what the target is, PAN provides where it is.

The detection head features two key innovations. First, the decoupled architecture separates the classification and regression branches after the shared convolution, each with independent convolutional sequences. The classification branch outputs an H×W×6 probability map; the regression branch outputs H×W×4 bounding box offsets. This resolves the inherent conflict in coupled heads where classification requires semantic features whereas regression demands boundary-sensitive features. Second, the Anchor-Free mechanism discards predefined anchor boxes, directly predicting whether a target center falls within each grid cell and regressing distances to the four bounding box sides. This is particularly advantageous for classroom behaviors: Leaning over the table presents a wide aspect ratio, whereas hand-raising shows a tall and narrow proportion, making fixed anchor boxes unable to adapt to both types of targets.

The training objective consists of three loss functions. CIoU Loss enhances conventional IoU by introducing a center distance penalty and an aspect ratio consistency penalty, which ensures effective gradient transmission even when the prediction box does not overlap with the ground-truth box and constrains the shape deviation of targets. Distribution Focal Loss (DFL) reformulates bounding box regression as discrete probability distribution modeling. It quantifies regression distances into equally spaced intervals and achieves sub-pixel localization precision through probability-weighted expectation, mitigating the adverse impact of annotation noise on small target detection. BCE Loss calculates the binary cross-entropy for each category of every candidate box. In the inference phase, Non-Maximum Suppression (NMS) with an IoU threshold of 0.7 is adopted to eliminate redundant detection boxes. The processed results are divided into two branches. The visualization branch adopts Plotly on the Streamlit platform to display pie charts, bar charts, confidence histograms, and comparison results of active and passive classroom behaviors. Hand-raising, reading and writing are defined as active learning behaviors, while head lowering, leaning over the desk and mobile phone usage are classified as passive distracted behaviors. The data export branch generates an Excel report with two worksheets, storing detailed detection records and category-level statistical summary data respectively.

III. Design and Implementation of the Classroom Behavior Analysis System

A. Overall System Design

To meet the multi-dimensional functional requirements of classroom behavior analysis, this paper elaborates on the overall Web application architecture of the system based on the

Streamlit framework, as well as the composition, workflow and specific functions of each core module.

Considering the functional requirements of front-end human-computer interaction (e.g., behavior visualization, data query), back-end business logic (e.g., YOLOv8 inference, statistical analysis), and data management (e.g., student behavior data storage), this system adopts a Streamlit-based Web architecture. The presentation layer builds interactive interfaces for interaction and visualization; the business logic layer, developed in Python, integrates video processing and behavior analysis; the data management layer uses Streamlit's session state for in-memory data management, realizing temporary storage and state retention [2]. As shown in (Fig. 2.).

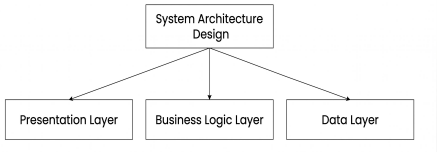
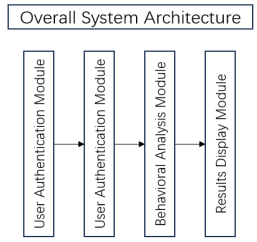
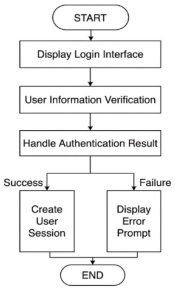


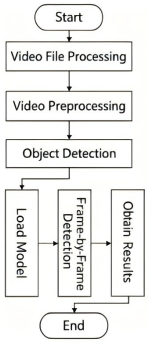
Fig. 2. Overall system architecture diagram



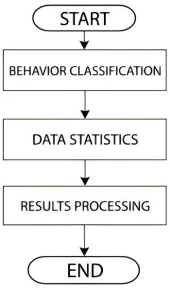
(a) System architecture diagram



(b) Workflow of the user authentication module



(c) Workflow of the video processing detection module.



(d) Workflow of the behavioral analysis module

Fig. 3. System Architecture and Workflows

This system adopts a modular structure [3] to realize classroom video detection, behavior analysis and data visualization (Fig. 3), including user authentication, video processing, behavior analysis and visualization modules. Users log in, upload videos for detection, and obtain visualized statistical results.

The authentication module verifies user accounts and permissions via database comparison. Valid login grants system access, while failed access will trigger prompts and page redirection.

Combined with detection results, the behavior analysis module classifies student behaviors, counts relevant frequency and proportion, and generates statistical reports. Visual charts are produced for intuitive viewing and real-time classroom behavior monitoring [6].

B. Implementation of System

This system is developed under stable hardware and software environments. The hardware satisfies the computation needs for model training and local deployment. Based on Python and the PyTorch framework, combined with OpenCV and NumPy, the lightweight YOLOv8n model is trained and deployed for image processing tasks.

The system adopts a modular structure, including input, detection inference, result processing and visualization modules. Each module works independently and collaboratively, ensuring excellent scalability and stability. It supports images, videos and real-time camera input, with automatic image preprocessing and format adaptation.

YOLOv8n is adopted as the core detection network. Input data is processed by scaling and normalization to achieve feature extraction, target localization and classification. Adaptive data augmentation and anchor box optimization are integrated to improve robustness in complex classroom environments.

After inference, detection frames, categories and confidence values are marked on images. Recognition results are presented in real time and saved locally for statistical analysis. Lightweight deployment effectively reduces hardware consumption and meets the real-time monitoring demands of classroom scenarios.

IV. Implementation of the YOLOv8-based Algorithm

A total of over 5,000 images depicting student classroom behaviors were collected through on-site photography and web scraping. These images were annotated using LabelImg [11], an open-source tool, and categorized into six classes: hand-raising, reading, writing, phone-using, head-down, and leaning on the desk. The dataset was saved in the YOLO format and partitioned into training and validation sets at a ratio of 80% and 20%, respectively; sample images are displayed in (Fig. 4.). As a widely utilized open-source annotation tool, LabelImg facilitates the addition of labeling information to target objects, which serves as critical ground truth for training computer vision models in tasks such as object detection and image classification. The annotation results were exported as .txt files, with the specific data format illustrated in (Fig. 5.).

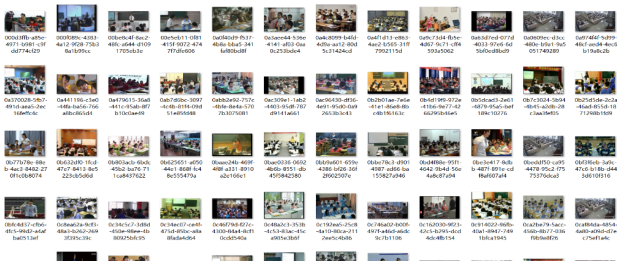


Fig. 4. Sample images of the dataset

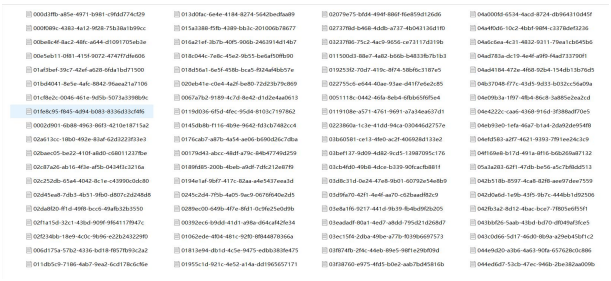


Fig. 5. Format of the .txt annotation files

This system adopts data augmentation at both image and label levels [4]. At the image level, pixel-wise transformations such as HSV adjustment, rotation, scaling, translation and flipping are applied to enrich feature diversity. HSV conversion and channel modulation simulate diverse lighting and color environments to enhance environmental robustness.

In addition, [5] Mosaic augmentation is used to splice four randomly cropped and scaled images into one sample. It effectively expands dataset volume, boosts small-object detection performance, and enables batch processing of multiple images to lower training memory consumption. The Mosaic augmentation workflow is shown in (Fig. 6).

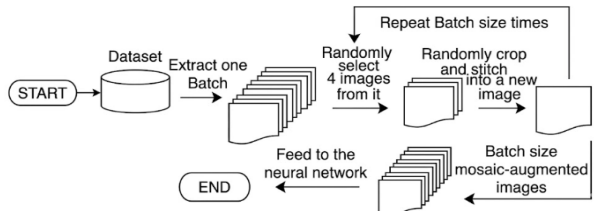


Fig. 6. Workflow of Mosaic data augmentation

The annotated dataset is split into training and validation sets. Transfer learning is applied [8] using five official YOLOv8 pre-trained models: YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l and YOLOv8x.

YOLOv8n is lightweight and fast for low-power devices; YOLOv8s balances speed and accuracy; YOLOv8m offers steady performance; YOLOv8l and YOLOv8x pursue higher accuracy at the expense of heavier computation.

Larger models achieve better precision but lower speed. Considering actual deployment needs, YOLOv8n.pt is adopted for training. The optimal trained weight is combined with the Streamlit framework to complete image and video detection.

V. System Testing and Comparative Experiments of YOLO Algorithms

A. System Testing

Functional testing mainly covers video processing, behavior analysis and result visualization. Video processing tests focus on file uploading and playback to verify format compatibility, file size limits, playback control and frame rate display. The system only supports PNG, JPEG and JPG formats, and prompts a format error when unsupported files are uploaded, as shown in (Fig. 7).

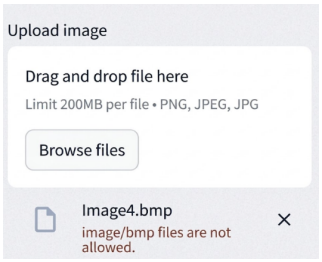


Fig. 7. Image format compatibility testing

Behavioral analysis testing consists of object detection and behavior classification to evaluate model accuracy, real-time performance, inference delay and false positive rate. The system can accurately identify classroom behaviors such as students raising hands [6], and the test results are presented in (Fig. 8).

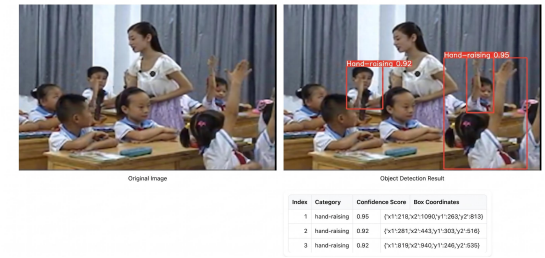


Fig. 8. Testing of behavioral recognition performance

Result visualization testing verifies chart display and real-time data updating, as illustrated in (Fig. 9). Through classroom image analysis, active learning behaviors account for 82.4%, while passive distracted behaviors account for 17.6%, which intuitively reflects the overall classroom behavior status of students.

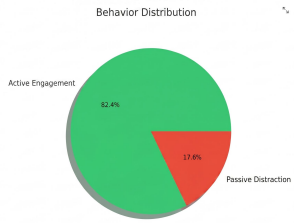


Fig. 9. Pie chart of behavioral distribution

B. Comparative Experiments of YOLO Algorithms

This study selects YOLOv3, YOLOv5s, YOLOv7 and YOLOv8n for comparison. YOLOv3, YOLOv5s and YOLOv7 are typical anchor-based detectors with distinct network designs, while the proposed YOLOv8n is a lightweight anchor-free model using C2f modules, decoupled heads and combined CIoU-DFL loss. All models are trained for 50 epochs with COCO pre-trained weights and full data augmentation, with input images resized to 640×640. Batch sizes are adjusted according to 8 GB VRAM limitation, and appropriate optimizers are adopted for each model. We adopt mAP@0.5, precision and recall as evaluation metrics, and further assess model complexity, inference speed and file size for practical deployment. The experimental results are presented in Table I.

TABLE I. PERFORMANCE COMPARISON OF DIFFERENT YOLO MODELS ON THE CLASSROOM BEHAVIOR DATASET

Model	mAP@0.5 (%)	Precision (%)	Recall(%)	FPS	Parameters (M)	FLOPs(G)
YOLOv3	45.2	48.5	46.1	45	104.0	282.69
YOLOv5s	52.8	56.4	53.2	210	7.2	16.2
YOLOv7	55.4	59.8	55.6	115	37.2	105.3
YOLOv8n	57.0	62.3	57.8	355	3.01	8.1

Experimental results demonstrate that YOLOv8n achieves an optimal balance between parameter volume and detection accuracy, exhibiting remarkable model efficiency advantages. YOLOv8n has only 3.01 million parameters, which is 34.6 times fewer than YOLOv3 (104.0M), 2.4 times fewer than YOLOv5s (7.2M), and 12.4 times fewer than YOLOv7 (37.2M). In terms of computational complexity, its floating-point operations amount to merely 8.1 GFLOPs, being 34.9 times, 2.0 times and 13.0 times lower than those of YOLOv3, YOLOv5s and YOLOv7 respectively.

The substantial reduction in model parameters and computational complexity mainly benefits from two major structural innovations. The C2f module enables more efficient gradient propagation than the C3 module without increasing parameter volume; the anchor-free detection head eliminates the dense prediction branch required by traditional anchor-based algorithms. Meanwhile, the lightweight model size of only 5.9 MB greatly enhances its deployment adaptability on edge devices and in bandwidth-limited scenarios.

YOLOv8n also presents a prominent advantage in inference speed, capable of stably realizing high-performance real-time detection.

The comprehensive comparative experiments draw a core conclusion: among the four comparative models, YOLOv8n delivers the best trade-off between accuracy and efficiency. The results fully verify that the design paradigm combining anchor-free mechanism and decoupled detection head can be well adapted to the real-time detection task of classroom behaviors, with excellent practicability and engineering deployment value. As shown in Table II.

TABLE II. SUMMARY COMPARISON OF MODEL ARCHITECTURAL PARADIGMS

Model	Paradigm Paradigm	Head Design	Feature Fusion	Key Innovation
YOLOv3	Anchor-Based	Coupled	FPN	Multi-scale prediction
YOLOv5s	Anchor-Based	Coupled	FPN+PAN	C3+auto-anchor
YOLOv7	Anchor-Based	Coupled + Auxiliary	E-ELAN	Deep supervision
YOLOv8n	Anchor-Free	Decoupled	FPN+PAN	C2f + CIoU+DFL

The architectural comparison reveals a clear trend from anchor-based to anchor-free, from coupled to decoupled detection heads, and from single-scale to multi-scale feature fusion. YOLOv8n embodies the culmination of these trends, combining them into a detector that achieves competitive accuracy at a fraction of the computational budget. For classroom behavior detection, where real-time multi-camera processing is an operational necessity, this efficiency advantage is decisive.

VI. CONCLUSION AND FUTURE WORK

This study successfully accomplishes the identification and analysis of fundamental classroom behaviors, providing a viable behavioral analysis solution for smart classrooms. Nevertheless, there remains significant potential for further multi-dimensional expansion, which can be explored through two primary avenues: algorithmic optimization and application ecosystems. Regarding algorithmic optimization, while the current model identifies six typical behaviors—such as hand-raising, reading, and writing—the granularity of classification requires further refinement. For instance, subdividing "reading" into sub-types such as silent reading, following-along, and interactive reading would allow specific behavioral manifestations to map accurately onto different levels of student cognitive engagement. To expand monitoring dimensions, future research could incorporate non-learning behavioral recognition modules, such as capturing classroom interactions like whispering or passing objects to evaluate instructional order. Additionally, detection of physiological signals, such as yawning or frequent posture adjustments, could be utilized to construct student attention-decay models [12]. This would lead to a comprehensive evaluation system encompassing learning states, classroom interactions, and physiological responses.

In terms of the application ecosystem, it is essential to transcend current offline limitations by designing lightweight models compatible with mainstream online education platforms [14], such as Tencent Meeting and Zoom. Leveraging edge computing devices for distributed behavioral analysis, combined with the establishment of a cloud-based federated learning framework, would enable the aggregation of data from diverse classrooms while preserving privacy. This approach would continuously enhance the model's generalization performance, ultimately creating an integrated online-offline intelligent educational observation system that provides cross-scenario unified data support for educational quality assessment.

ACKNOWLEDGMENT

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REFERENCES

[1] Ma X Y, Shi W P. How vocational college students learn: Behavior analysis based on classroom observation[J]. Research in Educational Development, 2023, 43(Z1): 65-73. DOI:10.14121/j.cnki.1008-3855.2023.z1.006.

[2] Gao R G. Design of student classroom behavior analysis system based on deep learning[J]. Wireless Internet Technology, 2024, 21(13): 80-83.

[3] Li H. Research on detection, recognition and application of students' classroom behaviors based on multi-object tracking[D]. Yunnan Normal University, 2024. DOI:10.27459/d.cnki.gynfc.2024.001063.

[4] He, Z. Q., Huang, W. H., Xiao, J. Y., Jiang, X. Z., & Hu, J. J. (2023). Design of Student Classroom Behavior Analysis System Based on YOLOv5. Computer Knowledge and Technology, 19(26), 19–22. https://doi.org/10.14004/j.cnki.ckt.2023.1348

[5] Zhang X Y, Zou X Q. Deep learning-based analysis of students' classroom behaviors and teaching reform[J]. Shanxi Youth, 2024, (18): 119-121.

[6] Hou F X. Research on student classroom behavior analysis algorithm based on video[D]. Hebei University of Science and Technology, 2023. DOI:10.27107/d.cnki.ghbku.2023.000552.

[7] Wu F, Wu B. Research on student classroom behavior analysis based on video[J]. Information & Computer, 2022, 34(20): 183-185+200.

[8] Qu X D. Research on student classroom behavior recognition algorithm in smart classroom based on deep learning[D]. North Minzu University, 2023. DOI:10.27754/d.cnki.gbfmz.2023.000368.

[9] Du J Q. Research on student classroom behavior recognition method based on deep learning[D]. Shandong Technology and Business University, 2022. DOI:10.27903/d.cnki.gsdsg.2022.000121.

[10] Xu J H. Research on intelligent recognition of students' classroom behaviors for video streams[D]. Fujian Normal University, 2022. DOI:10.27019/d.cnki.gfjsu.2022.000177.

[11] Zheng D. Research on student classroom behavior recognition based on two-stream convolutional neural network[D]. Shenyang Normal University, 2022. DOI:10.27328/d.cnki.gshsc.2022.000260.

[12] Hu L P. Research on student classroom behavior recognition model based on deep learning[D]. Hangzhou Normal University, 2021. DOI:10.27076/d.cnki.ghzsc.2021.000555.

[13] Chen J. Student classroom behavior analysis system based on computer vision[D]. East China Normal University, 2022. DOI:10.27149/d.cnki.ghdsu.2022.000360.

[14] Zhong G F. Research on online learning behavior analysis of cloud classroom students based on RFE model[J]. Computer Knowledge and Technology, 2021, 17(21): 65-66. DOI:10.14004/j.cnki.ckt.2021.2079.

Optimizing Dynamic Window Approach (DWA) Path Planning Parameters for Mobile Robots: An Experimental Study Using Design of Experiments in Static and Dynamic Environments

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***Abstract*—This paper presents an optimization of path planning parameters for the Dynamic Window Approach (DWA) path planning algorithms used in mobile robots across various environments both statically and dynamically. Throughout the research, multiple path planning algorithms were tested and explored to enhance the robot's navigation capabilities by leveraging the latest version of the Robot Operating System (ROS). A full factorial Design of Experiments (DOE) approach was employed to analyze the selected path planning algorithm, focusing on key trajectory scoring parameters such as Path Distance Bias, Goal Distance Bias, and Obstacle Distance Scale. The goal was to identify the optimal configuration for different levels of environmental complexity. The findings demonstrate significant improvements in navigation efficiency, offering a robust framework for selecting path planning parameters based on various environmental conditions. This optimization achieves up to a 15 second reduction in travel time across various environments using the Inno Robot Platform.**

Keywords—Mobile robot; Path Planning; Dynamic Window Approach; Design of Experiments, Robot Navigation.

I. Introduction

Mobile robots have advanced significantly over the past decade, driven by the increasing need for autonomous systems capable of navigating complex environments efficiently. At the core of this advancement lies effective path planning and localization methods, enabling robots to travel autonomously, avoid obstacles, and achieve their goals. Algorithms such as the Dynamic Window Approach (DWA), Time Elastic Band (TEB), Rapidly Exploring Random Tree (RRT), and A* are frequently used for mobile robotics due to their reliability in diverse environments [1-3]. These algorithms must maintain real-time responsiveness and adapt to environmental changes while optimizing the path and minimizing travel time, all without compromising the safety of the robot [4-6].

Navigating dynamic environments introduces unique challenges. As the complexity of an environment increases due to higher obstacle density, dynamic obstacle speeds, or uneven terrain navigation efficiency decreases. Robots may encounter path deviations, increased travel time, or even failure to reach the target goal [7, 8]. Adapting appropriate parameters for the chosen path planning algorithm is crucial to overcoming these challenges. For example, navigating through densely populated areas with large, dynamic obstacles requires different parameter configurations compared to open static environments. Thus, fine-tuning these parameters improves navigation efficiency and ensures that the robot adapts to environmental complexity [3, 9, 10].

This study focuses on optimizing the DWA algorithm using a full factorial design approach. The primary objective is to evaluate how parameter changes affect travel time in various complex environments. By identifying optimal parameters for specific environments, the study provides a selection framework to suggest suitable configurations based on the complexity of the environment, thereby enhancing adaptability and navigation efficiency [11, 12].

DWA is one of the most popular local path planning algorithms. It generates multiple velocity trajectories in real-time and selects the optimal one to avoid collisions while reaching the target goal. The effectiveness of DWA increases when combined with a global path planning algorithm, such as A*, which improves overall performance in unknown environments [13-15]. In DWA, parameters such as path distance, obstacle distance scale, and goal weighting play a critical role in determining navigation performance. These parameters must be carefully balanced to ensure safety and efficiency, especially in dynamically changing environments [1,

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3, 8]. While DWA focuses on real-time collision-free movement, the Time Elastic Band (TEB) algorithm optimizes both the path and the time intervals between waypoints, offering an optimal solution for complex environments where time and space need to be managed efficiently. In contrast, global planners like A* provide high-level solutions suitable for static environments, whereas local planners like DWA ensure dynamic obstacle avoidance, creating a complementary system for robust autonomous navigation [6, 9].

Simultaneous Localization and Mapping (SLAM) is widely used for robot localization, especially in dynamic environments. SLAM provides real-time maps and maintains the robot's location within them, making it essential for navigation through unexplored terrains. However, SLAM algorithms, such as Gmapping and Hector SLAM, can struggle with dynamic environments due to the constant changes in the map, leading to potential navigation detours or localization failures. In this study, Gmapping is employed as the primary SLAM technique, ensuring accurate mapping and localization despite environmental changes [12, 13, 15].

Local path planners like DWA and TEB must adapt quickly to changing environments to maintain safe navigation and support the mapping process. This study uses full factorial design to identify the most critical DWA parameters—such as path distance, obstacle scale, and goal weighting—that affect navigation efficiency. Testing multiple parameters across different environments provides valuable insights for achieving optimal settings based on environmental complexity. This approach enhances robot navigation performance and reduces travel time [1, 3, 5, 8].

In summary, this research highlights the importance of fine-tuning DWA parameters to optimize robot navigation across dynamic and complex environments. The proposed selection framework, based on a full factorial design, offers practical insights for identifying optimal parameter configurations. By ensuring real-time responsiveness and efficient path planning, autonomous mobile robots can achieve safer, faster, and more reliable navigation in diverse environments [7, 9, 14].

II. Methodology

A. System Architecture

First, Hardware Setup: Inno robot platform features advanced sensors (360° LIDAR (YDLIDAR G6), 3D stereo camera (RealSense D435), Bosch IMU(BNO055), 5 ultrasonic sensors (HC-SR04) and powerful drive motors, supporting up to 80kg payload. It moves with precision using 6 wheels (2 drive + 4 caster, 25cm turning radius) as shown in Fig. 1. Equipped with a 20AH battery, it operates for 10 hours and charges 4. Connectivity includes Wi-Fi, USB, and Ethernet, while interactions are handled via speakers and LED indicators. A high-performance i9-powered PC ensures optimal performance. Software setup: The robot operates on ROS Noetic, with a modular architecture for autonomous navigation and task execution. It processes real-time sensor data for decision-making and includes a user-friendly web application for remote control from any device (PC/tablet) that does not require specialized expertise

Fig. 1. Hardware Steup of the Inno robot platform.

B. Path Planning and Localization Techniques

This study focuses on the Dynamic Window Approach (DWA) as the primary path planning algorithm. DWA is a velocity-based local planner that computes the optimal collision-free velocity commands (linear and angular) for a robot to move towards its target. It evaluates possible velocities within a dynamic window of feasible actions, based on the robot's current motion and kinematic constraints. Key parameters of DWA include path distance bias, which determines how closely the robot should follow the global path, and obstacle distance scale, which affects the robot's ability to avoid obstacles. These parameters are critical to balancing efficiency and safety, especially in dynamic environments where real-time decision-making is required.

For localization, the study employs a combination of sensors and the Extended Kalman Filter (EKF), integrating data from an Inertial Measurement Unit (IMU) to enhance the accuracy of the robot's position. Traditional localization methods relying on wheel odometry, or GPS are prone to errors in dynamic or indoor environments. By fusing IMU data with odometry through EKF, the robot achieves more reliable state estimation, mitigating issues like drift and improving overall localization accuracy in real-time.

C. Experimental Design

This study employs a Full Factorial Design of Experiments (DOE) to systematically test and optimize key parameters of the Dynamic Window Approach (DWA) for path planning. The design includes three critical parameters: path distance bias, goal distance bias, and obstacle distance scale, each set at three levels to explore their effects on the robot's performance. The path distance bias was tested at values of 16, 32, and 48, with higher values encouraging the robot to follow the global path more closely. The goal distance bias was set at 10, 20, and 30, with increasing values emphasizing the goal over following the exact path. The obstacle distance scale, tested at 0.01, 0.02, and 0.03,

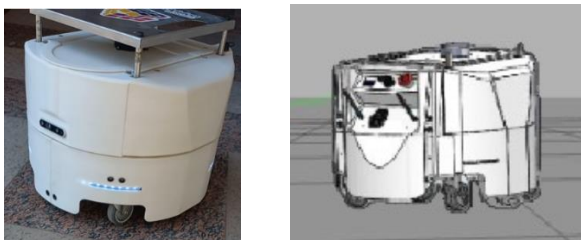
adjusted the robot's cautiousness around obstacles, impacting trajectory smoothness and safety.

The full factorial design resulted in 27 combinations of parameter settings, each tested in seven different environments. Each test was repeated three times to ensure reliable results, and the average time taken to reach the goal was recorded for each configuration. The environments were designed to simulate varying levels of complexity, with four static environments and three dynamic ones.

In the static environments, the only difference across the four setups was the number of stationary obstacles, providing insight into how DWA performs under increasing complexity. All environments had the same dimensions, ensuring that only the number of obstacles impacted performance. The dynamic environments introduced moving obstacles in addition to static ones, with each environment varying in the number of both static and dynamic elements. This setup allowed the study to assess DWA's adaptability and robustness in real-world scenarios where obstacles are not static.

III. Experiments and Results

A. Validation of Simulation Against Real-World



a) Inno Robot Actual Model ***b) Inno Robot CAD model***

Fig. 2. Actual Robot and its Model in Simulation

The simulation ensures the safety of the robot by allowing experimentation with various parameters. To confirm the effectiveness of the simulation in evaluating parameters and facilitating multiple trials, an identical robot model is designed within the Gazebo environment, replicating components such as motors, sensors, wheels, and other essential parts. The goal is to construct a virtual robot model under these assumptions, with the final design illustrated in Fig. 2. A simulation environment is then developed to closely mimic the real-world behavior of the robot as shown in Fig. 3. The experimental setting in the real world consisted of a 30-meter-long hallway, where the robot was tasked to reach a designated goal point at the end. A timer was employed to measure the robot's travel time from the starting point to the goal. To ensure consistency, the robot was provided with a map of the hallway in both the simulation and real-world environments. The robot underwent identical conditions in both environments to compare performance. In the simulation, the robot took an average of 100.3 seconds to travel the 30-meter distance, whereas in the real-world environment, the same task took 98.7 seconds on average. These results indicate that the simulation parameters were 98% identical to those of the real world, validating the reliability of the simulation for further experimentation and testing.

a) Real Experimental World ***b) Simulation World***

Fig. 3. Experimental Setup for Real and Simulation world

B. Optimization of Path Planning Parameters Using DOE

The Design of Experiments (DOE) approach was employed to systematically explore and optimize the Dynamic Window Approach (DWA) path planning parameters at a constant speed. A full factorial design was selected to test all possible combinations of three key parameters: path distance bias, goal distance bias, and obstacle distance scale. These parameters directly influence the robot's trajectory, obstacle avoidance, and efficiency in reaching the goal. Path distance bias was set to values of 16, 32, and 48, goal distance bias to 10, 20, and 30, and obstacle distance scale to 0.01, 0.02, and 0.03. This resulted in 27 unique configurations as shown in table 1.

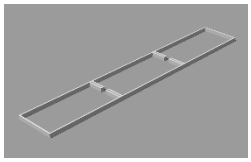
TABLE I. Configuration Combination for DWA

Path	Goal	Occident	**Config #**
16	30	0.02	1
32	30	0.03	2
48	30	0.03	3
16	30	0.01	4
~	~	~	~
~	~	~	~
~	~	~	~
32	20	0.02	24
16	30	0.03	25
48	20	0.01	26
32	10	0.01	27

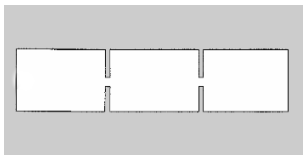
TABLE II. Environment Description and Layout

Features	**Environmental Description**						
	1	2	3	4	5	6	7
Static (x) Dynamic (√)	x	x	x	x	√	√	√
Static Obstacles #	0	8	10	14	8	10	14
Average Size (m^3)	0	0.6		0.5	0.6		0.5
Dynamic Obstacles #	N/A				3	4	7
Average Obstacle Velocity (m/s)	N/A				0.5		
Average Size (m^3)	N/A				0.343		

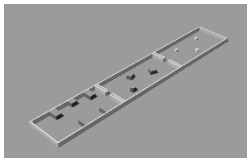
Each configuration was tested across the seven environments, that were divided into two sections, as shown in Table 2, Fig. 4 and Fig. 5. The first section consists of static environments; the first is obstacle-free, while the other three introduce increasing numbers of obstacles. The second section involves dynamic obstacles along with statical obstacles. The objective of the experiment was to identify the optimal parameter combinations that minimized the robot's travel time in each environment.



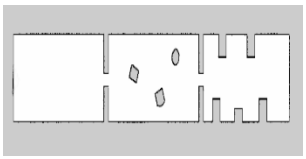
a) Environment 1 as level 1 complexity



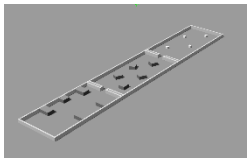
b) Results map from SLAM for Environment 1



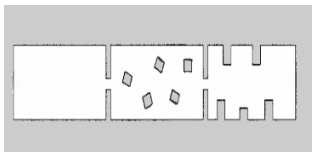
c) Environment 2 as level 2 complexity



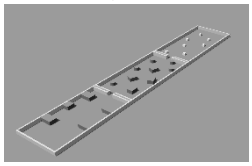
d) Results map from SLAM for Environment 2



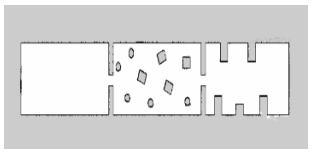
e) Environment 3 as level 3 complexity



f) Results map from SLAM for Environment 3

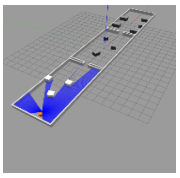


g) Environment 4 as level 4 complexity

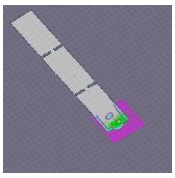


h) Results map from SLAM for Environment 4

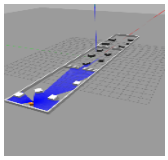
Fig. 4. Static Simulation world and resulted SLAM Maps



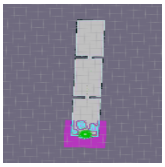
a) Environment 5 as level 5 complexity



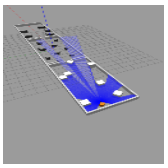
b) Real time SLAM for simulation environment 2



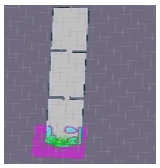
c) Environment 6 as level 6 complexity



d) Real time SLAM for simulation environment 3



e) Environment 7 as level 7 complexity



f) Real time SLAM for simulation environment 4

Fig. 5. Dynamic Simulation worlds

C. *Performance Analysis in Static and Dynamic Environments*

TABLE III. RESULTED TRAVEL TIME UNDER DIFFERENT CONFIGURATION IN STATIC ENVIRONMENTS.

Config #	Robot Travel Time across Environment (Sec)			
	Room 1	Room 2	Room 3	Room 4
1	98.89	98.9	97.46	110.38
2	99.97	104.97	100.62	100.4
3	98.99	98.06	104.82	98.64
4	96.03	97.84	98.88	97.92
~	~	~	~	~
24	112.11	101.43	98.55	101.67
25	96.88	99.98	97.43	97.83
26	97.65	99.64	97.84	101.69
27	98.56	99.43	101.49	101.79

TABLE IV. RESULTED TRAVEL TIME UNDER DIFFERENT CONFIGURATION IN DYNAMIC ENVIRONMENTS.

Config #	Robot Travel Time across environments (Sec)		
	Dynamic 1	Dynamic 2	Dynamic 3
1	106.8	103.76	117.95
2	104.35	102.39	104.87
3	99.77	101.12	101.76
4	99.58	98.57	116.73
~	~	~	~
~	~	~	~
~	~	~	~
24	108.21	102.47	115.32
25	106.87	99.18	112.67
26	104.74	98.57	101.95
27	99.21	95.85	109.11

The DOE results as shown in table 3 and table 4, highlight the differing impacts of environmental complexity on path planning performance. In static environments (Rooms 1–4), the

robot's performance varied based on the number of obstacles, with Configurations 20 and 22 delivering the most consistent results. As the number of obstacles increased, higher path distance bias values helped the robot maintain smoother trajectories, reducing travel time significantly. For example, in Room 4, which had the highest number of static obstacles, Configuration 23 achieved the shortest travel time due to its optimal balance between goal-seeking and obstacle avoidance.

In contrast, the dynamic environments (Dynamic 1–3) required more adaptive parameter tuning. Dynamic obstacles introduced variability in obstacle positions and movement patterns, necessitating a more cautious approach. Configuration 8 was the most effective in these environments, as it prioritized obstacle avoidance while still allowing the robot to maintain efficient progress toward the goal. In Dynamic 3, the robot faced the highest complexity with both static and dynamic obstacles, and Configuration 14 provided the best balance between path smoothness and safety.

The results demonstrated that parameter tuning significantly impacts the robot's navigation performance, with certain configurations showing a clear advantage in specific environments. For instance, Configuration 20 was optimal for simpler environments such as Room 1 and Room 3, while Configuration 8 performed best in more complex dynamic environments like Dynamic 2

IV. Conclusion

In conclusion, a selection framework was developed to identify the optimal parallel path-planning parameters for DWA based on environmental complexity, including the number, size, and average speed of obstacles. The results showed that higher paths and goals consistently provided stable performance, as seen in Configuration 3, which achieved good average speeds. Additionally, in specific scenarios, such as Configuration #20, the lowest travel time was recorded in long-term experiments within an empty environment. These configurations were tested and implemented on the actual robot, resulting in a reduction of travel time by up to 15 seconds when applied to the real platform under the same environmental conditions. Therefore, this analysis can be utilized and adapted for varying levels of environmental complexity. Several challenges were encountered during this work, such as sensor deviations and behavioral changes in the robot caused by interference or noise. To account for these inconsistencies, an average of five trials per run was conducted, ensuring that any interference was sufficiently incorporated into the analysis.

For future work, as environments become more complex, further parameter tuning will be required. A comparison between the Dynamic Window Approach (DWA) and the Time Elastic Band (TEB) algorithm is also recommended. This comparison will enable the robot to select the most suitable planning algorithm to achieve the optimum path way and the lowest possible travel time. Finally, Machine Learning techniques will be integrated to enable predictive modeling and continuous optimization of robot path planning in nonlinear and dynamic real environments.

References

[1] Zhai, X., Tian, J. & Li, J. A Real-time Path Planning Algorithm for Mobile Robots Based on Safety Distance Matrix and Adaptive Weight Adjustment Strategy. Int. J. Control Autom. Syst. 22, 1385–1399 (2024). https://doi.org/10.1007/s12555-022-1016-5

[2] Gambhir, H. S., & Tanwar, R. (2024). Autonomous mobile robot localization and mapping using sensor fusion. Robotics and Autonomous Systems, 172, 104484. https://doi.org/10.1016/j.robot.2024.104484

[3] Wang, S., & Liu, X. (2022). Path planning optimization for autonomous vehicles with multi-objective constraints. Procedia Computer Science, 200, 324–330. https://doi.org/10.1016/j.procs.2022.01.067

[4] Patil, D., & Gupta, S. (2016). Robot navigation based on artificial intelligence techniques. Procedia Computer Science, 89, 505–512. https://doi.org/10.1016/j.procs.2016.06.095

[5] Amin, N., & Zahid, M. (2020). A hybrid SLAM technique for real-time mobile robots. Journal of Physics: Conference Series, 1529(4), 042032. https://doi.org/10.1088/1742-6596/1529/4/042032

[6] Jiang, Z., Wang, W., Sun, W. et al. Path Planning Method for Mobile Robot Based on a Hybrid Algorithm. J Intell Robot Syst 109, 47 (2023). https://doi.org/10.1007/s10846-023-01985-1

[7] Zhang, P., & Xu, L. (2023). Machine learning-based motion planning for autonomous robots. Procedia Manufacturing, 61, 40–47. https://doi.org/10.1016/j.promfg.2023.01.005

[8] Huttenrauch, H., & Norman, D. A. (2019). Towards more social mobile robots: A survey of design methodologies and applications. International Journal of Advanced Robotic Systems, 16(3), 1–16. https://doi.org/10.1177/1729881419839596

[9] Kanade, T., & Fu, K. S. (1989). Techniques in image processing for autonomous mobile robots. In J. K. Aggarwal (Ed.), Autonomous Robot Vehicles (pp. 312–328). Springer. https://doi.org/10.1007/978-3-642-82153-0_19

[10] Kovac, M., & Muraoka, S. (2021). Introduction to autonomous mobile robot research and multi-cooperation systems. ResearchGate. https://www.researchgate.net/publication/353279416_Introduction_into_Autonomous_Mobile_Robot_Research_and_Multi_Cooperation

[11] Cox, I. J., & Leonard, J. J. (2017). Advances in robotic vision systems: SLAM and object recognition. IFAC-PapersOnLine, 50(1), 234–241. https://doi.org/10.1016/j.ifacol.2017.08.059

[12] Singh, P., & Sharma, R. (2015). Sensor-based path planning for autonomous robot navigation. Procedia Computer Science, 75, 504–511. https://doi.org/10.1016/j.procs.2015.08.011

[13] Bhat, S., & Khan, A. (2021). Design and simulation of an indoor localization system for mobile robots. Ain Shams Engineering Journal, 12(4), 3309–3316. https://doi.org/10.1016/j.asej.2021.04.005

[14] Yan, S., & Lee, D. (2018). Real-time data fusion for autonomous robot navigation systems. Procedia Computer Science, 135, 412–419. https://doi.org/10.1016/j.procs.2018.08.186

[15] Silva, P., & Rodrigues, J. (2023). A multi-robot control strategy for dynamic environments. Machines, 11(10), 980. https://doi.org/10.3390/machines11100980

Autonomous Driving Decision at Unsignalized Intersections Integrating Biased Attention and Game Reasoning

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***Abstract*—Autonomous driving decision-making at unsignalized intersections faces complex dynamic interaction challenges. Existing deep reinforcement learning methods often treat the environment as a passive responder, lack explicit inference of traffic participants' intentions, and may struggle to capture key risk features over long time horizons. To address these issues, this paper proposes a decision-making framework (SAC-GT-Attn) that integrates a biased attention mechanism with game-theoretic Level-k reasoning. First, the proposed method employs a Transformer encoder with a temporal distance bias to extract temporally correlated contextual interaction features. Subsequently, a Level-k game module estimates the influence of the ego vehicle's candidate actions on surrounding vehicles, thereby quantifying interaction uncertainties and potential risks. Finally, this risk is transformed into a dense reward penalty to guide the Soft Actor-Critic (SAC) agent in learning risk-aware collision avoidance strategies. SUMO simulation results indicate that, compared with learning-based baselines such as Standard SAC and SAC-LSTM, the proposed method achieves lower queue lengths and conflict rates. Compared with the conservative FCFS rule, it improves traffic efficiency while maintaining a low conflict rate under the tested traffic densities.**

***Index Terms*—Autonomous Driving, Deep Reinforcement Learning (DRL), Biased Attention Mechanism, Game Theory Reasoning, Risk Assessment, Unsignalized Intersections**

I. Introduction

With the acceleration of urbanization, autonomous vehicle (AV) technology is regarded as a promising approach to addressing traffic congestion and enhancing road safety [1]. Within urban road networks, unsignalized intersections pose one of the most challenging scenarios due to the absence of explicit right-of-way allocation signals [2]. In these areas, the right-of-way for vehicles crossing intersections is often not determined by traffic lights, but rather relies on implicit games, probing, and negotiation among drivers. This high level of uncertainty requires autonomous driving systems not only to "see" surrounding vehicles, but also to "understand" their intentions and anticipate the chain reactions their own actions may trigger [3].

Traditional rule-based approaches, such as the First-Come-First-Served (FCFS) policy [4], offer strong interpretability. However, they often exhibit excessive conservatism when confronted with high-density traffic flows or aggressive drivers, leading to the "Freezing Robot Problem" and reducing intersection throughput efficiency. On the other hand, Deep Reinforcement Learning (DRL)-based methods demonstrate remarkable adaptability by learning policies through trial-and-error interactions with the environment [5].

In recent years, research on decision-making at unsignalized intersections has made considerable progress. For example, recent studies have improved the modeling of complex trajectories by utilizing spatio-temporal graph neural networks or improved attention mechanisms. At the same time, interactive decision-making based on game theory (such as Stackelberg games and cognitive hierarchy models) has gradually been introduced to address multi-vehicle cooperative scheduling problems.

However, directly applying generic DRL algorithms to this scenario faces two core challenges: First, insufficient generalization in attention allocation [6]. Standard Transformer self-attention, without task-specific temporal bias, may not sufficiently reflect the driving prior that recent observations are usually more decision-relevant. This may cause critical sudden features (e.g., a leading vehicle's emergency braking) to be diluted by long-term historical information. Second, the lack of explicit interaction reasoning [7]. Existing end-to-end models typically treat the environment as static or simplistically dynamic obstacles, ignoring the vehicle's own actions' impact on the environment. This results in a lack

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of counterfactual reasoning capability—the ability to infer "if I accelerate, will the other vehicle slow down to yield or accelerate to overtake?"

To address these challenges, this paper proposes a decision-making framework named Soft Actor-Critic with Game-Theoretic Reasoning and Biased Attention (SAC-GT-Attn). It aims to improve risk perception and decision-making through the integration of spatio-temporal feature encoding and interaction reasoning. Inspired by the role of attention bias in risk-related decision-making [8], we employ a biased attention mechanism to introduce temporal weighting into the Transformer encoder. This encourages the model to focus on recent interaction features that are more relevant to the current decision and helps reduce the influence of less informative long-term observations.

Building upon this foundation, we introduce a game-theoretic response model to represent the possible chain reaction that "the ego vehicle's actions may trigger reactions from other vehicles" through Level-k thinking hierarchies [9], [10]. This design allows the agent to incorporate a simple form of opponent-response reasoning and converts interaction-related risks into dense reward penalties. In this way, the Soft Actor-Critic (SAC) agent is guided to learn strategies that balance safety margins with traffic efficiency during policy exploration.

II. Methods and Principles

This section details the proposed SAC-GT-Attn autonomous driving decision framework. This framework aims to address the prevalent "perception-game" disconnect in unsignalized intersection scenarios. Due to partial observability in traffic interactions, we formulate the policy input as a history-augmented decision process and use a POMDP-inspired representation [11]. The closed-loop system comprises a bias-attention encoder (for spatio-temporal feature extraction), a Level-k game response model (for interactive risk estimation) [9], [12], and a risk-shaped SAC controller (for policy execution).

A. Problem Formulation

We formalize the history-augmented decision-making process of an autonomous vehicle (Ego Vehicle, EV) in dynamic traffic flow as a tuple $\mathcal{M} = \langle S, A, P, \mathcal{R}, \gamma \rangle$.

a) State space S: Given the non-stationary nature of the environment, the state must contain not only the physical quantities at the current time but also historical temporal information. Define $s_t = [s_{ego}, \mathcal{H}_{obs}, s_{map}]$, where $\mathcal{H}_{obs} = \{\tau_1, ..., \tau_N\}$ represents the historical trajectory sequence of N traffic participants within the sensor's perception range. Each trajectory $\tau_i \in \mathbb{R}^{T \times d}$ contains the position and velocity vectors of the past T frames.

b) Action Space A: To ensure smooth vehicle dynamics, we employ a continuous action space. An action $a_t \in [a_{min}, a_{max}]$ represents the longitudinal acceleration command. The route-level maneuver, such as going straight or turning left, is assumed to be pre-defined by the simulator, and the learned policy focuses on longitudinal motion control for interaction-aware crossing.

c) Optimization Objective: The agent aims to learn a policy $\pi_\phi(a_t|s_t)$ that maximizes the cumulative expected reward while penalizing potential risks arising from game interactions:

$$\max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t (r_{task}(s_t, a_t) - \lambda \cdot r_{risk}(s_t, a_t)) \right] \quad (1)$$

In the computational implementation, the model uses 32-bit single-precision floating-point numbers (FP32) for numerical consistency, while simulation interactions and policy control employ discrete decision updates at a rate of 10 Hz.

B. Spatio-Temporal Encoding via Biased-Attention

When processing long-sequence trajectory data $\mathcal{H}_{obs}$, a standard Transformer with ordinary positional encoding may still lack a task-specific temporal prior for driving interactions. In driving games, temporal causality is important—a sudden brake by the car ahead 0.1 seconds ago usually has a greater impact on the current decision than steady driving 5 seconds ago.

To emphasize recent information and accommodate this temporal sensitivity, we introduce the Time-Decay Biased Attention mechanism [13]. For the input sequence X, we first generate query, key, and value matrices through linear transformations. Unlike standard computations, we inject a static bias matrix B before Softmax normalization:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B \right) V \quad (2)$$

Among these, the elements $B_{i,j}$ of the bias matrix $B \in \mathbb{R}^{T \times T}$ are designed as linear functions of the time distance:

$$B_{i,j} = \begin{cases} -\alpha \cdot |i - j|, & \text{if } i \geq j \text{ (Causal Mask)} \\ -\infty, & \text{if } i < j \text{ (Future Mask)} \end{cases} \quad (3)$$

Here, $\alpha > 0$ represents the decay coefficient. The index i denotes the query time step within the historical sequence, and the causal mask prevents each intermediate representation from attending to later positions in the sequence. In implementation, the final context vector is obtained from the last valid time step. This mechanism produces two effects:

- **Recency Bias:** When j approaches i (i.e., recent historical moments), $B_{i,j} \approx 0$ maintains high attention weights, encouraging the model to emphasize instantaneous acceleration or braking intentions.
- **Long-term Suppression:** As temporal distance increases, $B_{i,j}$ evolves into larger negative values, exponentially suppressing the weights of stale information. This helps reduce the influence of stale historical information and provides downstream game inference with a more compact context vector h_{ctx}.

C. Level-k Game-Theoretic World Model

This module serves as the interaction reasoning component of the framework. Traditional risk evaluators are often formulated as discriminative models $P(\text{collision}|s_t)$ and may not explicitly consider the influence of the ego vehicle's candidate

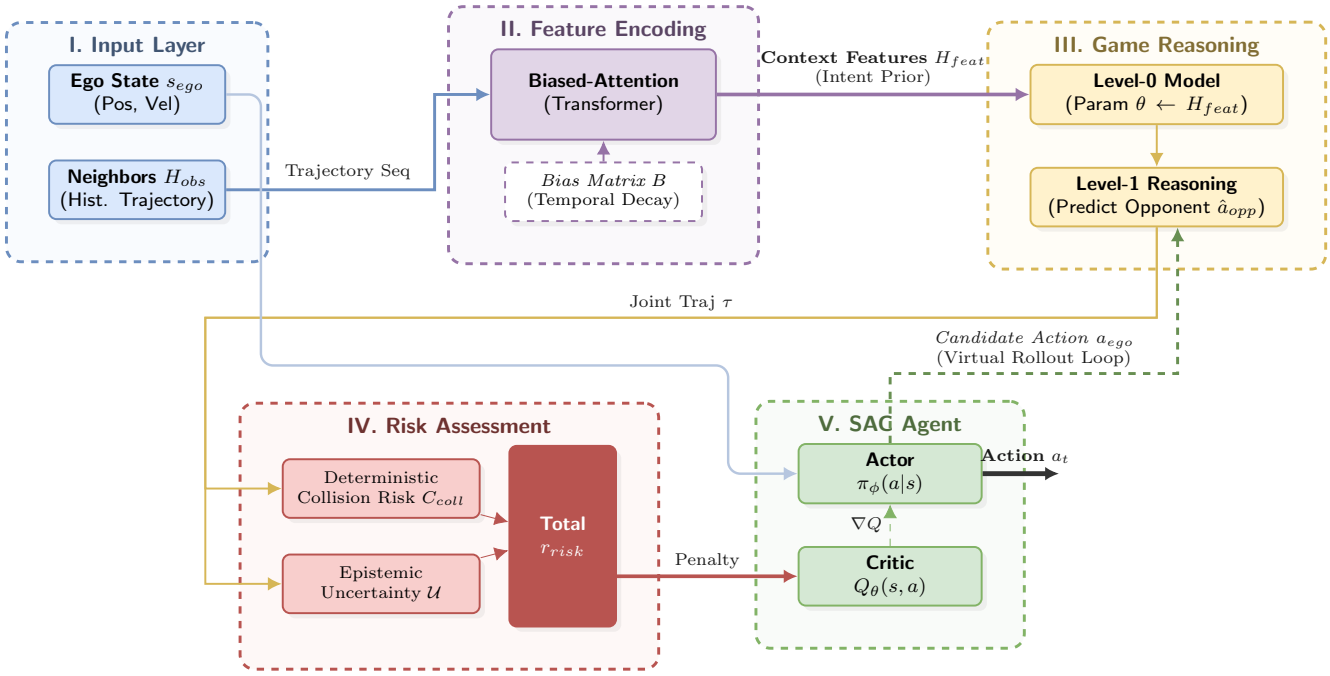

Fig. 1. Schematic Diagram of the Decision-Making Framework

actions on the environment. To provide the agent with a simple form of counterfactual reasoning, we feed h_{ctx} into a response model based on Cognitive Hierarchy Theory (CHT). This module does not directly output risk values but performs the following two-stage reasoning:

a) Strategy Inference Layer (Level-(k-1) Strategy Inference): We assume that other vehicles (Opponents) in the environment follow a Level-$(k-1)$ strategy. In actual algorithm deployment, we set the inference depth to $k = 1$, i.e., we assume the opponent follows a Level-0 response policy, which does not perform higher-order strategic reasoning but can react to the projected relative state induced by the ego vehicle's candidate action. This paper adopts $k = 1$ as an engineering simplification. Since the opponent is assumed to be a Level-0 agent following physical inertia or rule-based models (such as IDM), this module is essentially closer to efficient "single-step opponent response modeling" rather than full-fledged multi-layer game theory reasoning. Compared to higher-order reasoning, setting $k = 1$ can capture basic "action-reaction" relationships while avoiding the additional computational overhead associated with complex nesting. This is a practical design choice that supports efficient policy exploration in reinforcement learning.

Given the candidate action a_{ego} of the self-vehicle and the environmental context h_{ctx}, the model predicts the opponent's reaction action $\hat{a}_{opp}$:

$$\hat{a}_{opp} = \mathcal{F}_\theta^{L0}(s_{opp}, \text{Project}(s_{ego}, a_{ego})) \tag{4}$$

Here, Project$(\cdot)$ denotes forward state extrapolation based on a kinematic model. The key point of this step is that the future action $\hat{a}_{opp}$ of the other vehicle is conditioned on the ego vehicle's candidate action a_{ego} (Conditioned on Ego Action).

b) Potential Risk Valuation Layer: Based on the inferred joint action pair $(a_{ego}, \hat{a}_{opp})$, we utilize the world model to forward simulate the joint trajectory τ_{joint} for the next H steps and compute the Uncertainty-Aware Risk:

$$r_{risk}(s,a) = \left[\max_{t\in[1,H]} \underbrace{C_{coll}(\tau_{joint}^{(t)})}_{\substack{\text{Deterministic} \\ \text{Collision Risk}}} \right] + \eta \cdot \underbrace{U(\tau_{joint})}_{\substack{\text{Epistemic} \\ \text{Uncertainty}}} \tag{5}$$

Where:

- C_{coll}: Collision cost function based on physical distance (e.g., $1/\text{dist}^2$).
- U: Epistemic or predictive uncertainty, which can be approximated by the variance of the predicted opponent responses. A larger prediction variance indicates a more uncertain interaction state and therefore increases the risk penalty.

This design enables the agent to identify actions that, while currently at a safe physical distance, may trigger dangerous reactions from the other party (such as accelerating within their blind spot), thereby encouraging more cautious action selection.

D. Risk-Shaped SAC Policy Optimization

We integrate the aforementioned interaction risk signals into the Soft Actor-Critic (SAC) algorithm. SAC maintains

exploratory behavior by maximizing an entropy-regularized objective function [14]. In our framework, rather than adopting the form of hard constraints found in classical constrained reinforcement learning (such as CMDP or the Lagrange multiplier method), we treat r_{risk} as a risk penalty for reward shaping and incorporate it into the update objective of the Critic network.

a) Critic Network Update (Risk-Shaped Bellman Update):

$$J_Q(\theta) = \mathbb{E}_{(s,a,s')\sim\mathcal{D}}\Bigg[\frac{1}{2}\Big(Q_\theta(s,a) - (r_{task} - \lambda r_{risk} + \gamma V_{\bar{\theta}}(s'))\Big)^2\Bigg] \tag{6}$$

where $V_{\bar{\theta}}(s')$ denotes the entropy-regularized target value used in SAC.

b) Actor Network Update:

$$J_\pi(\phi) = \mathbb{E}_{s\sim\mathcal{D}}\left[\alpha \log \pi_\phi(a|s) - Q_\theta(s,a)\right] \tag{7}$$

In this manner, the Level-k world model acts as a "virtual referee." When an Actor attempts to execute high-risk interaction actions, the Critic network assigns a larger penalty, encouraging the policy distribution π_ϕ to move toward regions with lower game risk and higher passage efficiency. This helps alleviate the training inefficiencies and conservative policy tendencies of methods that mainly rely on sparse collision rewards.

III. Experimental Results

To evaluate the performance of the SAC-GT-Attn framework in interactive decision-making and dynamic risk management, we constructed a microscopic simulation environment for unsignalized intersections on the Simulation of Urban Mobility (SUMO) platform. This section first introduces the experimental setup and baseline algorithms, then analyzes the model's performance from two dimensions—throughput efficiency (queue length) and safety (conflict rate)—combined with convergence curves during training.

A. Experimental Setup

The experimental scenario is a typical two-way, four-lane intersection without traffic signals. Traffic flow consists of background vehicles (SV) whose behavior is controlled by an Intelligent Driver Model (IDM), with 20% aggressive driving noise injected to simulate human driving uncertainty. The ego vehicle must complete straight-through or left-turn maneuvers through perception and decision-making without guidance from right-of-way signals.

All reinforcement learning models were trained under the same environment random seed. The total number of training episodes was 230 (corresponding to approximately 230,000 simulation steps). Optimizer: Adam, with a learning rate set to 3×10^{-4}. Efficiency reward $\alpha_1 = 1.0$, risk penalty $\lambda = 5.0$.

To evaluate the performance of the proposed method, we introduce representative baseline strategies:

- **FCFS (First-Come-First-Served):** A classic rule-based scheduling strategy. Vehicles pass through the intersection strictly in the order of their timestamps at the stop line; if there is a conflict within the time window, they are forced to stop and wait.
- **Standard SAC:** A standard Soft Actor-Critic algorithm without game theory reasoning or the bias attention mechanism. Its state inputs are encoded using a standard MLP, and the risk penalty is based on instantaneous kinematic safety indicators, such as inter-vehicle distance or time-to-collision (TTC).
- **SAC-LSTM:** To provide a sequential reinforcement learning baseline, we introduce an algorithm that uses a Long Short-Term Memory (LSTM) network to encode state sequences instead of a standard MLP.

B. Throughput Efficiency Evaluation: Maximum Queue Length

Throughput efficiency serves as the core metric for evaluating the scheduling capability of unsignalized intersections. We recorded the maximum queue length across all lanes at each intersection during each training round, with results shown in Fig. 2.

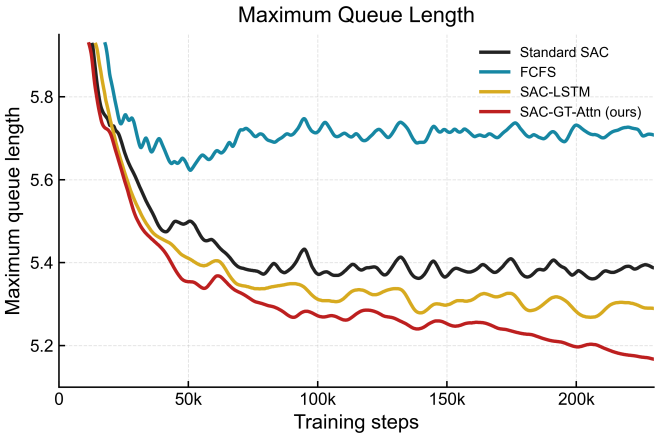

Fig. 2. Comparison of Dynamic Maximum Queue Length with Baseline Model

Result Analysis: Fig. 2 shows clear performance differences among the compared methods: the FCFS strategy maintains the largest queue length, with its queue length consistently remaining at a high level (approximately 5.7 vehicles) and exhibiting minimal fluctuation. This is because the FCFS strategy is inherently inflexible and unable to take advantage of fleeting gaps in traffic flow (Gap Acceptance). Even when an opportunity to pass arises due to an opponent's hesitation, FCFS forces the current vehicle to wait due to "timestamp sorting," resulting in wasted intersection capacity and increased vehicle backlog. Standard SAC outperforms the FCFS rule in terms of queue length, reducing the value to around 5.40. The SAC-LSTM baseline, which possesses temporal memory capabilities, further reduces it to around 5.25. However, SAC-GT-Attn achieves the lowest queue length

among the compared methods, ultimately stabilizing at around 5.15. This advantage is mainly related to the Biased-Attention mechanism, which filters out part of the noise in long historical sequences and emphasizes subtle changes in the preceding vehicle's intent, such as "starting to accelerate" or "yielding the right of way." Combined with Level-k game response reasoning, the agent can more easily infer a yielding trend from the other vehicle and select acceleration more decisively when the estimated risk is low. This increase in decision certainty may reduce unnecessary waiting time at intersections and help alleviate congestion.

C. Safety Assessment: Conflict Rate

Safety is a key requirement for autonomous driving [15]. We define the "conflict rate" as the frequency of hazardous proximity events (Time-to-Collision < 2s) occurring per unit time. Fig. 3 illustrates the conflict rate evolution during training for the compared learning-based methods.

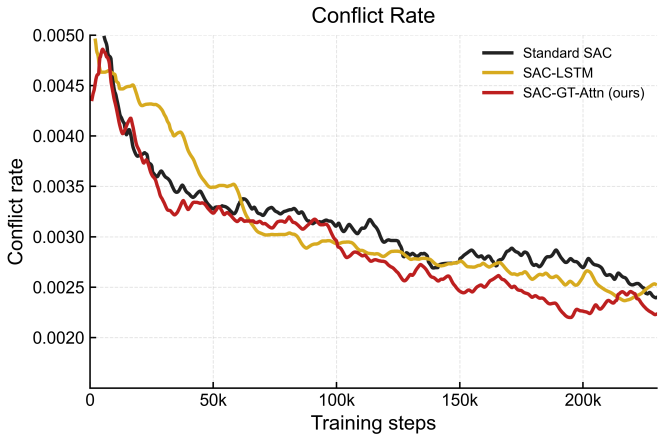

Fig. 3. Comparison of dynamic conflict rates with the baseline model

Result Analysis: As shown in Fig. 3, all three algorithms exhibit a downward trend after reaching a peak during the initial phase (0–20k steps). Among them, Standard SAC, which relies mainly on the current physical distance, shows the largest conflict rate and stronger oscillations in the later stages, with the conflict rate fluctuating between 0.0025 and 0.003; SAC-LSTM, leveraging temporal memory features, slightly reduces the conflict rate to approximately 0.0024. In contrast, the proposed SAC-GT-Attn shows a more stable downward trend and eventually stabilizes at around 0.0022 in this training run. This indicates that integrating bias attention with the Level-k game response model can help the agent form a more proactive risk-aware strategy, thereby reducing hazardous proximity events in the tested scenarios.

D. Robustness Across Traffic Densities

To verify the strategy's adaptability under varying levels of congestion, we set the test conditions to two representative traffic densities: Flow-1 (standard traffic) and Flow-2 (high-density traffic). The quantitative evaluation results for each algorithm after convergence are shown in Table I.

TABLE I
COMPARISON OF PERFORMANCE METRICS AT DIFFERENT TRAFFIC DENSITIES

	Method	MQL(vehicles)	CR (%)
Flow-1	FCFS	5.70	0.00
	Standard SAC	5.40	0.25
	SAC-LSTM	5.25	0.23
	SAC-GT-Attn(Ours)	5.15	0.22
Flow-2	FCFS	8.95	0.00
	Standard SAC	8.45	0.36
	SAC-LSTM	7.30	0.32
	SAC-GT-Attn(Ours)	6.10	0.28

As shown in Table I, under Flow-1 (standard traffic flow, corresponding to the training environment shown in Figs. 2 and 3), the SAC-LSTM with a temporal module shows improvement over the Standard SAC, while SAC-GT-Attn obtains the lowest queue length and conflict rate among the learning-based methods. When traffic density increases to Flow-2 (high-density traffic), Standard SAC shows noticeable degradation, with queue length increasing to 8.45 vehicles and the conflict rate rising to 0.36%. Although SAC-LSTM possesses some temporal processing capability, it still exhibits a conflict rate of 0.32% and a queue backlog of 7.30 vehicles under high-frequency multi-vehicle interactions. In contrast, SAC-GT-Attn keeps the queue length at around 6.10 vehicles, with the conflict rate at 0.28%. These results suggest that the proposed model has better adaptability under the tested dense-traffic setting, benefiting from the combination of "bias attention" for emphasizing key dynamics and "Level-k game response reasoning" for look-ahead risk assessment.

IV. CONCLUSION

This paper addresses the decision-making challenges of autonomous driving at unsignalized intersections by proposing a framework that integrates a biased attention mechanism with game-theoretic reasoning. By introducing the biased attention mechanism, this study helps address the issue of key interaction features being diluted in long-term trajectory data. Additionally, by augmenting the traditional passive risk predictor with a Level-k game response model, the framework mitigates the limitations of existing deep reinforcement learning algorithms in explicit intention inference.

Simulation results indicate that this method achieves favorable performance in highly dynamic interaction scenarios. Compared with learning-based baselines, the proposed framework reduces vehicle queue lengths and conflict rates at intersections. Compared with the conservative FCFS rule, it improves traffic efficiency while maintaining a low conflict rate in the tested simulation settings. This suggests that the learned policy qualitatively resembles a cautious yet efficient interaction strategy: proceeding decisively when the estimated interaction risk is low, and actively yielding during high-risk interactions.

Nevertheless, this study is conducted at the microscopic traffic-simulation level and currently relies on fixed assump-

tions about cognitive levels. The IDM model with aggressive noise used in the simulations is only a simplified abstraction of real human driving behavior. Future work will focus on two areas: first, exploring the introduction of complex human driver models trained on natural driving data (such as Gaussian mixture models, GMMs) to evaluate the proposed method in more realistic adversarial environments; second, investigating adaptive cognitive-level inference mechanisms to address more heterogeneous human driving styles, and studying the algorithm's generalization to mixed traffic flows involving vulnerable road users.

REFERENCES

[1] Hu, Jun, et al. "A survey of decision-making and planning methods for self-driving vehicles." Frontiers in Neurorobotics 19 (2025): 1451923.

[2] Park, Junho, and Maryam Zahabi. "A review of human performance models for prediction of driver behavior and interactions with in-vehicle technology." Human factors 66.4 (2024): 1249-1275.

[3] Strauss, Megan, and Stefan Mitsch. "Slow down, move over: a case study in formal verification, refinement, and testing of the responsibility-sensitive safety model for self-driving cars." International Conference on Tests and Proofs. Cham: Springer Nature Switzerland, 2023.

[4] Wu, Wei, et al. "Autonomous intersection management for connected and automated vehicles: A lane-based method." IEEE Transactions on Intelligent Transportation Systems 23.9 (2021): 15091-15106.

[5] Kiran B R, Sobh I, Talpaert V, et al. Deep reinforcement learning for autonomous driving: A survey[J]. IEEE transactions on intelligent transportation systems, 2021, 23(6): 4909-4926.

[6] Mineault, Patrick. "Is Attention All You Need?." From Human Attention to Computational Attention: A Multidisciplinary Approach. Cham: Springer Nature Switzerland, 2025. 297-314.

[7] Lefèvre S, Vasquez D, Laugier C. A survey on motion prediction and risk assessment for intelligent vehicles[J]. ROBOMECH journal, 2014, 1(1): 1.

[8] Hu, Mengchen, et al. "Attention biases the process of risky decision-making: Evidence from eye-tracking." PsyCh journal 13.2 (2024): 157-165.

[9] Xiao, Suyang, et al. "Cognitive Hierarchy Game-Based Method for Multi-Agent Lane Change Intention Recognition." 2025 IEEE 19th International Conference on Control & Automation (ICCA). IEEE, 2025.

[10] Hu D, Hu F, Yang L, et al. Game-Theoretic Risk-Shaped Reinforcement Learning for Safe Autonomous Driving[J]. arXiv preprint arXiv:2510.10960, 2025.

[11] Kamrani, Mohsen, et al. "Applying Markov decision process to understand driving decisions using basic safety messages data." Transportation Research Part C: Emerging Technologies 115 (2020): 102642.

[12] Guo, Fan, et al. "Game theory-based harmonious decision-making for autonomous bus lane change." IEEE Transactions on Intelligent Transportation Systems 26.4 (2025): 4934-4947.

[13] Dong C, Guo N. Biased-Attention Guided Risk Prediction for Safe Decision-Making at Unsignalized Intersections[J]. arXiv preprint arXiv:2510.12428, 2025.

[14] Igoe, Conor, et al. "Multi-alpha soft actor-critic: Overcoming stochastic biases in maximum entropy reinforcement learning." 2023 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2023.

[15] Garcıa J, Fernández F. A comprehensive survey on safe reinforcement learning[J]. Journal of Machine Learning Research, 2015, 16(1): 1437-1480.

Optimal Path Planning for a Robotic Manta Considering Its Kinematic Characteristics

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***Abstract*—To address the low path feasibility, poor path quality, and insufficient safety of a robotic manta in three-dimensional underwater environments caused by underactuation and non-holonomic constraints, this paper proposes the adaptive kinematics-constrained bidirectional rapidly-exploring random tree* (AKB-RRT*) algorithm. First, a constrained state space is constructed, where adaptive step-size adjustment and hybrid expansion are guided by relative Euclidean-distance feedback to satisfy the kinematic constraints. Second, parent reselection and rewiring are enhanced by a tortuosity penalty and depth-node optimization to reduce path backtracking and redundancy. Finally, an obstacle inflation mechanism and a stage-differentiated smoothing strategy are introduced to improve path safety and smoothness. Simulation results show that AKB-RRT* achieves a path feasibility rate of 100%, significantly outperforming the 60.04% of RRT* and the 61.40% of bidirectional-RRT*, while increasing the average minimum distance to obstacles to 1.07 m. Under full satisfaction of the kinematic constraints, the proposed algorithm plans safe paths with an average length of 60.34 m and reduces the average computation time by 39.50% compared with RRT*, effectively balancing computational cost and planning quality and meeting engineering application requirements.**

Keywords—robotic manta, path planning, RRT*, kinematic characteristics

I. INTRODUCTION

With the development of marine activities, bio-inspired underwater robots have attracted increasing attention. Among them, robotic manta, which mimic the flapping swimming of manta rays, are regarded as ideal platforms for long-duration and large-scale ocean monitoring due to their low noise, high propulsion efficiency, and long endurance [1] [2].

However, this type of robot relies on pectoral-fin flapping for forward thrust and caudal-fin deflection for pitching moment adjustment [3]. This results in significant underactuated and non-holonomic characteristics: the robot cannot translate laterally or move vertically, and its depth variation is strongly coupled with surge velocity and pitch angle [4]. These constraints limit its motion. Consequently, conventional path planning algorithms often generate trajectories that are difficult to track or suffer from severe oscillations, leading to mission failure or collision risks [5].

Planning safe, smooth, and kinematically feasible paths for non-holonomic underwater robots in complex 3D environments remains challenging. Existing methods have clear limitations. Graph-search algorithms (e.g., Dijkstra, A*) suffer from the "curse of dimensionality" in high-dimensional continuous spaces and struggle to incorporate constraints like curvature and pitch angle [6] [7]. Sampling-based path planning methods, such as Probabilistic Roadmap (PRM) and Rapidly-exploring Random Tree (RRT), handle high-dimensional spaces but often produce redundant searches in open regions, leading to slow convergence [8] [9]. Furthermore, their initial paths frequently exhibit abrupt turns and curvature mutations. Although bidirectional RRT* (Bi-RRT*) accelerates convergence, it still

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lacks consideration for non-holonomic constraints [10]. Artificial potential field and swarm-intelligence methods are easy to implement but prone to local minima or failure to satisfy strict kinematic constraints in complex 3D obstacle environments [11]. Without dedicated constraint optimization, generated trajectories significantly increase tracking difficulty and cause severe energy loss at turning points [12] [13].

To address these problems, this paper proposes an adaptive kinematics-constrained bidirectional RRT* (AKB-RRT*) algorithm, based on the kinematic characteristics of the robotic manta. The main improvements are as follows: (1) a constrained state space is constructed, and an adaptive hybrid sampling and expansion strategy is introduced to balance search efficiency and bidirectional tree connection accuracy; (2) a depth-node optimization-based parent reselection and rewiring strategy incorporating path tortuosity efficiency is developed to improve planning efficiency and reduce path tortuosity; (3) a path safety assurance and stage-differentiated smoothing strategy is introduced to ensure path safety while improving trajectory stability and smoothness.

The remainder of this paper is organized as follows. Section II introduces the kinematic characteristics of the robotic manta, the mathematical formulation of the path planning problem, and the related algorithms. Section III presents the improved strategies and core procedures of the proposed AKB-RRT* algorithm. Section IV verifies its superiority through comparative simulations in terms of path feasibility rate, minimum distance to obstacles, path length, and solving time. Finally, Section V concludes the paper and discusses future research directions.

II. Preliminaries and Problem Formulation

This section introduces the basic information of the robotic manta, together with the problem formulation of path planning, the RRT* algorithm, and the Bi-RRT* algorithm.

A. Overview of the Robotic Manta

To quantitatively evaluate the practical performance limits of the prototype (see in Fig. 1 and Table I), multi-dimensional pool experiments were conducted to record and analyze its motion behavior. The experiments focused on the key indicators, including the cruising speed, path pitch-angle limit, path deflection-angle limit. The path pitch-angle and deflection-angle limits were calculated from the 3D trajectories.

Fig. 1. The prototype of robotic manta.

TABLE I. The Related Information of The Robotic Manta

Items	Detailed information
Length × Width × Height	0.580m×0.955m×0.120m
Mass	13.451kg
Battery parameters	8.4-VDC13600mAhNi-H
Control mode	Ratio control
Attitude sensor	AHRS-2501
Cruising speed	-0.187m/s~0.466m/s
Path pitch-angle limit	-25.03°~20.97°
Path deflection-angle limit	-57.35°~57.35°
Limit of turning radius	0.096m~1.471m

B. Description of the Path Planning Problem

Let $Q \in R^3$ be the configuration space of the robot. Define the state vector as $p=[x, y, z, \theta, \varphi]^{\mathrm{T}}$, where (x, y, z) is the 3D position in the inertial frame, and (θ, φ) are the pitch and yaw angles of the robot. The objective of path planning is to find a continuous mapping $\sigma:[0,1]\rightarrow P_{free}$ that satisfies the boundary conditions $\sigma(0)=\boldsymbol{p}_{init}$ and $\sigma(1)=\boldsymbol{p}_{target}$. Constrained by the propulsion mechanism and physical characteristics, the path must satisfy the following kinematic constraints:

Non-holonomic constraints: Limited by the MPF (Median and/or Paired Fin) propulsion mode, the lateral velocity v and vertical velocity w in the body frame are zero. Changes in depth z are achieved through the coupling of surge velocity u and pitch angle θ; therefore, the trajectory must satisfy path pitch limits.

State space boundaries: The entire path must satisfy the physical limit envelope B calibrated from experiments. For $\forall \tau \in [0,1]: \sigma(\tau) \in P_{free}$, $|\theta_{path}| \le \theta_{path\text{-}max}$, and $|\omega_{path}| \le \omega_{path\text{-}max}$, where θ_{path} and $\theta_{path\text{-}max}$ are the path pitch-angle and its maximum limit, and ω_{path} and $\omega_{path\text{-}max}$ are the path deflection-angle and its maximum limit.

To simplify the constraints, the following parameters are adjusted based on water tank experimental data:

The basic planning step is 1 m; the safety distance constraint is one body width (1 m); the maximum path pitch-angle $\theta_{path\text{-}max}$=20° ; and the maximum path deflection-angle $\omega_{path\text{-}max}$=55°.

C. Foundational Concepts

1) Overview of RRT:* RRT* is an asymptotically optimal variant of the classic RRT algorithm. By introducing parent node reselection and path rewiring mechanisms, its path cost can converge to the global optimum as the iteration number increases. (see Fig. 2). When a new node x_{new} is inserted into the tree, the algorithm searches within its neighborhood of radius r and selects the neighboring node x_{near} that minimizes the accumulated cost from to the start point as its parent node. After the new node is inserted, the algorithm checks other nodes within its neighborhood. If using x_{new} as an intermediate node can reduce the path cost of these neighboring nodes, their parent nodes are changed to x_{new}.

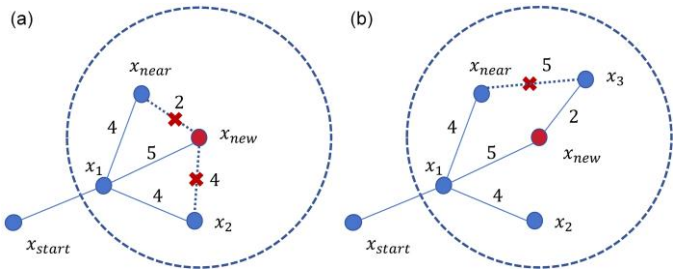

Fig. 2. Schematic diagram. (a) Reselect parent node. (b) Rewiring.

2) Overview of Bi-RRT:* Bi-RRT* is an asymptotically optimal extension of RRT* under a bidirectional search framework. It constructs two randomly expanding trees, T_{start} and T_{goal}, from the start and goal points, respectively, and connects them efficiently. This bidirectional structure enables faster convergence in complex environments while preserving asymptotic optimality (see TABLE II).

TABLE II. PSEUDOCODE FLOWCHART OF THE BI-RRT* ALGORITHM

Algorithm 1: $\mathbf{Bi-RRT}^*(x_{\text{start}}, x_{goal})$

$T_a.add(x_{start}), T_b.add(x_{goal}), i = 0$
while $i < N$
  $x_{rand} = Sampling(); i++$
  $x_{nearest} \leftarrow nearest_neighbor(T_a, x_{rand})$
  $x_{new} = Steer(x_{nearest}, x_{rand})$
  if not $(Grow(T_a, x_{new}) = Trapped)$ **then**
    if $(Connect(T_b, x_{new}) = Reached)$ **then**
      return $Path(T_a, T_b)$
  $Swap(T_a, T_b)$
return $Failure$

III. ADAPTIVE KINEMATICS-CONSTRAINED BI-RRT* PATH PLANNING METHOD

This section details three primary improvements of AKB-RRT* over Bi-RRT*. It also briefly describes path smoothing.

A. Adaptive Hybrid Sampling and Expansion under Non-holonomic Kinematic Constraints

To address the issues of local deadlock and kinematic mismatch caused by blind randomness of RRT* and its variants in 3D constrained spaces, this paper proposes an adaptive hybrid sampling and expansion mechanism for constrained state spaces that accommodates incomplete constraints.

Due to limitations in the robot's steady-state navigation performance, a fixed expansion step size causes coarse search during high-speed cruising and overshoot during low-speed docking. To address this issue, the algorithm introduces a continuous-state adaptive step size based on relative Euclidean distance feedback:

$$S_{step}=S_{min}+(S_{max}-S_{min})\cdot \min\left(\frac{\|P_{target}-P_{near}\|}{D_{th}},1\right) \tag{1}$$

where S_{min} and S_{max} are the minimum and maximum step boundaries, $\|P_{target}-P_{near}\|$ is the Euclidean distance between the target and the nearest node, and D_{th} is the threshold distance for triggering shrinkage.

Subsequently, a 3D constrained state space is constructed based on the path pitch-angle and deflection-angle limits identified from tank experiments. To avoid the blind randomness of uniform sampling and the deadlock risk of goal-biased sampling, an adaptive hybrid sampling strategy is proposed. When the projection of the goal node falls inside the constrained space, a three-level hybrid sampling mechanism is activated: goal-biased sampling with probability a, goal-quadrant uniform sampling with probability b, and adaptive Beta-informed sampling with probability $(1-a-b)$. Otherwise, the strategy switches to a two-level mechanism consisting of goal-quadrant boundary sampling and standard uniform sampling.

This part describes adaptive Beta-informed sampling. Using path deflection-angle sampling as an example, the target relative deflection demand ratio $m_\omega=|\Delta\omega_{target}|/\omega_{path\text{-}max}$ is defined as the expected mode of the Beta distribution. To maintain stable sampling concentration, a constant factor c is used to dynamically calculate the Beta distribution parameters α and β:

$$\begin{cases}\alpha=m_\omega\cdot c+1\\ \beta=(1-m_\omega)\cdot c+1\end{cases} \tag{2}$$

Based on these parameters, the final path deflection-angle increment is $\Delta\omega_{new}$:

$$\Delta\omega_{new}=Beta(a,b)\cdot\omega_{max}\cdot\mathrm{sgn}(\Delta\omega_{target}) \tag{3}$$

Similarly, the path pitch-angle increment is derived as $\Delta\theta_{new}$.

B. Heuristic Parent Reselection and Rewiring Based on Path Tortuosity and Depth-Node Optimization

In non-holonomic path planning, using Euclidean distance as the sole metric often leads to high-frequency steering and conflicts with the physical characteristics of the robot. This paper proposes a heuristic parent reselection and rewiring mechanism that integrates tortuosity efficiency and node depth optimization. This mechanism introduces a comprehensive cost evaluation system incorporating spatial kinematic penalties. For a given expansion node $P_{current}$ and its candidate nodes P_{near}, the single-step expansion cost C_{step} is:

$$\begin{cases}C_{step}=\|P_{current}-P_{near}\|\cdot\left(1+\lambda\cdot\left(\frac{\Delta\omega}{\omega_{path\text{-}max}}\right)^2\right)\\ \Delta\omega=\arccos\left(\frac{\vec{v}_{prev}-\vec{v}_{new}}{\|\vec{v}_{prev}\|\cdot\|\vec{v}_{new}\|}\right)\end{cases} \tag{4}$$

where $\|P_{current}-P_{near}\|$ is the Euclidean distance between two nodes; $\Delta\omega$ is the spatial deflection angle increment calculated from the previous path vector $\vec{v}_{prev}$ and the current path vector $\vec{v}_{new}$, and $\lambda=2$ is the tortuosity penalty weight.

The optimized mechanisms of parent reselection and rewiring are presented (see in Fig. 3 and Fig. 4). During the parent reselection stage, a depth-node optimization strategy based on the triangle inequality is introduced. After identifying the local optimal parent P_{opt}, the algorithm traces ancestor nodes up the tree. If a deepest ancestor P_{ance} can reach $P_{current}$ directly while remaining collision-free and satisfying kinematic constraints, the algorithm skips P_{opt} and intermediate nodes to reselect P_{ance} as the parent. This strategy effectively eliminates redundant topological branches, reducing cumulative path cost, backtracking, and node redundancy.

During the rewiring phase, the algorithm traverses the neighborhood node set of $P_{current}$. For each neighbor, if reconnecting to $P_{current}$ reduces the cumulative path cost, the algorithm verifies non-holonomic kinematic constraints and collision detection. Upon verification, the node's parent is updated to $P_{current}$, followed by a cascaded update of its descendants. Fig. 4 illustrates this using P_3 and P_5. P_3 is rejected because reconnecting to $P_{current}$ violates kinematic constraints. P_5 is successfully reconnected due to reduced path cost and valid verification, completing the cascaded update for child node P_6.

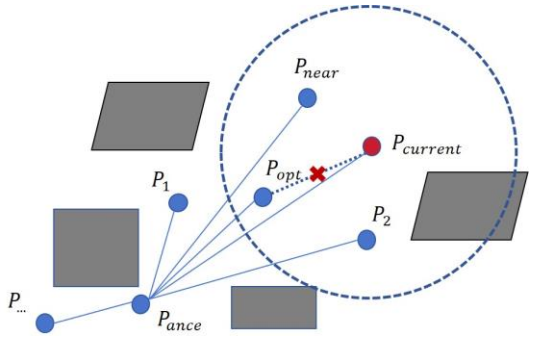

Fig. 3. Schematic diagram of optimized parent reselection.

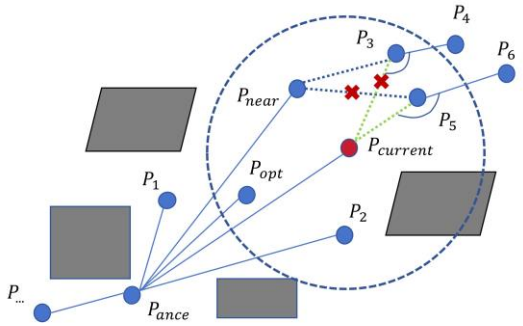

Fig. 4. Schematic diagram of optimized rewiring.

C. Path Safety Assurance and Stage-Differentiated Smoothing Strategy

To address collision risks, node redundancy, and curvature mutations in initial paths, the algorithm introduces an obstacle inflation mechanism and a phased differential smoothing strategy. For safety, obstacles undergo geometric inflation with a safety distance threshold R_{safe}. Collision detection verifies node penetration and a soft constraint $d_{pbs}>R_{safe}$ to prevent the robot from approaching obstacle boundaries.

Path smoothing employs a phased strategy combining global and local optimization. To address node sparsity and geometric redundancy, control points Q_1 and Q_2 are generated between P_i and P_{i+1} via trisection interpolation. A dynamic backtracking mechanism attempts to skip intermediate nodes for direct connections. If kinematic constraints or collisions occur, the algorithm reverts to the previous valid node to ensure the shortest executable path. To eliminate sharp nodes, quadratic Bezier curves reconstruct the turns. The actual turning radius d_{actual} is defined as:

$$d_{actual}=\min\left(R_{min}, \frac{\|P_i-P_{i-1}\|}{\gamma}, \frac{\|P_{i+1}-P_i\|}{\gamma}\right) \tag{5}$$

where γ=2.1 is the shrinkage factor, R_{min} is the minimum turning radius, P_i is the current node, P_{i-1} is the parent node, and P_{i+1} is the child node. Control points A and B are calculated as:

$$\begin{cases} A=P_i-d_{actual}\cdot\dfrac{(P_i-P_{i-1})}{\|P_i-P_{i-1}\|} \\ B=P_i+d_{actual}\cdot\dfrac{(P_{i+1}-P_i)}{\|P_{i+1}-P_i\|} \end{cases} \tag{6}$$

The parametric equation for the quadratic Bezier curve $C(t)$ is:

$$C(t)=(1-t)^2\cdot A+2t\cdot(1-t)\cdot P_i+t^2\cdot B,\ t\in[0,1] \tag{7}$$

This curve satisfies endpoint and tangent constraints, ensuring C^1 continuity at connection points. It eliminates curvature mutations while satisfying turning radius constraints.

IV. Simulation and Analysis

To evaluate the performance of the proposed algorithm, simulations were conducted in MATLAB. The hardware environment included an Intel Core i7-14650HX processor with 32 GB of RAM and 1024 GB of storage.

A. Simulation Setup

The simulation map dimensions are 40 m×40 m×20 m, with start and target coordinates set at [2,2,2] and [38,38,18], respectively. Obstacles include spheres, cylinders, and rectangular prisms. To ensure a fair comparison, 100 Monte Carlo trials were performed for each algorithm under identical conditions. Evaluation metrics include path feasibility rate, minimum distance from obstacles, path length and solving time. The final results are averaged over the 100 trials. Key parameters for AKB-RRT* are as follows: hybrid sampling probabilities a=0.5, b=0.2, and 1—a—b=0.3; Beta distribution factor c=4; and adaptive step boundaries S_{max}=3 m, S_{min}=1 m, and D_{th}=3 m.

B. Simulation Analysis

The optimal path of the proposed method and the explored path are shown in Fig. 5. It can be clearly seen that the exploration of the path is strictly limited within the constrained state spaces. After processing, the final path is smooth, safe, and suitable for the movement of the robot.

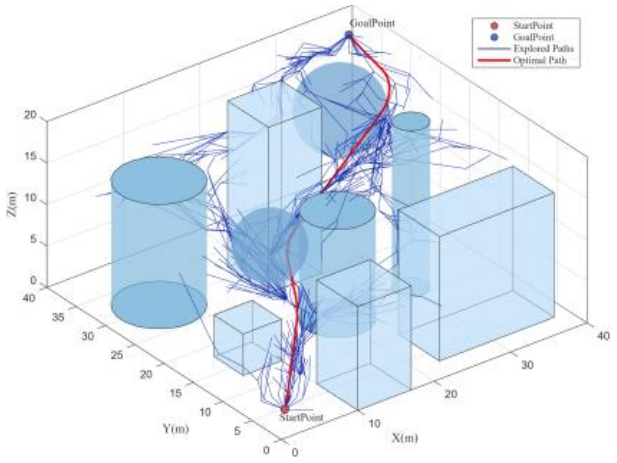

Fig. 5. The result of the proposed method (AKB-RRT*).

The optimal paths of the three algorithms within this environment are illustrated (see Fig. 6). Red and blue dots denote the start and goal, respectively. Green, blue, and red lines represent the paths of RRT*, Bi-RRT*, and AKB-RRT*. Path morphology and kinematic analysis show that AKB-RRT* offers significant advantages. The higher overall smoothness effectively suppresses control chattering caused by abrupt turns. Specifically, during the initial stage, the algorithm maintains a small path pitch-angle for a smooth acceleration transition. This low-impact, gradual motion aligns with the bionic kinematic mechanism of the manta ray, enhancing navigation stability while ensuring safety.

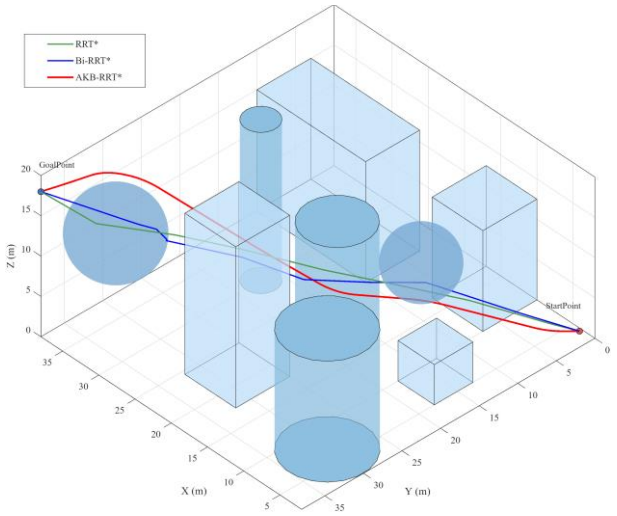

Fig. 6. Comparison of optimal paths.

To quantitatively analyze the four key performance metrics (see in Fig. 7 and Table III): path feasibility rate, minimum distance from obstacles, path length, and solving time. The data show that AKB-RRT* has significant advantages in path quality and safety. Its average path feasibility rate remains at 100%, far exceeding RRT* (60.04%) and Bi-RRT* (61.40%). Box plots reveal dispersed distributions and low medians for the comparison algorithms, proving the robustness of AKB-RRT* in complex constrained spaces. Regarding path length, AKB-RRT* averages 60.34 m, which is approximately 7% longer than RRT* (56.38 m). This increase is a necessary trade-off to satisfy kinematic constraints and safety margins. In contrast, shorter paths generated by RRT* frequently violate turning radius and path kinematic constraints, making them invalid. Therefore, AKB-RRT* balances safety and practicality by ensuring a 100% path feasibility rate with a slight increase in path length.

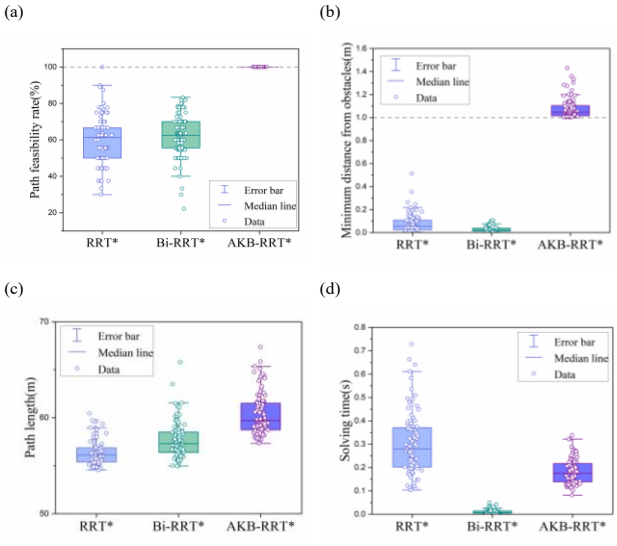

Fig. 7. Comparison of Performance. (a) Path feasibility rate. (b) Minimum distance from obstacles. (c) Path length. (d) Solving time.

However, these performance improvements benefit from strict state space constraints, heuristic parent reselection and rewiring mechanisms, and the phased smoothing strategy. This inevitably increases computational overhead. The solving time (average 0.18 s in the table) is higher than unconstrained Bi-RRT* (0.01 s) but lower than RRT* (0.30 s), which remains within an acceptable range.

In summary, AKB-RRT* achieves comprehensive improvements in safety (larger minimum distance from obstacles), path feasibility rate (100%), and path efficiency at a moderate computational cost. Although the solving time is higher than Bi-RRT*, it remains lower than RRT*. This performance best satisfies the rigorous engineering requirements for the robot in complex 3D environments, specifically trajectory smoothness, obstacle avoidance reliability, and kinematic feasibility.

TABLE III. AVERAGE PERFORMANCE INDICATORS

Items	Average path feasibility rate	Average minimum distance from obstacles	Average path length	Average solving time
RRT*	60.04%	0.08 m	56.38 m	0.30 s
Bi-RRT*	61.40%	0.03 m	57.65 m	0.01 s
AKB-RRT*	100.00%	1.07 m	60.34 m	0.18 s

V. CONCLUSION

To address the low path feasibility rate, poor path quality, and insufficient safety for the robot, this paper proposes AKB-RRT* based on the robot's kinematic characteristics. Its core contributions include: An adaptive hybrid sampling and expansion mechanism in constrained state spaces. This

improves sampling efficiency and path quality while balancing tree expansion and connection through adaptive step lengths. A heuristic parent reselection and rewiring mechanism integrating tortuosity efficiency and depth-node optimization strategy. It significantly reduces the probability of path circling or sharp turns while enhancing planning efficiency and smoothness. Path safety assurance and phased differential smoothing. An obstacle inflation mechanism ensures safety, and a phased smoothing strategy further improves final path smoothness and kinematic feasibility.

Simulation results indicate that AKB-RRT* generates safe, feasible paths for the robot in complex 3D underwater environments within a short solving time. This algorithm demonstrates strong potential for practical applications. Future work will investigate the balance between ocean currents and energy consumption, dynamic obstacle avoidance, and local map-based planning. Finally, the algorithm will be deployed on physical robots to address path planning challenges in real-world scenarios.

REFERENCES

[1] Q. Sun, J. Wu, C. Sheng, S. Hu, Z. Wang, and H. Huang, "Design and implementation of multi-level linkage mechanism bionic pectoral fin for manta ray robot," Ocean Eng., vol. 284, p. 115276, 2023.

[2] Y. Meng, Z. Wu, D. Chen, P. Zhang, M. Tan, and J. Yu, "Development and 3-D path-following control of an agile robotic manta with flexible pectoral fins," IEEE Trans. Cybern., vol. 54, no. 6, pp. 3727–3738, 2024.

[3] C. Xing, Y. Cao, Y. Cao, G. Pan, and Q. Huang, "Asymmetrical oscillating morphology hydrodynamic performance of a novel bionic pectoral fin," J. Mar. Sci. Eng., vol. 10, p. 289, 2022.

[4] Y. Xie, S. Ma, Y. Zhang, X. Liu, Y. Cao, and Y. Shi, "Dynamic modeling and tube-MPC based landing control for a robotic manta with flexible fins," IEEE/ASME Trans. Mechatronics, vol. 31, no. 1, pp. 151–161, 2026.

[5] J. Li, Z. Zhang, and H. Zhang, "Local path planning for AUV with fusion of DWA and RRT algorithms in a complex environment," CAAI Trans. Intell. Syst., vol. 19, no. 4, pp. 961–973, 2024.

[6] X. Xu, T. Li, Y. Dong, and H. Ren, "A review: Learning-based methods for UAV trajectory planning," J. Syst. Eng. Electron., vol. 35, no. 1, pp. 103–118, 2024.

[7] Z. Shen, Y. Wu, Y. Ma, and S. Guo, "An improved A* algorithm for UAV path planning under wind conditions based on the dubins curve," Front. Inf. Technol. Electron. Eng., vol. 25, no. 2, pp. 198–210, 2024.

[8] L. Zhang, Y. Zhang, and J. Zhang, "Path planning based on improved RRT* algorithm for AUV swarm," Chin. J. Ship Res., vol. 18, no. 1, pp. 43–51, 2023.

[9] C. Shi, T. Dao, H. Liu, and J. Bai, "A rapid planning repair method of three-dimensional path for AUV," Mob. Netw. Appl., vol. 29, no. 6, pp. 1951–1966, 2024.

[10] H. Qu, W. Pang, Z. Wu, and R. L. Batista, "An improved Bi-RRT* algorithm for robot path planning in automotive assembly line," Robot. Comput.-Integr. Manuf., vol. 71, p. 102165, 2021.

[11] S. P. Sahoo, B. Das, B. B. Pati, F. P. G. Márquez, and I. S. Ramírez, "Hybrid path planning using a bionic-inspired optimization algorithm for autonomous underwater vehicles," J. Mar. Sci. Eng., vol. 11, no. 4, pp. 761, 2023.

[12] C. Cheng, Q. Sha, B. He, G. Li, and Z. Zhang, "Motion planning for autonomous underwater vehicles considering temporal uncertainty," Ocean Eng., vol. 266, p. 112722, 2022.

[13] S. Guo, X. Zhang, Y. Zheng, J. Du, and X. Wang, "USV path planning based on the improved particle swarm optimization algorithm and the dynamic window approach," Front. Inf. Technol. Electron. Eng., vol. 25, no. 7, pp. 913–926, 2024.

A Local Obstacle Avoidance Method for Mobile Robots Based on Multi-Candidate Target Scoring and Tracking

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***Abstract*—To address the problems of local control fluctuation and reduced traversal efficiency of mobile robots in complex indoor environments, especially under obstacle-clinging motion, narrow-passage constraints, and persistent blockage conditions, this paper proposes a local obstacle avoidance method based on multi-candidate target scoring and speed regulation. On the basis of an initial algorithm built upon single look-ahead target tracking, multiple local candidate targets are generated under the constraint of the global path. The current local target is then selected by jointly considering cost information, heading deviation, and forward progression. In addition, a speed regulation mechanism is introduced to suppress control oscillation in constrained spaces. Pre-validation and scenario-based simulations show that the proposed method achieves better local control continuity and improves traversal efficiency and navigation stability.**

Keywords—mobile robot, local planning, path tracking, multi-candidate target scoring, complex indoor environment

I. INTRODUCTION

With the wider use of mobile robots in warehouse handling, indoor delivery and service scenarios, autonomous navigation in complex indoor environments has become an important research topic. Indoor environments usually contain narrow corridors, dense obstacles and frequent local disturbances, which place higher requirements on the robot's local obstacle avoidance ability [1]. In practical navigation tasks, the global path only provides the overall route, while the local planner directly determines whether the robot can pass through constrained areas smoothly [2].

Many local obstacle avoidance methods have been studied. The artificial potential field method and Bug algorithms have simple structures and are easy to implement [3]. VFH describes local obstacle information through a polar histogram and improves real-time direction selection [4]. DWA considers velocity constraints during local decision-making and has been widely used in engineering applications [5]. TEB improves trajectory smoothness through local trajectory optimization [6]. However, when the robot moves close to obstacles, enters narrow passages or faces persistent blockage, traditional methods may still suffer from unstable control output and reduced traversal efficiency [7].

To address these limitations, this paper first constructs an initial local obstacle avoidance method based on single look-ahead target tracking, and then improves it by introducing multi-candidate target scoring and speed regulation. Unlike DWA, which evaluates sampled velocity commands in the velocity space, and TEB, which optimizes a local trajectory band, the proposed method makes local decisions in the neighborhood of the global path. In each control cycle, several candidate targets are generated around the local reference point, filtered by costmap information, and scored by considering obstacle cost, heading deviation, and forward progression. The selected target is then used to generate velocity commands, while a speed regulation factor adjusts the linear velocity in constrained spaces. The main contributions of this work are the global-path-constrained candidate target generation strategy, the lightweight multi-factor scoring function, the velocity regulation mechanism, and the additional ablation and computation-overhead experiments used to support parameter selection and real-time feasibility.

II. INITIAL LOCAL OBSTACLE AVOIDANCE METHOD BASED ON LOOK-AHEAD TARGET TRACKING

A. Overall Design

The initial algorithm uses the global path as the reference for local control. In each control cycle, the planner searches forward along the global path and extracts one temporary target point. The robot then generates velocity commands according to the target position. At the same time, the planner checks the feasibility of the path segment ahead and switches to terminal pose adjustment when the robot approaches the goal. The whole process can be shown in "Fig. 1".

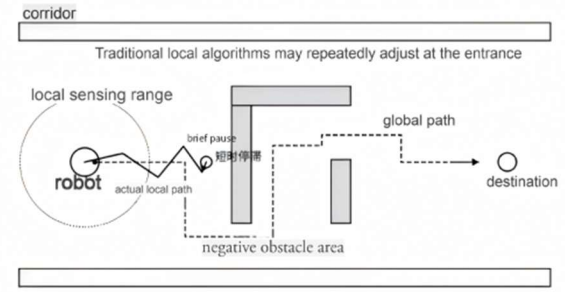

Fig. 1. Oscillation phenomenon of traditional obstacle avoidance methods.

This design gives the robot a stable tracking object during local motion. It avoids frequent direction switching caused by purely reactive obstacle avoidance and forms the basis for later improvement.

B. Look-Ahead Target Extraction

Let the i-th point on the global path be denoted by Pi. Its representation in the map frame and the robot body frame are denoted by P_i^m and P_i^b, respectively. Their relationship can be written as follows:

$$P_i^b = T_m^b P_i^m \tag{1}$$

where T_m^b is the transformation from the map frame to the robot body frame.

After transformation, the relative distance between the path point and the robot is calculated by

$$d_i = \sqrt{(x_i^b)^2 + (y_i^b)^2} \tag{2}$$

where x_i^b and y_i^b are the forward and lateral coordinates of the path point in the robot body frame.

The temporary target point is selected according to the look-ahead distance:

$$k = \min\{i | d_i > L_d\}, \quad P_t = P_k \tag{3}$$

where L_d is the look-ahead distance. In this paper, it is set to 0.3 m. A shorter look-ahead distance may cause frequent target switching between adjacent control cycles, while an excessively long distance weakens the response to local obstacle changes. Therefore, 0.3 m is adopted as a moderate value to maintain target stability and preserve local response ability.

As shown in "Fig. 2", the planner searches forward along the global path and selects the first path point satisfying the look-ahead condition as the temporary tracking target.

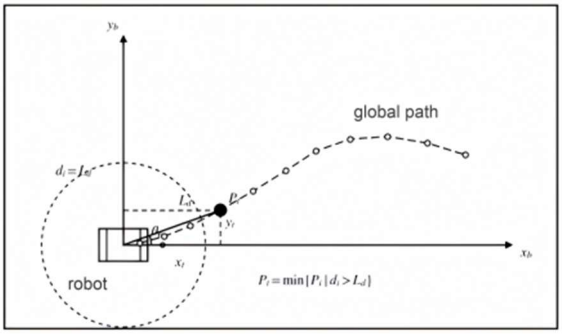

Fig. 2. Look-ahead target selection process.

C. Velocity Control and Path Feasibility Check

After the temporary target is obtained, the target position is mapped into velocity commands. If the target point in the robot body frame is (x_t, y_t), the linear velocity and angular velocity are given by:

$$v = k_x x_t \tag{4}$$

$$\omega = k_y y_t \tag{5}$$

where k_x and k_y are velocity control gains.

To avoid tracking a path segment that has entered a high-risk area, the planner checks the local cost value of several path points ahead. The path is regarded as infeasible when

$$C_i \geq C_{th} \tag{6}$$

C_i is the costmap value of the i-th path point. In this paper, C_{th} equals to 253, which means that the path point has entered the inscribed obstacle region or a higher-risk area.

As shown in "Fig. 3", the feasibility check is completed by mapping the forward path points into the local costmap.

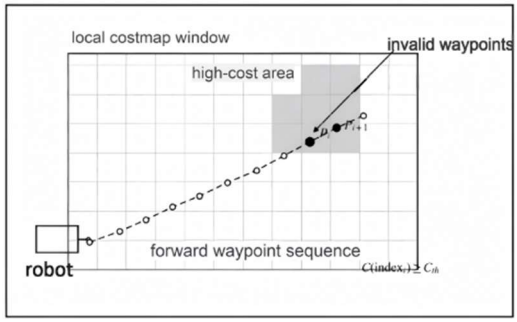

Fig. 3. Path feasibility checking based on the local costmap

D. Terminal Pose Adjustment

When the robot approaches the goal, the planner stops forward motion and adjusts the final orientation. The heading error is

$$e_\theta = \theta_{goal} - \theta_r \tag{7}$$

θ_{goal} is the target heading angle and θ_r is the current robot heading angle.

The angular velocity is calculated by

$$\omega_g = \text{sat}_{[-0.7,0.7]}(1.7 e_\theta) \tag{8}$$

where $\text{sat}_{[-0.7,0.7]}$ limits the angular velocity output.

III. Improved Local Obstacle Avoidance Method Based on Multi-Candidate Target Scoring

A. Improvement Strategy

The initial algorithm can complete basic path tracking, but its local decision object is limited to a single look-ahead target. In open areas, this strategy is usually sufficient because the available free space is relatively wide. However, in narrow passages, obstacle-clinging motion, or partially blocked routes, the selected target may become sensitive to small costmap changes and may not provide enough adjustment margin. To overcome this limitation, the improved method changes the target selection process from single-point tracking to multi-candidate target scoring. Several candidate points are generated around the local reference point under the constraint of the global path, filtered according to their relative positions and local cost values, and then scored by considering obstacle safety, heading consistency, and forward progression. In this way, the selected target remains close to the global path while being more suitable for the current local environment.

As shown in Fig. 4, the improved method consists of three main steps: candidate target generation, feasibility filtering, and scoring-based target selection.

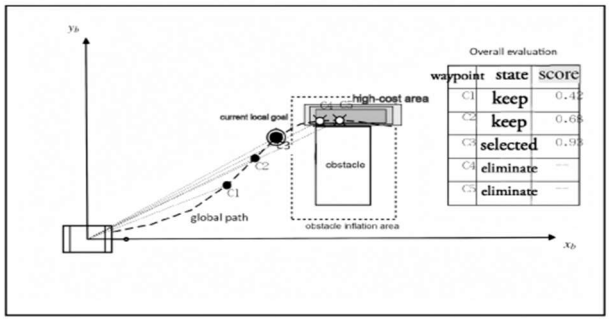

Fig. 4. Candidate target generation and scoring process

B. Candidate Target Generation

Let the local reference point on the global path be

$$P_f = P_{k_f} \tag{9}$$

Here, k_f denotes the index of the reference point. This reference point is still obtained from the global path, so the improved method does not change the overall navigation direction. It only expands the target selection range near the current local reference.

A candidate window is constructed around this point:

$$C_0=\{P_{k_f-m},...,P_{k_f},...,P_{k_f+m}\} \tag{10}$$

In this expression, m represents the half-width of the candidate window in this paper, m is set to 2, so five candidate points are generated in each control cycle. This setting provides a small local search range while keeping the candidates close to the current global path segment. The influence of this parameter is further examined in the parameter ablation experiment in Section IV.

Not all candidate points are suitable for control output. Points located behind the robot may cause backward tracking, and points in high-cost regions may lead the robot close to obstacles. Therefore, only candidate points located in front of the robot and outside high-risk regions are retained:

$$C=\{P_j \in C_0 | x_j>0, c_j<C_{th}\} \tag{11}$$

In Eq. (10), c_j is the cost value of candidate point P_j. After this filtering step, the planner obtains a feasible candidate set for the current control cycle.

C. Target Scoring and Speed Regulation

For each retained candidate point, the comprehensive score is calculated as

$$S_j=w_c S_{c,j}+w_h S_{h,j}+w_f S_{f,j} \tag{12}$$

The terms $S_{c,j}$, $S_{h,j}$, $S_{f,j}$ correspond to the cost score, heading score and forward progression score, respectively. The cost score suppresses candidates close to obstacles, the heading score reduces large lateral deviation, and the forward progression score encourages motion along the global path. In the implementation, the weights of the forward progression, cost, and heading terms are set to 0.45, 0.40, and 0.15, respectively. This setting gives slightly higher priority to forward motion while still preserving obstacle avoidance and heading consistency. The influence of the scoring weights is further examined in the parameter ablation experiment in Section IV too.

The candidate with the highest score is selected as the current local target:

$$P^*=\text{argmax}_{P_j \in C} S_j \tag{13}$$

Compared with the initial algorithm, this selection rule gives the planner a wider decision space. When one candidate becomes unsuitable due to obstacle cost or heading deviation, another candidate can still be selected, so the local control process becomes less dependent on a single point.

To further reduce velocity fluctuation in constrained spaces, a regulation factor is introduced:

$$\lambda_t=\text{sat}_{[\lambda_{\min},1]}\left(1-w_h f_h-w_c f_c-w_s f_s\right) \tag{14}$$

The variables f_h, f_c, f_s describe heading correction, cost constraint and short-distance shrinkage. When local risk increases, λ_t decreases to slow the robot; in clear space, it stays near 1 to maintain normal motion.

The final linear velocity is

$$v_t=k_v x_t \lambda_t \tag{15}$$

This speed regulation mechanism works together with target scoring. The scoring process determines a more suitable local target, while the regulation factor adjusts the motion intensity according to the current local risk. As a result, the improved algorithm can maintain better motion continuity in narrow passages and obstacle-dense areas.

The complete control process of the proposed method is summarized in “Fig.5”. The planner first searches the local reference point, generates and filters candidate targets, selects the best local target through scoring, and then outputs velocity commands with speed regulation.

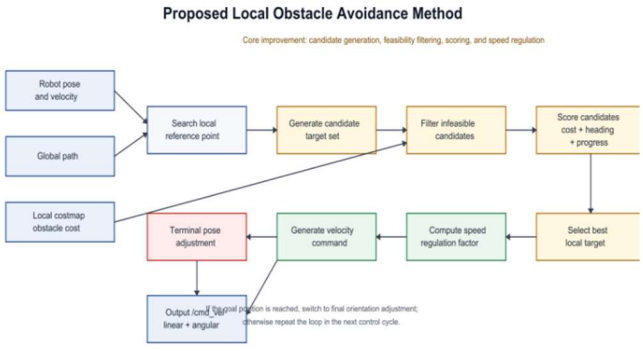

Fig. 5. Flowchart of the proposed local obstacle avoidance method.

The computational cost of the proposed method mainly comes from forward path feasibility checking and candidate target scoring. If Nc path points are checked ahead, the feasibility checking cost is O(Nc). The candidate scoring cost is O(2m+1), where m is the candidate-window half-width defined in Section III-B. With the adopted setting, only five candidates are evaluated in each control cycle. Therefore, under fixed parameters, the per-cycle computational cost is approximately constant and remains lightweight.

IV. Simulation Experiments and Analysis

A. *Pre-Validation*

MATLAB simulations are first conducted to verify the basic behavior of the proposed method. Two scenes are designed: a basic static scene and a dense static obstacle scene. The compared algorithms include DWA, TEB, the initial algorithm which names CV in the figure, and proposed Algorithm which names CV FINAL.

As shown in "Fig. 6", all four algorithms can complete the navigation task in the basic static scene.

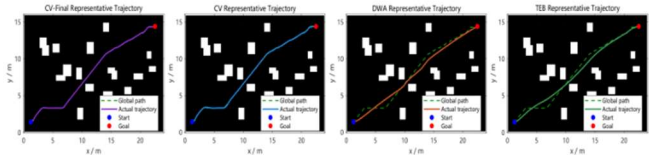

Fig. 6. Representative trajectories of four algorithms in the basic static scene

As shown in TABLE I, the improved algorithm achieves shorter travel time and smaller path deviation than the other algorithms. Its average navigation time is 38.64 s, and its average path deviation is 0.073477 m.

TABLE I. Core Evaluation Indices in the Basic Static Scene

Algorithm	*Success rate*	*Average travel time(s)*	Average path deviation
CV	100%	44.78	0.079
CV FINAL	100%	38.64	0.073
DWA	100%	50.22	0.38
TEB	100%	50.78	0.33

In the dense static obstacle scene, "Fig. 7" shows that DWA and TEB fail to pass the narrow passage in repeated trials, while the proposed algorithm reaches the goal.

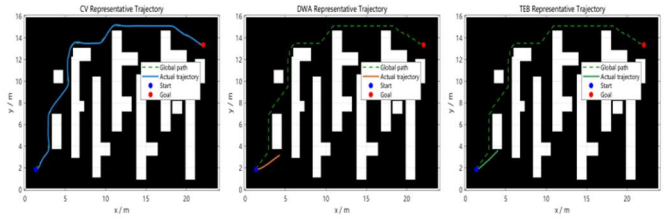

Fig. 7. Representative trajectories in the dense static obstacle scene

From "Fig. 8", the velocity curve of the improved algorithm which has orange color is smoother than that of the initial algorithm, with smaller peak-valley fluctuation and a more concentrated deceleration process near the goal.

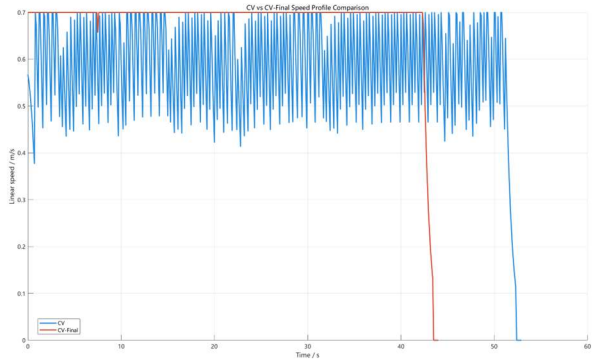

Fig. 8. Velocity curve comparison

The former results indicate that the improved method reduces unnecessary speed oscillation during local adjustment. After introducing multi-candidate target scoring and speed regulation, the planner can select a more suitable local target in constrained areas and adjust the motion intensity according to the surrounding risk. Therefore, the robot maintains better control continuity in complex static environments, and its traversal process becomes more stable and efficient.

Additionally, to explain the selection of key parameters, a parameter ablation experiment is conducted in the dense static scene. The tested parameters include the look-ahead distance, the candidate-window half-width, and the scoring weights. All parameter groups are tested under the same map, start and goal positions, global path, robot model, and velocity limits. The travel time values are expressed in decimal seconds. The result is shown in "Fig. 9".

Parameter Ablation Results

Group	Look-ahead (m)	Window	Weights F/C/H	Success rate	Time (s)	Deviation (m)	Stall ratio
P0 Current	0.3	2	0.45/0.40/0.15	100%	44.14	0.07796	0.01359
P1 Short look-ahead	0.2	2	0.45/0.40/0.15	100%	44.20	0.07775	0.01357
P2 Long look-ahead	0.4	2	0.45/0.40/0.15	100%	44.16	0.07794	0.01359
P3 Small window	0.3	1	0.45/0.40/0.15	100%	44.58	0.07471	0.01570
P4 Large window	0.3	3	0.45/0.40/0.15	100%	44.02	0.08014	0.01363
P5 Safety-biased	0.3	2	0.25/0.60/0.15	100%	44.68	0.07330	0.00895
P6 Progress-biased	0.3	2	0.65/0.20/0.15	100%	43.96	0.08137	0.01365

Fig. 9. Parameter Ablation Results

As shown in "Fig. 9", all parameter groups can complete the navigation task, so the parameter selection mainly concerns the balance among travel efficiency, path tracking, and motion continuity. A small candidate window increases the stall ratio, while a larger window reduces travel time but increases path deviation. A safety-biased weight setting reduces path deviation and stall ratio but increases travel time, whereas a progress-biased setting shortens travel time but causes larger path deviation. Therefore, the current setting, namely a look-ahead distance of 0.3 m, a candidate-window half-width of 2, and scoring weights of 0.45/0.40/0.15, is adopted as a balanced default setting in the following experiments.

B. *Formal Experiment*

Next, further experiments are carried out on the ROS platform. Compared with the two-dimensional MATLAB scene, the Gazebo simulation includes a more complete navigation process, such as map loading, local costmap updating, velocity command execution, sensor feedback and terminal pose adjustment. TABLE II shows the ROS Noetic environment configuration of the experimental platform used in this paper.

TABLE II. CONFIGURATION DETAILS OF HOST AND VIRTUAL MACHINE

Item	***Host***	***Virtual machine***
Operating System	Windows 11	Linux Ubuntu 20.04
Processor	Intel(R) Core(TM) i7-14650HX	8-core virtual CPU
Memory	8GB	9.4GB
Graphics card	NVIDIA GeForce RTX 4060 GPU	Integrated graphics of host
Disk space	1TB	59GB

The robot model used in the experiment is based on the differential-drive platform. The chassis is driven by velocity commands and can execute both linear velocity and angular velocity in Gazebo. The model is equipped with a 2D LiDAR and an odometry module. The LiDAR provides obstacle information for local costmap updating and local risk judgment, while the odometry module provides pose feedback for coordinate transformation and motion-state estimation.

As shown in "Fig. 10", the simulated robot model contains the basic chassis and sensor configuration required for local planning experiments.

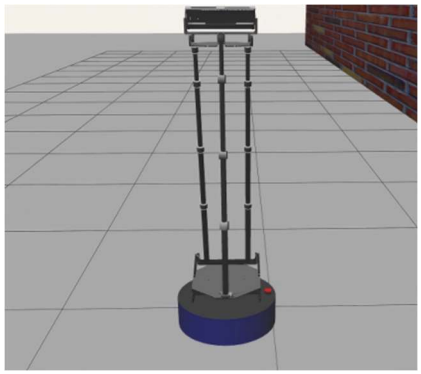

Fig. 10. Robot model figure

To further examine the performance of different algorithms, two representative Gazebo scenes are selected. The first scene is testworld5, which contains narrow passages and relatively dense static obstacles. It is mainly used to test the traversal efficiency and local stability of the robot in a constrained static environment. The second scene is testworld3, which is used to simulate persistent blockage. In this scene, an obstacle is placed at the end of one corridor, forcing the robot to switch to the other corridor. Then another blockage condition guides the robot back to the original route, so the recovery ability of the local planner can be observed more clearly.

As shown in "Fig. 11", testworld5 provides a constrained static scene for evaluating traversal efficiency and speed performance. Also it contains unknown obstacles which are marked by the red boxes. It also shows the route schematic in testworld5.

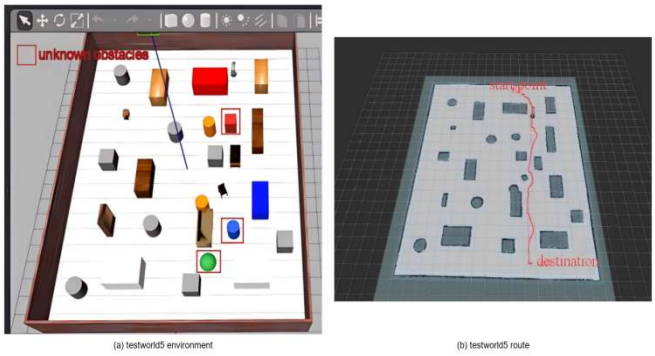

Fig. 11. testworld5 environment and route

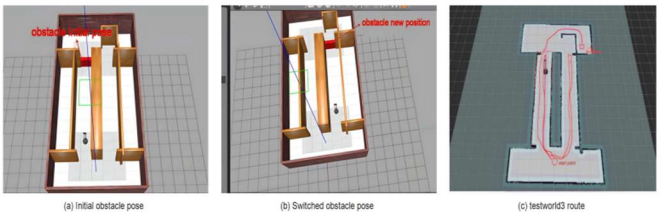

Fig. 12. testworld3 environment and route

As shown in "Fig. 12", testworld3 is used to construct a route-switching scenario under persistent blockage. The red block represents a switching unknown obstacle, which blocks the original corridor and forces the robot to leave the initial corridor and choose another available passage. It also shows the navigation route in testworld3. During navigation, the robot first moves along the original planned direction. When the corridor ahead is blocked, it turns into the adjacent corridor, follows the alternative passage to bypass the obstacle area, and then switches back toward the original target direction after passing the blocked section. This setting is used to evaluate whether the local planner can maintain safe and continuous navigation when the feasible corridor changes.

Each algorithm is tested three times in the two scenes under the same initial and target conditions. The compared methods include DWA, TEB, the initial algorithm and the improved algorithm. The main evaluation indices are success rate, average navigation time and average speed. All time-related metrics in this article are reported in decimal seconds and averaged arithmetically. The results are shown in TABLE III and TABLE IV.

TABLE III. TESTWORLD3 EXPERIMENTAL INDEX TABLE

Algorithm	***Success rate***	***Average travel time(s)***
CV	100%	117.99
CV FINAL	100%	72.866
DWA	0%	
TEB	66.7%	149.635

TABLE IV. TESTWORLD5 EXPERIMENTAL INDEX TABLE

Algorithm	*Average speed(m/s)*	*Average travel time(s)*
CV	0.428	45.232
CV FINAL	0.624	30.899
DWA	0.404	46.731
TEB	0.449	42.266

In the persistent blockage scene of testworld3, DWA fails in all trials, while TEB achieves a success rate of 66.7% with a long average navigation time of 149.635 s. Both the initial and improved algorithms complete all trials successfully. Compared with the initial algorithm, the improved algorithm reduces the average navigation time from 117.990 s to 72.866 s, showing better route recovery ability under blockage.

In the constrained static scene of testworld5, the improved algorithm also shows higher efficiency. Its average speed reaches 0.624 m/s, and the average navigation time is shortened to 30.899 s, outperforming DWA, TEB and the initial algorithm. Overall, the two tables indicate that the improved method maintains a higher success rate in difficult scenes and achieves better traversal efficiency through multi-candidate target scoring and speed regulation.

To further evaluate runtime performance, the computation time of each local planning cycle is recorded in ROS. The tested scenes are testworld5 and testworld3, and the control-cycle deadline is set to 100 ms. During each trial, the time consumed by one velocity-command computation is measured throughout the whole navigation process. “Fig. 13” shows the result.

Scene	Planner	Mean time (ms)	Max time (ms)	Deadline miss rate (%)
Testworld5	Proposed method	0.216	14.296	0.000
Testworld5	DWA	2.874	34.464	0.000
Testworld5	TEB	4.493	56.021	0.000
Testworld3	Proposed method	0.235	23.637	0.000
Testworld3	DWA	2.245	25.815	0.000
Testworld3	TEB	10.978	199.180	0.366

Fig. 13. Computation Overhead Comparison of Local Planners

As shown in “Fig. 13”, the mean time reflects the average per-cycle planning cost, the max time gives the largest observed single-cycle cost, and the deadline miss rate indicates the proportion of cycles exceeding the 100 ms deadline. The proposed method has the lowest mean time in both scenes, with 0.216 ms in testworld5 and 0.235 ms in testworld3. In contrast, DWA and TEB require higher average computation time, and TEB shows a 199.180 ms maximum time with a 0.366% miss rate in testworld3. These results show that the added candidate generation and scoring process remains lightweight and satisfies the real-time requirement of local planning.

V. CONCLUSION

This paper investigates local obstacle avoidance for mobile robots in complex indoor environments. An initial single look-ahead target tracking algorithm is developed, and an improved local planning method is further proposed by combining multi-candidate target scoring and velocity regulation. The improved method provides more local decision choices while maintaining consistency with the global path, which helps the robot achieve stable navigation in obstacle-clinging, narrow-passage, and persistent-blockage scenarios. Simulation results, parameter ablation, and runtime comparison with DWA and TEB show that the proposed method improves navigation adaptability while keeping low computational cost. Future work will include real-robot validation, adaptive parameter adjustment, dynamic obstacle prediction, and broader comparisons with more advanced local planning algorithms under diverse indoor scenarios.

REFERENCES

[1] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, Introduction to Autonomous Mobile Robots, 2nd ed. Cambridge, MA: MIT Press, 2011.

[2] S. M. LaValle, Planning Algorithms. Cambridge, U.K.: Cambridge University Press, 2006.

[3] O. Khatib, “Real-time obstacle avoidance for manipulators and mobile robots,” The International Journal of Robotics Research, vol. 5, no. 1, pp. 90–98, Spring 1986.

[4] J. Borenstein and Y. Koren, “The vector field histogram-fast obstacle avoidance for mobile robots,” IEEE Transactions on Robotics and Automation, vol. 7, no. 3, pp. 278–288, June 1991.

[5] D. Fox, W. Burgard, and S. Thrun, “The dynamic window approach to collision avoidance,” IEEE Robotics & Automation Magazine, vol. 4, no. 1, pp. 23–33, March 1997.

[6] C. Rösmann, W. Feiten, T. Wösch, F. Hoffmann, and T. Bertram, “Trajectory modification considering dynamic constraints of autonomous robots,” in Proc. ROBOTIK 2012, Munich, Germany, 2012, pp. 1–6.

[7] C. Wang, Y. Mei, Y. Zhang, and X. Liu, “Obstacle avoidance of mobile robots based on an improved DWA algorithm,” Machine Design and Research, vol. 40, no. 1, pp. 92–96, 2024

Research on Navigation Accuracy of Intelligent Mobile Robots During Movement

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***Abstract*—Intelligent mobile robots represent a specialized category of robotics that have established comprehensive assembly systems for flexible material handling across various industries. Serving as a model of intelligent modern logistics technology, these high-precision positioning logistics robots enable accurate material handling, enhance operational efficiency, and meet users' personalized requirements, thereby laying a solid foundation for the future development of intelligent mobile robot logistics systems. This paper focuses on "intelligent mobile logistics handling robots," employing a multi-sensor fusion approach to improve guidance and positioning accuracy, thereby advancing cargo transportation toward greater efficiency and intelligence.**

Keywords—Intelligent mobile robot, high precision, multi-sensor, guidance and positioning accuracy

I. INTRODUCTION

With the implementation of the "China Manufacturing 2025" national strategy, various industries are transitioning toward intelligent operations [1]. The logistics sector must also align with contemporary trends and increase its demand for smart logistics equipment. Traditional manual handling methods can no longer meet current operational requirements. Intelligent logistics handling solutions not only reduce costs but also optimize resource allocation. For smart mobile logistics robots, the primary functions are cargo loading, unloading, and transportation. Their high flexibility minimizes error rates, enhances work efficiency, reduces manual labor intensity, and advance toward an "intelligent + unmanned" model [2].

Fig. 1. Transportative embodied intelligent mobile robot

Funded by the "Humanoid Nanny-style Elderly Care Robot Based on Embodied Intelligence" project (derived from the "Elderly Care Technology Support" project of the 2025 High-quality Development Plan for Science and Technology Industry of the Shanghai Science and Technology Committee, project number: 25YL1901200).

Fig. 2. Service-oriented embodied intelligent mobile robot

II. DEFINITION AND TECHNOLOGY OF EMBODIED INTELLIGENT MOBILE ROBOTS

A. Definition and Characteristics of Embodied Intelligent Mobile Robots

In contemporary technological advancements, embodied intelligent mobile robots play a pivotal role. Equipped with electromagnetic or optical automatic guidance systems, they can autonomously navigate and perform various material transfer tasks under well-coordinated planning. Currently, these robots are most prevalent in manufacturing and logistics sectors. Their key features include unmanned operation, high flexibility, robust payload capacity, enhanced safety, high utilization efficiency, and compliance with environmental standards [3].

B. Introduction to Ambient Intelligent Mobile Robot Guidance Technology

Embodied intelligent mobile robots have been further advanced in the industrialization process due to their unique characteristics. Examples include automotive assembly lines, FMS systems, as well as applications in heavy machinery and specialized industries such as medical embodied intelligent mobile robot distribution systems and postal logistics sorting systems, where their functionality is more readily observable. Depending on the specific guidance route configuration of embodied intelligent mobile robots, it is essential to analyze both wired and wireless guidance methods based on practical requirements, ensure the fulfillment of predetermined tasks, and adhere to established construction standards. For detailed specifications, refer to Table 1 below. Upon receiving specific commands, the embodied intelligent mobile robot can operate effectively along its pre-set trajectory.

TABLE I. CLASSIFICATION OF NAVIGATION METHODS FOR MOBILE ROBOTS

classify	According to the guidance information source		In guided route format	
Guidance Method	Fixed path guidance outside the vehicle	Free Path Guidance	Wired	Wireless
	Electromagnetic Guidance	Coordinate Guidance	Electromagnetic Guidance	Ultrasound-guided
	Ultrasound-guided	Inertial guidance	Tape guiding	laser guidance
	Laser/Infrared Guidance	Self-guided	Color strip guidance	Coordinate Guidance
	Optical guidance		Grid Guidance	Inertial guidance
	Marked guidance		Marked guidance	Self-guided

III. LOGISTICS HANDLING PHYSICAL INTELLIGENT MOBILE ROBOT: ROBOT SYSTEM STRUCTURE

A. Overall Structure of the Embodied Intelligent Mobile Robot System

Under normal operating conditions, the vehicle must function properly within a specific environment with a flat road surface, primarily operating at low speeds, frequently stopping without experiencing jumps. It features a rigid body and optimal wheel configuration, with a reliable connection between the vehicle body and wheels to meet power requirements, ensure adequate power output for steering in intelligent mobile robots, and maximize reliability.

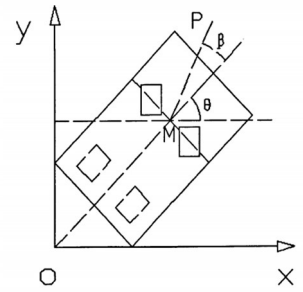

Fig. 3. Schematic diagram of motion analysis for an embodied intelligent mobile robot

The four-wheel configuration with dual-differential steering consists of an upper body and a lower body. It is primarily composed of multiple components including a stepper motor, drive unit, onboard control system, front wheels, wireless communication module, and a CD camera. The visual guidance system offers high flexibility, providing convenience while minimizing the impact of lighting conditions. Typically, the operation panel of an embodied intelligent mobile robot can be categorized as follows:

- Light alarm light.
- start button.
- scram button.
- It is composed of elements such as the display screen.

B. Composition and Principle of the Visual Guidance Physical Intelligent Mobile Robot Path Tracking System

First, from the perspective of visual applications in embodied intelligent mobile robots, humanoid visual recognition employs anthropomorphic visual bionic principles, implementing line image recognition technology to enhance the robot's emergency response capabilities. Second, the visual guidance pose discrimination principle requires tailored guidance strategies based on target requirements. Subsequently, under dynamic CCD camera acquisition, analog voltage signals are converted. After effective processing of the analog images, the vehicle's positional deviation for the next movement is determined. Based on this information, precise control commands are issued via computer and I/O interface units to achieve the intended travel path. Finally, regarding visual guidance image acquisition, routes are identified using an "electronic map," with the vehicle's exact position determined through multi-sensor perception [4].

C. Motion Control System for the Solid-High Motion Control Card-Based Embodied Intelligent Mobile Robot

Based on a standard PC, implement the user configuration system following the "PC + motion controller + servo motor" model, translating it into specific operational steps and motor actions to maximize precision and lay a solid foundation for future development. A DC servo motor is selected here; according to requirements, the maximum resistance F acting on the driving wheels during cart operation is given by F = S • FN. Detailed electrical parameters are provided in Table 2.

TABLE II. ELECTRICAL PARAMETERS

seat No	order number	power rating	rated speed	rated voltage	rated current	peak torque	armature resistance	Armature inductance
130SYX	02	1.5KW	3000RPM	48V	11.7A	40N*M	0.2Q	3.4MH

The rear wheel speed detection system primarily utilizes incremental optical encoders in coordinated integration with the GE-400.SV motion control card, enabling effective data feedback and ultimately achieving precise motion control for the embodied intelligent mobile robot. Both output and input components must be meticulously planned according to internal interconnections. The acceleration/deceleration control employs an S-curve mode, facilitating convenient modification of task position parameters. The embedded operating system demonstrates efficient resource utilization, ensuring real-time performance while enhancing system reliability. Additionally, the system maintains excellent openness and configurability. In practical implementation, two key components are executed: vehicle drive control and state parameter settings for traffic management. By leveraging digital maps and environmental conditions, the robot generates corresponding task commands to execute movements efficiently, maximizing automated control objectives. The control system adopts a multi-layer architecture for precise route navigation. Future modifications are designed for user-friendly operation. The robot model performs forward/retreat, left/right turns, and stop commands via wireless communication. Upon completion of initialization, the embodied intelligent mobile robot enters the main human-machine interaction interface, where users can perform

appropriate operations based on actual operational requirements. The robot's movement path is first determined by the system's electronic map to ensure route continuity, as illustrated in Figure 4. The system employs a network programming model—the client/server architecture—where one component serves as the server while the other functions as the client program. [5]

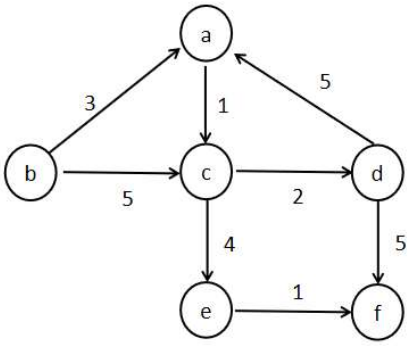

Fig. 4. Schematic diagram of the cart path

IV. Research on Positioning Precision Testing of An Embodied Intelligent Mobile Robot Vehicle for Guidance

Laboratory testing of the embodied intelligent mobile robot was conducted under zero-load conditions for the drive motor, first under left-turn conditions and then under right-turn conditions. Although errors occurred, these errors were inversely proportional to the motor's rotational speed and remained within a controllable range. Subsequently, during operation under the rated load condition, the robot underwent initialization: it moved forward initially followed by backward movement, maintaining the same inverse relationship. The maximum pulse count was 200, the maximum rotational speed was 6000 rpm, and the output current was 50 mA with short-circuit protection. Red line: +12 V; green line: OV; brown line: A; yellow line: B. By inputting the measurement values twice, the actual test distance for the robot's wheels could be calculated accordingly.

The trajectory data planned by the embodied intelligent mobile robot during its movement is transmitted to the evaluation system via the ROS system [5]. Real-time trajectory data measured by external motion capture devices is collected, and the trajectory planning data is compared with the actual travel trajectory data to assess its positioning accuracy. The specific experimental steps are as follows:

- Select a test site free of obstacles both indoors and outdoors, with five consecutive test points labeled A, B, C, D, and E, and ensure the distance between any two adjacent points exceeds 100 meters.
- The embodied intelligent mobile robot uses its own algorithm to plan the path trajectory and sequentially visits five points while maintaining a constant speed of 50% of its rated speed.
- Use high-precision RTK measurement equipment to determine its motion trajectory and positioning parameters at five fixed reference points.
- The computer analyzes and plots the motion trajectory of the RTK measurement data during movement, generating a trajectory graph.
- Analyze the motion trajectory graph and the planned path; use computer calculations to determine the overlap degree between the two trajectory curves and assess the corresponding overlap error

Define two trajectories: one planned trajectory and one actual motion trajectory, with lengths n and m respectively. The relationship between the planned trajectory and the actual trajectory is expressed by the following formula:

$$tr_1 =< p_1^1 \to p_1^2 \to \cdots p_1^n >$$

$$tr_2 =< p_2^1 \to p_2^2 \to \cdots p_2^m > \quad (1)$$

The Hausdorff distance is employed; it represents the maximum distance between the closest points of two trajectories. [6]

$$d_H(tr_1, tr_2) = max\{h(tr_1, tr_2), h(tr_1, tr_2)\} \quad (2)$$

The function $h(tr_1, tr_2)$ is defined as the unidirectional Hausdorff distance between tr_1, tr_2, with the following definition:

$$h(tr_1, tr_2)=\max_{p\in tr_1}\left\{\min_{q\in tr_2} d(p,q)\right\}$$

$$h(tr_2, tr_1)=\max_{p\in tr_2}\left\{\min_{q\in tr_1} d(p,q)\right\} \quad (3)$$

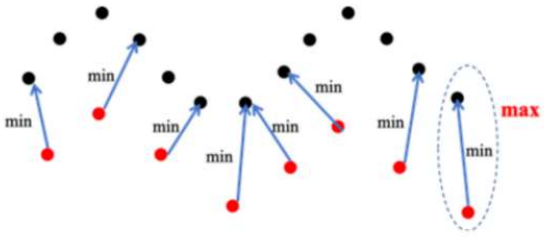

Fig. 5. Path Point

By comparing the trajectory data planned by the embodied intelligent mobile robot with the actual operational data, we can determine whether its accuracy meets the required positioning precision. The measurement results are evaluated using the RPG_evaluation metric to assess the positioning accuracy trajectory: blue represents the planned output trajectory, while green indicates the actual positioning trajectory, as shown in Figure 6 below.[7]

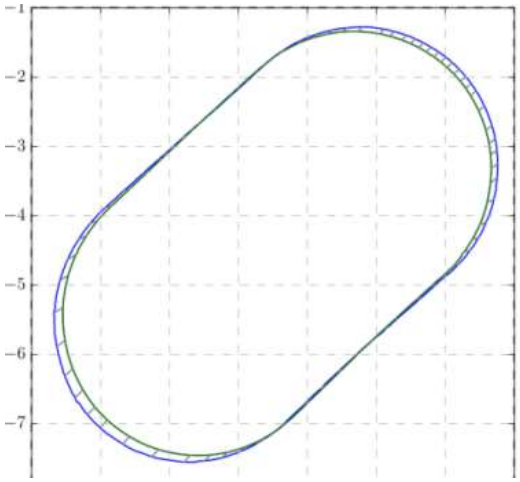

Fig. 6. Path trajectory of an embodied intelligent mobile robot

The trajectory positioning accuracy, as analyzed using software algorithms, meets the required precision standards, as shown in Figure 7 below.

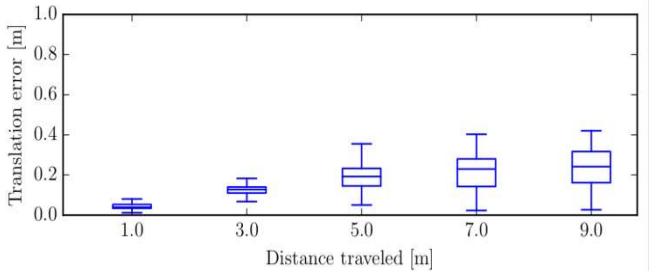

Fig. 7. Path positioning accuracy of an embodied intelligent mobile robot

Based on actual test data obtained under laboratory conditions, the intelligent mobile robot achieved an average positioning accuracy of 0.083 meters with a standard deviation of 0.012 meters during relatively constant-speed navigation, fully meeting the core requirement for sub-10 cm positioning accuracy in indoor navigation scenarios. Additionally, the measured Hausdorff distance reached a maximum value of 0.117 meters, further demonstrating that the robot's overall trajectory deviation remains within a low and controllable range, thereby confirming the robust stability of its positioning system.

V. Conclusion

In robot navigation and motion planning tasks, the Hausdorff distance can be employed to measure the similarity between the planned trajectory and the desired trajectory: if it falls below a predefined threshold, the planned trajectory is deemed sufficiently close to the target trajectory. Through analytical testing of navigation accuracy in embodied robots, this study demonstrates that such accuracy meets current application requirements.

With social development, embodied intelligent mobile robots have been widely adopted across various sectors in China. This paper focuses on robot navigation accuracy, analyzing the positioning precision of both intelligent mobile handling robots and intelligent mobile service robots. Such research can enhance overall robotic efficiency and reduce repetitive or inaccurate tasks. For logistics handling scenarios requiring high-precision positioning, research on positioning technologies for embodied intelligent mobile robots will become a key future direction, providing essential support for their subsequent design and development.

References

[1] Zhang Wang. Research on Local Path Planning Technology for Mobile Robots [D]. Nanjing University of Posts and Telecommunications, 2024. DOI: 10.27251/d.cnki.gnjdc. 2024.001472.Wu Weitao. Overall Solution and Key Technology Research of Mobile Robots for Logistics Handling [D]. Shenyang University of Technology.

[2] Nuyts, F. Autonomous delivery vehicles enable flexible assembly [J]. Modern Manufacturing, 2000,000(010):72-75.

[3] C.K. Wando, J.R. Schultz, Bram Young Jacobs Wanderflista, et al. Device and method for path determination using electronic maps: 2019.

[4] Xu Yaohui, Liu Zuoshi, Li Kaijie. Research on Multi-Point SLAM-Based Navigation Technology for Mobile Robots [J]. Sensors and Microsystems, 2026,45(4):7-11. DOI: 10.13873/J.1000-9787(2026)04-0007-05.

[5] Zhao Yaoyao. Research on the Interaction Interface Design of Hospital Drug Intelligent Delivery Robots [D]. Tianjin Polytechnic University, 2021. DOI: 10.27357/d.cnki.gtgyu. 2021.000978.

[6] Zhang B, Li G, Zhang J, et al. A reliable traversability learning method based on human-demonstrated risk cost mapping for mobile robots over uneven terrain[J]. Engineering Applications of Artificial Intelligence, 2024, 138(PA):109339-109339. DOI:10.1016/J.ENGAPPAI.2024.109339.

[7] Pan Chen. Research on visual SLAM algorithms for indoor dynamic scenarios [D]. Harbin Engineering University, 2025. DOI: 10.27060/d.cnki.ghbcu. 2025.000253.

Field Path Planning Based on an Improved A-star Algorithm and Local Window Replanning

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***Abstract*—Agricultural robots operating in unstructured farmland environments face challenges including difficulty in obtaining globally optimal paths under static conditions, high computational demand during dynamic obstacle avoidance, and limited battery capacity. To tackle these issues, a hybrid path planning framework is developed by combining an improved A-star algorithm, a collinearity-based keypoint extraction strategy, and a local window replanning mechanism. A multi-factor terrain cost model is constructed by incorporating soil softness, terrain slope, and water accumulation, together with direction-aware search constraints. In addition, a battery-adaptive dual-mode mechanism is designed to accommodate endurance limitations. Simulation experiments demonstrate that the proposed approach compresses key path nodes by 81.8%, reduces turning frequency by 66%, shortens local replanning time to 13.39 ms, and decreases return distance by 7.3% under low-power conditions. Overall, the method enhances path smoothness, computational efficiency, and operational reliability, providing a practical solution for agricultural robot navigation.**

Keywords—agricultural robot; A-star algorithm; keypoint extraction; local window replanning; hybrid path; battery-aware strategy.

I. Introduction

In autonomous robotic systems, path planning aims to generate a safe and efficient route between a given start position and a specified target while avoiding obstacles. Classical graph-search algorithms, such as Dijkstra's method, guarantee global optimality; however, their exhaustive exploration strategy often results in high computational complexity[1]. Sampling-based approaches, including RRT*, generate probabilistic search trees to explore feasible solutions, yet the resulting paths may exhibit limited smoothness and reduced efficiency in environments with strict constraints[2].

To enhance computational efficiency, heuristic-based graph search methods, especially A*[3] and its variant Theta*[4],have become prevalent in practical applications. Theta* introduces a line-of-sight (LOS) checking mechanism to generate any-angle paths, thereby improving path smoothness compared with the conventional A* algorithm.Nevertheless, this approach mainly emphasizes geometric optimality and rarely incorporates environmental risk factors or motion constraints specific to agricultural platforms.In contrast,this paper builds upon the A* framework by incorporating environmental cost modeling and direction-constrained search mechanisms to enhance adaptability in complex farmland environments. Recent studies have significantly enhanced the *A** algorithm by introducing five-neighborhood searches with dynamic weighting[5],utilizing rigorous maze-based mathematical models for dynamic route optimization[6], and employing bidirectional search combined with path smoothing techniques to reduce sharp turns and improve robustness[7].For dynamically changing environments, incremental heuristic algorithms have gained prominence. Algorithms like *D* Lite* [8]and Multi-Objective Path-Based *D* Lite*[9] reuse previous search information or prune dominated solutions, enabling robots to replan exponentially faster than searching from scratch.

Although substantial progress has been achieved, most heuristic planners derived from A* still prioritize distance optimality while giving limited consideration to terrain conditions and vehicle kinematics. In complex farmland environments, terrain conditions and nonholonomic constraints significantly affect robot performance. To overcome these limitations, the proposed method extends the conventional A* framework by integrating a multi-factor terrain cost model, direction-aware search constraints, and a localized replanning mechanism, thereby improving adaptability in complex farmland scenarios.

II. Farmland Environment Modeling

A. Grid-Based Environment Modeling

The farmland workspace is discretized using a rasterized grid representation. The field is modeled as a rectangular region characterized by length L and width W. Given grid resolution η, the area is divided into M and N grid cells along the length and width directions, respectively.

$$M = \frac{L}{\eta}, N = \frac{W}{\eta} \tag{1}$$

where L and W denote the length and width of the rectangular field, respectively, and η represents the grid resolution.

In the grid map, non-traversable regions correspond to crops or fixed obstacles and are set as forbidden nodes. Other regions are traversable and are assigned different traversal costs according to terrain and soil conditions, as shown in Fig. 1.

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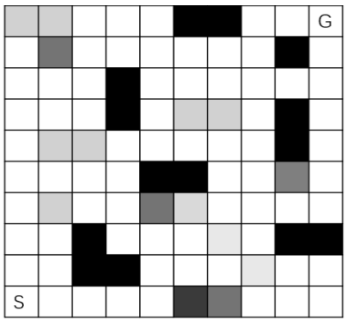


Fig. 1. Raster modeling diagram.

B. Construction of a Multi-Factor Environmental Cost Model

In existing path planning studies, traversable and obstacle regions are often manually defined, which may not reflect real road conditions. In this paper, the proposed multi-factor comprehensive cost model includes environmental factors with different sources, physical dimensions, and value ranges; thus, direct linear weighting is inappropriate. Therefore, soil softness, slope and water accumulation are uniformly normalized. For raw data $X(v)$ within $[X_{min}, X_{max}]$, the dimensionless normalized factor is defined as:

$$\overset{*}{X}(v) = \frac{X(v) - X_{\min}}{X_{\max} - X_{\min}} \tag{2}$$

where $X^*(v)$ denotes the normalized value of an environmental factor at grid node v; $X(v)$ is the original measured value of the factor; and X_{min} and X_{max} represent the minimum and maximum values within its range, respectively.

After normalization, each factor is mapped into [0, 1], where a larger value indicates poorer traversability. The weighted comprehensive environmental cost coefficient at node v is denoted as $M(v)$:

$$M(v) = \omega_{soil} \times S^*(v) + \omega_{slope} \times P^*(v) + \omega_{water} \times W^*(v) \tag{3}$$

where $\omega_{soil} = 0.4$, $\omega_{slope} = 0.3$, and $\omega_{water} = 0.3$ are empirically set weights; $S^*(v)$, $P^*(v)$, and $W^*(v)$ are the normalized soil softness, slope, and water accumulation values, and $\omega_{soil} + \omega_{slope} + \omega_{water} = 1$.

When the robot moves from the current grid node u to the next node v, the geometric distance is defined as:

$$Dist(u,v) = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2} \tag{4}$$

where x_u and y_u denote the coordinates of the current node, and $x_v$and y_v denote the coordinates of the next node.

By incorporating $M(v)$ into the distance-based cost, penalties are imposed for soft soil, steep slopes, and waterlogged areas, enabling avoidance of high-cost regions. The single-step cost is defined as:

$$\mathrm{Cos}t(u,v) = Dist(u,v) \times (1 + M(v)) \tag{5}$$

If grid v corresponds to crops or obstacles, Cost(u, v) is set to infinity so that the algorithm automatically avoids it and searches for the minimum-cost safe path in traversable regions.

III. Improvements to the Path Planning Algorithm

A. Improvements to the A* Algorithm

To meet farmland operational requirements, an improved A* algorithm combined with collinearity-based keypoint extraction is used to generate the globally optimal path based on the multi-factor cost model in Section II-B.

1) Improved Evaluation Function

To jointly consider operational efficiency and safety requirements, the evaluation function of A^* is reformulated as:

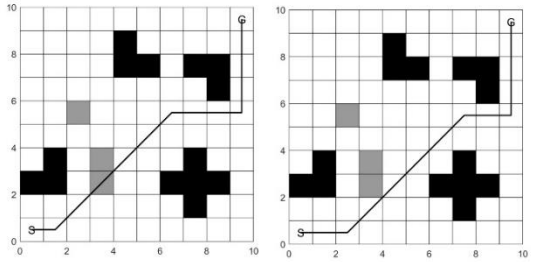
$$f(v) = g(v) + h(v) \tag{6}$$

where $g(v)$ is the accumulated actual cost from the start to node v, and $h(v)$ is the heuristic estimated cost from v to the goal G.

Let u be the current node and v be a neighboring node to be expanded. By incorporating the environmental risk factor $M(v)$, the accumulated cost is computed as:

$$g(v) = g(u) + Dist(u,v) \times [1 + M(v)] \tag{7}$$

This formulation jointly considers the geometric distance and environmental risk, enabling avoidance of high-risk regions. For the heuristic term h(v), the Euclidean distance is adopted to guide the search and improve efficiency, as illustrated in Fig. 2.



(a)standard A* (b) improved A*

Fig. 2. Multi-factor cost model

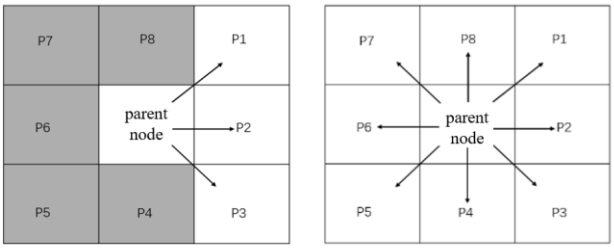
2) Direction-Constrained Grid Search Model

Direction constraints have been applied in mobile robot path planning studies.Traditional grid-based A^* uses an 8-neighborhood adjacency and allows free turning to any neighboring direction,ignoring the nonholonomic characteristics of agricultural robots. In this paper, the search state is extended to:

$$s = (x, y, \theta_k), \quad \theta_k \in \{0, 45^\circ, \ldots, 315^\circ\} \tag{8}$$

where x and y are grid coordinates and θ_k is the heading angle.

Each grid position thus has 8 discrete headings. For any state (x, y, θ_k), only three actions consistent with forward motion are permitted:(i) move straight forward one cell with heading θ_k;(ii) move forward with a small left turn, updating heading from θ_k to $\theta_{(k+1)}$; (iii) move forward with a small right turn, updating heading from θ_k to $\theta_{(k-1)}$This approximately captures the minimum turning radius constraint.



(a) 3-neighborhood search (b) 8-neighborhood search

Fig. 3. Direction constraint diagram.

3) Collinearity-Based Keypoint Extraction

In open areas, the *A** algorithm often produces paths consisting of many collinear nodes, which are redundant for control. Therefore, a post-processing redundancy removal strategy based on vector collinearity is introduced to retain only geometric turning points. Let the path node sequence output by *A** be $p = \{p_0, p_1, \ldots, p_n\}$, where p_0 and p_n are the start and goal.By traversing intermediate nodes and computing displacement vectors between adjacent nodes, collinear nodes are identified and removed:

$$\overrightarrow{v_1} = p_i - p_{i-1}, \quad \overrightarrow{v_2} = p_{i+1} - p_i \tag{9}$$

where p_i denotes the current node, $p_{(i-1)}$ denotes the previous node, and $p_{(i+1)}$ denotes the next node.

If v_1 and v_2 have the same direction：

$$\frac{y_i - y_{i-1}}{x_i - x_{i-1}} = \frac{y_{i+1} - y_i}{x_{i+1} - x_i} \tag{10}$$

where (x_i, y_i) represent the coordinates of the current node, $(x_{(i-1)}, y_{(i-1)})$ represent those of the previous node, and $(x_{(i+1)}, y_{(i+1)})$ represent those of the next node.

Then $p_{(i-1)}$, p_i, and $p_{(i+1)}$ are collinear and p_i is deemed a redundant node and removed. After pruning, the grid path is compressed into a key-node path consisting of the start, goal, and turning points.

B. Local Window Dynamic Replanning for Obstacle Avoidance

When environmental changes are frequent but localized, the global cost maintenance in *D* Lite* may still cause considerable computational overhead. Inspired by the local replanning concept of *D* Lite*, this paper introduces a local window dynamic replanning strategy on top of the static plan obtained by the improved *A** and keypoint extraction. Only a local region near obstacles is updated.

A local search window is defined by the robot's current position and the dynamic obstacle position, expanded by radius *r*. The window bounds in the *x* and *y* directions are:

$$\begin{aligned} x_{\min} &= \min(x_{\text{cur}}, x_{\text{obs}}) - r \\ x_{\max} &= \max(x_{\text{cur}}, x_{\text{obs}}) + r \\ y_{\min} &= \min(y_{\text{cur}}, y_{\text{obs}}) - r \\ y_{\max} &= \max(y_{\text{cur}}, y_{\text{obs}}) + r \end{aligned} \tag{11}$$

where x_{min}, x_{max}, y_{min}, and y_{max} define the boundaries of the local search window; r is the window expansion radius; (x_{cur}, y_{cur}) represent the current position; and(x_{obs}, y_{obs}) represent the position of the dynamic obstacle.

From the current position, the robot traverses forward along the original global path to find the last path point within the local window boundary, denoted as a temporary local goal p_{exit}. The current robot position is denoted as p_{cur}. The *A** search in Section III-A is reused to replan between p_{cur} and p_{exit} under the window constraint:

$$x_{\min} \le x_v \le x_{\max}, \quad y_{\min} \le y_v \le y_{\max} \tag{12}$$

where (x_v, y_v) represent the coordinates of the neighboring grid node.

The resulting local optimal path is then concatenated with the remaining segment of the original global path from p_{exit} to the final goal to form a new globally feasible path.

An adaptive window expansion strategy is adopted. The initial radius is set to $r = 2$. If no feasible path is found within the current window, r is increased by 1 up to a maximum of 4. If the maximum is reached and replanning still fails, a global replanning strategy is triggered.

IV. Battery-aware Dual-mode Path Planning and Workflow Design

A. Battery-Aware Dual-Mode Strategy

Agricultural robots are typically battery-powered. During long-duration field operations, if a detouring strategy is used under low battery conditions, the robot may fail to return. Based on the multi-factor environmental cost model, a battery-aware mechanism is introduced, and a dual-mode cost function is designed to adaptively switch with the battery state.

The normalized state of charge is defined as:

$$B_{soc} = \frac{B}{B_{\max}} \tag{13}$$

where *B* is the current battery energy, B_{max} is the maximum battery capacity, and B_{soc} denotes the battery state of charge SOC.

The cost function is defined as:

$$\text{Cost}(u,v) = \begin{cases} \text{Dist}(u,v), & B_{\text{soc}} < B_{\text{th}} \\ \text{Dist}(u,v)\left[1 + M(v)\right], & B_{soc} \ge B_{\text{th}} \end{cases} \tag{14}$$

where B_{th} represents the battery state-of-charge threshold.

When $B_{soc} \ge B_{th}$, the multi-factor cost model is used to guide the path away from muddy and steep regions. When $B_{soc} < B_{th}$, only the geometric distance is considered to reduce travel distance and extra energy consumption.

B. Overall Workflow

The system first builds a grid-based representation of the farmland environment, integrates the multi-factor cost model and the battery-aware mechanism, and improves the A* algorithm. After a complete global plan is generated and

compressed, local window replanning is performed for dynamic obstacle avoidance. The overall process is shown in Fig. 4.

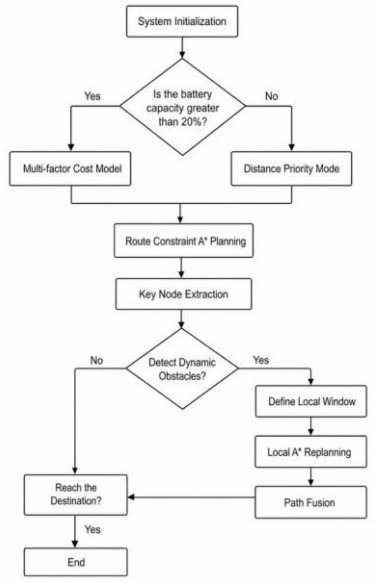


Fig. 4. System Flow Diagram.

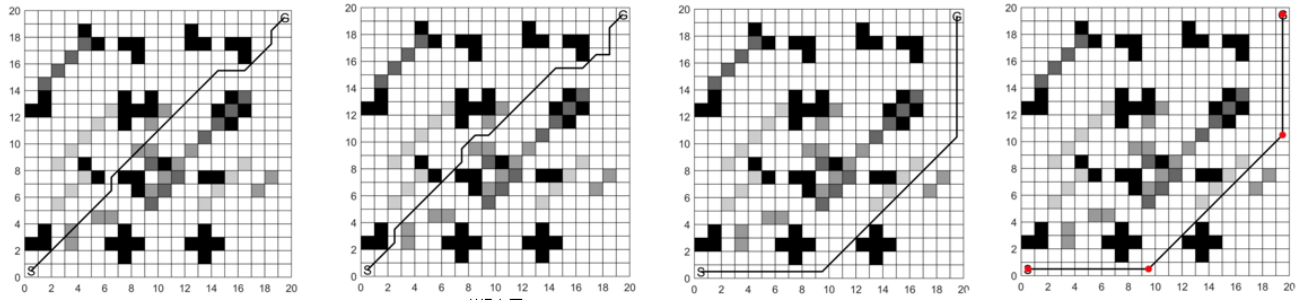
V. Experiments and Analysis

Simulations were conducted in MATLAB R2022a on a 20×20 grid map. Three experiments were designed: (i) a static global planning comprehensive comparison, (ii) a battery-aware dual-mode experiment, (iii) a local window replanning experiment. Black cells denote obstacles or crops; gray cells with different intensities indicate regions such as soft soil; and white cells are traversable. Each grid cell is assumed to be 1 m in side length. The start S is set at the lower-left corner and the goal G at the upper-right corner.

In the simulations, soil softness, terrain slope, and water accumulation were modeled using graded assignments. Each factor was divided into three risk levels (low/medium/high) with values 0.2, 0.5, and 0.8, respectively; a larger value indicates poorer traversability.

A. Static Global Planning Comprehensive Comparison

To evaluate the comprehensive performance in static environments, four methods were tested under the same start/goal settings: traditional *A** (Algorithm a), standard multi-factor *A** (Algorithm b), direction-constrained multi-factor *A** (Algorithm c), and the hybrid method with keypoint extraction (Algorithm d). The results are shown in Fig. 5 and Table I.



(a)Algorithm a (b)Algorithm b (c)Algorithm c (d)Algorithm d

Fig. 5. Static Planning Experimental Path.

TABLE I. Static Planning Results

Algorithm	Path Length (m)	Turns	Key Nodes	Time (ms)	Expanded Nodes
Algorithm A	28.04	6	22	10.73	82
Algorithm B	28.62	10	23	3.80	86
Algorithm C	32.14	2	29	12.36	226
Algorithm D	32.14	2	4	13.12	226

As shown in Fig. 5 and Table I,different improvement strategies have a significant impact.Traditional *A** aims at the shortest distance and yields the shortest path length,but it does not consider environmental risk or kinematic constraints, leading to frequent turns.Introducing the multi-factor environmental cost enables avoidance of high-risk areas such as soft soil and waterlogged regions; however, multiple polyline turns may occur where local costs vary.

The direction-constrained search model incorporates heading into the search state and restricts turning behaviors. The resulting path better matches the limited turning radius of agricultural robots: the number of turns is reduced from 6 to 2, and path smoothness is improved. A slightly longer path is traded for better executability and operational safety. After applying keypoint extraction, the number of path nodes is compressed by 81.8% without changing the geometric shape, substantially reducing storage and control computation burdens.

B. Battery-Aware Dual-Mode Experiment

To address limited endurance, this section validates the proposed battery-aware dual-mode switching strategy. The battery threshold was set to 20%. In high-battery mode, the multi-factor cost model is used; in low-battery mode, a distance-priority model is adopted. Under the same 20×20 map and direction constraints, high- and low-battery behaviors were simulated. Results are shown in Fig. 6 and Table II.

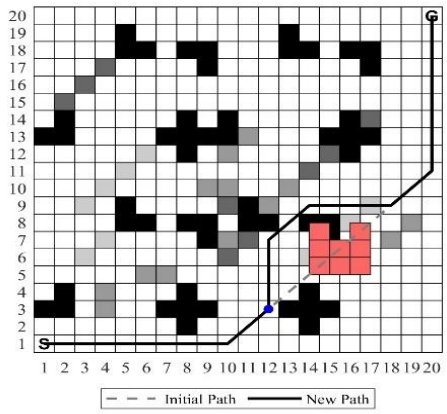


Fig. 6. Battery-Aware Experimental Path.

TABLE II. BATTERY LEVEL EXPERIMENT DATA

Mode	Path Length (m)	Turns	Key Nodes	Time (ms)	Expanded Nodes
High-Energy	32.14	2	29	6.10	226
Low-Energy	29.79	3	25	5.22	235

The results indicate that the algorithm can dynamically adjust planning strategies based on remaining battery. With sufficient battery, planning prioritizes the comprehensive environmental cost to actively avoid high-risk regions, and the number of turns is small, which benefits stable operation. When the battery drops below the threshold, the cost switches to a distance-priority mode, generating a path closer to the shortest geometric route. The path length is reduced by 7.3%, effectively lowering energy consumption during return. The node expansion count and planning time change only slightly, indicating that the mode switching does not introduce additional computational burden. Overall, the mechanism achieves a reasonable trade-off between safety and endurance, improving operation continuity in energy-limited scenarios.

C. Local Window Replanning Experiment

To verify responsiveness under contingencies, dynamic obstacles were introduced based on the global plan. When the robot reached the 12th node, a group of dynamic obstacles was generated to block the original path, triggering the local replanning mechanism with an initial window radius of $r = 2$. The results are shown in Fig. 7 and Table III.

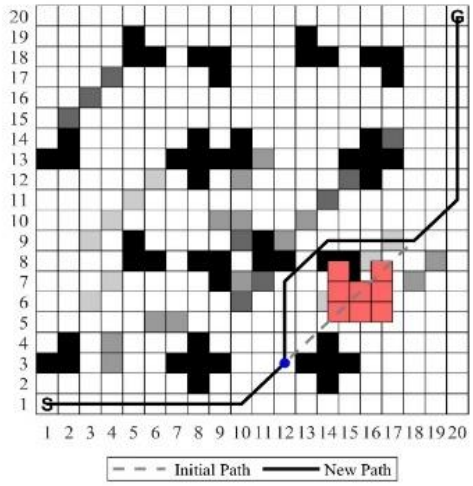


Fig. 7. Path replanning experiment.

TABLE III. DYNAMIC OBSTACLE RESULTS

Scenario	Path Length (m)	Time (ms)	Expanded Nodes
Static Global Path	32.14	348.71	223
Replanned Path	34.45	13.39	37

As shown in Table III, after a dynamic obstacle appears, the algorithm avoids recomputing the global path and instead performs local search only in the affected region. Compared with global planning, the number of expanded nodes is reduced to 16.6% of the original, and the replanning time is shortened to 13.39 ms, significantly reducing computational cost.

These results demonstrate that the local window mechanism effectively limits the search space so that the computational complexity of dynamic avoidance matches the obstacle's impact range. The replanned path is smoothly concatenated with the original global path at the window boundary, without introducing extra turns or abrupt changes, ensuring continuity

and executability. This strategy is suitable for dynamic field operation scenarios where obstacle locations are uncertain but the influence range is limited.

VI. Conclusion

This study develops and validates an integrated path planning framework tailored to agricultural robots operating in complex farmland environments. Comparative experiments demonstrate measurable improvements over the traditional A* algorithm, including reduced turning frequency, fewer path nodes, shorter local replanning time, and decreased return distance under low battery conditions, enabling stable task execution in complex farmland environments.

Future research will focus on real-world validation using physical agricultural platforms and integrating multi-sensor perception as well as task-level decision modules to enhance practical deployment capability. By incorporating multi-sensor perception and considering vehicle posture as well as tool coordination constraints, higher-level decision-making modules such as task planning and crop-yield maps will be integrated to improve practicality in large-scale and long-term agricultural production scenarios.

References

[1] H. Wang, Y. Yu, and Q. Yuan, “Application of Dijkstra algorithm in robot path-planning,” in Proc. 2011 2nd Int. Conf. Mechanic Automation and Control Engineering, 2011, pp. 1067–1069.

[2] I. Noreen, A. Khan, and Z. Habib, “Optimal path planning using RRT* based approaches: A survey and future directions,” International Journal of Advanced Computer Science and Applications, vol. 7, no. 11, pp. 97–107, 2016.

[3] P. E. Hart, N. J. Nilsson and B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," in IEEE Transactions on Systems Science and Cybernetics, vol. 4, no. 2, pp. 100-107, July 1968.

[4] K. Daniel, A. Nash, S. Koenig, and A. Felner, “Theta*: Any-angle path planning on grids,” J. Artif. Intell. Res., vol. 39, pp. 533–579, 2010.

[5] C. Han and B. Li, “Mobile robot path planning based on improved A* algorithm” in Proc. 2023 IEEE 11th Joint Int. Information Technology and Artificial Intelligence Conf. (ITAIC), 2023, pp. 672–676.

[6] Y. Huang, Y. Wang, and O. Zatarain, “Dynamic path optimization for robot route planning” in Proc. 2019 IEEE 18th Int. Conf. Cognitive Informatics & Cognitive Computing (ICCICC), 2019, pp. 47–53.

[7] H. Wang, S. Lou, J. Jing, et al., “The EBS-A* algorithm: An improved A* algorithm for path planning” PLOS ONE, vol. 17, no. 2, p. e0263841, 2022.

[8] W. Yue, J. Franco, W. Cao, et al., “ID* Lite: Improved D* Lite algorithm” in Proc. 2011 ACM Symp. Applied Computing, 2011, pp. 1364–1369.

[9] Z. Ren, S. Rathinam, M. Likhachev, et al., “Multi-objective path-based D* lite” IEEE Robotics and Automation Letters, vol. 7, no. 2, pp. 3318–3325, 2022.

Dual-Cone Generation Method and Simulation for Hob Rake Face Applied to Robot Jointsr

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***Abstract*—Planar double-enveloping toroidal worm drives are ideal transmission schemes for heavy-duty industrial robot joint reducers due to their multi-tooth contact and high load-bearing characteristics. To facilitate manufacturing, toroidal worm gear hobs typically employ straight flutes. However, this structure often leads to excessive negative rake angles on the cutting edges of multi-start hobs, severely degrading cutting performance and the manufacturing accuracy of robot joints[3]. To address this, this paper proposes a precise forming method for the rake face of helical flutes based on double-cone grinding wheel grinding. First, based on gear meshing theory, a mathematical model is established for generating the helical rake face using a double-cone generating surface, and the formula for calculating the rake angle at any point on the cutting edge is derived. Second, comparative calculations demonstrate that the helical flutes generated by this method optimize the rake angle distribution. The range is improved from −19.5° to 19.5° (straight flutes) to −8.1° to 7.3°, significantly reducing the absolute value of the negative rake angle. Finally, a specialized CNC machine model is constructed in the VERICUT environment for machining simulation and measurement. The measurement results align with theoretical calculations, validating the effectiveness of the proposed method in improving the manufacturing quality of tools for robot joint reducers.**

Keywords—Robot Joint; Hourglass Worm Gear Hob; Dual-Cone Generation; Rake Face; VERICUT Simulation

I. INTRODUCTION

With the rapid development of industrial robots, the performance requirements for their core component—the joint reducer—are continuously increasing. In heavy-duty applications, the planar double-enveloping hourglass worm drive has become an ideal solution for robot joint transmission due to its multi-tooth contact and high load-bearing characteristics[1]. However, the transmission accuracy of the reducer is highly dependent on the manufacturing quality of the worm gear pair, and the hourglass worm gear hob is precisely the key tool for machining high-precision worm gears[2].

For multi-thread hourglass worms commonly used in robot joints, their helix angle is relatively large[3]. If traditional hobs with straight flutes are used, it leads to excessively large negative rake angles on one side of the cutting edge[4]. Research indicates that large negative rake angles cause a sharp increase in cutting force and accelerated tool wear, severely affecting the tooth surface accuracy of the worm gear and, consequently, limiting the performance of the robot reducer[5].

Yang Jie et al. investigated the method for machining the rake face of an hourglass worm gear hob with helical chip flutes using a cylindrical generating surface[6]. S.L. Chang employed a method using a disc milling cutter to machine the rake face of a cylindrical hob, which improved machining efficiency and is suitable for the rough machining of helical chip flutes[7]. Machining the rake face of a hob with straight chip flutes using a plane grinding wheel is feasible. However, when grinding the rake face of a helical chip flute, a very pronounced central bulge occurs, and the problem of interference between the grinding wheel plane and the tool tooth is prone to arise[8]. To avoid the aforementioned issues, Liu Jiehua employed a plane grinding wheel with a modified axial profile to grind the helical chip flute rake face of a gear hob[9]. For cylindrical hobs and gear hobs, using a single grinding wheel axial profile is sufficient to grind the helical chip flute rake face[10] However, for an hourglass worm gear hob, the cutting edge curves on both sides of each tooth, as well as the helix angle at each point on these curves relative to the helical line of the hob's basic worm tooth surface, are all different. Machining each tooth would require a form grinding wheel with a different axial profile, which is very difficult to achieve in actual production[11].

To address the aforementioned issues, this paper proposes a precise modeling method for generating a helical rake face based on a dual-cone surface grinding wheel. By establishing the meshing model between the generating surface and the hob, the rake face equation and the calculation formula for the rake angle are derived, and the correctness of the method is verified in the VERICUT virtual simulation environment. This research provides a theoretical basis for enhancing the manufacturing level of specialized tools for domestically produced robot joint reducers.

II. MATHEMATICAL MODELING FOR MACHINING RAKE FACE USING DUAL-CONE GENERATING SURFACE

A. Machining Principle and Coordinate System Establishment

In order to machine ideal spiral flutes on a multi-start hourglass worm gear hob, this paper proposes a generating grinding method using a double-cone grinding wheel. The basic worm helical surface Σ_1 of the hob is obtained through a single

enveloping process with plane Σ_d as the generating surface; whereas the rake face Σ_3 of the hob is obtained through a single enveloping process with double-cone surface Σ_s as the generating surface. When machining the rake face, a right-handed Cartesian coordinate system is established as shown in Fig.1.

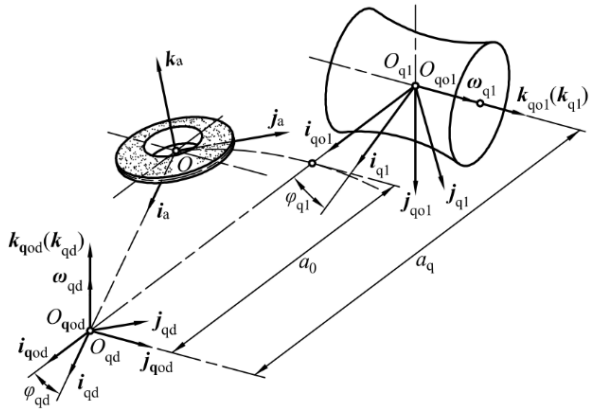

Fig. 1. Coordinate system for machining the rake face

Establish the right-handed Cartesian coordinate system shown in Fig.1. Take the fixed coordinate system $\sigma_{qo1}(O_{qo1};i_{qo1};j_{qo1};k_{qo1})$ attached to the machine frame, and the moving coordinate system $\sigma_{q1}(O_{q1};i_{q1};j_{q1};k_{q1})$ attached to the hob's basic worm. O_{q1} is the coordinate origin, O_{qo1} and O_{q1} coincide, and $k_{qo1}=k_{q1}$ coincides with the rotation axis of the hob's basic worm.

Take another fixed coordinate system, denoted as $\sigma_{qod}(O_{qod};i_{qod};j_{qod};k_{qod})$ attached to the machine frame, where $i_{qod}=i_{qo1}$, $k_{qod}=-j_{qo1}$. Take the moving coordinate system $\sigma_{qd}(O_{qd};i_{qd};j_{qd};k_{qd})$ attached to the tool post, where $k_{qd}=k_{qod}$, k_{qd} coincides with the tool post axis. The distance from the tool post axis center O_{qod} to point O is a_0.

B. Coordinate Transformation and Dual-Cone Generating Surface Equation

The coordinate transformation is performed using rotation matrices, for example, from σ_a to σ_{qd}, denoted as $\sigma_a \rightarrow \sigma_{qd}$. When machining the hob rake face, the following coordinate transformation rotation matrices are used:

$$S_a \rightarrow S_{qd}: R\left[i_{qd}, -\left(90^\circ - \beta_a\right)\right] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin\beta_a & \cos\beta_a \\ 0 & -\cos\beta_a & \sin\beta_a \end{bmatrix}$$

$$\sigma_{qd} \rightarrow \sigma_{qod}: R\left[k_{qod}, \varphi_{qd}\right] = \begin{bmatrix} \cos\varphi_{qd} & -\sin\varphi_{qd} & 0 \\ \sin\varphi_{qd} & \cos\varphi_{qd} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sigma_{qod} \rightarrow \sigma_{qo1}: R[i_{qo1}, 90^\circ] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\sigma_{qo1} \rightarrow \sigma_{q1}: R\left[k_{q1}, -\varphi_{q1}\right] = \begin{bmatrix} \cos\varphi_{q1} & \sin\varphi_{q1} & 0 \\ -\sin\varphi_{q1} & \cos\varphi_{q1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

When machining the rake face, the speed ratio between the tool post and the hob's basic worm is:

$$i_{qd1} = -\frac{i_{1d} \cdot {r_2}^2}{{r_1}^2} \tag{1}$$

r_1 is the pitch circle radius at the throat of the hob's basic worm, r_2 is the pitch arc radius of the hob. When machining the basic worm of the hob, its blank is fixed to σ_{q1}. The equation of the hob's basic worm helical surface in σ_{q1} is given by (2).

$$\begin{cases} (r_d)_{q1} = r_d(u,v,\varphi_d) \\ u = u(v,\varphi_d) \end{cases} \tag{2}$$

As shown in Fig.2, the dual-cone generating surface Σ_{qd} for generating the hob's rake face is located to the right of the grinding wheel's mid-plane. The coordinate system $\sigma_a(o;i_a,j_a,k_a)$ is fixed to the grinding wheel; k_a coincides with the grinding wheel axis, and i_a and j_a lie on the grinding wheel's mid-plane.

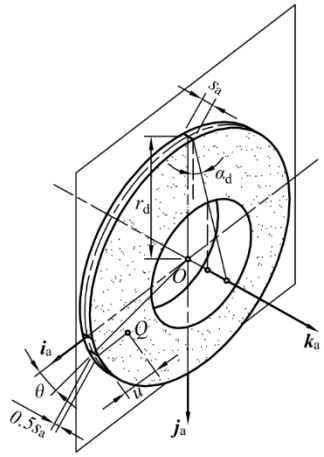

Fig. 2. Dual-Cone generating surface

Using the circular vector function and the spherical vector function, the vector expression of the right generating conical surface Σ_{qd} in σ_a is:

$$(r_a)_a = r_d e_a(\theta) + 0.5 s_a k_a - u n_a(\theta, \alpha_d) \tag{3}$$

The unit normal vector at any point Q on the right generating conical surface Σ_{qd} is:

$$(n)_a = m_a(\theta, \alpha_d) \tag{4}$$

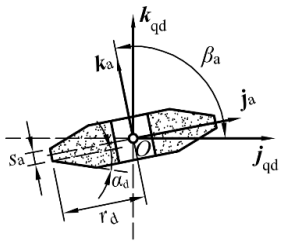

Fig. 3. The Position of the Dual-Cone Grinding Wheel in the Tool Post's Fixed Coordinate System

The position of the grinding wheel on the tool post is shown in Fig.3. By rotating the generating conical surface counterclockwise around the i_{qd} axis by an angle β_a, and then performing the following four consecutive coordinate transformations on (3): $\sigma_a \rightarrow \sigma_s$, $\sigma_s \rightarrow \sigma_{os}$, $\sigma_{os} \rightarrow \sigma_{o3}$, and $\sigma_{o3} \rightarrow \sigma_3$, the vector expression of the generating conical surface Σ_{qd} in the moving coordinate system σ_{q1} is obtained as:

$$\begin{aligned}(r_s)_3 &= r_s(u_s, \theta_s, \varphi_3) \\ &= R[k_3, -\varphi_3]\{R[i_{o3}, 90°]R[k_{os}, \varphi_3]\{-a_0 i_s \\ &\quad + R[i_s, -(90° - \beta_s)](r_a)_a\} + a i_{o3}\} \\ &= [Au_s + D, Bu_s + E, Cu_s + F]^T \end{aligned} \tag{5}$$

where $A = -cos\alpha_d cos\theta$

$$B = -\cos\alpha_d \sin\beta_a \sin\theta + \sin\alpha_d \cos\beta_a$$

$$C = \cos\alpha_d \cos\beta_a \sin\theta + \sin\alpha_d \sin\beta_a$$

$$D = r_d \cos\theta - a_0$$

$$E = r_d \sin\beta_a \sin\theta + 0.5 s_a \cos\beta_a$$

$$F = -r_d \cos\beta_a \sin\theta + 0.5 s_a \sin\beta_a$$

Performing the coordinate transformations $\sigma_a \rightarrow \sigma_{qd}$, $\sigma_{qd} \rightarrow \sigma_{qod}$, $\sigma_{qod} \rightarrow \sigma_{qo1}$, and $\sigma_{qo1} \rightarrow \sigma_{q1}$ on (4), the unit normal vector at any point Q on the right generating conical surface Σ_{qd} in the moving coordinate system σ_{q1} is obtained as:

$$\begin{aligned}(n_{qd})_{q1} &= abcdm_a(\theta, \alpha_d) \\ &= \left[n_{xqd}, n_{yqd}, n_{zqd}\right]^T \end{aligned} \tag{6}$$

where $a = R\left[k_{q1}, -\varphi_{q1}\right]$

$$b = R\left[i_{qo1}, 90°\right]$$

$$c = R\left[k_{qod}, \varphi_{qd}\right]$$

$$d = R\left[i_{qd}, -(90° - \beta_a)\right]$$

$$n_{xqd} = -sin\alpha_d cos\theta$$

$$n_{yqd} = -sin\alpha_d sin\theta sin\beta_a - cos\alpha_d cos\beta_a$$

$$n_{zqd} = sin\alpha_d sin\theta cos\beta_a - cos\alpha_d sin\beta_a$$

C. The Equation of the Hob Rake Face in σ_{q1}

In the σ_{qod} coordinate system, the relative velocity between the dual-cone generating surface and the rake face is $(V_{qd1})_{qod}$ Based on gear meshing principles, the conjugate condition function for the dual-cone generating surface and the rake face is:

$$\emptyset_{qod} = \left(V_{qd1}\right)_{qod} \cdot \left(n_{q1}\right)_{qod} \tag{7}$$

From $\phi_{qod} = 0$, the first enveloping conjugate condition equation for the dual-cone generating surface is obtained as:

$$u = \frac{H(\theta, \varphi_{qd})}{G(\theta, \varphi_{qd})} \tag{8}$$

By combining (5) and (8), the equation of the hob rake face Σ_2 in σ_{q1} can be obtained as:

$$\begin{cases} \left(r_{qd}\right)_{q1} = r_{qd}(u, \theta, \varphi_{qd}) \\ u = \dfrac{H(\theta, \varphi_{qd})}{G(\theta, \varphi_{qd})} \end{cases} \tag{9}$$

D. Expression for the Rake Angle V_q

By combining the (2) for the hob's basic worm helical surface Σ_1 and (9) for the rake face Σ_2 into a system of equations, the cutting edge line of the hob tooth can be obtained.

In the σ_{q1} coordinate system, the expression for the rake angle V_q at any point on the cutting edge line is obtained from the normal vector $(n_{qd})_{q1}$ on the rake face and the normal vector $(n_d)_{q1}$ on the helical surface at that point, as follows:

$$V_q = \cos^{-1}\left[(n_{qd})_{q1} \cdot (n_d)_{q1}\right] - 90° \tag{10}$$

where $\cos^{-1}\left[(n_{qd})_{q1} \cdot (n_d)_{q1}\right]$ is the included angle W_q at that point.

III. Case Study Analysis and VERICUT Simulation Verification

A. Case Study and Results Analysis

This paper addresses the case study provided in Table 1. By varying the values of each parameter of the dual-cone grinding wheel to calculate the corresponding rake angles, a set of relatively reasonable dual-cone grinding wheel parameters is selected. The selected parameters for the dual-cone grinding wheel are as follows: dual-cone grinding wheel radius $r_d = 50mm$, Dual-cone grinding wheel tip width $s_a = 6mm$, dual-cone grinding wheel profile angle $\alpha_d = 15°$, and dual-cone grinding wheel axis inclination angle $\beta_a = 116.5°$. The rake angles for both straight-flute and helical-flute hobs are solved separately. Setting the rotation angle of the hob's basic worm about its axis $\varphi_{q1} = 0$ and the speed ratio with the tool post $i_{qd1} = 0$, the corresponding rake angles of the straight-flute hob on the pitch toroidal surface from the exit to the entrance are obtained as shown in Table 2. When the speed ratio $i_{qd1} = -1.45$, the corresponding rake angles of the helical-flute hob on the pitch toroidal surface from the exit to the entrance are obtained as shown in Table 3.

TABLE I. Basic Parameters of the Planar Double-Enveloping Hourglass Worm Gear Hob and the Dual-Cone Grinding Wheel

No.	Name	Symbol	Value
1	Center distance	a/mm	160
2	Transmission ratio	i_{12}	10
3	Number of hob threads	z_1	4
4	Worm gear pitch circle pressure angle	$\alpha/°$	21.8667
5	Worm gear pitch circle diameter	d_2/mm	255
6	Worm pitch circle diameter	d_1/mm	65

7	Hob tooth tip arc radius	R_{a1}/mm	122.24
8	Hob tooth root arc radius	R_{f1}/mm	133.24
9	Hob working length	L_w/mm	90
10	Lead angle at hob throat pitch circle	$\gamma_m/°$	21.4205
11	Inclination angle of generating plane axis	$\beta/°$	22.5
12	Master base circle diamete	d_b/mm	95

TABLE II. RAKE ANGLES ON THE LEFT AND RIGHT SIDES OF EACH TOOTH ON THE PITCH TOROIDAL SURFACE CORRESPONDING TO THE STRAIGHT FLUTE RAKE FACE

Rake Angle Type	*Tooth 1*	*Tooth 2*	*Tooth 3*	*Tooth 4*	*Tooth 5*
Left Side Rake Angle/(°)	-15.72	-18.19	-19.53	-19.01	-17.01
Axial Position$Z_i/(mm)$	-43.5	-24,5	-4.71	15.4	35.8
Right Side Rake Angle/(°)	16.99	19.01	19.53	18.19	15.72
Axial Position$Z_i/(mm)$	-35.8	-15.4	4.71	24.5	43.5

TABLE III. RAKE ANGLES ON THE LEFT AND RIGHT SIDES OF EACH TOOTH ON THE PITCH TOROIDAL SURFACE CORRESPONDING TO THE HELICAL FLUTE RAKE FACE

Rake Angle Type	*Tooth 1*	*Tooth 2*	*Tooth 3*	*Tooth 4*	*Tooth 5*
Left Side Rake Angle/(°)	5.56	6.66	7.25	1.62	-1.34
Axial Position$Z_i/(mm)$	-37.66	-21.21	-4.18	13.35	31.15
Right Side Rake Angle/(°)	-5.48	3.06	6.2	0.75	-8.11
Axial Position$Z_i/(mm)$	-30.61	-13.32	3.99	21.42	38.35

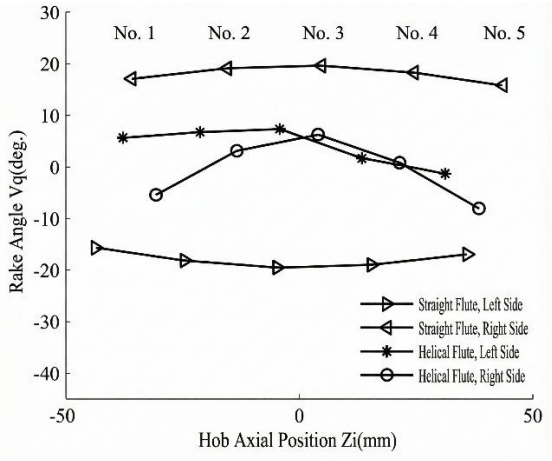

Fig. 4. The change of rake angles on both sides

The calculation results show that the rake angle distribution of the straight-flute hob is extremely uneven (approximately ±20°). The large negative rake angles and left-right asymmetry result in poor cutting conditions. In contrast, the helical-flute hob generated by the dual-cone generation method proposed in this paper reduces the peak rake angle to ±8.1° (a reduction of approximately 60%), and the left and right rake angles of the core throat teeth tend to be balanced. This optimization significantly improves the cutting force state, making it more suitable for machining high-precision robot worm gears.

B. VERICUT Simulation Machining

Taking the 4-thread hob from Table 1 as an example, a dedicated CNC machine tool for hourglass worms with B-axis and C-axis linkage was established in the VERICUT software. The simulation machining of the hob's basic worm helical surface and the hob's helical chip flute was then carried out.

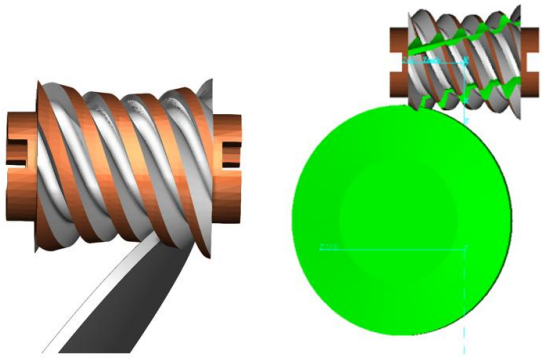

Fig. 5. Simulated Machining of Hob Helical Surface and Chip Flute

After the simulated machining of the helical flute hob is completed, the included angle W_q on both sides of the tool teeth on the pitch toroidal surface is measured, as shown in Fig.6.

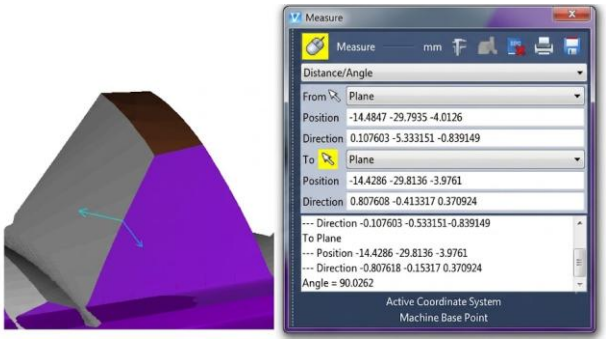

Fig. 6. Measuring the Included Angle

From $V_q = W_q - 90°$, the rake angle V_q on both sides of the tool teeth is calculated, as shown in Table 4.

TABLE IV. MEASURING THE RAKE ANGLES ON THE LEFT AND RIGHT SIDES OF EACH TOOTH ON THE PITCH TOROIDAL SURFACE CORRESPONDING TO THE HELICAL-FLUTE RAKE FACE.

Rake Angle Type	*Tooth 1*	*Tooth 2*	*Tooth 3*	*Tooth 4*	*Tooth 5*
Left Side Rake Angle/(°)	5.64	6.3	7.13	1.61	-1.04
Axial Position$Z_i/(mm)$	-37.67	-21.21	-4.18	13.35	31.14

Right Side Rake Angle/(°)	-5.38	2.51	5.64	0.17	-8.31
Axial PositionZ_i/ (mm)	-30.6	-13.32	3.99	21.42	38.35

Comparing the measured results in Table 4 with the theoretically calculated results in Table 3 shows that the measurements contain errors. The maximum error occurs at the right-side rake angle of tooth No. 4, which is 0.5699°. The minimum error occurs at the left-side rake angle of tooth No. 4, which is 0.0114°. The measurement errors are primarily caused by inaccuracies in the selection of measurement point locations.

IV. CONCLUSION

To address the issues of the large absolute value of the negative rake angle and the unbalanced cutting conditions on the left and right sides of the tool teeth in straight-flute hobs, a method for grinding the helical flute rake face of an hourglass worm gear hob using a dual-cone grinding wheel is proposed.

By establishing a grinding coordinate system, a mathematical model for generating the helical chip flute rake face of a planar double-enveloping hourglass worm gear hob using a dual-cone generating surface is derived. This allows for the calculation of the rake angle value for each tooth along the pitch toroidal helical line. The calculation results indicate that, compared to the straight chip flute hob, the corresponding rake angle of the helical chip flute hob is reduced by 60%, effectively decreasing the absolute value of the negative rake angle and facilitating balanced cutting conditions on both sides of the tool teeth.

The helical chip flute of the hob is simulated and machined, and the rake angles are measured within the simulation software. The measurement results show an error of less than 0.6° compared to the theoretical calculations, verifying the correctness of the design method for the rake face of the hourglass worm gear hob.

ACKNOWLEDGMENT

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REFERENCES

[1] DONG X Z. Design and modification of enveloping worm gearing[M].Beijing: Mechanical Industry Press, 2004:14-15(in Chinese).

[2] W. Luo, Y. Chen, and G. Zhang, "Helix error detection and tracing on tooth surface of planar enveloping toroidal worm," *J. Southwest Jiaotong Univ.*, vol. 50, no. 2, pp. 279–285, Apr. 2015. (in Chinese)

[3] ZHOU L Y. The modification principle and manufacturing technology of hourglass worm [M]. Changsha: National University of Defense Technology press, 2005:558-559(in Chinese).

[4] YANG J, LI H T, RUI C J. Research on a generating method of spiral flutes of hourglass worm gear hob[C]// Proceedings of the ASME Design Engineering Technical Conference. American Society of Mechanical Engineers (ASME) press, 2017,10.

[5] GUNAY M. Investigation of the effect of rake angle on main cutting force [J]. International Journal of Machine Tools & Manufacture, 2004 (44):953-959.

[6] YANG J, LI H T, RUI C J, DONG X Z. Design and correct method of spiral flutes rake face of an hourglass worm gear hob [J]. Journal of Beijing University of Aeronautics and Astronautics.

[7] CHANG S L. Helix gash of hob cutter manufactured by milling [J]. Journal of Materials Processing Technology, 2003(142): 569-575.

[8] Shu Q Y. Design and research of dual-cone doublehourglass worm gear pair and its key manufacturing equipment [D]. Zhejiang: Zhejiang University, 2015:21-30(in Chinese).

[9] Liu J (1996) Dressing calculation of the profile of the grinding wheel on the rake face of the hob with zero rake angle. Grinding machine and grinding 04:25-26+72.

[10] CHEN J, LIU F L, KANG Q. An optimization method for hob relief grinding[J]. China Mechanical Engineering, 2014,25(14):1883-1887(in Chinese).

[11] F.L. Litvin. Theory of Gearing[M]. NASA Reference Publication, 1989.

Experimental Investigation on Hydrodynamic Interactions in a Tandem Formation of Bionic Manta Ray Robots

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***Abstract*—This study experimentally examines hydrodynamic interactions between two bionic manta ray robots in tandem. Thrust, input power, and wake flow were measured in a recirculating water channel while longitudinal spacing and relative flapping phase were varied. A three-stage protocol comprising an isolated-robot baseline, static coupling, and fully dynamic coupling was used to isolate interaction effects. Relative phase governed whether coupling was beneficial or detrimental, whereas spacing mainly controlled its magnitude. Under anti-phase flapping, both robots obtained hydrodynamic benefits over 0.2–0.9 BL, and the follower achieved nearly 40% thrust gain at Dx < 0.2 BL. In contrast, in-phase flapping at Dx = 0.1 BL caused about 32% thrust loss for the follower. PIV results show that beneficial interactions arise from co-rotating vortex coalescence and enhanced jet momentum, whereas detrimental interactions result from vortex collision and breakdown. These findings provide experimental guidance for energy-efficient formation control of manta-ray-inspired robots.**

***Keywords*—Robotic manta ray, Tandem formation, Vortex interaction, Hydrodynamic experiment.**

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I. INTRODUCTION

Underwater biomimetic robots—particularly those inspired by the locomotion mechanisms of manta rays—have garnered significant interest for applications in marine exploration, environmental monitoring, and defense, owing to their superior maneuverability, propulsive efficiency, and acoustic concealment [1]–[3]. However, the operational capability of an isolated robot remains insufficient for complex oceanic tasks. Inspired by the remarkable energy-saving cohesion observed in natural fish schools [4]–[6], employing multi-robot formations to harness hydrodynamic interactions for enhanced collective endurance, speed, and payload capacity has emerged as a pivotal research direction.

The physical principle underlying this collective benefit is interactive flow-field coupling. In nature, trailing fish synchronize their body kinematics with the vortex wake shed by leaders, extracting energy from or reducing drag against oncoming flows [7], [8]. Experimental evidence demonstrates energy savings of up to 79% for schooling zebrafish in turbulent flows [9], leading to the proposition of mechanisms such as "turbulence sheltering" [10]. This bio-inspiration sets a clear objective for robotic collectives: strategically exploiting or mitigating flow-field disturbances to optimize group performance. Consequently, research into the hydrodynamic mechanisms of formation swimming has advanced on multiple fronts. Fundamental studies on isolated manta-ray-inspired propulsors indicate that thrust generation is primarily governed by tip and leading-edge vortices, whose structures are highly sensitive to kinematic parameters [11]–[13]. Furthermore, a body of high-fidelity numerical studies on Multiple Flapping Foils (MFF) has elucidated the intrinsic mechanisms of

formation-based hydrodynamic coupling. For tandem configurations, the relative flapping phase difference and longitudinal spacing are identified as the two key parameters dictating performance. Precise phase adjustment can shift wake interactions from detrimental vortex collision to beneficial vortex coalescence, markedly enhancing both thrust and efficiency [14], [15]. Research has since expanded to three-dimensional configurations and multi-agent systems [15]–[17].

Despite these advances, a critical gap persists between mechanistic insights and their application to real-world robotic systems. The majority of existing studies rely on numerical simulations or experiments utilizing idealized biological profiles and simplified two-dimensional geometries. While invaluable for elucidating fundamental physics, these approaches inevitably abstract away critical physical details inherent to practical robotic platforms, such as complex three-dimensional morphology, distributed structural compliance, and the nonlinear dynamics of multi-servo actuation systems. Consequently, the vortex dynamics and mutual interference generated by physical robots likely differ substantially from predictions based on smoothed or rigid models. There is a notable scarcity of controlled experimental studies that systematically quantify the hydrodynamic coupling between authentic manta-ray-inspired robotic entities, specifically regarding the simultaneous effects of spacing and phase. This lack of empirical data hinders the translation of fluid dynamic principles into a reliable foundation for guiding practical formation control strategies. To bridge this gap, this study presents a comprehensive experimental investigation into the hydrodynamic interactions between two autonomous bionic manta ray robots swimming in tandem. Using a custom-developed robotic platform in a recirculating water channel, we simultaneously measure the thrust, power consumption, and wake vortex topologies of both robots under systematically varied longitudinal spacings and relative flapping phase differences, integrating a six-axis force/torque sensor with Particle Image Velocimetry (PIV). A comparative experimental protocol—comprising a single-robot baseline, static coupling, and fully dynamic coupling—is employed to accurately isolate pure dynamic interaction effects. The objectives of this work are threefold: (1) to experimentally reveal the quantitative laws governing how spacing and phase modulate hydrodynamic coupling in a physical robotic system; (2) to verify and refine existing vortex-based interaction mechanisms (e.g., coalescence vs. collision) via direct flow visualization; and (3) to provide foundational experimental data and physical insights for the development of high-precision, flow-aware formation control algorithms for biomimetic underwater collectives.

II. Design Of The Bio-inspired Manta Ray Robot

A. Mechanical Design of the Bio-inspired Manta Ray Robot

This bio-inspired manta ray robot platform adheres to the bionic principle of integrating "morphology-motion-function". It is designed to replicate the highly efficient propulsion mechanism of manta rays, thereby providing a controllable experimental platform for subsequent research on collective hydrodynamic interactions. Based on morphological data obtained from CT scans of real manta rays (Fig. 1a), the platform adopts a modular architecture. Utilizing integrated 3D printing technology, it integrates a rigid main body, bionic flexible pectoral fins, a caudal fin, and a distributed actuation system (Fig. 1b, c).

The pectoral fin serves as the core actuator for efficient propulsion and employs a structure-function gradient design to mimic the mechanical distribution of biological fins, transitioning from high stiffness at the root to high compliance at the tip. At its core is an anisotropic flexible skeleton (Fig. 1c). By incorporating specific grooves in the chordwise direction and enhancing the flexibility of the leading edge and fin tip, the fin achieves controllable passive bending in both the spanwise and chordwise directions. This skeleton is encapsulated with a super-elastic silicone rubber skin to form the bionic pectoral fin, enabling it to simulate the complex three-dimensional large deformations of a biological fin surface during motion.

The platform is driven by five waterproof servo motors, four of which are symmetrically arranged to independently drive the bilateral pectoral fins. A planar five-bar mechanism serves as the core transmission device (Fig. 1e). This mechanism converts the rotational input from two driving links into a specific trajectory at the fin connection point through linkage coupling.

B. Control System Design of the Bio-inspired Manta Ray Robot

To realize efficient and coordinated bio-inspired motion of the pectoral fins and precisely study the hydrodynamic coupling during dual-robot collaboration, this study follows the method proposed by Cao et al. [18], constructing a control system based on an improved Central Pattern Generator (CPG). The CPG method is widely used in bio-inspired robot control due to its inherent advantages in generating rhythmic motion patterns. The CPG network adopted in this study is constructed based on a nonlinear oscillator model, capable of producing smooth, stable periodic signals and effectively overcoming the limitations of traditional trajectory tracking methods in dynamic adjustment.

As shown in Fig. 1e, f, the output signal of the CPG controller is mapped to the rotational input of the two driving links (Link 1 and Link 5) of the five-bar mechanism. The variation of their output angles over time can be expressed as:

$$\begin{cases} \theta_1 = A_1 \sin(\omega_1 t + \varphi_1) + X_1 \\ \theta_2 = A_2 \sin(\omega_2 t + \varphi_2) + X_2 \end{cases} \quad (1)$$

where $\omega_1 = \omega_2 = 2\pi f$, ensuring the synchronized fundamental frequency of bilateral motion; A_1 , A_2 are the flapping amplitudes; φ_1, φ_2 are the phase angles. When the two inputs are in-phase, the pectoral fin primarily generates pure flapping perpendicular to the body axis; when a specific phase difference $\Delta\varphi_{ij}$ is introduced, the mechanism drives the pectoral fin to produce a periodic twisting-oscillating motion with changes in the angle of attack. This control system provides a reliable and precise experimental foundation for the subsequent systematic study of the coupled effects of spacing and relative phase on hydrodynamic interactions.

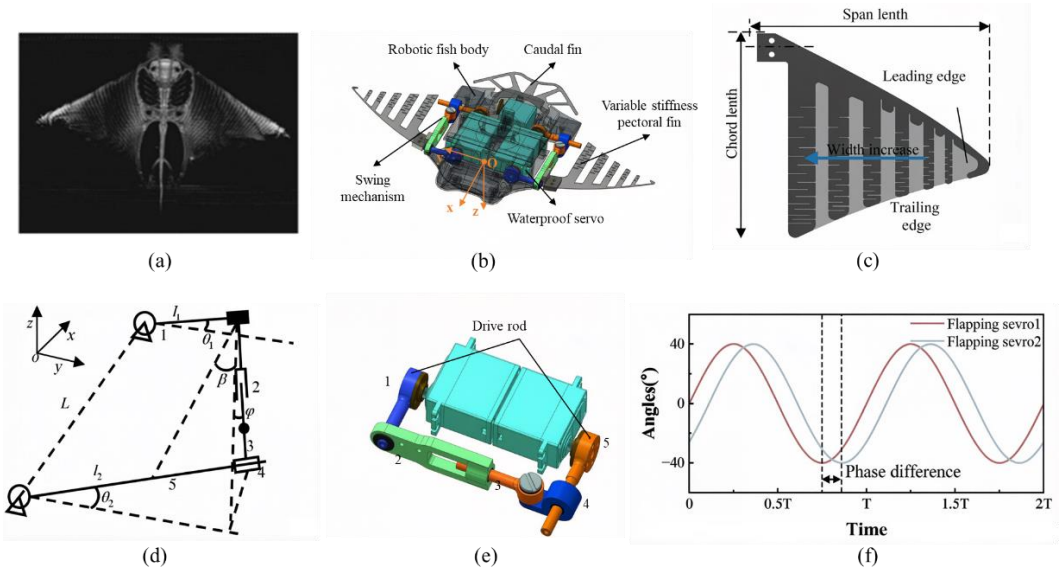


Fig. 1. Design of the bio-inspired manta ray robot prototype. (a) CT scan of the ray skeleton; (b) Structure of the bio-inspired manta ray robot; (c) Design of the flexible pectoral fin; (d) Kinematic diagram of the 2-degree-of-freedom (2-DOF) mechanism; (e) Schematic of the 2-DOF driving mechanism; (f) Sinusoidal locomotion generated by the 2-DOF drive mechanism

III. Hydrodynamic Experimental Platform

A. Integrated Hydrodynamic Testing Platform for Manta-Ray-Inspired Robotic Fish

To quantify propulsion and inter-robot flow interactions, an integrated platform was built, combining a low-turbulence water channel, force measurement, and flow visualization (Fig. 2). Two robotic fish (Fig. 3) were tested in a tandem configuration (aligned along the streamwise X-axis) to study longitudinal interactions. To minimize wall-interference effects, the prototypes are mounted in the center of the channel's test section. Two six-axis force/torque sensors, mounted via rigid adapters on a custom frame (Fig. 3), measured instantaneous forces on the leading and following robots, while a synchronized DAQ system acquired thrust and power data. Furthermore, a 2D PIV system captured the flow field, utilizing a laser sheet to illuminate the mid-plane (X-Z) and synchronized high-speed cameras recording at 400 Hz to analyze vortex dynamics in the wake. The integrated platform enables synchronized, high-fidelity acquisition of both quantitative force data and qualitative flow-field structures for the tandem robotic fish, providing a reliable experimental basis for systematically elucidating the hydrodynamic mechanisms governed by spacing and phase.

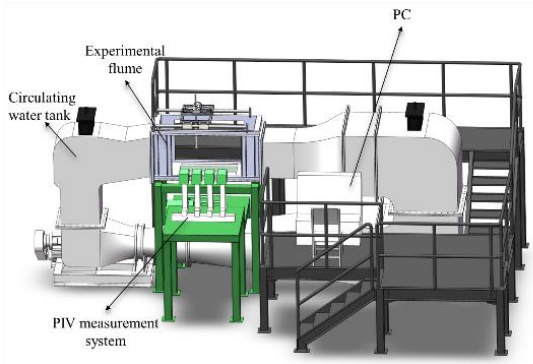


Fig. 2. Schematic of the hydrodynamic experimental platform.

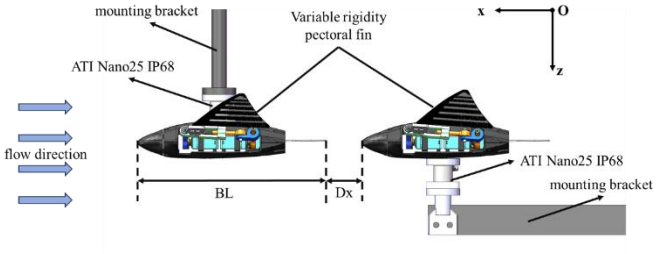


Fig. 3. Experimental setup of the robotic swarm and definition of formation parameters.

B. Experimental Design for Collective Hydrodynamics

To systematically study the hydrodynamic interactions between two tandem robotic fish, controlled experiments were conducted to quantify the coupled effects of longitudinal spacing (Dx) and relative flapping phase on thrust, power, and flow fields.

1) Kinematic Parameters: A high-efficiency propulsion state was used: Strouhal number St=1.3, wave number W=0.4, and non-dimensional flapping amplitude A=0.7. This generates a wake with distinct vortex structures, providing a basis for studying vortex interactions[11]. Experiments employed a tandem configuration (Fig. 3), with Dx(distance from leader's trailing edge to follower's leading edge) varied from 0.1 to 1.0 body lengths (BL). Both in-phase and anti-phase motions were tested. Additionally, we define φ as the motion phase difference between the front and rear fish.

2) Experimental Protocol: A "single-robot baseline—static coupling—fully dynamic coupling" protocol was designed to isolate the purely dynamic interaction. This involved measuring:

a) Baseline Value: baseline thrust/power (F_base, P_base) for an isolated robot.

b) Static Interference: static coupling forces (F_static,leader, F_static,follower) with one robot stationary to quantify wake sheltering and pressure rebound effects.

c) Fully Dynamic Coupling: total forces (F_total, P_total) for both robots moving synchronously.

Based on this, the purely dynamic interaction force and power can be obtained by differential calculation:

$$\begin{cases} \Delta F_{dynamic,i} = F_{total,i} - F_{static,i} \\ \Delta P_{dynamic,i} = P_{total,i} - P_{static,i} \end{cases} \quad (2)$$

where $i \in \{leader, follower\}$.

Hydrodynamic Performance Evaluation: Propulsive performance is evaluated using the propulsive efficiency (η) based on the Froude efficiency definition, calculated as follows:

$$\eta_i = \frac{\bar{F}_i v_{tip}}{P_i} = \frac{\bar{F}_i 2\pi fr}{P_i} \quad (3)$$

where $\bar{F}$ is the average thrust over one flapping cycle, v_{tip} is the fin tip speed, r is the distance from the fin tip to the flapping axis, and $P_i$is the average power consumption over one flapping cycle, with $i \in \{leader, follower\}$. By comparing the baseline efficiency of a single robot with the efficiencies under different coupling conditions, the gains and losses due to hydrodynamic interactions can be quantitatively assessed. This comparative experimental protocol, by separating static background interference from dynamic interactions, provides a reliable data foundation for subsequent quantitative analysis of the mechanisms by which spacing and phase affect the hydrodynamic coupling between the two robots.

IV. Analysis of Results

A. Hydrodynamic Performance with Varying Streamwise Spacing and Phase Difference

Under in-phase motion, the longitudinal spacing significantly modulates the system's unsteady hydrodynamic response, revealing the asymmetric roles between the leader and follower in flow coupling (Fig. 4a). Both vehicles exhibit significant spacing-dependent thrust gains accompanied by asymmetric oscillations about the zero line, indicating the spatial selectivity of energy transfer and vortex interactions.

As the primary disturbance recipient, the follower shows intense thrust fluctuations. At minimal spacing (Dx ≈ 0.1 BL), it experiences about 32% thrust loss. The gain turns positive as spacing increases, peaking at Dx ≈ 0.6 BL (~29% gain) with a secondary peak near 0.9 BL (~18%). This suggests moderate spacing improves the follower's force environment, though gains decay with further spacing increase. Conversely, the leader—as the primary vortex generator—shows smaller thrust fluctuations (max gain ≈10%) and greater stability. Its faint multi-peak thrust curve implies detectable reverse modulation from the follower, confirming the in-phase energy transfer is predominantly leader-to-follower. Notably, within Dx ≈ 0.4–0.7 BL, both robots gain thrust simultaneously, indicating a synergistic window. Propulsion efficiency analysis (Fig. 4b) quantifies this synergy. The follower's efficiency mirrors its thrust response, forming a "high-efficiency zone" where it notably exceeds the single-robot baseline (e.g., Dx ≈ 0.5–0.6 BL). The leader's efficiency remains slightly but consistently above baseline at most spacings (Dx > 0.2 BL), showing it maintains effective self-propulsion while providing a favorable wake for the follower. Thus, optimizing spacing in in-phase mode can simultaneously improve thrust and efficiency within a specific range, showing energy-saving potential.

Under anti-phase motion (φ=π), spacing modulation shows distinct features. While a multi-peak structure persists, the interaction exhibits stronger constructive interference with fundamentally changed modulation.

The follower's response shifts qualitatively to sustained net positive gain across all spacings—a hallmark "all-distance positive" characteristic (Fig. 4a). Its maximum gain migrates to very close spacing (Dx < 0.2 BL), reaching nearly 40%, highlighting that thrust gain is strongly modulated by phase, not just spacing.

The leader also shows a clear multi-peak response, reaching its global maximum (~18% gain) at Dx ≈ 0.5 BL, significantly surpassing its in-phase performance. This confirms the follower provides positive hydrodynamic feedback, breaking the unidirectional energy transfer. The spatial lag between the leader's and follower's optimal spacings further reveals inherent asymmetry in their response mechanisms.

The propulsion efficiency map shows both robots operate above the single-robot baseline across a broad spacing range (Dx ≈ 0.3 – 1 BL), forming a "high-efficiency plateau" (Fig. 4b). Compared to in-phase mode, anti-phase coupling offers superior peak efficiency and a wider efficient operating range, establishing relative phase as the core variable for regulating formation energy efficiency.

B. Analysis of Vortex Structures

To delve deeply into the wake interference mechanisms of the two tandem manta-ray-inspired robotic fish, this study employed PIV (Particle Image Velocimetry) technology to capture instantaneous flow fields in the X-Z plane, systematically analyzing the coupled effects of center spacing (Dx) and phase difference (φ) on thrust generation and propulsive efficiency.

Figure 5 focuses on comparing the wake of a single robot with the wake of the leader under the anti-phase condition with Dx=0.5 BL. As shown in Fig. 5a, in the single-body motion mode, the manta-ray-inspired robot generates two dominant pairs of counter-rotating vortices through the up-and-down flapping of its pectoral fins, accompanied by a strong axial jet flow. This jet, while being expelled backward, produces a forward reaction force constituting the main source of thrust. When a tandem pair is introduced and set to anti-phase motion (Dx=0.5 BL, Fig. 5b), the vortex pairs in the leader's wake are subjected to the induced velocity field of the follower's leading-edge vortex, resulting in a notable phenomenon of vortex attraction. This fluid dynamic coupling effect causes the axis of the leader's jet to deflect, aligning more closely with the direction of the robots' motion, thereby reducing energy dissipation sideways. Macroscopically, this manifests as a significant improvement in propulsive efficiency.

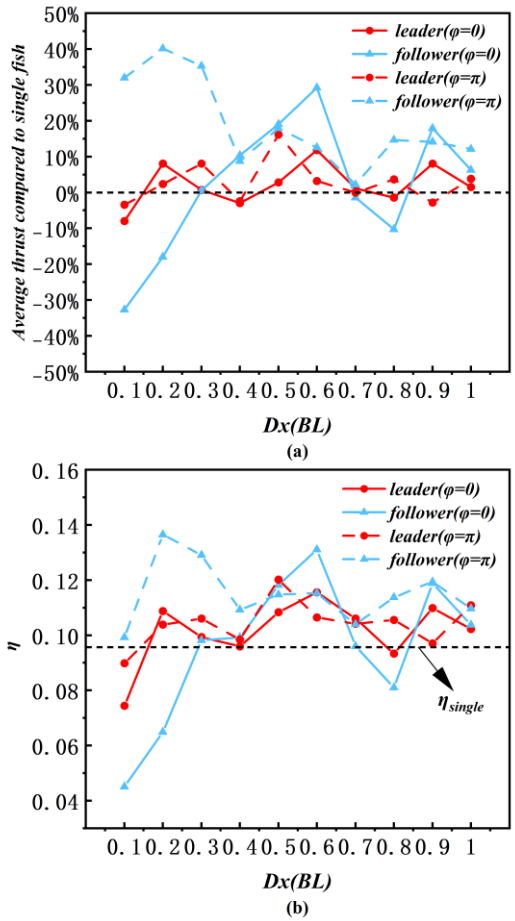
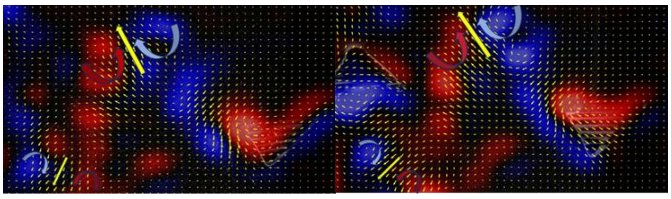


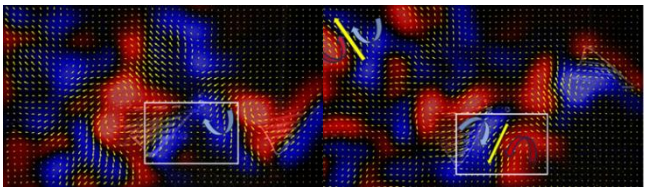
Fig. 4. Propulsion performance of manta rays in tandem formation: (a) Average thrust gain relative to the single-robot baseline, (b) Propulsive efficiency.



(a) (b)

Fig. 5. Comparative vortex structures generated by flapping motions: (a) single robot; (b) leading robot in a tandem configuration (Dx = 0.6 BL).

Figure 6 further reveals the differences in vortex dynamics underlying the extreme points of propulsive efficiency, comparing the worst-case (Dx=0.1 BL, in-phase) and optimal (Dx=0.2 BL, anti-phase) conditions. As shown in Fig. 6a, at close spacing with in-phase motion, the vortex rings shed from the leader impact and sweep directly over the follower's pectoral fin surface during their early development. This severe flow disturbance leads to an unsteady breakdown of the leader's vortex structures, preventing the establishment of a coherent low-pressure region and an effective jet in the wake, causing the leader's thrust to drop sharply. Simultaneously, the follower's own vortex generation is inhibited by the turbulent wake of the leader, failing to form an ordered thrust jet. The mechanism of destructive interference under this condition is the fundamental reason for the overall system efficiency falling below that of a single robot, as seen in Fig. 4. In contrast, under the optimal condition shown in Fig. 6b (Dx=0.2 BL, anti-phase), a typical constructive interference phenomenon is observed. The jet vortex pairs formed by the leader (particularly the lower blue vortex) and the leading-edge vortex (LEV) being generated by the follower in space undergo coalescence of co-rotating vortices. This merging process not only acts to "suction and compress" the leader's wake in terms of mass and vorticity, thereby strengthening the jet momentum, but the resulting stronger vortex structure also significantly accelerates the formation and accumulation of the follower's leading-edge vortex. Consequently, the jet velocity and momentum flux generated by the follower far exceed those in the single-body state. However, it is noteworthy that despite benefiting from jet enhancement, the leader experiences weakened direct gain from its own wake due to the follower's upward flapping motion inducing a lift of the upper vortex structures. Therefore, under this condition, the follower obtains a higher thrust gain, reflected in its superior propulsive efficiency compared to the leader.



(a) (b)

Fig. 6. Comparison of vortex structures under conditions of worst and best propulsive efficiency: (a) Dx = 0.1 BL, in-phase; (b) Dx = 0.2 BL, anti-phase.

V. Conclusion

This study systematically investigated the hydrodynamic coupling effects within closely spaced tandem formations of robotic manta rays. Using an integrated experimental platform combining high-precision six-axis force/torque measurements with Particle Image Velocimetry (PIV), we quantitatively evaluated the coupled effects of longitudinal spacing (Dx) and relative flapping phase difference (φ), revealing the underlying vortex dynamics. The principal conclusions are as follows:

1) Relative phase dictates the fundamental nature of the interaction, with the anti-phase mode yielding significant synergistic advantages. Our findings demonstrate that phase is the primary determinant of hydrodynamic coupling quality. Under anti-phase motion ($\varphi = \pi$), the system exhibits robust constructive interference across a broad range of spacings (Dx ≈ 0.2–0.9 BL), simultaneously enhancing thrust and efficiency for both robots. Notably, the follower achieved a nearly 40% thrust increase at minimal spacing (Dx < 0.2 BL). Conversely, in-phase interactions ($\varphi = 0$) are weaker and highly unstable, inducing destructive interference at Dx ≈ 0.1 BL that causes the follower to lose approximately 32% of its thrust.

2) Longitudinal spacing nonlinearly modulates interaction intensity, with the optimal spacing being highly phase-dependent. The specific spacing required to maximize performance gains shifts drastically based on the phase condition. In the in-phase mode, the follower attains its maximum thrust gain (~29%) at Dx ≈ 0.6 BL. In stark contrast,

under the anti-phase mode, the follower's peak gain (~40%) migrates to an extremely close proximity (Dx < 0.2 BL), while the leader's maximum gain (~18%) occurs at Dx ≈ 0.4 BL. This spatial discrepancy underscores the asymmetric roles the leader and follower play during flow-field energy transfer.

3) Vortex interaction mechanisms provide the physical basis for the observed macroscopic performance. PIV flow-field analysis offers a definitive mechanistic explanation for the quantitative trends. Constructive interference is driven by the "coalescence of co-rotating vortices," a merging process that strengthens jet momentum and directly correlates with the high thrust gains observed (18% and 40% for the leader and follower, respectively). Destructive interference, conversely, results from "direct vortex collision and breakdown," which destabilizes the ordered wake and accelerates energy dissipation (corresponding to the follower's 32% thrust penalty). This establishes a clear mechanistic link transitioning from kinematic parameters to flow-field topologies, and ultimately to macroscopic performance.

By utilizing a physical bionic system, this research comprehensively establishes the core principle that relative phase dictates the quality of hydrodynamic interaction while spacing modulates its magnitude. Elucidating these physical mechanisms through flow visualization represents a crucial step toward translating bio-inspired hydrodynamic theory into practical, cooperative control strategies for robotic swarms.

VI. Future Work

While this study focused on a planar tandem configuration, future research will expand to three-dimensional formations and multi-agent systems to explore more complex hydrodynamic interaction networks. A critical next step involves integrating these empirically derived quantitative laws and flow mechanisms into collective motion models, paving the way for hydrodynamic model-predictive controllers capable of orchestrating efficient autonomous formations.

Acknowledgment

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References

[1] A. Liu, C. Xing, W. Cao, B. Liu, S. Wang, J. Jin et al., "Bio-inspired intermittent locomotion: a novel energy-saving strategy for manta ray-inspired underwater vehicles," Ocean Eng., vol. 341, p. 122425, Dec. 2025.

[2] C. Xing, Z. Yin, H. Xu, Y. Cao, Y. Qu, Q. Huang, et al., "Experimental investigation on the hydrodynamic performance of a bioinspired manta-ray underwater vehicle in various forward propulsion modes," Ocean Eng., vol. 312, p. 119039, Nov. 2024.

[3] B. Liu, Y. Zhao, Y. Hao, S. Wang, K. Luo, J. Jin et al., "Vortex phase matching for energy efficiency enhancement in staggered formations of manta ray-inspired AUVs," 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO), Chengdu, China, 2025, pp. 2119-2124.

[4] M. Ligman, J. Lund, and M. Fürth, "A comprehensive review of hydrodynamic studies on fish schooling," Bioinspiration Biomimetics, vol. 19, no. 1, p. 11002, Jan. 2024.

[5] G. V. Lauder, C. F. White, D. Ramesh, Y. Pan, J. Chaumel, and Y. Zhang, "Six questions in fish locomotor biology and the technical approaches needed to address them," J. Exp. Biol., vol. 229, no. 7, p. jeb251375, Apr. 2026, doi: 10.1242/jeb.251375.

[6] S. Ito and N. Uchida, "Vortex phase matching of a self-propelled model of fish with autonomous fin motion," Phys. Fluids, vol. 35, no. 11, p. 111902, Nov. 2023

[7] X. Mao, Z. Xin, X. Mao, X. Li, J. Wang, and J. Deng, "A coupled framework for modeling fish schooling," Bioinspiration Biomimetics, pp. 0–22, Jan. 2026.

[8] L. Li, M. Nagy, J. M. Graving, J. Bak-Coleman, G. Xie, and I. D. Couzin, "Vortex phase matching as a strategy for schooling in robots and in fish," Nat. Commun., vol. 11, no. 1, Oct. 2020.

[9] Y. Zhang and G. V. Lauder, "Energy conservation by collective movement in schooling fish," eLife, vol. 12, p. RP90352, Feb. 2024.

[10] Y. Zhang, H. Ko, M. A. Calicchia, R. Ni, and G. V. Lauder, "Collective movement of schooling fish reduces the costs of locomotion in turbulent conditions," PLOS Biol., vol. 22, no. 6, p. e3002501, Jun. 2024.

[11] D. Zhang, Q.-G. Huang, G. Pan, L.-M. Yang, and W.-X. Huang, "Vortex dynamics and hydrodynamic performance enhancement mechanism in batoid fish oscillatory swimming," J. Fluid Mech., vol. 930, p. A28, Jan. 2022.

[12] T.-Y. Gao, P.-N. Sun, X.-T. Huang, J.-L. Zhao, Y. Xu, and S.-Y. Zhong, "Numerical simulation of manta ray swimming using a smoothed-particle hydrodynamics method and investigation of the vortical structures in the wake," Phys. Fluids, vol. 36, no. 9, p. 91915, Sep. 2024.

[13] H. Safari, M. Abbaspour, and M. Darbandi, "Numerical study to evaluate the important parameters affecting the hydrodynamic performance of manta ray's in flapping motion," Appl. Ocean Res., vol. 109, p. 102559, Apr. 2021.

[14] P. Gao, Q. Huang, and G. Pan, "The effect of phase difference on the group hydrodynamic performance of three typical formations," Phys. Fluids, vol. 37, no. 9, p. 91922, Sep. 2025.

[15] P. Gao, X. Tian, Q. Huang, G. Pan, Y. Chu, and J. Bai, "Group hydrodynamic analysis of two manta rays under spacing and phase-difference coupling," Phys. Fluids, vol. 36, no. 9, Sep. 2024.

[16] P. Gao, B. Song, Q. Huang, X. Tian, G. Pan, Y. Chu et al., "Hydrodynamic performance of two manta rays swimming in staggered groups with different vertical spacing," Ocean Eng., vol. 313, p. 119415, Dec. 2024.

[17] P. Gao, Q. Huang, G. Pan, Y. Chu, and J. Bai, "Formation effects on the group propulsion performance of manta rays," Phys. Fluids, vol. 37, no. 5, May 2025.

[18] Y. Cao, Y. Lu, Y. Cai, S. Bi, and G. Pan, "CPG-fuzzy-based control of a cownose-ray-like fish robot," Ind. Robot: int. j. robot. res. appl., vol. 46, no. 6, pp. 779–791, Nov. 2019.

Magnetically-controlled Bearingless Motor for Driving Blood Pumps in Artificial Hearts

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***Abstract*—This paper addresses the limitations of traditional artificial heart blood pump drive motors, such as limited torque output capacity and insufficient suspension support capabilities during high-speed operation, by proposing a magnetically modulated bearingless motor designed specifically for driving artificial heart blood pumps. This motor utilizes the magnetic field modulation effect of a segmented rotor—composed of magnetic rotor segments and a non-magnetic support structure—to improve the air gap magnetic field distribution characteristics. Based on this, the paper details the operating mechanism of the motor and verifies its performance using three-dimensional finite element Maxwell transient electromagnetic simulation. Simulation results indicate that this structure achieves minimal interference between the torque windings and the suspension windings during concurrent operation. Compared to existing same-polarity stator permanent magnet bearingless motors, the average suspension force is increased by approximately 45.3%, and the output torque is increased by approximately 10.5%, validating the motor's advantages in torque output and suspension support capability.**

Keywords—Bearingless motors, Egmented rotors, Permanent-magnet stators, Halbach arrays, Blood pump applications

I. Introduction

As a core component of ventricular assist devices, artificial heart blood pumps place high demands on the suspension and support capabilities of their drive systems, as well as their long-term operational reliability[1]. Currently, most artificial heart blood pumps utilize bearingless permanent magnet synchronous motors or bearingless reluctance motors as their drive motors [2]. However, traditional bearingless motors typically employ a rotor permanent magnet structure[3,4], which, under the high-speed and high-cleanliness operating conditions required for artificial hearts, faces challenges such as limited rotor heat dissipation[5], a high risk of permanent magnet demagnetization, and high replacement costs[6]. To address these issues, researchers both domestically and internationally have shifted their focus in recent years to stator-permanent-magnet bearingless motors[7]. By relocating the permanent magnets to the stator side, this type of motor significantly improves the heat dissipation conditions of traditional bearingless motors, reduces the risk of permanent magnet demagnetization, and enhances rotor replaceability[8]. Researchers such as Gruber[9,10], and Noh [11] have verified the feasibility of this type of motor in achieving decoupled control of suspension and drive. However, existing research has largely focused on proposing basic topologies and verifying underlying mechanisms; there remains room for further research in enhancing torque output capacity, improving levitation support capability, and increasing axial passive stiffness.

To address the issues of limited torque output capacity and insufficient suspension support capacity found in conventional bearingless motors, this paper proposes a magnetically modulated bearingless motor. This design incorporates a Halbach permanent magnet array on the stator side to enhance the air gap magnetic field distribution, and employs a segmented rotor on the rotor side, consisting of segmented magnetic rotor blocks and a non-magnetic support frame. By leveraging the significant magnetic field modulation effect of the segmented rotor in conjunction with the Halbach array on the stator side, this design effectively enhances both torque output and levitation support capabilities.

II. Structural Design of a Bearingless Motor with Magnetic Control

This paper investigates a magnetically controlled bearingless motor designed for driving an artificial heart blood pump. The objective is to integrate torque output with radial active suspension within a single electromagnetic structure, while providing a certain degree of axial passive stiffness through a permanent magnet bias circuit to meet the requirements for long-term stable operation, low wear, and minimal blood damage in artificial heart blood pumps. The overall structure of the motor designed in this paper is shown in Figure 2-1. It primarily consists of a stator, a segmented rotor, stator windings, and a Halbach permanent magnet array.

A. Stator and Winding Structure

The stator consists of stator teeth and a stator yoke, with 12 stator teeth evenly distributed around the circumference. To simultaneously achieve both drive and suspension functions, a dual-winding structure is arranged within the stator slots: the outer layer consists of four pole pairs of torque windings, which are primarily responsible for generating torque, while the inner layer consists of one pole pair of suspension windings, which primarily generate the radial force required to maintain the rotor's stable suspension through interaction with the magnetic flux density in the air gap. Through the synergistic action of the dual windings, the motor is capable of actively suspending and supporting the rotor while simultaneously outputting torque.

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The overall layout of the stator and windings is shown in Figure 2-1.

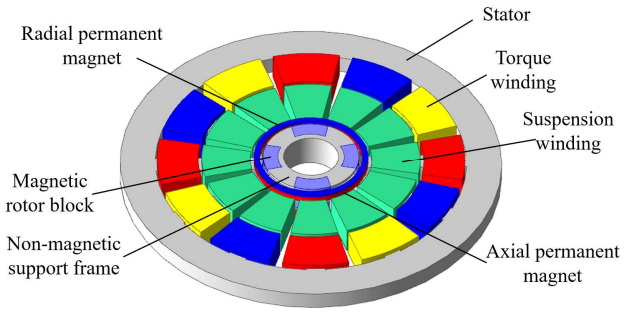

Fig. 2-1 Schematic diagram of a bearingless motor with magnetic control

B. Segmented Rotor Structure

The rotor features a segmented structure consisting of four magnetic rotor segments and a non-magnetic rotor support frame. Each magnetic rotor core is uniformly embedded within the non-magnetic rotor support frame, with one magnetic rotor segment secured between every two adjacent non-magnetic support teeth, forming a total of four evenly distributed rotor segments. The outer surface of the magnetic rotor blocks features a three-layer pole structure, comprising upper and lower layers of protrusions and a rotor pole tooth located at the axial center of the rotor block that is the same thickness as the stator teeth. All three layers of poles protrude radially outward from the magnetic rotor body and point toward the stator teeth. Their function is to establish a biased magnetic flux path and participate in the modulation of the air gap magnetic field. The specific structure of the segmented rotor is shown in Figure 2-2.

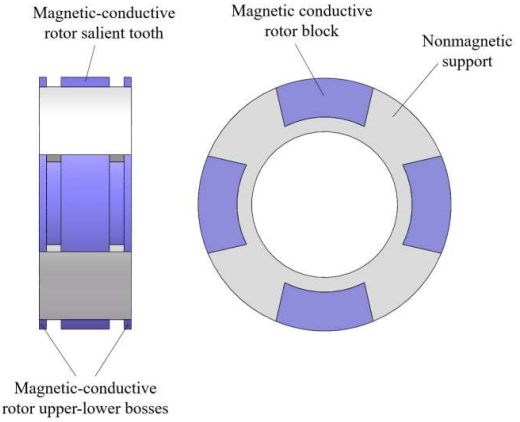

Fig. 2-2 Schematic diagram of a segmented rotor

C. Halbach Permanent Magnet Array Structure

To establish a permanent magnet bias circuit and increase axial passive stiffness, Halbach permanent magnet arrays are uniformly arranged circumferentially along the stator's inner diameter on the upper and lower end faces of the stator teeth. At each end, the two sets of permanent magnet arrays consist of permanent magnets with different magnetization directions, arranged uniformly in a fixed sequence. Specifically, the axial permanent magnets are magnetized axially outward and form a complete ring composed of multiple arcs that adhere to the stator end face, while the radial permanent magnets are magnetized radially inward and form a complete ring composed of multiple arcs that adhere to the outer edge of the axial ring. The design objective of this array is to strengthen the magnetic field on the active side of the radial magnetic field and weaken the field on the non-active side, thereby effectively improving permanent magnet utilization and enhancing magnetic field distribution characteristics. Furthermore, the thickness ratio between the axial and radial permanent magnets satisfies a specific ratio, namely, the thickness of the axially magnetized permanent magnets is approximately twice that of the radially magnetized permanent magnets, ensuring that the Halbach array achieves optimal magnetic field distribution characteristics[12,13]. The structure of the permanent magnet array is shown in Figure 2-3.

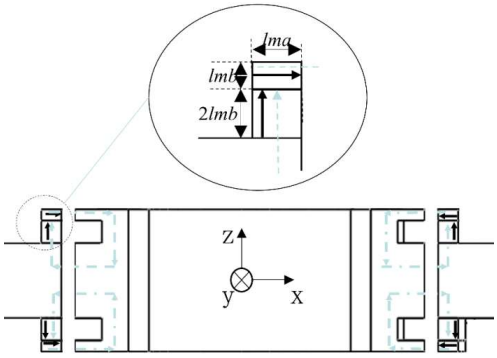

Fig. 2-3 Schematic diagram of the Halbach permanent magnet array structure

In this configuration, the biasing magnetic flux generated by the permanent magnets originates at the stator end, passes sequentially through the axial and radial sets of permanent magnets, and then traverses the air gap to enter the upper and lower flanges of the rotor. The magnetic flux lines are guided into the rotor core via the upper and lower flanges of the magnetic rotor blocks, and return to the stator end through the pole teeth of the magnetic rotor blocks across the air gap, thereby forming a complete closed loop for the permanent magnet biasing circuit. This bias magnetic circuit not only provides the fundamental magnetic field for active suspension but also enhances the motor's axial passive stability.

Taking into account manufacturing realities and tolerances, this paper proposes a larger radial air gap of 2 mm to accommodate certain tolerances. Meanwhile, the magnetic rotor segments can be fabricated from solid blocks of 20 steel, and the stator-side Halbach array can be directly assembled from two sets of standard arc-shaped permanent magnet blocks.

III. MECHANISM OF OPERATION

Since the drive and suspension of magnetically controlled bearingless motors are based on the air-gap magnetic field distribution, it is necessary to first clarify the formation characteristics and configuration of the air-gap magnetic field when analyzing their operating mechanisms. This motor employs a segmented rotor structure, where the radial magnetic resistance differs between the magnetic rotor segments and the non-magnetic support frame. Since the magnetic resistance of the segmented magnetic rotor segments is lower, the permanent magnet bias flux generated by the permanent magnet array is not uniformly distributed along the circumference of the air gap.

Instead, it concentrates primarily through the segmented magnetic rotor segments, causing the magnetic flux density in the air gap to exhibit periodic fluctuations along the circumference[14]. The distribution characteristics of the air gap magnetic flux density around the circumference of the segmented rotor are shown in Figure 3-1. It can be observed that the air gap magnetic flux density is significantly enhanced in the regions corresponding to the magnetic rotor segments. This indicates that the segmented rotor structure exerts a distinct modulating effect on the air gap magnetic field, causing the motor's air gap magnetic field to exhibit a non-uniform distribution that differs from that of traditional continuous rotor structures.

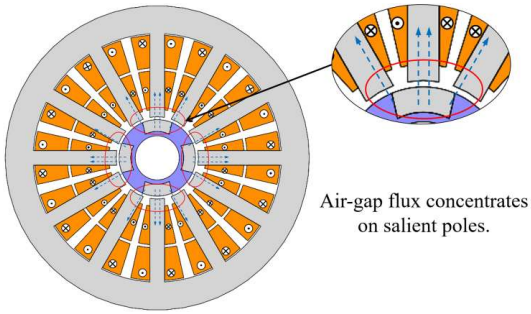

Fig. 3-1 Map of the magnetic flux density distribution in the air gap along the circumference of a segmented rotor

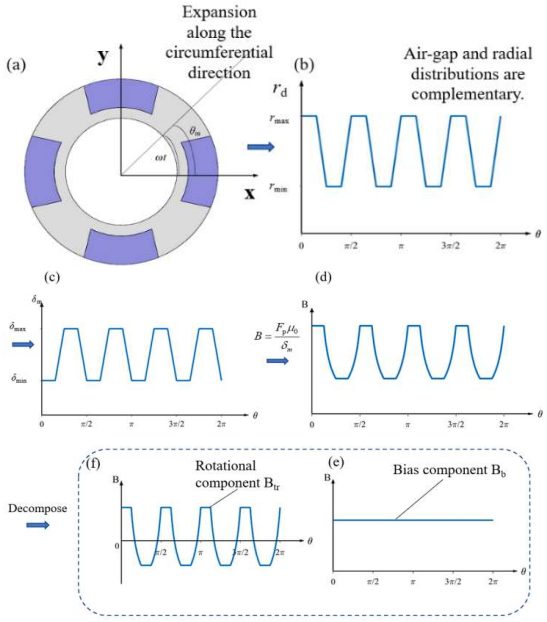

Fig. 3-2 Schematic diagram of air-gap flux density decomposition

To clarify the distinct roles of the air gap magnetic field in driving and suspension, it is necessary to decompose the air gap magnetic flux density distribution under the segmented rotor conditions shown in Figure 3-1. The objective is to isolate the magnetic field components from the air gap magnetic field that respectively govern the suspension and driving effects. Based on the segmented rotor reference coordinate system and its circumferential unfolding model, the magnetic field established by the permanent magnets in the air gap can be decomposed into two parts: the bias magnetic field component B_b and the rotating magnetic field component B_{tr}. To carry out this decomposition, we first establish a segmented rotor coordinate system and perform a circumferential unfolding to characterize the periodic variation of the air gap length along the circumference. Based on this, by combining the relationship between the air gap permeability distribution and the magnetic field, we obtain the analytical distribution characteristics of the air gap magnetic flux density; further, we can extract the bias magnetic field component and the rotating magnetic field component from this, as shown in Figure 3-2.

The above analysis shows that the segmented rotor structure results in a non-uniform distribution of the air-gap magnetic field, which can be further decomposed into a stationary magnetic field component and a rotating magnetic field component. Under conditions where the core magnetic resistance, leakage flux effects, and local magnetic saturation are neglected, the air-gap magnetic flux density B can be expressed as a function of the magnetic field and the equivalent air-gap length.

$$B = \frac{F_p \mu_0}{\delta_m} \qquad (3-1)$$

In Equation (3-1), F_p and δ_m represent the magnetic field of the permanent magnet and the equivalent air gap length, respectively; μ_0 is the magnetic permeability of free space.

When the suspension winding is energized, the suspension magnetic flux it generates interacts with the bias magnetic field component B_b to produce a controllable radial suspension force. The rotating magnetic field generated by the torque winding interacts with the rotating component B_{tr}, thereby driving the rotor to rotate in sync and producing steady-state torque.

IV. Finite Element Analysis and Performance Validation

For magnetically tuned bearingless motor designs, independent control of torque and levitation force is essential for stable operation. However, since the torque windings and levitation windings are housed within the same set of stator slots and share the main magnetic circuit modulated by the segmented rotor, they are prone to mutual interference during actual operation. Therefore, finite element simulation is required to analyze the torque and levitation force outputs and verify the feasibility of this design.

In this paper, a three-dimensional transient electromagnetic field simulation model of the motor was developed using Ansys Maxwell software. Through transient electromagnetic field solver and analysis, the interference between the motor's torque windings and suspension windings during simultaneous operation was verified.

The 3D electromagnetic field model established includes key structures such as the stator, segmented rotor, windings, and Halbach permanent magnet arrays; the relevant geometric parameters are set according to Table 4-1.

The simulation was solved using the transient magnetic field analysis method. To improve the accuracy of the magnetic field calculations, local mesh refinement was performed in the air gap, the regions adjacent to the permanent magnets, and areas where the magnetic field varies significantly. The total number of mesh elements was 128,527.

Regarding material parameters, the stator core uses DW270-35 silicon steel sheets, the magnetic rotor blocks use 20 steel, the non-magnetic rotor support frame uses aluminum, the winding material uses copper, and the permanent magnet material uses N40SH. The demagnetization curve is set based on its actual residual magnetic flux density and coercive force. Magnetic-insulating boundary conditions are applied to the outer boundaries of the calculation domain. The motion domain is configured as a band structure with the "Rotation" state, and a "Force" is set in the "Parameters" for the motion domain to simulate the levitation force. The time step is set to 0.0000375 s, and the total simulation duration is 0.00375 s. Under these conditions, characteristic parameters such as motor torque and levitation force are extracted to verify the validity of the constructed model and the results of the winding analysis.

TABLE 4-1. KEY PARAMETERS OF THE SIMULATION MODEL

Parameter Name	Value	Parameter Name	Value
Stator inner diameter D_{si}/mm	56	Stator outer diameter D_{so}/mm	207
Armature length l_{ef}/mm	19	Stator yoke width l_{sy}/mm	19
Stator tooth width l_t/mm	10	Air gap length δ_m/mm	2
Inner diameter of the rotor d_{ri}/mm	30	Rotor outer diameter d_{ro}/mm	52
Rotor axial length l_r/mm	19	Outer diameter of the rotor body d_{rbo}/mm	46
Axial thickness of the rotor flange l_{rs}/mm	1.5	Axial thickness of the rotor's salient pole teeth l_{rt}/mm	10
Floating winding current /A	1	Torque-winding current /A	1.5
Inner pole arc angle of the rotor /deg	54.07	Radial permanent magnet thickness l_{rmb}/mm	1.5
Outer pole arc angle of the rotor /deg	45.24	Permanent magnet width l_{ma}/mm	3
Thickness of axial permanent magnets l_{zmb}/mm	3	Number of turns per phase in the torque winding /N_r	220
Number of turns in the floating winding /N_s	270		

To perform finite element simulation analysis, the 3D model of the motor was first meshed, as shown in Figure 4-1. Key areas such as the air gap and permanent magnets were locally refined.

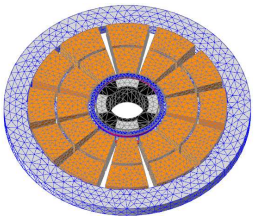

Fig. 4-1 Mesh generation of motor simulation model

As shown in the simulation results in Figure 4-2, the maximum magnetic flux density in the motor air gap reaches 1938.31 mT. The magnetic field is concentrated in the regions where the stator teeth and the magnetic rotor blocks are in relative alignment, exhibiting an overall non-uniform periodic distribution. This is consistent with the magnetic field modulation effect of the segmented rotor described in the operating mechanism in Chapter III, thereby verifying the modulation effect of the segmented rotor on the air gap magnetic field. At the same time, the bias magnetic circuit formed by the Halbach permanent magnet array on the stator side is stable, with a continuous magnetic field distribution and no significant distortion. This provides a solid magnetic foundation for the cooperative operation of the torque windings and the suspension windings, and ensures the stable generation of torque output and radial suspension force in subsequent stages.

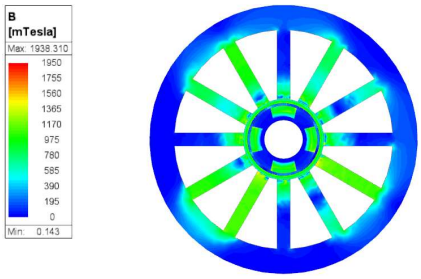

Fig. 4-2 Simulated cloud map of magnetic flux density

Given that the torque windings and the suspension windings are located within the same set of stator slots and share the main magnetic circuit modulated by the segmented rotor, it is essential to ensure that they do not interfere with one another during operation. Therefore, further analysis of the motor's torque output and suspension force output is required.

A. Verification of Torque and Lift Performance

In actual operation, the two windings may interfere with each other when operating simultaneously. Therefore, further verification is required to compare the motor output waveforms under different excitation conditions.

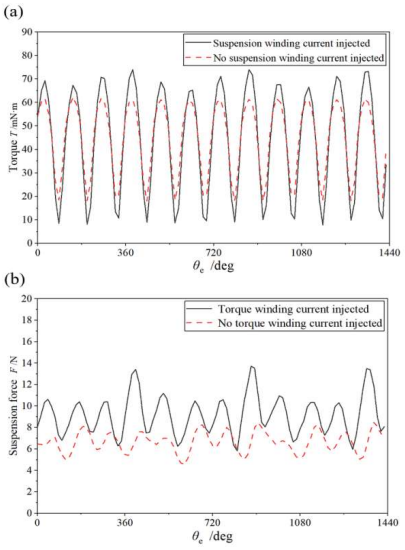

Fig. 4-3. Simulation and verification of motor performance decoupling. (a) Comparison of torque waveforms with and without suspension winding current; (b) Comparison of suspension force waveforms with and without torque winding current

Figure 4-3 shows the waveforms of the motor's output torque and levitation force under different excitation conditions. As shown in Figure 4-3(a), when a levitation current is applied to the torque winding, it causes a slight distortion in the air-gap magnetic field, resulting in a slight increase in the amplitude of torque fluctuations; however, the average torque does not deviate significantly. Similarly, Figure 4-3(b) shows that when a torque current is applied while the suspension winding is in its rated operating state, although the transient peaks and troughs of the suspension force exhibit some fluctuation, the output average and the fundamental waveform remain stable.

The results above indicate that there is minimal interference between the torque windings and the suspension of the magnetically controlled bearingless motor when they operate in concert; the simulation results validate the validity of the established model.

To further illustrate the torque output and levitation capabilities of the magnetically controlled bearingless motor designed in this paper, it was compared with an existing same-polarity stator permanent magnet bearingless motor [14]. The detailed comparison results are shown in Table 4-2.

TABLE 4-2. COMPREHENSIVE PERFORMANCE COMPARISON PARAMETERS OF THE MOTORS

Comparison Model	Main text	Xiong Feng, author of Reference[14]
Stator outer diameter	207mm	207mm
Outer diameter of the rotor body	46mm	44mm
Inner diameter of the rotor	30mm	36mm
Rotor axial length	19mm	19mm
Suspension force	9.21N	6.34N
Output torque	44.28mN·m	40.06 mN·m

As shown in the comparative data in Table 4-2, under similar electromagnetic dimensions and operating conditions, the magnetically tuned bearingless motor designed in this paper achieves an average levitation force approximately 45.3% higher and an average torque approximately 10.5% higher than the motor described in Reference[14]. These results demonstrate that this structural design effectively enhances both torque output and levitation support capability.

V. CONCLUSION

To address the limited torque output capacity and insufficient suspension support capabilities of the drive motor in artificial heart blood pumps during high-speed operation, a magnetically controlled bearingless motor with a segmented rotor structure has been proposed.

This paper provides a detailed description of the structure of this magnetically controlled bearingless motor. Based on this structure, the operating mechanism is described, explaining the operating principle of this magnetically controlled bearingless motor.

Maxwell finite element simulation results confirm that there is minimal interference between the torque windings and the suspension windings of this motor when they operate in tandem. Under comparable dimensional conditions, the motor designed in this study achieves an average suspension force approximately 45.3% higher and an average torque approximately 10.5% higher than the reference model. This structure effectively enhances both average torque and average suspension force, offering a new solution for artificial heart blood pumps.

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REFERENCES

[1] WANG M W, Liu Ze, Miao Yu, et al. Development and control of Maglevmotor prototype in artificial heart[J].Transducer and Microsystem Technologies,2023,42(01):105-108.(In Chinese)

[2] Kant K, Weinreb B S, Hegy M, et al. A novel bearingless interior permanent magnet slice motor driven blood pump[J].Mechatronics,2023,90(4):102946.

[3] ZHAO H N,ZHANG A,ZHANG X Y,et al. Topology optimization and performance simulation analysis of magnetic bearings for artificial heart pump[J].Bearing，2025(8):86-92.(In Chinese)

[4] Li G, Ojeda J, Hoang E, et al. Thermal–electromagnetic analysis for driving cycles of embedded flux-switching permanent-magnet motors[J].IEEE ransactions on vehicular technology,2011,61(1):140-151.

[5] Lin C Y, Liu X J, Xu C Y, et al. Hematological, Biochemical, and End-organ effects of the CH-VAD in Ovine Model[C].World Congress on Medical Physics and Biomedical Engineering.2015:305-308.(In Chinese)

[6] Ni Y K.Research on Structure Design and Control System of Bearingless Permanent Magnet slice Motor[D].Jiangsu University,2016.(In Chinese)

[7] CHEN X D.Analysis Of Internal Flow Field Characteristics And Rotor Stability Control Strategy Of Bearingless Direct-Drive Centrifugal Pump[D].Jiangsu University,2023.(In Chinese)

[8] LU J,SUN C,ZHU H Q.et al.Research status and development overview of key technologies on bearingless permanent magnet slice motor[J].Bearing.2025(8):1-7.(In Chinese)

[9] Gruber W, Rothbock M, Schob R T. Design of a Novel Homopolar Bearingless Slice Motor With Reluctance Rotor[J].IEEE Transactions on Industry Applications,2015,51(2):1456-1464.

[10] Gruber, Wolfgang, Bauer, et al. Comparison of homopolar and heteropolar bearingless reluctance slice motor prototypes[J].SAGE Publications,2017,231(5):339-347.

[11] Noh M. Homopolar bearingless slice motors with magnet-free rotors for extracorporeal life support[D].Massachusetts Institute of Technology,2018.

[12] Liu J, Shang M, Gong C. Analytical modeling and analysis of Halbach array permanent magnet synchronous motor[J]. World Electric Vehicle Journal, 2025, 16(8): 413.

[13] Wang X Y,Yin C X,Li T Y.Magnetic field analysis for axial flux permanent magnet motor with novel Halbach array[J].Electric Machines and Control,2024,28(02):21-31.(In Chinese)

[14] XION F.Research on the Homopolar Stator Permanent Magnet Bearingless Slice Motor[D].Nanjing University of Aeronautics and Astronautics.2020.(In Chinese)

Command-Conditioned AMP Reference Sampling for Multi-Skill Humanoid Locomotion

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***Abstract*—Multi-skill humanoid locomotion requires a controller to follow velocity commands while maintaining coherent whole-body motion style. Adversarial Motion Priors (AMP) provide demonstration-based style supervision, but random reference sampling can become inconsistent when the reference buffer contains walking, running, lateral, and turning motions. This paper presents a command-conditioned AMP reference sampling strategy for multi-skill humanoid locomotion. The method partitions retargeted reference motions into skill-related subsets and samples reference transitions according to normalized command similarity and direction consistency. By changing the reference-transition distribution rather than the policy architecture, discriminator architecture, or reward formulation, the proposed sampler reduces command-style mismatch in AMP supervision. Qualitative Sim2Sim and Sim2Real demonstrations on a 23-DOF humanoid robot show improved consistency between commanded skills and reference motion styles across walking, running, lateral motion, and turning. A sampling-level diagnostic further shows that the command-conditioned sampler assigns substantially higher probability to command-matched reference subsets than random sampling.**

Keywords—humanoid locomotion; reinforcement learning; adversarial motion priors; motion imitation; command-conditioned sampling; Sim2Real

I. INTRODUCTION

Humanoid locomotion controllers are increasingly expected to handle multiple commanded skills with a single policy. In the setting considered here, the robot must respond to commands for forward walking, faster locomotion, lateral movement, and turning while maintaining balance and coordinated whole-body motion. This requirement is challenging because command tracking and motion style are coupled through the pelvis, legs, torso, and arms.

Task rewards can specify target motion objectives, but they do not fully characterize the movement style expected from a humanoid robot. A policy may satisfy a desired velocity while producing stiff or poorly coordinated motion. Reference-motion learning addresses this limitation by shaping the distribution of generated transitions using demonstrations.

Adversarial Motion Priors (AMP) provide a compact mechanism for incorporating demonstration-based style supervision. Instead of tracking a reference trajectory frame by frame, AMP trains a discriminator to distinguish reference transitions from policy-generated transitions and converts the discriminator output into an auxiliary style reward. The task reward can then guide command following while the adversarial prior encourages reference-like motion.

However, a multi-skill reference buffer introduces a sampling problem. If walking, running, lateral motion, and turning are stored together, a randomly selected reference transition may not match the active command. For example, a lateral command may be paired with forward-walking reference data, and a turning command may be paired with straight-line locomotion. This command-style mismatch can make the AMP style reward less specific to the intended behavior.

This paper addresses the mismatch by conditioning AMP reference sampling on the current command. The proposed sampler changes the reference-transition distribution rather than the policy architecture, discriminator architecture, or task reward. Other legged-robot studies are used only as background and do not form the technical focus of this work.

Related work provides the algorithmic and data-driven basis for this study. PPO is used as the policy-optimization framework [1], prior legged and bipedal studies motivate learned feedback control and Sim2Real-aware gait design [2], [3], and DeepMimic, adversarial imitation learning, and AMP motivate demonstration-based style supervision [4], [5], [6], [7]. Recent humanoid locomotion studies further highlight the importance of whole-body coordination [19].

The data-processing pipeline is also central to the proposed method. GPU-accelerated simulation frameworks support efficient policy training [8], [9], while Sim2Real studies caution that limited hardware demonstrations should not be interpreted as complete transfer validation [10]. The reference motions used in this work are derived from AMASS, CMU, ACCAD, and self-recorded videos, with GVHMR and GMR used for motion recovery and humanoid retargeting [11], [12], [13], [14], [15], [16], [17], [18].

The contributions of this paper are summarized as follows.

1．First, the paper formulates the command-style mismat ch problem caused by random AMP reference sampling in mult i-skill humanoid locomotion.

2．Second, it proposes a command-conditioned reference sampling rule based on normalized velocity similarity and direc tion consistency.

3．Third, it builds a multi-skill reference-motion processi ng pipeline for humanoid AMP training, combining public moti on-capture data, self-recorded videos, GVHMR motion recover y, GMR retargeting, and skill slicing.

4．Fourth, it demonstrates the resulting command-style c onsistency in Sim2Sim and Sim2Real examples and adds a sam pling-level quantitative diagnostic to support the qualitative co mparison.

II. METHODOLOGY

A. Problem Formulation

AMP training in this paper involves a policy, a critic, and a discriminator. The discriminator observes consecutive state pairs and learns to classify whether each pair originates from the reference motion data or from the policy rollout. Its inputs are motion-related quantities, including root and joint information, so that the learned signal emphasizes body style rather than external task commands.

Let D_{ref} denote reference transitions and D_{π} denote transitions generated by the current policy. Using the least-squares AMP discriminator objective, the discriminator is trained as

$$min_D E_{(s,s')\sim D_{ref}}[(D(s,s')-1)^2] + E_{(s,s')\sim D_{\pi}}[(D(s,s'))^2] \quad (1)$$

The discriminator score is then mapped to a motion-style reward:

$$r_{style}(s_t, s_{t+1}) = max[0, 1 - 0.25(D((s_t, s_{t+1})) - 1)^2] \quad (2)$$

The total reward used for policy optimization combines the task term and this style term:

$$r_t = r_{task}(s_t, a_t) + \lambda_{style} r_{style}(s_t, s_{t+1}) \quad (3)$$

With this reward, AMP acts as a training-time source of style supervision. At deployment, the policy does not need the discriminator or the reference motion buffer, because action selection depends only on the policy observation.

Standard AMP implementations often draw reference transitions without considering the command that produced the current rollout transition. This default can be reasonable for a single-skill dataset, where most reference samples describe the same behavior family. It becomes less appropriate when one buffer stores demonstrations with different movement directions, speeds, and sequence lengths.

The resulting failure mode is a mismatch between command and style sample. The policy is conditioned on a command c, such as desired forward velocity, lateral velocity, or yaw rate, but the discriminator may be trained against reference transitions from a different behavior. In that case, the style reward still measures resemblance to the buffer, yet the buffer sample may not express the behavior requested by c.

Omnidirectional locomotion makes this problem easy to see. The same humanoid controller may be asked to walk, run, move right, or turn right. If reference samples are selected randomly, side-stepping commands can be supervised by forward-walking data, and turning commands can be supervised by straight-line clips. The available qualitative results indicate that conditioning the sampler on commands makes the demonstrated skills visually more distinct.

B. Command-Conditioned AMP Sampling

The proposed strategy changes the reference sampler while leaving the rest of AMP intact. Motion clips are first divided into subsets associated with primary skills. This division is performed after data collection or retargeting. The reported setup further notes that GMR was extended with a slicing function so that clips containing multiple behaviors, such as walking and turning, could be separated before training.

For command-conditioned sampling, the command vector is defined as $c = [v_x, v_y, \omega_z]^T$, where v_x and v_y are desired linear velocities and ω_z is the desired yaw rate. Each motion subset i is represented by $v_i = [v_{\{x,i\}}, v_{\{y,i\}}, \omega_{\{z,i\}}]^T$, computed from the average linear and angular velocities of the corresponding retargeted motion subset. Small average linear velocities below 0.2 m/s are set to zero to reduce the influence of minor oscillations. To avoid directly mixing linear velocity and angular velocity units, the command-subset mismatch is computed as a normalized weighted distance:

$$d_i = \sqrt{\sum_{k\in\{v_x,v_y,\omega_z\}} w_k \left(\frac{c_k - v_{i,k}}{\sigma_k}\right)^2} \quad (4)$$

Here, w_k controls the relative importance of each command dimension and σ_k is the normalization scale for that dimension. The mismatch is transformed into a positive similarity value through an exponential kernel:

$$s_i = e^{-\alpha d_i} \quad (5)$$

The coefficient alpha > 0 controls the sharpness of the sampling distribution. A larger alpha concentrates probability on the closest subset, while a smaller value preserves more diversity among related subsets.

A direction-consistency term further down-weights subsets whose dominant motion direction conflicts with the command. In practice, consistency is determined by the sign agreement of the dominant nonzero velocity component, or equivalently by a positive directional similarity in the command space. The penalty coefficient satisfies 0 < beta < 1:

$$\begin{cases} s_i' = s_i, if\ direction - consistent \\ s_i' = \beta s_i,\ otherwise \end{cases} \quad (6)$$

The final sampling distribution is obtained by normalizing the adjusted scores:

$$p_i = \frac{s_i'}{\sum_j s_j} \quad (7)$$

During AMP training, this distribution determines which subset provides the next reference transition. Subsets close to the current command are sampled more frequently, whereas directionally inconsistent subsets remain possible but receive lower probability. This keeps limited reference diversity while reducing the influence of behaviorally irrelevant samples.

Relative to random sampling, the key change is the connection between command space and motion-data space. The method does not add a new evaluation metric, baseline, reward term, or robot architecture. Its contribution is a different distribution over existing AMP reference transitions.

C. Training Framework

The workflow begins with selecting human motions from public motion-capture sources and self-recorded videos. When video clips are used, GVHMR recovers world-grounded human motion. GMR then retargets the recovered or selected motions to the humanoid. The resulting sequences are sliced into skill-specific subsets, and each subset receives an average velocity descriptor. During training, the sampler chooses AMP reference transitions from these subsets according to the current command.

This procedure remains compatible with the usual AMP loop. RL rollouts provide policy transitions, and the command-conditioned sampler provides reference transitions. The discriminator learns to separate the two distributions, the discriminator reward is added to the task reward, and the optimizer updates the policy. Once training is complete, neither the reference sampler nor the discriminator is required for policy execution.

The command vector c provides a compact description of the intended locomotion behavior. The subset descriptor v_i summarizes whether the reference subset primarily represents forward walking, faster forward locomotion, lateral movement, or turning. The normalized distance in Eq. (4) therefore compares the active command and the reference subset in a common scaled command space.

Eq. (4) computes the normalized weighted command-subset mismatch. Eq. (5) maps this mismatch to a positive similarity score. Eq. (6) applies the direction-consistency penalty, and Eq. (7) normalizes the adjusted scores into sampling probabilities.

This view clarifies why the method should be interpreted as sampling control rather than reward design. The task reward and the AMP reward equations are unchanged. What changes is the source of the reference transitions used to train the discriminator. The discriminator is therefore exposed more often to examples that correspond to the command being practiced.

Random sampling implicitly treats every reference transition as equally useful for every rollout transition. That assumption is defensible when demonstrations describe one behavior family, but it weakens when the buffer contains several skills. In addition, longer subsets may dominate simply because they contain more transitions, even if their motion direction is irrelevant to the current command.

The command-conditioned rule replaces command-agnostic reference selection with a probability distribution derived from the relation between the current command and the motion subset descriptors. The resulting AMP supervision is more closely aligned with the intended behavior while remaining data-driven.

The method still depends on clean motion organization. A subset that mixes several behaviors can yield an average descriptor that is not representative of its transitions. The data pipeline therefore uses GMR processing and slicing to form clearer skill segments before command-conditioned sampling is applied.

III. Experiments

A. Simulation and Robot Platform

The experimental platform is a ROBO_Origin humanoid robot with 23 degrees of freedom. Training is performed with an NVIDIA RTX 4090 GPU. In the reported implementation, multi-skill locomotion training on ROBO_Origin required approximately 3 h using the available RTX 4090 setup. The policy is evaluated in MuJoCo for Sim2Sim validation and on the physical robot for Sim2Real demonstration.

TABLE I. Implementation details for the humanoid locomotion experiments.

Item	Reported setting
Robot platform	RoboOrigin humanoid robot
Degrees of freedom	23
Simulator	MuJoCo
Computing hardware	NVIDIA RTX 4090
Motion sources	AMASS, CMU Motion Capture Dataset, ACCAD Motion Capture Dataset, and self-recorded videos
Motion processing tools	GVHMR for video-based motion recovery; GMR for humanoid motion retargeting
Evaluated skills	Walk, run, move right, and turn right
Robot-level repeated-trial metrics	Not available; sampling-level diagnostic is reported in Tables IV-V.

B. Motion Data and Retargeting

The reference-motion set combines public motion-capture data with self-recorded video demonstrations. CMU and ACCAD provide public whole-body motion examples, while self-recorded directional motions are processed by GVHMR and retargeted to the RoboOrigin humanoid robot with GMR. These sources provide reference motions for AMP-based motion imitation and command-conditioned sampling.

Table II summarizes the motion sources in a category-level form rather than as raw file names. This organization is appropriate because the sampling strategy depends on skill categories and representative command directions. Random AMP sampling and command-conditioned sampling use the same AMP training framework, policy structure, discriminator role, task reward, and reference-motion pool; the controlled difference is the distribution used to select reference transitions.

TABLE II. Motion data sources for reference motion prior construction.

Motion category	Source	Representative motions	Usage in this work
Walking and transition	CMU Motion	run-to-walk and walking transition	Reference motions for forward

motions	Capture Dataset; ACCAD Motion Capture Dataset	clips	locomotion style learning and AMP motion-prior construction
Turning motions	ACCAD Motion Capture Dataset	Walk-turn-left/right clips with 45 deg and 90 deg turns; walk-turn-around; run-turn-left	Reference motions for directional skill learning and command-conditioned sampling
Standing/posture motions	ACCAD Motion Capture Dataset	A1-Stand-stageii	Static posture reference within the retargeted humanoid motion set
Direction-changing motions	ACCAD Motion Capture Dataset	Run-change-direction stageii	Reference examples for non-straight locomotion transitions
Self-recorded directional motions	Self-recorded videos processed by GVHMR and retargeted with GMR	Move $_{back}$, move$_{l}$, move$_{r}$, turn$_{l}$, turn$_{r}$	Supplementary command-related reference motions for AMP sampling and humanoid retargeting demonstrations

GVHMR is used for monocular video inputs to recover world-grounded human motion, which gives the self-recorded directional clips a consistent pose and trajectory representation. GMR then transfers the recovered human motion to the humanoid morphology through retargeting. Together, these steps make the self-recorded directional motions usable as supplementary reference examples for the AMP motion prior.

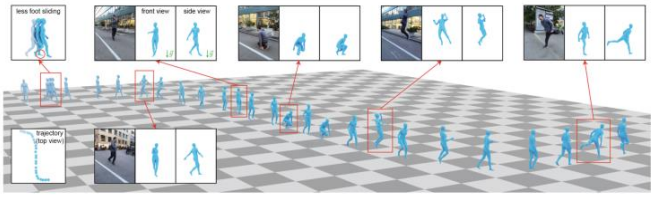

Fig. 1. Overview of GVHMR for world-grounded human motion recovery.

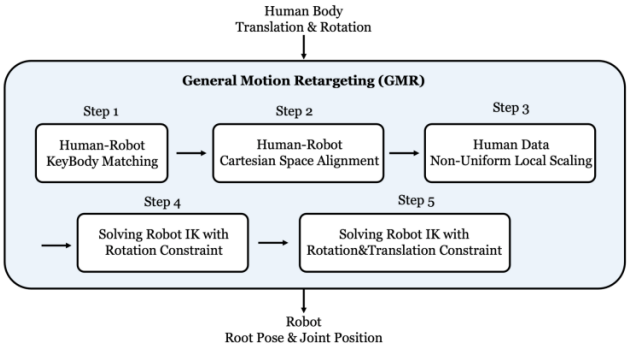

Fig. 2. Workflow of GMR for retargeting human motion to the humanoid robot.

The retargeted motions are not treated as quantitative experimental results. They serve as reference-motion inputs and as qualitative evidence that the motion-processing pipeline supports command-related humanoid skills. Figs. 1-4 illustrate the recovery and retargeting stages used to obtain these motions.

Fig. 3. Self-recorded move-backward motion processed using GVHMR.

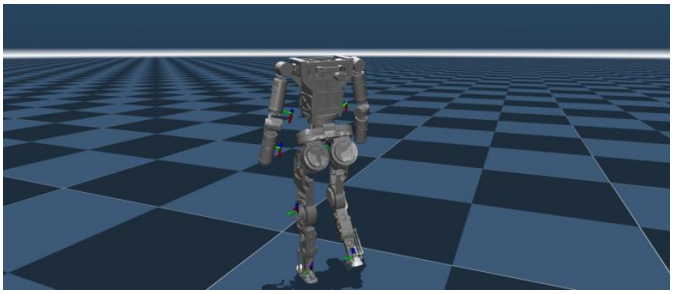

Fig. 4. Retargeted move-backward motion on the robot using GMR.

IV. Analysis and Discussion

A. Multi-Skill Locomotion Results

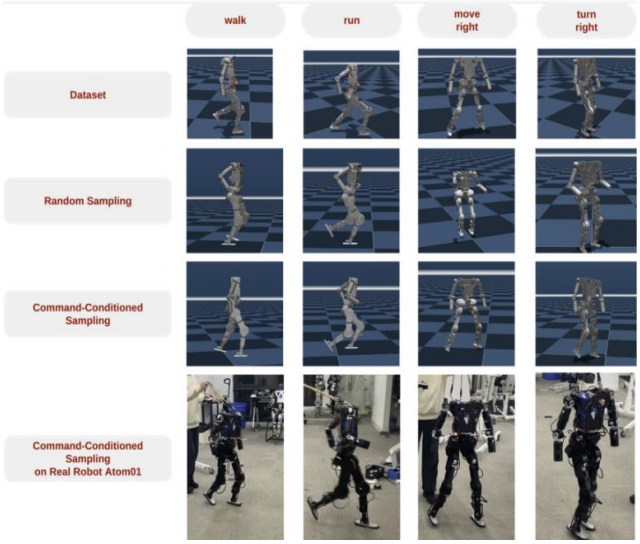

Fig. 5. Qualitative comparison between random AMP sampling and command-conditioned AMP sampling under walk, run, move-right, and turn-right commands. The comparison focuses on whether the sampled reference transitions are behaviorally consistent with the active command. Compared with random sampling, the proposed sampler selects reference subsets that better match the commanded forward, lateral, or turning motion, thereby reducing command-style mismatch in AMP supervision.

Fig. 5 presents the main qualitative comparison between random AMP sampling and command-conditioned AMP sampling under different locomotion commands. The figure is intended to show whether reference transitions used for AMP supervision are behaviorally consistent with the active command. Under random sampling, a lateral or turning

command can still be paired with forward-locomotion references. Under command-conditioned sampling, reference subsets with closer forward, lateral, or yaw-rate descriptors are sampled more frequently. The comparison therefore visualizes command-style consistency rather than a final performance ranking.

The qualitative observations are consistent with the sampler mechanism. Forward commands shift the reference distribution toward walking or running subsets, while lateral and turning commands shift it toward subsets with compatible lateral velocity or yaw-rate descriptors. This command-aware distribution is the main difference from random sampling: the AMP style signal is biased toward reference motions that express the skill currently being practiced.

Table III summarizes the problem addressed by the sampler, the expected command-conditioned effect, and the available evidence. To strengthen the table beyond visual description, Tables IV and V provide a sampling-level diagnostic computed from the proposed sampling rule. These diagnostics evaluate the reference-selection mechanism itself and should be interpreted separately from repeated-trial robot performance metrics.

TABLE III. QUALITATIVE AND SAMPLING-LEVEL SUMMARY OF MULTI-SKILL LOCOMOTION DEMONSTRATIONS

Skill	Random AMP sampling issue	Command-conditioned sampling effect	Available evidence
Walk	May sample lateral or turning references under a forward command.	Increases probability of forward-walking references.	Fig. 5 visual comparison; Table V matched sampling probability.
Run	May mix walking and running styles without considering commanded speed.	Assigns higher probability to faster forward-motion references.	Fig. 5 visual comparison; Table V run-related probability.
Move right	May pair a lateral command with forward-locomotion references.	Shifts sampling toward lateral-motion references.	Fig. 5 sequence; Table V lateral-subset probability.
Turn right	May pair a yaw-rate command with straight-line references.	Shifts sampling toward turning references.	Fig. 5 sequence; Table V turning-subset probability.

TABLE IV. SAMPLING PARAMETERS USED FOR THE COMMAND-CONDITIONED DIAGNOSTIC.

Parameter	Meaning	Value used in diagnostic
w_{vx}, w_{vy}, w_{ω}	Weights for forward, lateral, and yaw-rate dimensions	1, 1, 1
σ_{vx}, σ_{vy}, σ_{ω}	Normalization scales for command dimensions	unit-normalized descriptors
alpha	Sampling sharpness coefficient in Eq. (5)	3.0
beta	Direction-inconsistency penalty in Eq. (6)	0.25

TABLE V. SAMPLING-LEVEL DIAGNOSTIC COMPARING RANDOM AND COMMAND-CONDITIONED REFERENCE SELECTION.

Command	Random matched rate	Proposed matched rate	Average distance reduction
Walk	25.0%	80.6%	84.6%
Run	25.0%	81.3%	88.1%
Move right	25.0%	98.4%	98.0%
Turn right	25.0%	98.4%	98.0%

The sampling-level diagnostic supports the central claim that the proposed sampler makes command-matched reference subsets more likely than random sampling. Large-scale tracking-error, success-rate, and multi-seed statistics remain future evaluation targets and are discussed as limitations.

The diagnostic uses four unit-normalized subset descriptors for walk, run, move right, and turn right. It verifies the sampling mechanism: random sampling assigns 25.0% probability to the command-matched subset, whereas Eq. (4)-Eq. (7) concentrate probability on subsets with smaller command-space distance and compatible direction.

B. Sim2Sim and Sim2Real Observations

The evaluation includes MuJoCo Sim2Sim and physical ROBO_Origin demonstrations. These demonstrations show that the same command-conditioned sampling pipeline can be inspected in simulation and on hardware. Together with the sampling-level diagnostic, they support the mechanism-level conclusion that command-conditioned reference selection reduces behaviorally irrelevant AMP supervision compared with random sampling.

For walking and running, the sampler increases the probability of selecting forward-motion references under forward commands. For rightward motion and rightward turning, the sampler shifts probability toward lateral or yaw-rate-compatible subsets. This explains why the optimized examples in Fig. 5 show clearer stylistic separation among the four commanded skills.

For walking, Fig. 5 shows a visually recognizable forward walking behavior under the selected command, and the diagnostic indicates that the walking subset receives 80.6% matched probability rather than the 25.0% produced by uniform random sampling.

For running, the faster forward subset receives 81.3% matched probability in the diagnostic, which is consistent with using command-space similarity to distinguish faster forward locomotion from ordinary walking.

For rightward lateral movement, the available sequence suggests clearer lateral style and fewer visibly unnecessary arm motions in the command-conditioned result. The diagnostic assigns 98.4% probability to the lateral subset under the rightward command, illustrating how irrelevant forward references are suppressed at the sampling level.

For rightward turning, the yaw-rate-compatible subset also receives 98.4% probability in the diagnostic. This supports the interpretation that command-conditioned sampling can make AMP supervision more relevant to turning behavior than command-agnostic random sampling.

C. Limitations and Discussion

A limitation of the current study is that repeated-trial tracking errors, stable-duration measurements, fall-rate comparisons, and multi-seed training statistics are not yet available. Future work will conduct repeated simulation and hardware trials to quantify improvements in tracking accuracy, robustness, and Sim2Real transfer performance.

The method itself remains lightweight because it does not require an additional learned selector, a new discriminator architecture, or a modified policy network. Its practical cost is mainly in organizing, slicing, and assigning command-space descriptors to reference motions, which makes it compatible with existing AMP training pipelines.

V. Conclusion

This paper presents a command-conditioned AMP reference sampling method for multi-skill humanoid locomotion control. The method addresses command-style mismatch in multi-skill AMP training by partitioning motion demonstrations into skill-related subsets, constructing command-space descriptors, and sampling AMP reference transitions according to normalized command similarity and direction consistency.

The results suggest that conditioning AMP reference sampling on command-space similarity is a lightweight way to improve the relevance of style supervision without modifying the policy or discriminator architecture. Sim2Sim and Sim2Real demonstrations show improved consistency between commanded skills and motion styles, and the sampling-level diagnostic verifies that the proposed sampler assigns substantially higher probability to command-matched reference subsets than random sampling. Future work will extend the evaluation with velocity-tracking errors, success rates, multi-seed training, and repeated hardware trials.

References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv preprint arXiv:1707.06347, 2017.

[2] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," Science Robotics, vol. 4, no. 26, Art. no. eaau5872, 2019.

[3] J. Siekmann, Y. Godse, A. Fern, and J. Hurst, "Sim-to-real learning of all common bipedal gaits via periodic reward composition," in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2021, pp. 7309–7315.

[4] X. B. Peng, P. Abbeel, S. Levine, and M. van de Panne, "DeepMimic: Example-guided deep reinforcement learning of physics-based character skills," ACM Trans. Graph., vol. 37, no. 4, Art. no. 143, 2018.

[5] J. Ho and S. Ermon, "Generative adversarial imitation learning," arXiv preprint arXiv:1606.03476, 2016.

[6] X. B. Peng, Z. Ma, P. Abbeel, S. Levine, and A. Kanazawa, "AMP: Adversarial motion priors for stylized physics-based character control," ACM Trans. Graph., vol. 40, no. 4, pp. 1–20, 2021.

[7] A. Escontrela, X. B. Peng, W. Yu, T. Zhang, A. Iscen, K. Goldberg, and P. Abbeel, "Adversarial motion priors make good substitutes for complex reward functions," in Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), 2022, pp. 25–32.

[8] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac Gym: High performance GPU-based physics simulation for robot learning," arXiv preprint arXiv:2108.10470, 2021.

[9] M. Mittal et al., "Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning," arXiv preprint arXiv:2511.04831, 2025.

[10] J. Truong, M. Rudolph, N. H. Yokoyama, S. Chernova, D. Batra, and A. Rai, "Rethinking Sim2Real: Lower fidelity simulation leads to higher Sim2Real transfer in navigation," in Proc. 6th Conf. Robot Learn. (CoRL), Proc. Mach. Learn. Res., vol. 205, 2023, pp. 859–870.

[11] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black, "AMASS: Archive of motion capture as surface shapes," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2019, pp. 5442–5451.

[12] Carnegie Mellon University, "CMU motion capture dataset." [Online]. Available: http://mocap.cs.cmu.edu/. Accessed: Apr. 29, 2026.

[13] The Ohio State University Advanced Computing Center for the Arts and Design, "ACCAD motion capture dataset." [Online]. Available: https://accad.osu.edu/research/motion-lab/mocap-system-and-data. Accessed: Apr. 29, 2026.

[14] Z. Shen, H. Pi, Y. Xia, Z. Cen, S. Peng, Z. Hu, H. Bao, R. Hu, and X. Zhou, "World-grounded human motion recovery via gravity-view coordinates," in Proc. SIGGRAPH Asia Conf., 2024.

[15] Y. Ze, J. P. Araújo, J. Wu, and C. K. Liu, "GMR: General motion retargeting," GitHub repository, 2025. [Online]. Available: https://github.com/YanjieZe/GMR. Accessed: Apr. 29, 2026.

[16] J. P. Araújo, Y. Ze, P. Xu, J. Wu, and C. K. Liu, "Retargeting matters: General motion retargeting for humanoid motion tracking," arXiv preprint arXiv:2510.02252, 2025.

[17] Y. Ze, Z. Chen, J. P. Araújo, Z.-a. Cao, X. B. Peng, J. Wu, and C. K. Liu, "TWIST: Teleoperated whole-body imitation system," in Proc. 9th Conf. Robot Learn. (CoRL), Proc. Mach. Learn. Res., vol. 305, 2025, pp. 2143–2154.

[18] Y. Ze, S. Zhao, W. Wang, A. Kanazawa, R. Duan, P. Abbeel, G. Shi, J. Wu, and C. K. Liu, "TWIST2: Scalable, portable, and holistic humanoid data collection system," arXiv preprint arXiv:2511.02832, 2025.

[19] S. Zhu, Z. Zhuang, M. Zhao, K.-Y. Lee, and H. Zhao, "Hiking in the wild: A scalable perceptive parkour framework for humanoids," arXiv preprint arXiv:2601.07718, 2026.